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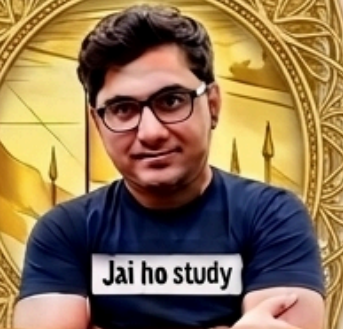


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# अंगद सामान्य अध्ययन

प्रतियोगी  
परीक्षाओं हेतु

सफलता  
की सही दिशा



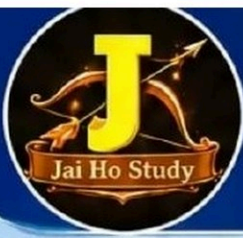
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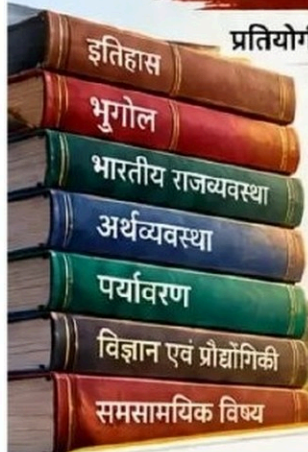


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# अंगद

## सामान्य अध्ययन

प्रतियोगी परीक्षाओं के लिए समग्र एवं अद्यतन अध्ययन सामग्री



**Utkarsh Dwivedi सर**  
Founder & Author

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परीक्षा केंद्रित  
सामग्री



सरल एवं  
सटीक भाषा



अद्यतन  
तथ्यों का समावेश



चार्ट, तालिका  
एवं चित्रों सहित



सफलता की  
दिशा में एक मजबूत कदम

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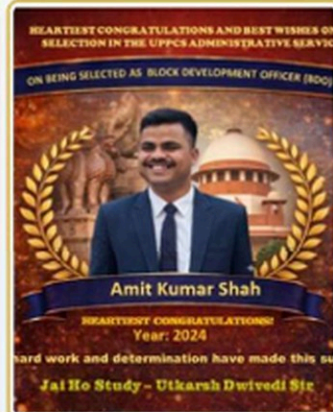
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हजारों घंटों की रिसर्च, सटीक नोट्स और परीक्षा के पैटर्न पर पैनी नजर — यही जय हो स्टडी को अलग बनाती है। हमारे सफल छात्र मेहनत, सही रणनीति और भरोसेमंद मार्गदर्शन का प्रमाण हैं।



हम वादों में नहीं, परिणामों में विश्वास रखते हैं।  
अगली बारी आपकी है — बस सही दिशा में पहला कदम बढ़ाएं।

#### हमारे संचालित बैच और कार्यक्रम

- यूपीपीसीएस प्रारंभिक परीक्षा - प्रतिष्ठा बैच 2026
- यूपीपीसीएस मुख्य परीक्षा - समाधान बैच 2026
- एएफसी आरओ / एआरओ परीक्षा तैयारी - 2026
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- स्वास्थ्य शिक्षा अधिकारी परीक्षा तैयारी
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- एकदिवसीय परीक्षाओं हेतु संपूर्ण तैयारी बैच

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Edition  
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01

History

UPPCS PRELIMINARY EXAMINATION (PRELIMS): HISTORY

1 Periods of History

| Period                | Main Characteristic                                       | Principal Example / Era                               |
|-----------------------|-----------------------------------------------------------|-------------------------------------------------------|
| Prehistoric Period    | No written evidence is available.                         | Stone Age                                             |
| Proto-historic Period | Evidence of a script exists, but it remains undeciphered. | Indus Valley Civilisation                             |
| Historical Period     | Written evidence is available and has been deciphered.    | Begins from the 6th century BCE (Mahajanapada period) |

2 Stone Age: Tools, Characteristics and Major Sites

| Period              | Tools and Main Characteristics                                                                                | Major Sites (India/Pakistan)                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Lower Palaeolithic  | Heavy, crude stone tools; <b>handaxe</b> , cleaver                                                            | Pallavaram and Attirampakkam (Tamil Nadu), Hunsgi-Isampur (Karnataka), Soan Valley (Pakistan), Narmada Valley (MP) |
| Middle Palaeolithic | <b>Flake-based tools</b> ; scrapers and other small tools                                                     | Nevasa (Maharashtra), Bhimbetka (MP), <b>Belan Valley (UP)</b> , Didwana (Rajasthan)                               |
| Upper Palaeolithic  | <b>Blades and blade-based tools</b> ; use of bone implements                                                  | Bhimbetka (MP), Patne (Maharashtra), Kurnool Caves (Andhra Pradesh)                                                |
| Mesolithic          | <b>Microliths</b> (tiny stone tools); main occupation: hunting and food-gathering                             | Bagor (Rajasthan), Langhnaj (Gujarat), <b>Sarai Nahar Rai, Mahadaha, Damdama (UP)</b> , Adamgarh (MP)              |
| Neolithic           | Ground and polished stone tools; development of <b>agriculture, animal husbandry and permanent settlement</b> | Mehrgarh, Burzahom, Gufkral, <b>Koldihwa, Mahagara, Lahuradewa (UP)</b> , Chirand, Sanganakallu-Piklihal           |



### 3 Major Neolithic and Mesolithic Sites (UPPCS Special)

| Site                      | State/Region           | Evidence Found and Exam Facts                                                                                                                            |
|---------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lahuradewa                | Uttar Pradesh          | <b>Earliest evidence of rice (paddy) and agriculture</b> (as per the latest findings).                                                                   |
| Koldihwa                  | Uttar Pradesh          | On the bank of the Belan river (south of Allahabad/Prayagraj); <b>early evidence of paddy cultivation</b> .                                              |
| Mahadaha                  | Uttar Pradesh          | Circular huts, cultivation of paddy and barley, animal husbandry; also important for <b>Mesolithic burials</b> .                                         |
| Sarai Nahar Rai, Damdama  | Uttar Pradesh          | Important Mesolithic sites of Uttar Pradesh.                                                                                                             |
| Chopani Mando             | Uttar Pradesh          | Located in the Belan Valley; evidence of microliths and <b>handmade pottery</b> (a Mesolithic–Neolithic transition site).                                |
| Burzahom                  | Kashmir                | Famous for <b>pit dwellings</b> ; evidence of <b>dogs buried with their masters</b> in graves.                                                           |
| Gufkral                   | Kashmir                | Bone tools, grey pottery and grinding stones (silbatta).                                                                                                 |
| Mehrgarh                  | Baluchistan (Pakistan) | <b>Earliest evidence of wheat</b> , cultivation of two varieties of barley; the earliest site of <b>settled life</b> , agriculture and animal husbandry. |
| Chirand                   | Bihar                  | Especially famous for <b>Neolithic bone implements/tools</b> .                                                                                           |
| Sanganakallu and Piklihal | Karnataka              | Evidence of <b>ash mounds</b> , ash heaps and habitation sites (Neolithic sites of South India).                                                         |
| Sarutaru and Daojali      | Assam                  | Primitive axes, painted pottery (Sarutaru) and evidence of agriculture and animal husbandry (Daojali).                                                   |

### 4 Indus Valley Civilisation (IVC): Geographical Extent and Coastal Cities

| Boundary/Direction | Principal Site                             | River Bank    | Geographical Position                  |
|--------------------|--------------------------------------------|---------------|----------------------------------------|
| North              | <b>Manda</b> (Jammu)                       | Chenab river  | Northernmost point of the civilisation |
| South              | <b>Daimabad</b> (Ahmednagar, Maharashtra)  | Pravara river | Southernmost point of the civilisation |
| East               | <b>Alamgirpur</b> (Meerut, Uttar Pradesh)  | Hindon river  | Easternmost point of the civilisation  |
| West               | <b>Sutkagendor</b> (Baluchistan, Pakistan) | Dasht river   | Westernmost point of the civilisation  |

**Geographical Extent Facts:** The Indus Valley Civilisation was **triangular** in shape and covered a total area of about **1.3 million sq km (13 lakh sq km)**. Its extent included Punjab, Sindh, Baluchistan, Rajasthan, Gujarat and western Uttar Pradesh.

### > Major Ports / Coastal Towns of the Indus Civilisation

- **Lothal (Gujarat):** World-famous trading centre known for its **dockyard**.
- **Sutkagendor (Baluchistan):** Principal trading centre of the western frontier of the Indus Civilisation.
- **Nageshwar (Gujarat):** Especially famous for its **shell (conch) industry**.
- **Balakot (Baluchistan):** Coastal Harappan site; centre of **marine resources and the shell industry**.

## 5 Major Harappan Sites, Excavations and Important Finds

| Site (River Bank / Location)                                              | Excavator (Year)                                               | Main Evidence and Remains Found (Exam Points)                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Harappa</b><br>(left bank of the Ravi, Montgomery, Punjab–Pakistan)    | <b>Daya Ram Sahni</b><br>(1921)                                | <ul style="list-style-type: none"> <li>• Great Granary</li> <li>• <b>Cemetery R-37</b></li> <li>• Broken torso in a dancing posture</li> <li>• Rectangular seal showing <b>a plant growing out of a woman's womb</b></li> <li>• Bronze human figure and seals with the one-horned animal (unicorn)</li> </ul> |
| <b>Mohenjo-daro</b><br>(right bank of the Indus, Larkana, Sindh–Pakistan) | <b>Rakhal Das Banerji</b> (1922)                               | <ul style="list-style-type: none"> <li>• <b>Great Bath</b></li> <li>• Great Granary</li> <li>• <b>Bronze statue of the Dancing Girl</b></li> <li>• The famous <b>Pashupati Seal</b> (a three-faced yogi seated in padmasana)</li> </ul>                                                                       |
| <b>Lothal</b><br>(Bhogava river, Ahmedabad, Gujarat)                      | <b>S.R. Rao</b> (1955)                                         | <ul style="list-style-type: none"> <li>• <b>Dockyard</b></li> <li>• <b>Three paired (joint) burials</b></li> <li>• Bead-making factories</li> <li>• <b>Ivory scale</b> / measuring rod and fire altars</li> </ul>                                                                                             |
| <b>Dholavira</b><br>(Rann of Kutch, Gujarat)                              | <b>J.P. Joshi</b> (1967–68)<br>and <b>R.S. Bisht</b> (1990–91) | <ul style="list-style-type: none"> <li>• <b>Advanced water-management system</b> (rock-cut tanks / huge reservoirs)</li> <li>• <b>Earliest inscription (signboard) with 10 signs</b></li> </ul>                                                                                                               |
| <b>Kalibangan</b><br>(Hanumangarh, Rajasthan)                             | <b>B.B. Lal</b> and <b>B.K. Thapar</b> (1961)                  | <ul style="list-style-type: none"> <li>• <b>Evidence of a ploughed field</b></li> <li>• Buildings of mud bricks</li> <li>• <b>Earliest evidence of an earthquake</b></li> <li>• Wells and fire altars in almost every house</li> </ul>                                                                        |



### > Other Harappan Sites and Important Objects:

- **Banawali (Haryana): Toy model of a plough (terracotta plough)**, beads of lapis lazuli and carnelian, seals and weights.
- **Daimabad (Maharashtra):** Bronze figures of **a man driving a chariot**, a bull and a rhinoceros; a seal bearing the Indus script and pottery.
- **Nausharo (near Mehrgarh):** Evidence of **vermilion (sindoor) in the hair-parting** of women.
- **Balakot (Pakistan):** Fragments of **bangles made of shell** (shell industry).
- **Manda (Jammu):** Bronze **spiral-headed pin** (northernmost site).
- **Hulas (UP):** A **stamp/sealing** bearing the Indus script.
- **Sutkagendor (Pakistan):** Remains of clay bangles.

## 6 Vedic Literature: Vedas, Priests and Upavedas

| Veda                                                        | Priest (Ritvik)                                            | Upaveda       | Nature of Compilation / Main Subject                                                |
|-------------------------------------------------------------|------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------|
| Rigveda                                                     | Hota (Hotri)                                               | Ayurveda      | <b>Collection of richas (hymns/mantras).</b>                                        |
| Yajurveda<br>(two parts: <i>Shukla</i> and <i>Krishna</i> ) | Adhvaryu                                                   | Dhanurveda    | Hymns and rules related to <b>sacrificial rites</b> (in both prose and verse).      |
| Samaveda                                                    | Udgata                                                     | Gandharvaveda | <b>Collection of songs (chants);</b> regarded as the <b>father of Indian music.</b> |
| Atharvaveda                                                 | Brahma<br>(the senior priest who supervises the sacrifice) | Shilpaveda    | Contains <b>charms and spells, magic, remedies against disease</b> and medicines.   |

## 1 Evolution of Vedic Literature

| Stage (Sequence) | Main Characteristic and Definition                                                                                           |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1. Samhita       | Comprises the four Vedas: <b>Rigveda, Yajurveda, Samaveda and Atharvaveda.</b>                                               |
| 2. Brahmana      | Prose texts written as <b>commentaries (exegesis)</b> on the Samhitas.                                                       |
| 3. Aranyaka      | Appendices to the Brahmana texts, composed in the <b>forests (aranya).</b>                                                   |
| 4. Upanishad     | The special and concluding part of the Aranyakas, regarded as ' <b>Vedanta</b> ' and the <b>source of Indian philosophy.</b> |

## 2 Vedas, Shakhas, Brahmanas, Aranyakas and Upanishads (Complete Compendium)

| Veda      | Principal Shakhas (Recensions)                                                                                                                                                                        | Related Brahmanas                                        | Related Aranyakas                      | Related Upanishads                                                                  | Special Historical Facts (Exam Facts)                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rigveda   | <i>Not mentioned</i>                                                                                                                                                                                  | Aitareya, Kaushitaki (Shankhayana)                       | Aitareya, Kaushitaki                   | Aitareya, Kaushitaki                                                                | <ul style="list-style-type: none"> <li>• <b>Aitareya Brahmana:</b> first description of the Rajasuya sacrifice and of the four yugas (<b>Satyuga, Treta, Dvapara, Kaliyuga</b>).</li> </ul>                                                                                                                                                                                                                                               |
| Yajurveda | <ul style="list-style-type: none"> <li>• <b>Krishna Yajurveda:</b> Kathaka, Kapishtala, Maitrayani, Taittiriya</li> <li>• <b>Shukla Yajurveda:</b> Kanva (Vajasaneyi Samhita), Madhyandina</li> </ul> | Taittiriya, Shatapatha                                   | Taittiriya, Maitrayani, Brihadaranyaka | Taittiriya, Kathopanishad, Maitrayani, Shvetashvatara, Ishopanishad, Brihadaranyaka | <ul style="list-style-type: none"> <li>• <b>Shatapatha Brahmana:</b> Yajnavalkya–Maitreyi dialogue, the story of the Great Flood, the story of Videgha Madhava, the Pururavas–Urvashi legend.</li> <li>• <b>Brihadaranyaka Upanishad:</b> the saying "<b>Tamaso ma jyotirgamaya</b>" and the <b>Yajnavalkya–Gargi</b> dialogue.</li> <li>• <b>Kathopanishad:</b> the famous <b>Yama–Nachiketa</b> dialogue.</li> </ul>                    |
| Samaveda  | Kauthuma, Ranayaniya, Jaiminiya                                                                                                                                                                       | Panchavimsha (Tandya), Chhandogya, Shadvimsha, Jaiminiya | Chhandogya, Jaiminiya                  | Chhandogya, Kena                                                                    | <ul style="list-style-type: none"> <li>• <b>Chhandogya Upanishad:</b> the oldest Upanishad; first mention of 'Krishna'; the story of Satyakama Jabala, the dialogue of Uddalaka Aruni and Shvetaketu.</li> <li>• <b>Three Ashramas:</b> Brahmacharya, Grihastha and Vanaprastha are mentioned in the Chhandogya.</li> <li>• <b>Taittiriya Upanishad:</b> gives women the right to <b>wear the sacred thread (yajnopavita)</b>.</li> </ul> |



| Veda        | Principal Shakhas (Recensions) | Related Brahmanas | Related Aranyakas  | Related Upanishads                 | Special Historical Facts (Exam Facts)                                                                                                                                                                          |
|-------------|--------------------------------|-------------------|--------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atharvaveda | Shaunakiya, Pippalada          | Gopatha           | <i>No Aranyaka</i> | Mundakopanishad, Mandukya, Prashna | <ul style="list-style-type: none"> <li>• <b>Mundakopanishad:</b> India's national motto '<b>Satyameva Jayate</b>' is taken from here.</li> <li>• <b>Mandukya Upanishad:</b> the shortest Upanishad.</li> </ul> |

### 3 Mandalas of the Rigveda and their Composers/Features

| Mandala | Composer (Rishi)                       | Main Feature / Evidence                                                                                                                          |
|---------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| First   | Madhuchhanda, Medhatithi, Gotama, etc. | Early mandala                                                                                                                                    |
| Second  | Gritsamada                             | -                                                                                                                                                |
| Third   | <b>Vishvamitra</b>                     | Contains the famous <b>Gayatri Mantra</b> (dedicated to the deity Savitri/Sun).                                                                  |
| Fourth  | Vamadeva                               | Mentions mantras related to <b>agriculture</b> .                                                                                                 |
| Fifth   | Atri                                   | -                                                                                                                                                |
| Sixth   | Bharadvaja                             | -                                                                                                                                                |
| Seventh | <b>Vasishtha</b>                       | Describes the famous <b>Dasharajna War (Battle of Ten Kings)</b> .                                                                               |
| Eighth  | Kanva and Angiras                      | -                                                                                                                                                |
| Ninth   | Various rishis                         | Called the ' <b>Soma Mandala</b> ' (dedicated to the deity Soma).                                                                                |
| Tenth   | Vimada, Indra, Shachi and others       | Contains the <b>Purusha Sukta</b> , which first gives the theory of the origin of the <b>four varnas</b> (Brahmana, Kshatriya, Vaishya, Shudra). |

### 4 Principal Deities of the Rigvedic Period

| Deity             | Importance / Feature in the Rigvedic Period                                     |
|-------------------|---------------------------------------------------------------------------------|
| Indra (Purandara) | The most prominent deity of the Rigveda; <b>leader in war and god of rain</b> . |
| Agni              | <b>Intermediary between gods and men.</b>                                       |
|                   |                                                                                 |

| Deity   | Importance / Feature in the Rigvedic Period                                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Varuna  | Creator of the sky, earth and sun; god of the ocean; controller of the world; sustainer of the gods; <b>upholder of cosmic order (Ritasya Gopa)</b> , symbol of truth; agent of the change of seasons and of day and night. |
| Dyaus   | <b>God of the sky</b> (the oldest deity).                                                                                                                                                                                   |
| Soma    | <b>God of plants (vegetation).</b>                                                                                                                                                                                          |
| Ashvins | Deities who <b>remove calamities</b> (gods of medicine).                                                                                                                                                                    |
| Pushan  | <b>God of cattle</b> (became the god of the Shudras in the Later Vedic period).                                                                                                                                             |
| Usha    | <b>Goddess of progress and rising (dawn).</b>                                                                                                                                                                               |

5 Major Vedic Sacrifices

| Name of Sacrifice | Main Purpose and Description                                                                                       |
|-------------------|--------------------------------------------------------------------------------------------------------------------|
| Ashvamedha        | <b>The most important sacrifice</b> ; performed by the king to extend the boundaries of his kingdom (sovereignty). |
| Rajasuya          | Sacred sacrifice related to the king's <b>coronation</b> .                                                         |
| Vajapeya          | Sacrifice related to the king's <b>display of valour</b> and a chariot race (entertainment).                       |
| Agnishtoma        | Sacrifice in which <b>animals were offered</b> to Agni and Soma juice was drunk to please the gods.                |

6 Regional Systems of Rule Described in the Aitareya Brahmana

| Direction/Region      | Type of Rule | Title of the Ruler |
|-----------------------|--------------|--------------------|
| East                  | Samrajya     | <b>Samrat</b>      |
| West                  | Svarajya     | <b>Svarat</b>      |
| North                 | Vairajya     | <b>Virat</b>       |
| South                 | Bhojya       | <b>Bhoja</b>       |
| Madhya Desha (Centre) | Rajya        | <b>Raja</b>        |



## 7 Ancient and Modern Names of Vedic Rivers (Match the Following Special)

| Ancient Name | Modern Name                        | Ancient Name | Modern Name                            |
|--------------|------------------------------------|--------------|----------------------------------------|
| Vitasta      | Jhelum                             | Kubha        | Kabul                                  |
| Parushni     | Ravi                               | Krumu        | Kurram                                 |
| Asikni       | Chenab                             | Suvastu      | Swat                                   |
| Vipasha      | Beas                               | Gomati       | Gomal                                  |
| Shutudri     | Sutlej                             | Sarayu       | Ghaghara                               |
| Sadanira     | Gandak                             | Yamuna       | Yamuna                                 |
| Drishadvati  | Associated with the Ghaggar region | Marudvridha  | Associated with the Ravi/Chenab region |

## 8 Vedic Administration and Principal Officials (Ratnins/Officials)

| Administrative Post (Officer) | Department / Main Function                                                                        |
|-------------------------------|---------------------------------------------------------------------------------------------------|
| Kulapati                      | Head of the family (kula); officer of family administration.                                      |
| Gramani                       | Head of the village; officer concerned with village administration and military leadership.       |
| Spasha                        | <b>Spy / secret agent</b> ; messenger who gathered information for the king.                      |
| Vrajapati                     | Chief officer in charge of <b>pastures and livestock</b> .                                        |
| Bhagadugha                    | <b>Revenue collector (tax collector)</b> ; collected taxes for the king.                          |
| Jivagribha                    | <b>Police officer</b> ; caught criminals and maintained law and order.                            |
| Akshavapa                     | <b>Accountant</b> ; the king's partner in the game of dice and officer of income and expenditure. |
| Adhipati                      | <b>Chief Justice</b> ; highest officer of the judicial system.                                    |
| Sangrihitri                   | <b>Treasurer</b> ; officer in charge of managing the royal treasury.                              |
| Takshan                       | Carpenter; skilled artisan who made objects from wood.                                            |
| Palagala                      | Messenger; carried royal messages from one place to another.                                      |
| Govikartana                   | <b>Keeper of forests and of the hunt (game)</b> .                                                 |

## 9 Important Facts Related to the Puranas

- **Composers:** Sage Lomaharsha and his son Ugrashrava.
- **Total number:** 18 Puranas.
- **Period of redaction:** The Puranas were recast into their present form during the **Gupta period**.
- **Social rule:** Women and Shudras were not permitted to read the Vedas, but they could listen to the Puranas.

### > Principal Puranas and the Dynasties/Details Associated with Them:

| Name of Purana | Historical Importance / Associated Dynasty / Details                                                                                                                     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Matsya Purana  | <ul style="list-style-type: none"> <li>• <b>The oldest and most authentic</b> of the Puranas.</li> <li>• Historical account of the <b>Satavahana dynasty</b>.</li> </ul> |
| Vishnu Purana  | <ul style="list-style-type: none"> <li>• Source of information on the history of the <b>Maurya dynasty</b>.</li> </ul>                                                   |
| Vayu Purana    | <ul style="list-style-type: none"> <li>• Mentions the history and genealogy of the <b>Gupta dynasty</b>.</li> </ul>                                                      |
| Agni Purana    | <ul style="list-style-type: none"> <li>• Important Purana that gives <b>details of agriculture</b> along with polity (monarchy).</li> </ul>                              |

#### Note

**Note:** Of the 18 Puranas, the genealogies of kings are found in only **five Puranas** (Matsya, Vayu, Vishnu, Brahmanda and Bhagavata Purana). The best systematic account of Indian historical traditions is found in the Puranas.

## 1 Principal Mottos / Sayings and their Authentic Sources

| Motto / Saying          | Authentic Source                            | Special Exam Note (Exam Warning/Fact)                                                                                                           |
|-------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Satyameva Jayate        | Mundakopanishad                             | • India's <b>national motto</b> .                                                                                                               |
| Gayatri Mantra          | Rigveda (Third Mandala)                     | <ul style="list-style-type: none"> <li>• Composer: <b>Sage Vishvamitra</b>.</li> <li>• Dedicated to: the deity <b>Savitri (Sun)</b>.</li> </ul> |
| Tamaso ma jyotirgamaya  | Brihadaranyaka Upanishad (1.3.28) / Rigveda | • A principal great saying (mahavakya) of the Upanishad.                                                                                        |
| Aham Brahmasmi          | Brihadaranyaka Upanishad                    | • A principal Vedic mahavakya.                                                                                                                  |
| Sarve bhavantu sukhinah | Brihadaranyaka                              | • A popular peace prayer; do not write it as the <b>original mantra</b> of the Brihadaranyaka Upanishad.                                        |
| Yama–Nachiketa dialogue | Kathopanishad                               | • Nachiketa receives from Yama the <b>knowledge of the soul and of death</b> .                                                                  |

| Motto / Saying                             | Authentic Source                                    | Special Exam Note (Exam Warning/Fact)                                                       |
|--------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| Vasudhaiva Kutumbakam                      | Maha Upanishad (6.71-75)                            | • The whole earth is one family.                                                            |
| Janani janmabhumishcha svargadapi gariyasi | Ramayana (Yuddha Kanda)                             | • Mother and motherland are greater even than heaven.                                       |
| Karmanyevadhikaraste...                    | Shrimad Bhagavad Gita                               | • Message of doing one's duty without worrying about the fruits.                            |
| Vidya dadati vinayam                       | Hitopadesha                                         | • <b>Note:</b> do not write it as a line from the Gita; it is from the <b>Hitopadesha</b> . |
| Atithi Devo Bhava                          | Taittiriya Upanishad (1.11.2) / Taittiriya Brahmana | • Instruction to treat the guest as equal to a god.                                         |
| Tat Tvam Asi                               | Chhandogya Upanishad                                | • The famous dialogue of <b>Sage Uddalaka and Shvetaketu</b> .                              |
| Satyam jnanam anantam Brahma               | Taittiriya Upanishad                                | • <b>Note:</b> do not write 'Satyam Shivam Sundaram' as a line from the Kathopanishad.      |

## 2 Indian Philosophies and their Founders (Shad-darshana and Philosophical Schools)

| Philosophy / Philosophical School       | Founder                     |
|-----------------------------------------|-----------------------------|
| Sankhya                                 | Sage Kapila                 |
| Yoga                                    | Maharishi Patanjali         |
| Nyaya                                   | Sage Gautama                |
| Vaisheshika                             | Maharishi Kanada (or Uluka) |
| Purva Mimamsa                           | Maharishi Jaimini           |
| Uttara Mimamsa                          | Maharishi Badarayana        |
| Charvaka                                | Maharishi Charvaka          |
| Advaita (Non-dualism)                   | Adi Shankaracharya          |
| Dvaita (Dualism)                        | Madhvacharya                |
| Dvaitadvaita (Dualistic non-dualism)    | Nimbarkacharya              |
| Vishishtadvaita (Qualified non-dualism) | Ramanujacharya              |
| Shuddhadvaita (Pure non-dualism)        | Vallabhacharya              |



### 3 The 16 Mahajanapadas: Capitals and Present Geographical Location

| Mahajanapada | Capital                                                                                                                               | Present Geographical Region                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Kashi        | Varanasi                                                                                                                              | Eastern Uttar Pradesh (Varanasi region)              |
| Koshala      | <ul style="list-style-type: none"> <li>• <b>North Koshala:</b> Shravasti</li> <li>• <b>South Koshala:</b> Saketa / Ayodhya</li> </ul> | Awadh and eastern Uttar Pradesh                      |
| Malla        | Kushinara (Kushinagar) and Pava                                                                                                       | Eastern Uttar Pradesh (Kushinagar–Gorakhpur region)  |
| Vatsa        | Kaushambi                                                                                                                             | Region around Prayagraj                              |
| Kuru         | Indraprastha ( <i>other city: Hastinapur</i> )                                                                                        | Delhi, Haryana and western Uttar Pradesh             |
| Panchala     | <ul style="list-style-type: none"> <li>• <b>North Panchala:</b> Ahichchhatra</li> <li>• <b>South Panchala:</b> Kampilya</li> </ul>    | Western Uttar Pradesh (Bareilly–Farrukhabad region)  |
| Shurasena    | Mathura                                                                                                                               | Braj region (western Uttar Pradesh)                  |
| Chedi        | Shuktimati / Sotthivati                                                                                                               | Bundelkhand (Uttar Pradesh and Madhya Pradesh)       |
| Anga         | Champa                                                                                                                                | Eastern Bihar (Bhagalpur–Munger region)              |
| Magadha      | Girivraja / Rajagriha ( <i>later Pataliputra</i> )                                                                                    | Southern Bihar (Patna–Gaya–Nalanda region)           |
| Vajji        | Vaishali                                                                                                                              | Northern Bihar (Vaishali–Muzaffarpur region)         |
| Matsya       | Viratnagar                                                                                                                            | Eastern Rajasthan (Jaipur–Alwar region)              |
| Avanti       | <ul style="list-style-type: none"> <li>• <b>North Avanti:</b> Ujjayini</li> <li>• <b>South Avanti:</b> Mahishmati</li> </ul>          | Malwa and western Madhya Pradesh                     |
| Gandhara     | Taxila (or Pushkalavati)                                                                                                              | North-western Pakistan (Peshawar–Rawalpindi)         |
| Kamboja      | Rajapura (or Hataka)                                                                                                                  | North-western Pakistan and north-eastern Afghanistan |
| Ashmaka      | Potana / Potali (or Paithan)                                                                                                          | Godavari valley (Maharashtra–Telangana region)       |

### 4 Jainism: Life of Mahavira Swami (Fact Sheet)

| Event of Life | Details and Exam Facts                                                             |
|---------------|------------------------------------------------------------------------------------|
| Birth         | Born in <b>599 BCE</b> (traditional) at <b>Kundagrama / Kundalpur (Vaishali)</b> . |

| Event of Life                      | Details and Exam Facts                                                                                                         |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Childhood name                     | <b>Vardhamana.</b>                                                                                                             |
| Clan/Family                        | Father <b>Siddhartha</b> (head of the Jnatrika Kshatriya clan).                                                                |
| Mother                             | <b>Trishala / Videhadatta</b> (sister of Chetaka, chief of the Lichchhavi republic of Vaishali).                               |
| Wife and daughter                  | Wife: <b>Yashoda</b> (a girl of the Kaundinya gotra). Daughter: <b>Anojja / Priyadarshana.</b>                                 |
| Attainment of Kaivalya (knowledge) | Under a <b>sal tree</b> on the bank of the <b>Rijupalika river</b> , near Jrimbhikagrama.                                      |
| New names after enlightenment      | Came to be called <b>Kevalin, Arhat</b> (the worthy one), <b>Jina</b> (the conqueror) and <b>Nirgrantha</b> (free from bonds). |
| First disciple                     | His son-in-law <b>Jamali</b> (husband of Priyadarshana).                                                                       |
| Philosophical doctrine propounded  | <b>Anekantavada / Syadvada</b> (Saptabhangi Naya).                                                                             |
| Mahaparinirvana / death            | At <b>Pavapuri</b> near Rajgir, according to the traditional date of <b>527 BCE.</b>                                           |
| Meaning of the word Tirthankara    | One who shows the way to cross the ocean of worldly existence (related to Jainism).                                            |
| Place in the order of Tirthankaras | Mahavira Swami was the <b>24th and last Tirthankara</b> of Jainism.                                                            |

## 5 Important Facts Related to the Puranas (Quick Revision Link)

- **Total number:** 18
- **Composers:** Sage Lomaharsha and his son Ugrashrava.
- **Redaction:** Took place in the Gupta period. Women and Shudras had the right to listen to them, but not to read them.
- **Principal dynastic links:**
  - **Vishnu Purana** → Maurya dynasty
  - **Vayu Purana** → Gupta dynasty
  - **Matsya Purana** → Satavahana dynasty (the oldest and most authentic Purana)
  - **Agni Purana** → Account of agriculture along with polity (monarchy).

## 1 Jainism: Chaturyama Teaching, Pancha Mahavrata and the Tri-ratna

| Vow / Teaching  | Meaning / Description                        | Special Exam Facts                                                                                                                                       |
|-----------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Satya        | Always speaking the truth.                   | • <b>Chaturyama teaching:</b> these first four vows were propounded by the <b>23rd Tirthankara, Parshvanatha</b> .                                       |
| 2. Ahimsa       | Complete abstention from violence.           |                                                                                                                                                          |
| 3. Asteya       | Not stealing (theft forbidden).              |                                                                                                                                                          |
| 4. Aparigraha   | Not accumulating wealth/property.            |                                                                                                                                                          |
| 5. Brahmacharya | Control of the senses (the fifth great vow). | • This fifth vow was added by the <b>24th Tirthankara, Mahavira Swami</b> , after which they came to be called the 'Pancha Mahavrata' (Five Great Vows). |

### > Tri-ratna (Three Jewels) of Jainism

- **Samyak Darshana (Right Faith):** Belief in truth (faith).
- **Samyak Jnana (Right Knowledge):** Real knowledge free from doubt.
- **Samyak Acharana (Right Conduct):** Equanimity (detachment) towards the pleasures and pains arising from worldly objects.

### > Jainism: Ananta Chatushtaya (the four infinite powers attained after moksha)

1. Ananta Jnana (Infinite Knowledge) | 2. Ananta Darshana (Infinite Perception) | 3. Ananta Virya (Infinite Energy/Power) | 4. Ananta Sukha (Infinite Bliss)

## 2 The 24 Tirthankaras of Jainism and their Emblems

(Most important for 'Match the Following' questions in UPPCS Prelims)

| No. | Name of Tirthankara     | Emblem (Lanchhana)          | No. | Name of Tirthankara | Emblem (Lanchhana)  |
|-----|-------------------------|-----------------------------|-----|---------------------|---------------------|
| 1   | Rishabhadeva (Adinatha) | Bull                        | 13  | Vimalanatha         | Boar                |
| 2   | Ajitanatha              | Elephant                    | 14  | Anantanatha         | Falcon (hawk)       |
| 3   | Sambhavanatha           | Horse                       | 15  | Dharmanatha         | Thunderbolt (vajra) |
| 4   | Abhinandana             | Monkey                      | 16  | Shantinatha         | Deer                |
| 5   | Sumatinatha             | Chakravaka (ruddy shelduck) | 17  | Kunthunatha         | Goat                |



| No. | Name of Tirthankara           | Emblem (Lanchhana) | No. | Name of Tirthankara                | Emblem (Lanchhana) |
|-----|-------------------------------|--------------------|-----|------------------------------------|--------------------|
| 6   | Padmaprabha                   | Lotus              | 18  | Aranatha                           | Fish               |
| 7   | Suparshvanatha                | Swastika           | 19  | <b>Mallinatha</b>                  | Pitcher (kalasha)  |
| 8   | Chandraprabhu                 | Crescent moon      | 20  | Munisuvrata                        | Tortoise           |
| 9   | Pushpadanta<br>(Suvidhinatha) | Crocodile          | 21  | Naminatha                          | Blue lotus         |
| 10  | Shitalanatha                  | Shrivatsa          | 22  | <b>Neminatha<br/>(Arishtanemi)</b> | Conch              |
| 11  | Shreyamsanatha                | Rhinoceros         | 23  | <b>Parshvanatha</b>                | Serpent (snake)    |
| 12  | Vasupujya                     | Buffalo            | 24  | <b>Mahavira Swami</b>              | Lion               |

### 3 Principal Jain Tirthankaras and the Places of their Nirvana (Moksha)

| Name of Tirthankara      | Place of Nirvana  | Present Geographical Location / Special Fact |
|--------------------------|-------------------|----------------------------------------------|
| Rishabhanatha (Adinatha) | <b>Ashtapada</b>  | Also associated with Mount Kailash.          |
| Vasupujya                | <b>Champapuri</b> | Present-day Bhagalpur region of Bihar.       |
| Neminatha (Arishtanemi)  | <b>Urjayanta</b>  | Present-day Girnar hills, Gujarat.           |
| Mahavira Swami           | <b>Pavapuri</b>   | Present-day Bihar (near Rajgir).             |

### 4 Jain Councils

| Council               | Time (Year)       | Place                      | President                      | Main Outcome / Work                                                                                                                                                            |
|-----------------------|-------------------|----------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First Council</b>  | 322 BCE / 300 BCE | <b>Pataliputra</b> (Bihar) | <b>Sthulabhadra</b>            | <ul style="list-style-type: none"> <li>The <b>12 Angas</b> of Jainism were compiled.</li> <li>The Jain order split into two sects—<b>Shvetambara and Digambara</b>.</li> </ul> |
| <b>Second Council</b> | 513 / 526 CE      | <b>Vallabhi</b> (Gujarat)  | <b>Devardhi Kshamashramana</b> | <ul style="list-style-type: none"> <li>Scattered Jain texts were collected and the <b>11 Angas</b> were finally committed to writing.</li> </ul>                               |

### 5 Fundamental Doctrines of Jain Philosophy

| Philosophical Doctrine | Main Meaning and Feature                                                             |
|------------------------|--------------------------------------------------------------------------------------|
| <b>Anekantavada</b>    | Doctrine of many-sidedness/plurality of reality (an object has infinite attributes). |

| Philosophical Doctrine             | Main Meaning and Feature                                                                        |
|------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Syadvada (Saptabhangi Naya)</b> | Doctrine of relativity (the law of the relativity of knowledge).                                |
| <b>Anishvaravada (Non-theism)</b>  | Denial of any role of God in the creation of the world (creation is beginningless and eternal). |
| <b>Atmavada / Rebirth</b>          | Acceptance of the existence of the soul and belief in rebirth according to the fruits of karma. |

## 6 Principal Jain Texts and their Authors

| Jain Text                 | Author              | Main Theme of the Text                                                                                  |
|---------------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| <b>Kalpasutra</b>         | Bhadrabahu          | Related to the life histories of the Tirthankaras (especially Mahavira and Parshvanatha) (in Sanskrit). |
| <b>Parishishta Parvan</b> | Hemachandra         | Important epic related to Jain history, kings and Jain tradition.                                       |
| <b>Tattvartha Sutra</b>   | Umasvami            | A highly important, systematic and authoritative sutra text of Jain philosophy.                         |
| <b>Syadvadamanjari</b>    | Mallisena           | Detailed exposition of the Syadvada and Anekantavada of Jain philosophy.                                |
| <b>Nyayavatara</b>        | Siddhasena Divakara | Early and foundational text of Jain Nyaya (logic).                                                      |
| <b>Pravachanasara</b>     | Kundakunda          | A great text of Digambara Jain philosophy (original language: <b>Prakrit</b> ).                         |
| <b>Dravya Sangraha</b>    | Nemichandra         | Scientific treatment of substance (dravya), qualities (guna) and modes (paryaya) in Jain philosophy.    |
| <b>Nyaya Dipika</b>       | Dharmabhushana      | Related to the Jain Nyaya tradition and logic.                                                          |
| <b>Panchavimshatika</b>   | Padmanandi          | Related to Jain philosophical literature and spirituality.                                              |
| <b>Shloka Vartika</b>     | Vidyananda Swami    | Related to Jain logic and the refutation of Mimamsa.                                                    |

## 7 Buddhism: Life of Mahatma Buddha (The Light of Asia)

| Event / Fact of Life           | Details and Exam Evidence                                                       |
|--------------------------------|---------------------------------------------------------------------------------|
| <b>Birth</b>                   | Born in <b>563 BCE</b> at <b>Lumbini</b> (near Kapilavastu, present-day Nepal). |
| <b>Original/childhood name</b> | <b>Siddhartha</b> (gotra name - <b>Gautama</b> ).                               |

| Event / Fact of Life                    | Details and Exam Evidence                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Parents and clan                        | Father: <b>Shuddhodana</b> (king of the <b>Shakya</b> republic of Kapilavastu).<br>Mother: <b>Mahamaya</b> (of the <b>Koliya</b> republic). |
| Upbringing                              | As his mother died after his birth, he was brought up by his maternal aunt <b>Mahaprajapati Gautami</b> .                                   |
| Family details                          | Wife: <b>Yashodhara</b> ( <i>other names: Bimba, Gopa, Bhaddakachchha</i> ).<br>Son: <b>Rahula</b> .                                        |
| Mahabhinishkramana (Great Renunciation) | Left home at the age of <b>29</b> . Horse: <b>Kanthaka</b> ; charioteer: <b>Channa</b> .                                                    |
| First teacher (Vaishali)                | <b>Alara Kalama</b> (from whom Siddhartha received initiation in <b>Sankhya philosophy</b> ).                                               |
| Second teacher (Rajgir)                 | <b>Rudraka Ramaputta</b> (from whom he gained knowledge of yoga).                                                                           |
| Enlightenment (Bodh Gaya)               | At the age of <b>35</b> , on the bank of the <b>Niranjana river (Punpun/Phalgu)</b> , under a <b>peepal tree (Bodhi tree)</b> .             |
| Dharmachakra Pravartana (First Sermon)  | Gave his first sermon to five Brahmin ascetics at <b>Rishipatana (Sarnath)</b> .                                                            |
| Most sermons                            | Delivered at <b>Shravasti</b> , capital of Koshala.                                                                                         |
| Last sermon                             | Given at Kushinagar to his disciple <b>Subhadda</b> .                                                                                       |
| Mahaparinirvana (death)                 | In <b>483 BCE</b> at <b>Kushinagar</b> , capital of the Malla republic, at the age of <b>80</b> .                                           |

## 8 Buddhist Symbols Related to the Life of the Buddha

| Event / Place of Life   | Buddhist Symbol            | Specific Name of the Event / Place                             |
|-------------------------|----------------------------|----------------------------------------------------------------|
| Birth                   | <b>Lotus</b> (and bull)    | Place: Lumbini                                                 |
| Renunciation of home    | <b>Horse</b>               | Event: <b>Mahabhinishkramana</b> (name of the horse: Kanthaka) |
| Enlightenment           | <b>Peepal (Bodhi tree)</b> | Place: Bodh Gaya                                               |
| First sermon            | <b>Wheel</b>               | Event: <b>Dharmachakra Pravartana</b> (place: Sarnath)         |
| Nirvana (footprints)    | <b>Footprints</b>          | Symbol of liberation from the bonds of life.                   |
| Death (Mahaparinirvana) | <b>Stupa</b>               | Place: Kushinagar                                              |

## 9 The 'Ashtangika Marga' (Eightfold Path) of Buddhism

The Eightfold Path is related to the fourth Noble Truth of Buddhism, '**Dukkha Nirodha Gamini Patipada**' (the path leading to the cessation of suffering). It is also called the '**Madhyama Pratipada**' (Middle Path) and comprises the following 8 steps (stages):

| Step of the Eightfold Path              | Real Meaning and Practical Form                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------|
| 1. Samyak Drishti (Right View)          | Understanding an object in its real (true) form.                                     |
| 2. Samyak Sankalpa (Right Resolve)      | Holding pure thoughts free from greed, hatred and violence.                          |
| 3. Samyak Vak (Right Speech)            | Complete renunciation of unpleasant, harsh and false words.                          |
| 4. Samyak Karmanta (Right Action)       | Always following good deeds in life.                                                 |
| 5. Samyak Ajiva (Right Livelihood)      | Earning a just livelihood marked by good conduct and honesty.                        |
| 6. Samyak Vyayama (Right Effort)        | Continuous wholesome effort for mental and physical well-being.                      |
| 7. Samyak Smriti (Right Mindfulness)    | Always maintaining a pure and alert state of mind (keeping away from false notions). |
| 8. Samyak Samadhi (Right Concentration) | Concentration of the mind (attaining the higher state of meditation).                |

## 1 Fundamental Doctrines of Buddhist Philosophy

| Philosophical Doctrine       | Main Meaning and Practical Form                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Anishvaravada (Non-theism)   | No belief in a divine being (God as creator).                                                                                  |
| Kshanikavada (Momentariness) | The world is changing; nothing here is permanent or stable.                                                                    |
| Anatmavada (No-soul)         | Denial of the eternal existence of the soul, while laying the greatest emphasis on self-awareness.                             |
| Shunyavada (Emptiness)       | All things in the world are relative and without inherent existence (empty of own-nature).                                     |
| Rebirth                      | <b>Note:</b> Although Buddhism does not accept the soul, it still <b>believes in rebirth</b> (rebirth of consciousness/karma). |

## 2 Buddhist Councils: Time, Place, Ruler, President and Main Work

*(This table falls in the 'most important' category for UPPCS Prelims)*



| Council (Time)                         | Place / Cave                          | Ruler (Dynasty)                          | President / Vice-President                                                                                                                     | Main Work and Historical Outcome (Exam Facts)                                                                                                                                                                                                                                                              |
|----------------------------------------|---------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First</b><br>(483 BCE)              | <b>Rajagriha</b><br>(Saptaparni Cave) | <b>Ajatashatru</b><br>(Haryanka dynasty) | <ul style="list-style-type: none"> <li>• <b>President:</b> Mahakashyapa (Mahakassapa)</li> <li>• <b>Principal associate:</b> Ananda</li> </ul> | <ul style="list-style-type: none"> <li>• The tradition of compiling the Buddha's teachings began.</li> <li>• <b>Ananda</b> compiled/recited the <b>Sutta Pitaka</b> (teachings) and <b>Upali</b> the <b>Vinaya Pitaka</b> (rules).</li> </ul>                                                              |
| <b>Second</b><br>(383 BCE)             | <b>Vaishali</b>                       | <b>Kalashoka</b><br>(Shishunaga dynasty) | <ul style="list-style-type: none"> <li>• <b>President:</b> Sabakami (Sabbakami / Sabakami)</li> </ul>                                          | <ul style="list-style-type: none"> <li>• Main dispute among monks over 10 rules of the Vinaya.</li> <li>• First split of the Buddhist Sangha into the <b>Sthaviras (Theravada)</b> and the <b>Mahasanghikas</b>.</li> </ul>                                                                                |
| <b>Third</b><br>(250 BCE)              | <b>Pataliputra</b>                    | <b>Ashoka</b><br>(Maurya dynasty)        | <ul style="list-style-type: none"> <li>• <b>President:</b> Moggaliputta Tissa</li> </ul>                                                       | <ul style="list-style-type: none"> <li>• The <b>Abhidhamma Pitaka</b> was compiled and the '<b>Kathavatthu</b>' was composed.</li> <li>• Attempt to remove unorthodox views from the Sangha.</li> <li>• Missionaries were sent abroad (<b>Mahendra and Sanghamitra were sent to Sri Lanka</b>).</li> </ul> |
| <b>Fourth</b><br>(72 CE / 1st century) | <b>Kundalavana</b><br>(Kashmir)       | <b>Kanishka</b><br>(Kushana dynasty)     | <ul style="list-style-type: none"> <li>• <b>President:</b> Vasumitra</li> <li>• <b>Vice-President:</b> Ashvaghosha</li> </ul>                  | <ul style="list-style-type: none"> <li>• Final, complete division of Buddhism into <b>Hinayana</b> (basically Sthaviravadin) and <b>Mahayana</b> (basically Mahasanghika).</li> <li>• <b>Use of the Sanskrit language</b> increased in Buddhist literature.</li> </ul>                                     |

### 3 The 'Tripitaka' and Other Important Texts of Buddhism

| Tripitaka / Text         | Original Language | Subject Matter / Internal Division (Core Content)                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sutta Pitaka</b>      | Pali              | <ul style="list-style-type: none"> <li>• Contains the <b>principal teachings of the Buddha and the Dhamma</b>.</li> <li>• <b>Divided into 5 Nikayas:</b> 1. Digha Nikaya, 2. Majjhima Nikaya, 3. Samyutta Nikaya, 4. Anguttara Nikaya, 5. Khuddaka Nikaya.</li> <li>• <i>(Note: the principal texts belonging to the Khuddaka Nikaya are the <b>Dhammapada, Jatakas, Suttanipata, Theragatha and Therigatha</b>).</i></li> </ul> |
| <b>Vinaya Pitaka</b>     | Pali              | <ul style="list-style-type: none"> <li>• <b>Rules of conduct, discipline and of the Sangha</b> applicable to the monastic life of monks and nuns.</li> </ul>                                                                                                                                                                                                                                                                     |
| <b>Abhidhamma Pitaka</b> | Pali              | <ul style="list-style-type: none"> <li>• <b>Philosophical analysis and systematisation of the doctrines</b> taught to monks (contains a total of <b>7 texts</b>).</li> </ul>                                                                                                                                                                                                                                                     |

| Tripitaka / Text                                       | Original Language | Subject Matter / Internal Division (Core Content)                                                                                                     |
|--------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Milindapanho</b><br>( <i>Questions of Milinda</i> ) | Pali              | • Famous philosophical dialogue between the Indo-Greek ruler <b>Milinda (Menander)</b> and the Kashmiri Sarvastivadin Buddhist monk <b>Nagasena</b> . |
| <b>Other texts</b>                                     | Pali/Sanskrit     | • The <b>Divyavadana</b> , <b>Dipavamsa</b> , <b>Mahavamsa</b> , etc. are other important Buddhist texts.                                             |

#### 4 Principal Buddhist Texts and their Authors

| Buddhist Text                                      | Author       |
|----------------------------------------------------|--------------|
| Milindapanho                                       | Nagasena     |
| Buddhacharita, Saundarananda, Shariputra Prakarana | Ashvaghosha  |
| Madhyamakakarika                                   | Nagarjuna    |
| Visuddhimagga                                      | Buddhaghosha |
| Abhidharmakosha                                    | Vasubandhu   |

#### 5 Principal Dynasties of Ancient India: Founders and Last Rulers

| Dynasty                                        | Founder                                 | Last Ruler                           | Special Exam Note (UPPCS Facts)                                                                            |
|------------------------------------------------|-----------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------|
| Nanda dynasty                                  | <b>Mahapadma Nanda</b><br>(or Ugrasena) | <b>Dhanananda</b>                    | Chandragupta Maurya defeated Dhanananda and laid the foundation of the Maurya Empire.                      |
| Maurya dynasty                                 | <b>Chandragupta Maurya</b>              | <b>Brihadratha</b>                   | The commander <b>Pushyamitra Shunga</b> killed Brihadratha and founded the Shunga dynasty.                 |
| Shunga dynasty                                 | <b>Pushyamitra Shunga</b>               | <b>Devabhuti</b>                     | Brahmin dynasty established over Magadha immediately after the end of Mauryan power.                       |
| Satavahana dynasty                             | <b>Simuka</b>                           | <b>Yajna Shri Satakarni</b>          | Also called the Andhra dynasty; Yajnashri was among the prominent last rulers.                             |
| Gupta dynasty                                  | <b>Shri Gupta</b>                       | <b>Kumaragupta III / Vishnugupta</b> | After Shrigupta came Ghatotkacha, then Chandragupta I ascended the throne. (Last known ruler: Vishnugupta) |
| Chalukya dynasty<br>( <i>Vatapi / Badami</i> ) | <b>Pulakeshin I</b>                     | <b>Kirtivarman II</b>                | Capital: <b>Vatapi (Badami)</b> .                                                                          |

## 6 Ancient Rulers / Personalities and their Other Names (Titles)

| Real Name           | Other Names in Greek Sources / Inscriptions   | Specific Meaning / Description                                                       |
|---------------------|-----------------------------------------------|--------------------------------------------------------------------------------------|
| Chandragupta Maurya | <b>Sandrokottos</b> / Androkottos             | The Greek writer <b>Justin</b> called Chandragupta Maurya 'Androkottos'.             |
| Bindusara           | <b>Amitraghata</b> / Amitrochates             | Literally means ' <b>slayer of enemies</b> '.                                        |
| Ashoka              | <b>Piyadassi (Priyadarshi)</b> / Devanampriya | The title " <b>Devanampiya</b> " (beloved of the gods) is found in the inscriptions. |
| Chanakya            | <b>Vishnugupta</b> / Kautilya                 | Prime Minister of Chandragupta Maurya and author of the famous text 'Arthashastra'.  |

## 1 Maurya Dynasty: The Founder, Chandragupta Maurya and Bindusara

| Ruler / Personality                        | Important Events / Historical Evidence (Exam Facts)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chandragupta Maurya<br>( <i>founder</i> )  | <ul style="list-style-type: none"> <li>• <b>Teacher and Prime Minister:</b> Kautilya (Chanakya).</li> <li>• <b>Religious end:</b> In the last days of his life he became a disciple of the Jain monk <b>Bhadrabahu</b>. In <b>298 BCE</b> he gave up his body by the <b>Sallekhana method</b> at <b>Shravanabelagola</b> (Karnataka).</li> <li>• <b>War (305 BCE):</b> War and treaty with Seleucus Nicator. Seleucus ceded the regions of <b>Kabul, Kandahar, Herat and Makran</b>; in return Chandragupta gave <b>500 war elephants</b>.</li> </ul> |
| Megasthenes<br>( <i>Greek ambassador</i> ) | <ul style="list-style-type: none"> <li>• Sent by Seleucus to the court of Chandragupta Maurya. Wrote the famous book '<b>Indica</b>' (lost in its original form; fragments survive in the works of later Greco-Roman writers).</li> <li>• <b>7 Indian castes:</b> according to him society had 7 castes—1. philosophers, 2. cultivators, 3. herdsmen, 4. artisans, 5. soldiers, 6. overseers (inspectors), 7. councillors.</li> </ul>                                                                                                                 |
| Bindusara<br>( <i>297–273 BCE</i> )        | <ul style="list-style-type: none"> <li>• Name in Greek sources: <b>Amitrochates</b> / <b>Amitraghata</b> (slayer of enemies).</li> <li>• Asked the Syrian ruler <b>Antiochus</b> for 3 things: 1. sweet wine, 2. dried figs, 3. a philosopher.<br/>(<i>Note: Antiochus refused to send a philosopher because Greek law did not permit philosophers to be sold</i>).</li> </ul>                                                                                                                                                                        |

## 2 Kautilya's 'Arthashastra' and the Saptanga Theory

- **Subject matter:** Detailed and practical description of the state, administration, economy, justice and the espionage system.

- **Saptanga theory:** For the efficient running of the state, Kautilya described 7 limbs (constituents) of the state:

| Limb of the State      | Symbol in the Human Body (Analogy)   | Limb of the State                  | Symbol in the Human Body (Analogy)    |
|------------------------|--------------------------------------|------------------------------------|---------------------------------------|
| 1. Raja (Svami – King) | Like the <b>head</b>                 | 5. Janapada (territory and people) | The <b>thighs (legs)</b> of the state |
| 2. Amatya (Minister)   | The <b>eye</b> of the state          | 6. Danda (Army)                    | The <b>brain (mind)</b> of the state  |
| 3. Kosha (Treasury)    | The <b>mouth</b> of the state        | 7. Mitra (Ally)                    | The <b>ears</b> of the state          |
| 4. Durga (Fort)        | The <b>arms (hands)</b> of the state |                                    |                                       |

### 3 Emperor Ashoka: Life and Basic Facts about his Inscriptions

| Fact / Position              | Details and Exam Evidence (Core Facts)                                                                                                               |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parents                      | Father: <b>Bindusara</b> . Mother: <b>Subhadrangi</b> (chief queen of Bindusara, also called 'Dharma').                                              |
| Coronation                   | In <b>269 BCE</b> (4 years after his accession to the throne).                                                                                       |
| Kalinga War                  | <b>261 BCE</b> (in the 8th/9th year after his coronation). After this Ashoka adopted the <b>policy of Dhamma Vijaya</b> (conquest by righteousness). |
| Language of the inscriptions | Basically the <b>Prakrit</b> language. (Exception: the <b>Kandahar inscription is bilingual—Greek and Aramaic</b> ).                                 |
| Script of the inscriptions   | <b>Brahmi</b> (in most rock edicts), <b>Kharoshthi</b> (in the north-western region), <b>Aramaic</b> and <b>Greek</b> .                              |

### 4 Names and Titles of Ashoka in the Sources

| Title / Name | Source / Inscription                                                            | Historical Importance / Special Note                                                                           |
|--------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Ashoka       | • <b>Maski, Gujarra, Nittur, Udegolam</b> minor rock edicts.                    | • The name first occurs in the <b>Maski edict</b> . " <b>Devanampiya Asoka</b> " is clearly engraved at Maski. |
| Priyadarshi  | • <b>Bhabru (Bairat) edict, Kandahar inscription</b> and the <b>Dipavamsa</b> . | • Generally used in the inscriptions as " <b>Devanampriya Priyadarshi</b> ".                                   |
| Devanampriya | • In the <b>12th and 13th</b> major rock edicts.                                | • It means 'beloved of the gods'.                                                                              |



| Title / Name                | Source / Inscription                                          | Historical Importance / Special Note                                                                             |
|-----------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Devanampriya Priyadarshi    | • <b>Girnar, Kalsi, Erragudi</b> and the Nigliva inscription. | • <b>Note:</b> the <b>Kharoshthi script</b> is used in the Shahbazgarhi and Mansehra inscriptions.               |
| Ashokavardhana              | • In the <b>Vishnu Purana</b> .                               | • This name occurs in the Buddhist and Puranic traditions.                                                       |
| Magadhadhiraja              | • In the <b>Bhabru (Bairat) edict</b> .                       | • This inscription is the greatest proof of Ashoka's <b>firm faith in Buddhism</b> .                             |
| <i>Buddha Shakya (Note)</i> | <i>Mentioned in the Maski inscription.</i>                    | <b>Exam warning:</b> "Buddha Shakya" is not a title of Ashoka; it is a reference to Mahatma Buddha (Shakyamuni). |

## 5 Discoverers of Ashoka's Important Inscriptions / Pillar Edicts

| Rock Edict / Pillar Edict | Discoverer (Year)              | Present Geographical Location / Special Fact                                                                         |
|---------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Girnar Rock Edict         | <b>Colonel Tod</b> (1822)      | Junagadh, <b>Gujarat</b> .                                                                                           |
| Shahbazgarhi Rock Edict   | <b>General Court</b> (1836)    | Peshawar, <b>Pakistan</b> (in the Kharoshthi script).                                                                |
| Dhauli Rock Edict         | <b>Kittoe</b> (1837)           | Puri, <b>Odisha</b> (principal rock edict of the Kalinga region).                                                    |
| Jaugada Rock Edict        | <b>Walter Elliot</b> (1850)    | Ganjam, <b>Odisha</b> (site of the Separate Kalinga Edicts).                                                         |
| Kalsi Rock Edict          | <b>Forrest</b> (1860)          | Dehradun, <b>Uttarakhand</b> (includes the text of the 13th rock edict).                                             |
| Mansehra Rock Edict       | <i>Year of discovery: 1889</i> | Hazara, <b>Pakistan</b> ( <b>Note:</b> naming "General Cunningham" as its discoverer is incorrect; do not write it). |

## 6 Subject Matter of Ashoka's 14 Major Rock Edicts (Core Themes)

| Rock Edict | Main Theme Propounded (Core Content)                                                                                                                                               |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (First)  | <b>Prohibition of the killing of living beings</b> ; ban on animal sacrifice and festive gatherings, and strict control over animal slaughter.                                     |
| 2 (Second) | Mentions <b>medical arrangements for men and animals</b> in the kingdoms of the Cholas, Pandyas, Satiyaputras, Keralaputras, Tamraparni (Sri Lanka) and the Yavana king Antiochus. |
|            |                                                                                                                                                                                    |

| Rock Edict      | Main Theme Propounded (Core Content)                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 (Third)       | Order to royal officials ( <b>Yuktas, Rajjukas and Pradeshikas</b> ) to tour the country every 5th year.                                                                                                                                                                                                                                                                                                 |
| 4 (Fourth)      | <b>The sound of the war-drum (bherighosha) replaced by the sound of Dhamma (dhammaghosha)</b> (moral/religious conquest in place of war).                                                                                                                                                                                                                                                                |
| 5 (Fifth)       | Mentions the appointment, in the 13th year of the coronation, of new officers called ' <b>Dhamma Mahamattas</b> '.                                                                                                                                                                                                                                                                                       |
| 6 (Sixth)       | Teaching of self-control and order to the <b>Prativedakas (reporters)</b> to report the condition of the people at all times.                                                                                                                                                                                                                                                                            |
| 7 (Seventh)     | <b>Religious tolerance</b> ; people of all sects may reside peacefully everywhere.                                                                                                                                                                                                                                                                                                                       |
| 8 (Eighth)      | <b>Beginning of Ashoka's Dhamma Yatras</b> (e.g., the journey to Bodh Gaya).                                                                                                                                                                                                                                                                                                                             |
| 9 (Ninth)       | True courtesy and <b>declaring 'Dhamma Mangala' (ceremony of Dhamma) the best</b> ; avoiding extravagance and useless ceremonies.                                                                                                                                                                                                                                                                        |
| 10 (Tenth)      | Condemnation of fame and glory; emphasis on the <b>superiority of the policy of Dhamma</b> .                                                                                                                                                                                                                                                                                                             |
| 11 (Eleventh)   | Explanation of <b>Dhamma Dana</b> (the gift of Dhamma is the best gift).                                                                                                                                                                                                                                                                                                                                 |
| 12 (Twelfth)    | <b>Religious tolerance and appointment of Stri Mahamattas (women's officers)</b> ; emphasis on the growth of the essence (core elements) of all sects.                                                                                                                                                                                                                                                   |
| 13 (Thirteenth) | <ul style="list-style-type: none"> <li>• Moving description of the terrible war of the <b>Kalinga conquest (261 BCE)</b> and of Ashoka's change of heart.</li> <li>• Mentions the <b>5 Yavana kings</b> and the southern kingdoms (Cholas, Pandyas, Satiyaputras, Keralaputras, Tamraparni).</li> <li>• It is the <b>largest and conclusive in character</b> among all the major rock edicts.</li> </ul> |
| 14 (Fourteenth) | Inspiration to lead a religious life and the <b>main purpose</b> of having the edicts engraved.                                                                                                                                                                                                                                                                                                          |

## 7 Find-spots of the 14 Major Rock Edicts (Geographical Locations)

| Find-spot | District / Present State | Country |
|-----------|--------------------------|---------|
| Kalsi     | Dehradun, Uttarakhand    | India   |
| Girnar    | Junagadh, Gujarat        | India   |
| Sopara    | Thane, Maharashtra       | India   |
| Dhauili   | Puri, Odisha             | India   |
| Jaugada   | Ganjam, Odisha           | India   |
|           |                          |         |

| Find-spot    | District / Present State     | Country                      |
|--------------|------------------------------|------------------------------|
| Erragudi     | Kurnool, Andhra Pradesh      | India                        |
| Sannati      | Gulbarga, Karnataka          | India                        |
| Shahbazgarhi | Peshawar, Khyber Pakhtunkhwa | Pakistan (Kharoshthi script) |
| Mansehra     | Hazara, Khyber Pakhtunkhwa   | Pakistan (Kharoshthi script) |

8

The Five Yavana Kings Mentioned in the 13th Rock Edict (Contemporary Hellenic Kings)

| Name in the Inscription (Pali/Prakrit) | Greek / Modern Name     | Related Region / Country (Kingdom) |
|----------------------------------------|-------------------------|------------------------------------|
| Antiyoka                               | Antiochus II Theos      | Syria                              |
| Turamaya                               | Ptolemy II Philadelphus | Egypt                              |
| Antikina                               | Antigonus Gonatas       | Macedonia                          |
| Maga                                   | Magas                   | Cyrene                             |
| Alikasudara                            | Alexander               | Epirus                             |

1

Hierarchical Units of Mauryan Administration (Administrative Units)

| Administrative Unit | Head / Officer               | Administrative Level / Feature (Core Facts)                          |
|---------------------|------------------------------|----------------------------------------------------------------------|
| Empire (Samrajya)   | King                         | Supreme central authority.                                           |
| Province            | Viceroy (Kumara / Aryaputra) | • <b>Principal provinces:</b> Taxila, Ujjayini, Tosali, Suvarnagiri. |
| Mandala             | Pradeshta                    | Officer concerned with the judicial and penal (criminal) system.     |
| Ahara or Vishaya    | Vishayapati                  | The principal <b>district-level</b> unit of administration.          |
| Sthaniya            | -                            | A group of <b>800 villages</b> .                                     |
| Dronamukha          | -                            | A group of <b>400 villages</b> .                                     |
| Kharvatika          | -                            | A group of <b>200 villages</b> .                                     |
| Sangrahana          | -                            | A group of <b>10 villages</b> .                                      |
|                     |                              |                                                                      |

| Administrative Unit | Head / Officer | Administrative Level / Feature (Core Facts)                              |
|---------------------|----------------|--------------------------------------------------------------------------|
| Grama (Village)     | Gramika        | The <b>smallest unit</b> of Mauryan administration; head of the village. |

## 2 Central Administration: The 18 Tirthas (Highest Administrative Officers)

(According to Kautilya's Arthashastra, the top-ranking officials were called 'Tirthas' or 'Mahamatras')

| No. | Tirtha (Officer)                 | Department / Main Function (Core Responsibility)                                          |
|-----|----------------------------------|-------------------------------------------------------------------------------------------|
| 1   | <b>Purohita / Prime Minister</b> | Chief adviser and administrative counsellor of the king.                                  |
| 2   | <b>Yuvaraja</b>                  | Heir to the king (crown prince).                                                          |
| 3   | <b>Samaharta</b>                 | <b>Revenue minister / collector of income</b> (supreme head of the revenue department).   |
| 4   | <b>Sannidhata</b>                | <b>Treasurer</b> (supreme head of the royal treasury, treasury-house and storehouses).    |
| 5   | <b>Senapati</b>                  | Minister of war / supreme military officer of the army.                                   |
| 6   | <b>Pradeshta</b>                 | Judge / commissioner of the <b>criminal courts</b> (Kantakashodhana).                     |
| 7   | <b>Vyavaharika</b>               | Chief judge of the <b>civil courts</b> (Dharmasthiya).                                    |
| 8   | <b>Dandapala</b>                 | Officer who collected the army's provisions, materials and military resources.            |
| 9   | <b>Antapala</b>                  | Head in charge of the frontier (border) regions and forts.                                |
| 10  | <b>Karmantika</b>                | Chief minister of the country's industries and factories.                                 |
| 11  | <b>Durgapala</b>                 | Head in charge of the forts inside the country (internal forts).                          |
| 12  | <b>Nagaraka</b>                  | Supreme head of city administration and civic arrangements.                               |
| 13  | <b>Prashasta</b>                 | Head who preserved royal documents/edicts (registrar-general).                            |
| 14  | <b>Nayaka</b>                    | Chief of city defence / officer leading the army in the battlefield.                      |
| 15  | <b>Atavika</b>                   | Supreme head of the forest department.                                                    |
| 16  | <b>Mantriparishad Adhyaksha</b>  | President of the council of ministers.                                                    |
| 17  | <b>Sena</b>                      | Direct commander of the army.                                                             |
| 18  | <b>Antarvamshika</b>             | In charge of the king's <b>bodyguard</b> and the security of the royal household (harem). |



### 3 Principal 'Adhyakshas' (Heads of Departments) of the Mauryan Period

| Adhyaksha (Superintendent)                     | Department / Portfolio                                                                                    |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Akaradhyaksha                                  | Supreme superintendent of <b>mines and minerals</b> (mining).                                             |
| Lakshanadhyaksha                               | <b>Superintendent of the mint</b> ; supervised the minting of coins and the royal currency.               |
| Sitadhyaksha                                   | Supervised the <b>royal agricultural land</b> (sita land) and headed the agriculture department.          |
| Vivitadhyaksha                                 | Head in charge of <b>pastures and the protection of cattle</b> and of uninhabited tracts.                 |
| Pautavadhyaksha                                | <b>Superintendent of weights and measures</b> ; officer for the standardisation of weights and measures.  |
| Manadhyaksha                                   | Officer who fixed the standards of time, distance, direction and measurement.                             |
| Suradhyaksha                                   | <b>Superintendent of liquor</b> (sale of liquor and excise control).                                      |
| Panyadhyaksha                                  | <b>Superintendent of commerce and trade</b> ; controlled the market system.                               |
| Shulkadhyaksha                                 | Officer who collected <b>tolls and customs</b> duties.                                                    |
| Navadhyaksha                                   | Supreme officer of <b>shipping</b> and waterways.                                                         |
| Mudradhyaksha                                  | Officer for royal passports/seals and permits (passes) for movement.                                      |
| Koshthagaradhyaksha                            | Chief superintendent of the royal granary and storehouse.                                                 |
| Ashvadhyaksha / Hastyadhyaksha / Rathadhyaksha | Chief officers of horses (cavalry), elephants (elephant corps) and chariots (chariot corps) respectively. |
| Ganikadhyaksha                                 | Officer who supervised the department concerning courtesans (ganikas).                                    |

### 4 Kautilya's 'Saptanga Theory' (Organic Theory of State)

Kautilya regarded the state as **equivalent to a human body**. He called these 7 limbs "**Prakritis**", of which the **king (Svami)** is regarded as the supreme and principal limb of the state:

| Limb (Saptanga)      | Symbol in the Human Body (Analogy) | Special UPPCS Exam Note (Core Conceptual Facts)                                                               |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1. Svami (King)      | Head                               | • The supreme, principal and most important limb of the state; supports the monarchical system.               |
| 2. Amatya (Minister) | Eyes                               | • A body of qualified officials; the king's principal associates in administration.                           |
| 3. Janapada          | Thighs (legs)                      | • Kautilya used the word " <b>Janapada</b> " for it (in notes 'Rashtra' is given as the symbol of this limb). |
| 4. Durga (Fort)      | Arms                               | • The main basis of the security, defence and military fortification of the state.                            |
| 5. Kosha (Treasury)  | Mouth                              | • The basis of the economic strength of the state (royal treasury).                                           |
| 6. Danda (Army)      | Brain (mind)                       | • Force/army relates to the punitive and defensive power of government.                                       |
| 7. Mitra (Ally)      | Ears                               | • External associates or friendly states who help in times of calamity.                                       |

## 5 The Most Important Inscriptions of Ancient India (Rock Edicts and Prashastis)

| Name of Inscription                   | Related Ruler (Dynasty)                           | Composer / Specific Evidence and Historical Importance (Exam Points)                                                        |
|---------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Hathigumpha inscription               | <b>Kharavela, king of Kalinga</b> (Chedi dynasty) | • Located at Udayagiri in Odisha; the principal and systematic source of information on the events of Kharavela's reign.    |
| Junagadh (Girnar) inscription         | <b>Rudradaman</b> (Shaka Kshatrapa)               | • Located in <b>Gujarat</b> ; one of the first long and authentic inscriptions engraved in the <b>Sanskrit language</b> .   |
| Nasik inscription                     | <b>Gautami Balashri</b> (Satavahana)              | • Detailed description of the military victories and achievements of the Satavahana emperor <b>Gautamiputra Satakarni</b> . |
| Prayag (Allahabad) Pillar inscription | <b>Samudragupta</b> (Gupta dynasty)               | • Composer: Samudragupta's court poet <b>Harishena</b> (a famous prashasti composed in the champu style).                   |
| Aihole inscription                    | <b>Pulakeshin II</b> (Chalukya dynasty)           | • Composer: <b>Ravikirti</b> ; it gives evidence of Pulakeshin II defeating Harshavardhana.                                 |
| Mandsaur inscription                  | <b>Yashovarman, king of Malwa</b>                 | • Historical mention of Yashodharman defeating the famous <b>Huna ruler Mihirakula</b> .                                    |

| Name of Inscription               | Related Ruler (Dynasty)                              | Composer / Specific Evidence and Historical Importance (Exam Points)                                                   |
|-----------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Gwalior inscription               | <b>Bhoja, the Pratihara king</b> (Gurjara-Pratihara) | • Important evidence relating to the history of the Gurjara-Pratihara dynasty and the achievements of King Bhoja.      |
| Bhitari and Junagadh inscriptions | <b>Skandagupta</b> (Gupta dynasty)                   | • Authentic mention of the barbaric invasion of the Hunas and of Skandagupta's great successes/victories against them. |
| Deopara inscription               | <b>Vijayasena</b> (Sena dynasty)                     | • Historical account of the achievements of Vijayasena, the powerful ruler of the Sena dynasty of Bengal.              |

## 6 Other Important Inscriptions of the Post-Mauryan Period

| Name of Inscription  | Related Ruler / Personality           | Historical Evidence Useful for the Exam (Core Facts)                                                                                                      |
|----------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ayodhya inscription  | <b>Dhanadeva</b>                      | • The first written evidence of the history of the Shunga dynasty and of <b>Pushyamitra Shunga</b> and the <b>Ashvamedha sacrifices</b> performed by him. |
| Besnagar inscription | <b>Heliodorus</b> (Yavana ambassador) | • Garuda pillar located at Vidisha (MP); the oldest archaeological evidence of Heliodorus adopting the <b>Bhagavata (Vaishnava) religion</b> .            |
| Shinkot inscription  | <b>Menander</b> (Milinda)             | • Epigraphic evidence relating to the famous Indo-Greek ruler Menander.                                                                                   |
| Naneghat inscription | <b>Naganika</b> (Satavahana queen)    | • Important information on the history of the Satavahana dynasty and on Queen Naganika's royal acts and sacrifices (land grants).                         |
| Panjtar inscription  | <b>Kushanas</b>                       | • Important epigraphic evidence relating to the expansion of the Kushana empire and its administration in India.                                          |

## 1 Foreign Invaders of the Post-Mauryan Period: Sequence, Rulers and Original Homeland

| No. | Invading Dynasty     | Most Famous Ruler                                                                                           | Capital / Important Facts                                                                                                                                                                                                                            | Original Homeland    |
|-----|----------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1   | Indo-Greeks          | Menander / Milinda                                                                                          | <ul style="list-style-type: none"> <li>• <b>Capital:</b> Shakala (present-day Sialkot).</li> <li>• Philosophical dialogue with the Kashmiri Buddhist monk Nagasena (text: Milindapanho).</li> </ul>                                                  | Bactria, Afghanistan |
| 2   | Shakas (Scythians)   | Nahapana (in western India)<br><i>(Note: Rudradaman was also a Shaka ruler, whose capital was Ujjayini)</i> | <ul style="list-style-type: none"> <li>• A historical mention of Nahapana is found in Satavahana inscriptions.</li> <li>• Rudradaman had the famous Sanskrit inscription of Junagadh engraved.</li> </ul>                                            | Scythia, Iran        |
| 3   | Pahlavas (Parthians) | Gondophernes                                                                                                | <ul style="list-style-type: none"> <li>• Evidence of their coins has been found at <b>Begram (Afghanistan)</b>.</li> </ul>                                                                                                                           | Parthia, Iran        |
| 4   | Kushanas (Yuezhi)    | Kanishka<br><i>(real founder: Kujula Kadphises)</i>                                                         | <ul style="list-style-type: none"> <li>• <b>Capitals:</b> Purushapura (present-day Peshawar) and Mathura (two capitals of the empire).</li> <li>• Started the <b>Shaka Samvat (Shaka era)</b> in 78 CE (used by the Government of India).</li> </ul> | Chinese Turkestan    |

## 2 Important Rulers of the Kushana Empire

| Kushana Ruler    | Reign (approx.) | Important Historical Facts and Evidence (Exam Facts)                                                                                                                                                                                                                                                                                                     |
|------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kujula Kadphises | 30 CE - 80 CE   | <ul style="list-style-type: none"> <li>• Regarded as the <b>real founder</b> of Kushana power in India; he was the father of Vima Kadphises.</li> </ul>                                                                                                                                                                                                  |
| Vima Kadphises   | 95 CE - 127 CE  | <ul style="list-style-type: none"> <li>• The first Kushana ruler to issue <b>pure gold coins</b>. His coins depict <b>Shiva and Nandi</b>.</li> <li>• <b>Rabatak inscription evidence:</b> according to it he was the son of Sadashkana and the father of Kanishka.</li> </ul>                                                                           |
| Kanishka         | 127 CE - 150 CE | <ul style="list-style-type: none"> <li>• The most famous Kushana ruler and a great patron of Buddhism (convened the Fourth Buddhist Council).</li> <li>• Started the <b>Shaka Samvat</b> calendar in <b>78 CE</b>.</li> <li>• Built the Kanishka stupa at Purushapura. <b>Gandhara and Mathura art</b> developed unprecedentedly in his time.</li> </ul> |

### 3 Distinctive Titles of Post-Mauryan Kings

| Ruler (Dynasty)                                | Titles Held                                                                                                                                                                       | Historical Sources and Evidence Useful for the Exam                                                                                                                                                                                                              |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pushyamitra Shunga<br>(Shunga dynasty)         | <ul style="list-style-type: none"> <li>'Senani' (commander)</li> <li>'Audbhijja' (one who rises suddenly)</li> </ul>                                                              | <ul style="list-style-type: none"> <li>The <b>Harivamsha Purana</b> calls him '<b>Ashvamedha Yajin</b>' (performer of two Ashvamedha sacrifices).</li> <li>An authentic mention of his sacrifices is found in Dhanadeva's <b>Ayodhya inscription</b>.</li> </ul> |
| Kharavela<br>(Chedi dynasty, Kalinga)          | <ul style="list-style-type: none"> <li>'Rajarshi', 'Lemaraja', 'Vridddharaja'</li> <li>'Bhikshuraja', 'Dharmaraja', 'Mahavijayaraja'</li> </ul>                                   | <ul style="list-style-type: none"> <li>The only source of information on all these titles and on his life is the <b>Hathigumpha inscription</b> at Udayagiri.</li> </ul>                                                                                         |
| Satakarni I<br>(Satavahana dynasty)            | <ul style="list-style-type: none"> <li>'Lord of Dakshinapatha'</li> <li>'Wielder of the unchecked wheel (of sovereignty)'</li> <li>'Pashchimapati', 'Pratishthanapati'</li> </ul> | <ul style="list-style-type: none"> <li>An important ruler who established Satavahana power.</li> </ul>                                                                                                                                                           |
| Gautamiputra Satakarni<br>(Satavahana dynasty) | <ul style="list-style-type: none"> <li>'Advitiya Brahmana' (the unique Brahmana)</li> <li>'Refuge of the Vedas' (Dvijakulavardhana)</li> </ul>                                    | <ul style="list-style-type: none"> <li>These titles are clearly mentioned in the Nasik prashasti (composed by Gautami Balashri).</li> </ul>                                                                                                                      |

### 4 History of Coins (Currency) in Ancient India

| Type of Coins / Metal          | Issuing Dynasty / Ruler                                  | Special Exam Note (Exam Warning)                                                                                                                                                |
|--------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First gold coins (gold)        | By the <b>Indo-Greek</b> rulers                          | <ul style="list-style-type: none"> <li>The earliest evidence of gold coinage in India comes from here.</li> </ul>                                                               |
| First pure gold coins          | By the <b>Kushana rulers</b> (by <b>Vima Kadphises</b> ) | <ul style="list-style-type: none"> <li>Kushana gold coins bear images of Greek, Iranian and Indian deities.</li> </ul>                                                          |
| First inscribed coins          | By the <b>Greek (Indo-Greek)</b> rulers                  | <ul style="list-style-type: none"> <li>Started the tradition of writing the names, portraits and titles of kings on coins.</li> </ul>                                           |
| Largest number of gold coins   | By the <b>Gupta rulers</b>                               | <ul style="list-style-type: none"> <li><b>Note:</b> the Guptas issued them in the largest number, but Chandragupta I was the first Gupta ruler to issue a gold coin.</li> </ul> |
| Largest number of copper coins | By the <b>Kushana rulers</b>                             | <ul style="list-style-type: none"> <li>Kushana copper coins have been found in large numbers, especially in rural areas.</li> </ul>                                             |
| Lead and tin (potin) coins     | By the <b>Satavahana rulers</b>                          | <ul style="list-style-type: none"> <li>Lead was the most prominent metal in the trade currency of the Satavahanas.</li> </ul>                                                   |



| Type of Coins / Metal                     | Issuing Dynasty / Ruler      | Special Exam Note (Exam Warning)                                                      |
|-------------------------------------------|------------------------------|---------------------------------------------------------------------------------------|
| Largest number of coins (on a wide scale) | By the <b>Mauryan rulers</b> | • <b>Punch-marked (ahata)</b> coins were widely in circulation in the Mauryan period. |

## 5 Gupta Dynasty: Sequence, Historical Facts and Queens

- **Gupta rule:** from about 275 CE to 550 CE.
- **Successive family tree of the rulers:** Shrigupta (founder) → Ghatotkacha → Chandragupta I → Samudragupta → Ramagupta → Chandragupta II → Kumaragupta I → Skandagupta → Purugupta → Kumaragupta II → Budhagupta → ... → Vishnugupta (last known Gupta ruler).

### > Gupta Rulers, their Queens and Principal Facts

| Gupta Ruler                              | Queen / Marriage Alliance                                                                | Titles and Principal Exam Facts (Core Facts)                                                                                                                                                                                                                       |
|------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chandragupta I                           | <b>Kumaradevi</b><br>( <i>Lichchhavi princess</i> )                                      | <ul style="list-style-type: none"> <li>• The <b>real founder</b> of the Gupta dynasty.</li> <li>• The first Gupta ruler to issue a gold coin. Started the '<b>Gupta Samvat</b>' (<b>Gupta era</b>) in <b>319 CE</b>.</li> </ul>                                    |
| Samudragupta                             | <b>Dattadevi</b>                                                                         | <ul style="list-style-type: none"> <li>• Title: '<b>Napoleon of India</b>' (given by the historian Vincent Smith).</li> <li>• His conquests are mentioned in Harishena's '<b>Prayag Prashasti</b>'. Issued gold coins of the Ashvamedha type.</li> </ul>           |
| Chandragupta II                          | <b>Kuberanaga</b> ( <i>Naga dynasty</i> ) and <b>Dhruvasvamini</b> ( <i>Dhruvadevi</i> ) | <ul style="list-style-type: none"> <li>• Title: '<b>Vikramaditya</b>'. The contemporary Chinese traveller <b>Fa-Hien</b> came to India.</li> <li>• The first Gupta ruler to issue <b>silver coins</b>. The Mehrauli iron pillar is associated with him.</li> </ul> |
| Kumaragupta I                            | <b>Anantadevi</b>                                                                        | <ul style="list-style-type: none"> <li>• Title: '<b>Mahendraditya</b>'. <b>Founder of the famous Nalanda University</b> in Bihar.</li> </ul>                                                                                                                       |
| Rudrasena II<br>( <i>Vakataka king</i> ) | <b>Prabhavati Gupta</b><br>( <i>daughter of Chandragupta II</i> )                        | <ul style="list-style-type: none"> <li>• <b>Note:</b> he was not a Gupta ruler but a Vakataka king contemporary with the Gupta empire, with whom Chandragupta II formed a matrimonial alliance.</li> </ul>                                                         |

### > The 'Navaratnas' (Nine Gems) of the Court of Chandragupta II (Vikramaditya)

1. **Kalidasa** (great poet and writer) | 2. **Dhanvantari** (famous physician) | 3. **Varahamihira** (astronomer) | 4. **Amarasimha** (lexicographer - author of the Amarakosha) | 5. **Kshapanaka** (astrologer) | 6. **Shanku** (architect) | 7. **Vetala Bhatta** (magician) | 8. **Ghatakarpara** (skilled diplomat) | 9. **Vararuchi** (great grammarian)

## > Principal Kings of the Gupta Period and Related Inscriptions

| King / Ruler    | Related Inscription                                   | Excavator / Composer / Historical Evidence Found                                                                                                         |
|-----------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Samudragupta    | <b>Eran inscription</b><br>(and the Prayag Prashasti) | • The Eran inscription describes Samudragupta as benevolent and just, <b>like Kubera</b> , the god of wealth, and <b>Yama</b> , the god of death.        |
| Chandragupta II | <b>Udayagiri Cave inscription</b>                     | • Engraved by his minister <b>Virasena (Shaba)</b> . (The Mehrauli iron pillar of Delhi is also associated with him).                                    |
| Kumaragupta I   | <b>Mandsaur inscription</b>                           | • Composer: <b>Vatsabhathi</b> . This inscription gives important information on the activities of the <b>guild of silk weavers</b> in the Gupta period. |
| Patika          | <b>Taxila inscription</b>                             | • <b>Note:</b> he was not a Gupta ruler; this is an authentic copper-plate inscription of the Kushana/Shaka period.                                      |

## 7 Principal Administrative Officers of the Gupta Period and their Offices

*(The most important table for the administrative terminology of UPPCS Prelims)*

| Administrative Post (Officer) | Department / Main Area of Work (Core Responsibility)                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| Kumaramatyā                   | The <b>highest administrative officer</b> of the Gupta empire (equivalent to the IAS).                      |
| Mahadandanayaka               | <b>Chief Justice</b> ; a post carrying high responsibilities, both military and judicial.                   |
| Maha-sandhivigrahika          | <b>Minister of war, peace and foreign affairs</b> (the court poet <b>Harishena</b> held this post).         |
| Maha-baladhikrita             | <b>Commander-in-chief</b> / supreme head of the military department.                                        |
| Maha-akshapatalika            | <b>Supreme head of the accounts department</b> / superintendent of records and accounts.                    |
| Pratihara                     | Guard of the security of only the <b>inner part (harem/antahpura)</b> of the royal palace.                  |
| Mahapratihara                 | High officer in charge of the <b>security of the whole palace and of the king</b> (head of the bodyguards). |
| Ashvapati / Mahashvapati      | Officer in charge/head of the cavalry (horses).                                                             |
| Pilupati                      | Supreme officer in charge of the elephant corps ( <b>army of elephants</b> ).                               |
| Narapati                      | Chief military officer concerned with the <b>infantry</b> .                                                 |

| Administrative Post (Officer) | Department / Main Area of Work (Core Responsibility)                                                         |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|
| Ranabhandagarika              | Officer in charge of <b>military supplies (provisions)</b> and stores of military material.                  |
| Nagara Shreshthi              | Leading merchant who supervised city administration; representative of the mercantile community.             |
| Gramadhyaksha / Gramika       | <b>Village headman</b> ; supreme head of rural administration.                                               |
| Dutakarma / Dutaka            | Officer who conveyed the king's royal orders, gifts and <b>land-grant charters</b> to the persons concerned. |

## 1 Administrative Units of the Gupta Period

Arranged in *ascending order*

> Grama (smallest) → Petha → Vithi → Vishaya → Bhukti → Desha or Rashtra (largest)

| Administrative Unit            | Head / Administrator | Special Exam Facts                                                                    |
|--------------------------------|----------------------|---------------------------------------------------------------------------------------|
| Grama ( <i>smallest unit</i> ) | Gramika              | The highest officer of village-level administration.                                  |
| Petha                          | —                    | A group of several villages (a unit larger than the village).                         |
| Vithi                          | —                    | An administrative unit larger than the Petha and smaller than the Vishaya (district). |
| Vishaya ( <i>district</i> )    | Vishayapati          | The principal district-level unit of Gupta administration.                            |
| Bhukti ( <i>province</i> )     | Uparika              | The largest administrative division (province) of the empire.                         |
| Desha or Rashtra               | King / Emperor       | The supreme administrative and central unit of the Gupta empire.                      |

## 2 Administrative Terminology Related to Land in the Gupta Period

| Type of Land   | Real Meaning and Use (Core Meaning)                     |
|----------------|---------------------------------------------------------|
| Kshetra        | Land <b>fit for cultivation</b> (suitable for farming). |
| Vastu          | Land <b>fit for habitation</b> (for building houses).   |
| Charagaha land | Grasslands suitable for the grazing of cattle.          |

| Type of Land | Real Meaning and Use (Core Meaning)                               |
|--------------|-------------------------------------------------------------------|
| Khila        | Land that was <b>not fit for ploughing</b> (barren or infertile). |
| Aprahata     | Land that was <b>forest or uncultivated (unploughed)</b> .        |

### 3 Important Literary Works of the Gupta Period and of Ancient India

| Literary Work (Book)                                                                                                                                                                                                | Author                       | Specific Details / Exam Note                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Abhijnanashakuntalam,</li> <li>Malavikagnimitram,</li> <li>Vikramorvashiyam</li> <li>• Raghuvamsham,</li> <li>Kumarasambhavam</li> <li>• Meghaduta, Ritusamhara</li> </ul> | <b>Kalidasa</b>              | <ul style="list-style-type: none"> <li>• The Meghaduta has two parts: <b>Purvamegha and Uttaramegha</b>.</li> <li>• Malavikagnimitram, Abhijnanashakuntalam and Vikramorvashiyam are his famous plays.</li> </ul> |
| <ul style="list-style-type: none"> <li>• Mudrarakshasa</li> <li>• Devichandraguptam</li> </ul>                                                                                                                      | <b>Vishakhadatta</b>         | <ul style="list-style-type: none"> <li>• Mudrarakshasa throws light on Mauryan history and Devichandraguptam on Gupta history.</li> </ul>                                                                         |
| <ul style="list-style-type: none"> <li>• Brihatsamhita</li> <li>• Panchasiddhantika</li> </ul>                                                                                                                      | <b>Varahamihira</b>          | <ul style="list-style-type: none"> <li>• Famous Gupta-period texts on astronomy and astrology.</li> </ul>                                                                                                         |
| <b>Brahmasiddhanta</b>                                                                                                                                                                                              | <b>Brahmagupta</b>           | <ul style="list-style-type: none"> <li>• He is also called the Newton of India.</li> </ul>                                                                                                                        |
| <ul style="list-style-type: none"> <li>• Aryabhatiyam</li> <li>• Suryasiddhanta</li> </ul>                                                                                                                          | <b>Aryabhata</b>             | <ul style="list-style-type: none"> <li>• The great mathematician and astronomer of ancient India.</li> </ul>                                                                                                      |
| <b>Nyayavatara</b>                                                                                                                                                                                                  | <b>Siddhasena (Divakara)</b> | <ul style="list-style-type: none"> <li>• The early systematic text of Jain Nyaya philosophy.</li> </ul>                                                                                                           |
| <b>Panchatantra</b>                                                                                                                                                                                                 | <b>Vishnu Sharma</b>         | <ul style="list-style-type: none"> <li>• A world-famous collection of moral fables.</li> </ul>                                                                                                                    |
| <ul style="list-style-type: none"> <li>• Nitisara</li> <li>• Nitishastra</li> </ul>                                                                                                                                 | <b>Kamandaka</b>             | <ul style="list-style-type: none"> <li>• Text based on Gupta-period government and politics.</li> </ul>                                                                                                           |
| <b>Mrichchhakatikam</b>                                                                                                                                                                                             | <b>Shudraka</b>              | <ul style="list-style-type: none"> <li>• Literally means 'The Little Clay Cart' (depicts the social system of the time).</li> </ul>                                                                               |
| <b>Kavyadarsha</b><br><b>Dashakumaracharita</b>                                                                                                                                                                     | <b>Dandin</b>                | <ul style="list-style-type: none"> <li>• Court poet of Narasimhavarman II (Pallava dynasty).</li> </ul>                                                                                                           |
| <b>Svapnavasavadatta</b><br><b>Charudatta</b>                                                                                                                                                                       | <b>Bhasa</b>                 | <ul style="list-style-type: none"> <li>• A great dramatist of ancient India.</li> </ul>                                                                                                                           |
| <b>Kiratarjuniyam</b>                                                                                                                                                                                               | <b>Bharavi</b>               | <ul style="list-style-type: none"> <li>• Court poet of Simhavishnu (Pallava dynasty).</li> </ul>                                                                                                                  |
| <b>Ravanavadha</b>                                                                                                                                                                                                  | <b>Bhatti</b>                | <ul style="list-style-type: none"> <li>• Also known as the '<b>Bhattikavya</b>'.</li> </ul>                                                                                                                       |

| Literary Work (Book) | Author              | Specific Details / Exam Note                                                |
|----------------------|---------------------|-----------------------------------------------------------------------------|
| Amarakosha           | Amarasimha          | • One of the Navaratnas of Chandragupta II (famous Sanskrit lexicographer). |
| Visuddhimagga        | Buddhaghosha        | • A highly important text of Buddhism.                                      |
| Kamasutra            | Vatsyayana          | • The authoritative text of ancient Indian erotics (Kamashastra).           |
| Charaka Samhita      | Charaka             | • Famous Ayurvedic text related to the Kushana period (Kanishka).           |
| Ramayana             | Maharishi Valmiki   | • Sanskrit epic.                                                            |
| Mahabharata          | Maharishi Vedavyasa | • The largest epic of the world.                                            |
| Nitishataka          | Bhartrihari         | • Famous Sanskrit shataka (century of verses) poem.                         |
| Yogachara            | Asanga              | • Related to the Vijnanavada/Yogachara school of Buddhist philosophy.       |

#### 4 Gupta Architecture: Principal Temples and their Location

| Name of Temple                     | Location       | District and State                            |
|------------------------------------|----------------|-----------------------------------------------|
| Dashavatara Temple of Deogarh      | Deogarh        | Lalitpur district, Uttar Pradesh (UP Special) |
| Krishna/Brick Temple of Bhitargaon | Bhitargaon     | Kanpur district, Uttar Pradesh (UP Special)   |
| Vishnu Temple of Eran              | Eran           | Sagar district, Madhya Pradesh                |
| Shiva Temple of Bhumara            | Bhumara        | Satna district, Madhya Pradesh                |
| Vishnu Temple of Pipariya          | Pipariya       | Satna district, Madhya Pradesh                |
| Vishnu Temple of Tigawa            | Tigawa         | Jabalpur, Madhya Pradesh                      |
| Parvati Temple of Nachna-Kuthara   | Nachna-Kuthara | Panna, Madhya Pradesh                         |
| Lakshmana Temple of Sirpur         | Sirpur         | Raipur district, Chhattisgarh                 |

#### 5 Vardhana Dynasty (Pushyabhuti Dynasty)

- **Founder:** Pushyabhuti.
- **Sequence of rulers:** Naravardhana → Rajyavardhana I → Adityavardhana → Prabhakaravardhana → Rajyavardhana II → Harshavardhana.



## > Emperor Harshavardhana (606–647 CE)

- **Family background:** Father—Prabhakaravardhana, Mother—Yashomati.
- **Other name and title:** Other name—**Shiladitya**. Title—**Sakalottarapathanatha** (Lord of the whole of North India).
- **Change of capital:** His capital was first **Thaneshwar**, which he later shifted to **Kannauj (Mahodayashri, Uttar Pradesh)**.
- **Literary works:** He himself was the author of three famous Sanskrit plays—1. **Priyadarshika**, 2. **Ratnavali**, 3. **Nagananda**.
- **Aihole inscription evidence (historic battle):** According to the Aihole inscription composed by Ravikirti, court poet of the Chalukya king **Pulakeshin II**, a fierce battle took place between Harsha and Pulakeshin II on the bank of the Narmada, in which **Pulakeshin II was victorious** and Harsha's southern conquest was halted.

## 6 Pallava Dynasty: Founder, Rulers and Cultural Contribution

- **Founder:** **Simhavishnu** (some historians also regard Bappadeva as the founder).
- **Capital:** **Kanchi (Kanchipuram)**.
- **Last ruler:** **Aparajitavarman**.

## > Principal Pallava Rulers and their Works

| Pallava Ruler (Period)                            | Titles / Court Scholars                                                                                                       | Architecture and Historical Achievements (Exam Facts)                                                                                                                                                                                                                 |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simhavishnu</b><br>(575-600 CE)                | <ul style="list-style-type: none"> <li>● <b>Bharavi</b>, author of the '<b>Kirtarjuniyam</b>', lived at his court.</li> </ul> | <ul style="list-style-type: none"> <li>● Pioneer of the revival of Pallava power; extended the kingdom up to the Chola country (Kaveri river).</li> </ul>                                                                                                             |
| <b>Mahendravarman I</b><br>(600-630 CE)           | <ul style="list-style-type: none"> <li>● He himself wrote the famous Sanskrit play '<b>Mattavilasa Prahasana</b>'.</li> </ul> | <ul style="list-style-type: none"> <li>● Began <b>rock-cut (cut out of hills) cave-temple architecture</b> (monolithic temples) in South India.</li> </ul>                                                                                                            |
| <b>Narasimhavarman I</b><br>(630-668 CE)          | <ul style="list-style-type: none"> <li>● Titles: '<b>Vatapikonda</b>' and '<b>Mahamalla</b>'.</li> </ul>                      | <ul style="list-style-type: none"> <li>● In 642 CE he <b>captured Vatapi (Badami)</b>, the capital of the Chalukyas.</li> <li>● Began the construction of the famous '<b>Ratha temples</b>' (<b>Seven Pagodas</b>) of <b>Mahabalipuram (Mamallapuram)</b>.</li> </ul> |
| <b>Mahendravarman II and Parameshvaravarman I</b> | —                                                                                                                             | <ul style="list-style-type: none"> <li>● Period of continuous conflict with the Chalukya ruler Vikramaditya I and of Pallava–Chalukya wars.</li> </ul>                                                                                                                |

| Pallava Ruler (Period)                                         | Titles / Court Scholars                                                                       | Architecture and Historical Achievements (Exam Facts)                                                                                                                                                                         |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Narasimhavarman II</b><br>( <i>Rajasimha</i> ) (695-722 CE) | • <b>Dandin</b> , the famous author of the ' <b>Dashakumaracharita</b> ', lived at his court. | <ul style="list-style-type: none"> <li>Used the new '<b>Rajasimha style</b>' of temple building.</li> <li>Built the famous '<b>Kailashanatha Temple</b>' of Kanchi and the '<b>Shore Temple</b>' of Mahabalipuram.</li> </ul> |
| <b>Parameshvaravarman II</b><br>(722-731 CE)                   | —                                                                                             | <ul style="list-style-type: none"> <li>The final period of the decline of the Pallava dynasty and of intense succession struggles.</li> </ul>                                                                                 |

## 7 Administrative and Local Self-Government System of the Chola Period

### > Administrative Divisions of the Chola Empire (in descending order of seniority)

Mandalam (province) → Valanadu (commissionery) → Nadu (district) → Kurram / Grama (smallest unit)

### > Principal Chola Officials and Bodies

| Administrative Post / Body      | Department / Actual Responsibility (Core Responsibility)                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Olai</b>                     | Principal duty of implementing the king's oral and written orders on the ground.                                  |
| <b>Tirumandira Olai Nayagam</b> | The chief high officer who <b>recorded on palm leaf the oral orders</b> given by the king in the public assembly. |
| <b>Perundaram</b>               | Administrative officers of the <b>upper grade (first class)</b> of the Chola central administration.              |
| <b>Sirudaram</b>                | Administrative officers of the <b>lower grade (second class)</b> of the Chola administration.                     |
| <b>Viceroy</b>                  | Governors/leaders of the provinces, who were usually members of the royal family or close to the king.            |
| <b>Nattar</b>                   | Leading persons who headed the local administrative assembly of the Nadu (district).                              |
| <b>Nattuvagai Nadu Council</b>  | Local council supervising revenue, justice and administration at the level of the Nadu (district).                |
| <b>Nattuvais</b>                | Officials connected with local agriculture, land survey and rural agricultural administration.                    |

### > Local Self-Government of the Chola Period (Village Assembly and Committees)

(According to the famous **Uttaramerur inscription**, the working of Chola village government was as follows)

| Village Assembly / Committee | Practical Form and Method of Selection (Functional Domain)                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Ur                           | The basic assembly of the common residents of ordinary (non-Brahmin) villages.                                     |
| Sabha or Mahasabha           | The central assembly of learned Brahmins of <b>Brahmadeya villages (villages granted to Brahmins)</b> .            |
| Nagaram                      | Mainly the <b>administrative assembly of merchants, artisans and trading towns</b> .                               |
| Kudavolai                    | The ' <b>lottery method</b> ' ( <b>system of slips</b> ) used to select the members of the committees (variya).    |
| Variyam                      | <b>Special working committees</b> of men formed to run the various administrative functions of village government. |
| ↳ <i>Eri Variyam</i>         | Committee for the maintenance of <b>irrigation and tanks</b> .                                                     |
| ↳ <i>Totta Variyam</i>       | Committee for the management of <b>gardens (orchards)</b> .                                                        |
| ↳ <i>Pon Variyam</i>         | Committee that examined <b>gold, currency and finance (revenue)</b> .                                              |

### > Chola Dynasty: Rulers, Titles and Historical Facts

- **Founder:** Vijayalaya.
- **Last ruler:** Rajendra III.
- **Capital:** Thanjavur (Tanjore) → later **Gangaikonda Cholapuram**.

| Chola Ruler   | Titles           | Important Facts and Evidence Useful for the Exam (Exam Facts)                                                                                                                                                                                            |
|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parantaka I   | • 'Maduraikonda' | <ul style="list-style-type: none"> <li>• Completely eliminated the Pallavas.</li> <li>• <b>Battle of Takkolam (949 CE):</b> defeated by the Rashtrakuta ruler Krishna III.</li> </ul>                                                                    |
| Rajaraja I    | —                | <ul style="list-style-type: none"> <li>• Built the famous <b>Brihadishvara Temple</b> (Rajarajeshvara Temple) at Thanjavur.</li> </ul>                                                                                                                   |
| Rajendra I    | • 'Gangaikonda'  | <ul style="list-style-type: none"> <li>• Defeated the Pala ruler Mahipala and the Eastern Ganga ruler in order to bring Ganga water.</li> <li>• Founded the new capital <b>Gangaikonda Cholapuram</b> and built a Brihadishvara temple there.</li> </ul> |
| Kulottunga II | —                | <ul style="list-style-type: none"> <li>• Had the <b>image of Govindaraja (Vishnu)</b> removed from the Chidambaram Nataraja temple and thrown into the sea.</li> </ul>                                                                                   |

### > The Principal Feature of Chola Administration: Autonomy of Village Administration

Committees (**variya**) were set up to supervise the activities of the villages:

- **Eri Variyam:** Tank committee (the notes say 'Uri Variyam'; the standard form is 'Eri')
- **Pon Variyam:** Gold committee (financial system)
- **Totta Variyam:** Garden/orchard committee (the notes say 'Toh Variyam'; the standard form is 'Totta')
- **Udasina Variyam:** Committee to supervise royal directions/rules

## 2 Principal Sun Temples of India and their Builders

| Name of Sun Temple | Location   | Builder / Ruler (Dynasty)                      | Special Historical Fact                                       |
|--------------------|------------|------------------------------------------------|---------------------------------------------------------------|
| Martand Sun Temple | Kashmir    | <b>Lalitaditya Muktapida</b> (Karkota dynasty) | Famous ancient temple of Kashmir.                             |
| Konark Sun Temple  | Odisha     | <b>Narasimhadeva I</b> (Eastern Ganga dynasty) | Also called the ' <b>Black Pagoda</b> '.                      |
| Modhera Sun Temple | Gujarat    | <b>Bhimadeva I</b> (Solanki/Chalukya dynasty)  | India's <b>largest Surya Kund (sun tank)</b> is located here. |
| Tanjore Sun Temple | Tamil Nadu | Related to the Chola period                    | <b>The only</b> sun temple <b>of South India</b> .            |

## 3 Outline of the Three Sangams (Tamil Assemblies)

### Note

**Note:** All three Sangams were held under the royal patronage of the **Pandya kings**.

| Sangam | Venue                        | President                                                      | Members / Literature Compiled                                                                                                                 |
|--------|------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| First  | <b>Madurai</b>               | <b>Sage Agastya</b>                                            | <ul style="list-style-type: none"> <li>• 549 members.</li> <li>• Texts compiled: Akattiyam, Mudunarai, Paripadal, Mudukurukku.</li> </ul>     |
| Second | <b>Kapatapuram</b> (Alaivai) | <b>Sage Agastya</b><br><i>(later his disciple Tolkappiyar)</i> | <ul style="list-style-type: none"> <li>• 59 members.</li> <li>• Literature compiled: <b>Tolkappiyam</b> (a text on Tamil grammar).</li> </ul> |
| Third  | <b>Madurai</b>               | <b>Nakkirar</b>                                                | <ul style="list-style-type: none"> <li>• 49 members.</li> <li>• Works compiled: Nedunthogai, Kurunthogai, Ainkurunuru, Paripadal.</li> </ul>  |

## 4 Important Sangam Literature and their Authors

| Sangam Text                                                | Author                                 | Subject Matter / Internal Feature (Core Themes)                                                                                                                                                                                                              |
|------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Silappadikaram</b><br>( <i>The Tale of the Anklet</i> ) | <b>Ilango Adigal</b>                   | <ul style="list-style-type: none"> <li>One of the 5 great epics of Tamil literature.</li> <li>The love story of the hero Kovalan, the heroine Kannagi and the dancer Madhavi. Later the '<b>Kannagi (Pattini) cult</b>' developed in South India.</li> </ul> |
| <b>Manimekalai</b>                                         | <b>Sittalai Sattanar</b>               | <ul style="list-style-type: none"> <li>A Tamil epic related to <b>Buddhism</b>.</li> <li>It is the sequel to the Silappadikaram; its heroine is <b>Manimekalai (daughter of Kovalan and Madhavi)</b>.</li> </ul>                                             |
| <b>Tolkappiyam</b>                                         | <b>Tolkappiyar</b>                     | <ul style="list-style-type: none"> <li>The only surviving work of the Second Sangam; the oldest text on <b>Tamil grammar and poetics</b>.</li> <li>It has three parts—letters (ezhuttu), words (sol) and meaning (porul).</li> </ul>                         |
| <b>Tirukkural (or Kural)</b>                               | <b>Tiruvalluvar</b>                    | <ul style="list-style-type: none"> <li>The 'Fifth Veda' of Tamil literature; related to <b>philosophy</b> (1330 couplets, 133 chapters).</li> <li>It has three parts—Aram (dharma/virtue), Porul (artha/wealth) and Inbam (kama/love).</li> </ul>            |
| <b>Jivaka Chintamani</b>                                   | <b>Tiruttakkadevar</b>                 | <ul style="list-style-type: none"> <li>Counted among the great epics of Tamil literature; related to <b>Jainism</b> (the author was a Jain ascetic of Madurai).</li> </ul>                                                                                   |
| <b>Kundalakesi</b>                                         | <b>Nathakuthanar</b><br>(Nagakuthanar) | <ul style="list-style-type: none"> <li>A <b>Buddhist Tamil epic</b>; the tragic love story of Kundalakesi, a girl of the merchant caste, and the Buddhist criminal Kalan.</li> </ul>                                                                         |
| <b>Ahananuru (Ahanar)</b>                                  | <b>Rudrasharma</b>                     | <ul style="list-style-type: none"> <li>A collection of famous love poems of the Sangam age.</li> </ul>                                                                                                                                                       |
| <b>Murugarruppadai</b>                                     | <b>Nakkirar</b>                        | <ul style="list-style-type: none"> <li>Religious text in praise of Lord Murugan (Kartikeya).</li> </ul>                                                                                                                                                      |

## 5 Principal Coastal and Administrative Towns of the Sangam Age

| Name of Town                  | Dynasty       | Historical Position and Geographical Importance (Exam Points)                                                                                       |
|-------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Uraiyur</b>                | <b>Chola</b>  | <ul style="list-style-type: none"> <li>Early inland capital of the Cholas; in ancient times a <b>very large centre of cotton cloth</b>.</li> </ul>  |
| <b>Puhar (Kaveripattinam)</b> | <b>Chola</b>  | <ul style="list-style-type: none"> <li>Second and coastal capital of the Cholas; the famous <b>international port</b> of the Sangam age.</li> </ul> |
| <b>Madurai</b>                | <b>Pandya</b> | <ul style="list-style-type: none"> <li>Main capital of the Pandya rulers and the central venue of the Sangam assemblies.</li> </ul>                 |
| <b>Korkai</b>                 | <b>Pandya</b> | <ul style="list-style-type: none"> <li>Second and <b>coastal capital</b> of the Pandya dynasty (famous for pearls).</li> </ul>                      |



| Name of Town      | Dynasty | Historical Position and Geographical Importance (Exam Points)   |
|-------------------|---------|-----------------------------------------------------------------|
| Vanji (or Karur)  | Chera   | • Main administrative capital of the Chera dynasty.             |
| Mushiri (Muziris) | Chera   | • A highly famous and busy <b>sea port</b> of the Chera period. |

## 6 Principal Texts and Musicians of Medieval Indian Music

| Musician           | Music Text (Book)                                                                                                                       | Philosophical Feature of the Text / System Propounded (Core Concept)                                                                                                                                                                                                                      |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pundarika Vitthala | <ul style="list-style-type: none"> <li>• <b>Shadragachandrodaya</b></li> <li>• <b>Ragamala</b></li> <li>• <b>Ragamanjari</b></li> </ul> | <ul style="list-style-type: none"> <li>• He composed the '<b>Ragamanjari</b>' at the instance of Raja Man Singh.</li> <li>• The Ragamanjari contains the first classification in Indian music of <b>15 ragas of Persian origin</b>. The 'Sadragachandrodaya' is also his work.</li> </ul> |
| Shrikantha         | • <b>Rasa Kaumudi</b>                                                                                                                   | <ul style="list-style-type: none"> <li>• He was a worthy pupil of Pundarika Vitthala and court poet of Satrusalya.</li> <li>• This text deals with vocal music, instrumental music and <b>dance</b>.</li> </ul>                                                                           |
| Somanatha          | • <b>Raga-vibodha</b>                                                                                                                   | <ul style="list-style-type: none"> <li>• He was a Telugu Brahmin of Rajahmundry.</li> <li>• This text contains a scientific treatment of the '<b>Janya-Janaka</b>' or '<b>Thaat-Raga system</b>' for classifying ragas. It has a scheme of <b>23 melas (thaats)</b>.</li> </ul>           |
| Venkatamakhin      | <ul style="list-style-type: none"> <li>• <b>Chaturdandiprakashika</b><br/>(composed c. 1660 CE)</li> </ul>                              | <ul style="list-style-type: none"> <li>• He was a great musicologist of the Carnatic music tradition.</li> <li>• He introduced the '<b>scientific system of 72 melas (thaats)</b>' to organise the ragas of Carnatic music, which is still regarded as authoritative.</li> </ul>          |

## 1 Important Foreign Travellers and Ambassadors

| Foreign Traveller | Period      | Country of Origin | Contemporary Indian Ruler             | Travel Account / Book and Main Facts                                                                                                                                                       |
|-------------------|-------------|-------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Megasthenes       | 302-298 BCE | Greece            | <b>Chandragupta Maurya</b>            | <ul style="list-style-type: none"> <li>• Was the ambassador of Seleucus Nicator.</li> <li>• Famous book: '<b>Indica</b>'.</li> </ul>                                                       |
| Fa-Hien           | 399-414 CE  | China             | <b>Chandragupta II (Vikramaditya)</b> | <ul style="list-style-type: none"> <li>• The <b>first Chinese Buddhist monk</b> to come to India.</li> <li>• Famous book: '<b>Fo-Guo-Ji</b>' (Record of the Buddhist Kingdoms).</li> </ul> |

| Foreign Traveller                  | Period                        | Country of Origin | Contemporary Indian Ruler                                              | Travel Account / Book and Main Facts                                                                                                                                      |
|------------------------------------|-------------------------------|-------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hiuen Tsang<br>( <i>Xuanzang</i> ) | 630-645 CE<br>(or 636-645 CE) | China             | <b>Harshavardhana</b><br>(also at the court of <i>Bhaskaravarman</i> ) | <ul style="list-style-type: none"> <li>Famous book: '<b>Si-Yu-Ki</b>' (Record of the Western Regions).</li> <li>Studied at <b>Nalanda University</b>.</li> </ul>          |
| I-Tsing                            | 671-695 CE                    | China             | Post-Gupta period                                                      | <ul style="list-style-type: none"> <li>Came to India by the <b>sea route</b>.</li> <li>Studied at Nalanda; book: '<i>Nanhai Jigui Neifa Zhuan</i>'.</li> </ul>            |
| Matwajin                           | —                             | China             | Chalukya ruler                                                         | <ul style="list-style-type: none"> <li>Mentioned the <b>cultural relations</b> between the Chalukya rulers and China.</li> </ul>                                          |
| Al-Masudi                          | 915 CE                        | Arabia            | Rashtrakuta/Gurjara-Pratihara period                                   | <ul style="list-style-type: none"> <li>Famous book: '<b>Muruj-ul-Zahab</b>'.</li> <li>He is also called the '<b>Herodotus of the Arabs</b>'.</li> </ul>                   |
| Al-Biruni<br>( <i>Abu Raihan</i> ) | 1024-1030 CE                  | Khwarizm (Arabia) | <b>Mahmud of Ghazni</b> (came with him)                                | <ul style="list-style-type: none"> <li>Famous book: '<b>Kitab-ul-Hind</b>' (<b>Tahqiq-i-Hind</b>).</li> <li>The first Muslim scholar to <b>study Sanskrit</b>.</li> </ul> |
| Ma Huan                            | 1451 CE                       | China             | Medieval period (Sayyid/Lodi period)                                   | <ul style="list-style-type: none"> <li>A Chinese traveller who visited India in the medieval period.</li> </ul>                                                           |

## 2 Principal Historical Battles of Ancient India

(*Most important for matching dates and victorious sides in UPPCS Prelims*)

| Name of Battle                       | Time (Period)           | Fought Between (Contenders)                 | Victorious Side            | Special UPPCS Note (Core Facts)                                               |
|--------------------------------------|-------------------------|---------------------------------------------|----------------------------|-------------------------------------------------------------------------------|
| Dasharajna War (Battle of Ten Kings) | Rigvedic period         | King Sudas vs. a confederacy of 10 kings    | <b>King Sudas</b>          | Fought on the bank of the Parushni (Ravi) river.                              |
| Battle of the Hydaspes               | 326 BCE                 | Alexander vs. King Porus                    | <b>Alexander</b>           | Fought on the plain of the Jhelum river (Vitasta).                            |
| Chandragupta–Seleucus War            | 303 BCE<br>(or 305 BCE) | Chandragupta Maurya vs. Seleucus Nicator    | <b>Chandragupta Maurya</b> | As a result of the treaty, four provinces were added to the Maurya Empire.    |
| Kalinga War                          | 261 BCE                 | Emperor Ashoka vs. Kalinga (Padmanabha)     | <b>Emperor Ashoka</b>      | In the 8th year after Ashoka's coronation (described in the 13th rock edict). |
| Kanishka–China War                   | 1st century             | Kanishka vs. the Chinese emperor (Pan Chao) | <b>Kanishka</b>            | After defeat in the first phase, Kanishka was ultimately victorious.          |

| Name of Battle                                 | Time (Period) | Fought Between (Contenders)                                                   | Victorious Side           | Special UPPCS Note (Core Facts)                                                                     |
|------------------------------------------------|---------------|-------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------|
| Skandagupta–Huna War                           | 466 CE        | Skandagupta vs. the barbarian Huna invaders                                   | <b>Skandagupta</b>        | This victory is described in detail in the <b>Bhitari pillar inscription</b> .                      |
| Battle of the Narmada                          | 630 CE        | Harshavardhana vs. Pulakeshin II                                              | <b>Pulakeshin II</b>      | The Chalukya king Pulakeshin II halted Harsha's southern conquest (Aihole inscription).             |
| Battles of Suramara, Parisala and Manimangalai | 641-642 CE    | Narasimhavarman I vs. Pulakeshin II                                           | <b>Narasimhavarman I</b>  | The Pallava ruler Narasimhavarman I killed Pulakeshin II and took the title ' <b>Vatapikonda</b> '. |
| Battle of Sindh                                | 712 CE        | Raja Dahir Sen vs. Muhammad bin Qasim                                         | <b>Muhammad bin Qasim</b> | The first successful Arab/Muslim invasion of India.                                                 |
| Battle of Venni (Venuka)                       | Chola period  | Chola king vs. 11 kings including the Cheras and Pandyas                      | <b>Chola ruler</b>        | The early supremacy of the Cholas was established.                                                  |
| Battle of Takkolam                             | 949 CE        | Parantaka I (Chola) vs. the Rashtrakutas (Krishna III) and the Western Gangas | <b>Parantaka Chola</b>    | The Chola crown prince Rajaditya was killed in this battle against the Rashtrakuta Krishna III.     |
| Battle of Vellore                              | 955 CE        | Parantaka I (Chola) vs. the Pandyas and Sri Lanka                             | <b>Parantaka Chola</b>    | Parantaka I defeated the combined army of the Pandyas and Simhala (Sri Lanka).                      |
| Battle of Tilakaid                             | —             | Ganga dynasty vs. the Chalukyas (Vishnuvardhana)                              | <b>Vishnuvardhana</b>     | The Chalukya ruler of Vengi, Vishnuvardhana, was victorious.                                        |
| Battle of Koppam                               | 1052 CE       | Rajadhiraja (Chola) vs. Someshvara (Chalukya)                                 | <b>Chola ruler</b>        | When Rajadhiraja was killed on the battlefield, Virarajendra led the Cholas to victory.             |

### 3 Principal Historical Eras (Important Eras/Calendars)

*(The favourite UPPCS Prelims section for arranging in 'chronology', i.e. **chronological order**)*

| Name of Era              | Starting Year | Founder / Ruler (Dynasty)                     | Special Historical Exam Facts                                                                                                                 |
|--------------------------|---------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Vikrama Samvat        | 57 BCE        | The traditional King Vikramaditya of Ujjain   | • Also called the <b>Krita Samvat</b> or <b>Malava Samvat</b> ; its association with Vikramaditya is traditional.                             |
| 2. Shaka Samvat          | 78 CE         | The Kushana king Kanishka                     | • Associated with the victory over the Shakas or with Kanishka's accession. <b>The official national calendar of the Government of India.</b> |
| 3. Kalachuri Samvat      | 248 CE        | The Abhira king Ishvarasena                   | • Historically also known as the ' <b>Chedi Samvat</b> '.                                                                                     |
| 4. Gupta Samvat          | 319–320 CE    | Chandragupta I (Gupta dynasty)                | • Started to commemorate the accession of Chandragupta I.                                                                                     |
| 5. Harsha Samvat         | 606 CE        | Harshavardhana (Vardhana/Pushyabhuti dynasty) | • Associated with the accession of Harshavardhana, ruler of Kannauj.                                                                          |
| 6. Hijri Samvat          | 622 CE        | Prophet Muhammad                              | • In memory of the Prophet Muhammad's <b>migration from Mecca to Medina (Hijrat)</b> (622 CE = 1 Hijri).                                      |
| 7. Kollam Samvat         | 825 CE        | Adi Shankaracharya (Malabar)                  | • In the regional language it is also called the ' <b>Malayalam Samvat</b> '.                                                                 |
| 8. Simha / Siddha Samvat | 1113 CE       | Jayasimha Siddharaja (Chalukya/Solanki)       | • Issued by Jayasimha Siddharaja, the powerful Chalukya ruler of Gujarat.                                                                     |
| 9. Lakshmana Samvat      | 1190 CE       | Lakshmana Sena (Sena dynasty, Bengal)         | • Issued in the Mithila/Bengal region by the ruler of the Sena dynasty of Bengal.                                                             |
| 10. Ilahi Samvat         | 1583 CE       | Mughal emperor Akbar                          | • A solar calendar issued after the 'Din-i-Ilahi' propounded by Akbar.                                                                        |
| 11. Rajashaka Samvat     | 1673 CE       | Chhatrapati Shivaji Maharaj                   | • Began on the occasion of the <b>coronation</b> of Chhatrapati Shivaji Maharaj of the Maratha empire.                                        |

## 1 Delhi Sultanate: Dynasties and Rulers (1206 - 1526)

| Dynasty (Tenure)                   | Rulers (in sequence)                                                                            |
|------------------------------------|-------------------------------------------------------------------------------------------------|
| Slave (Mamluk) dynasty (1206-1290) | Qutbuddin Aibak → Aram Shah → Iltutmish → Razia Sultan → Bahram Shah → Balban → Kayumars        |
| Khilji dynasty (1290-1320)         | Jalaluddin Khilji → Alauddin Khilji → Shihabuddin Umar → Qutbuddin Mubarak → Nasiruddin Khusrau |
|                                    |                                                                                                 |

| Dynasty (Tenure)            | Rulers (in sequence)                                                                                                                                                                                                              |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tughlaq dynasty (1320-1412) | Ghiyasuddin Tughlaq → Muhammad bin Tughlaq → Firoz Shah Tughlaq → Ghiyasuddin Tughlaq II → Muhammad Shah III<br><i>Note: at the end there were two independent rulers — Nasiruddin Mahmud (Delhi) and Nusrat Shah (Ferozabad)</i> |
| Sayyid dynasty (1414-1451)  | Khizr Khan → Mubarak Shah → Muhammad Shah → Alauddin Shah                                                                                                                                                                         |
| Lodi dynasty (1451-1526)    | Bahlol Lodi → Sikandar Lodi → Ibrahim Lodi                                                                                                                                                                                        |

- **Additional fact (Tughlaq dynasty):** It was a Muslim dynasty of Turkic origin. It was founded by Ghazi Malik (Ghiyath al-Din Tughluq) in 1320.

## 2 Buildings of the Delhi Sultanate Period and their Builders

| Ruler / Builder         | Principal Buildings / Architecture                                                  |
|-------------------------|-------------------------------------------------------------------------------------|
| Qutbuddin Aibak         | Adhai Din ka Jhonpra (Ajmer), Quwwat-ul-Islam Mosque (Delhi)                        |
| Iltutmish               | Tomb of Iltutmish, Jama Masjid, Atarkin ka Darwaza, Sultan Ghari                    |
| Balban                  | Lal Mahal (Red Palace)                                                              |
| Alauddin Khilji         | Jamat Khana Masjid, Alai Darwaza, Hazar Sutun (palace of a thousand pillars)        |
| Ghiyasuddin Tughlaq     | Tughlaqabad city / fort                                                             |
| Muhammad bin Tughlaq    | Jahanpanah city, tomb of Sheikh Nizamuddin Auliya                                   |
| Firoz Shah Tughlaq      | Tomb of Firoz Shah Tughlaq ( <i>at Hauz Khas, New Delhi, near Alauddin's tank</i> ) |
| Junan Shah Khan-i-Jahan | Tomb of Firoz Shah, Khirki Masjid, Kali Masjid                                      |
| Miyan Bhuwa             | Moth ki Masjid                                                                      |

- **Special note:** Firoz Shah Tughlaq (reign: 1351-1388) is also called the **Shah Jahan of the Sultanate period** because of his vast contribution to architecture. He wrote his autobiography, the '**Futuhat-i-Firozshahi**'.

## 3 Principal Books of Medieval India and their Authors

| Name of Book | Author / Composer |
|--------------|-------------------|
| Akbarnama    | Abul Fazl         |
|              |                   |



| Name of Book   | Author / Composer |
|----------------|-------------------|
| Shahnama       | Firdausi          |
| Baburnama      | Babur             |
| Humayunnama    | Gulbadan Begum    |
| Buddhacharitam | Ashvaghosha       |
| Kavyamimamsa   | Rajashekhara      |
| Harshacharita  | Banabhatta        |
| Meghaduta      | Kalidasa          |
| Mudrarakshasa  | Vishakhadatta     |
| Rajatarangini  | Kalhana           |
| Indica         | Megasthenes       |
| Kamasutra      | Vatsyayana        |
| Vinaya Patrika | Tulsidas          |
| Gita Govinda   | Jayadeva          |
| Hitopadesha    | Narayana Pandit   |
| Malgudi Days   | R.K. Narayan      |
| My Truth       | Indira Gandhi     |

#### 4 Books Written by Emperors and their Language

| Ruler / Emperor      | Name of Book                        | Language            |
|----------------------|-------------------------------------|---------------------|
| Harshavardhana       | Ratnavali, Priyadarshika, Nagananda | Sanskrit            |
| Shri Krishnadevaraya | Amuktamalyada                       | Telugu              |
| Jahangir             | Tuzuk-i-Jahangiri                   | Persian             |
| Babur                | Baburnama (Tuzuk-i-Baburi)          | Chaghatai (Turkish) |

## 5 Principal Rulers and their Wives / Begums

| Name of Begum / Wife                   | Ruler / Husband |
|----------------------------------------|-----------------|
| Shah Turkan                            | Itutmish        |
| Malika-i-Jahan                         | Alauddin Khilji |
| Hamida Banu Begum                      | Humayun         |
| Mariam-uz-Zamani (Hirabai / Jodha Bai) | Akbar           |
| Nur Jahan (Mehrunnisa)                 | Jahangir        |
| Mumtaz Mahal (Arjumand Banu Begum)     | Shah Jahan      |

## 6 Principal Battles of the Medieval Period (Chronology)

| Year | Name of Battle           | Between Whom and Result                                                                                                                    |
|------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1556 | Second Battle of Panipat | The Mughal emperor <b>Akbar</b> defeated the Hindu king <b>Hemu</b> .                                                                      |
| 1567 | Battle of Thanesar       | <b>Akbar</b> defeated two groups of sanyasis (ascetics).                                                                                   |
| 1575 | Battle of Tukaroi        | <b>Akbar</b> defeated the Sultanate of Bihar and Bengal.                                                                                   |
| 1576 | Battle of Haldighati     | <b>Akbar</b> (Man Singh) defeated the valiant <b>Maharana Pratap</b> .                                                                     |
| 1671 | Battle of Saraighat      | The Ahom commander <b>Lachit Borphukan</b> defeated the Mughal army (Ram Singh).                                                           |
| 1739 | Battle of Karnal         | The Iranian ruler <b>Nadir Shah</b> defeated the Mughal emperor <b>Muhammad Shah</b> (looted the Kohinoor diamond and the Peacock Throne). |

## 7 Principal Battles Fought by Humayun (Special Table)

| Year | Name of Battle                      | Place / River Bank             | Result / Special Fact                                                                                                                       |
|------|-------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1532 | <b>Battle of Dauhariya (Dadrah)</b> | On the bank of the Gomti river | Humayun defeated the combined army of Mahmud Lodi and Sher Shah.                                                                            |
| 1539 | <b>Battle of Chausa (26 June)</b>   | Buxar, Bihar                   | Sher Shah Suri defeated Humayun. A water-carrier (bhishti) named Nizam saved Humayun's life. Sher Khan took the title ' <b>Sher Shah</b> '. |

| Year | Name of Battle                       | Place / River Bank | Result / Special Fact                                                          |
|------|--------------------------------------|--------------------|--------------------------------------------------------------------------------|
| 1540 | <b>Battle of Kannauj (Bilgram)</b>   | Kannauj, U.P.      | Humayun suffered a decisive defeat and Sher Shah captured Delhi.               |
| 1555 | <b>Battle of Machhiwara</b> (15 May) | Punjab             | Humayun defeated the army of Sikandar Sur.                                     |
| 1555 | <b>Battle of Sirhind</b> (22 June)   | Punjab             | Humayun again defeated Sikandar Sur and recaptured Delhi in <b>July 1555</b> . |

## 8 Principal Religious Schools (Philosophies) and their Founders

(Asked repeatedly in UPPCS Prelims for matching)

| School / Philosophy         | Founder        |
|-----------------------------|----------------|
| Advaitavada                 | Shankaracharya |
| Vishishtadvaitavada         | Ramanujacharya |
| Dvaitadvaitavada            | Nimbarkacharya |
| Shuddhadvaitavada           | Vallabhacharya |
| Dvaitavada                  | Madhvacharya   |
| Bhedabhedavada              | Bhaskaracharya |
| Shaiva Vishishtadvaita      | Shrikantha     |
| Achintya Bhedabhedavada     | Baladeva       |
| Vira Shaiva Vishishtadvaita | Shripati       |

## 9 Mughal Architecture and Buildings

| Building / Monument | Place                       | Builder / Ruler |
|---------------------|-----------------------------|-----------------|
| Kabuli Bagh Mosque  | Panipat                     | Babur           |
| Jami Masjid         | Sambhal (Rohilkhand)        | Babur           |
| Mosque of Agra      | Agra (inside the Lodi fort) | Babur           |
| Mosque of Fatehabad | Punjab (Hisar)              | Humayun         |

| Building / Monument                                                                                                                                                                   | Place                                   | Builder / Ruler                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------|
| Dinpanah city                                                                                                                                                                         | Delhi                                   | Humayun                            |
| Qila-i-Kuhna Mosque                                                                                                                                                                   | Delhi (inside the Old Fort/Purana Qila) | Sher Shah Suri                     |
| Shersur city                                                                                                                                                                          | Kannauj                                 | Sher Shah Suri                     |
| Rohtasgarh Fort                                                                                                                                                                       | North-West Frontier Province            | Sher Shah Suri                     |
| Tomb of Sher Shah                                                                                                                                                                     | Sasaram (Bihar)                         | Sher Shah Suri                     |
| Humayun's Tomb                                                                                                                                                                        | Delhi                                   | Haji Begum<br>(Akbar's stepmother) |
| Agra Fort                                                                                                                                                                             | Agra                                    | Akbar                              |
| Jahangiri Mahal and Akbari Mahal                                                                                                                                                      | Agra Fort                               | Akbar                              |
| Buildings of Fatehpur Sikri<br>(Diwan-i-Am, Diwan-i-Khas, Panch Mahal, Palace of the Turkish Sultana, Mariam Mahal, Buland Darwaza, Tomb of Sheikh Salim Chishti, Tomb of Islam Shah) | Fatehpur Sikri                          | Akbar                              |
| Lahore Fort                                                                                                                                                                           | Lahore                                  | Akbar                              |
| Fort with 40 pillars (the last building of Akbar's period)                                                                                                                            | Allahabad (Prayagraj)                   | Akbar                              |
| Akbar's Tomb                                                                                                                                                                          | Sikandra (Agra)                         | Jahangir                           |
| Tomb of Itimad-ud-Daulah                                                                                                                                                              | Agra                                    | Nur Jahan                          |
| Jahangir's Tomb                                                                                                                                                                       | Shahdara (Lahore)                       | Nur Jahan                          |
| Tomb of Rabia-ud-Daurani (Bibi ka Maqbara)                                                                                                                                            | Aurangabad                              | Aurangzeb                          |
| Badshahi Mosque                                                                                                                                                                       | Lahore                                  | Aurangzeb                          |
| Moti Masjid (Pearl Mosque) (inside the Red Fort)                                                                                                                                      | Delhi                                   | Aurangzeb                          |

## 1 The Ashtadiggajas of Krishnadevaraya (Vijayanagara Empire)

(UPPCS asks directly, giving names, which of these were or were not among the Ashtadiggajas)

- **Allasani Peddana** (also called 'Andhra Kavita Pitamaha' – grandfather of Telugu poetry)
- **Nandi Timmana**
- **Bhattumurti**

- Pingali Surana
- Dhurjati
- Tenali Ramakrishna

## 2 The Ten Gurus of the Sikhs and their Important Works (Chronology)

| No. | Name of the Guru (Tenure)             | Important Works / Exam Facts                                                                                                    |
|-----|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1.  | <b>Guru Nanak Dev</b> (1469-1539)     | Founder of Sikhism, the first Guru.                                                                                             |
| 2.  | <b>Guru Angad (Lehna)</b> (1504-1552) | Invented / introduced the <b>Gurmukhi script</b> .                                                                              |
| 3.  | <b>Guru Amar Das</b> (1479-1574)      | Made the langar system permanent; established 22 gaddis (seats).                                                                |
| 4.  | <b>Guru Ram Das</b> (1534-1581)       | Akbar granted 500 bighas of land, on which the <b>city of Amritsar</b> was founded. Made the office of Guru <b>hereditary</b> . |
| 5.  | <b>Guru Arjan Dev</b> (1563-1606)     | Founded the town of Tarn Taran. Compiled the <b>Adi Granth (Guru Granth Sahib)</b> . Sentenced to death by <b>Jahangir</b> .    |
| 6.  | <b>Guru Hargobind</b> (1594-1644)     | Transformed the Sikhs into a martial community. Built the <b>Akal Takht</b> .                                                   |
| 7.  | <b>Guru Har Rai</b> (1630-1661)       | Had blessed Dara Shikoh.                                                                                                        |
| 8.  | <b>Guru Har Krishan</b> (1656-1664)   | Died young of smallpox.                                                                                                         |
| 9.  | <b>Guru Tegh Bahadur</b> (1621-1675)  | Executed in Delhi by <b>Aurangzeb</b> (Sis Ganj Gurudwara).                                                                     |
| 10. | <b>Guru Gobind Singh</b> (1666-1708)  | Founded the <b>Khalsa Panth (1699)</b> at Anandpur. The last Guru.                                                              |

## 3 Delhi Sultanate: Descending Order of Length of Rule (Chronology)

(A pet question of UPPCS Prelims - from longest to shortest rule)

| Dynasty            | Duration of Rule (Years) | Period of Rule (CE) |
|--------------------|--------------------------|---------------------|
| 1. Tughlaq dynasty | <b>94 years</b>          | 1320 - 1414 CE      |
| 2. Slave dynasty   | <b>84 years</b>          | 1206 - 1290 CE      |

| Dynasty           | Duration of Rule (Years) | Period of Rule (CE) |
|-------------------|--------------------------|---------------------|
| 3. Lodi dynasty   | 76 years                 | 1451 - 1526 CE      |
| 4. Sayyid dynasty | 36 years                 | 1414 - 1451 CE      |
| 5. Khilji dynasty | 30 years                 | 1290 - 1320 CE      |

- The last three rulers of the Sayyid dynasty (in sequence): Mubarak Shah (1421-34) → Muhammad Shah (1434-45) → Alauddin Alam Shah (1445-51).

#### 4 Administrative Units and Officers of the Mughal Period

| Administrative Unit | Name of Officer              | Main Function / Department                                          |
|---------------------|------------------------------|---------------------------------------------------------------------|
| Province (Suba)     | Sipahsalar / Subedar / Nazim | Chief executive (Sipahsalar under Akbar).                           |
|                     | Diwan                        | In charge of the revenue department (finance head of the suba).     |
|                     | Bakhshi                      | In charge of the military department.                               |
| District (Sarkar)   | Faujdar                      | Administrative head (maintenance of peace).                         |
|                     | Amal / Amalguzar             | Revenue collection (tax collection) officer.                        |
|                     | Kotwal                       | Law and order, disposal of criminal cases, price control.           |
| Pargana             | Shiqdar                      | Administrative head (his duties also included those of the kotwal). |
|                     | Amin / Qanungo               | Revenue officer (land records and tax assessment).                  |
| Village             | Muqaddam                     | Village headman.                                                    |
|                     | Patwari                      | Accountant (maintained the village land/revenue accounts).          |
|                     | Chaukidar                    | Security (watchman).                                                |

#### 5 Principal Books of Medieval History and their Authors

| Name of Book  | Author       | Special Details                                    |
|---------------|--------------|----------------------------------------------------|
| Taj-ul-Maasir | Hasan Nizami | The first official history of the Delhi Sultanate. |
|               |              |                                                    |



| Name of Book         | Author             | Special Details                                                             |
|----------------------|--------------------|-----------------------------------------------------------------------------|
| Tabaqat-i-Nasiri     | Minhaj-us-Siraj    | Account of the period of Iltutmish and Nasiruddin Mahmud.                   |
| Fawa'id-ul-Fuad      | Amir Hasan Dehlavi | Collection of the sayings of the Sufi saint Nizamuddin Auliya.              |
| Tarikh-i-Firozshahi  | Ziauddin Barani    | An important source for the period of Alauddin and the Tughlaqs.            |
| Futuhat-i-Firozshahi | Firoz Shah Tughlaq | <b>Autobiography</b> of Sultan Firoz Shah Tughlaq.                          |
| Futuh-us-Salatin     | Abdul Malik Isami  | History of the founding of the Bahmani kingdom and of the Sultanate period. |

## 6 Principal Works of Amir Khusrau and their Subject Matter

(Highly important for Match the Following)

| Name of Work     | Main Subject Matter (what it describes)                                                           |
|------------------|---------------------------------------------------------------------------------------------------|
| Qiran-us-Sadain  | Description of the historic meeting of Bughra Khan and his son Kaiqubad.                          |
| Miftah-ul-Futuh  | Description of the military campaigns and victories of Jalaluddin Khilji.                         |
| Khazain-ul-Futuh | Description of the reign of Alauddin Khilji and his conquest of the South.                        |
| Nuh Sipihr       | Description of the period of Mubarak Khilji, and of the climate of India and its praise.          |
| Ashiqi           | The love story of Deval Rani, daughter of King Karan of Gujarat, and Khizr Khan, son of Alauddin. |
| Tughlaqnama      | Depiction of the victory of Ghiyasuddin Tughlaq and his struggle with Khusrau Shah.               |

## 7 Principal Sufi Silsilas (Orders) of India and their Founders

| Sufi Silsila       | Founder (in India)                            |
|--------------------|-----------------------------------------------|
| Chishti Silsila    | Khwaja Muinuddin Chishti (Dargah: Ajmer)      |
| Suhrawardi Silsila | Bahauddin Zakariya                            |
| Qadiri Silsila     | Sayyid Makhdum Muhammad Jilani                |
| Shattari Silsila   | Shah Abdullah Shattari                        |
| Firdausi Silsila   | Sheikh Badruddin Samarqandi (active in Bihar) |

| Sufi Silsila       | Founder (in India)                                     |
|--------------------|--------------------------------------------------------|
| Naqshbandi Silsila | Baqi Billah (Aurangzeb was a follower of this silsila) |

## 8 Administrative System of the Delhi Sultanate (Departments and Sultans)

| Name of Department | Related Sultan / Creator | Type / Function of the Department                       |
|--------------------|--------------------------|---------------------------------------------------------|
| Diwan-i-Arz        | Balban                   | Military department                                     |
| Diwan-i-Mustakhraj | Alauddin Khilji          | Finance department (for realising arrears of revenue)   |
| Diwan-i-Riyasat    | Alauddin Khilji          | Market control department / regulation of trade         |
| Diwan-i-Amir Kohi  | Muhammad bin Tughlaq     | Agriculture department (for improvement in agriculture) |
| Diwan-i-Bandagan   | Firoz Shah Tughlaq       | Department of slaves                                    |
| Diwan-i-Khairat    | Firoz Shah Tughlaq       | Charity department (for helping orphans and widows)     |
| Diwan-i-Istihqaq   | Firoz Shah Tughlaq       | Pension department                                      |

## 9 Principal Ministers / Officers of the Sultanate Period

| Name of Officer   | Related Department / Function                                                             |
|-------------------|-------------------------------------------------------------------------------------------|
| Ariz-i-Mumalik    | Head of the military department (head of the Diwan-i-Arz).                                |
| Barid-i-Mumalik   | Chief officer of the intelligence department.                                             |
| Qazi-ul-Quzat     | Highest officer of the judicial department (after the Sultan).                            |
| Mushrif-i-Mumalik | Accountant-General (kept accounts of income and expenditure received from the provinces). |
| Wakil-i-Dar       | Looked after the Sultan's personal services and the royal karkhanas (workshops).          |
| Khazin            | Treasurer.                                                                                |

## 10 Tax (Revenue) System of the Delhi Sultanate

| Name of Tax | Type of Tax / From Whom it was Collected             |
|-------------|------------------------------------------------------|
| Ushr        | Land tax collected from <b>Muslims</b> .             |
| Kharaj      | Land tax collected from <b>non-Muslims</b> (Hindus). |

| Name of Tax | Type of Tax / From Whom it was Collected                             |
|-------------|----------------------------------------------------------------------|
| Zakat       | Religious tax on <b>Muslims</b> (one-fortieth of property, or 2.5%). |
| Jizya       | Religious / protection tax collected from <b>non-Muslims</b> .       |

## 11 Foreign Travellers who Came in the Time of Shah Jahan

| Name of Traveller       | Country of Origin | Period (CE)    | Special Details (UPPCS Factory)                                                                                  |
|-------------------------|-------------------|----------------|------------------------------------------------------------------------------------------------------------------|
| Peter Mundy             | Italy (Italian)   | 1630 - 1634 CE | Described the severe famine that occurred in the time of Shah Jahan.                                             |
| Niccolao Manucci        | Italy (Italian)   | 1653 - 1708 CE | Was first an <b>artilleryman in the army of Dara Shikoh</b> , later became a physician. (Book: Storia do Mogor). |
| Jean-Baptiste Tavernier | France (French)   | 1638 - 1663 CE | A <b>jeweller</b> by profession. <b>Visited India 6 times</b> .                                                  |
| François Bernier        | France (French)   | 1656 - 1717 CE | A <b>physician</b> by profession. Eyewitness of the war of succession.                                           |

## 12 Projects of Muhammad bin Tughlaq (MBT) (Chronology)

(UPPCS asks for their correct chronological order every second year)

- 1. Increase of taxation in the Doab (1325 CE)** - The very first scheme, which failed because of famine.
- 2. Transfer of capital (1326-27 CE)** - Shifted the capital from Delhi to Devagiri (Daulatabad).
- 3. Introduction of token currency (1330 CE)** - Issued copper/bronze coins.
- 4. Khurasan expedition (1332 CE)** - Gave the soldiers 1 year's salary in advance, but the expedition was called off.
- 5. Qarachil expedition (1337 CE)** - Sent an army into the Kumaon hills (Uttarakhand), which failed with heavy destruction.

## 13 Principal Events of Babur's Time (Chronology)

| Date / Year     | Name of Battle / Event         | Against Whom / Result                                             |
|-----------------|--------------------------------|-------------------------------------------------------------------|
| 21 April 1526   | <b>First Battle of Panipat</b> | Defeated Ibrahim Lodi (foundation of the Mughal Empire in India). |
| 16 March 1527   | <b>Battle of Khanwa</b>        | Defeated Rana Sanga of Mewar (gave the call of jihad).            |
| 29 January 1528 | <b>Battle of Chanderi</b>      | Defeated Medini Rai.                                              |

| Date / Year      | Name of Battle / Event   | Against Whom / Result                                                  |
|------------------|--------------------------|------------------------------------------------------------------------|
| 05 May 1529      | <b>Battle of Ghaghra</b> | Defeated the Afghans (Mahmud Lodi) (fought both on water and on land). |
| 26 December 1530 | <b>Death of Babur</b>    | Died at Agra (later buried at Kabul).                                  |

## 14 Principal Military Invasions of Alauddin Khilji

(UPPCS mainly asks for matching the king and the year)

| Year (CE) | Place of Invasion          | Contemporary Ruler / King                                           |
|-----------|----------------------------|---------------------------------------------------------------------|
| 1296      | Devagiri (first invasion)  | Ramachandra Deva                                                    |
| 1299      | Gujarat                    | Rai Karan (Karnadeva Baghela)                                       |
| 1301      | Ranthambhore               | Hammir Deva                                                         |
| 1303      | Chittor                    | Raja Ratan Singh                                                    |
| 1305      | Malwa                      | Mahlak Deva                                                         |
| 1308      | Devagiri (second invasion) | Ramachandra Deva                                                    |
| 1310      | Warangal                   | Rudradeva (Prataparudra Deva - the Kohinoor was obtained from here) |
| 1311      | Dwarasamudra               | Vira Ballala III (Hoysala dynasty)                                  |

## 15 Administrative Units of the Vijayanagara Empire (in descending order)

- 1. Province (Mandala)** - The largest unit (under a governor)
- 2. Kottam or Valanadu** - District-level unit
- 3. Nadu** - Pargana/tehsil-level unit
- 4. Melagrama** - A group of 50 villages
- 5. Uragrama** - The smallest unit (village)

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## Independent States that Broke away from the Bahmani Kingdom (Deccan Sultanates) and their Dynasties

| Name of State | Name of Dynasty | Founder (additional information) |
|---------------|-----------------|----------------------------------|
| Bijapur       | Adil Shahi      | Yusuf Adil Shah                  |
| Ahmednagar    | Nizam Shahi     | Malik Ahmad                      |
| Berar         | Imad Shahi      | Fathullah Imad Shah              |
| Golconda      | Qutb Shahi      | Quli Qutb Shah                   |
| Bidar         | Barid Shahi     | Amir Ali Barid                   |

## 1 Wars of Succession (Aurangzeb's Period) – Chronology

| No. | Name of Battle      | Time / Date   | Main Result                                                             |
|-----|---------------------|---------------|-------------------------------------------------------------------------|
| 1.  | Battle near Banaras | February 1658 | Defeat of Shah Shuja.                                                   |
| 2.  | Battle of Dharmat   | April 1658    | Aurangzeb and Murad defeated (the army of) Dara Shikoh (Jaswant Singh). |
| 3.  | Battle of Samugarh  | May 1658      | Aurangzeb decisively defeated Dara Shikoh.                              |
| 4.  | Battle of Khajwa    | January 1659  | Aurangzeb finally defeated Shah Shuja.                                  |
| 5.  | Battle of Deorai    | April 1659    | Aurangzeb's final decisive victory over Dara Shikoh.                    |

## 2 Later Mughal Rulers (Tenure and Principal Facts)

| Name of Ruler   | Reign          | Important Exam Facts / Nicknames                                                                                                 |
|-----------------|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| Bahadur Shah I  | 1707 – 1712 CE | Son of Aurangzeb (Shah Alam I); met the Sikh Guru Gobind Singh. Nickname: ' <b>Shah-i-Bekhabar</b> ' (the heedless king).        |
| Jahandar Shah   | 1712 – 1713 CE | Married Lal Kunwar; Zulfiqar Khan was influential. Nickname: ' <b>Lampat Murkh</b> ' (the profligate fool).                      |
| Farrukhsiyar    | 1713 – 1719 CE | Got the throne with the help of the Sayyid brothers (king-makers). Nickname: ' <b>Ghrinit Kayar</b> ' (the contemptible coward). |
| Rafi-ud-Darajat | 1719 CE        | The Mughal emperor with the shortest reign; influence of the Sayyid brothers.                                                    |
| Rafi-ud-Daula   | 1719 CE        | Took the title 'Shah Jahan II'; brother of Rafi-ud-Darajat.                                                                      |

| Name of Ruler            | Reign          | Important Exam Facts / Nicknames                                                                    |
|--------------------------|----------------|-----------------------------------------------------------------------------------------------------|
| Muhammad Shah 'Rangeela' | 1719 – 1748 CE | In his time <b>Nadir Shah invaded in 1739</b> (the Kohinoor diamond was looted).                    |
| Ahmad Shah Bahadur       | 1748 – 1754 CE | Son of Muhammad Shah.                                                                               |
| Alamgir II               | 1754 – 1759 CE | Ascended the throne after Ahmad Shah Bahadur (ruler at the time of the Battle of Plassey).          |
| Shah Alam II             | 1759 – 1806 CE | In his time the <b>Battle of Buxar (1764)</b> and the <b>Treaty of Allahabad (1765)</b> took place. |
| Akbar II                 | 1806 – 1837 CE | Sent Rammohan Roy to England after giving him the <b>title of 'Raja'</b> .                          |

### 3 Early Medieval Period: The Tripartite (Triangular) Struggle

| Main Point               | Description                                                                         |
|--------------------------|-------------------------------------------------------------------------------------|
| Dynasties involved       | The <b>Gurjara-Pratiharas</b> , the <b>Rashtrakutas</b> and the <b>Pala dynasty</b> |
| Cause of the struggle    | To establish <b>control over Kannauj</b> , the principal centre of North India      |
| Duration of the struggle | Lasted about <b>100 years</b>                                                       |
| Final victory            | <b>Of the Gurjara-Pratiharas</b>                                                    |

### 4 Principal Dynasties of the Tripartite Struggle

| Dynasty                   | Founder                     | Capital                     | Principal Exam Facts                                                                                                                               |
|---------------------------|-----------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Gurjara-Pratihara dynasty | <b>Nagabhata I</b> (730 CE) | Bhinmal (Jalore, Rajasthan) | * The triangular struggle began in the time of <b>Vatsaraja</b> .<br>* At the time of Mahmud of Ghazni's invasion the ruler was <b>Rajyapala</b> . |
| Rashtrakuta dynasty       | <b>Dantidurga</b> (752 CE)  | Manyakheta                  | * <b>Krishna I</b> built the famous <b>Kailasha Temple</b> of Ellora.<br>* The last ruler was <b>Karka II</b> .                                    |
| Pala dynasty              | <b>Gopala</b> (750 CE)      | Munger                      | * <b>Dharmapala</b> founded <b>Vikramashila University</b> .<br>* The last ruler was <b>Govindapala</b> .                                          |



## 5 Early Muslim Invaders of India

| Invader            | Time / Period | Principal Historical Facts                                                                                                                                     |
|--------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muhammad bin Qasim | 712 CE        | The <b>first Muslim ruler</b> to invade India (Sindh).                                                                                                         |
| Mahmud of Ghazni   | 998–1030 CE   | * Son of Subuktigin; <b>invaded India 17 times</b> (1000–1027 CE).<br>* Made his famous attack on the <b>Somnath Temple in 1026 CE</b> .                       |
| Muhammad Ghori     | 1175–1206 CE  | * Ruler of the province of Ghor; first invasion on Multan in 1175 CE.<br>* In 1178 CE he was defeated by <b>Bhima II (Mularaja II)</b> , the ruler of Gujarat. |

### > Court Scholars of Mahmud of Ghazni (asked directly):

- **Al-Biruni:** Kitab-ul-Hind
- **Utbi:** Kitab-ul-Yamini
- **Firdausi:** Shahnama
- **Baihaqi:** Tarikh-i-Subuktigin

## 6 Principal Battles of Muhammad Ghori

| Year    | Name of Battle                 | Against Whom       | Result / Special Fact                                            |
|---------|--------------------------------|--------------------|------------------------------------------------------------------|
| 1191 CE | <b>First Battle of Tarain</b>  | Prithviraj Chauhan | Prithviraj Chauhan was victorious (Ghori lost).                  |
| 1192 CE | <b>Second Battle of Tarain</b> | Prithviraj Chauhan | Muhammad Ghori victorious (foundation of Muslim power in India). |
| 1194 CE | <b>Battle of Chandawar</b>     | Raja Jaichand      | Muhammad Ghori victorious (Jaichand was killed).                 |

## 7 Delhi Sultanate: Principal Dynasties and the Works of the Sultans

### > (a) Slave Dynasty (1206 - 1290 CE)

| Sultan                 | Reign        | Principal Works and Important Facts                                                                                                                                                                                   |
|------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Qutbuddin Aibak</b> | 1206–1210 CE | * Founder of the Slave dynasty; nicknamed ' <b>Lakh Bakhsh</b> ' (giver of lakhs) and ' <b>Quran Khan</b> '.<br>* Constructions: Quwwat-ul-Islam Mosque (Delhi), Adhai Din ka Jhonpra (Ajmer); began the Qutub Minar. |

| Sultan              | Reign        | Principal Works and Important Facts                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Iltutmish</b>    | 1211–1236 CE | <ul style="list-style-type: none"> <li>* Moved the capital from Lahore to Delhi; completed the Qutub Minar.</li> <li>* Started the <b>iqta system</b>; formed the '<b>Turkan-i-Chahalgani</b>' of 40 nobles.</li> <li>* Issued the silver '<b>tanka</b>' and the copper '<b>jital</b>' coins.</li> </ul>                                                                             |
| <b>Razia Sultan</b> | 1236–1240 CE | <ul style="list-style-type: none"> <li>* The <b>only woman ruler</b> to become Sultan of Delhi.</li> <li>* Altunia rebelled against her; Razia later married him.</li> </ul>                                                                                                                                                                                                         |
| <b>Balban</b>       | 1266–1286 CE | <ul style="list-style-type: none"> <li>* Real name: Bahauddin; ended the '<b>Turkan-i-Chahalgani</b>'.</li> <li>* Took the title '<b>Zill-i-Ilahi</b>' (shadow of God); called himself 'Niyabat-i-Khudai'.</li> <li>* Introduced the Iranian customs of '<b>sijda</b>' and '<b>paibos</b>' and the <b>Nauroz festival</b>. Adopted the policy of '<b>blood and iron</b>'.</li> </ul> |

### > (b) Khilji Dynasty (1290 - 1320 CE)

| Sultan                         | Reign        | Principal Works and Important Facts                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jalaluddin Firoz Khilji</b> | 1290–1296 CE | <ul style="list-style-type: none"> <li>* Founded the Khilji dynasty after killing Kayumars. The revolt of <b>Malik Chhajju</b> took place in his time.</li> </ul>                                                                                                                                                                                                                                                                                                         |
| <b>Alauddin Khilji</b>         | 1296–1316 CE | <ul style="list-style-type: none"> <li>* Took the title '<b>Sikandar II (Sikandar-i-Sani)</b>'; obtained a slave named <b>Malik Kafur</b> from Gujarat.</li> <li>* <b>Works:</b> created the '<b>Diwan-i-Mustakhraj</b>' for land revenue based on measurement of land; implemented the policy of <b>market control (price control)</b>; started the practice of <b>branding horses (dagh)</b> and <b>recording the descriptive roll (huliya)</b> of soldiers.</li> </ul> |

- **Last ruler:** Nasiruddin Khusrau Shah.

### All the Military Campaigns of Alauddin Khilji (North and South India) Correct Chronological Order

| Campaign / Region      | Year           | Ruler                                  | Location of the Region  |
|------------------------|----------------|----------------------------------------|-------------------------|
| <b>Gujarat</b>         | 1299 CE        | Karnadeva Vaghela (Rai Karan)          | North India             |
| <b>Ranthambhore</b>    | 1301 CE        | Hamir Deva Chauhan                     | North India (Rajputana) |
| <b>Mewar (Chittor)</b> | 1303 CE        | Rana Ratan Singh                       | North India (Rajputana) |
| <b>Malwa</b>           | 1305 CE        | Mahlak Deva Parmar                     | North India             |
| <b>Devagiri</b>        | 1307 - 1308 CE | Ramachandra Deva (Yadava dynasty)      | South India             |
| <b>Warangal</b>        | 1309 - 1310 CE | Prataparudradeva II (Kakatiya dynasty) | South India             |
| <b>Dwarasamudra</b>    | 1310 - 1311 CE | Vira Ballala III (Hoysala dynasty)     | South India             |
| <b>Jalore</b>          | 1311 CE        | Kanhadadeva Chauhan                    | North India (Rajputana) |

(Note: before becoming Sultan, Alauddin Khilji had also made a successful raid on Devagiri in **1296 CE**, but his formal southern campaign as Sultan began in **1307 CE** under the leadership of Malik Kafur.)

### > (c) Tughlaq Dynasty (1320 - 1412 CE)

| Sultan               | Reign        | Principal Works and Important Facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ghiyasuddin Tughlaq  | 1320–1325 CE | <ul style="list-style-type: none"> <li>* Founder of the Tughlaq dynasty; founded the city of <b>Tughlaqabad</b>.</li> <li>* He was the <b>first Sultan</b> of Delhi to have <b>canals constructed</b>.</li> </ul>                                                                                                                                                                                                                                                                                           |
| Muhammad bin Tughlaq | 1325–1351 CE | <ul style="list-style-type: none"> <li>* Real name: Ulugh Khan (Juna Khan). Created the <b>agriculture department 'Diwan-i-Amir Kohi'</b>.</li> <li>* <b>5 schemes (in order)</b>: increase of taxation in the Doab (1325) → transfer of capital (1326) → token currency (1330) → Khurasan and Qarachil expeditions.</li> <li>* In 1333 CE the Moroccan traveller <b>Ibn Battuta</b> came to India (book: '<b>Rehla</b>'); he was appointed Qazi of Delhi.</li> </ul>                                       |
| Firoz Shah Tughlaq   | 1351–1388 CE | <ul style="list-style-type: none"> <li>* Imposed only 4 Sharia taxes—Kharaj, Khums, Jizya, Zakat. Imposed <b>jizya on Brahmins</b>.</li> <li>* Issued the '<b>Shashgani</b>' coin; the first Sultan to levy an irrigation tax (<b>Haq-i-Sharb</b>).</li> <li>* First created a '<b>Public Works Department (PWD)</b>'. Founded towns: Fatehabad, Hisar, Firozpur, Jaunpur, Firozabad.</li> <li>* New departments: Diwan-i-Khairat (charity), Dar-ul-Shafa (hospital), Diwan-i-Bandagan (slaves).</li> </ul> |

- **Last ruler of the Tughlaq dynasty**: was **Nasiruddin Mahmud (1394-1412 CE)**. In his time Malik Sarwar made **Jaunpur** independent and **Timur (Tamerlane)** invaded **Delhi in 1398 CE**.

### > (d) Sayyid and Lodi Dynasties

| Sultan / Dynasty            | Reign        | Principal Works and Important Facts                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Khizr Khan (Sayyid dynasty) | 1414–1421 CE | <ul style="list-style-type: none"> <li>* Founder of the Sayyid dynasty; he was a commander of Timur. (Last ruler: Alauddin Alam Shah).</li> </ul>                                                                                                                                                                                                                         |
| Bahlol Lodi (Lodi dynasty)  | 1451–1489 CE | <ul style="list-style-type: none"> <li>* Founder of the Lodi dynasty; issued a <b>coin called the 'Bahloli'</b>. Annexed Jaunpur to the Delhi Sultanate.</li> </ul>                                                                                                                                                                                                       |
| Sikandar Lodi               | 1489–1517 CE | <ul style="list-style-type: none"> <li>* Founded the <b>city of 'Agra' in 1504 CE</b> and made it the capital. Introduced the '<b>Gaz-i-Sikandari</b>' for land measurement.</li> <li>* Wrote Persian poetry under the pen-name '<b>Gulrukhi</b>'. Persian translations: '<b>Farhang-i-Sikandari</b>' (Ayurveda) and '<b>Lazzat-i-Sikandar Shahi</b>' (music).</li> </ul> |

| Sultan / Dynasty | Reign            | Principal Works and Important Facts                                                                                                                                                                           |
|------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ibrahim Lodi     | 1517–<br>1526 CE | <p>* The last ruler of the Delhi Sultanate. Lost the <b>Battle of Khatoli in 1518 CE</b> to Rana Sanga.</p> <p>* Was defeated and killed by Babur in the <b>First Battle of Panipat on 21 April 1526</b>.</p> |

## 8 Principal Administrative Officers of the Sultanate Period and their Functions

(Extremely important for UPPCS Prelims 'Match the Following')

| Designation (Officer)           | Related Department / Administrative Function                       |
|---------------------------------|--------------------------------------------------------------------|
| Wazir                           | Head of the finance department (prime minister).                   |
| Ariz-i-Mumalik                  | Head of the military department.                                   |
| Diwan-i-Insha / Dabir-i-Risalat | Chief officer of the department of royal correspondence.           |
| Diwan-i-Risalat                 | Head of the Foreign Department.                                    |
| Sadr-us-Sudur                   | Chief officer of the department of religion and charity.           |
| Qazi-ul-Quzat                   | Chief officer of the judicial department (chief justice).          |
| Barid-i-Mumalik                 | Chief officer of the intelligence department (intelligence chief). |
| Wakil-i-Hajib                   | Chief officer in charge of court etiquette.                        |
| Sar-i-Jandar                    | Head of the Sultan's personal bodyguards.                          |
| Mustaufi-i-Mumalik              | Auditor General (who examined expenditure).                        |
| Mushrif-i-Mumalik               | Accountant General (who kept account of income).                   |

## 1 Principal Administrative Departments of the Delhi Sultanate and their Founders

| Name of Department | Type / Function of the Department       | Founder (Sultan) |
|--------------------|-----------------------------------------|------------------|
| Naib-i-Mamlikat    | Running of the entire administration    | Bahram Shah      |
| Diwan-i-Arz        | Military department                     | Balban           |
| Diwan-i-Riyasat    | Market control / regulation of commerce | Alauddin Khilji  |
| Diwan-i-Mustakhraj | Realisation of arrears of revenue (tax) | Alauddin Khilji  |
|                    |                                         |                  |

| Name of Department | Type / Function of the Department                          | Founder (Sultan)     |
|--------------------|------------------------------------------------------------|----------------------|
| Diwan-i-Amir Kohi  | Agriculture department (for agricultural improvement)      | Muhammad bin Tughlaq |
| Diwan-i-Bandagan   | Care of slaves                                             | Firoz Shah Tughlaq   |
| Diwan-i-Khairat    | Charity department (for orphans and widows)                | Firoz Shah Tughlaq   |
| Dar-ul-Shafa       | Department of free medical treatment (charitable hospital) | Firoz Shah Tughlaq   |

2 Dynasties of the Vijayanagara Empire (Chronology)

| Dynasty            | Time / Period | Principal Rulers (in sequence)                                                                                         |
|--------------------|---------------|------------------------------------------------------------------------------------------------------------------------|
| 1. Sangama dynasty | 1336–1485 CE  | Harihara I → Bukka I → Harihara II → Virupaksha I → Bukka II → Devaraya I → Devaraya II → Mallikarjuna → Virupaksha II |
| 2. Saluva dynasty  | 1485–1505 CE  | Narasimha Saluva → Narasa Nayaka                                                                                       |
| 3. Tuluva dynasty  | 1505–1565 CE  | Vira Narasimha → <b>Krishnadevaraya</b> → Achyuta Deva Raya → Venkata I → Sadashiva Raya                               |
| 4. Aravidu dynasty | 1570–1647 CE  | Tirumala → Sriranga I → Venkata II → Sriranga III → Ramadeva → Venkata III → Sriranga III                              |

3 Extremely Important Exam Facts Related to the Vijayanagara Empire

| Subject / Event                         | Ruler / Character | Details and Exam Facts                                                                 |
|-----------------------------------------|-------------------|----------------------------------------------------------------------------------------|
| First ruler                             | Harihara I        | The first ruler of the Vijayanagara Empire (founded in 1336 CE by Harihara and Bukka). |
| Title: Veda Marga Pratishthapaka        | Bukka I           | Assumed this title for promoting the Vedic religion.                                   |
| The war of the 'goldsmith's daughter'   | Devaraya I        | Fought between Devaraya I and the Bahmani ruler Firoz Shah.                            |
| Title: Gajabetekara and Immadi Devaraya | Devaraya II       | <b>Gajabetekara</b> means 'hunter of elephants'. He is also called Immadi Devaraya.    |
|                                         |                   |                                                                                        |

| Subject / Event                                                                | Ruler / Character                                 | Details and Exam Facts                                                                                                                                                                                   |
|--------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title: Andhra Bhoja and Yavanarajya Sthapanacharya</b>                      | Krishnadevaraya                                   | The most powerful ruler of the Tuluva dynasty.                                                                                                                                                           |
| <b>Literary works: Amuktamalyada and Jambavati Kalyanam</b>                    | Krishnadevaraya                                   | Wrote the Amuktamalyada (in Telugu) and the Jambavati Kalyanam (in Sanskrit).                                                                                                                            |
| <b>Construction of temples and towns</b>                                       | Krishnadevaraya                                   | Built the <b>Hazara Ramaswamy</b> and <b>Vitthala Swamy</b> temples; founded the town of ' <b>Nagalapura</b> ' in the name of his mother.                                                                |
| <b>Text: Madhura Vijayam</b>                                                   | Gangadevi                                         | Composed by Gangadevi, wife of Kumara Kampana, to celebrate his conquest of Madurai.                                                                                                                     |
| <b>Battle of Talikota</b><br><i>(Battle of Rakshasi-Tangadi / Banni Hatti)</i> | <b>Rama Raya</b><br><i>(leading Vijayanagara)</i> | <b>Took place in January 1565.</b> The Muslim confederacy of the Deccan (Bijapur, Ahmednagar, Golconda, Bidar) defeated and destroyed Vijayanagara. <i>Note: Berar was not part of this confederacy.</i> |

- **Administrative units of the Vijayanagara Empire (in descending order):** Kottam / Valanadu (district) → Nadu (pargana) → Melagrama (a group of 50 villages) → Ur (village).

#### 4 Foreign Travellers who Came to Vijayanagara (a pet question for Match the Following)

| Foreign Traveller | Country of Origin | Contemporary Vijayanagara Ruler |
|-------------------|-------------------|---------------------------------|
| Nicolo Conti      | Italy             | Devaraya I                      |
| Nikitin           | Russia            | Devaraya I                      |
| Abdur Razzaq      | Persia            | Devaraya II                     |
| Duarte Barbosa    | Portugal          | Krishnadevaraya                 |
| Domingo Paes      | Portugal          | Krishnadevaraya                 |
| Fernao Nuniz      | Portugal          | Achyuta Devaraya                |
| Caesar Frederick  | Italy             | Sadashiva Raya                  |

#### 5 History of Medieval India: The Jizya Tax (Chronology)

| Year / Period | Name of Ruler             | Historical Details Regarding the Jizya Tax                |
|---------------|---------------------------|-----------------------------------------------------------|
| 712 CE        | <b>Muhammad bin Qasim</b> | The first ruler to impose the jizya tax in India (Sindh). |



| Year / Period | Name of Ruler            | Historical Details Regarding the Jizya Tax                            |
|---------------|--------------------------|-----------------------------------------------------------------------|
| After 1351 CE | Firoz Shah Tughlaq       | The first Sultan of Delhi to impose jizya <b>even on Brahmins</b> .   |
| 1564 CE       | Akbar                    | The ruler who completely <b>abolished</b> jizya in the Mughal Empire. |
| 1674 CE       | Aurangzeb                | The ruler who <b>reimposed</b> jizya in the Mughal period.            |
| 1713 CE       | Farrukhsiyar             | Abolished jizya once again.                                           |
| 1717 CE       | Farrukhsiyar             | Reimposed jizya.                                                      |
| 1719 CE       | Muhammad Shah 'Rangeela' | <b>Finally (completely) abolished</b> jizya in India.                 |

## 6 Principal Sufi Silsilas and their Founders (with reference to India)

| Sufi Silsila       | Founder (in India / original)                                         |
|--------------------|-----------------------------------------------------------------------|
| Chishti Silsila    | Khwaja Muinuddin Chishti (Dargah: Ajmer)                              |
| Suhrawardi Silsila | Shihabuddin Suhrawardi / Bahauddin Zakariya (made it famous in India) |
| Qadiri Silsila     | Sheikh Abdul Qadir Jilani                                             |
| Shattari Silsila   | Shah Abdullah Shattari                                                |
| Firdausi Silsila   | Sheikh Badruddin (most active in the Bihar region)                    |
| Naqshbandi Silsila | Baqi Billah                                                           |

## 7 The Four Principal Peethas (Mathas) Established by Adi Shankaracharya

*Note: according to the source, the Sharada Peetha has been given here in place of the fourth peetha; remember it in the order of the states.*

| Name of Peetha    | Site / Place | State     | Associated Goddess/God  |
|-------------------|--------------|-----------|-------------------------|
| Shringeri Peetha  | Mysore       | Karnataka | Shiva                   |
| Govardhana Peetha | Puri         | Odisha    | Balabhadra and Subhadra |
| Sharada Peetha    | Dwarka       | Gujarat   | Krishna                 |

## 8 Principal Battles Fought by the Mughals – Chronology

| Name of Battle              | Exact Date         | Between Whom and Result                                             |
|-----------------------------|--------------------|---------------------------------------------------------------------|
| First Battle of Panipat     | 21 April 1526 CE   | Babur defeated Ibrahim Lodi.                                        |
| Battle of Khanwa            | 16 March 1527 CE   | Babur defeated Rana Sanga.                                          |
| Battle of Chanderi          | 29 January 1528 CE | Babur defeated Medini Rai.                                          |
| Battle of Ghaghra           | 5 May 1529 CE      | Babur defeated the Afghans (Mahmud Lodi).                           |
| Battle of Chausa            | 26 June 1539 CE    | Sher Shah Suri defeated Humayun.                                    |
| Battle of Kannauj (Bilgram) | 17 May 1540 CE     | Sher Shah Suri defeated Humayun again and took possession of Delhi. |

## 9 The Five Independent States Formed by the Disintegration of the Bahmani Kingdom

| Name of State | Year of Foundation | Founder         | Name of Dynasty |
|---------------|--------------------|-----------------|-----------------|
| Bijapur       | 1489 CE            | Yusuf Adil Shah | Adil Shahi      |
| Ahmednagar    | 1490 CE            | Malik Ahmad     | Nizam Shahi     |
| Berar         | 1490 CE            | Fathullah       | Imad Shahi      |
| Golconda      | 1518 CE            | Qutb Shah       | Qutb Shahi      |
| Bidar         | 1527 CE            | Amir Ali Barid  | Barid Shahi     |

### > Special Exam Facts of the Adil Shahi Dynasty (Bijapur):

- **Ibrahim Adil Shah II:** He was called '**Abla Baba**' for helping the poor and '**Jagat Guru**' for patronising scholars. He composed the '**Kitab-i-Nauras**', a collection of Hindi songs.
- **Tarikh-i-Firishta:** The historian Firishta composed this work under the patronage of Ibrahim Adil Shah II.
- **Muhammad Adil Shah:** He built the world-famous '**Gol Gumbaz**' at Bijapur.

## 10 Medieval Historical Literature (Books and Authors)

*(This table has been prepared by combining both the literary sections given by you, so that revision is easy)*

| Name of Book / Text           | Author / Composer | Main Subject / Special Fact                        |
|-------------------------------|-------------------|----------------------------------------------------|
| Kitab-ul-Hind / Tahqiq-i-Hind | Al-Biruni         | A view of India in the period of Mahmud of Ghazni. |

| Name of Book / Text                                                                   | Author / Composer        | Main Subject / Special Fact                                         |
|---------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------|
| Tarikh-i-Yamini                                                                       | Utbi                     | History of Mahmud of Ghazni.                                        |
| Shahnama                                                                              | Firdausi                 | He is also called 'the Homer of the East' (Ghaznavid period).       |
| Taj-ul-Maasir                                                                         | Hasan Nizami             | Early administrative information on the Delhi Sultanate.            |
| Tabaqat-i-Nasiri                                                                      | Minhaj-us-Siraj          | The period of Iltutmish and Nasiruddin Mahmud.                      |
| Siyasatnama                                                                           | Nizam-ul-Mulk Tusi       | A book of administrative rules.                                     |
| Tarikh-i-Firozshahi                                                                   | Ziauddin Barani          | Detailed account of the period of Alauddin Khilji and the Tughlaqs. |
| Fatawa-i-Jahandari                                                                    | Ziauddin Barani          | Political philosophy of the Sultanate period.                       |
| Futuh-i-Firozshahi                                                                    | Firoz Shah Tughlaq       | <b>Autobiography</b> of Sultan Firoz Shah Tughlaq.                  |
| Kitab-ul-Rehla (Rehla)                                                                | Ibn Battuta              | Travel account of the Moroccan traveller (MBT period).              |
| Tarikh-i-Mubarakshahi                                                                 | Yahya bin Ahmad Sirhindi | The only authentic source for the history of the Sayyid dynasty.    |
| Works of Amir Khusrau<br>( <i>Khazain-ul-Futuh, Nuh Sipahr, Ashiqa, Tughlaqnama</i> ) | Amir Khusrau             | History of various rulers of the Khilji and Tughlaq dynasties.      |
| Futuh-us-Salatin                                                                      | Isami / Malik Isami      | Important history related to the Bahmani kingdom.                   |
| Tarikh-i-Shershahi                                                                    | Abbas Khan Sherwani      | History of the administration and life of Sher Shah Suri.           |
| Akbarnama                                                                             | Abul Fazl                | Official history of the Mughal emperor Akbar.                       |
| Tabaqat-i-Akbari                                                                      | Nizamuddin Ahmad         | History of Akbar's time.                                            |
| Muntakhab-ut-Tawarikh                                                                 | Abdul Qadir Badauni      | A critical account of Akbar's policies.                             |
| Tuzuk-i-Baburi (Baburnama)                                                            | Babur                    | <b>Autobiography</b> of Babur (in the Chaghatai Turkish language).  |
| Humayunnama                                                                           | Gulbadan Begum           | Biography written by Humayun's sister.                              |
|                                                                                       |                          |                                                                     |

| Name of Book / Text | Author / Composer                    | Main Subject / Special Fact                                  |
|---------------------|--------------------------------------|--------------------------------------------------------------|
| Tuzuk-i-Jahangiri   | Jahangir, Mutamid Khan and Hadi Khan | Autobiography of Jahangir (later completed by Mutamid Khan). |
| Alamgirnama         | Mirza Muhammad Kazim                 | History of Aurangzeb's reign.                                |
| Fatawa-i-Alamgiri   | Sheikh Nizamuddin Ahmad              | A famous compilation of Islamic law of Aurangzeb's period.   |
| Nuskha-i-Dilkusha   | Bhimsen                              | History of the Deccan (South India) in Aurangzeb's time.     |
| Tarikh-i-Rashidi    | Mirza Haidar Dughlat                 | History of Central Asia and Kashmir/the Mughals.             |
| Rajatarangini       | Kalhana                              | A systematic history of Kashmir.                             |
| Gaudavaho           | Vakpati                              | Description of the conquests of King Yashovarman of Kannauj. |
| Ramacharita         | Sandhyakara Nandi                    | History of Ramapala, ruler of the Pala dynasty.              |
| Ballalacharita      | Ananda Bhatta                        | History of Ballalasena, ruler of the Sena dynasty of Bengal. |

## 1 Bhakti Movement: Saints, Schools and Sects

| Saint / Philosopher                     | Philosophical School / Doctrine |
|-----------------------------------------|---------------------------------|
| Shankaracharya                          | Advaitavada                     |
| Ramanujacharya                          | Vishishta Advaitavada           |
| Madhvacharya                            | Dvaitavada                      |
| Nimbarkacharya ( <i>Nimbakacharya</i> ) | Dvaitadvaitavada                |

### > The Bhakti Tradition in South India

- **Alvar saints:** They were devotees of **Vishnu** (Vaishnava).
- **Nayanar saints:** They were devotees of **Shiva** (Shaiva).

### > The Vaishnava Bhakti Tradition in Maharashtra

- **Varkari sect:** Worship of **Vitthala or Vithoba** (a form of Shri Krishna).
- **Dharkari sect:** Worship of **Shri Rama**.

## 2 Sufi Movement: Categories and Silsilas

### > Main Categories of Sufism

- **Ba-shara (those who followed the Islamic Sharia law):** Chishti, Suhrawardi, Qadiri, Naqshbandi, Firdausi.
- **Be-shara (free from the bonds of Islamic law):** Qalandar, Madari.

### > Principal Sufi Silsilas and their Founders in India

| Sufi Silsila | Founder in India / Principal Saint                                      |
|--------------|-------------------------------------------------------------------------|
| Chishti      | Sheikh Muinuddin Chishti (Dargah: Ajmer)                                |
| Suhrawardi   | Sheikh Bahauddin Zakariya ( <i>originally: Shihabuddin Suhrawardi</i> ) |
| Qadiri       | Abdul Qadir Jilani                                                      |
| Naqshbandi   | Khwaja Baqi Billah ( <i>originally: Khwaja Bahauddin Naqshband</i> )    |
| Firdausi     | Sheikh Jalaluddin Surkh                                                 |
| Shattari     | Abdullah Shattari                                                       |

## 3 Maratha Empire: Chhatrapati Shivaji Maharaj and the Ashtapradhan

| Event of Life          | Year / Date    | Historical Details and Exam Facts                                                |
|------------------------|----------------|----------------------------------------------------------------------------------|
| Birth                  | April 1627     | At Shivneri Fort ( <i>Mother: Jijabai, Father: Shahji Bhonsle</i> )              |
| Guru / Guardian        | —              | Guardian-teacher: <b>Dadaji Kondadev</b> ; spiritual guru: <b>Samarth Ramdas</b> |
| Killing of Afzal Khan  | 1659 CE        | Killed Afzal Khan (Abdullah Bhatari), the commander of Bijapur.                  |
| Defeat of Shaista Khan | 1663 CE        | Defeated the Mughal subedar Shaista Khan.                                        |
| First sack of Surat    | 1664 CE        | First plunder of Surat, the trading city of the Mughals.                         |
| Treaty of Purandar     | <b>1665 CE</b> | Between the Mughal commander <b>Mirza Raja Jai Singh</b> and <b>Shivaji</b> .    |
| Escape from Agra       | 1666 CE        | Succeeded in escaping from Aurangzeb's captivity (Jaipur Bhawan, Agra).          |
| Second sack of Surat   | 1670 CE        | Plundered the city of Surat a second time.                                       |

| Event of Life       | Year / Date             | Historical Details and Exam Facts                             |
|---------------------|-------------------------|---------------------------------------------------------------|
| Sons and coronation | June and September 1674 | Crowned at Raigad Fort. ( <i>Sons: Sambhaji and Rajaram</i> ) |
| Death               | 12 April 1680           | Death of Chhatrapati Shivaji Maharaj.                         |

### › Shivaji's Ashtapradhan (Council of Ministers):

(Matching of the departments of administrative officers comes in the exam every time)

| Designation                | Administrative Function / Department                                     |
|----------------------------|--------------------------------------------------------------------------|
| 1. Peshwa                  | Prime Minister (general administration and acting in the king's absence) |
| 2. Amatya (Majumdar)       | Finance Minister (revenue and accounts of income and expenditure)        |
| 3. Mantri (Waqianavis)     | Kept a record of the king's daily activities and court proceedings       |
| 4. Sachiv                  | Supervised the department of correspondence                              |
| 5. Sumant (Dabir)          | Foreign Minister (supervised foreign affairs and ambassadors)            |
| 6. Senapati (Sar-i-Naubat) | Military Minister (recruitment, organisation and discipline of the army) |
| 7. Panditrao               | Dharmadhikari (head of religious affairs, charity and grants)            |
| 8. Nyayadhis               | Head of Justice (highest officer for civil and criminal cases)           |

## 4 Successors of Shivaji Maharaj (Chronology)

| No. | Ruler (Chhatrapati) | Reign (CE)     |
|-----|---------------------|----------------|
| 1.  | Sambhaji            | 1680 – 1689 CE |
| 2.  | Rajaram             | 1689 – 1700 CE |
| 3.  | Tarabai             | 1700 – 1707 CE |
| 4.  | Shahuji             | 1707 – 1749 CE |
| 5.  | Rajaram II          | 1749 – 1777 CE |

## 5 Areas of Influence / Control of the Maratha Chiefs

| Maratha Dynasty / Chief | Area of Influence (Headquarters) |
|-------------------------|----------------------------------|
| Peshwa                  | Poona (Pune)                     |



| Maratha Dynasty / Chief | Area of Influence (Headquarters) |
|-------------------------|----------------------------------|
| Scindia                 | Gwalior                          |
| Holkar                  | Indore                           |
| Bhonsle                 | Nagpur                           |
| Gaekwad                 | Baroda (Vadodara)                |

## 6 Anglo-Maratha Wars and Important Treaties (Chronology)

### > Anglo-Maratha Wars:

- **First Anglo-Maratha War:** 1775 – 1782 CE
- **Second Anglo-Maratha War:** 1803 – 1805 CE
- **Third Anglo-Maratha War:** 1817 – 1818 CE

### > Historic Treaties and their Signatories (Match the Following):

| Name of Treaty            | Year (CE)        | Between Whom (Signatories)                         |
|---------------------------|------------------|----------------------------------------------------|
| Treaty of Purandar        | 1665 CE          | Mughal commander Jai Singh and Shivaji             |
| Treaty of Sangola         | 1750 CE          | Peshwa Balaji Bajirao and Rajaram II               |
| Treaty of Surat           | 1775 CE          | Raghunath Rao (Raghoba) and the East India Company |
| Second Treaty of Purandar | 1776 CE          | Peshwa Madhav Rao Narayan and the English          |
| Convention of Wadgaon     | 1779 CE          | Peshwa Madhav Rao Narayan and the English          |
| Treaty of Salbai          | 1782 CE          | Peshwa Madhav Rao Narayan and the English          |
| Treaty of Bassein         | 1802 CE          | Bajirao II and the English                         |
| Treaty of Deogaon         | 17 December 1803 | Raghuji Bhonsle II and the English                 |
| Treaty of Surji-Anjangaon | 30 December 1803 | Scindia and the English                            |
| Treaty of Rajpurghat      | 1805 CE          | Holkar and the English                             |
| Treaty of Poona           | June 1817        | Bajirao II and the English                         |
| Treaty of Gwalior         | November 1817    | Daulatrao Scindia and the English                  |
| Treaty of Mandsaur        | January 1818     | Malharrao Holkar and the English                   |

## 7 Mughal Empire: Babur, Humayun, Sher Shah and Akbar (Battles and Policies)

### > (a) Babur (1526 - 1530 CE) – Founder of the Mughal Dynasty

- **Family details:** Father: Umar Sheikh Mirza; Mother: Qutlugh Nigar Khanum. Sons: Humayun, Kamran, Askari and Hindal.
- **Autobiography:** 'Tuzuk-i-Baburi' or 'Baburnama'.

| Name of Battle          | Exact Date / Year | Between Whom           | Victor |
|-------------------------|-------------------|------------------------|--------|
| First Battle of Panipat | 21 April 1526     | Ibrahim Lodi and Babur | Babur  |
| Battle of Khanwa        | 16 March 1527     | Babur and Rana Sanga   | Babur  |
| Battle of Chanderi      | 29 January 1528   | Babur and Medini Rai   | Babur  |
| Battle of Ghaghra       | 5 May 1529        | Babur and Mahmud Lodi  | Babur  |

### > (b) Humayun (1530 - 1556 CE)

- **Life facts:** Born: 1508 CE at Kabul. Crowned at Agra in 1530 CE. Died: in 1556 CE.
- **Literature and architecture:** 'Humayun's Tomb' in Delhi was built by his wife **Haji Begum**. The biography 'Humayunnama' was written by his step-sister **Gulbadan Begum**.

| Name of Battle              | Year (CE) | Between Whom                       | Victor    |
|-----------------------------|-----------|------------------------------------|-----------|
| Battle of Devra (Dauhariya) | 1532      | Humayun and the Afghan Mahmud Lodi | Humayun   |
| Battle of Chausa            | 1539      | Humayun and Sher Shah Suri         | Sher Shah |
| Battle of Kannauj (Bilgram) | 1540      | Humayun and Sher Shah Suri         | Sher Shah |
| Battle of Sirhind           | 1555      | Humayun and Sikandar Sur           | Humayun   |

### > (c) Sher Shah Suri (1540 - 1545 CE)

- **Childhood name:** Farid Khan. The subedar of Bihar, '**Muhammad Bahar Khan Lohani**', gave him the title '**Sher Khan**'.
- **Throne and death:** Ascended the throne of Delhi in 1540. Died in 1545 during the **conquest of Kalinjar**.
- **Administrative work:** Introduced the '**Qabuliyat and Patta system**' to help the peasants.

### > (d) Mughal Emperor Akbar (1556 - 1605 CE)

- **Life facts:** Born: 1542 CE in the palace of Raja Virsal of Amarkot. Parents: Hamida Banu and Humayun. Childhood name: **Jalal**.
- **Coronation and death:** Crowned in 1556 CE at **Kalanaur** in Punjab. Guardian: **Bairam Khan**. Died: in 1605. Tomb: located at **Sikandra** (near Agra).

### > Principal Battles of Akbar's Period:

| Name of Battle           | Exact Date      | Between Whom              | Result / Victor      |
|--------------------------|-----------------|---------------------------|----------------------|
| Second Battle of Panipat | 5 November 1556 | Akbar and Hemu            | Akbar was victorious |
| Battle of Haldighati     | 18 June 1576    | Akbar and Maharana Pratap | Akbar was victorious |

### > Principal Policy / Administrative Measures of Akbar (Chronology - extremely important for UPPCS):

| Administrative Measure / Reform                  | Year (CE) |
|--------------------------------------------------|-----------|
| Abolition of slavery                             | 1562 CE   |
| Abolition of the pilgrimage tax                  | 1563 CE   |
| Abolition of jizya                               | 1564 CE   |
| Capital shifted from Agra to Fatehpur Sikri      | 1571 CE   |
| Construction of the Ibadatkhana                  | 1575 CE   |
| Doors of the Ibadatkhana opened to all religions | 1578 CE   |
| Proclamation of the Mahzar                       | 1579 CE   |
| Ibadatkhana closed                               | 1582 CE   |
| Founding of the Din-i-Ilahi                      | 1582 CE   |

## 1 Akbar's Navaratnas (Nine Gems)

- Abul Fazl
- Faizi
- Tansen
- Birbal
- Todar Mal
- Raja Man Singh
- Abdul Rahim Khan-i-Khanan
- Faqir Azizuddin
- Mulla Do Piyaza

## 2 Mughal Emperors: Jahangir, Shah Jahan and Aurangzeb

| Name of Ruler (Tenure)              | Real Name and Parents                                                    | Wives and Sons                                                                                                      | Important Exam Facts / Architecture                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jahangir</b><br>(1605–1627 CE)   | * <b>Real name:</b> Salim<br>* <b>Parents:</b> Harkha Bai - Akbar        | * <b>Wives:</b> Man Bai, Jagat Gosain, Mehrunnisa (Nur Jahan)<br>* <b>Sons:</b> Khusrau, Parvez, Shahryar, Jahandar | * His reign is called the ' <b>golden age of painting</b> '.<br>* Introduced the <b>do-aspa and sih-aspa</b> system.<br>* <b>Foreign travellers:</b> Hawkins, Sir Thomas Roe and Francisco Pelsaert came to his court.<br>* <b>Tomb:</b> at Shahdara (Lahore).<br>* <b>Architecture:</b> Akbar's Tomb (Sikandra).                                                                                                                  |
| <b>Shah Jahan</b><br>(1628–1657 CE) | * <b>Real name:</b> Khurram<br>* <b>Parents:</b> Jagat Gosain - Jahangir | * <b>Wife:</b> Arjumand Banu Begum (Mumtaz Mahal)<br>* <b>Sons:</b> Dara Shikoh, Aurangzeb, Murad, Shah Shuja       | * His reign is called the ' <b>golden age of architecture</b> '.<br>* In 1638 CE shifted the capital <b>from Agra to Delhi</b> .<br>* Gave Dara Shikoh the title ' <b>Shah Buland Iqbal</b> '.<br>* Dara Shikoh translated the Upanishads into Persian under the name ' <b>Sirr-i-Akbar</b> '.<br>* <b>Tomb:</b> Agra (Taj Mahal).<br>* <b>Architecture:</b> Jama Masjid and Red Fort of Delhi; Moti Masjid and Taj Mahal of Agra. |
| <b>Aurangzeb</b><br>(1658–1707 CE)  | * <b>Parents:</b> Mumtaz Mahal - Shah Jahan                              | * <b>Wife:</b> Dilras Banu Begum (Rabia-ud-Daurani)<br>* <b>Sons:</b> Muhammad Azam, Muhammad Akbar, Kam Bakhsh     | * <b>Crowned twice:</b> first on 31 July 1658 (Agra), second on 15 June 1659 (Delhi).<br>* <b>Death:</b> February 1707.<br>* <b>Tomb:</b> Khuldabad / Aurangabad (Maharashtra).<br>* <b>Architecture:</b> <b>Moti Masjid</b> inside the Red Fort (Delhi).                                                                                                                                                                          |

## 3 Wars of Succession in the Time of Shah Jahan (Exact Chronology)

(The favourite question of UPPCS Prelims - for arranging in correct order by date)

| No. | Name of Battle              | Exact Date       | Between Whom                                                                                    | Victor               |
|-----|-----------------------------|------------------|-------------------------------------------------------------------------------------------------|----------------------|
| 1.  | <b>Battle of Bahadurpur</b> | 24 February 1658 | Between Shah Shuja and the imperial army                                                        | <b>Imperial army</b> |
| 2.  | <b>Battle of Dharmat</b>    | 15 April 1658    | Combined army of Aurangzeb + Murad Bakhsh vs. the imperial army of Jaswant Singh and Qasim Khan | <b>Aurangzeb</b>     |
| 3.  | <b>Battle of Samugarh</b>   | 29 May 1658      | Combined army of Aurangzeb + Murad Bakhsh vs. Dara Shikoh                                       | <b>Aurangzeb</b>     |

| No. | Name of Battle          | Exact Date       | Between Whom                      | Victor                     |
|-----|-------------------------|------------------|-----------------------------------|----------------------------|
| 4.  | <b>Battle of Khajwa</b> | 5 January 1659   | Between Aurangzeb and Shah Shuja  | <b>Aurangzeb</b>           |
| 5.  | <b>Battle of Deorai</b> | 12-14 April 1659 | Between Dara Shikoh and Aurangzeb | <b>Aurangzeb (finally)</b> |

#### 4 Administrative Divisions of the Mughal Empire and Principal Officers

| Administrative Unit            | Principal Officer / Person Responsible |
|--------------------------------|----------------------------------------|
| Suba (province)                | Subedar or Nazim                       |
| Sarkar (district)              | Faujdar                                |
| Pargana or Mahal               | Shiqdar                                |
| Mauza / Diha / Grama (village) | Khut, Muqaddam or Chaudhari            |

#### 5 Principal Literature / Books of the Mughal Period and their Authors

| Name of Book / Text        | Author / Composer   |
|----------------------------|---------------------|
| Baburnama (Tuzuk-i-Baburi) | Babur               |
| Humayunnama                | Gulbadan Begum      |
| Tabaqat-i-Akbari           | Nizamuddin Ahmad    |
| Tarikh-i-Sher Shahi        | Abbas Khan Sherwani |
| Alamgirnama                | Muhammad Kazim      |
| Qanun-i-Humayuni           | Khwandamir          |
| Nuskha-i-Dilkusha          | Bhimsen             |
| Siyar-ul-Mutakhkherin      | Ghulam Husain       |
| Iqbalnama-i-Jahangiri      | Mutamid Khan        |
| Futuhāt-i-Alamgiri         | Ishwar Das Nagar    |
| Khulasat-ut-Tawarikh       | Sujan Rai Bhandari  |

## 6 Important Policy Measures of Akbar (Chronology)

| Year (CE) | Important Measure Taken by Akbar                                             |
|-----------|------------------------------------------------------------------------------|
| 1562      | Abolition of slavery                                                         |
| 1562      | Akbar's complete freedom from the 'harem party' (petticoat government)       |
| 1563      | Abolition of the pilgrimage tax                                              |
| 1564      | Abolition of jizya                                                           |
| 1571      | Founding of Fatehpur Sikri and transfer of the capital from Agra to Fatehpur |
| 1575      | Establishment of the Ibadatkhana (Fatehpur Sikri)                            |
| 1579      | Proclamation of the Mahzar (supremacy of the emperor in religious matters)   |
| 1580      | ' <b>Dahsala system</b> ' (settlement system by Todar Mal) implemented       |
| 1582      | Proclamation of the Din-i-Ilahi (Tauhid-i-Ilahi)                             |

## 7 Important Administrative and Economic Terminology of Medieval History

(A question asking the meaning of a single term, or a matching question, comes directly from here in UPPCS Prelims)

### > (a) Land Revenue, Tax and Economic Terminology

| Term / Terminology | Administrative or Economic Meaning                                                             |
|--------------------|------------------------------------------------------------------------------------------------|
| Iqta               | In the Delhi Sultanate, a special right to collect revenue (tax) from land.                    |
| Iqtadar            | The officer or provincial noble who received an iqta.                                          |
| Jagir              | A tract of land assigned for revenue collection in the Mughal period.                          |
| Jagirdar           | The Mughal mansabdar/officer who received a jagir.                                             |
| Khalisa            | Land whose revenue went directly to the <b>royal treasury</b> rather than to any intermediary. |
| Madad-i-Maash      | <b>Revenue-free land</b> or grant given to scholars or religious persons.                      |
| Kharaj             | Land revenue tax levied on the land of non-Muslims.                                            |
| Ushr               | Land tax collected from Muslim cultivators.                                                    |
| Jizya              | Protection/religious tax levied on non-Muslim subjects (zimmi).                                |



| Term / Terminology | Administrative or Economic Meaning                                                                  |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <b>Zakat</b>       | Religious tax compulsorily levied on Muslims.                                                       |
| <b>Khums</b>       | Share of the booty obtained in war or of buried treasure (of which 1/5 went to the royal treasury). |
| <b>Hundi</b>       | A written credit instrument used in the medieval period for the transfer of money or payment.       |
| <b>Banjara</b>     | A nomadic community that transported and traded grain and other goods over long distances.          |
| <b>Waqf</b>        | Immovable property dedicated to religious, educational or charitable purposes.                      |

### > (b) Military and Royal Control Terminology

| Term / Terminology  | Administrative or Military Meaning                                                                                     |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Mansab</b>       | The rank of an officer in the Mughal administration, denoting his status, pay and military obligation (zat and sawar). |
| <b>Dagh</b>         | Official mark branded on horses in the army to prevent their substitution.                                             |
| <b>Chehra</b>       | System of recording the descriptive roll of soldiers to prevent corruption.                                            |
| <b>Sulh-i-Kul</b>   | Akbar's policy of peace and tolerance towards all religions (universal peace).                                         |
| <b>Din-i-Ilahi</b>  | A moral-religious ideology/code of conduct introduced by Akbar.                                                        |
| <b>Farman</b>       | The highest royal order issued by the Mughal emperor or ruler.                                                         |
| <b>Sanad</b>        | A royal document (certificate) confirming a right, post or land grant.                                                 |
| <b>Muhar (Seal)</b> | Official mark (seal) used on royal orders or important documents for authentication.                                   |
| <b>Tughra</b>       | A special artistic/calligraphic style of writing the Sultan's name or royal titles.                                    |
| <b>Fatwa</b>        | A religious ruling given by a mufti or qazi on the basis of the Islamic Sharia law.                                    |
| <b>Ulema</b>        | The learned scholars of theology and Islamic law in Muslim society.                                                    |

### > (c) Classes of Peasants and Methods of Land Revenue Assessment

| Term / Terminology | Meaning of the Type of Peasant or Land Revenue Method         |
|--------------------|---------------------------------------------------------------|
| <b>Raiyat</b>      | The ordinary peasant or the subjects who cultivated the land. |

| Term / Terminology | Meaning of the Type of Peasant or Land Revenue Method                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Khudkasht</b>   | Peasants who cultivated their own land in their own village.                                                    |
| <b>Pahikasht</b>   | Peasants who went temporarily to other villages and cultivated land on a sharecropping basis.                   |
| <b>Muzara</b>      | A landless peasant or sharecropping labourer who cultivated another's land.                                     |
| <b>Batai</b>       | System of dividing the harvested crop/produce between the cultivator and the landowner in a fixed ratio.        |
| <b>Dahsala</b>     | System of fixing land revenue on the basis of the average produce and average prices of the previous ten years. |
| <b>Zabti</b>       | System of collecting tax on the basis of actual measurement of land and a fixed cash rate.                      |
| <b>Kankut</b>      | Method of fixing rent by merely estimating (notional assessment of) the standing crop in the fields.            |
| <b>Nasaq</b>       | Method of fixing land revenue on the basis of a previous assessment or old records, without fresh measurement.  |

## 1 Literature of the Sultanate Period, Regional Dynasties and the Vijayanagara Empire

| No. | Name of Text / Book | Author / Composer         | Special Details Useful for the Exam (where available)  |
|-----|---------------------|---------------------------|--------------------------------------------------------|
| 1.  | Tabaqat-i-Nasiri    | Minhaj-us-Siraj           | Period of Iltutmish and Nasiruddin Mahmud              |
| 2.  | Taj-ul-Maasir       | Hasan Nizami              | The first official history of the Delhi Sultanate      |
| 3.  | Khazain-ul-Futuh    | Amir Khusrau              | Description of Alauddin Khilji's conquest of the South |
| 4.  | Tarikh-i-Alai       | Amir Khusrau              | History of the reign of Alauddin Khilji                |
| 5.  | Ashiqi              | Amir Khusrau              | Love story of Khizr Khan and Deval Rani                |
| 6.  | Nuh Sipahr          | Amir Khusrau              | Period of Mubarak Khilji and praise of India           |
| 7.  | Tughlaqnama         | Amir Khusrau              | Depiction of the victory of Ghiyasuddin Tughlaq        |
| 8.  | Futuh-us-Salatin    | Isami (Abdul Malik Isami) | History of the Bahmani kingdom                         |
| 9.  | Tarikh-i-Firozshahi | Ziauddin Barani           | Period of Alauddin Khilji and the Tughlaqs             |
| 10. | Fatawa-i-Jahandari  | Ziauddin Barani           | Political philosophy of the Sultanate period           |

| No. | Name of Text / Book    | Author / Composer                    | Special Details Useful for the Exam (where available) |
|-----|------------------------|--------------------------------------|-------------------------------------------------------|
| 11. | Futuhāt-i-Firozshahi   | Firoz Shah Tughlaq                   | <b>Autobiography</b> of Sultan Firoz Shah Tughlaq     |
| 12. | Tarikh-i-Firozshahi    | Shams-i-Siraj Afif                   | Continued Barani's history                            |
| 13. | Tarikh-i-Mubarakshahi  | Yahya bin Ahmad Sirhindi             | The only authentic source on the Sayyid dynasty       |
| 14. | Rihla (Kitab-ul-Rehla) | Ibn Battuta                          | Travel account of the Moroccan traveller (MBT period) |
| 15. | Chachnama              | Ali bin Hamid Kufi                   | History of the Arab conquest of Sindh                 |
| 16. | Kitab-ul-Hind          | Al-Biruni                            | A view of India in the period of Mahmud of Ghazni     |
| 17. | Madhuravijayam         | Gangadevi                            | Description of Kumara Kampana's conquest of Madurai   |
| 18. | Amuktamalyada          | Krishnadevaraya                      | An administrative text written in the Telugu language |
| 19. | Jambavati Kalyanam     | Krishnadevaraya                      | A drama written in the Sanskrit language              |
| 20. | Manucharitam           | Allasani Peddana                     | Principal among the Ashtadiggajas of Krishnadevaraya  |
| 21. | Panduranga Mahatmyam   | Tenali Ramakrishna                   | Famous poet included among the Ashtadiggajas          |
| 22. | Varadambika Parinayam  | Tirumalamba                          | Woman writer of the Vijayanagara period               |
| 23. | Achyutarayabhyudayam   | Rajanatha Dindima                    | Biography of King Achyutadeva Raya                    |
| 24. | Saluvabhyudayam        | Rajanatha Dindima                    | History of the Saluva dynasty                         |
| 25. | Krishnaraya Vijayam    | Kumara Dhurjati                      | Description of the victories of Krishnadevaraya       |
| 26. | Rayavachakamu          | Unknown author                       | A Telugu prose source on the history of Vijayanagara  |
| 27. | Tarikh-i-Firishta      | Muhammad Qasim Hindu Shah (Firishta) | Dedicated to Ibrahim Adil Shah of Bijapur             |
| 28. | Kitab-i-Nauras         | Ibrahim Adil Shah II                 | Collection of Hindi songs (ruler of Bijapur)          |
| 29. | Burhan-i-Maasir        | Sayyid Ali bin Azizullah Tabatabai   | History of Ahmednagar (Nizam Shahi)                   |
| 30. | Adab-ul-Harb           | Fakhr-i-Mudabbir                     | Book on the art of war and military organisation      |
| 31. | Dalail-i-Firozshahi    | Azizuddin Khan                       | Persian translation of Sanskrit philosophical texts   |

| No. | Name of Text / Book | Author / Composer          | Special Details Useful for the Exam (where available)       |
|-----|---------------------|----------------------------|-------------------------------------------------------------|
| 32. | Mirat-i-Sikandari   | Sikandar bin Muhammad Arif | History of the Gujarat Sultanate                            |
| 33. | Riyaz-us-Salatin    | Ghulam Husain Salim        | History of Bengal                                           |
| 34. | Riyaz-ul-Insha      | Mahmud Gawan               | Collection of the letters of the Bahmani wazir Mahmud Gawan |

## 2 History of the Mughal Period and the Later Mughal Period

| No. | Name of Text / Book   | Author / Composer     | Special Details Useful for the Exam (where available)             |
|-----|-----------------------|-----------------------|-------------------------------------------------------------------|
| 35. | Baburnama             | Babur                 | <b>Autobiography</b> of Babur (in the Chaghatai Turkish language) |
| 36. | Humayunnama           | Gulbadan Begum        | Biography written by Humayun's sister                             |
| 37. | Tarikh-i-Shershahi    | Abbas Khan Sarwani    | Principal source for the history of Sher Shah Suri                |
| 38. | Akbarnama             | Abul Fazl             | Official history of Akbar                                         |
| 39. | Ain-i-Akbari          | Abul Fazl             | The third and statistical part of the Akbarnama itself            |
| 40. | Muntakhab-ut-Tawarikh | Abdul Qadir Badauni   | A critical work on Akbar's religious policies                     |
| 41. | Tabaqat-i-Akbari      | Nizamuddin Ahmad      | General history of India in the Mughal period                     |
| 42. | Tarikh-i-Akbari       | Arif Qandhari         | Account of Akbar's early years                                    |
| 43. | Tuzuk-i-Jahangiri     | Jahangir              | <b>Autobiography</b> of Jahangir (Persian language)               |
| 44. | Iqbalnama-i-Jahangiri | Mutamid Khan          | History of Jahangir (completed the autobiography)                 |
| 45. | Padshahnama (I)       | Abdul Hamid Lahori    | Official history of Shah Jahan's reign                            |
| 46. | Padshahnama (II)      | Muhammad Amin Qazvini | Another authentic text of Shah Jahan's period                     |
| 47. | Amal-i-Salih          | Muhammad Salih Kamboh | Account of the whole life of Shah Jahan                           |
| 48. | Alamgirnama           | Mirza Muhammad Kazim  | History of the first 10 years of Aurangzeb's reign                |
| 49. | Maasir-i-Alamgiri     | Saqi Mustaid Khan     | Also called the ' <b>Gazetteer of the Mughal State</b> '          |
| 50. | Futuh-i-Alamgiri      | Ishwar Das Nagar      | An important source for the history of Aurangzeb's period         |
| 51. | Nuskha-i-Dilkusha     | Bhimsen               | History of the Deccan (South) in Aurangzeb's time                 |
| 52. | Muntakhab-ul-Lubab    | Khafi Khan            | History of Aurangzeb's period (written secretly)                  |

| No. | Name of Text / Book   | Author / Composer       | Special Details Useful for the Exam (where available)         |
|-----|-----------------------|-------------------------|---------------------------------------------------------------|
| 53. | Mirat-i-Alam          | Bakhtawar Khan          | Account of Aurangzeb's early reign                            |
| 54. | Ruqaat-i-Alamgiri     | Aurangzeb               | Collection of Aurangzeb's letters                             |
| 55. | Qanun-i-Humayuni      | Khwandamir              | Description of Humayun's administrative rules and inventions  |
| 56. | Siyar-ul-Mutakhkherin | Ghulam Husain Tabatabai | History of the later Mughal period and of Bengal              |
| 57. | Shahjahannama         | Inayat Khan             | History of Shah Jahan's reign                                 |
| 58. | Chahar Chaman         | Chandrabhan Brahman     | Description of Mughal court life and of Shah Jahan's period   |
| 59. | Khulasat-ut-Tawarikh  | Sujan Rai Bhandari      | Geographical and economic facts of India in the Mughal period |

### 3 Maratha History, the Rajput Period and Other Regional Literature

| No. | Name of Text / Book | Author / Composer         | Special Details Useful for the Exam (where available)     |
|-----|---------------------|---------------------------|-----------------------------------------------------------|
| 60. | Shivabharata        | Paramananda               | Sanskrit epic on the life of Shivaji Maharaj              |
| 61. | Sabhasad Bakhar     | Krishnaji Anant Sabhasad  | The oldest and most authentic bakhar of Shivaji's period  |
| 62. | Chitnis Bakhar      | Malhar Ramrao Chitnis     | A source for later Maratha history                        |
| 63. | Shivadigvijay       | Unknown author            | Bakhar based on the victories of Shivaji Maharaj          |
| 64. | Ajnapatra           | Ramchandrapant Amatya     | A text on Maratha administrative and political principles |
| 65. | Prithvirajavijaya   | Jayanaka                  | Sanskrit poem on the victories of Prithviraj Chauhan      |
| 66. | Prithviraj Raso     | Chand Bardai              | The first epic of Hindi (of Prithviraj Chauhan's period)  |
| 67. | Hammira Mahakavya   | Nayachandra Suri          | History of Hammir Deva, the Chauhan ruler of Ranthambhore |
| 68. | Hammir Raso         | Jodhraj                   | Description of the heroic battles of Hammir Deva          |
| 69. | Rajatarangini       | Kalhana                   | The first systematic historical text of Kashmir           |
| 70. | Prabandhachintamani | Merutunga                 | History of the Solanki/Chalukya kings of Gujarat          |
| 71. | Prabandhakosha      | Rajashekhara Suri         | Biographies of medieval kings and Jain acharyas           |
| 72. | Tutinama            | Khwaja Ziauddin Nakhshabi | Persian translation of Sanskrit tales (Shukasaptati)      |

## 4 Bhakti Movement, Sufi Saints and Cultural Literature

| No. | Name of Text / Book            | Author / Composer     | Nature of the Bhakti/Sufi Tradition                             |
|-----|--------------------------------|-----------------------|-----------------------------------------------------------------|
| 73. | Ramcharitmanas                 | Tulsidas              | Written in the Awadhi language (contemporary of Akbar)          |
| 74. | Sursagar                       | Surdas                | The principal text of Krishna bhakti in Braj Bhasha             |
| 75. | Bijak                          | Kabir                 | Collection of the teachings/sakhis of Kabirdas                  |
| 76. | Guru Granth Sahib (Adi Granth) | Guru Arjan Dev        | The holy book of Sikhism (compiled in 1604 CE)                  |
| 77. | Chaurasi Vaishnavan ki Varta   | Gokulnath             | Account of the Vaishnava saints of the Vallabha sect            |
| 78. | Bhaktamal                      | Nabhadas              | Biographies of medieval Vaishnava devotees and saints           |
| 79. | Premvatika                     | Raskhan               | Poetry based on Krishna bhakti                                  |
| 80. | Fawaid-ul-Fuad                 | Amir Hasan Sijzi      | Collection of the sayings of the Sufi saint Nizamuddin Auliya   |
| 81. | Padmavat                       | Malik Muhammad Jayasi | Written in the Awadhi language (contemporary of Sher Shah Suri) |

## 1 Arrival of the European Companies in India, their Establishment and Trading Centres

| European Company   | Year of Arrival/Establishment                                    | First Factory in India (Year) | Principal Trading Centres / Headquarters          |
|--------------------|------------------------------------------------------------------|-------------------------------|---------------------------------------------------|
| 1. Portuguese      | 1498 (Vasco da Gama came to Calicut) / company established: 1505 | Cochin (1503)                 | Goa, Diu, Daman                                   |
| 2. Dutch           | 1596 (arrival) / company established: 1602 (VOC)                 | Masulipatnam (1605 CE)        | Pulicat, Nagapattinam, Chinsura                   |
| 3. English         | 1608 (arrival) / company established: 1600                       | Surat (1613 CE)               | Surat, Madras, Bombay, Calcutta                   |
| 4. Danes (Denmark) | 1616                                                             | Tranquebar (1620, Tamil Nadu) | Tranquebar (Tamil Nadu), Serampore (Bengal)       |
| 5. French          | 1664                                                             | Surat (1668 CE)               | Pondicherry, Chandernagore, Mahe, Karaikal, Yanam |

- **Correct order of arrival of the European powers:** Portuguese → Dutch → English → Danes → French.

## 2 Two Historic Battles that Decided the Destiny of India

| Name of Battle    | Exact Date      | Between Whom                                                                                                                                                           | Result and Historical Importance                                                                                  |
|-------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Battle of Plassey | 23 June 1757    | The Nawab of Bengal, <b>Siraj-ud-Daulah</b> , vs. the English commander <b>Robert Clive</b>                                                                            | The English were victorious. This battle laid the <b>foundation of the British Empire</b> in India.               |
| Battle of Buxar   | 22 October 1764 | The combined front of <b>Mir Qasim</b> (Bengal), <b>Shuja-ud-Daulah</b> (Awadh) and <b>Shah Alam II</b> (Mughal emperor) vs. the English commander <b>Hector Munro</b> | The English were victorious. This battle established the <b>real and sovereign power</b> of the English in India. |

## 3 Principal Social Reform Acts of the British Period

| Act / Law                    | Year (CE)     | Governor-General Responsible                               | Main Subject / Legal Provision                                                                                                      |
|------------------------------|---------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Prohibition of infanticide   | 1795 and 1802 | Sir John Shore and Lord Wellesley                          | Complete ban on the killing of newborn infants (especially girls).                                                                  |
| Prohibition of Sati          | 1829 CE       | Lord William Bentinck                                      | Regulation XVII completely banned the practice of Sati and declared it a punishable offence.                                        |
| Prohibition of slavery       | 1843 CE       | Lord Ellenborough                                          | Under the provisions of the Charter Act of 1833, slavery was declared illegal in India in 1843.                                     |
| Hindu Widows' Remarriage Act | 1856 CE       | Lord Canning ( <i>draft prepared in Dalhousie's time</i> ) | Gave legal recognition to widow remarriage and treated the children born of it as legitimate (effort of Ishwar Chandra Vidyasagar). |
| Native Marriage Act          | 1872 CE       | Lord Northbrook                                            | Gave legal recognition to inter-caste marriages.                                                                                    |
| Age of Consent Act           | 1891 CE       | Lord Lansdowne                                             | The minimum age of marriage for girls was fixed at <b>12 years</b> .                                                                |
| Sarda Act                    | 1929 CE       | Lord Irwin                                                 | Minimum age of marriage fixed: <b>14 years</b> for girls and <b>18 years</b> for boys.                                              |

## 4 Principal Historic Treaties of Modern India (1639 - 1846)

| Name of Treaty       | Year (CE) | Between Whom (Parties)         | Main Terms / Result of the Treaty                                                                        |
|----------------------|-----------|--------------------------------|----------------------------------------------------------------------------------------------------------|
| Treaty of Asurar Ali | 1639 CE   | Mughal Empire and Ahom kingdom | The boundary between the Mughal and Ahom kingdoms was fixed; Mughal attempts to conquer the Ahoms ended. |



| Name of Treaty              | Year (CE)          | Between Whom (Parties)                                                                 | Main Terms / Result of the Treaty                                                                                                                                                                                                                                                     |
|-----------------------------|--------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Treaty of Purandar (First)  | 1665 CE            | Mughal commander <b>Jai Singh I</b> and <b>Chhatrapati Shivaji Maharaj</b>             | After the siege of Purandar fort, Shivaji Maharaj had to hand over 23 forts to the Mughals.                                                                                                                                                                                           |
| Treaty of Alinagar          | 1757<br>(February) | Nawab <b>Siraj-ud-Daulah</b> and <b>Robert Clive</b>                                   | The English were permitted to fortify Calcutta and to trade in Bengal without taxes.                                                                                                                                                                                                  |
| First Treaty of Allahabad   | 12 August<br>1765  | <b>Robert Clive</b> and the Nawab of Awadh, <b>Shuja-ud-Daulah</b>                     | <ul style="list-style-type: none"> <li>* The Nawab gave the districts of Allahabad and Kara to the Mughal emperor Shah Alam II.</li> <li>* Paid the Company an indemnity of ₹50 lakh.</li> <li>* The jagir of Balwant Singh, zamindar of Banaras, was restored.</li> </ul>            |
| Second Treaty of Allahabad  | 16 August<br>1765  | <b>Robert Clive</b> and the Mughal emperor <b>Shah Alam II</b>                         | <ul style="list-style-type: none"> <li>* The Company obtained the <b>permanent Diwani of Bengal, Bihar and Orissa</b>.</li> <li>* An annual pension of ₹26 lakh was fixed for the emperor.</li> <li>* The emperor was to live at Allahabad under the Company's protection.</li> </ul> |
| Treaty of Madras            | 1769 CE            | <b>Hyder Ali</b> (Mysore) and the English                                              | The First Anglo-Mysore War ended; conquered territories were returned and a defensive agreement was made.                                                                                                                                                                             |
| Treaty of Purandar (Second) | 1776 CE            | The Marathas (Nana Phadnavis) and the English (Calcutta Council)                       | Made in the first phase of the First Anglo-Maratha War to annul the Treaty of Surat.                                                                                                                                                                                                  |
| Convention of Wadgaon       | 1779 CE            | The Marathas and the English                                                           | A humiliating treaty after the defeat of the English during the First Anglo-Maratha War.                                                                                                                                                                                              |
| Treaty of Salbai            | 1782 CE            | The Marathas and the English (mediation of Mahadji Scindia)                            | <b>The First Anglo-Maratha War ended;</b> peace was established for the next 20 years.                                                                                                                                                                                                |
| Treaty of Seringapatam      | 1792 CE            | <b>Tipu Sultan</b> and the English ( <b>Lord Cornwallis</b> + Marathas + Nizam)        | <b>The Third Anglo-Mysore War ended;</b> Tipu had to give up half his kingdom and pay an indemnity of ₹3 crore.                                                                                                                                                                       |
| Treaty of Lahore            | 1846<br>(March)    | Governor-General <b>Henry Hardinge</b> and Maharaja <b>Dalip Singh</b> (Lahore Darbar) | <b>The First Anglo-Sikh War ended;</b> a heavy indemnity was imposed on the Sikhs and a British Resident was appointed.                                                                                                                                                               |
| Treaty of Amritsar          | 1846<br>(March)    | The British East India Company and <b>Maharaja Gulab Singh</b>                         | The English <b>sold Kashmir to the Dogra Raja Gulab Singh for ₹75 lakh</b> (ruled until 1947).                                                                                                                                                                                        |

## 5 Chronology of the Peshwas (Rise of the Maratha Empire)

| No. | Name of Peshwa            | Tenure         | Special Exam Points                                                                                                                |
|-----|---------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | <b>Balaji Vishwanath</b>  | 1713 – 1720 CE | He is also called 'the second founder of the Maratha Empire'.                                                                      |
| 2.  | <b>Bajirao I</b>          | 1720 – 1740 CE | 'The fighting Peshwa'; gave the principle of Hindu Pad-Padshahi; seized Salsette and Bassein from the Portuguese.                  |
| 3.  | <b>Balaji Bajirao</b>     | 1740 – 1761 CE | Famous as ' <b>Nana Saheb</b> '; the <b>Third Battle of Panipat took place on 14 January 1761</b> in his time.                     |
| 4.  | <b>Madhavrao I</b>        | 1761 – 1772 CE | Re-established Maratha power after the defeat at Panipat.                                                                          |
| 5.  | <b>Narayan Rao</b>        | 1772 – 1773 CE | His uncle Raghunath Rao (Raghoba) had him murdered.                                                                                |
| 6.  | <b>Madhav Narayan Rao</b> | 1774 – 1795 CE | Minor Peshwa; the 'Barbhai Council' (Nana Phadnavis) ran the government.                                                           |
| 7.  | <b>Bajirao II</b>         | 1795 – 1818 CE | The last Peshwa; signed the Treaty of Bassein (1802) with the English. The office was abolished after the Third Anglo-Maratha War. |

## 6 Anglo-Mysore Wars (Chronology and Treaties)

| War                     | Duration (Years) | Ruler of Mysore                                                    | English Governor / Leadership | Ended by which Treaty / Result                                                                                                       |
|-------------------------|------------------|--------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| First Anglo-Mysore War  | 1767 – 1769 CE   | <b>Hyder Ali</b>                                                   | Lord Verelst                  | <b>Treaty of Madras (1769)</b> : the English were defeated.                                                                          |
| Second Anglo-Mysore War | 1780 – 1784 CE   | <b>Hyder Ali</b> ( <i>after his death in 1782 Tipu took over</i> ) | <b>Warren Hastings</b>        | <b>Treaty of Mangalore (1784)</b> : both sides returned each other's conquered territories.                                          |
| Third Anglo-Mysore War  | 1790 – 1792 CE   | <b>Tipu Sultan</b>                                                 | <b>Lord Cornwallis</b>        | <b>Treaty of Seringapatam (1792)</b> : heavy defeat of Tipu; half his kingdom went to the Company and its allies.                    |
| Fourth Anglo-Mysore War | <b>1799 CE</b>   | <b>Tipu Sultan</b>                                                 | <b>Lord Wellesley</b>         | <b>No treaty</b> : Tipu Sultan died fighting. The independent power of Mysore fell and the throne was handed to the Wodeyar dynasty. |

## 7 Principal Civil, Peasant and Tribal Revolts of Modern India (Chronology)

(UPPCS Prelims asks direct matching of years or the correct chronological order of 4 revolts)

| No. | Name of Revolt / Movement                    | Exact Year          | Geographical Area / Main Feature                                                                                               |
|-----|----------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1.  | <b>Sanyasi Revolt</b>                        | <b>1763 CE</b>      | Bengal; the plot of Bankim Chandra Chatterjee's novel 'Anandamath' is based on it.                                             |
| 2.  | <b>Paika Rebellion</b>                       | 1817 CE             | Odisha; a strong revolt against the English under the leadership of Bakshi Jagabandhu.                                         |
| 3.  | <b>Bhil Revolt</b>                           | 1818 CE             | Western Ghats (Khandesh/Maharashtra); under the leadership of Sewaram, against agrarian hardships.                             |
| 4.  | <b>Ramosi Revolt</b>                         | 1822 CE             | Western Ghats (Maharashtra); under the leadership of Chittur Singh.                                                            |
| 5.  | <b>Ahom Revolt</b>                           | 1828 CE             | Assam; under the leadership of Gomdhar Konwar, because the English did not vacate the region.                                  |
| 6.  | <b>Wahabi Movement</b>                       | 1831 CE             | In India Sayyid Ahmad of Rae Bareilly was its chief propounder (main centre: Patna).                                           |
| 7.  | <b>Kol Revolt</b>                            | 1831 CE             | Chotanagpur region (Jharkhand); under the leadership of Buddho Bhagat.                                                         |
| 8.  | <b>Chotanagpur Revolt</b>                    | 1831-32 CE          | Collective land discontent of the tribals in the Chotanagpur plateau region.                                                   |
| 9.  | <b>Kuka Movement</b>                         | 1840 CE             | Punjab; a religio-political movement started by Bhagat Jawahar Mal (Sian Sahib) and Baba Ram Singh Kuka.                       |
| 10. | <b>Santhal Rebellion</b>                     | <b>1855-56 CE</b>   | Daman-i-Koh (Bhagalpur to the Rajmahal hills); a major revolt under the leadership of <b>Sidhu and Kanhu</b> .                 |
| 11. | <b>Munda Rebellion</b><br>( <i>Ulgulan</i> ) | <b>1899-1900 CE</b> | Ranchi and Khunti (Jharkhand); under the leadership of <b>Birsa Munda</b> ('Dharti Aba') to save collective farming.           |
| 12. | <b>Tana Bhagat Movement</b>                  | 1914 CE             | Chotanagpur (Jharkhand); a cultural reform movement under the leadership of Jatra Bhagat and Balram Bhagat.                    |
| 13. | <b>Eka Movement</b>                          | 1921 CE             | Awadh (Hardoi, Bahraich, Sitapur); under the leadership of <b>Madari Pasi</b> , against the increase in rent.                  |
| 14. | <b>Moplah Rebellion</b>                      | <b>1921 CE</b>      | Malabar coast (Kerala); revolt of Muslim tenants against Hindu landlords and the English under the leadership of Ali Musliyar. |
| 15. | <b>Rampa Rebellion</b>                       | 1922 CE             | Andhra Pradesh (Godavari region); under the leadership of <b>Alluri Sitarama Raju</b> , against the forest laws.               |

## 8 Important Political Organisations Established before the Founding of the Congress (1885)

| Name of the Organisation                      | Year of Establishment | Place of Establishment | Principal Founders                                                                                      |
|-----------------------------------------------|-----------------------|------------------------|---------------------------------------------------------------------------------------------------------|
| East India Association                        | 1866 CE               | London                 | Dadabhai Naoroji (to acquaint the British public with Indian issues)                                    |
| Poona Sarvajanik Sabha                        | 1870 CE               | Poona                  | S.H. Chiplunkar, G.V. Joshi and Mahadev Govind (M.G.) Ranade                                            |
| Indian Society                                | 1872 CE               | London                 | Anand Mohan Bose                                                                                        |
| Indian Association<br>( <i>Bharat Sangh</i> ) | 1876 CE               | Kolkata                | Surendranath (S.N.) Banerjee and Anand Mohan Bose (an important precursor organisation of the Congress) |
| Madras Mahajan Sabha                          | 1884 CE               | Chennai (Madras)       | G.S. Iyer, M. Viraraghavachari and P. Ananda Charlu                                                     |
| Bombay Presidency Association                 | 1885 CE               | Mumbai (Bombay)        | Pherozeshah Mehta, K.T. Telang and Badruddin Tyabji                                                     |

## 1 The Revolt of 1857: Principal Centres, Leadership and Date of Outbreak

(UPPCS Prelims asks for their chronological order or direct matching)

| Centre of the Revolt | Date of Outbreak of the Revolt | Principal Indian Leader                                             |
|----------------------|--------------------------------|---------------------------------------------------------------------|
| Barrackpore          | 29 March 1857                  | <b>Mangal Pandey</b> (the first spark of the revolt)                |
| Meerut               | <b>10 May 1857</b>             | <b>Kadam Singh</b> (open revolt by the soldiers and march to Delhi) |
| Lucknow              | <b>4 June 1857</b>             | <b>Begum Hazrat Mahal</b>                                           |
| Jhansi               | June 1857                      | <b>Rani Lakshmi Bai</b>                                             |
| Kanpur               | <b>5 June 1857</b>             | <b>Nana Saheb</b>                                                   |

## 2 Raja Rammohan Roy: Chronology of Life and Reform Work

| Year (CE) | Historical Event / Social Reform Work                                              |
|-----------|------------------------------------------------------------------------------------|
| 1809      | Wrote the book ' <b>A Gift to Monotheists</b> ' (Tuhfat-ul-Muwahhidin) in Persian. |
| 1815      | Founded the <b>Atmiya Sabha</b> for religious and social discussion.               |

| Year (CE) | Historical Event / Social Reform Work                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| 1817      | Founded the <b>Hindu College</b> in Calcutta with the help of David Hare.                                                      |
| 1820      | Wrote the book ' <b>The Precepts of Jesus</b> '.                                                                               |
| 1825      | Founded the <b>Vedanta College</b> to synthesise Western and Indian learning.                                                  |
| 1828      | Founded the <b>Brahma Sabha</b> , which later became the <b>Brahmo Samaj</b> .                                                 |
| 1829      | Brought about the <b>complete abolition of Sati</b> (Regulation XVII) with the help of Governor-General Lord William Bentinck. |

### 3 The Kisan Sabha Movement (especially with reference to Uttar Pradesh)

*(Extremely important for both the 'UP Special' and 'Modern History' sections of UPPCS)*

| Kisan Sabha / Movement       | Year of Establishment | Place of Establishment | Principal Founders and Associated Leaders                                                                                                                |
|------------------------------|-----------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| United Provinces Kisan Sabha | 1918 CE               | Lucknow                | Indra Narayan Dwivedi, Madan Mohan Malaviya and Gauri Shankar Mishra.                                                                                    |
| Awadh Kisan Sabha            | 1920 CE               | Pratapgarh             | <b>Baba Ramchandra</b> and Jawaharlal Nehru.                                                                                                             |
| All India Kisan Sabha        | 1936 CE               | <b>Lucknow</b>         | * <b>President:</b> Swami Sahajanand Saraswati.<br>* <b>Secretary:</b> N.G. Ranga.<br>* <b>Other leaders:</b> E.M.S. Namboodiripad, Pandit Narendra Dev. |

### 4 Important Sessions of the Indian National Congress (INC) and their Features

*(A coded matching question from here is asked in UPPCS Prelims almost every year)*

| Year (CE) | Venue of the Session | Name of the President     | Historical Feature / Event Occurring for the First Time                    |
|-----------|----------------------|---------------------------|----------------------------------------------------------------------------|
| 1887      | Madras               | <b>Badruddin Tyabji</b>   | Became the <b>first Muslim President</b> of the Congress.                  |
| 1888      | Allahabad            | <b>George Yule</b>        | Became the <b>first English/European President</b> of the Congress.        |
| 1896      | Kolkata              | Rahimtulla M. Sayani      | The national song ' <b>Vande Mataram</b> ' was sung for the first time.    |
| 1906      | Kolkata              | Dadabhai Naoroji          | The word ' <b>Swaraj</b> ' was used from the platform for the first time.  |
| 1911      | Kolkata              | Pandit Bishan Narayan Dar | The national anthem ' <b>Jana Gana Mana</b> ' was sung for the first time. |

| Year (CE) | Venue of the Session     | Name of the President | Historical Feature / Event Occurring for the First Time              |
|-----------|--------------------------|-----------------------|----------------------------------------------------------------------|
| 1917      | Kolkata                  | Annie Besant          | Became the <b>first woman President</b> of the Congress.             |
| 1923      | Delhi ( <i>special</i> ) | Abul Kalam Azad       | Became the <b>youngest President</b> in the history of the Congress. |
| 1925      | Kanpur                   | Sarojini Naidu        | Became the <b>first Indian woman President</b> of the Congress.      |
| 1929      | Lahore                   | Jawaharlal Nehru      | The historic resolution of ' <b>Purna Swaraj</b> ' was passed.       |
| 1931      | Karachi                  | Vallabhbhai Patel     | Resolution demanding <b>Fundamental Rights</b> was passed.           |
| 1937      | Faizpur                  | Jawaharlal Nehru      | <b>The first session of the Congress held in a village.</b>          |

## 5 Principal 'First' Newspapers and Journals of India

| Newspaper / Journal | Year of Founding | Place of Publication | Founder / Editor     | Language and Special Exam Facts                       |
|---------------------|------------------|----------------------|----------------------|-------------------------------------------------------|
| Bengal Gazette      | 1780 CE          | Kolkata              | James Augustus Hicky | <b>India's first newspaper</b> (English language).    |
| Madras Courier      | 1784 CE          | Madras               | —                    | The first newspaper published from Madras (Chennai).  |
| Samachar Darpan     | 1818 CE          | Kolkata              | William Carey        | The <b>first Bengali weekly</b> newspaper.            |
| Mirat-ul-Akhbar     | 1822 CE          | Kolkata              | Raja Rammohan Roy    | India's <b>first Persian journal</b> .                |
| Jam-i-Jahan Numa    | 1822 CE          | Kolkata              | —                    | The <b>first newspaper in the Urdu language</b> .     |
| Udant Martand       | 1826 CE          | Kolkata              | Jugal Kishore Shukla | The <b>first newspaper in the Hindi language</b> .    |
| Bombay Samachar     | 1830 CE          | Mumbai               | —                    | The <b>first newspaper in the Gujarati language</b> . |

## 6 Principal Social Reformers and their Organisational Work / Posts

| Name of Social Reformer                   | Related Important Work / Administrative Post |
|-------------------------------------------|----------------------------------------------|
| D.K. Karve ( <i>Dhondo Keshav Karve</i> ) | Secretary, Widow Remarriage Association.     |
|                                           |                                              |

| Name of Social Reformer                                  | Related Important Work / Administrative Post                                                                                                      |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| J.E.D. Bethune ( <i>John Elliot Drinkwater Bethune</i> ) | Founded the famous ' <b>Bethune Girls' School' (1849)</b> in Calcutta.                                                                            |
| Ishwar Chandra Vidyasagar                                | Was <b>Principal of the famous Sanskrit College</b> of Calcutta.                                                                                  |
| B.M. Malabari ( <i>Behramji M. Malabari</i> )            | Parsi social reformer; began an intensive struggle <b>against child marriage</b> (the Age of Consent Act of 1891 came about through his efforts). |

## 7 Principal Education Commissions and Committees of the British Period (Chronology)

(The favourite section of UPPCS Prelims for 'Match the Following' and chronology)

| Education Commission / Policy                               | Year | Chairman / Governor                  | Main Recommendations and Exam Facts                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Macaulay's Minute                                           | 1835 | Lord Macaulay (Bentinck's period)    | The <b>foundation of English education</b> was laid in India and a fixed sum was allotted for education. (Downward Filtration Theory).                                                                                                                                                |
| Wood's Despatch                                             | 1854 | Charles Wood (Dalhousie's period)    | * It is called the ' <b>Magna Carta</b> ' of English education in India.<br>* Recommended English as the medium for primary, secondary and graduate-level education, and the opening of education departments in the provinces and of universities on the model of London University. |
| Hunter Commission                                           | 1882 | William Hunter (Ripon's period)      | To study the progress of education since 1854. It laid <b>special emphasis on primary education</b> .                                                                                                                                                                                 |
| Universities Commission                                     | 1902 | Thomas Raleigh (Curzon's period)     | To implement improvements in the condition of Indian universities. The <b>Indian Universities Act, 1904</b> was passed on its recommendation.                                                                                                                                         |
| Sadler Commission ( <i>Calcutta University Commission</i> ) | 1917 | Michael Sadler (Chelmsford's period) | Study of the problems of Calcutta University. It emphasised <b>separating secondary education from university education</b> .                                                                                                                                                         |
| Hartog Committee                                            | 1929 | Philip Hartog (Irwin's period)       | Study of the democratisation of primary education and of the problem of <b>wastage and stagnation</b> in education.                                                                                                                                                                   |
| Sargent Plan                                                | 1944 | John Sargent (Wavell's period)       | A long-term (40-year) target was set for implementing universal, free and compulsory primary education in India.                                                                                                                                                                      |

## 8 Principal Trade Unions (Central Labour Organisations) and their Related Political Parties

(Highly important for current political coded-matching questions)



| Name of the Trade Union (Labour Organisation) | Related Parent Political Party             |
|-----------------------------------------------|--------------------------------------------|
| Bharatiya Mazdoor Sangh (BMS)                 | Bharatiya Janata Party (BJP)               |
| Indian National Trade Union Congress (INTUC)  | Indian National Congress (INC)             |
| All India Trade Union Congress (AITUC)        | Communist Party of India (CPI)             |
| United Trade Union Congress (UTUC)            | Communist Party of India - Marxist (CPI-M) |

## 9 Principal National Newspapers / Books of Modern India and their Authors

| Name of Book / Newspaper                                  | Author / Editor / Composer                                                                                  |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Commonweal                                                | Annie Besant                                                                                                |
| New India                                                 | Annie Besant                                                                                                |
| Comrade                                                   | Mohammad Ali                                                                                                |
| Hamdard                                                   | Mohammad Ali                                                                                                |
| Som Prakash                                               | Ishwar Chandra Vidyasagar ( <i>the first newspaper against which the Vernacular Press Act was applied</i> ) |
| Independent                                               | Motilal Nehru (newspaper)                                                                                   |
| Sambad Kaumudi                                            | Raja Rammohan Roy                                                                                           |
| Free Hindustan                                            | Tarakanath Das                                                                                              |
| Krantikari                                                | Sachindranath Sanyal                                                                                        |
| Amrita Bazar Patrika                                      | Sisir Kumar Ghosh ( <i>converted into English overnight to escape the Vernacular Press Act</i> )            |
| Satyarth Prakash                                          | Dayanand Saraswati                                                                                          |
| Gulamgiri                                                 | Jyotiba Phule                                                                                               |
| The Indian Struggle                                       | Subhas Chandra Bose                                                                                         |
| 'India is for the Indians'<br>( <i>slogan/statement</i> ) | Chittaranjan Das (C.R. Das)                                                                                 |

## 1 Important Events of Indian History (1911 – 2013)

(UPPCS Prelims asks every year from this section to arrange 4 events in correct chronological order)

### > (a) Pre-Independence Period (1911 – 1947)

| Year | Important Historical Event / Exam Facts                                                                                                                                                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1911 | <ul style="list-style-type: none"> <li>* Arrival in India of the British Emperor <b>King George V</b>.</li> <li>* Announcement of the <b>transfer of the capital from Calcutta to Delhi</b>.</li> <li>* '<b>Jana Gana Mana</b>' was sung for the first time at the Calcutta session of the INC.</li> <li>* India's (and the world's) <b>first airmail service</b> began between Bamrauli and Allahabad.</li> </ul> |
| 1919 | <ul style="list-style-type: none"> <li>* <b>Dyarchy</b> was introduced in the provinces by the Government of India Act, 1919.</li> <li>* The repressive <b>Rowlatt Act 1919</b> was enforced.</li> <li>* The <b>Jallianwala Bagh tragedy</b> took place on 13 April.</li> </ul>                                                                                                                                    |
| 1920 | <ul style="list-style-type: none"> <li>* Beginning of the <b>Khilafat Movement</b> and of the <b>Non-Cooperation Movement</b> by Mahatma Gandhi.</li> </ul>                                                                                                                                                                                                                                                        |
| 1922 | <ul style="list-style-type: none"> <li>* The <b>Chauri Chaura incident (outburst)</b> took place at Gorakhpur in Uttar Pradesh.</li> <li>* Gandhiji <b>withdrew (suspended) the Non-Cooperation Movement</b>.</li> </ul>                                                                                                                                                                                           |
| 1928 | <ul style="list-style-type: none"> <li>* Visit of the <b>Simon Commission</b> to India (nationwide protest).</li> <li>* <b>Death of Lala Lajpat Rai</b> from a lathi-charge during a protest demonstration.</li> </ul>                                                                                                                                                                                             |
| 1929 | <ul style="list-style-type: none"> <li>* <b>Historic resolution of complete independence (Swaraj)</b> at the Lahore session of the INC (President: Nehru).</li> </ul>                                                                                                                                                                                                                                              |
| 1930 | <ul style="list-style-type: none"> <li>* Beginning of the <b>Dandi March</b> (Salt Satyagraha) and the <b>Civil Disobedience Movement</b> by Gandhiji.</li> </ul>                                                                                                                                                                                                                                                  |
| 1931 | <ul style="list-style-type: none"> <li>* The <b>Gandhi-Irwin Pact</b> was concluded.</li> <li>* <b>Bhagat Singh, Sukhdev and Rajguru were hanged</b> on 23 March.</li> </ul>                                                                                                                                                                                                                                       |
| 1935 | <ul style="list-style-type: none"> <li>* The <b>Government of India Act, 1935</b> was passed (dyarchy at the centre, provincial autonomy).</li> </ul>                                                                                                                                                                                                                                                              |
| 1942 | <ul style="list-style-type: none"> <li>* Visit of the Cripps Mission to India.</li> <li>* The <b>Quit India Movement</b> began in August.</li> <li>* <b>Formation of the Azad Hind Fauj</b> by Captain Mohan Singh and Rash Behari Bose.</li> </ul>                                                                                                                                                                |
| 1946 | <ul style="list-style-type: none"> <li>* Visit of the three-member <b>Cabinet Mission</b> to India (formation of the Constituent Assembly).</li> </ul>                                                                                                                                                                                                                                                             |
| 1947 | <ul style="list-style-type: none"> <li>* <b>15 August</b>: Partition of India and attainment of independence.</li> </ul>                                                                                                                                                                                                                                                                                           |

### > (b) Post-Independence Period (1948 – 2013)

| Year | Important National and International Events                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1948 | <ul style="list-style-type: none"> <li>* <b>Assassination of Mahatma Gandhi</b> on 30 January.</li> <li>* <b>Pakistan's first attack on India</b> over the Kashmir issue.</li> </ul> |
| 1950 | <ul style="list-style-type: none"> <li>* With the Constitution coming into force on 26 January, <b>India became a sovereign republic</b>.</li> </ul>                                 |
|      |                                                                                                                                                                                      |

| Year | Important National and International Events                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1951 | <ul style="list-style-type: none"> <li>* Beginning of the <b>First Five-Year Plan (1951-56)</b>.</li> <li>* The <b>first Asian Games</b> were held in New Delhi.</li> </ul>                                                                                                                                                       |
| 1952 | <ul style="list-style-type: none"> <li>* The <b>first general elections</b> in independent India were held.</li> </ul>                                                                                                                                                                                                            |
| 1954 | <ul style="list-style-type: none"> <li>* India and China signed the historic <b>Panchsheel Agreement</b>.</li> </ul>                                                                                                                                                                                                              |
| 1956 | <ul style="list-style-type: none"> <li>* <b>Reorganisation of Indian states on a linguistic basis</b> by the States Reorganisation Act.</li> </ul>                                                                                                                                                                                |
| 1957 | <ul style="list-style-type: none"> <li>* Introduction of the <b>decimal system</b> in the Indian currency system.</li> </ul>                                                                                                                                                                                                      |
| 1959 | <ul style="list-style-type: none"> <li>* <b>India's first television service</b> began experimentally in New Delhi.</li> </ul>                                                                                                                                                                                                    |
| 1961 | <ul style="list-style-type: none"> <li>* <b>Liberation of Goa from the Portuguese</b> through Operation Vijay.</li> </ul>                                                                                                                                                                                                         |
| 1962 | <ul style="list-style-type: none"> <li>* <b>Sino-Indian War</b> (China's sudden attack on India).</li> </ul>                                                                                                                                                                                                                      |
| 1964 | <ul style="list-style-type: none"> <li>* <b>Death of Jawaharlal Nehru</b>, India's first Prime Minister.</li> </ul>                                                                                                                                                                                                               |
| 1965 | <ul style="list-style-type: none"> <li>* <b>Second Indo-Pak War</b> (over the Kashmir border dispute).</li> </ul>                                                                                                                                                                                                                 |
| 1966 | <ul style="list-style-type: none"> <li>* <b>Death of Lal Bahadur Shastri</b>, the then Prime Minister, during the Tashkent Agreement.</li> </ul>                                                                                                                                                                                  |
| 1969 | <ul style="list-style-type: none"> <li>* <b>Split in the Indian National Congress</b> due to the Syndicate dispute.</li> <li>* <b>Nationalisation of 14 major banks</b> by the Indira Gandhi government.</li> <li>* Commercial operation began at India's first nuclear power station at <b>Tarapur (Maharashtra)</b>.</li> </ul> |
| 1971 | <ul style="list-style-type: none"> <li>* <b>Third Indo-Pak War</b>, which resulted in the birth of Bangladesh.</li> </ul>                                                                                                                                                                                                         |
| 1972 | <ul style="list-style-type: none"> <li>* Signing of the historic <b>Shimla Agreement</b> between India and Pakistan.</li> </ul>                                                                                                                                                                                                   |
| 1974 | <ul style="list-style-type: none"> <li>* India's <b>first successful nuclear test</b> at Pokhran (Rajasthan) on 18 May (codename: Smiling Buddha).</li> </ul>                                                                                                                                                                     |
| 1975 | <ul style="list-style-type: none"> <li>* India's very first artificial satellite '<b>Aryabhata</b>' was launched into space.</li> <li>* <b>Proclamation of Emergency</b> on the ground of internal disturbance in the country.</li> </ul>                                                                                         |
| 1977 | <ul style="list-style-type: none"> <li>* After defeat in the general election, <b>the Congress lost power at the Centre for the first time</b> (Morarji Desai became Prime Minister).</li> </ul>                                                                                                                                  |
| 1980 | <ul style="list-style-type: none"> <li>* <b>Return of the Congress to power</b> with a huge majority in the general election.</li> <li>* <b>Six more commercial banks were nationalised</b>.</li> </ul>                                                                                                                           |
| 1982 | <ul style="list-style-type: none"> <li>* Beginning of <b>colour television</b> and national broadcasting in India.</li> <li>* The <b>ninth Asian Games</b> were held in New Delhi.</li> </ul>                                                                                                                                     |
| 1984 | <ul style="list-style-type: none"> <li>* <b>Assassination of Smt. Indira Gandhi</b>, the then Prime Minister, on 31 October.</li> <li>* The Indian Army established full control over the Siachen Glacier under <b>Operation Meghdoot</b>.</li> </ul>                                                                             |
| 1991 | <ul style="list-style-type: none"> <li>* <b>Assassination of former Prime Minister Rajiv Gandhi</b> at Sriperumbudur in Tamil Nadu.</li> <li>* Beginning of <b>economic liberalisation (LPG reforms)</b> in India under the Rao-Manmohan model.</li> </ul>                                                                        |

| Year | Important National and International Events                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1992 | * <b>Demolition of the disputed Babri Masjid structure</b> at Ayodhya.<br>* Constitutional beginning of <b>Panchayati Raj and urban local bodies</b> through the 73rd and 74th Constitutional Amendments. |
| 1995 | * <b>Commercial launch of internet service</b> in India by VSNL.                                                                                                                                          |
| 1998 | * India's <b>second successful nuclear test</b> at Pokhran (under the codename <b>Operation Shakti</b> ).                                                                                                 |
| 1999 | * <b>Kargil War</b> against Pakistani infiltrators (India's victory through Operation Vijay).                                                                                                             |
| 2000 | * <b>India's population crossed 1 billion (100 crore)</b> officially.                                                                                                                                     |
| 2001 | * <b>January:</b> devastating earthquake at Bhuj in Gujarat.<br>* <b>December: terrorist attack by Lashkar/Jaish on the Indian Parliament</b> in New Delhi.                                               |
| 2002 | * The <b>Godhra incident</b> in Gujarat and the riots that broke out afterwards.                                                                                                                          |
| 2004 | * India's coastal areas were affected by the <b>devastating tsunami in the Indian Ocean</b> caused by the Sumatra earthquake.                                                                             |
| 2008 | * India's first spacecraft to the Moon, <b>Chandrayaan-1</b> , was successfully sent.                                                                                                                     |
| 2013 | * ISRO sent India's first <b>Mars Orbiter Mission (MOM) 'Mangalyaan'</b> towards Mars.                                                                                                                    |

## 2 States that Accepted Lord Wellesley's Subsidiary Alliance

(UPPCS Prelims repeatedly asks for the correct chronological order in which the states accepted the alliance)

| No. | Name of State                                                                 | Year / Time of Acceptance of the Alliance                                                   |
|-----|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1.  | <b>Hyderabad</b>                                                              | 1798 CE ( <i>revised treaty: 1800 CE</i> )                                                  |
| 2.  | <b>Mysore</b>                                                                 | 1799 CE ( <i>after the Fourth Anglo-Mysore War</i> )                                        |
| 3.  | <b>Tanjore</b>                                                                | 1799 CE ( <i>helpful for chronology in addition to the states mentioned in the source</i> ) |
| 4.  | <b>Awadh</b>                                                                  | November 1801 CE                                                                            |
| 5.  | <b>Peshwa (Maratha)</b>                                                       | December 1801 CE ( <i>Treaty of Bassein</i> )                                               |
| 6.  | <b>Bhonsle of Berar</b>                                                       | December 1803 CE ( <i>Treaty of Deogaon</i> )                                               |
| 7.  | <b>Scindia</b>                                                                | February 1804 CE ( <i>Treaty of Surji-Anjangaon</i> )                                       |
| 8.  | <b>Rajput states</b><br>( <i>Jodhpur, Jaipur, Macheri, Bundi, Bharatpur</i> ) | 1818 CE                                                                                     |

### 3 Principal Administrative Committees and Commissions of the British Period

(Extremely important for the matching questions of the UPPCS examination)

| Name of Committee / Commission | Year of Formation | Governor-General / Viceroy of the Time | Main Subject / Area of Inquiry of the Committee                                                      |
|--------------------------------|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------|
| Scott-Moncrieff Commission     | 1901 CE           | Lord Curzon                            | For improvement of the <b>irrigation</b> system                                                      |
| Hilton Young Commission        | 1926 CE           | Lord Reading                           | To advise on the <b>Indian currency and financial system</b> (basis of the establishment of the RBI) |
| Babington Smith Commission     | 1919 CE           | Lord Chelmsford                        | Reform of <b>currency</b> and the exchange rate                                                      |
| Hunter Committee Report        | 1919 CE           | Lord Chelmsford                        | <b>Punjab disturbances</b> (to inquire into the Jallianwala Bagh massacre)                           |
| Muddiman Committee             | 1924 CE           | Lord Reading                           | To <b>examine the working of dyarchy</b> introduced under the Montagu-Chelmsford reforms             |
| Butler Commission              | 1927 CE           | Lord Irwin                             | To examine the relations between the <b>British Crown and the Indian princely states</b>             |
| Simon Commission               | 1928 CE           | Lord Irwin                             | To review the Act of 1919 and suggest new steps of administrative reform                             |
| Whitley Commission             | 1929 CE           | Lord Irwin                             | To study <b>labour</b> conditions in industrial establishments                                       |
| Sapru Commission               | 1935 CE           | Lord Linlithgow                        | To inquire into the causes of <b>unemployment</b> in the country                                     |
| Chatfield Commission           | 1939 CE           | Lord Linlithgow                        | For the modernisation of the <b>Indian Army</b> and assessment of military expenditure               |
| Floud Commission               | 1940 CE           | Lord Linlithgow                        | <b>Tenancy in Bengal</b> and the land revenue system (basis of the Tebhaga movement)                 |

### 4 Important Schemes Related to Education in Modern India (Post-Independence)

(UPPCS Prelims asks for the correct matching of the years of these national schemes)

| No. | Name of the Education Scheme / Mission | Year / Session of Commencement |
|-----|----------------------------------------|--------------------------------|
| 1.  | Non-formal Education Scheme            | 1975 CE                        |
| 2.  | Operation Blackboard                   | 1987 CE                        |
| 3.  | Total Literacy Campaign                | 1991 CE                        |

| No. | Name of the Education Scheme / Mission         | Year / Session of Commencement |
|-----|------------------------------------------------|--------------------------------|
| 4.  | Mid-Day Meal Scheme                            | 1995 CE                        |
| 5.  | Kalp Shiksha Yojana                            | 1998 CE                        |
| 6.  | Shiksha Mitra Yojana                           | 2000 - 01 CE                   |
| 7.  | Sarva Shiksha Abhiyan (SSA)                    | 2001 CE                        |
| 8.  | Vidya Vahini Project                           | 2003 CE                        |
| 9.  | Kasturba Gandhi Balika Vidyalaya Scheme (KGBV) | 2004 CE                        |
| 10. | Saakshar Bharat Mission                        | 2009 CE                        |

## 5 Principal Peasant Movements and Satyagrahas of Modern India

| Name of the Movement (Year)    | Principal Leader                            | Place / Region of the Movement | Main Cause and Exam-Useful Result                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|---------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Champaran Satyagraha (1917 CE) | Mahatma Gandhi                              | Champaran, Bihar               | <p>* <b>Cause:</b> against the '<b>Tinkathia</b>' system imposed by the English (compulsory cultivation of indigo on 3/20 of the land).</p> <p>* <b>Result:</b> Gandhiji's <b>first successful satyagraha</b> in India; the British government had to abolish the Tinkathia system.</p>                                                                                    |
| Kheda Satyagraha (1918 CE)     | Mahatma Gandhi and Sardar Vallabhbhai Patel | Kheda district, Gujarat        | <p>* <b>Cause:</b> forcible collection of land revenue (rent) by the British government despite a severe famine.</p> <p>* <b>Result:</b> a successful satyagraha; the government had to suspend tax collection and the peasants got great relief.</p>                                                                                                                      |
| Moplah Rebellion (1921 CE)     | Ali Musliyar and local leaders              | Malabar region, Kerala         | <p>* <b>Cause:</b> discontent of the Muslim Moplah tenants against Hindu landlords and the exploitative taxes of the British government.</p> <p>* <b>Result:</b> the movement later turned into a communal/religious conflict and was harshly suppressed by the British army.</p>                                                                                          |
| Bardoli Satyagraha (1928 CE)   | Sardar Vallabhbhai Patel                    | Bardoli, Gujarat               | <p>* <b>Cause:</b> a sudden <b>22% increase in rent (land tax)</b> on the peasants by the provincial government.</p> <p>* <b>Result:</b> the peasants, united, refused to pay the tax; the government had to withdraw the increase. On the success of this movement, Gandhiji, on behalf of the women of Bardoli, gave Vallabhbhai Patel the <b>title of "Sardar"</b>.</p> |

| Name of the Movement (Year)          | Principal Leader                             | Place / Region of the Movement | Main Cause and Exam-Useful Result                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tebhaga Movement (1946-47 CE)</b> | Sharecropping peasants and communist leaders | Bengal province                | <p>* <b>Cause:</b> the sharecroppers (tenants) demanded to give the landlords only one-third of the crop instead of half.</p> <p>* <b>Result:</b> largely successful; under its pressure major reforms were made in the landlord and tenancy laws of Bengal.</p> |

## 1 Principal Tribal Revolts: Causes, Leadership and Results

| Name of the Revolt (Year)                 | Geographical Region (Place)            | Principal Leader                   | Main Cause of the Revolt and Exam-Useful Result                                                                                                                                                                                                                                                                 |
|-------------------------------------------|----------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ho Revolt (1820 - 1827 CE)</b>         | Chotanagpur (Jharkhand)                | Local Ho tribals                   | <p>* <b>Cause:</b> economic exploitation of the tribals by British colonialism and local landlords.</p> <p>* <b>Result:</b> the British army suppressed it harshly, but it inspired later movements.</p>                                                                                                        |
| <b>Kol Revolt (1831 - 1832 CE)</b>        | Chotanagpur, Jharkhand                 | <b>Budhu Bhagat</b> and associates | <p>* <b>Cause:</b> seizure by the English of the traditional land of the Kol tribals and the oppression of outside landlords (dikus).</p> <p>* <b>Result:</b> the revolt was suppressed, but the English had to think of a policy of reducing outside interference in the tribal areas.</p>                     |
| <b>Santhal Rebellion (1855 - 1856 CE)</b> | Santhal Parganas (Bihar and W. Bengal) | <b>Sidho and Kanhu Murmu</b>       | <p>* <b>Cause:</b> excessive tax policies, the interest trap of moneylenders and intense economic exploitation by landlords.</p> <p>* <b>Result:</b> suppressed after widespread violent resistance. As a reform, <b>the 'Santhal Pargana' was constituted</b> for administrative convenience.</p>              |
| <b>Bhil Revolt (1818 - 1831 CE)</b>       | Rajasthan, Gujarat and Madhya Pradesh  | <b>Govind Guru</b>                 | <p>* <b>Cause:</b> crisis in the traditional forest rights and economic life of the Bhils due to the new forest and tax policies of the British government.</p> <p>* <b>Result:</b> suppressed by the British army after a long struggle, but the movement became a symbol of the defence of tribal rights.</p> |

## 2 Evolution of the Highest Office of British India (Governor-General and Viceroy)

| Act / Law                     | Changed Designation (New Name)                         | First Holder           | Special Exam Details                                                                       |
|-------------------------------|--------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| <b>Regulating Act of 1773</b> | Governor of Bengal → <b>Governor-General of Bengal</b> | <b>Warren Hastings</b> | The Bombay and Madras Presidencies were placed under Bengal (beginning of centralisation). |



| Act / Law                       | Changed Designation (New Name)                                | First Holder                 | Special Exam Details                                                                                                       |
|---------------------------------|---------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Charter Act of 1833             | Governor-General of Bengal → <b>Governor-General of India</b> | <b>Lord William Bentinck</b> | A single Governor-General was fixed for the whole of British India. He became the <b>first Governor-General of India</b> . |
| Government of India Act of 1858 | Governor-General of India → <b>Viceroy of India</b>           | <b>Lord Canning</b>          | Company rule ended and power passed into the hands of the British Crown. Canning became the <b>first Viceroy</b> of India. |

- **Note:** In **1948** the office of Governor-General/Viceroy of India was completely abolished. **Lord Mountbatten** was the **last Viceroy** of India.

### 3 The First Anglo-Maratha War (1775-82) and Maratha Treaties

#### > Main Details of the War:

- **Main cause:** An internal war of succession for the office of Peshwa. **Raghunath Rao (Raghoba)**, eager to become Peshwa, sought help from the English (Bombay Council), which sparked the war.
- **Principal parties involved:** Raghunath Rao, Nana Phadnavis (Maratha confederacy) and Warren Hastings (British Governor-General).
- **Treaty and conclusion:** The **Treaty of Purandar in 1776** (between Nana Phadnavis and the English) was signed. Finally the war ended with the **Treaty of Salbai in 1782**, under which the English abandoned Raghoba and peace prevailed between the two sides for the next two decades (20 years).

#### > Principal Treaties of the Anglo-Maratha Conflict (exact chronology - a UPPCS pet question):

*(Their year-matching or chronology is certainly asked in the exam)*

| No. | Name of Treaty               | Exact Year (CE) | No. | Name of Treaty                   | Exact Year (CE)           |
|-----|------------------------------|-----------------|-----|----------------------------------|---------------------------|
| 1.  | <b>Treaty of Surat</b>       | 1775 CE         | 7.  | <b>Treaty of Surji-Anjangaon</b> | 1803 CE                   |
| 2.  | <b>Treaty of Purandar</b>    | 1776 CE         | 8.  | <b>Treaty of Rajpurghat</b>      | 1804 CE                   |
| 3.  | <b>Convention of Wadgaon</b> | 1779 CE         | 9.  | <b>Treaty of Nagpur</b>          | 1816 CE                   |
| 4.  | <b>Treaty of Salbai</b>      | 1782 CE         | 10. | <b>Treaty of Gwalior</b>         | 1817 CE <i>(November)</i> |
| 5.  | <b>Treaty of Bassein</b>     | 1802 CE         | 11. | <b>Treaty of Poona</b>           | 1817 CE <i>(June)</i>     |
| 6.  | <b>Treaty of Deogaon</b>     | 1803 CE         | 12. | <b>Treaty of Mandsaur</b>        | 1818 CE                   |

### 4 Principal National Writers, Books and Journals of Modern India

*(A complete integrated compilation of both the literary sources given - the biggest scoring section of UPPCS Prelims)*

| Writer / National Leader                    | Their Principal Books / Newspapers / Journals                                                                                                                                    |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mahatma Gandhi</b>                       | The Story of My Experiments with Truth (autobiography), Young India, Harijan, Navajivan, Hind Swaraj                                                                             |
| <b>B.R. Ambedkar</b>                        | Mook Nayak, Bahishkrit Bharat, Annihilation of Caste, Pakistan or the Partition of India, Who were the Shudras, The Untouchables, Buddha or Karl Marx, The Buddha and his Dhamma |
| <b>Jawaharlal Nehru</b>                     | The Discovery of India, Glimpses of World History                                                                                                                                |
| <b>Lala Lajpat Rai</b>                      | The Story of My Deportation, Unhappy India                                                                                                                                       |
| <b>Bal Gangadhar Tilak</b>                  | Gita Rahasya, Kesari (in Marathi), Mahratta (in English)                                                                                                                         |
| <b>Surendranath Banerjee</b>                | A Nation in Making, Bengalee (newspaper)                                                                                                                                         |
| <b>Maulana Abul Kalam Azad</b>              | India Wins Freedom, Al-Hilal (journal)                                                                                                                                           |
| <b>Motilal Nehru</b>                        | Independent (newspaper)                                                                                                                                                          |
| <b>Dadabhai Naoroji</b>                     | Poverty and Un-British Rule in India, Rast Goftar (journal)                                                                                                                      |
| <b>Aurobindo Ghosh</b>                      | Karmayogin, Yugantar, Savitri, The Life Divine, Essays on the Gita, Bande Mataram                                                                                                |
| <b>Bankim Chandra Chattopadhyay</b>         | Anandamath (Vande Mataram is in it), Devi Chaudhurani                                                                                                                            |
| <b>Madan Mohan Malaviya</b>                 | Abhyudaya, Hindustan, Leader                                                                                                                                                     |
| <b>Bhagat Singh</b>                         | Why I am an Atheist, An Introduction to the Dreamland                                                                                                                            |
| <b>Rajendra Prasad</b>                      | India Divided                                                                                                                                                                    |
| <b>Rabindranath Tagore</b>                  | Gitanjali, The Home and the World                                                                                                                                                |
| <b>Ram Manohar Lohia</b>                    | Guilty Men of India's Partition                                                                                                                                                  |
| <b>Sachindranath Sanyal</b>                 | Bandi Jeevan ( <i>regarded as a textbook of the revolutionaries</i> )                                                                                                            |
| <b>Bhagwati Charan Vohra</b>                | The Philosophy of the Bomb                                                                                                                                                       |
| <b>Valentine Chirol</b>                     | Indian Unrest ( <i>it was he who called Tilak 'the father of Indian unrest'</i> )                                                                                                |
| <b>Lady Katherine Mayo</b>                  | Mother India                                                                                                                                                                     |
| <b>Larry Collins and Dominique Lapierre</b> | Freedom at Midnight                                                                                                                                                              |
| <b>Gopal Krishna Gokhale</b>                | Nation                                                                                                                                                                           |

| Writer / National Leader | Their Principal Books / Newspapers / Journals  |
|--------------------------|------------------------------------------------|
| Keshab Chandra Sen       | Indian Mirror                                  |
| Sir Syed Ahmad Khan      | Tahzib-ul-Akhlaq                               |
| Dinabandhu Mitra         | Nil Darpan (a play based on the Indigo Revolt) |
| Bharatendu Harishchandra | Bharat Durdasha                                |
| R.C. Dutt                | The Economic History of India                  |
| Bibhutibhushan Banerjee  | Pather Panchali                                |
| Lord Curzon              | Problems of the Far East                       |

### ➤ Special Books Related to the Revolt of 1857:

| Name of Book                                       | Author / Composer |
|----------------------------------------------------|-------------------|
| The Indian War of Independence, 1857               | V.D. Savarkar     |
| The Sepoy Mutiny and the Revolt of 1857            | R.C. Majumdar     |
| Civil Rebellion in the Indian Mutinies (1857-1859) | S.B. Chaudhuri    |
| 1857, The Great Rebellion                          | Asoka Mehta       |

## 5 The Nawabs of Bengal and the Establishment of British Rule (Important Facts and Chronology)

*(Political changes of the Nawabs, especially useful for UPPCS)*

| Year / Date       | Historical Event and Exam Facts                                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1717 CE           | The Mughal emperor <b>Farrukhsiyar</b> appointed <b>Murshid Quli Khan</b> as <b>Subedar (Nawab)</b> of <b>Bengal</b> .                                                           |
| Change of capital | <b>Murshid Quli Khan</b> shifted the capital <b>from Dhaka to Murshidabad</b> , while later <b>Mir Qasim</b> moved the capital <b>from Murshidabad to Munger</b> .               |
| 1740 CE           | <b>Alivardi Khan</b> became Nawab of Bengal after killing Sarfaraz Khan in the <b>Battle of Giria</b> (he stopped paying revenue to the Mughals).                                |
| 1756 CE           | After the death of Alivardi, <b>Siraj-ud-Daulah</b> became Nawab of Bengal.                                                                                                      |
| 20 June 1756      | The famous ' <b>Black Hole Tragedy</b> ' took place in Calcutta, described by J.Z. Holwell (according to the source, the reference is to an account mentioned by George Orwell). |

| Year / Date            | Historical Event and Exam Facts                                                                                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>23 June 1757</b>    | The famous <b>Battle of Plassey</b> was fought between Nawab Siraj-ud-Daulah and Robert Clive (the Nawab was defeated through Mir Jafar's treachery).                                                                              |
| <b>Puppet rulers</b>   | After Plassey the English made first <b>Mir Jafar</b> and then, in 1760, <b>Mir Qasim</b> the puppet Nawab of Bengal.                                                                                                              |
| <b>22 October 1764</b> | The <b>Battle of Buxar</b> was fought between the combined front of Mir Qasim, Shah Alam II (Mughal emperor) and Shuja-ud-Daulah (Nawab of Awadh) and the English commander <b>Hector Munro</b> . Complete victory of the English. |
| <b>12 August 1765</b>  | <b>First Treaty of Allahabad</b> — between Robert Clive and the Mughal emperor Shah Alam II (the English got the Diwani of Bengal, Bihar and Orissa).                                                                              |
| <b>16 August 1765</b>  | <b>Second Treaty of Allahabad</b> — concluded between Robert Clive and the Nawab of Awadh, Shuja-ud-Daulah.                                                                                                                        |
| <b>1765 – 1772 CE</b>  | The period of <b>Dual Government</b> in Bengal, which was ended by Warren Hastings.                                                                                                                                                |

## 6 The Three Historic Round Table Conferences of the British Period

| Conference                           | Date of Commencement    | Viceroy of the Time    | Main Purpose and Reason for the Failure of Talks (Exam Facts)                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|-------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First Round Table Conference</b>  | <b>12 November 1930</b> | Lord Irwin             | <ul style="list-style-type: none"> <li>* <b>Purpose:</b> held in London to consider the recommendations of the Simon Commission.</li> <li>* <b>Outcome:</b> the Congress did not take part in it, as a result of which the talks failed completely.</li> </ul>                                                                                                                                     |
| <b>Second Round Table Conference</b> | <b>7 September 1931</b> | <b>Lord Willingdon</b> | <ul style="list-style-type: none"> <li>* <b>Purpose:</b> under the Gandhi-Irwin Pact, <b>Mahatma Gandhi took part in it as the sole representative of the Congress.</b></li> <li>* <b>Outcome:</b> the talks failed because of Gandhiji's disagreement with Dr. B.R. Ambedkar's <b>demand for a separate electorate for the Dalits.</b></li> </ul>                                                 |
| <b>Third Round Table Conference</b>  | <b>17 November 1932</b> | Lord Willingdon        | <ul style="list-style-type: none"> <li>* <b>Purpose:</b> discussion of the final draft of the constitutional reforms.</li> <li>* <b>Outcome:</b> the conference was completely boycotted by the Congress. After the White Paper of this conference, the Government of India Bill was introduced in the British Parliament, which later became the <b>Government of India Act, 1935.</b></li> </ul> |

## 1 Principal Women Related to the History of Modern India and their Work

| Woman Revolutionary / Leader    | Main Exam Facts Related to the National Movement, Organisations and History                                                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Madame Bhikaji Cama</b>      | <ul style="list-style-type: none"> <li>* Parsi revolutionary; founded the '<b>Paris Indian Society</b>'.</li> <li>* In 1907 at Stuttgart (Germany) she <b>hoisted the first version of the national flag</b> (she is also called 'the mother of the Indian revolution').</li> </ul>                                                               |
| <b>Annie Besant</b>             | <ul style="list-style-type: none"> <li>* First president of the '<b>Women's Indian Association</b>'.</li> <li>* Founder of the '<b>Indian Home Rule League</b>' (1916).</li> <li>* Made a central contribution to the development of the Theosophical Society.</li> </ul>                                                                         |
| <b>Captain Lakshmi Sahgal</b>   | <ul style="list-style-type: none"> <li>* Was a leading officer of the '<b>Rani of Jhansi Regiment</b>' of the Azad Hind Fauj.</li> <li>* Founder member of the '<b>All India Democratic Women's Association</b>' (AIDWA).</li> </ul>                                                                                                              |
| <b>Kamaladevi Chattopadhyay</b> | <ul style="list-style-type: none"> <li>* Inspired by Gandhian ideas; founder of the '<b>Indian Cooperative Union</b>'.</li> <li>* Was actively associated with the <b>All India Women's Conference</b> (AIWC).</li> </ul>                                                                                                                         |
| <b>Kasturba Gandhi</b>          | <ul style="list-style-type: none"> <li>* Wife of Mahatma Gandhi; struggled for the rights of Indian immigrants in South Africa.</li> <li>* Took an active part in the <b>Borsad Civil Disobedience Movement</b> in Gujarat.</li> </ul>                                                                                                            |
| <b>Aruna Asaf Ali</b>           | <ul style="list-style-type: none"> <li>* <b>Hoisted the Indian national flag</b> at the Gowalia Tank Maidan in Mumbai during the <b>Quit India Movement (1942)</b>.</li> <li>* After independence she became the <b>first Mayor of Delhi</b>.</li> </ul>                                                                                          |
| <b>Sarojini Naidu</b>           | <ul style="list-style-type: none"> <li>* Became the <b>first Indian woman President of the Indian National Congress (Kanpur session, 1925)</b>.</li> <li>* Went to London with Mahatma Gandhi to take part in the <b>Second Round Table Conference (1931)</b>.</li> </ul>                                                                         |
| <b>Usha Mehta</b>               | <ul style="list-style-type: none"> <li>* During the Quit India Movement she began operating an <b>underground (secret) radio programme</b> to spread national consciousness.</li> </ul>                                                                                                                                                           |
| <b>Shanti Ghosh</b>             | <ul style="list-style-type: none"> <li>* A great young revolutionary heroine of the freedom struggle.</li> <li>* Together with her classmate <b>Suniti Chaudhuri</b>, in December 1931 she <b>shot dead C.G.B. Stevens, the oppressive British Collector of Tripura (Comilla)</b>. The youngest woman revolutionary pair in the world.</li> </ul> |

- *Note: According to the original source, there is a partial mention of the Chittagong Armoury Raid (1930) in the accounts of Aruna Asaf Ali and Usha Mehta, but historically those mainly associated with the Chittagong Armoury Raid, along with Surya Sen, were **Pritilata Waddedar** and **Kalpana Dutt**.*

## 2 Principal Revolutionary and Political Organisations of the Freedom Movement (Chronology)

(UPPCS Prelims asks every year for their year-wise order or direct coded matching)

| Name of the Organisation / Samiti                 | Year of Founding | Place of Founding               | Principal Founders / Leaders                                                                                              |
|---------------------------------------------------|------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Mitra Mela                                        | 1899 CE          | Poona                           | The Savarkar brothers (Ganesh Savarkar and V.D. Savarkar)                                                                 |
| Anushilan Samiti<br>(Calcutta and Dhaka)          | 1902 CE          | Bengal                          | Barindra Kumar Ghosh, Jatindranath Banerjee, Pramatha Nath Mitra (P. Mitra) and Pulin Das                                 |
| Anushilan Samiti<br>(Midnapore)                   | 1902 CE          | Midnapore (Bengal)              | Jnanendra Nath Bose                                                                                                       |
| Abhinav Bharat Society                            | <b>1904 CE</b>   | Poona                           | Vinayak Damodar (V.D.) Savarkar ( <i>a transformed form of the Mitra Mela</i> )                                           |
| Bharat Mata Samiti                                | 1904 CE          | South India                     | J.M. Chatterjee, Nilakantha Brahmachari and Vanchi Iyer                                                                   |
| Swadesh Bandhab Samiti                            | 1905 CE          | Barisal (Bengal)                | Ashwini Kumar Dutt (formed during the Swadeshi Movement)                                                                  |
| Bharat Mata Society                               | 1907 CE          | Punjab                          | Sufi Amba Prasad and Sardar Ajit Singh ( <i>Bhagat Singh's uncle</i> )                                                    |
| Ghadar Dal (Ghadar Party)                         | <b>1913 CE</b>   | San Francisco (USA)             | Lala Har Dayal and Sohan Singh Bhakna                                                                                     |
| Hindi Association of America                      | 1913 CE          | America                         | Sohan Singh Bhakna ( <i>the Ghadar movement developed out of it</i> )                                                     |
| Hindustan Republican Association (HRA)            | <b>1924 CE</b>   | <b>Kanpur</b>                   | Sachindranath Sanyal, Ramprasad Bismil and Jogesh Chandra Chatterjee ( <i>supported by Bhagat Singh and Udham Singh</i> ) |
| Hindustan Socialist Republican Association (HSRA) | <b>1928 CE</b>   | <b>Delhi</b> (Feroz Shah Kotla) | <b>Chandrashekhar Azad</b> and Bhagat Singh                                                                               |
| Hindu Dharma Sangh                                | —                | Maharashtra                     | The Chapekar brothers (Damodar and Balkrishna Chapekar)                                                                   |
| Arya Bandhav Samaj                                | —                | Maharashtra                     | Bal Gangadhar Tilak                                                                                                       |
| Atmonnati Samiti                                  | —                | Bengal                          | Bipin Behari Ganguly                                                                                                      |
| Bengal Volunteers                                 | —                | Bengal                          | Hemchandra Ghosh and Leela Nag (Leela Roy)                                                                                |
| Indian Republican Army (IRA)                      | 1930 CE          | Chittagong (Bengal)             | <b>Surya Sen</b> ('Master Da')                                                                                            |

### 3 Principal Ashrams of Mahatma Gandhi and their Years of Founding (Chronology)

(UPPCS Prelims repeatedly asks for their correct chronological order)

| No. | Name of the Ashram                       | Place of Founding                             | Year of Founding                              |
|-----|------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| 1.  | <b>Phoenix Farm (Phoenix Settlement)</b> | Durban (South Africa)                         | <b>1904 CE</b>                                |
| 2.  | <b>Tolstoy Farm</b>                      | Johannesburg (South Africa)                   | <b>1910 CE</b>                                |
| 3.  | <b>Satyagraha Ashram</b>                 | Kochrab, Ahmedabad (Gujarat)                  | <b>1915 CE</b> <i>(May)</i>                   |
| 4.  | <b>Sabarmati Ashram</b>                  | On the bank of the Sabarmati river, Ahmedabad | <b>1915 CE</b> <i>(later shifted in 1917)</i> |
| 5.  | <b>Anasakti Ashram</b>                   | Kausani (Uttarakhand)                         | <b>1929 CE</b>                                |
| 6.  | <b>Sevagram Ashram</b>                   | Wardha (Maharashtra)                          | <b>1936 CE</b>                                |

#### 4 Principal Historic Titles Conferred on Mahatma Gandhi

| Title / Name                              | Person / Group who Gave the Title                 | On what Occasion / Reason                                                                                                         |
|-------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Kaiser-i-Hind</b>                      | British government                                | For cooperation in the British war effort at the time of the First World War.                                                     |
| <b>Recruiting Sergeant</b>                | Indian public/critics                             | For encouraging Indians to join the British army at the time of the First World War.                                              |
| <b>Coolie Barrister</b>                   | English magistrate of South Africa                | As a racial and racist remark in a South African court.                                                                           |
| <b>Half-naked fakir / seditious fakir</b> | <b>Winston Churchill</b>                          | Said on seeing him climbing the steps of St James's Palace during the Second Round Table Conference (1931).                       |
| <b>Malang Baba</b>                        | The tribesmen of the North-West Frontier Province | To express their deep reverence for him during the Civil Disobedience Movement.                                                   |
| <b>Bapu</b>                               | C.F. Andrews and Jawaharlal Nehru                 | Because of his simple nature and fatherly protection.                                                                             |
| <b>King of Beggars</b>                    | Pandit Madan Mohan Malaviya                       | For his life of sacrifice and his mass awakening for the nation.                                                                  |
| <b>Father of the Nation</b>               | <b>Subhas Chandra Bose</b>                        | First said on 6 July 1944 from Singapore Radio while seeking his blessings for the soldiers of the Azad Hind Fauj.                |
| <b>One Man Boundary Force</b>             | <b>Lord Mountbatten</b>                           | Said on seeing his courage in single-handedly calming the riots at Noakhali (Bengal) at the time of the partition of the country. |



| Title / Name                         | Person / Group who Gave the Title | On what Occasion / Reason                                                              |
|--------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------|
| <b>Mahatma</b>                       | <b>Rabindranath Tagore</b>        | Gave this title after the historic success of the <b>Champaran Satyagraha (1917)</b> . |
| <b>Ajatashatru of the Modern Age</b> | Dr. Rajendra Prasad               | To show his highest respect for Gandhiji.                                              |

## 5 Principal Educational Institutions Established in the British Period

*(Extremely important for both the 'UP Special' and general history sections of UPPCS Prelims)*

| Name of the Educational Institution      | Year of Founding | Place of Founding  | Founder / Governor         | Main Purpose / Feature of Founding                                                                                  |
|------------------------------------------|------------------|--------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>First Madrasa in Calcutta</b>         | <b>1781 CE</b>   | Kolkata            | Warren Hastings            | To promote the study of Muslim laws and the Arabic-Persian languages.                                               |
| <b>Asiatic Society of Bengal</b>         | <b>1784 CE</b>   | Kolkata            | <b>Sir William Jones</b>   | To encourage oriental studies, the understanding of Indian history and culture, and translation work.               |
| <b>Banaras Sanskrit College</b>          | <b>1791 CE</b>   | Varanasi (Banaras) | <b>Jonathan Duncan</b>     | For the systematic study of Hindu religion, philosophy and laws (U.P. special fact).                                |
| <b>Fort William College</b>              | <b>1801 CE</b>   | Kolkata            | <b>Lord Wellesley</b>      | To train the new civil servants (ICS) of the East India Company in Indian languages, culture and local customs.     |
| <b>Mayo College</b>                      | 1875 CE          | Ajmer (Rajasthan)  | Lord Mayo                  | To give higher education on Western lines to the children of Indian princes and nobles.                             |
| <b>Muhammadan Anglo-Oriental College</b> | <b>1875 CE</b>   | Aligarh            | <b>Sir Syed Ahmad Khan</b> | To spread modern Western education among Muslims (it later became <b>Aligarh Muslim University / AMU in 1920</b> ). |

## 6 Principal Wars and Historic Treaties of Modern India (1746 – 1849 CE)

*(The biggest core area of the history section of UPPCS Prelims - the best reference for chronology and coded matching)*

| Name of War                    | Period / Time          | Historical Background, Important Facts, Treaty and Result                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First Carnatic War</b>      | 1746 - 1748 CE         | <ul style="list-style-type: none"> <li>* The famous <b>Battle of St. Thome</b> was fought between the French and the English.</li> <li>* The war ended with the <b>Treaty of Aix-la-Chapelle (1748)</b>; the English got back Madras.</li> </ul>                                                                                                                                                                                                                  |
| <b>Second Carnatic War</b>     | 1749 - 1754 CE         | <ul style="list-style-type: none"> <li>* Internal dispute over succession in the Deccan and the Carnatic.</li> <li>* The war ended with the <b>Treaty of Pondicherry (1754)</b>; Dupleix was recalled to France.</li> </ul>                                                                                                                                                                                                                                       |
| <b>Third Carnatic War</b>      | 1758 - 1763 CE         | <ul style="list-style-type: none"> <li>* An extension of the Seven Years' War of Europe. Under it was fought the famous <b>Battle of Wandiwash (22 January 1760)</b>, in which Sir Eyre Coote decisively defeated the French.</li> <li>* The war ended with the <b>Treaty of Paris (1763)</b>.</li> </ul>                                                                                                                                                         |
| <b>Battle of Plassey</b>       | <b>23 June 1757</b>    | <ul style="list-style-type: none"> <li>* Between the Nawab of Bengal, Siraj-ud-Daulah, and Robert Clive.</li> <li>* On the Nawab's side, <b>Mir Jafar and Diwan Rai Durlabh committed treachery</b>.</li> <li>* The Nawab's loyal commanders <b>Mir Madan and Mohanlal</b> died fighting bravely.</li> </ul>                                                                                                                                                      |
| <b>Battle of Buxar</b>         | <b>22 October 1764</b> | <ul style="list-style-type: none"> <li>* On one side was the combined front of Nawab Mir Qasim (Bengal), Shuja-ud-Daulah (Awadh) and Shah Alam II (Mughal emperor), and on the other the English commander <b>Hector Munro</b>.</li> <li>* Mir Qasim's main army was led by <b>Gurgin Khan</b>. Complete victory of the English.</li> </ul>                                                                                                                       |
| <b>First Anglo-Mysore War</b>  | 1767 - 1769 CE         | <ul style="list-style-type: none"> <li>* The English began the war against Hyder Ali after making a triple alliance with the Marathas and the Nizam of Hyderabad, but the clever Hyder Ali won the Nizam over to his side.</li> <li>* The war ended with the <b>Treaty of Madras (1769)</b> (the English were defeated).</li> </ul>                                                                                                                               |
| <b>Second Anglo-Mysore War</b> | 1780 - 1784 CE         | <ul style="list-style-type: none"> <li>* <b>Battle of Porto Novo (1781)</b>: the British commander <b>Sir Eyre Coote</b> defeated Hyder Ali.</li> <li>* In 1782 Hyder Ali died of wounds received during the war.</li> <li>* His son Tipu Sultan continued the war and finally it ended with the <b>Treaty of Mangalore (1784)</b>. Warren Hastings was the Governor-General at this time.</li> </ul>                                                             |
| <b>Third Anglo-Mysore War</b>  | 1790 - 1792 CE         | <ul style="list-style-type: none"> <li>* Governor-General <b>Lord Cornwallis</b>, with the cooperation of the Nizam of Hyderabad and the Marathas, laid siege to Tipu's capital Seringapatam.</li> <li>* The war ended with the <b>Treaty of Seringapatam (1792)</b> (Tipu had to lose half his kingdom).</li> <li>* <b>Famous statement of Cornwallis</b>: <i>"We have effectually crippled our enemy without making our friends too formidable."</i></li> </ul> |

| Name of War                     | Period / Time         | Historical Background, Important Facts, Treaty and Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fourth Anglo-Mysore War</b>  | <b>1799 CE</b>        | <ul style="list-style-type: none"> <li>* On 4 May 1799 <b>Tipu Sultan</b> died a hero's death at the gate of the fort of Seringapatam, fighting bravely against the combined English army of Lord Wellesley.</li> <li>* The English placed the minor boy Krishnaraja III of the old <b>Wodeyar dynasty</b> on the throne of Mysore and imposed the <b>Subsidiary Alliance</b>.</li> </ul>                                                                                                                                                                     |
| <b>First Anglo-Maratha War</b>  | 1775 - 1782 CE        | <ul style="list-style-type: none"> <li>* The <b>Treaty of Purandar in 1776</b> was made between the English government of Calcutta and the Poona Darbar (Nana Phadnavis), but the Bombay Council opposed it.</li> <li>* The <b>Convention of Wadgaon (1779)</b> was made during the war.</li> <li>* Finally the war ended with the <b>Treaty of Salbai (1782)</b> and there was peace for 20 years between the Marathas and the English.</li> </ul>                                                                                                           |
| <b>Second Anglo-Maratha War</b> | 1803 - 1805 CE        | <ul style="list-style-type: none"> <li>* Background: Peshwa Bajirao II accepted the subsidiary alliance by signing the <b>Treaty of Bassein (1802)</b> with the English.</li> <li>* <b>Lord Wellesley</b> (Arthur Wellesley in the Deccan and General Lake in North India) conducted successful campaigns against the Maratha chiefs.</li> <li>* <b>Important treaties:</b> the <b>Treaty of Deogaon (1803)</b> with the Bhonsle, the <b>Treaty of Surji-Anjangaon (1803)</b> with Scindia and the <b>Treaty of Rajpurghat (1805)</b> with Holkar.</li> </ul> |
| <b>Third Anglo-Maratha War</b>  | 1816 - 1819 CE        | <ul style="list-style-type: none"> <li>* The <b>repressive military action against the Pindaris</b> by Governor-General Lord Hastings hurt the self-respect of the Marathas and the war broke out.</li> <li>* <b>Treaties:</b> the <b>Treaty of Nagpur (1816)</b> with the Bhonsle, the <b>Treaty of Gwalior (1817)</b> with Scindia, the <b>Treaty of Poona (1817)</b> with Peshwa Bajirao II and the <b>Treaty of Mandsaur (1818)</b> with Holkar. The office of Peshwa was completely abolished and he was sent to Bithoor.</li> </ul>                     |
| <b>First Anglo-Sikh War</b>     | 1845 - 1846 CE        | <ul style="list-style-type: none"> <li>* Governor-General of the time: <b>Lord Hardinge</b>; English army chief: <b>Lord Gough</b>.</li> <li>* Decisive battle: <b>Battle of Sobraon (February 1846)</b>.</li> <li>* The war ended with the <b>Treaty of Lahore (March 1846)</b> (the territories south of the Sutlej went to the English). This was followed by the <b>Treaty of Bhyrowal (December 1846)</b>, under which a British army was stationed at Lahore until Raja Dalip Singh came of age.</li> </ul>                                             |
| <b>Second Anglo-Sikh War</b>    | 1848 - 1849 CE        | <ul style="list-style-type: none"> <li>* Principal battles: <b>Battle of Ramnagar</b> (November 1848) → <b>Battle of Chillianwala</b> (January 1849) → <b>Battle of Gujrat</b> (February 1849 - also called 'the battle of the guns').</li> <li>* The Sikh army was completely defeated under the leadership of General <b>Charles Napier</b>.</li> <li>* <b>In March 1849 Lord Dalhousie completely annexed Punjab to the British Empire.</b> The Kohinoor diamond was sent to the Queen.</li> </ul>                                                         |
| <b>Anglo-Nepal War</b>          | <b>1814 - 1816 CE</b> | <ul style="list-style-type: none"> <li>* Fought in the time of Lord Hastings. The <b>Treaty of Sugauli (1816)</b> ended the border dispute with Nepal and the Gurkhas joined the British army.</li> </ul>                                                                                                                                                                                                                                                                                                                                                     |

| Name of War                  | Period / Time  | Historical Background, Important Facts, Treaty and Result                                                                                                                                               |
|------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anglo-Burmese War            | 1824 - 1826 CE | * The First Anglo-Burmese War took place in the time of Lord Amherst and ended with the <b>Treaty of Yandabo (1826)</b> .                                                                               |
| Anglo-Afghan Wars            | 1839 - 1880 CE | * <b>First war (1839-42)</b> : fought in the time of Lord Auckland; it was disastrous for the English.<br>* <b>Second war (1878-80)</b> : fought in the time of Lord Lytton (under the Forward Policy). |
| Complete annexation of Sindh | 1843 CE        | * In 1843, during the tenure of Governor-General <b>Lord Ellenborough</b> , Sindh was completely annexed to the British Empire under the military leadership of <b>Charles Napier</b> .                 |

## 7 Principal Historical Events of the 18th Century (Chronology)

(UPPCS Prelims often asks for the direct chronological order of these historical dates)

1. **Black Hole incident (Black Hole Tragedy)**: 20 June 1756 CE
2. **Treaty of Alinagar**: 9 February 1757 CE
3. **Battle of Plassey**: 23 June 1757 CE
4. **Battle of Buxar**: 22 October 1764 CE
5. **First Treaty of Allahabad**: 12 August 1765 CE
6. **Second Treaty of Allahabad**: 16 August 1765 CE
7. **Treaty of Banaras**: the year 1773 CE

## 1 Governors-General and Viceroys of British India (Chronology and Important Events)

| Governor-General / Viceroy (Tenure) | Status of the Office             | Historical Events and Important Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warren Hastings (1773-1785 CE)      | First Governor-General of Bengal | <ul style="list-style-type: none"> <li>* The <b>Regulating Act</b> of 1773 was passed.</li> <li>* Associated with the '<b>Ring Fence Policy</b>'.</li> <li>* Establishment of the <b>Supreme Court</b> at Calcutta.</li> <li>* The First Anglo-Maratha War (1775-82), the Second Anglo-Mysore War (1780-84) and the <b>Treaty of Salbai (1782)</b> took place.</li> <li>* Regarded as the <b>father of the judicial service</b> in India.</li> </ul> |

| Governor-General / Viceroy (Tenure)         | Status of the Office                                                       | Historical Events and Important Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lord Cornwallis (1786–1793 CE)</b>       | Governor-General of Bengal                                                 | <ul style="list-style-type: none"> <li>* The Third Anglo-Mysore War (1790-92) and the <b>Treaty of Seringapatam (1792)</b> took place.</li> <li>* Introduced the <b>Permanent Settlement (1793 CE)</b>.</li> <li>* <b>Father of the civil service and the police service</b> in India.</li> <li>* <b>Special fact:</b> he died on 5 October 1805 at <b>Ghazipur (Uttar Pradesh)</b>, where his tomb is located.</li> </ul>                                                |
| <b>Lord Wellesley (1798–1805 CE)</b>        | Governor-General of Bengal                                                 | <ul style="list-style-type: none"> <li>* Introduced the <b>Subsidiary Alliance (1798)</b>.</li> <li>* The Fourth Anglo-Mysore War (1799) and the <b>Treaty of Bassein (1802)</b> took place.</li> <li>* The Second Anglo-Maratha War (1803-05) was fought.</li> <li>* Founded <b>Fort William College</b> in Calcutta (1801).</li> </ul>                                                                                                                                  |
| <b>Lord Hastings (1813–1823 CE)</b>         | Governor-General of Bengal                                                 | <ul style="list-style-type: none"> <li>* After the Third Anglo-Maratha War (1817-19) he adopted the <b>policy of non-intervention</b> and <b>abolished the office of Peshwa in 1818</b>.</li> <li>* <b>Anglo-Nepal War (1814-16)</b> and the Treaty of Sugauli.</li> <li>* The <b>Ryotwari system</b> was introduced in Bombay and Madras by Thomas Munro (1820).</li> <li>* Approval of the <b>Mahalwari system</b> in Awadh and the North-Western Provinces.</li> </ul> |
| <b>Lord William Bentinck (1828–1835 CE)</b> | <b>Last Governor-General of Bengal and first Governor-General of India</b> | <ul style="list-style-type: none"> <li>* By the <b>Charter Act of 1833</b> the Governor-General of Bengal was made the <b>Governor-General of India</b> (thus he became the first Governor-General of India).</li> <li>* <b>Abolished Sati in 1829 by Regulation XVII</b>.</li> <li>* <b>Suppressed thuggee in 1830</b> with the help of Colonel Sleeman.</li> </ul>                                                                                                      |
| <b>Lord Ellenborough (1842–1844 CE)</b>     | Governor-General of India                                                  | <ul style="list-style-type: none"> <li>* The First Anglo-Afghan War ended.</li> <li>* <b>In 1843 Sindh was completely annexed to the British Empire</b>.</li> <li>* <b>Complete abolition of slavery by Act V of 1843</b>.</li> </ul>                                                                                                                                                                                                                                     |
|                                             |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Governor-General / Viceroy (Tenure)  | Status of the Office                                    | Historical Events and Important Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lord Dalhousie (1848–1856 CE)</b> | Governor-General of India                               | <ul style="list-style-type: none"> <li>* The Second Anglo-Sikh War (1848-49) and the <b>annexation of Punjab on 29 March 1849</b> (the Kohinoor diamond was sent to the Queen).</li> <li>* Annexation of states under the <b>Doctrine of Lapse</b> (see the order below).</li> <li>* <b>Father of the railways in India</b>; the <b>first train ran in 1853</b> (Bombay to Thane, 34 km) and the second train in 1854 (Calcutta to Raniganj).</li> <li>* Established the <b>P.W.D. (Public Works Department)</b> for the first time in India.</li> </ul>     |
| <b>Lord Canning (1856–1862 CE)</b>   | <b>Last Governor-General of India and first Viceroy</b> | <ul style="list-style-type: none"> <li>* The <b>First War of Independence (Revolt) of 1857</b> took place.</li> <li>* By the <b>Act of 1858, Company rule came to an end</b> and the <b>office of the Mughal emperor was abolished</b>.</li> <li>* In 1857 three new universities were founded at Calcutta, Bombay and Madras.</li> </ul>                                                                                                                                                                                                                    |
| <b>Lord Mayo (1869–1872 CE)</b>      | Viceroy of India                                        | <ul style="list-style-type: none"> <li>* The <b>Archaeological Survey of India (ASI)</b> was set up in 1871 CE.</li> <li>* The <b>first (non-synchronous) census of India in 1872 CE</b> was held in his tenure.</li> </ul>                                                                                                                                                                                                                                                                                                                                  |
| <b>Lord Lytton (1876–1880 CE)</b>    | Viceroy of India                                        | <ul style="list-style-type: none"> <li>* Appointment of the <b>first Famine Commission</b> in 1878 under the chairmanship of Richard Strachey.</li> <li>* <b>Vernacular Press Act (1878) passed</b>; Surendranath Banerjee called it '<i>a thunderbolt from the blue</i>'.</li> <li>* The Second Anglo-Afghan War (1878-80) took place and the <b>Statutory Civil Service</b> was constituted in 1878.</li> <li>* Reduced the maximum age for entry into the civil service examination <b>from 21 to 19 years</b>.</li> </ul>                                |
| <b>Lord Ripon (1880–1884 CE)</b>     | Viceroy of India                                        | <ul style="list-style-type: none"> <li>* Introduced the <b>regular (decennial) census in India in 1881 CE</b>.</li> <li>* The <b>resolution on local self-government (1882)</b> was passed; hence he is called '<b>the father of local self-government</b>'.</li> <li>* Constitution of the <b>William Hunter Education Commission (1882)</b> related to primary education.</li> <li>* <b>Repealed the Vernacular Press Act (1882)</b>.</li> <li>* The <b>Ilbert Bill controversy (1883)</b> took place, also known as the '<b>White Mutiny</b>'.</li> </ul> |

| Governor-General / Viceroy (Tenure) | Status of the Office | Historical Events and Important Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lord Dufferin<br>(1884–1888 CE)     | Viceroy of India     | <ul style="list-style-type: none"> <li>* <b>Founding of the Indian National Congress (INC) in 1885 CE</b> by A.O. Hume.</li> <li>* <b>Allahabad University was founded in 1887 CE</b> (U.P. special fact).</li> </ul>                                                                                                                                                                                                                         |
| Lord Minto II<br>(1905–1910 CE)     | Viceroy of India     | <ul style="list-style-type: none"> <li>* <b>Split in the Congress at the Surat session of 1907</b> (Moderates and Extremists).</li> <li>* Under the <b>Morley-Minto Reforms (1909)</b>, <b>separate electorates were established</b> for Muslims for the first time.</li> </ul>                                                                                                                                                               |
| Lord Hardinge II<br>(1910–1916 CE)  | Viceroy of India     | <ul style="list-style-type: none"> <li>* <b>Delhi Durbar (1911)</b>: announcement of the <b>annulment of the Partition of Bengal</b> and of the transfer of the capital from Calcutta to Delhi.</li> <li>* <b>In 1912 Delhi became the official capital of India.</b></li> <li>* A bomb was thrown at him in Delhi on 23 December 1912 (<b>Delhi Conspiracy Case</b>), in which Bhai Bal Mukund was hanged.</li> </ul>                        |
| Lord Chelmsford<br>(1916–1921 CE)   | Viceroy of India     | <ul style="list-style-type: none"> <li>* The August Declaration of 20 August 1917, famous as the '<b>Montagu-Chelmsford Declaration</b>'.</li> </ul>                                                                                                                                                                                                                                                                                          |
| Lord Reading<br>(1921–1926 CE)      | Viceroy of India     | <ul style="list-style-type: none"> <li>* The <b>only Jewish Viceroy</b> of India.</li> <li>* <b>In 1922 CE the I.C.S. examination was held for the first time</b> simultaneously in London and <b>Allahabad</b>.</li> <li>* The '<b>Public Service Commission (PSC)</b>' was established on 1 October 1926.</li> </ul>                                                                                                                        |
| Lord Irwin<br>(1926–1931 CE)        | Viceroy of India     | <ul style="list-style-type: none"> <li>* Appointment of the <b>Harcourt Butler Committee (1927)</b> regarding the princely states.</li> <li>* <b>Arrival of the Simon Commission in India</b> on 3 February 1928.</li> <li>* The <b>Gandhi-Irwin Pact</b> was concluded on 5 March 1931.</li> </ul>                                                                                                                                           |
| Lord Linlithgow<br>(1936–1944 CE)   | Viceroy of India     | <ul style="list-style-type: none"> <li>* The <b>first provincial general elections (1936-37)</b> were held; the Congress formed governments in 8 of the 11 provinces (full majority in 6, coalition in 2).</li> <li>* <b>The Cripps Mission came to India in 1942</b>; Gandhiji called it "<i>a post-dated cheque</i>".</li> <li>* The <b>Quit India Movement (August Revolution)</b> began on 8 August 1942 (slogan 'Do or Die').</li> </ul> |



| Governor-General / Viceroy (Tenure)  | Status of the Office                                                                 | Historical Events and Important Exam Facts                                                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lord Wavell<br>(1944–1947 CE)        | Viceroy of India                                                                     | <ul style="list-style-type: none"> <li>* The <b>Sargent Plan (1944 CE)</b> for educational reform was announced.</li> <li>* The <b>Wavell Plan / Simla Conference (1945 CE)</b> was presented.</li> <li>* On 20 February 1947 the British PM Attlee announced the transfer of power to India by June 1948.</li> </ul> |
| Lord Mountbatten<br>(1947–1948 CE)   | <b>Last Viceroy of British India and first Governor-General of independent India</b> | <ul style="list-style-type: none"> <li>* The <b>Mountbatten Plan (Dickie Bird Plan)</b>, containing the outline of the partition of India, was announced on 3 June 1947.</li> <li>* After independence on 15 August 1947 he became the <b>first Governor-General of independent India</b>.</li> </ul>                 |
| C. Rajagopalachari<br>(1948–1950 CE) | Governor-General of independent India                                                | <ul style="list-style-type: none"> <li>* He was the <b>last Governor-General</b> of India and the <b>first and only Indian Governor-General</b> of independent India.</li> </ul>                                                                                                                                      |

## 2 States Annexed under Dalhousie's 'Doctrine of Lapse' – Chronology

(UPPCS Prelims asks for its correct chronological order every second year)

1. **Satara** — 1848 CE
2. **Jaitpur** — 1849 CE
3. **Sambalpur** — 1849 CE
4. **Baghat** — 1850 CE
5. **Udaipur** — 1852 CE
6. **Jhansi** — 1853 CE
7. **Nagpur** — 1854 CE
8. **Awadh** — 1856 CE (*Special: it was annexed not under the Doctrine of Lapse but on the charge of misgovernment (Outram Report)*)

## 3 Principal Commissions Constituted during Lord Curzon's Tenure (1899 - 1905 CE)

(The favourite coded matching question of UPPCS)

| Year (CE) | Name of the Commission       | Name of the Chairman      | Main Purpose of Formation / Area of Inquiry                       |
|-----------|------------------------------|---------------------------|-------------------------------------------------------------------|
| 1900 CE   | <b>Famine Commission</b>     | Sir Antony MacDonnell     | To review famine relief works and frame policy.                   |
| 1901 CE   | <b>Irrigation Commission</b> | Sir Colin Scott-Moncrieff | To increase agricultural productivity and for irrigation reforms. |

| Year (CE) | Name of the Commission         | Name of the Chairman | Main Purpose of Formation / Area of Inquiry                                                                         |
|-----------|--------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| 1901 CE   | <b>Railway Commission</b>      | Thomas Robertson     | To reform the administrative and financial structure of the railways.                                               |
| 1902 CE   | <b>Police Commission</b>       | Sir Andrew Fraser    | To recommend efficiency and reforms in police administration.                                                       |
| 1902 CE   | <b>Universities Commission</b> | Sir Thomas Raleigh   | To improve the condition of higher education/universities (on its basis came the <b>Universities Act of 1904</b> ). |

- **Special note:** On **19 July 1905 CE** Curzon made the '**announcement of the Partition of Bengal**' at Shimla; it was approved by the British sovereign in September 1905 and came into effect on 16 October 1905.

## 4 Principal Land Revenue Systems of the British Period

*(The most important comparative table for the UPPCS Prelims examination)*

| Feature / Point            | 1. Permanent Settlement (Zamindari)                                                | 2. Ryotwari System                                                                         | 3. Mahalwari System                                                                                   |
|----------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Year of introduction       | <b>1793 CE</b>                                                                     | <b>1820 CE</b>                                                                             | <b>1822 CE</b>                                                                                        |
| Principal leader / father  | <b>Lord Cornwallis</b> <i>(and John Shore)</i>                                     | <b>Thomas Munro</b> <i>(and Captain Read)</i>                                              | <b>Holt Mackenzie</b> <i>(and Martin Bird)</i>                                                        |
| Geographical area          | Bengal, Bihar, Orissa, Uttar Pradesh (Banaras division), northern Karnataka.       | Bombay, Madras, Assam.                                                                     | Uttar Pradesh (United Provinces), Central Provinces, Punjab.                                          |
| Total area (percentage)    | <b>19% of the area</b> of British India                                            | <b>51% of the area</b> of British India <i>(the largest)</i>                               | <b>30% of the area</b> of British India                                                               |
| Real owner of the land     | The <b>zamindar</b> was regarded as owner.                                         | The <b>peasant (raiyyat)</b> was regarded as owner.                                        | The <b>village (mahal / collective community)</b> was regarded as owner.                              |
| Responsibility for revenue | The responsibility for paying tax to the government lay with the <b>zamindar</b> . | The responsibility for paying tax to the government lay directly with the <b>peasant</b> . | The responsibility for paying tax to the government lay with the <b>village headman (lambardar)</b> . |

## 1 Principal Commissions and Committees of the British Period

| Name of Commission / Committee | Year of Establishment | Viceroy of the Time | Main Subject / Purpose of Inquiry of the Committee |
|--------------------------------|-----------------------|---------------------|----------------------------------------------------|
| <b>Inam Commission</b>         | 1852 CE               | Lord Dalhousie      | To examine the jagirs and titles of landholders.   |

| Name of Commission / Committee | Year of Establishment | Viceroy of the Time | Main Subject / Purpose of Inquiry of the Committee                                                 |
|--------------------------------|-----------------------|---------------------|----------------------------------------------------------------------------------------------------|
| Strachey Commission            | 1880 CE               | Lord Lytton         | The first Famine Commission; for a policy of providing relief to famine victims.                   |
| Hunter Commission              | 1882 CE               | Lord Ripon          | For a <b>review of the progress of education</b> after Wood's Despatch of 1854.                    |
| Herschell Committee            | 1893 CE               | Lord Lansdowne      | To give suggestions on the <b>mint/currency</b> and finance.                                       |
| Lyll Commission                | 1897 CE               | Lord Elgin II       | To study the report of the Famine Commission of 1880.                                              |
| MacDonnell Commission          | 1900 CE               | Lord Curzon         | The second Famine Commission; to give its suggestions on the report of the Strachey Commission.    |
| Moncrieff Commission           | 1901 CE               | Lord Curzon         | To plan <b>irrigation expenditure</b> and outline the development of canals.                       |
| Raleigh Commission             | 1902 CE               | Lord Curzon         | For reforms and work relating to <b>university education</b> .                                     |
| Fraser Commission              | 1902 CE               | Lord Curzon         | To examine the working and efficiency of <b>police administration</b> .                            |
| Sadler Commission              | 1917 CE               | Lord Chelmsford     | To examine the working of <b>Calcutta University</b> .                                             |
| Butler Committee               | 1927 CE               | Lord Irwin          | Suggestions for improving the relations between the British Crown and the <b>princely states</b> . |
| Hartog Committee               | 1929 CE               | Lord Irwin          | For the development and improvement of primary and missionary education.                           |
| Whitley Commission             | 1929 CE               | Lord Irwin          | For <b>labour reforms</b> in industrial workshops, mills and plantations.                          |
| Sapru Committee                | 1934 CE               | Lord Willingdon     | Study of the causes of <b>unemployment</b> in the United Provinces (Uttar Pradesh).                |
| Woodhead Commission            | 1943-44 CE            | Lord Wavell         | To inquire into the causes of the <b>severe Bengal Famine of 1943</b> .                            |

## 2 The Great Revolt of 1857: Principal Exam Facts and the Cycle of Suppression

### > Important Historical Points:

- The first spark of the revolt:** On **29 March 1857 Mangal Pandey** of the 34th Native Infantry at Barrackpore (West Bengal, Hooghly/Parganas district) rebelled. He fired at his Sergeant-Major Hewson and the Adjutant. Mangal Pandey was hanged on **8 April 1857**.

- **Collective beginning of the revolt:** Began on **10 May 1857 at Meerut** with the soldiers' open revolt and march to Delhi.
- **Top office-holders of the time:** At the time of the revolt the Governor-General of India was **Lord Canning**, and the symbolic Mughal ruler of Delhi was **Bahadur Shah II 'Zafar'**.

### > Principal Centres of the Revolt of 1857 and the British Suppressors (Match the Following):

| Centre of the Revolt  | Date of Outbreak | Principal Indian Leader (Hero)                             | Principal British Suppressor          | Date of Suppression / Control |
|-----------------------|------------------|------------------------------------------------------------|---------------------------------------|-------------------------------|
| Delhi                 | 11 May 1857      | Bahadur Shah II (real command: <b>Bakht Khan</b> )         | John Nicholson, Hodson                | 20 September 1857             |
| Lucknow               | 30 May 1857      | <b>Begum Hazrat Mahal</b> , Birjis Qadr                    | Colin Campbell                        | March 1858                    |
| Jhansi                | 4 June 1857      | <b>Rani Lakshmibai</b>                                     | General Hugh Rose                     | 17 June 1858                  |
| Kanpur                | 5 June 1857      | <b>Nana Saheb</b> (Tatya Tope)                             | Colin Campbell                        | 6 December 1857               |
| Jagdishpur (Bihar)    | 12 June 1857     | <b>Kunwar Singh</b> (and Amar Singh)                       | Major William Taylor and Vincent Eyre | December 1858                 |
| Faizabad              | June 1857        | <b>Maulvi Ahmadullah</b> ( <i>gave the call of jihad</i> ) | Colin Campbell                        | 5 June 1858                   |
| Bareilly              | June 1857        | <b>Khan Bahadur Khan</b>                                   | Colin Campbell                        | 1858                          |
| Allahabad (Prayagraj) | June 1857        | <b>Liaqat Ali</b>                                          | Colonel Neill                         | 1858                          |
| Fatehpur              | 1857             | <b>Azimullah</b>                                           | General Renaud                        | 1858                          |

## 3 Pioneers of the Socio-Religious Reform Movements

### > (a) Raja Rammohan Roy and the Brahmo Samaj

| Subject Section    | Historical Details and Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Life               | Born on <b>22 May 1772</b> at Radhanagar, Hooghly (Bengal). Died on 27 September 1833 at Bristol (England).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Titles and honours | <ul style="list-style-type: none"> <li>* <b>Raja:</b> given by the Mughal ruler <b>Akbar II</b>.</li> <li>* <b>Yugadoot (Messenger of the Age):</b> given by Subhas Chandra Bose.</li> <li>* <b>Father of the constitutional agitation:</b> so called by Surendranath Banerjee.</li> <li>* <b>Other names:</b> herald/father of the Indian Renaissance, father of modern India, father of Indian nationalism, morning star of the new dawn, bridge between the past and the future, father of the Indian awakening, messiah of new India, the first modern man and man of the age.</li> </ul> |

| Subject Section                | Historical Details and Exam Facts                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Newspapers published           | <ul style="list-style-type: none"> <li>* <b>Sambad Kaumudi</b> (Bengali language) — 1821 CE</li> <li>* <b>Mirat-ul-Akhbar</b> (Persian language) — 1822 CE</li> </ul>                                                                                                                  |
| Principal books / works        | <ul style="list-style-type: none"> <li>* <b>Tuhfat-ul-Muwahhidin</b> (A Gift to Monotheists) — 1809 CE</li> <li>* <b>The Precepts of Jesus</b> — 1820 CE</li> </ul>                                                                                                                    |
| Principal institutions founded | <ul style="list-style-type: none"> <li>* <b>Atmiya Sabha</b> (1814-15 CE) — Calcutta</li> <li>* <b>Brahmo Samaj</b> (1828 CE) — Calcutta</li> <li>* <b>Hindu College and David Hare College</b> (1817 CE) — Calcutta</li> <li>* <b>Vedanta College</b> (1825 CE) — Calcutta</li> </ul> |

### > (b) Swami Dayanand Saraswati and the Arya Samaj

| Subject Section                 | Historical Details and Exam Facts                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Life                            | Born on 12 February 1824 at Tankara (Gujarat). His childhood name was <b>Mool Shankar</b> . His guru Purnananda named him 'Swami Dayanand Saraswati'.                                                                                                                            |
| Title                           | He is called ' <b>the Martin Luther of India</b> '.                                                                                                                                                                                                                              |
| Institution founded and slogans | <ul style="list-style-type: none"> <li>* Founded the <b>Arya Samaj in 1875 CE at Bombay</b> (headquarters later shifted to Lahore).</li> <li>* <b>Slogans: "Go back to the Vedas", "India is for the Indians"</b>.</li> </ul>                                                    |
| Principal movements             | Started the <b>Shuddhi movement</b> (return of those who had become non-Hindus) and the <b>cow-protection movement</b> .                                                                                                                                                         |
| Principal books / works         | <ul style="list-style-type: none"> <li>* <b>Satyarth Prakash</b> (Hindi/Sanskrit medium) — <b>1874 CE</b> (<i>it is called the Bible of the Arya Samaj</i>).</li> <li>* <b>Pakhand Khandan</b> (1866 CE) and <b>Advaitamat ka Khandan</b> (1873 CE).</li> </ul>                  |
| Special exam facts              | <ul style="list-style-type: none"> <li>* He was the first to use the word 'Swaraj' and <b>the first person to accept Hindi as the national language</b>.</li> <li>* His famous statement is: <i>"The worst indigenous rule is better than the best foreign rule."</i></li> </ul> |

## 4 Hindu Socio-Religious Movements and Organisations (Chronology)

| Name of the Organisation / Movement | Year of Founding | Principal Founder / Propounder | Place of Founding |
|-------------------------------------|------------------|--------------------------------|-------------------|
| Asiatic Society of Bengal           | 1784 CE          | Sir William Jones              | Calcutta (Bengal) |
| Young Bengal Movement               | 1826 CE          | Henry Louis Vivian Derozio     | Calcutta (Bengal) |
| Brahmo Samaj                        | 1828 CE          | Raja Rammohan Roy              | Calcutta (Bengal) |
| Tattvabodhini Sabha                 | 1839 CE          | Debendranath Tagore            | Calcutta (Bengal) |

| Name of the Organisation / Movement | Year of Founding | Principal Founder / Propounder              | Place of Founding                 |
|-------------------------------------|------------------|---------------------------------------------|-----------------------------------|
| Paramahansa Mandali                 | 1840 CE          | Gopal Hari Deshmukh ( <i>Lokahitavadi</i> ) | Bombay                            |
| Rahnumai Mazdayasnan Sabha          | 1851 CE          | Dadabhai Naoroji, S.S. Bengalee             | Bombay (Parsi reform)             |
| Prarthana Samaj                     | <b>1867 CE</b>   | Dr. Atmaram Pandurang, M.G. Ranade          | Bombay                            |
| Arya Samaj                          | <b>1875 CE</b>   | Swami Dayanand Saraswati                    | Bombay                            |
| Theosophical Society                | <b>1875 CE</b>   | Madame H.P. Blavatsky, Colonel Olcott       | New York (centre in India: Adyar) |
| Deccan Education Society            | 1884 CE          | G.G. Agarkar, Bal Gangadhar Tilak           | Poona                             |
| Indian National Social Conference   | 1887 CE          | Mahadev Govind (M.G.) Ranade                | Bombay                            |
| Dev Samaj                           | 1887 CE          | Shivnarayan Agnihotri                       | Lahore                            |
| Ramakrishna Mission                 | <b>1897 CE</b>   | <b>Swami Vivekananda</b>                    | Belur (Bengal)                    |
| Servants of India Society           | <b>1905 CE</b>   | <b>Gopal Krishna Gokhale</b>                | Bombay                            |
| Poona Seva Sadan                    | 1909 CE          | Smt. Ramabai Ranade                         | Poona                             |
| Social Service League               | 1911 CE          | N.M. Joshi                                  | Bombay                            |
| Seva Samiti                         | 1914 CE          | Hriday Nath (H.N.) Kunzru                   | Allahabad (Prayagraj)             |
| Hindu Mahasabha                     | <b>1915 CE</b>   | <b>Pandit Madan Mohan Malaviya</b>          | Haridwar                          |
| Women's Indian Association          | 1917 CE          | Lady Sadasiva Iyer, Annie Besant            | Madras                            |
| Vishva Bharati                      | 1921 CE          | Rabindranath Tagore                         | Calcutta                          |

## 5 Dalit, Backward and Lower-Caste Movements / Organisations

| Name of the Movement / Organisation   | Year of Founding | Principal Founder / Leader                                             | Place of Founding    |
|---------------------------------------|------------------|------------------------------------------------------------------------|----------------------|
| Satyashodhak Samaj                    | <b>1873 CE</b>   | <b>Jyotiba Phule</b> ( <i>book: Gulamgiri</i> )                        | Maharashtra          |
| Aruvippuram Movement                  | 1888 CE          | Sree Narayana Guru ( <i>slogan: one caste, one religion, one God</i> ) | Aruvippuram (Kerala) |
| Sree Narayana Dharma Paripalana Yogam | 1903 CE          | Sree Narayana Guru, Dr. Palpu and Kumaran Asan                         | Kerala               |

| Name of the Movement / Organisation | Year of Founding | Principal Founder / Leader                 | Place of Founding    |
|-------------------------------------|------------------|--------------------------------------------|----------------------|
| Depressed Classes Mission Society   | 1906 CE          | V.R. Shinde                                | Bombay               |
| Bahujan Samaj                       | 1910 CE          | Mukundrao Patil                            | Satara (Maharashtra) |
| Justice Party Movement              | 1915-16 CE       | C.N. Mudaliar, T.M. Nair and P. Theagaraya | Madras               |
| Bahishkrit Hitakarini Sabha         | 1924 CE          | Dr. Bhimrao Ambedkar                       | Bombay               |

## 6 Muslim Socio-Religious Movements and Organisations

| Name of the Movement / Organisation | Year of Founding | Principal Founder / Leader                                                         | Main Centre / Place        |
|-------------------------------------|------------------|------------------------------------------------------------------------------------|----------------------------|
| Darul-Uloom (Deoband Movement)      | 1866 CE          | Muhammad Qasim Nanautavi and Rashid Ahmad Gangohi                                  | Deoband (Saharanpur, U.P.) |
| Aligarh Movement                    | 1875 CE          | Sir Syed Ahmad Khan                                                                | Aligarh (U.P.)             |
| Ahmadiya Movement                   | 1889-90 CE       | Mirza Ghulam Ahmad ( <i>called himself Masih-ul-Mauood, the promised Messiah</i> ) | Qadian / Faridkot (Punjab) |
| Nadwatul Ulama Movement             | 1894-95 CE       | Maulana Shibli Numani                                                              | Lucknow (U.P.)             |

## 7 Political Organisations before the Founding of the Indian National Congress (1885)

(The favourite section of UPPCS Prelims for coded matching and chronology)

| Name of the Organisation                                                                        | Principal Founder           | Year of Founding | Headquarters / Place |
|-------------------------------------------------------------------------------------------------|-----------------------------|------------------|----------------------|
| Landholders' Society ( <i>Zamindari Association</i> )                                           | Dwarkanath Tagore           | 1838 CE          | Calcutta             |
| British India Society                                                                           | William Adams               | 1839 CE          | London / Calcutta    |
| Bengal British India Society                                                                    | George Thompson             | 1843 CE          | Calcutta             |
| British Indian Association ( <i>merger of the Landholders' + Bengal British India Society</i> ) | Debendranath Tagore         | 1851 CE          | Calcutta             |
| Bombay Association                                                                              | Dadabhai Naoroji            | 1852 CE          | Bombay               |
| Madras Native Association                                                                       | Gazulu Lakshminarasu Chetty | 1852 CE          | Madras               |



| Name of the Organisation                                                   | Principal Founder                                | Year of Founding | Headquarters / Place |
|----------------------------------------------------------------------------|--------------------------------------------------|------------------|----------------------|
| East India Association<br>(merger of the London India Committee + Society) | Dadabhai Naoroji                                 | 1866 CE          | London               |
| Poona Sarvajanik Sabha                                                     | Ganesh Vasudev Joshi, M.G. Ranade                | 1870 CE          | Poona                |
| India Society                                                              | Anand Mohan Bose                                 | 1872 CE          | London               |
| Indian Association (Bharat Sangh)                                          | Anand Mohan Bose, Surendranath Banerjee          | 1876 CE          | Calcutta             |
| Native Press Association                                                   | Surendranath Banerjee                            | 1877 CE          | Delhi                |
| Madras Mahajan Sabha                                                       | P. Ananda Charlu, M. Viraraghavachari            | 1884 CE          | Madras               |
| Bombay Presidency Association                                              | Badruddin Tyabji, Pherozeshah Mehta, K.T. Telang | 1885 CE          | Bombay               |

## 1 Principal Sessions of the Indian National Congress (INC) and Historical Facts

(UPPCS Prelims often asks for matching the 'President' and 'special fact' from this table)

| Venue     | Year (CE) | President of the Session | Historical Feature / Exam Fact                                                              |
|-----------|-----------|--------------------------|---------------------------------------------------------------------------------------------|
| Bombay    | 1885 CE   | Womesh Chandra Bonnerjee | The first session of the Congress; <b>72 delegates</b> from all over the country took part. |
| Calcutta  | 1886 CE   | Dadabhai Naoroji         | The Indian National Association (Conference) merged with the Congress.                      |
| Madras    | 1887 CE   | <b>Badruddin Tyabji</b>  | Became the <b>first Muslim President</b> of the Congress.                                   |
| Allahabad | 1888 CE   | <b>George Yule</b>       | Became the <b>first European (English) elected President</b> of the Congress.               |
| Calcutta  | 1896 CE   | Rahimtulla M. Sayani     | The national song ' <b>Vande Mataram</b> ' was sung for the first time.                     |
| Banaras   | 1905 CE   | Gopal Krishna Gokhale    | Support for the <b>Swadeshi and Boycott Movement</b> against the Partition of Bengal.       |
| Surat     | 1907 CE   | Rash Behari Ghosh        | The historic <b>split of the Congress into Moderates and Extremists</b> .                   |
| Calcutta  | 1911 CE   | Bishan Narayan Dar       | The national anthem ' <b>Jana Gana Mana</b> ' was sung for the first time.                  |

| Venue    | Year (CE) | President of the Session | Historical Feature / Exam Fact                                                                                                        |
|----------|-----------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Lucknow  | 1916 CE   | Ambika Charan Majumdar   | ' <b>Lucknow Pact</b> ' — the Moderates and Extremists reunited and an agreement was made between the Congress and the Muslim League. |
| Calcutta | 1917 CE   | <b>Annie Besant</b>      | Became the <b>first woman President</b> in the history of the Congress.                                                               |
| Delhi    | 1923 CE   | Maulana Abul Kalam Azad  | Special session; became the <b>youngest President</b> at the age of 35.                                                               |
| Belgaum  | 1924 CE   | <b>Mahatma Gandhi</b>    | The <b>only session</b> held under the presidency of Mahatma Gandhi.                                                                  |
| Kanpur   | 1925 CE   | <b>Sarojini Naidu</b>    | Became the <b>first Indian woman President</b> of the Congress (U.P. special fact).                                                   |
| Lahore   | 1929 CE   | Pandit Jawaharlal Nehru  | On the bank of the Ravi the resolution <b>demanding 'Purna Swaraj'</b> was passed for the first time.                                 |
| Karachi  | 1931 CE   | Sardar Vallabhbhai Patel | <b>Fundamental Rights</b> were demanded from the Congress platform for the first time.                                                |
| Faizpur  | 1937 CE   | Jawaharlal Nehru         | <b>The first session held in a village (rural area).</b>                                                                              |

## 2 Exact Chronology of the Principal Events of the Freedom Movement

*(The most core area of the UPPCS Prelims examination; a question arranging events by date certainly comes from here)*

### > (a) First Phase: From the Partition of Bengal to the Non-Cooperation Movement (1905 – 1922 CE)

| No. | Name of the Historical Event                                                                                 | Exact Date / Period |
|-----|--------------------------------------------------------------------------------------------------------------|---------------------|
| 1.  | Formal proclamation of the Swadeshi Movement                                                                 | 7 August 1905 CE    |
| 2.  | Partition of Bengal comes into effect (day of mourning throughout Bengal)                                    | 16 October 1905 CE  |
| 3.  | Founding of the Muslim League at Dhaka                                                                       | 30 December 1906 CE |
| 4.  | Madame Bhikaji Cama hoists the first Indian flag at Stuttgart (Germany)                                      | 1907 CE             |
| 5.  | Split of the Congress (Surat session)                                                                        | 1907 CE             |
| 6.  | Founding of a branch of the Muslim League in London (by Syed Ameer Ali)                                      | 1908 CE             |
| 7.  | Holding of the Delhi Durbar (announcement of the annulment of the Partition of Bengal and change of capital) | 12 December 1911 CE |

| No. | Name of the Historical Event                                                                  | Exact Date / Period           |
|-----|-----------------------------------------------------------------------------------------------|-------------------------------|
| 8.  | Founding of the Ghadar Party at San Francisco                                                 | 1 November 1913 CE            |
| 9.  | Komagata Maru ship incident                                                                   | 1914 CE                       |
| 10. | Mahatma Gandhi's permanent return to India from South Africa (Pravasi Bharatiya Divas)        | 9 January 1915 CE             |
| 11. | British government confers the title 'Kaiser-i-Hind' on Gandhiji                              | 1915 CE                       |
| 12. | Bal Gangadhar Tilak founds the Home Rule League at Poona                                      | April 1916 CE                 |
| 13. | Annie Besant starts the Home Rule League at Madras                                            | September 1916 CE             |
| 14. | Lucknow Pact (Congress-League Pact)                                                           | December 1916 CE              |
| 15. | Champaran Satyagraha (Gandhiji's first satyagraha in India)                                   | 10 April 1917 CE              |
| 16. | 'Montagu Declaration' by the Secretary of State Edwin Montagu                                 | 20 August 1917 CE             |
| 17. | Ahmedabad mill workers' movement (Gandhiji's first hunger strike)                             | February - March 1918 CE      |
| 18. | Kheda Satyagraha (movement in favour of the peasants of Gujarat)                              | March 1918 CE                 |
| 19. | Enforcement of the repressive Rowlatt Act (black law)                                         | March 1919 CE                 |
| 20. | <b>Jallianwala Bagh massacre</b> (Amritsar tragedy)                                           | <b>13 April 1919 CE</b>       |
| 21. | Formation of the Khilafat Committee at Bombay                                                 | September 1919 CE             |
| 22. | All India Khilafat Conference at Delhi (Gandhiji became its president)                        | November 1919 CE              |
| 23. | <b>Beginning of the Non-Cooperation Movement and death of Bal Gangadhar Tilak</b>             | <b>1 August 1920 CE</b>       |
| 24. | Special Congress session at Calcutta (resolution on non-cooperation moved for the first time) | September 1920 CE             |
| 25. | Resolution on the Non-Cooperation Movement finally passed at the Nagpur annual session        | December 1920 CE              |
| 26. | <b>Chauri Chaura incident</b> (Gorakhpur, Uttar Pradesh)                                      | <b>4 - 5 February 1922 CE</b> |
| 27. | Suspension (withdrawal) of the Non-Cooperation Movement by the Bardoli resolution             | 12 February 1922 CE           |
| 28. | Gandhiji's arrest by the English after the withdrawal of the movement                         | 10 March 1922 CE              |

## > (b) Second Phase: From the Swaraj Party to the Attainment of Independence (1922 – 1946 CE)

| No. | Name of the Historical Event                                                                    | Exact Date / Period          |
|-----|-------------------------------------------------------------------------------------------------|------------------------------|
| 29. | Peshawar Communist (leftist) Conspiracy Case                                                    | 1922 - 1923 CE               |
| 30. | <b>Founding of the Swaraj Party</b> at Allahabad by C.R. Das and Motilal Nehru                  | January 1923 CE              |
| 31. | Kanpur Bolshevik (Communist) Conspiracy Case                                                    | 1924 CE                      |
| 32. | Formation of the Hindustan Republican Association (HRA) at Kanpur                               | 1924 CE                      |
| 33. | Agreement between Gandhi and Chittaranjan Das (Gandhi-Das Pact)                                 | November 1924 CE             |
| 34. | <b>Kakori Train Action</b> (the changed name of the Kakori train incident, Lucknow)             | <b>9 August 1925 CE</b>      |
| 35. | Vithalbhai Patel becomes President of the Central Legislative Assembly                          | 1925 CE                      |
| 36. | Arrival of the Simon Commission in India (Bombay)                                               | 3 February 1928 CE           |
| 37. | Reorganisation of the Hindustan Socialist Republican Association (HSRA) at Delhi                | 1928 CE                      |
| 38. | Presentation of Motilal Nehru's 'Nehru Report' by the All-Parties Committee                     | August 1928 CE               |
| 39. | Bardoli Satyagraha in the Surat district of Gujarat                                             | October 1928 CE              |
| 40. | Killing of the British officer Saunders by revolutionaries at Lahore                            | 17 December 1928 CE          |
| 41. | Central Assembly bomb case by Bhagat Singh and Batukeshwar Dutt                                 | 8 April 1929 CE              |
| 42. | Lahore Conspiracy Case conducted against the revolutionaries                                    | 1929 CE                      |
| 43. | Gandhiji begins the historic <b>Dandi March</b> from Sabarmati                                  | 12 March 1930 CE             |
| 44. | Beginning of the <b>Civil Disobedience Movement</b> by breaking the salt law at Dandi           | 6 April 1930 CE              |
| 45. | First Round Table Conference in London (without the participation of the Congress)              | 1930 - 1931 CE               |
| 46. | The historic <b>Gandhi-Irwin Pact</b> at Delhi                                                  | 5 March 1931 CE              |
| 47. | <b>Bhagat Singh, Rajguru and Sukhdev hanged in Lahore jail</b>                                  | <b>23 March 1931 CE</b>      |
| 48. | Second Round Table Conference (in which Gandhiji represented the Congress)                      | September - December 1931 CE |
| 49. | Announcement of separate electorates for the Dalits by the British PM ( <b>Communal Award</b> ) | 16 August 1932 CE            |

| No. | Name of the Historical Event                                                                             | Exact Date / Period         |
|-----|----------------------------------------------------------------------------------------------------------|-----------------------------|
| 50. | The historic <b>Poona Pact</b> between Gandhiji and Dr. Ambedkar                                         | 24 September 1932 CE        |
| 51. | Third Round Table Conference (complete boycott by the Congress)                                          | November - December 1932 CE |
| 52. | Formation of the Congress Socialist Party by Jayaprakash Narayan and Acharya Narendra Dev                | 1934 CE                     |
| 53. | Formation of the <b>Forward Bloc</b> by Subhas Chandra Bose                                              | May 1939 CE                 |
| 54. | The Muslim League celebrates ' <b>Deliverance Day</b> ' after the resignation of the Congress ministries | 22 December 1939 CE         |
| 55. | Official 'demand for Pakistan' at the Lahore session of the Muslim League                                | March 1940 CE               |
| 56. | 'August Offer' by Viceroy Linlithgow for cooperation in the war                                          | 8 August 1940 CE            |
| 57. | Arrival in India of the Cripps Mission for wartime constitutional reforms                                | March 1942 CE               |
| 58. | Formal beginning of the <b>Quit India Movement</b> from the Gowalia Tank Maidan, Bombay                  | 9 August 1942 CE            |
| 59. | Holding of the Simla Conference to remove the deadlock over the Wavell Plan                              | 1945 CE                     |
| 60. | The historic <b>Royal Indian Navy Mutiny (RIN Mutiny)</b> by the sailors of the warship Talwar at Bombay | 1946 CE                     |

### 3 Important Social Reform Acts of British India

| Year          | Social Reform Act / Law               | Governor-General Responsible               | Legal Subject / Main Social Reform                                                                                 |
|---------------|---------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1795 and 1804 | <b>Prohibition of Infanticide Act</b> | Lord Wellesley <i>(and Sir John Shore)</i> | Complete legal ban on killing/murdering newborn infants (especially girls).                                        |
| 1829          | <b>Prohibition of Sati Act</b>        | Lord William Bentinck                      | Regulation XVII declared Sati illegal and punishable (through the intensive efforts of <b>Raja Rammohan Roy</b> ). |
| 1829          | Prohibition of Child Marriage Act     | Lord William Bentinck                      | An initial attempt to prohibit the marriage of boys below 18 years of age.                                         |
| 1837          | Hindu Women's / Property Act          | Lord Auckland                              | Hindu women were legally given a limited right to property.                                                        |
| 1843          | <b>Prohibition of Slavery Act</b>     | Lord Ellenborough                          | In keeping with the spirit of the Charter Act of 1833, slavery was completely prohibited in India.                 |

| Year | Social Reform Act / Law                                   | Governor-General Responsible | Legal Subject / Main Social Reform                                                                                                                         |
|------|-----------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1856 | <b>Hindu Widows' Remarriage Act</b>                       | Lord Canning                 | Widow remarriage was declared legally valid and the children received inheritance (through the efforts of <b>Ishwar Chandra Vidyasagar</b> ).              |
| 1872 | <b>Native Marriage Act</b> ( <i>Brahmo Marriage Act</i> ) | Lord Northbrook              | Legality of inter-caste marriages and an attempt to stop polygamy (with the cooperation of <b>Keshab Chandra Sen</b> ).                                    |
| 1891 | <b>Age of Consent Act</b>                                 | Lord Lansdowne               | The minimum age of consent/marriage for girls was raised from 10 to <b>12 years</b> (through the efforts of the Parsi reformer <b>Behramji Malabari</b> ). |
| 1930 | <b>Sarda Act</b>                                          | Lord Irwin                   | To stop child marriage the minimum age of marriage was fixed at <b>18 years</b> for boys and <b>14 years</b> for girls.                                    |
| 1931 | Child Marriage Restraint Act                              | Lord Irwin                   | Child marriage was legally prohibited in order to enforce the provisions of the Sarda Act strictly.                                                        |

#### 4 Principal Peasant and Civil Movements of Modern India

(The most important and highest-scoring table for the 'Match the Following' section of UPPCS Prelims)

| Name of the Revolt / Movement                        | Affected Geographical Area | Principal Leaders                          | Main Cause and Result of the Revolt                                                                                                                                                                                                                                                     |
|------------------------------------------------------|----------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moplah Rebellion</b> (1836-1854) and (1921-22 CE) | Malabar coast (Kerala)     | K.N. Das, Sayyid Qazi, <b>Ali Musliyar</b> | <p>* <b>Cause:</b> oppression by local Hindu landlords (jenmis), the repressive rent policies of the English government and the influence of the Khilafat movement.</p> <p>* <b>Result:</b> turned into violent communal riots; brutally suppressed by the British army.</p>            |
| <b>Pagalpanthi Revolt</b>                            | North Bengal               | <b>Karam Shah and Tipu Shah</b>            | <p>* <b>Cause:</b> revolt of the peasants of the semi-religious Pagalpanthi sect against the excessive rent and oppression of the local landlords.</p> <p>* <b>Result:</b> Tipu Shah tried to set up an independent administrative system; later brought under control by the army.</p> |
| <b>Faraizi Movement</b> (1838 CE)                    | Faridpur (Bengal)          | <b>Haji Shariatullah and Dudu Miyan</b>    | <p>* <b>Cause:</b> protection of the religio-social order of the Faraizi sect and strong opposition to the illegal taxes imposed by the landlords.</p>                                                                                                                                  |

| Name of the Revolt / Movement                    | Affected Geographical Area                           | Principal Leaders                                        | Main Cause and Result of the Revolt                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Indigo Revolt (1859 - 1860 CE)</b>            | Bengal (Nadia district)                              | <b>Bishnu Biswas and Digambar Biswas</b>                 | <p>* <b>Cause:</b> forcible, loss-making cultivation of indigo imposed on the peasants by European planters (dadni system).</p> <p>* <b>Result:</b> a highly organised and successful movement; the government had to set up the Indigo Commission. <i>(It is accurately depicted in Dinabandhu Mitra's play 'Nil Darpan').</i></p>                                     |
| <b>Pabna Revolt (1873 - 1876 CE)</b>             | Bengal                                               | Ishan Chandra Roy, Keshab Chandra Roy, Shambhunath Pal   | <p>* <b>Cause:</b> attempts by landlords to illegally seize the land of tenants/artisans, eviction and excessive increases in rent.</p> <p>* <b>Result:</b> a peaceful and legal struggle; it was under its pressure that the <b>Bengal Tenancy Act of 1885</b> later came.</p>                                                                                         |
| <b>Deccan Riots (1874 - 1875 CE)</b>             | Poona, Ahmednagar, Sholapur and Satara (Maharashtra) | Baba Saheb Deshmukh and local leaders                    | <p>* <b>Cause:</b> heavy increase in rent even after the slump in cotton, and the eviction of peasants from their land by outside Marwari/Gujarati moneylenders after trapping them in debt.</p> <p>* <b>Result:</b> the moneylenders' account books were burnt; the government passed the <b>Deccan Agriculturists' Relief Act (1879)</b> to protect the peasants.</p> |
| <b>Bijolia Movement (1905, 1913, 1916, 1927)</b> | Mewar / Rajasthan                                    | Sadhu Sitaram Das, Vijay Singh Pathik, Manikya Lal Verma | <p>* <b>Cause:</b> 84 different kinds of illegal cesses (lag-bag) imposed on the peasants of the Bijolia jagir by the feudal administration.</p> <p>* <b>Result:</b> the longest-running non-violent peasant movement of India; finally there was a large reduction in taxes.</p>                                                                                       |
| <b>Champaran Satyagraha (1917 CE)</b>            | Champaran, Motihari (Bihar)                          | <b>Mahatma Gandhi</b> , Rajendra Prasad, Rajkumar Shukla | <p>* <b>Cause:</b> against the exploitative '<b>Tinkathia system</b>' imposed by European indigo planters (compulsion to sow indigo on 3/20 of the land).</p> <p>* <b>Result:</b> Gandhiji's <b>first successful movement</b> in India; the Tinkathia system was abolished and 25% of the illegal collection was refunded.</p>                                          |
| <b>Kheda Satyagraha (1918 CE)</b>                | Kheda (Gujarat)                                      | <b>Mahatma Gandhi</b> and Vallabhbhai Patel              | <p>* <b>Cause:</b> forcible collection of tax, contrary to the revenue rules, despite the whole crop being destroyed by severe famine.</p> <p>* <b>Result:</b> the government bowed down and secretly ordered that tax be taken only from those peasants who were able to pay.</p>                                                                                      |



| Name of the Revolt / Movement         | Affected Geographical Area                           | Principal Leaders               | Main Cause and Result of the Revolt                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Awadh Kisan Movement (1920 CE)</b> | Pratapgarh, Rae Bareli, Sultanpur (Uttar Pradesh)    | <b>Baba Ramchandra</b>          | <p>* <b>Cause:</b> illegal rent collection (nazrana) by taluqdars and landlords and enforcement of eviction rules (U.P. special fact).</p> <p>* <b>Result:</b> unprecedented unity was established among the peasants; the <b>Awadh Kisan Sabha</b> was formed.</p>                                                                             |
| <b>Eka Movement (1921-22 CE)</b>      | Barabanki, Sitapur, Hardoi, Bahraich (Uttar Pradesh) | <b>Madari Pasi</b> and Sahdev   | <p>* <b>Cause:</b> against a <b>rent increase of 50% above the recorded rate</b> and forcible tax collection by thekedars (contractors) (U.P. special fact).</p> <p>* <b>Result:</b> a militant movement of small peasants and sharecroppers; later quelled by repression.</p>                                                                  |
| <b>Bardoli Satyagraha (1928 CE)</b>   | Bardoli taluka of Surat (Gujarat)                    | <b>Sardar Vallabhbhai Patel</b> | <p>* <b>Cause:</b> a sudden <b>22% increase in rent</b> imposed on the peasants by the Bombay government.</p> <p>* <b>Result:</b> the movement was completely successful; the rent increase was reduced to 6.03%. Vallabhbhai received the title "<b>Sardar</b>" on behalf of the women of Bardoli.</p>                                         |
| <b>Tebhaga Movement (1946 CE)</b>     | Bengal province                                      | Kamparam, Bhawan Singh          | <p>* <b>Cause:</b> on the basis of the recommendation of the provincial Flood Commission, the sharecroppers decided that instead of giving 50% of the crop to the landlords they would give only <b>one-third (1/3 or 33%)</b>.</p> <p>* <b>Result:</b> major social and legal improvements in the condition of the sharecropping peasants.</p> |

## 1 First and Second Phases of the Revolutionary Movement (Important Events and Coded Facts)

| Revolutionary Event / Organisation   | Exact Year / Date | Principal Mastermind / Leader          | Special Details Useful for the Exam (UPPCS Coded Facts)                                              |
|--------------------------------------|-------------------|----------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Mitra Mela and Abhinav Bharat</b> | 1899 CE / 1904 CE | <b>Vinayak Damodar (V.D.) Savarkar</b> | The 'Mitra Mela' founded in 1899 itself became the secret society ' <b>Abhinav Bharat</b> ' in 1904. |
| <b>Anushilan Samiti (Bengal)</b>     | 1902 CE           | <b>Pramatha Nath (P.) Mitra</b>        | Formed to spread revolutionary nationalism in Bengal.                                                |
|                                      |                   |                                        |                                                                                                      |

| Revolutionary Event / Organisation                | Exact Year / Date | Principal Mastermind / Leader                         | Special Details Useful for the Exam (UPPCS Coded Facts)                                                                                                                                           |
|---------------------------------------------------|-------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muzaffarpur bomb case                             | 30 April 1908     | Khudiram Bose and Prafulla Chaki                      | * Threw a bomb at the carriage of Judge Kingsford (he escaped).<br>* Prafulla Chaki <b>committed suicide</b> to avoid arrest.<br>* Khudiram Bose was <b>hanged</b> in 1908 at the age of only 15. |
| Delhi Hardinge bomb case                          | 23 December 1912  | Rash Behari Bose and Sachin Sanyal                    | A bomb was thrown at the procession of Viceroy Lord Hardinge II at Chandni Chowk.                                                                                                                 |
| Hindustan Republican Association (HRA)            | October 1924      | Sachindra Sanyal, Ramprasad Bismil, Jogesh Chatterjee | Place of founding: <b>Kanpur</b> (Uttar Pradesh).                                                                                                                                                 |
| Kakori Train Action ( <i>Kakori incident</i> )    | 9 August 1925     | Revolutionaries of the HRA                            | * The <b>8 Down train</b> was looted near Lucknow for the government treasury.<br>* 29 revolutionaries were tried; Chandrashekhar Azad remained absconding.                                       |
| Hindustan Socialist Republican Association (HSRA) | 1928 CE           | Chandrashekhar Azad and Bhagat Singh                  | Place of founding: <b>Feroz Shah Kotla ground, Delhi</b> (new form of the HRA).                                                                                                                   |
| Saunders murder (Lahore)                          | 17 December 1928  | Bhagat Singh, Chandrashekhar Azad, Rajguru            | To avenge the death of Lala Lajpat Rai caused by a lathi-charge.                                                                                                                                  |
| Central Assembly bomb case                        | 8 April 1929      | Bhagat Singh and Batukeshwar Dutt                     | Threw bombs on the empty benches of the Central Legislature in Delhi "to make the deaf hear".                                                                                                     |
| Alfred Park encounter                             | 27 February 1931  | Chandrashekhar Azad                                   | While fighting the police in Alfred Park, Allahabad (Prayagraj), he shot himself with the last bullet.                                                                                            |
| Lahore jail hanging                               | 23 March 1931     | Bhagat Singh, Rajguru and Sukhdev                     | Hanged after being found guilty in the Saunders murder (Lahore Conspiracy Case).                                                                                                                  |

### ➤ Places of Hanging of the Martyrs of the Kakori Train Action (1925) (Match the Following):

- Ram Prasad Bismil → Gorakhpur Jail
- Rajendra Lahiri → Gonda Jail
- Roshan Lal → Naini Jail (Allahabad)
- Ashfaqulla Khan → Faizabad Jail (Ayodhya)

## > The Jallianwala Bagh Massacre (13 April 1919) and the Cycle of Repression

| Stage of Events                   | Date / Time          | Principal Character / Officer                       | Historical Details and Exam Facts                                                                                                                   |
|-----------------------------------|----------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Passing of the Rowlatt Act        | 18 March 1919        | —                                                   | A repressive 'black law' of no vakil (lawyer), no appeal, no daleel (argument).                                                                     |
| Arrest of popular leaders         | 9 April 1919         | Dr. Satyapal and Dr. Saifuddin Kitchlew             | Arrested on the orders of the Governor of Punjab, <b>Michael O'Dwyer</b> ; great public anger.                                                      |
| Handing over of military command  | 10 April 1919        | <b>Brigadier-General R. Dyer</b>                    | The administration and security of the city of Amritsar were handed over to the military officer R. Dyer.                                           |
| Jallianwala Bagh massacre         | <b>13 April 1919</b> | General R. Dyer                                     | <b>On the day of Baisakhi</b> the British army fired indiscriminately on a huge peaceful gathering.                                                 |
| Renunciation of title / post      | After the massacre   | <b>Rabindranath Tagore</b> and <b>Sankaran Nair</b> | * Rabindranath Tagore renounced the title of ' <b>Knighthood</b> ' (Sir).<br>* Sankaran Nair resigned from the <b>Viceroy's Executive Council</b> . |
| Formation of an inquiry committee | After the massacre   | <b>Pandit Madan Mohan Malaviya</b>                  | The Congress formed a ' <b>Tahqiqat Committee</b> ' (inquiry committee) to investigate this heinous massacre.                                       |

## 3 The Non-Cooperation Movement (1920-22) and the Chauri Chaura Incident

| Stage of the Event            | Date / Period        | Principal Mastermind / Character   | Important Exam Facts Related to the Movement                                                                                                                                                                          |
|-------------------------------|----------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special session at Calcutta   | September 1920       | <b>Lala Lajpat Rai</b> (president) | Initial approval was given for launching the Non-Cooperation Movement for the attainment of Swaraj.                                                                                                                   |
| Beginning of the movement     | <b>1 August 1920</b> | <b>Mahatma Gandhi</b>              | The movement began on this day; unfortunately, <b>Bal Gangadhar Tilak died</b> on the same day.                                                                                                                       |
| Nagpur annual session         | December 1920        | <b>C.R. Das</b> (mover)            | The resolution on the Non-Cooperation Movement was finally passed; C.R. Das was its mover.                                                                                                                            |
| Formation of a financial fund | 1921 CE              | —                                  | The ' <b>Tilak Swaraj Fund</b> ' was created for the financial support of the movement.                                                                                                                               |
| Renunciation of titles        | During the movement  | Mahatma Gandhi and Jamnalal Bajaj  | * Gandhiji renounced the title ' <b>Kaiser-i-Hind</b> '.<br>* Jamnalal Bajaj renounced the title ' <b>Rai Bahadur</b> '.<br>* <b>Note: Rabindranath Tagore called the burning of foreign cloth 'heartless waste'.</b> |

| Stage of the Event                 | Date / Period       | Principal Mastermind / Character  | Important Exam Facts Related to the Movement                                                                               |
|------------------------------------|---------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Chauri Chaura incident (Gorakhpur) | 4 - 5 February 1922 | Militant peasant crowd and police | Because the police fired on the demonstrators, the crowd set fire to the police station; <b>22 policemen were killed</b> . |
| Suspension of the movement         | 12 February 1922    | Mahatma Gandhi                    | Saddened by the violent incident of Chauri Chaura, Gandhiji <b>withdrew</b> the Non-Cooperation Movement.                  |

#### 4 Formation of the Swaraj Party (March 1923 CE)

| Main Point         | Details and Exam Facts                                                                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founders and posts | <b>C.R. Das</b> (President) and <b>Motilal Nehru</b> (General Secretary).                                                                                       |
| Place of founding  | <b>Allahabad</b> (Prayagraj), Uttar Pradesh.                                                                                                                    |
| Main objective     | To oppose the policies of the British government from within by contesting the elections to the councils / legislatures.                                        |
| Historic success   | Through the party's efforts, <b>Vithalbhai Patel was elected in 1925 CE</b> as the <b>first Indian President (Speaker)</b> of the Central Legislative Assembly. |

#### 5 Mahatma Gandhi: The Historic Dandi March (Civil Disobedience Movement)

| Main Point / Feature            | Statistical and Historical Details (UPPCS Facts)                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| Date of commencement            | <b>12 March 1930</b>                                                                               |
| Place of commencement           | <b>Sabarmati Ashram</b> , Ahmedabad (Gujarat)                                                      |
| Number of followers             | A total of <b>78 chosen followers</b> accompanied Gandhiji.                                        |
| Date and place of arrival       | Reached <b>Dandi</b> (Navsari district, Gujarat) on <b>5 April 1930</b> .                          |
| Date of breaking the salt law   | On <b>6 April 1930</b> he began the Civil Disobedience Movement by making salt on the seashore.    |
| Total distance and time         | Covered a total distance of <b>390 kilometres</b> (about 241 miles) in a march of <b>24 days</b> . |
| Accompanying foreign journalist | The American journalist <b>Webb Miller</b> was with Gandhiji on this historic march.               |

## 6 The Three Historic Round Table Conferences

### > Exact Period of the Conferences:

- **First Round Table Conference:** from November 1930 to January 1931. *(Congress absent)*
- **Second Round Table Conference:** from 7 September 1931 to 1 December 1931. *(Congress present)*
- **Third Round Table Conference:** from 17 November 1932 to 24 December 1932. *(boycotted by the Congress)*

### > Special Coded Facts Related to the Second Round Table Conference:

- **British PM and Viceroy:** At this time the Prime Minister of Britain was **Ramsay MacDonald** and the Viceroy of India was **Lord Willingdon**.
- **Gandhiji's journey:** As the sole representative of the Congress, Gandhiji went to London on 29 August 1931 by the ship '**S.S. Rajputana**'.
- **Other prominent Indian personalities who attended:** Mahadev Desai, Madan Mohan Malaviya, Devdas Gandhi, Ghanshyam Das Birla, Mirabehn, Annie Besant and Sarojini Naidu.
- **Participant in all three conferences:** **Dr. Bhimrao Ambedkar** was the only prominent leader who took part in all three Round Table Conferences and raised the demand for a separate electorate for the Dalits.

## 7 The Communal Award and the Historic Poona Pact (1932 CE)

| Event / Stage                                 | Exact Date               | Principal Mastermind / Signatory                              | Historical Details and Exam Facts                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|--------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Communal Award</b>                         | <b>16 August 1932</b>    | Ramsay MacDonald (British PM)                                 | Treating the depressed classes as separate from the Hindus, a <b>separate electorate</b> was announced for them.                                                                                                                                                                                                                                           |
| <b>Gandhiji's fast</b>                        | 20 September 1932        | Mahatma Gandhi                                                | In protest against this divisive decision Gandhiji began a fast unto death in <b>Yerwada Jail (Poona)</b> .                                                                                                                                                                                                                                                |
| <b>Mediation</b>                              | September 1932           | Madan Mohan Malaviya, Dr. Rajendra Prasad, C. Rajagopalachari | These leaders played the main role in bringing about an agreement between Gandhiji and Dr. Ambedkar.                                                                                                                                                                                                                                                       |
| <b>Poona Pact</b><br><i>(Poona Agreement)</i> | <b>24 September 1932</b> | <b>Dr. B.R. Ambedkar</b> and <b>Madan Mohan Malaviya</b>      | <p>* <b>Signatories:</b> Dr. Ambedkar signed on behalf of the depressed classes and Madan Mohan Malaviya on behalf of the Hindus.</p> <p>* <b>Special:</b> the <b>final draft of the agreement was prepared by C. Rajagopalachari</b>. The separate electorate for the Dalits was abolished and the seats in the provincial assemblies were increased.</p> |

## 8 Formation and Development of the Azad Hind Fauj (Indian National Army)

| Stage of Development            | Exact Date / Year  | Principal Mastermind and Commander                     | Organisational Facts and Features                                                                                                           |
|---------------------------------|--------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Original idea                   | —                  | Giani Pritam Singh and Fujiwara                        | The original idea of forming the Azad Hind Fauj first came to their minds.                                                                  |
| Initial formation               | March 1942         | <b>Captain Mohan Singh</b>                             | Formation of the ' <b>Indian National Army</b> ' began in Malaya (Singapore island) under Mohan Singh's military leadership.                |
| Formation of the first division | 1 September 1942   | Captain Mohan Singh                                    | The first division of the army was formally formed; Mohan Singh became its <b>first commander</b> .                                         |
| Transfer of command             | <b>4 July 1943</b> | <b>Rash Behari Bose</b> and <b>Subhas Chandra Bose</b> | Rash Behari Bose officially handed over the supreme command of the Azad Hind Fauj in Tokyo/Singapore to <b>Netaji Subhas Chandra Bose</b> . |

- *Note: Among post-independence developments, the non-violent "**Bhoodan Movement**" for land reform in the country was successfully led by **Acharya Vinoba Bhave** in 1951 (Pochampalli).*

## 1 Important Events and Coded Facts of the Freedom Movement

| Historical Event / Mission                                 | Exact Date / Period      | Principal Leader / President   | Important UPPCS Exam Facts (Statement Facts)                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------|--------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chittagong Armoury Raid                                    | <b>18 April 1930</b>     | <b>Surya Sen ('Master Da')</b> | The Chittagong branch of the Indian Republican Army stormed and captured the armoury. <b>Pritilata Waddedar and Kalpana Dutt</b> were also involved. Master Da was hanged on 12 January 1934.                                                                                                                    |
| Arrival of the Simon Commission                            | 3 February 1928          | Sir John Simon                 | Reached Bombay; there was intense protest throughout the country with slogans of "Simon Go Back".                                                                                                                                                                                                                |
| Pakistan Resolution                                        | 22-24 March 1940         | <b>Mohammad Ali Jinnah</b>     | The 'Pakistan Resolution' was officially passed at the <b>Lahore annual session</b> of the Muslim League.                                                                                                                                                                                                        |
| August Offer                                               | <b>8 August 1940</b>     | <b>Lord Linlithgow</b>         | A declaration made by the Viceroy to obtain the cooperation of Indians in the Second World War.                                                                                                                                                                                                                  |
| Individual Satyagraha<br>( <i>Delhi Chalo Satyagraha</i> ) | Began on 17 October 1940 | Mahatma Gandhi (initiator)     | <ul style="list-style-type: none"> <li>* Started by the Congress after the complete rejection of the August Offer.</li> <li>* <b>First satyagrahi: Acharya Vinoba Bhave</b></li> <li>* <b>Second satyagrahi: Pandit Jawaharlal Nehru</b></li> <li>* <b>Third satyagrahi: Sardar Vallabhbhai Patel</b></li> </ul> |

| Historical Event / Mission | Exact Date / Period | Principal Leader / President | Important UPPCS Exam Facts (Statement Facts)                                                                                                |
|----------------------------|---------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Cripps Mission             | March 1942          | Sir Stafford Cripps          | A one-member mission led by a Labour leader/Member of Parliament, sent by the British PM Winston Churchill to break the political deadlock. |

## 2 The Quit India Movement (1942) and Parallel Governments

### > Sequence of Events of the Movement (Chronology):

- **7 August 1942:** The annual meeting of the Congress was held at the historic **Gowalia Tank Maidan** in Bombay under the presidency of **Abul Kalam Azad**.
- **Moving of the resolution:** At this meeting **Jawaharlal Nehru** moved the historic resolution of the "Quit India Movement".
- **Drafters:** The draft of the movement was prepared jointly by **Mahatma Gandhi, Jawaharlal Nehru and Abul Kalam Azad**.
- **8 August 1942:** The Quit India resolution was passed by a full majority; at this time Gandhiji gave the historic slogan "**Do or Die**".
- **9 August 1942 (morning):** Under "**Operation Zero Hour**" the British government arrested Gandhiji and all the other top leaders.
- **Accompanying journalist:** During the movement the American journalist **Louis Fischer** (biographer of Mahatma Gandhi) was with him.

### > Principal Parallel Governments Established during the Movement (Match the Following):

(UPPCS Prelims asks a direct matching question from here)

| Place / Region             | Leader                              | Duration and Special Facts                                                                             |
|----------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------|
| Ballia (Uttar Pradesh)     | Chittu Pande                        | The very first parallel government (U.P. special fact).                                                |
| Tamluk (Midnapore, Bengal) | Local leader Sati Samanta and party | It was known as the " <b>Jatiya Sarkar</b> ". It remained active from December 1942 to September 1944. |
| Satara (Maharashtra)       | Y.V. Chavan and Nana Patil          | It was the <b>longest-running government</b> , functioning until 1945.                                 |

## 3 First Phase of the Revolutionary Movement in India (1897 – 1915 CE)

(An infallible table for UPPCS Prelims 'Match the Following' and for fixing the correct year of events)



| Year (CE) | Revolutionary Event / Organisation        | Principal Revolutionaries                                  | Special Details Useful for the Exam (Coded Facts)                                                                                                                                    |
|-----------|-------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1897      | <b>Murder of Plague Commissioner Rand</b> | <b>Chapekar brothers</b> ( <i>Damodar and Balkrishna</i> ) | Plague Commissioner W.C. Rand was shot dead at Poona (Maharashtra). Beginning of the first phase in India.                                                                           |
| 1899      | <b>Mitra Mela</b>                         | Vinayak Damodar (V.D.) Savarkar                            | Founded in Maharashtra as a secret revolutionary organisation.                                                                                                                       |
| 1902      | <b>Anushilan Samiti</b>                   | Pramathanath (P.) Mitra                                    | The first secret organisation for conducting revolutionary activities in Bengal.                                                                                                     |
| 1904      | <b>Abhinav Bharat</b>                     | V.D. Savarkar                                              | The reorganised name of the 'Mitra Mela' of 1899 (Poona).                                                                                                                            |
| 1906      | Yugantar organisation and paper           | Barindra Kumar Ghosh and associates                        | Developed to spread revolutionary ideas after the Partition of Bengal (1905).                                                                                                        |
| 1908      | <b>Muzaffarpur bomb case</b>              | <b>Khudiram Bose</b> and <b>Prafulla Chaki</b>             | * Attempt to kill Judge Kingsford.<br>* Prafulla Chaki <b>committed suicide</b> to avoid arrest.<br>* Khudiram Bose was <b>hanged</b> at the youngest age on <b>11 August 1908</b> . |
| 1908      | <b>Alipore bomb case</b>                  | Barindra Ghosh and Aurobindo Ghosh                         | Many revolutionaries were tried after a bomb factory was caught at Maniktola in Kolkata.                                                                                             |
| 1909      | <b>Murder of Curzon Wylie</b>             | <b>Madan Lal Dhingra</b>                                   | Sir Curzon Wylie, political adviser to the Secretary of State for India, was shot dead in London.                                                                                    |
| 1909      | <b>Jackson murder case</b> (Nasik)        | <b>Anant Laxman Kanhere</b>                                | A.M.T. Jackson, Collector of Nasik, was shot dead.                                                                                                                                   |
| 1912      | <b>Delhi Hardinge bomb case</b>           | <b>Rash Behari Bose</b> and Sachin Sanyal                  | On <b>23 December 1912</b> a bomb was thrown at Chandni Chowk on the arrival of Viceroy Lord Hardinge in Delhi.                                                                      |
| 1913      | <b>Ghadar Movement (Ghadar Party)</b>     | <b>Lala Har Dayal</b> and Sohan Singh Bhakna               | Organisation of the Ghadar movement in America (San Francisco) and Canada.                                                                                                           |
| 1914      | <b>Komagata Maru incident</b>             | Baba Gurdit Singh                                          | Revolutionary nationalism flared up abroad because the Canadian government stopped the ship of Indian emigrants.                                                                     |
| 1915      | Failure of the armed revolt               | Rash Behari Bose and the Ghadar revolutionaries            | The <b>first phase ended</b> with the failure of the Ghadar armed revolt planned during the First World War.                                                                         |

## 4 Second Phase of the Revolutionary Movement (1920 – 1934 CE)

(Complete chronological summary of the Lahore Conspiracy, the Kakori Action and the Assembly bomb case)

| Revolutionary Organisation / Event                | Exact Date / Year | Principal Masterminds / Martyred Revolutionaries                           | Socialist / Revolutionary Philosophy and UPPCS Exam Facts                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|-------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hindustan Republican Association (HRA)            | October 1924      | Sachindranath Sanyal, Ramprasad Bismil, Jogesh Chandra Chatterjee          | <p>* <b>Place:</b> Kanpur (U.P.).</p> <p>* <b>Objective:</b> to end British rule through armed revolution and establish a <b>federal republic</b> in India.</p>                                                                                                                                                                                                                                                                                                                                                 |
| Kakori Train Action                               | 9 August 1925     | Revolutionaries of the HRA (Bismil, Ashfaq, Roshan, Lahiri, Azad)          | <p>* <b>Place:</b> Kakori near Lucknow. The 8 Down train was stopped and the government treasury was looted.</p> <p>* <b>Dates of hanging (1927):</b></p> <ol style="list-style-type: none"> <li>1. <b>17 December:</b> Rajendranath Lahiri (<b>Gonda Jail</b>)</li> <li>2. <b>19 December:</b> Ramprasad Bismil (<b>Gorakhpur</b>), Ashfaqulla Khan (<b>Faizabad</b>), Roshan Singh (<b>Naini/Prayagraj</b>).</li> </ol> <p>* <b>Note:</b> Chandrashekhar Azad remained absconding. Heavy loss to the HRA.</p> |
| Hindustan Socialist Republican Association (HSRA) | 1928 CE           | Chandrashekhar Azad, Bhagat Singh, Sukhdev, Rajguru, Bhagwati Charan Vohra | <p>* <b>Place:</b> Feroz Shah Kotla ground, Delhi (new form of the HRA).</p> <p>* <b>Philosophy:</b> included <b>socialist ideas</b> and the establishment of an exploitation-free society among its goals.</p>                                                                                                                                                                                                                                                                                                 |
| Lahore Saunders murder                            | 17 December 1928  | Bhagat Singh, Rajguru and Chandrashekhar Azad                              | Killed Assistant Superintendent of Police J.P. Saunders at Lahore to avenge the <b>death of Lala Lajpat Rai (17 November 1928)</b> , who had been injured in a lathi-charge during the anti-Simon protest on 30 October 1928.                                                                                                                                                                                                                                                                                   |
| Central Legislative Assembly bomb case            | 8 April 1929      | Bhagat Singh and Batukeshwar Dutt                                          | <p>* <b>Place:</b> Central Assembly, Delhi.</p> <p>* <b>Objective:</b> not to kill anyone, but <i>'to make the deaf government hear'</i> against the repressive Trade Disputes Bill and Public Safety Bill. Both deliberately courted arrest.</p>                                                                                                                                                                                                                                                               |
| Ultimate martyrdom (final phase)                  | 1931 CE           | Chandrashekhar Azad, Bhagat Singh, Sukhdev, Rajguru                        | <p>* <b>27 February 1931:</b> during a police encounter in <b>Alfred Park</b>, Allahabad, Chandrashekhar Azad shot himself.</p> <p>* <b>23 March 1931:</b> Bhagat Singh, Sukhdev and Rajguru were <b>hanged</b> under the <b>Lahore Conspiracy Case</b> (murder of Saunders).</p>                                                                                                                                                                                                                               |

## 1 Sequence of Events of the Transfer of Power and Independence (1945 – 1947 CE)

(UPPCS Prelims asks every year from this historical chronology for arranging events directly by date)

| Historical Event / Mission                | Exact Date / Period | Principal Mastermind / Viceroy                                                           | Important Exam Facts (Code-based Facts)                                                                                                                                                                                                                               |
|-------------------------------------------|---------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wavell Plan                               | 14 June 1945        | Lord Wavell                                                                              | A plan presented by the Conservative government of British PM Winston Churchill to resolve the constitutional deadlock.                                                                                                                                               |
| Arrival of the Cabinet Mission            | 24 March 1946       | * <b>Chairman:</b> Pethick-Lawrence<br>* <b>Members:</b> A.V. Alexander, Stafford Cripps | A three-member mission constituted by British PM Clement Attlee. Under it the <b>elections to the Constituent Assembly were held in July 1946</b> . ( <i>Note: the idea of a Constituent Assembly first came up at the Lucknow session of the Congress in 1936</i> ). |
| Invitation to form the Interim Government | 12 August 1946      | Viceroy and Jawaharlal Nehru                                                             | The Viceroy sent an invitation to Nehru to form the government, which he accepted. The Muslim League strongly opposed it.                                                                                                                                             |
| Direct Action Day                         | 16 August 1946      | Muslim League                                                                            | At the League's call terrible communal bloodshed took place in the country (especially in Bengal). Mahatma Gandhi undertook a padayatra at <b>Noakhali</b> to restore peace.                                                                                          |
| Formation of the Interim Government       | 2 September 1946    | Pt. Jawaharlal Nehru and other members                                                   | Under Nehru's leadership the cabinet ministers took the oath of office and secrecy (the list of portfolios is given below).                                                                                                                                           |
| Attlee's historic declaration             | 20 February 1947    | Clement Attlee (British PM)                                                              | Declared in the House of Commons that the English would transfer power to Indians <b>by June 1948</b> . <b>Mountbatten</b> was appointed the new Viceroy in place of Wavell.                                                                                          |
| Lord Mountbatten Plan                     | 3 June 1947         | Lord Mountbatten ( <i>34th Governor-General</i> )                                        | After mutual agreement among Indian leaders, the British government presented the <b>final plan for the partition of India</b> .                                                                                                                                      |
| Introduction of the Independence Bill     | 4 July 1947         | British Parliament                                                                       | The Indian Independence Bill was introduced in the British Parliament.                                                                                                                                                                                                |
| Royal assent to the Independence Bill     | 18 July 1947        | British sovereign                                                                        | The Bill received royal assent. According to it, two dominions named India and Pakistan were to be established on 15 August.                                                                                                                                          |
| Attainment of independence                | 14 - 15 August 1947 | —                                                                                        | Pakistan on <b>14 August</b> and India on <b>15 August</b> became independent as sovereign nations.                                                                                                                                                                   |

## 2 The Interim Government (1946 CE): At a Glance

(The most important and indispensable table for UPPCS Prelims 'Match the Following')

| Name of the Cabinet Minister | Administrative Department / Ministry Allotted                                   |
|------------------------------|---------------------------------------------------------------------------------|
| Pt. Jawaharlal Nehru         | Vice-President (Executive Council), External Affairs and Commonwealth Relations |
| Sardar Vallabhbhai Patel     | Home, Information and Broadcasting, and matters relating to the princely states |
| Dr. Rajendra Prasad          | Minister of Food and Agriculture                                                |
| Baldev Singh                 | Minister of Defence                                                             |
| Liaquat Ali Khan             | Minister of Finance ( <i>from the Muslim League's quota</i> )                   |
| Jogendra Nath Mandal         | Minister of Law ( <i>from the Muslim League's quota</i> )                       |
| Aruna Asaf Ali               | Minister of Railways                                                            |
| C. Rajagopalachari           | Minister of Education                                                           |
| Jagjivan Ram                 | Minister of Labour                                                              |
| John Mathai                  | Minister of Industries and Supplies                                             |
| I.I. Chundrigar              | Minister of Commerce                                                            |
| Abdur Rab Nishtar            | Minister of Communications                                                      |
| Ghazanfar Ali Khan           | Minister of Health                                                              |
| C.H. Bhabha                  | Minister of Works, Mines and Ports                                              |

### 3 Netaji Subhas Chandra Bose: Brief Introduction and Titles

| Subject Section                                | Historical Details and Coded Exam Facts                                                                                                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Birth and early life                           | Born on <b>23 January 1897</b> at Cuttack (Odisha). In 1920 he secured <b>fourth rank</b> in the Indian Civil Service (ICS), but resigned to serve the country.                           |
| Political guru and beginning                   | Joined the Congress in 1921 inspired by Gandhiji. His political guru was <b>Deshbandhu Chittaranjan Das (C.R. Das)</b> . In 1927 he formed the 'Independence League' together with Nehru. |
| Resignation from the Congress and organisation | <b>In 1939 (after the Tripuri crisis)</b> he resigned from the presidency of the Congress and formed a new organisation named the <b>Forward Bloc</b> .                                   |
| Title: 'Netaji'                                | He was first addressed by the name 'Netaji' by <b>the people of Germany/Hitler</b> .                                                                                                      |
| Title: 'Desh Nayak'                            | <b>Rabindranath Tagore</b> honoured him with the title 'Desh Nayak'.                                                                                                                      |

| Subject Section              | Historical Details and Coded Exam Facts                                     |
|------------------------------|-----------------------------------------------------------------------------|
| Title: 'Patriot of patriots' | <b>Mahatma Gandhi</b> called Subhas Chandra Bose 'the patriot of patriots'. |

## 4 Principal National Newspapers, Books of Modern India and their Authors

*(An infallible integrated compilation of all 41 literary sources given by you - the best reference for one-hundred-per-cent matching in the exam)*

| Writer / National Leader / Historian | Their Principal Books / Newspapers / Journals                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Mahatma Gandhi</b>                | Young India, Hind Swaraj, Gokhale: My Political Guru, My Experiments with Truth (autobiography)              |
| <b>Dr. Rajendra Prasad</b>           | India Divided <i>(Note: with partial correction of the source list, India Divided is his principal work)</i> |
| <b>Jawaharlal Nehru</b>              | Glimpses of World History, The Discovery of India                                                            |
| <b>Lala Lajpat Rai</b>               | The Story of My Deportation, Unhappy India                                                                   |
| <b>Bal Gangadhar Tilak</b>           | Gita Rahasya, Kesari (Marathi), Mahratta (English)                                                           |
| <b>Abul Kalam Azad</b>               | India Wins Freedom, Al-Hilal                                                                                 |
| <b>Annie Besant</b>                  | Commonweal, New India                                                                                        |
| <b>Aurobindo Ghosh</b>               | The Life Divine, Karmayogin, Essays on the Gita                                                              |
| <b>Bankim Chandra Chatterjee</b>     | Anandamath (based on the Sanyasi Revolt), Durgeshnandini                                                     |
| <b>Raja Rammohan Roy</b>             | The Precepts of Jesus, Sambad Kaumudi, Mirat-ul-Akhbar                                                       |
| <b>B.R. Ambedkar</b>                 | Annihilation of Caste, Mook Nayak, Bahishkrit Bharat                                                         |
| <b>Bhagat Singh</b>                  | Why I am an Atheist, An Introduction to the Dreamland                                                        |
| <b>Sachindra Nath Sanyal</b>         | Bandi Jeevan <i>(the textbook of the revolutionaries)</i>                                                    |
| <b>Bhagwati Charan Vohra</b>         | The Philosophy of the Bomb                                                                                   |
| <b>Dr. Ram Manohar Lohia</b>         | Guilty Men of India's Partition                                                                              |
| <b>Surendra Nath Banerjee</b>        | Bengalee (newspaper), A Nation in Making                                                                     |
| <b>Subhas Chandra Bose</b>           | The Indian Struggle                                                                                          |
| <b>Sarojini Naidu</b>                | The Broken Wing                                                                                              |
| <b>Jayaprakash Narayan</b>           | Towards Struggle                                                                                             |

| Writer / National Leader / Historian         | Their Principal Books / Newspapers / Journals                     |
|----------------------------------------------|-------------------------------------------------------------------|
| Michael Madhusudan Dutt                      | The Captive Ladie                                                 |
| Manubehn Gandhi                              | Bapu: My Mother                                                   |
| Dr. Rafiq Zakaria                            | The Man Who Divided India                                         |
| Durga Das                                    | India from Curzon to Nehru and After                              |
| Valentine Chirol                             | Indian Unrest ( <i>called Tilak the father of Indian unrest</i> ) |
| W.W. Hunter                                  | Our Indian Musalmans                                              |
| Katherine Mayo                               | Mother India                                                      |
| Louis Fischer ( <i>American journalist</i> ) | The Life of Mahatma Gandhi                                        |
| Bipan Chandra ( <i>historian</i> )           | India's Struggle for Independence                                 |
| Vinayak Damodar (V.D.) Savarkar              | The Indian War of Independence, 1857                              |
| R.C. Majumdar ( <i>historian</i> )           | The Sepoy Mutiny and the Revolt of 1857                           |
| Asoka Mehta                                  | The Great Rebellion: 1857                                         |
| S.B. Chaudhuri                               | Civil Rebellion in the Indian Mutinies (1857-1859)                |
| Rudrangshu Mukherjee                         | Awadh in Revolt: 1857-1858                                        |
| Syed Ahmad Khan                              | Tahzib-ul-Akhlaq (journal), Asbab-i-Baghawat-i-Hind               |
| Hugh Toye                                    | The Springing Tiger (biography of Subhas Chandra Bose)            |
| Dilip Mukherjee                              | Terrorist                                                         |

- *Note: The 'Amrita Bazar Patrika' was originally edited by Sisir Kumar Ghosh and Motilal Ghosh, and according to the source list the name 'Moti Lal' is associated with it.*

02

# World Geography

WORLD GEOGRAPHY

## > Principal Geographers and their Theories/Concepts

| Name of the Theory / Concept | Name of the Propounder / Geographer (Direct Match Standard) |
|------------------------------|-------------------------------------------------------------|
| Continental Drift Theory     | Alfred Wegener                                              |
| Theory of Plate Tectonics    | Parker and Morgan (and Harry Hess)                          |
| Concept of Isostasy          | Pratt and Airy                                              |
| Convection Current Theory    | Arthur Holmes                                               |

## > Principal Theories Related to the Origin of the Earth (Cosmogony)

| Name of the Theory / Hypothesis | Name of the Propounder / Scientist (Direct Match Standard) |
|---------------------------------|------------------------------------------------------------|
| Gaseous Mass Hypothesis         | Immanuel Kant                                              |
| Nebular Hypothesis              | Pierre Laplace                                             |
| Planetesimal Hypothesis         | Chamberlin and Moulton                                     |
| Tidal Hypothesis                | James Jeans and Jeffreys                                   |
| Binary Star Hypothesis          | Henry Russell                                              |
| Big Bang Theory                 | Georges Lemaître                                           |

## > Principal Theories Related to Mountain Building and their Propounders

| Name of the Theory / Hypothesis                      | Name of the Propounder / Scholar (Direct Match) |
|------------------------------------------------------|-------------------------------------------------|
| Geosynclinal (landslide) theory of mountain building | Kober                                           |
| Thermal Contraction Theory                           | Jeffreys                                        |
| Continental Displacement Theory                      | Alfred Wegener                                  |
| Convection Current Theory                            | Arthur Holmes                                   |
| Plate Tectonic Theory                                | Morgan, Harry Hess, McKenzie, Parker            |
| Radioactivity Theory                                 | Joly                                            |



| Name of the Theory / Hypothesis | Name of the Propounder / Scholar (Direct Match) |
|---------------------------------|-------------------------------------------------|
| Continental Sliding Theory      | Daly                                            |

## > Major Tectonic Plates of the Earth

| Name of the Major Plate | Geographical Parts / Regions Included                                        |
|-------------------------|------------------------------------------------------------------------------|
| Eurasian Plate          | Europe + Asia + part of the eastern Atlantic Ocean.                          |
| Indo-Australian Plate   | India + Australia + Indian Ocean.                                            |
| African Plate           | African continent + eastern Atlantic Ocean + western Indian Ocean.           |
| Pacific Plate           | The entire Pacific Ocean floor (it is mainly an oceanic plate).              |
| North American Plate    | North-western Atlantic Ocean + North American continent.                     |
| South American Plate    | South American continent + south-western Atlantic Ocean.                     |
| Antarctic Plate         | Antarctic continent + the oceanic plate surrounding Antarctica on all sides. |

## > Internal Structure of the Earth: Discontinuities

| Name of the Discontinuity | Between which layers is it located? (correct order from top to bottom)            |
|---------------------------|-----------------------------------------------------------------------------------|
| 1. Conrad                 | • Between the <b>upper crust</b> and the <b>lower crust</b> .                     |
| 2. Mohorovičić / Moho     | • Between the <b>lower crust</b> and the <b>upper mantle</b> .                    |
| 3. Repetti                | • Between the <b>upper mantle</b> and the <b>lower mantle</b> .                   |
| 4. Gutenberg              | • Between the <b>lower mantle</b> and the <b>outer core</b> .                     |
| 5. Lehmann                | • Between the <b>outer core</b> and the <b>inner core</b> ( <i>the deepest</i> ). |

## > Geological Time Scale: Details of the Principal Eras

| Era            | Principal Geological Events and Development of Life (Core Pointers)                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Precambrian | <ul style="list-style-type: none"> <li>• After the formation of the oceans, the <b>first life was formed</b>.</li> <li>• In India the <b>Aravalli mountains</b> and the <b>rocks of the Dharwar system</b> were formed at this time.</li> </ul> |
| 2. Palaeozoic  | <ul style="list-style-type: none"> <li>• The period of primary life; <b>marine life</b> developed abundantly and some reptiles were born on land.</li> </ul>                                                                                    |
| 3. Mesozoic    | <ul style="list-style-type: none"> <li>• It is called the period of 'middle life'. It was mainly the <b>age of the dinosaurs</b>.</li> <li>• It includes the Triassic, Jurassic and Cretaceous periods.</li> </ul>                              |

| Era         | Principal Geological Events and Development of Life (Core Pointers)                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Cenozoic | <ul style="list-style-type: none"> <li>This is the <b>present era</b>, called the '<b>Age of Mammals</b>'.</li> <li>During it new fold mountain ranges such as the <b>Rockies</b> of North America and the <b>Andes</b> of South America were formed.</li> </ul> |

### > The Four Main Positions of the Earth's Revolution

| Position / Event           | Exact Date / Time | Position of the Sun (Vertical Latitude) | Condition of the Hemispheres and Features (UPPCS Pointers)                                                                                                                                                                                                                               |
|----------------------------|-------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Summer Solstice         | 20-21 June        | Tropic of Cancer (23½° N latitude)      | <ul style="list-style-type: none"> <li><b>Longest day in the Northern Hemisphere</b> and the shortest night (it is summer here).</li> <li>Winter in the Southern Hemisphere; <b>continuous day for 6 months at the North Pole</b>.</li> </ul>                                            |
| 2. Autumnal Equinox        | 22-23 September   | Equator (0° latitude)                   | <ul style="list-style-type: none"> <li>The sun shines equally on both hemispheres. <b>Day and night are equal (12 hours each) all over the world</b>.</li> <li>Autumn north of the Equator and spring to the south.</li> </ul>                                                           |
| 3. Winter Solstice         | 21-22 December    | Tropic of Capricorn (23½° S latitude)   | <ul style="list-style-type: none"> <li>The sun's rays fall vertically on the Tropic of Capricorn. <b>Longest day in the Southern Hemisphere</b> and the shortest night (it is summer here).</li> <li>Winter in the Northern Hemisphere, where days are short and nights long.</li> </ul> |
| 4. Vernal / Spring Equinox | 20-21 March       | Equator (0° latitude)                   | <ul style="list-style-type: none"> <li>The sun shines equally on both hemispheres. <b>Day and night are equal</b>.</li> <li>Spring north of the Equator and autumn to the south.</li> </ul>                                                                                              |

### > Important Facts Related to the Solar System

| Component of the Solar System / Question | Exact Answer and Special Explanation (Core Facts)                                                   |
|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Brightest planet                         | <b>Venus:</b> regarded as the brightest planet seen from the Earth.                                 |
| Earth's twin                             | <b>Venus:</b> because it is similar to the Earth in size and mass.                                  |
| Rotation in the opposite direction       | <b>Venus:</b> unlike the other planets, it rotates from east to west (clockwise).                   |
| Morning / evening star                   | <b>Venus:</b> it is also called the 'morning star' or the 'evening star'.                           |
| Hottest planet                           | <b>Venus:</b> because of its dense CO <sub>2</sub> atmosphere (maximum temperature: <b>462°C</b> ). |
| Planet with the longest day              | <b>Venus:</b> one of its days equals <b>243 Earth days</b> .                                        |
| Planets with no satellites               | <b>Mercury and Venus:</b> they have no natural moon.                                                |

| Component of the Solar System / Question | Exact Answer and Special Explanation (Core Facts)                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------|
| Planet with prominent rings              | <b>Saturn:</b> distinct ring systems are found around it.                                    |
| Planet with the shortest day             | <b>Jupiter:</b> it rotates fastest on its axis (one rotation in <b>9 hours 55 minutes</b> ). |
| Coldest planet                           | <b>Uranus:</b> its effective temperature stays at about <b>-216°C</b> .                      |
| Nearest star to the Earth                | <b>Proxima Centauri:</b> the nearest star to the Earth after the Sun.                        |

### > Layers of the Atmosphere and their Main Features (Atmospheric Layers)

| Name of the Layer | Main Features and Atmospheric Phenomena (Core Pointers)                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Troposphere    | • All kinds of <b>climatic and weather phenomena</b> (clouds, rain, storms) occur in this layer.                                                   |
| 2. Stratosphere   | • The <b>ozone layer</b> is found in it. Being calm in weather, it is ideal for <b>flying aircraft</b> .                                           |
| 3. Mesosphere     | • Greenhouse gases are absent. <b>Meteors / shooting stars</b> burn up in this layer as soon as they enter from space.                             |
| 4. Ionosphere     | • Electrically charged particles are found. <b>Radio waves transmitted from the Earth</b> are reflected back from here.                            |
| 5. Exosphere      | • The outermost layer of the atmosphere. Owing to the weak force of gravity, <b>gases are present in extremely rarefied/small</b> quantities here. |

### > El Niño versus La Niña: Climatic Phenomena (Comparative Table)

| Criterion / Effect           | El Niño                                                                                                                                                                       | La Niña                                                                                                                                                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic concept                | • Sea surface in the central and eastern tropical Pacific Ocean becoming <b>warmer than average</b> .                                                                         | • Sea surface in the central and eastern Pacific Ocean becoming <b>colder than average</b> .                                                                                                                                    |
| Effect on the Indian monsoon | • <b>The Indian monsoon becomes weak</b> (drought conditions arise in India).                                                                                                 | • <b>The Indian monsoon becomes very strong</b> (good rainfall/floods in India).                                                                                                                                                |
| Temperature and rainfall     | <ul style="list-style-type: none"> <li>• Heavy rain/floods in the Horn of Africa and in southern America.</li> <li>• Drought conditions in northern South America.</li> </ul> | <ul style="list-style-type: none"> <li>• Heavy rain and floods in the north-western part of the Pacific Ocean and in Canada.</li> <li>• Drought in southern America; colder than normal in North America and Canada.</li> </ul> |

| Criterion / Effect                 | El Niño                                                                                         | La Niña                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hurricanes</b>                  | • More hurricanes are encouraged in the eastern and central Pacific region.                     | • More severe than average hurricanes in the Atlantic Ocean, while fewer than average in the eastern Pacific.                                               |
| <b>Agriculture and marine life</b> | • A heavy <b>decline/reduction</b> in the number of marine organisms such as fish and plankton. | • Cold-water species (salmon, squid) come to the California coast ( <b>increase in marine life</b> ).<br>• Yields of wheat and soybean may fall in America. |

### › Correct Order of the Continents of the World (by Area and Population)

| Rank | Descending Order of Continents by Area | Descending Order of Continents by Population   |
|------|----------------------------------------|------------------------------------------------|
| 1    | <b>Asia</b> ( <i>largest</i> )         | <b>Asia</b> ( <i>most populous</i> )           |
| 2    | <b>Africa</b>                          | <b>Africa</b>                                  |
| 3    | <b>North America</b>                   | <b>Europe</b>                                  |
| 4    | <b>South America</b>                   | <b>North America</b>                           |
| 5    | <b>Antarctica</b>                      | <b>South America</b>                           |
| 6    | <b>Europe</b>                          | <b>Australia</b>                               |
| 7    | <b>Australia</b> ( <i>smallest</i> )   | <b>Antarctica</b> ( <i>uninhabited/least</i> ) |

### › Countries of the World with the Longest Coastlines (in Descending Order)

| Rank | Name of the Country | UPPCS Special Note (Coastline)                              |
|------|---------------------|-------------------------------------------------------------|
| 1    | <b>Canada</b>       | <i>The country with the longest coastline in the world.</i> |
| 2    | <b>Norway</b>       | —                                                           |
| 3    | <b>Indonesia</b>    | —                                                           |
| 4    | <b>Russia</b>       | —                                                           |
| 5    | <b>Philippines</b>  | —                                                           |

**Position of India:** With a total coastline of **11,098 kilometres** (on the basis of compiled estimates including the island territories), India ranks **13th** in the world.

> Highest Mountain Peaks by Continent (Quick Pointers)

| Name of the Continent | Name of the Highest Peak                                                                                     | Country / Specific Geographical Location                                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Asia                  | Mount Everest                                                                                                | Nepal (the highest mountain of the world and of Asia)                              |
| Europe                | Mount Elbrus                                                                                                 | Russia (the highest mountain of Europe)                                            |
| Africa                | Mount Kilimanjaro                                                                                            | Tanzania (the highest mountain of the African continent)                           |
| North America         | Mount McKinley / Denali                                                                                      | United States of America (USA)                                                     |
| South America         | Ojos del Salado / Aconcagua                                                                                  | Andes range <i>(Note: in the data both names are described as principal peaks)</i> |
| Antarctica            | Vinson Massif                                                                                                | Antarctic continent                                                                |
| Australia             | Mount Kosciuszko                                                                                             | Australian mainland                                                                |
| Other famous peaks    | <ul style="list-style-type: none"><li>• Jura Mountains: France</li><li>• Mount Titlis: Switzerland</li></ul> | —                                                                                  |

> Principal Mountain Ranges of the World and their Highest Points

| Name of the Mountain Range | Geographical Location (Place/Country) | Highest Point / Peak                                               |
|----------------------------|---------------------------------------|--------------------------------------------------------------------|
| Great Dividing Range       | Eastern Australia                     | Kosciuszko                                                         |
| Transantarctic             | Antarctica                            | Mount Vinson Massif                                                |
| Caucasus Mountains         | Russia / Georgia                      | Mount Elbrus <i>(the highest peak of Europe)</i>                   |
| Alaska Range               | USA (Alaska)                          | Mount McKinley (Denali) <i>(the highest peak of North America)</i> |
| Sierra Madre Oriental      | Mexico                                | Orizaba                                                            |
| Atlas Range                | North-west Africa                     | Mount Toubkal                                                      |
| Tien Shan Range            | South-central Asia                    | Pik Pobeda                                                         |
| Altai Mountains            | Central Asia (Russia/Mongolia)        | Gora Belukha                                                       |
| Zagros Mountains           | Iran                                  | Zard Kuh                                                           |
| Elburz Mountains           | Iran                                  | Damavand                                                           |
|                            |                                       |                                                                    |

| Name of the Mountain Range | Geographical Location (Place/Country) | Highest Point / Peak                        |
|----------------------------|---------------------------------------|---------------------------------------------|
| Appalachian Mountains      | Eastern USA and Canada                | Mount Mitchell                              |
| Alps                       | Central Europe                        | Mont Blanc                                  |
| Cascade Range              | USA and Canada                        | Mount Rainier                               |
| Apennine Mountains         | Italy                                 | Corno Grande                                |
| Drakensberg                | South-east Africa                     | The highest range of South Africa           |
| Pyrenees                   | Border of France and Spain            | <i>A fold mountain of the Alpine system</i> |
| Ural / Verkhoyansk         | Central Russia / Eastern Russia       | —                                           |
| Pennines / Cambrian        | England                               | —                                           |

### > Major Plateaus of the World

| Name of the Plateau       | Location / Related Country                              | Name of the Plateau   | Location / Related Country                                  |
|---------------------------|---------------------------------------------------------|-----------------------|-------------------------------------------------------------|
| Pamir or Tibetan Plateau  | Tibet (China) <i>(the largest plateau in the world)</i> | Antarctic Plateau     | Antarctica <i>(the second largest plateau in the world)</i> |
| Djado Plateau             | Niger                                                   | Libyan Plateau        | Libya and Egypt                                             |
| Laurentian Plateau        | Canada                                                  | Mongolian Plateau     | Mongolia and China                                          |
| Taklamakan / Loess        | China                                                   | Anatolia / Asia Minor | Turkey                                                      |
| Malwa / Deccan            | India                                                   | Abyssinian Plateau    | Ethiopia                                                    |
| Chhota Nagpur / Meghalaya | India                                                   | Columbia / Colorado   | United States of America (USA)                              |
| Iranian Plateau           | Iran                                                    | Ozark / Piedmont      | United States of America (USA)                              |
| Ahaggar / Tassili         | Algeria / Libya                                         | Great Basin Plateau   | United States of America (USA)                              |
| Kimberley Plateau         | Australia                                               | Patagonian Plateau    | Argentina                                                   |
| Ubangi Plateau            | Central Africa                                          | Guiana Highlands      | Venezuela, Guyana, Suriname, French Guiana                  |
|                           |                                                         |                       |                                                             |

| Name of the Plateau | Location / Related Country | Name of the Plateau     | Location / Related Country           |
|---------------------|----------------------------|-------------------------|--------------------------------------|
| Bié Plateau         | Angola                     | Mato Grosso / Brazilian | Brazil                               |
| Adamawa Plateau     | Nigeria                    | Bolivian Plateau        | Bolivia (Andes range)                |
| Shan Plateau        | Myanmar                    | Potwar / Khorat         | Pakistan (Potwar), Thailand (Khorat) |
| Meseta Plateau      | Iberian Peninsula (Spain)  | Chiapas Plateau         | Southern Mexico                      |
| Alaska / Yukon      | Canada                     |                         |                                      |

## > Major Deserts of the World

| Name of the Desert              | Geographical Location / Country (UPPCS Matching Special)                                          |
|---------------------------------|---------------------------------------------------------------------------------------------------|
| Sahara Desert                   | Algeria, Chad, Egypt, Libya, Mali, Mauritania, Morocco, Niger, Sudan, Tunisia and Western Sahara. |
| Nubian / Bayuda Desert          | Sudan (northern/north-eastern region)                                                             |
| Namib Desert                    | Angola, Namibia and South Africa                                                                  |
| Kalahari Desert                 | Namibia, Botswana and South Africa                                                                |
| Gobi Desert                     | China and Mongolia ( <i>temperate desert</i> )                                                    |
| Atacama Desert                  | Northern Chile and southern Peru                                                                  |
| Al Nafud / Rub al Khali / Hamad | Saudi Arabia (part of the Arabian Desert)                                                         |
| Dasht-e-Lut / Dasht-e-Kavir     | Iran                                                                                              |
| Lop / Taklamakan                | China                                                                                             |
| Kyzyl Kum                       | Kazakhstan, Uzbekistan, Turkmenistan                                                              |
| Karakum                         | Turkmenistan, Kazakhstan                                                                          |
| Mojave / Red Desert             | United States of America (USA)                                                                    |
| Sonoran                         | USA and Mexico                                                                                    |
| Thar / Kharan                   | India-Pakistan (Thar), Pakistan (Kharan)                                                          |
| Dasht-e-Margo / Libyan          | Afghanistan (Margo), Libya (Libyan)                                                               |



## > Various Seas, Oceans and the Salinity of Water

- **Measurement of salinity:** It is expressed as the amount of salt dissolved per thousand grams of water (‰).

| Sea / Ocean / Lake             | Salinity (‰) | UPPCS Core Exam Fact                                                                    |
|--------------------------------|--------------|-----------------------------------------------------------------------------------------|
| Lake Van of Türkiye            | <b>330‰</b>  | The lake with the <b>highest salinity in the world</b> .                                |
| Dead Sea                       | <b>238‰</b>  | Because of its very high density a person does not sink in it.                          |
| Great Salt Lake (USA)          | <b>220‰</b>  | The most saline lake of North America.                                                  |
| Atlantic Ocean                 | 36‰          | Its average salinity is slightly higher among the oceans.                               |
| Arabian Sea                    | 36‰          | More saline than the Bay of Bengal because of higher evaporation.                       |
| Indian Ocean                   | 35‰          | Equal to the global average.                                                            |
| Average salinity of the oceans | <b>35‰</b>   | The standard point of the <b>global oceanic average</b> .                               |
| Pacific Ocean                  | 31 to 35‰    | Regional variation in salinity because of its vast extent.                              |
| Bay of Bengal                  | <b>30‰</b>   | Low salinity because of the large supply of fresh water by rivers (Ganga, Brahmaputra). |

## > Principal Salts Dissolved in Ocean Water

| Name of the Salt (Chemical Compound)       | Percentage of Total Amount in Ocean Water |
|--------------------------------------------|-------------------------------------------|
| 1. Sodium chloride (NaCl)                  | <b>77.80%</b> ( <i>largest amount</i> )   |
| 2. Magnesium chloride (MgCl <sub>2</sub> ) | <b>10.90%</b>                             |
| 3. Magnesium sulphate                      | 4.70%                                     |
| 4. Calcium sulphate                        | 3.60%                                     |
| 5. Potassium sulphate                      | 2.50%                                     |
| 6. Calcium carbonate                       | 0.30%                                     |
| 7. Magnesium bromide                       | 0.20%                                     |
| Total                                      | <b>100%</b>                               |

## > Principal Lakes of the World by Continent (Regional Lakes Mapping)

| Continent     | Names of the Principal Lakes                                                                                                                                                                                                                                                                        | Littoral Countries / Specific Geographical Location (UPPCS Matching Special)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Asia          | <ul style="list-style-type: none"> <li>• <b>Caspian Sea</b></li> <li>• <b>Lake Baikal</b></li> <li>• <b>Aral Sea</b></li> <li>• Lake Van</li> <li>• Lake Balkhash</li> <li>• Toba / Pangong Lake</li> <li>• Tonle Sap / Lop Nur</li> <li>• Issyk Kul / Urmia</li> <li>• Onega and Ladoga</li> </ul> | <ul style="list-style-type: none"> <li>• Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan (<i>the largest lake in the world</i>)</li> <li>• Russia (Siberia) (<i>the deepest and oldest lake in the world</i>)</li> <li>• Kazakhstan and Uzbekistan (<i>now drying up rapidly</i>)</li> <li>• Türkiye (<i>most saline</i>)</li> <li>• Kazakhstan (Balkhash)</li> <li>• Indonesia (Toba), India-China (Pangong)</li> <li>• Cambodia (Tonle Sap), in the Tarim Basin of China (Lop Nur)</li> <li>• Kyrgyzstan (Issyk Kul), Iran (Urmia)</li> <li>• European part of Russia (<i>also included in the Europe mapping</i>)</li> </ul> |
| Africa        | <ul style="list-style-type: none"> <li>• <b>Lake Victoria</b></li> <li>• <b>Lake Tanganyika</b></li> <li>• Lake Chad</li> <li>• Lake Malawi (Nyasa)</li> <li>• Lake Nasser / Tana</li> <li>• Volta / Kariba / Albert</li> </ul>                                                                     | <ul style="list-style-type: none"> <li>• Kenya, Uganda, Tanzania (<i>the Equator passes through it</i>)</li> <li>• Tanzania, Zaire (DRC), Burundi, Zambia</li> <li>• Cameroon, Nigeria, Chad, Niger</li> <li>• Malawi, Mozambique and Tanzania</li> <li>• Egypt and Sudan (Nasser - artificial), Ethiopia (Tana)</li> <li>• Ghana (Volta), Zambia-Zimbabwe (Kariba - artificial), Uganda-Congo (Albert)</li> </ul>                                                                                                                                                                                                                 |
| North America | <ul style="list-style-type: none"> <li>• <b>Great Lakes</b></li> <li>• Great Bear / Great Slave</li> <li>• Athabasca / Winnipeg / Reindeer</li> <li>• Great Salt Lake / Lake of the Woods</li> </ul>                                                                                                | <ul style="list-style-type: none"> <li>• <b>A group of 5 lakes (H.O.M.E.S):</b> Superior, Huron, Ontario, Michigan (<i>only in the USA</i>), Erie. (<i>Canada + USA border</i>)</li> <li>• Canada (<i>the Arctic Circle passes through Great Bear</i>)</li> <li>• Other principal freshwater sources of Canada</li> <li>• United States of America (USA); Canada-USA border (Lake of the Woods)</li> </ul>                                                                                                                                                                                                                         |
| South America | <ul style="list-style-type: none"> <li>• <b>Lake Titicaca</b></li> <li>• Lake Maracaibo</li> <li>• Lake Poopó / Red Lagoon</li> </ul>                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Located between Bolivia and Peru (<i>the highest navigable lake in the world</i>)</li> <li>• Located in the north of Venezuela (famous for petroleum reserves)</li> <li>• Principal lakes located in Bolivia</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| Europe        | <ul style="list-style-type: none"> <li>• Lake Constance</li> <li>• Ladoga / Onega</li> <li>• Vättern and Vänern</li> </ul>                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• It is the <b>meeting point of Germany, Austria and Switzerland.</b></li> <li>• The largest lakes of Europe, located in Russia.</li> <li>• Principal lakes located in Sweden.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                   |
| Australia     | <ul style="list-style-type: none"> <li>• <b>Lake Eyre</b></li> <li>• Other lakes</li> </ul>                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• <b>The lowest point of Australia</b> (main catchment basin).</li> <li>• Lake Disappointment, Lake Austin and Lake Barlee.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## > Major Canals of the World

| Name of the Canal      | Country / Region         | Which Regions / Seas / Lakes does it Connect? (UPPCS Mapping Special)                                                                                                                                                                                                                                                                                   |
|------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suez Canal             | Egypt                    | <ul style="list-style-type: none"> <li>• Connects the <b>Mediterranean Sea with the Red Sea</b> (the largest artificial canal in the world).</li> <li>• <b>Northern end:</b> Port Said; <b>southern end:</b> Port Suez.</li> <li>• <b>Order of lakes from north to south:</b> Timsah → Great Bitter → Little Bitter (<i>chronology fact</i>)</li> </ul> |
| Panama Canal           | Panama                   | <ul style="list-style-type: none"> <li>• Connects the <b>Pacific Ocean with the Atlantic Ocean</b>.</li> </ul>                                                                                                                                                                                                                                          |
| Welland Canal          | Canada                   | <ul style="list-style-type: none"> <li>• Connects <b>Lake Erie with Lake Ontario</b>.</li> </ul>                                                                                                                                                                                                                                                        |
| Erie Canal             | United States of America | <ul style="list-style-type: none"> <li>• Connects <b>Lake Erie with the Hudson River</b>.</li> </ul>                                                                                                                                                                                                                                                    |
| Soo Canal              | United States of America | <ul style="list-style-type: none"> <li>• Connects <b>Lake Superior with Lake Huron</b>.</li> </ul>                                                                                                                                                                                                                                                      |
| Kiel Canal             | Germany                  | <ul style="list-style-type: none"> <li>• Connects the <b>Baltic Sea with the North Sea</b>.</li> </ul>                                                                                                                                                                                                                                                  |
| New Waterway Canal     | Germany                  | <ul style="list-style-type: none"> <li>• Connects the <b>North Sea with Rotterdam</b>.</li> </ul>                                                                                                                                                                                                                                                       |
| North Sea Canal        | Germany                  | <ul style="list-style-type: none"> <li>• Located between Amsterdam and the North Sea.</li> </ul>                                                                                                                                                                                                                                                        |
| Ludwig Canal           | Germany                  | <ul style="list-style-type: none"> <li>• Used as an inland waterway in southern Germany.</li> </ul>                                                                                                                                                                                                                                                     |
| Göta Canal             | Sweden                   | <ul style="list-style-type: none"> <li>• Connects Gothenburg with Stockholm.</li> </ul>                                                                                                                                                                                                                                                                 |
| Manchester Canal       | Great Britain (UK)       | <ul style="list-style-type: none"> <li>• Connects Liverpool with Manchester.</li> </ul>                                                                                                                                                                                                                                                                 |
| Volga-Don Canal        | Russia                   | <ul style="list-style-type: none"> <li>• Connects Stalingrad with Rostov.</li> </ul>                                                                                                                                                                                                                                                                    |
| White Sea Canal        | Russia                   | <ul style="list-style-type: none"> <li>• Connects the <b>White Sea with Lake Onega</b>.</li> </ul>                                                                                                                                                                                                                                                      |
| Albert Canal           | Belgium                  | <ul style="list-style-type: none"> <li>• Connects Antwerp with Liège.</li> </ul>                                                                                                                                                                                                                                                                        |
| K.P. Canal             | India                    | <ul style="list-style-type: none"> <li>• Connects <b>Tamil Nadu with Andhra Pradesh</b> (<i>Indian geography special</i>).</li> </ul>                                                                                                                                                                                                                   |
| Sault Ste. Marie Canal | Canada                   | <ul style="list-style-type: none"> <li>• A principal canal located in Canada.</li> </ul>                                                                                                                                                                                                                                                                |

> Major Waterfalls of the World

| Name of the Waterfall          | Related River / Type      | Location / Country (Direct Match)                | Special UPPCS Fact                                     |
|--------------------------------|---------------------------|--------------------------------------------------|--------------------------------------------------------|
| Salto Angel (Angel Falls)      | Rio Caroní / Churún river | Venezuela                                        | It is <b>the highest waterfall in the world</b> .      |
| Victoria Falls                 | Zambezi river             | Border of Zambia and Zimbabwe                    | The most famous waterfall of Africa.                   |
| Niagara Falls                  | St. Lawrence river system | United States of America (USA) and Canada border | Located near the Great Lakes region.                   |
| Tugela Falls                   | —                         | South Africa                                     | The second highest waterfall of the African continent. |
| Kukenán                        | —                         | Venezuela                                        | —                                                      |
| Mardalsfossen / Tyssestrengene | —                         | Norway                                           | Principal waterfalls of Europe.                        |
| Yosemite / Ribbon              | —                         | California (United States of America)            | —                                                      |
| Gavarnie Falls                 | —                         | France                                           | —                                                      |
| Sutherland                     | —                         | New Zealand                                      | —                                                      |

> Principal Dams and Rivers of the World

| Name of the Dam  | Related River       | Location / Country (Direct Match Standard)                  |
|------------------|---------------------|-------------------------------------------------------------|
| Rogun Dam        | —                   | Tajikistan (one of the highest dams in the world)           |
| Nurek Dam        | —                   | Tajikistan                                                  |
| Aswan Dam        | Nile river          | Egypt                                                       |
| Owen Falls Dam   | White Nile river    | Uganda ( <i>outflow region of the White Nile</i> )          |
| Sennar Dam       | Blue Nile river     | Sudan                                                       |
| Cahora Bassa Dam | Zambezi river       | Mozambique                                                  |
| Kariba Dam       | Zambezi river       | Zambia - Zimbabwe border ( <i>data reference: Zambezi</i> ) |
| Inga Dam         | Congo / Zaire river | Democratic Republic of the Congo (DRC)                      |
| Kainji Dam       | Niger river         | Nigeria ( <i>data reference: Niger</i> )                    |
| Akosombo Dam     | Volta river         | Ghana                                                       |

## > Principal Local Winds of the World

### A. Cold Local Winds

| Name of the Wind | Region / Country of Flow (UPPCS Matching Special) | Name of the Wind | Region / Country of Flow (UPPCS Matching Special) |
|------------------|---------------------------------------------------|------------------|---------------------------------------------------|
| Mistral          | Spain and France                                  | Bora             | Adriatic Sea coast (Europe)                       |
| Norther / Norte  | United States of America and Mexico               | Tramontane       | Central Europe                                    |
| Williwaw         | Alaska (USA)                                      | Levanter         | Spain                                             |
| Pampero          | Argentina and Uruguay                             | Gregale          | Mediterranean region of southern Europe           |
| Papagayo         | Mexico                                            | Joran            | Jura Mountains (Switzerland)                      |
| Purga / Buran    | Tundra region / Russia (Siberia)                  | Seistan          | Eastern Iran                                      |
| Bise             | Eastern France, Switzerland                       | Southerly Buster | New South Wales (Australia)                       |
| Haboob           | Sudan                                             | Cape Doctor      | Republic of South Africa                          |

### B. Warm Local Winds

| Name of the Wind | Region / Mountain / Country of Flow               | Name of the Wind | Region / Mountain / Country of Flow                |
|------------------|---------------------------------------------------|------------------|----------------------------------------------------|
| Chinook          | Rocky Mountains (USA) ( <i>snow-eating wind</i> ) | Santa Ana        | Southern California (USA)                          |
| Foehn            | Alps (Switzerland)                                | Shamal / Samoon  | Iraq (Shamal), Iran (Samoon/Garmsil)               |
| Sirocco          | Italy ( <i>wind that causes 'blood rain'</i> )    | Khamsin / Ghibli | Egypt (Khamsin), Libya (Ghibli)                    |
| Simoom           | Arabian desert                                    | Berg / Leste     | South Africa (Berg), Madeira and Canary (Leste)    |
| Brickfielder     | Australia                                         | Sukhovey / Zonda | Russia-Kazakhstan (Sukhovey), Andes region (Zonda) |
| Black Roller     | Plains of North America                           | Karaburan        | Tarim Basin of Central Asia                        |
| Loo              | Northern India                                    | Yamo / Kumbang   | Japan (Yamo), Java-Indonesia (Kumbang)             |
| Aiala / Solano   | France (Aiala), south-eastern Spain (Solano)      | Temporal / Chili | Central America (Temporal), Tunisia (Chili)        |

## > Principal Islands Located in the Oceans (Islands by Ocean)

| Name of the Island      | Ocean / Sea where Located                               | Name of the Island | Ocean / Sea where Located  |
|-------------------------|---------------------------------------------------------|--------------------|----------------------------|
| Greenland               | Arctic Ocean ( <i>the largest island in the world</i> ) | Baffin Island      | Northern Arctic Ocean      |
| New Guinea              | Western Pacific Ocean                                   | Victoria Island    | North Polar (Arctic) Ocean |
| Honshu Island           | North Pacific Ocean ( <i>main island of Japan</i> )     | Ellesmere Island   | North Polar (Arctic) Ocean |
| North and South Islands | South-west Pacific Ocean ( <i>New Zealand</i> )         | Britain            | North Atlantic Ocean       |
| Tasmania                | S.W. Pacific Ocean ( <i>Australia</i> )                 | Iceland / Ireland  | North Atlantic Ocean       |
| Borneo / Madagascar     | Indian Ocean                                            | Quebec / Cuba      | Caribbean Sea              |
| Java / Sumatra          | Indian Ocean ( <i>Indonesia</i> )                       | Sri Lanka          | Indian Ocean               |
| Sulawesi Island         | Indian Ocean                                            |                    |                            |

## > Principal Cold and Warm Ocean Currents

### > Ocean Currents of the Pacific Ocean

| Warm Currents of the Pacific Ocean                                                                                                                                                                                                                                                                                                                                                        | Cold Currents of the Pacific Ocean                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Kuroshio current (<i>the black current of Japan</i>)</li> <li>• Alaska current</li> <li>• El Niño current (<i>temporary warm current</i>)</li> <li>• Tsushima current</li> <li>• East Australian current</li> <li>• North Equatorial current / South Equatorial current</li> <li>• North Pacific current / Counter Equatorial current</li> </ul> | <ul style="list-style-type: none"> <li>• Humboldt or Peru current (<i>El Niño forms when it weakens</i>)</li> <li>• California current</li> <li>• Kurile / Oyashio current</li> <li>• Okhotsk current</li> <li>• Antarctic current</li> </ul> |

### > Currents of the Atlantic and Indian Oceans

| Ocean          | Warm Currents                                                                                                                                                                                                                                                                                                              | Cold Currents                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atlantic Ocean | <ul style="list-style-type: none"> <li>• <b>Gulf Stream</b> (<i>the blanket of Europe</i>)</li> <li>• <b>Brazil current</b></li> <li>• <b>Florida current</b></li> <li>• Norwegian current / Irminger current</li> <li>• North Atlantic current / Guinea current</li> <li>• North and South Equatorial currents</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Canary current</b></li> <li>• <b>Benguela current</b></li> <li>• <b>Labrador current</b></li> <li>• Falkland current</li> <li>• East Greenland current</li> <li>• Antarctic current</li> </ul> |
| Indian Ocean   | <ul style="list-style-type: none"> <li>• <b>Mozambique current</b></li> <li>• <b>Agulhas current</b></li> <li>• South Equatorial current</li> </ul>                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• <b>West Australian current</b></li> <li>• West Wind Drift</li> </ul>                                                                                                                              |

### > Principal Local Vegetation Types and their Natural Regions

| Name of the Vegetation | Adapted Natural Region / Country (Direct Match Standard)                                     |
|------------------------|----------------------------------------------------------------------------------------------|
| Maquis                 | Shrubby vegetation found in the coastal region of the Mediterranean Sea.                     |
| Chaparral              | Vegetation of the Mediterranean climate of <b>California (USA)</b> .                         |
| Matorral               | Vegetation of <b>Chile</b> (South America).                                                  |
| Fynbos                 | The distinctive vegetation of the Cape province of <b>South Africa</b> .                     |
| Baobab                 | Characteristic of the <b>savanna grasslands</b> ( <i>the upside-down tree/bottle tree</i> ). |
| Epiphytes              | Vegetation of the <b>equatorial (tropical rainforest)</b> region.                            |
| Cedar (Deodar)         | Vegetation of the Mediterranean mountain regions.                                            |

### > Major Grasslands of the World

| Type of Grassland       | Name of the Grassland                                                                                                                                                 | Related Country / Geographical Region (UPPCS Matching Special)                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1. Tropical Grasslands  | <ul style="list-style-type: none"> <li>• <b>Savanna</b></li> <li>• <b>Campos</b></li> <li>• <b>Llanos</b></li> </ul>                                                  | Africa<br>Brazil<br>Venezuela and Colombia                                                                          |
| 2. Temperate Grasslands | <ul style="list-style-type: none"> <li>• <b>Prairies</b></li> <li>• <b>Pampas</b></li> <li>• <b>Veld</b></li> <li>• <b>Downs</b></li> <li>• <b>Steppes</b></li> </ul> | USA and Canada<br>Argentina<br>South Africa<br>Australia<br>Asia, Ukraine, Russia and the Manchuria region of China |



## > Major Valleys of the World

| Name of the Valley         | Geographical Location / Country | Special UPPCS Exam Pointer                                                         |
|----------------------------|---------------------------------|------------------------------------------------------------------------------------|
| Valley of the Kings        | Egypt                           | A historical site located on the western bank of the Nile.                         |
| Silicon Valley             | Northern California (USA)       | The <b>computer-chip production region</b> located south of the San Francisco Bay. |
| Grand Valley               | United States of America (USA)  | Located between the Rocky Mountains and the Sierra Nevada.                         |
| Taylor Valley              | Antarctica                      | A dry valley; it was discovered by <b>Robert Falcon Scott</b> .                    |
| Panjshir Valley            | Afghanistan                     | A strategic valley located north of Kabul.                                         |
| Chumbi Valley              | India, Bhutan and Tibet border  | A strategic triangle near the Doklam area of Sikkim (India).                       |
| Po Valley                  | Italy                           | Famous for rice production and industrial development.                             |
| Loire Valley               | France                          | Famous for its historic castles (châteaux).                                        |
| Sacred Valley              | Peru                            | A sacred site of the Inca civilisation.                                            |
| Chitwan / Kathmandu Valley | Nepal                           | Famous for Chitwan National Park.                                                  |
| Rabbit / Pleasant Valley   | United States of America (USA)  | —                                                                                  |
| Humber Valley              | Canada                          | —                                                                                  |
| Ebro Valley                | Spain                           | —                                                                                  |
| Blackwood / King / Wolgan  | Australia                       | Principal Australian valleys.                                                      |

## > Other Principal Volcanoes and their Location

| Name of the Volcano | Location (Country) | UPPCS Special Fact                                         |
|---------------------|--------------------|------------------------------------------------------------|
| Mount Kinabalu      | Malaysia           | The highest point on the island of Borneo.                 |
| Alborz (Elburz)     | Iran               | Mountain/volcanic region located south of the Caspian Sea. |

| Name of the Volcano | Location (Country)             | UPPCS Special Fact                                                |
|---------------------|--------------------------------|-------------------------------------------------------------------|
| Colima              | Mexico                         | A highly active volcano of Mexico.                                |
| Mount Rainier       | United States of America (USA) | A large stratovolcano located in Washington state.                |
| Taal Volcano        | Philippines                    | A small but active volcano located within a lake on Luzon island. |

> Major Oceanic Trenches of the World

| Trench                                            | Related Ocean  | Location / Feature                                                                                      | Maximum Depth |
|---------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------|---------------|
| Mariana Trench                                    | Pacific Ocean  | • Located east of the Philippines, it is <b>the deepest point in the world.</b>                         | 11,034 metres |
| Tonga Trench                                      | Pacific Ocean  | • Located east of the Tonga islands in the South Pacific Ocean.                                         | —             |
| Sunda Trench<br><i>(former name: Java Trench)</i> | Indian Ocean   | • Located south of the island of Java in Indonesia, it is <b>the deepest point of the Indian Ocean.</b> | 7,725 metres  |
| Puerto Rico Trench                                | Atlantic Ocean | • Located on the boundary of the Caribbean Sea and the Atlantic Ocean.                                  | —             |

> Principal Rivers of the North American and African Continents

| Name of the River    | Continent / Location | UPPCS Special Main Features (Core Concept Pointers)                                                                                                                                                                                    |
|----------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mississippi-Missouri | North America        | <ul style="list-style-type: none"><li>• They are two separate rivers which together form the <b>fourth largest river system</b> in the world.</li><li>• Its delta is shaped like a <b>'bird's foot' (bird's-foot delta)</b>.</li></ul> |
| Rio Grande           | North America        | <ul style="list-style-type: none"><li>• This river forms the natural international boundary <b>between the United States of America (USA) and Mexico.</b></li></ul>                                                                    |
| St. Lawrence         | North America        | <ul style="list-style-type: none"><li>• Forms the <b>busiest inland water transport route</b> in the world.</li><li>• The world-famous <b>'Niagara Falls'</b> is located on this river (in the lake-connecting region).</li></ul>      |
| Colorado             | North America        | <ul style="list-style-type: none"><li>• Flows in the western part of the continent.</li><li>• The world-famous <b>'Grand Canyon'</b> and the <b>'Hoover Dam'</b> are on this river.</li></ul>                                          |
|                      |                      |                                                                                                                                                                                                                                        |

| Name of the River | Continent / Location | UPPCS Special Main Features (Core Concept Pointers)                                                                                           |
|-------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Niger             | Africa               | • Falls into the Gulf of Guinea and is famous in West Africa as the 'Oil River'.                                                              |
| Zambezi           | Africa               | • The famous <b>Victoria Falls</b> and the <b>Kariba Dam</b> are located on this river.                                                       |
| Congo / Zaire     | Africa               | • The only river in the world that <b>crosses the Equator twice</b> .<br>• The famous <b>Stanley and Livingstone Falls</b> are located on it. |
| Limpopo           | Africa               | • Rises in the Highveld of South Africa and <b>crosses the Tropic of Capricorn twice</b> .                                                    |

### ➤ Principal Rivers of Europe and their Features

| Name of the River | Source                  | Where it Falls (Mouth) | Main Features (Quick Revision Notes)                                                                                                                    |
|-------------------|-------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volga             | Valdai Hills of Russia  | Caspian Sea            | • It is <b>the longest river of Europe</b> .<br>• Connects the Baltic Sea with the Caspian Sea; the lifeline of Russian culture and trade.              |
| Danube            | Black Forest of Germany | Black Sea              | • It is <b>the second longest river of Europe</b> (~2,850 km).<br>• It passes through <b>ten countries</b> . Its delta is a UNESCO World Heritage Site. |
| Rhine             | Swiss Alps              | North Sea              | • It is <b>one of the busiest and most important waterways of Europe</b> (1,233 km).<br>• Separates Germany from the Netherlands.                       |
| Loire             | French Cévennes range   | Atlantic Ocean         | • It is <b>the longest river of France</b> .                                                                                                            |
| Po                | Italian Cottian Alps    | Adriatic Sea           | • It is <b>the longest river of Italy</b> (more than 652 kilometres).                                                                                   |

### ➤ Countries Located on the Principal Geographical Lines (Equator, Cancer, Capricorn, GMT)

| Name of the Line                                            | Countries Located (Continent-wise Classification - UPPCS Special)                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equator</b><br><i>(through 13 countries/territories)</i> | <ul style="list-style-type: none"> <li>• <b>Africa:</b> São Tomé and Príncipe, Gabon, Republic of the Congo, Democratic Republic of the Congo (DRC), Uganda, Kenya, Somalia.</li> <li>• <b>Asia:</b> Maldives, Indonesia, Kiribati.</li> <li>• <b>South America:</b> Ecuador, Colombia, Brazil.</li> </ul> |
|                                                             |                                                                                                                                                                                                                                                                                                            |

| Name of the Line          | Countries Located (Continent-wise Classification - UPPCS Special)                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tropic of Cancer          | <ul style="list-style-type: none"> <li>• <b>North America:</b> Hawaii islands (USA), Bahamas, Mexico.</li> <li>• <b>Africa:</b> Western Sahara, Mauritania, Algeria, Libya, Egypt, Mali, Niger, Chad.</li> <li>• <b>Asia:</b> India, Myanmar, Oman, Saudi Arabia, United Arab Emirates (UAE), Bangladesh, China, Taiwan.</li> </ul> |
| Tropic of Capricorn       | <ul style="list-style-type: none"> <li>• <b>South America:</b> Chile, Argentina, Paraguay, Brazil.</li> <li>• <b>Africa:</b> Namibia, Botswana, South Africa, Mozambique, Madagascar.</li> <li>• <b>Oceania:</b> Australia (<i>your original data describes only 2 continents</i>).</li> </ul>                                      |
| Prime Meridian (GMT line) | <ul style="list-style-type: none"> <li>• <b>Europe:</b> United Kingdom (UK), Spain, France.</li> <li>• <b>Africa:</b> Ghana, Algeria, Mali, Togo, Burkina Faso.</li> <li>• <b>Other:</b> Antarctica.</li> </ul>                                                                                                                     |

### > List of Countries with the Most Time Zones

| Rank | Name of the Country            | Total Number of Time Zones                                        |
|------|--------------------------------|-------------------------------------------------------------------|
| 1    | France                         | 12 ( <i>the most, owing to its overseas territories/islands</i> ) |
| 2    | Russia                         | 11                                                                |
| 3    | Antarctica                     | 10                                                                |
| 4    | United States of America (USA) | 9                                                                 |
| 5    | Australia                      | 8                                                                 |
| 6    | Denmark                        | 5                                                                 |

### > Principal Geographical Triangles/Regions of the World

| Name of the Region | Main Feature                                                         | Extent / Related Countries (UPPCS Core Map Point)                           |
|--------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Bermuda Triangle   | • A mysterious region of the north-west Atlantic Ocean.              | <b>Bermuda → Miami (southern Florida, USA) → San Juan (Puerto Rico).</b>    |
| Golden Triangle    | • The principal <b>opium-producing</b> region of South-East Asia.    | <b>Myanmar → Laos → Thailand.</b>                                           |
| Golden Crescent    | • The notorious <b>opium-producing</b> region of south-central Asia. | <b>Iran → Afghanistan → Pakistan.</b>                                       |
| Coral Triangle     | • A marine region with extremely rich <b>biodiversity/corals</b> .   | <b>Philippines → Indonesia → Malaysia → Papua New Guinea → Timor-Leste.</b> |

### > Countries Sharing a Border with the Red Sea

The UPPCS special short trick to remember the countries touching the Red Sea is "**DESSEY**":

- **List of countries:** 1. Egypt 2. Sudan 3. Eritrea 4. Djibouti 5. Saudi Arabia 6. Yemen (*Note: your data mainly describes Egypt, Saudi Arabia, Yemen, Sudan and Djibouti*).

### > Countries Included in the Arabian Peninsula

The Arabian Peninsula is the largest peninsula in the world. The countries included in it are as follows:

- **List of countries:** 1. Saudi Arabia 2. Yemen 3. Oman 4. United Arab Emirates (UAE) 5. Qatar 6. Kuwait 7. Jordan 8. Iraq (southern part).

### > Nordic Countries

The Nordic countries are a geographical and cultural region located in northern Europe and the North Atlantic.

- **Countries included:** 1. Iceland 2. Norway 3. Denmark 4. Sweden 5. Finland

### > The 6 Countries Sharing a Border with the Black Sea

The UPPCS special short trick to remember the countries touching the Black Sea is "**BURGER-T**":

**Bulgaria → Ukraine → Russia → Georgia → Romania → Turkey**

### > Principal Tribes of the World and their Habitats

| Name of the Tribe | Geographical Habitat / Country (Extremely Important for UPPCS Matching)                                  |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Pygmy             | In the rainforests of the <b>Congo Basin</b> (live in huts built on trees).                              |
| Bushmen           | <b>Kalahari Desert</b> (Botswana/South Africa) ( <i>termites are called the 'rice of the Bushmen'</i> ). |
| Masai             | Plateau regions of <b>Kenya, northern Tanzania</b> and eastern Uganda in East Africa.                    |
| Eskimo            | Arctic/icy regions of <b>Alaska, Canada and Greenland</b> ( <i>igloo dwellings</i> ).                    |
| Bora              | Western Amazon Basin (Brazil, Peru and Colombia).                                                        |
| Kirghiz           | Kyrgyzstan and adjoining areas in Central Asia ( <i>yurt dwellings</i> ).                                |
| Semang / Sakai    | Forest regions of <b>Malaysia</b> .                                                                      |
| Maori             | The original inhabitants of <b>New Zealand</b> .                                                         |
| Afridi            | North-western mountainous region of <b>Pakistan</b> .                                                    |
| Veddas / Ainu     | <b>Sri Lanka</b> (Veddas), <b>Japan</b> (Ainu).                                                          |
| Berber / Nuba     | Morocco (Berber), Sudan (Nuba).                                                                          |

| Name of the Tribe          | Geographical Habitat / Country (Extremely Important for UPPCS Matching)                             |
|----------------------------|-----------------------------------------------------------------------------------------------------|
| Zulu / Bantu / Ikatha      | <b>South Africa</b> ( <i>the Bantu are also spread over the whole of central/southern Africa</i> ). |
| Papuans / Lapps            | Papua New Guinea (Papuans), the Sami region of Norway, Sweden and Finland (Lapps).                  |
| Yupik / Yukaghir / Tartar  | Eastern Russia and Alaska (Yupik), north-eastern Siberia (Yukaghir), Siberia (Tartar/Kamasin).      |
| Bedouin / Budh             | Nomadic herders of the Arabian desert.                                                              |
| Maya / Red Indians         | Central America (Maya), the North American continent (Red Indians).                                 |
| Hadza                      | Tanzania.                                                                                           |
| Fulani / Hausa / Fula      | West Africa and Nigeria (Fulani/Hausa), the Mediterranean region of Africa (Fula).                  |
| Kurds                      | The border mountain region of Iran, Iraq, Türkiye and Armenia.                                      |
| Oromo                      | Ethiopia.                                                                                           |
| Luo and Baganda            | The region around <b>Lake Victoria</b> .                                                            |
| Kikuyu and Kamba           | The mountainous region of Kenya.                                                                    |
| Aborigines / Black Fellows | The original inhabitants of <b>Australia</b> .                                                      |
| Fellah                     | <b>Agricultural labourers of Egypt</b> .                                                            |
| Lai / Uighur               | Myanmar (Lai), the Tarim Basin/Xinjiang of China (Uighur).                                          |

### > Classification of the Principal Vegetation of the World (Vegetation Types)

| Name of the Vegetation | Adapted Geographical Region / Natural Environment (Direct Match Standard)                |
|------------------------|------------------------------------------------------------------------------------------|
| Xerophyte              | • Thorny vegetation growing in <b>tropical desert (arid)</b> regions.                    |
| Hydrophyte             | • Vegetation living in water or in <b>waterlogged</b> areas.                             |
| Hygrophyte             | • Vegetation of <b>marshy</b> areas and of equatorial hot-humid (high-moisture) regions. |
| Mesophyte              | • Vegetation of the <b>temperate zone</b> (normal rainfall and temperature).             |
| Cryophyte              | • Vegetation of the <b>tundra</b> and extremely cold (icy) regions.                      |
| Halophyte              | • Vegetation found in <b>salty / saline</b> soils or coastal areas.                      |

| Name of the Vegetation | Adapted Geographical Region / Natural Environment (Direct Match Standard)   |
|------------------------|-----------------------------------------------------------------------------|
| Lithophyte             | • Vegetation growing on <b>hard rocks</b> or stones.                        |
| Tropophyte             | • Grasses and vegetation of the <b>tropical climate</b> (changing seasons). |

### > Principal Industrial Cities of the World and their Related Industries

| Industrial City           | Country / Region                      | Principal Industry (UPPCS Matching Special)                              |
|---------------------------|---------------------------------------|--------------------------------------------------------------------------|
| Detroit                   | United States of America (USA)        | <b>Automobile (motor car) industry</b> ( <i>extremely important</i> )    |
| Turin                     | Italy                                 | <b>Automobile / motor car industry</b>                                   |
| Nagoya                    | Japan                                 | <b>Automobile industry</b>                                               |
| Chicago                   | United States of America (USA)        | <b>Meat processing</b> and the iron and steel industry                   |
| Buenos Aires              | Argentina                             | <b>Meat and dairy industry</b>                                           |
| Kinshasa                  | Congo (DRC)                           | <b>Meat industry</b>                                                     |
| Pittsburgh                | United States of America (USA)        | <b>Iron and steel industry</b> ( <i>the steel capital of the world</i> ) |
| Hamilton                  | Canada                                | Iron and steel industry                                                  |
| Essen / Tula              | Germany (Essen), Russia (Tula)        | Iron and steel industry                                                  |
| Kobe / Anshan             | Japan (Kobe), China (Anshan)          | Iron and steel industry                                                  |
| Magnitogorsk / Krivoy Rog | Russia (Magnitogorsk), Ukraine region | Iron and steel industry                                                  |
| Los Angeles               | United States of America (USA)        | <b>Hollywood</b> , oil refining and aircraft                             |
| New Orleans / Houston     | United States of America (USA)        | <b>Oil refining</b> (New Orleans), oil and natural gas (Houston)         |
| San Francisco             | United States of America (USA)        | Oil refining, computers and ships                                        |
| Maracaibo / Abadan        | Venezuela (Maracaibo), Iran (Abadan)  | <b>Oil refining industry</b>                                             |
| Mosul                     | Iraq                                  | Mineral oil industry                                                     |



| Industrial City         | Country / Region                           | Principal Industry (UPPCS Matching Special)                              |
|-------------------------|--------------------------------------------|--------------------------------------------------------------------------|
| Manchester / Osaka      | Great Britain (Manchester), Japan (Osaka)  | <b>Cotton textile industry / textile industry</b>                        |
| Shanghai / Cairo        | China (Shanghai), Egypt (Cairo)            | Textiles and machinery (Shanghai), textile industry (Cairo)              |
| Milan / Lyons           | Italy (Milan), France (Lyons)              | <b>Silk textile industry</b>                                             |
| Rio de Janeiro          | Brazil                                     | Textiles and the <b>coffee industry</b>                                  |
| São Paulo               | Brazil                                     | <b>Coffee industry</b> ( <i>the largest coffee market in the world</i> ) |
| Havana                  | Cuba                                       | <b>Cigar industry</b>                                                    |
| Antwerp                 | Belgium                                    | <b>Diamond industry (diamond cutting/trade)</b>                          |
| Kimberley               | South Africa                               | <b>Diamond mining industry</b>                                           |
| Johannesburg            | South Africa                               | <b>Gold mining industry</b>                                              |
| Montreal / Belfast      | Canada (Montreal), Great Britain (Belfast) | <b>Shipbuilding industry</b>                                             |
| Rotterdam / Amsterdam   | Netherlands                                | Shipbuilding                                                             |
| Glasgow / Hamburg       | Great Britain (Glasgow), Germany (Hamburg) | Shipbuilding industry                                                    |
| Stockholm / Vladivostok | Sweden (Stockholm), Russia (Vladivostok)   | Shipbuilding                                                             |
| Ottawa                  | Canada                                     | <b>Paper industry</b>                                                    |
| Dundee                  | Scotland / Great Britain                   | <b>Jute industry</b>                                                     |
| Frankfurt / Zurich      | Germany (Frankfurt), Switzerland (Zurich)  | Engineering and transport (Frankfurt), engineering (Zurich)              |
| Paris                   | France                                     | Transport and aircraft                                                   |
| Vienna / Venice         | Austria (Vienna), Italy (Venice)           | <b>Glass industry</b>                                                    |
| Copenhagen              | Denmark                                    | <b>Dairy industry</b>                                                    |
| Santiago                | Chile                                      | <b>Wine industry</b>                                                     |
| Multan                  | Pakistan                                   | <b>Pottery</b>                                                           |

| Industrial City        | Country / Region | Principal Industry (UPPCS Matching Special) |
|------------------------|------------------|---------------------------------------------|
| Wuhan                  | China            | Ships and iron and steel                    |
| Pelialia / Chelyabinsk | Russian region   | Metal and military machinery industry       |

## > Research Institutes

### > National Research Institutes

| Name of the Research Institute        | Headquarters / Location (Place)                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Institutes of Dehradun (Uttarakhand)  | <ul style="list-style-type: none"> <li>• Wildlife Institute of India (WII)</li> <li>• Forest Research Institute (FRI)</li> <li>• Forest Survey of India (FSI)</li> <li>• Wadia Institute of Himalayan Geology (WIHG)</li> <li>• Indian Council of Forestry Research and Education (ICFRE)</li> <li>• Indian Institute of Soil and Water Conservation (IISWC)</li> </ul> |
| Institutes of Lucknow (Uttar Pradesh) | <ul style="list-style-type: none"> <li>• National Botanical Research Institute (NBRI)</li> <li>• Birbal Sahni Institute of Palaeosciences (BSIP)</li> </ul>                                                                                                                                                                                                             |
| Institutes of Kolkata (West Bengal)   | <ul style="list-style-type: none"> <li>• Botanical Survey of India (BSI)</li> <li>• Zoological Survey of India (ZSI)</li> </ul>                                                                                                                                                                                                                                         |
| Institutes of New Delhi               | <ul style="list-style-type: none"> <li>• Indian Agricultural Research Institute (IARI)</li> <li>• National Institute of Plant Genome Research (NIPGR)</li> <li>• National Research Centre on Plant Biotechnology</li> <li>• National Environmental Science Academy (NESA)</li> <li>• National Physical Laboratory (NPL)</li> </ul>                                      |
| Institutes of Hyderabad (Telangana)   | <ul style="list-style-type: none"> <li>• National Geophysical Research Institute (NGRI)</li> <li>• Institute of Forest Biodiversity (IFB)</li> </ul>                                                                                                                                                                                                                    |
| Institutes of Chennai (Tamil Nadu)    | <ul style="list-style-type: none"> <li>• National Institute of Ocean Technology (NIOT)</li> <li>• National Centre for Coastal Research (NCCR)</li> <li>• National Centre for Sustainable Coastal Management (NCSCM)</li> </ul>                                                                                                                                          |
| Institutes of Maharashtra             | <ul style="list-style-type: none"> <li>• National Environmental Engineering Research Institute (NEERI) — Nagpur (<i>a favourite UPPCS question</i>)</li> <li>• Indian Institute of Tropical Meteorology (IITM) — Pune</li> <li>• National Chemical Laboratory (NCL) — Pune</li> </ul>                                                                                   |
| Institutes of Goa                     | <ul style="list-style-type: none"> <li>• National Institute of Oceanography (NIO) — Panaji</li> <li>• National Centre for Polar and Ocean Research (NCPOR) — Vasco da Gama</li> </ul>                                                                                                                                                                                   |

| Name of the Research Institute | Headquarters / Location (Place)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Institutes of Jharkhand        | <ul style="list-style-type: none"> <li>• <b>Central Institute of Mining and Fuel Research (CIMFR)</b> — Dhanbad</li> <li>• <b>National Metallurgical Laboratory (NML)</b> — Jamshedpur</li> <li>• <b>Institute of Forest Productivity (IFP)</b> — Ranchi</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Jodhpur / Jaipur (Rajasthan)   | <ul style="list-style-type: none"> <li>• <b>Central Arid Zone Research Institute (CAZRI)</b> — Jodhpur</li> <li>• <b>Arid Forest Research Institute (AFRI)</b> — Jodhpur</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Institutes of Haryana          | <ul style="list-style-type: none"> <li>• <b>Central Soil Salinity Research Institute (CSSRI)</b> — Karnal</li> <li>• <b>National Dairy Research Institute (NDRI)</b> — Karnal</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Madhya Pradesh / Chhattisgarh  | <ul style="list-style-type: none"> <li>• <b>Tropical Forest Research Institute (TFRI)</b> — Jabalpur</li> <li>• <b>National Institute of Biotic Stress Management (NIBSM)</b> — Raipur</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Other principal institutes     | <ul style="list-style-type: none"> <li>• G.B. Pant National Institute of Himalayan Environment — <b>Almora</b> (UK)</li> <li>• National Institute of Hydrology (NIH) — <b>Roorkee</b> (UK)</li> <li>• Himalayan Forest Research Institute (HFRI) — <b>Shimla</b> (HP)</li> <li>• Rain Forest Research Institute (RFRI) — <b>Jorhat</b> (Assam)</li> <li>• Institute of Wood Science and Technology (IWST) — <b>Bengaluru</b> (KN)</li> <li>• Institute of Forest Genetics and Tree Breeding (IFGTB) — <b>Coimbatore</b> (TN)</li> <li>• Institute of Minerals and Materials Technology (IMMT) — <b>Bhubaneswar</b> (OD)</li> <li>• North East Institute of Science and Technology (NEIST) — <b>Jorhat</b> (Assam)</li> <li>• National Institute for Interdisciplinary Science and Technology (NIIST) — <b>Thiruvananthapuram</b> (Kerala)</li> </ul> |

## > Principal Global Environment and Water Resource Organisations

| Name of the Organisation / Institution                | Headquarters / Location (City, Country) | UPPCS Special Fact / Importance                    |
|-------------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| United Nations Environment Programme (UNEP)           | <b>Nairobi, Kenya</b>                   | <i>The headquarters asked most often in UPPCS.</i> |
| International Union for Conservation of Nature (IUCN) | <b>Gland, Switzerland</b>               | Issues the Red Data Book.                          |
| World Wide Fund for Nature (WWF)                      | <b>Gland, Switzerland</b>               | Its emblem is the 'Giant Panda'.                   |
| World Meteorological Organization (WMO)               | Geneva, Switzerland                     | —                                                  |
| World Climate Research Programme (WCRP)               | Geneva, Switzerland                     | —                                                  |
| World Resources Institute (WRI)                       | Washington D.C., USA                    | —                                                  |

| Name of the Organisation / Institution                         | Headquarters / Location (City, Country) | UPPCS Special Fact / Importance |
|----------------------------------------------------------------|-----------------------------------------|---------------------------------|
| World Water Council (WWC)                                      | Marseille, France                       | —                               |
| International Water Management Institute (IWMI)                | Colombo, Sri Lanka                      | —                               |
| International Institute for Environment and Development (IIED) | London, Britain                         | —                               |
| International Institute for Applied Systems Analysis (IIASA)   | Laxenburg, Austria                      | —                               |
| International Geosphere-Biosphere Programme (IGBP)             | Stockholm, Sweden                       | —                               |
| Stockholm Resilience Centre                                    | Stockholm, Sweden                       | —                               |
| European Environment Agency (EEA)                              | Copenhagen, Denmark                     | —                               |
| Potsdam Institute for Climate Impact Research (PIK)            | Potsdam, Germany                        | —                               |
| Smithsonian Environmental Research Center (SERC)               | Edgewater, USA                          | —                               |

### > Global Energy, Maritime and Space Organisations

| Name of the Organisation / Institution                  | Headquarters / Location (City, Country) |
|---------------------------------------------------------|-----------------------------------------|
| International Energy Agency (IEA)                       | Paris, France                           |
| Intergovernmental Oceanographic Commission (IOC-UNESCO) | Paris, France                           |
| European Space Agency (ESA)                             | Paris, France                           |
| International Renewable Energy Agency (IRENA)           | Abu Dhabi, UAE                          |
| International Atomic Energy Agency (IAEA)               | Vienna, Austria                         |
| International Maritime Organization (IMO)               | London, Britain                         |

### > Principal Polar, Oceanographic and Meteorological Institutions (America and Europe)

| Name of the Organisation / Institution                 | Headquarters / Location (City, Country) |
|--------------------------------------------------------|-----------------------------------------|
| National Oceanic and Atmospheric Administration (NOAA) | Silver Spring, USA                      |

| Name of the Organisation / Institution                               | Headquarters / Location (City, Country) |
|----------------------------------------------------------------------|-----------------------------------------|
| NASA Goddard Institute for Space Studies (GISS)                      | New York, USA                           |
| US Geological Survey (USGS)                                          | Reston, USA                             |
| National Center for Atmospheric Research (NCAR)                      | Boulder, USA                            |
| Woods Hole Oceanographic Institution (WHOI)                          | Woods Hole, USA                         |
| International Polar Foundation (IPF)                                 | Brussels, Belgium                       |
| Alfred Wegener Institute (AWI)                                       | Bremerhaven, Germany                    |
| Scott Polar Research Institute (SPRI)                                | Cambridge, Britain                      |
| British Antarctic Survey (BAS)                                       | Cambridge, Britain                      |
| Commonwealth Scientific and Industrial Research Organisation (CSIRO) | Canberra, Australia                     |
| Australian Institute of Marine Science (AIMS)                        | Townsville, Australia                   |

### > Principal International Agricultural and Forestry Research Institutes

| Name of the Organisation / Institution                                          | Headquarters / Location (City, Country) | UPPCS Special / Indian Context                                                                                            |
|---------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)      | Patancheru (Hyderabad), India           | <b>Extremely important:</b> the only international agricultural institute located in India that is included in your data. |
| International Rice Research Institute (IRRI)                                    | Los Baños, Philippines                  | International rice research institute (a regional centre of it has also been set up at Varanasi, UP).                     |
| Center for International Forestry Research and World Agroforestry (CIFOR-ICRAF) | Nairobi, Kenya                          | —                                                                                                                         |

### Miscellaneous

## > Principal Geographical Sobriquets

### A. International Geographical Sobriquets

| Geographical Sobriquet (Epithet)                                             | Related City / Country / Place (UPPCS Direct One-liner)                |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Key to the Mediterranean / Pillars of Hercules                               | <b>Strait of Gibraltar</b> ( <i>extremely important mapping fact</i> ) |
| Sugar Bowl of the World / Pearl of the Antilles                              | <b>Quebec / Cuba</b> ( <i>Cuba</i> )                                   |
| Bread Basket of the World / Never-Never Land                                 | <b>Prairies / Prairies of the North</b> (North America)                |
| Sick Man of Europe / Gateway of the West                                     | <b>Turkey (Istanbul)</b>                                               |
| Playground of Europe                                                         | <b>Switzerland</b>                                                     |
| Cockpit of Europe                                                            | <b>Belgium</b>                                                         |
| Powder Keg of Europe                                                         | <b>Balkan region</b>                                                   |
| Garden of Lakes / Land of a Thousand Lakes                                   | <b>Finland</b>                                                         |
| Empire City / City of Skyscrapers                                            | <b>New York</b> (United States of America)                             |
| Smoky City / Windy City / Garden City                                        | <b>Chicago</b> (United States of America)                              |
| Oil Capital of Europe / Granite City                                         | <b>Aberdeen</b> (Scotland)                                             |
| Queen of the Adriatic / City of Canals                                       | <b>Venice</b> (Italy)                                                  |
| Babylon of the West                                                          | <b>Rome</b> (Italy)                                                    |
| Venice of the World                                                          | <b>Stockholm</b> (Sweden)                                              |
| Venice of the North                                                          | <b>St. Petersburg</b> (Russia)                                         |
| Pearl of Siberia                                                             | <b>Lake Baikal</b> (Russia)                                            |
| Land of the Rising Sun / Britain of the East / Land of Delicate Flowers      | <b>Japan</b>                                                           |
| Manchester of the East                                                       | <b>Osaka</b> (Japan)                                                   |
| Mistress of the Eastern Seas / Pearl of the East / Pearl of the Indian Ocean | <b>Sri Lanka</b>                                                       |
| Land of White Elephants / Paris of Asia                                      | <b>Thailand</b>                                                        |
| Land of the Sunset / Land of the Midnight Sun                                | <b>Norway</b>                                                          |
| Land of the Morning Calm / Hermit Kingdom                                    | <b>Korea</b>                                                           |

| Geographical Sobriquet (Epithet)                            | Related City / Country / Place (UPPCS Direct One-liner)                       |
|-------------------------------------------------------------|-------------------------------------------------------------------------------|
| Land of Kangaroos / Land of the Golden Fleece (golden wool) | <b>Australia</b>                                                              |
| Britain of the South                                        | <b>New Zealand</b>                                                            |
| Land of the Thunderbolt                                     | <b>Bhutan</b>                                                                 |
| Land of Golden Pagodas                                      | <b>Myanmar</b>                                                                |
| Land of Canals / Land of Rivers                             | <b>Pakistan</b> (of canals), <b>Bangladesh</b> (of rivers)                    |
| Gate of Tears                                               | <b>Bab-el-Mandeb Strait</b>                                                   |
| Island of Cloves                                            | <b>Zanzibar</b> (Tanzania)                                                    |
| City of Eternal Springs                                     | <b>Quito</b> (Ecuador)                                                        |
| Land of a Million Elephants                                 | <b>Laos</b>                                                                   |
| Loneliest Island in the World                               | <b>Tristan da Cunha</b>                                                       |
| Land of Promise                                             | <b>Mindanao</b> (Philippines)                                                 |
| City of Dreaming Spires / Garden of England                 | <b>Oxford</b> (England), <b>Kent</b> (England)                                |
| Sorrow of China                                             | <b>Hwang Ho river (Yellow River)</b>                                          |
| Emerald Isle / Land of the Lily                             | <b>Ireland</b> (Emerald), <b>Canada</b> (Land of the Lily)                    |
| Pamir Plateau                                               | <b>Roof of the World</b>                                                      |
| White Man's Grave                                           | <b>Guinea Coast</b> (West Africa)                                             |
| Land of Windmills                                           | <b>Netherlands</b>                                                            |
| Pearl of the Orient / City of the Golden Gate               | <b>Singapore</b> (Pearl), <b>San Francisco</b> (Golden Gate)                  |
| Quaker City / Forbidden City                                | <b>Philadelphia</b> (Quaker), <b>Lhasa-Tibet</b> (Forbidden)                  |
| Land of Cakes / Island of Pearls / City of Monuments        | <b>Scotland</b> (Cakes), <b>Bahrain</b> (Pearls), <b>Vienna</b> (Monuments)   |
| Blue Grass State / Paradise of the World                    | <b>Kentucky-USA</b> (Blue Grass), <b>Paris-France</b> (Paradise of the World) |



## B. Indian Geographical Sobriquets

| Geographical Sobriquet                          | Related City / State / River (UP Special and India Special) |
|-------------------------------------------------|-------------------------------------------------------------|
| Backbone of the economy of North India          | <b>Ganga river</b>                                          |
| Sorrow of Bengal / Sorrow of Bihar              | <b>Damodar river</b> (Bengal), <b>Kosi river</b> (Bihar)    |
| Land of Five Rivers / City of the Golden Temple | <b>Punjab</b> (land), <b>Amritsar</b> (city)                |
| Manchester of India                             | <b>Ahmedabad</b> (Gujarat)                                  |
| Gateway of India / City of Seven Islands        | <b>Mumbai</b> (Maharashtra)                                 |
| Queen of the Arabian Sea / Venice of the East   | <b>Kochi / Cochin</b> (Kerala)                              |
| Spice Garden of India                           | <b>Kerala</b>                                               |
| Switzerland of India                            | <b>Kashmir</b>                                              |
| Twin Cities                                     | <b>Hyderabad - Secunderabad</b>                             |
| Queen of Chhota Nagpur                          | <b>Netarhat</b> (Jharkhand)                                 |
| Queen of the Hills (India)                      | <b>Mussoorie</b> (Uttarakhand)                              |

### > Principal Straits of the World and their Connections

| Name of the Strait         | Which Seas/Gulfs does it Connect?              | Which Land Masses does it Separate? (Data Reference) |
|----------------------------|------------------------------------------------|------------------------------------------------------|
| <b>Strait of Gibraltar</b> | Mediterranean Sea and Atlantic Ocean           | Separates Spain (Europe) and Morocco (Africa).       |
| <b>Bering Strait</b>       | Bering Sea and Chukchi Sea / Arctic            | Separates Alaska (USA) and Russia (Asia).            |
| <b>Strait of Dover</b>     | English Channel and North Sea / Atlantic       | Separates Britain and France.                        |
| <b>Strait of Malacca</b>   | Andaman Sea / Bay of Bengal and Java Sea       | Malaya (Malaysia) and Sumatra (Indonesia).           |
| <b>Palk Strait</b>         | Bay of Bengal and Arabian Sea (Gulf of Mannar) | Separates India and Sri Lanka.                       |
| <b>Strait of Florida</b>   | Gulf of Mexico and Atlantic Ocean              | Florida (USA) and the West Indies.                   |
| <b>Strait of Hormuz</b>    | Persian Gulf and Gulf of Oman                  | Oman/UAE and the Iran region.                        |
| <b>Bab-el-Mandeb</b>       | Red Sea and Arabian Sea (Gulf of Aden)         | Yemen (Asia) and Djibouti (Africa).                  |

| Name of the Strait        | Which Seas/Gulfs does it Connect?      | Which Land Masses does it Separate? (Data Reference) |
|---------------------------|----------------------------------------|------------------------------------------------------|
| <b>Bosphorus Strait</b>   | Black Sea and Sea of Marmara           | The two parts of Istanbul/Turkey.                    |
| <b>Dardanelles Strait</b> | Sea of Marmara and Aegean Sea          | The Turkish region.                                  |
| <b>Sunda Strait</b>       | Java Sea and Indian Ocean              | The islands of Sumatra and Java.                     |
| <b>Bass Strait</b>        | Tasman Sea and Southern Sea            | The Australian mainland and Tasmania.                |
| <b>Davis Strait</b>       | Baffin Bay and Atlantic Ocean          | Greenland and Canada.                                |
| <b>Denmark Strait</b>     | North Atlantic and Arctic Ocean        | Greenland and Iceland.                               |
| <b>Torres Strait</b>      | Arafura Sea and Coral Sea              | Papua New Guinea and Australia.                      |
| <b>Makassar Strait</b>    | Java Sea and Celebes Sea               | The islands of Borneo and Sulawesi.                  |
| <b>Yucatán Strait</b>     | Gulf of Mexico and Caribbean Sea       | Mexico and Cuba.                                     |
| <b>Strait of Otranto</b>  | Adriatic Sea and Ionian Sea            | Italy and Albania.                                   |
| <b>North Channel</b>      | Irish Sea and Atlantic Ocean           | Ireland and Scotland.                                |
| <b>Luzon Strait</b>       | South China Sea and Philippine Sea     | Taiwan and Luzon island (Philippines).               |
| <b>Hudson Strait</b>      | Hudson Bay and Atlantic Ocean          | Baffin Island of Canada and the mainland.            |
| <b>Korea Strait</b>       | Sea of Japan and East China Sea        | South Korea and Japan.                               |
| <b>Strait of Magellan</b> | Pacific Ocean and South Atlantic Ocean | The Chilean mainland and Tierra del Fuego.           |

### > Important International Boundary Lines

| Name of the Boundary Line | Between which Countries / Regions?        |
|---------------------------|-------------------------------------------|
| <b>McMahon Line</b>       | India and China                           |
| <b>Radcliffe Line</b>     | India and Pakistan                        |
| <b>Durand Line</b>        | Pakistan and Afghanistan                  |
| <b>38th Parallel</b>      | South Korea and North Korea               |
| <b>49th Parallel</b>      | United States of America (USA) and Canada |
| <b>17th Parallel</b>      | Vietnam (North and South)                 |

| Name of the Boundary Line   | Between which Countries / Regions?    |
|-----------------------------|---------------------------------------|
| 31st Parallel               | Iran and Iraq                         |
| 20th / 22nd Parallel        | Sudan-Libya / Sudan-Egypt             |
| 25th Parallel               | Mauritania and Mali                   |
| Hindenburg / Oder-Neisse    | Poland and Germany                    |
| Maginot / Siegfried Line    | France and Germany                    |
| Mannerheim Line             | Russia and Finland                    |
| Blue Line                   | Lebanon and Israel                    |
| Green Line / UN Buffer Zone | Republic of Cyprus and Turkish Cyprus |

### > Principal Global Cities Located on the Banks of Rivers

| Name of the City | Name of the River on which Located                    | Name of the City | Name of the River on which Located |
|------------------|-------------------------------------------------------|------------------|------------------------------------|
| New York         | Hudson                                                | Hamburg          | Elbe                               |
| Lahore           | Ravi                                                  | Kyiv             | Dnieper                            |
| Cairo            | Nile                                                  | Rotterdam        | Rhine                              |
| Khartoum         | At the confluence of the White Nile and the Blue Nile | Quebec City      | St. Lawrence                       |
| Perth            | Swan                                                  | Warsaw           | Vistula                            |

### > Classification of Soil Particles and their Diameter Sizes

- **Special note:** The Uttar Pradesh Public Service Commission (UPPSC) asks direct questions to arrange the sizes of these particles in **ascending** or **descending** order.

| Name of the Soil Particle | Exact Diameter Size of the Particles                                                    |
|---------------------------|-----------------------------------------------------------------------------------------|
| 1. Gravel                 | Particles with a diameter of <b>more than 5.000 mm</b> ( <i>the largest particles</i> ) |
| 2. Fine gravel            | From 2.000 mm to 5.000 mm                                                               |
| 3. Coarse sand            | From 0.200 mm to 2.000 mm                                                               |
| 4. Fine sand              | From 0.020 mm to 0.200 mm                                                               |
| 5. Silt                   | From 0.002 mm to 0.020 mm                                                               |

| Name of the Soil Particle | Exact Diameter Size of the Particles                                                                                      |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 6. Clay                   | Particles with a diameter of <b>less than 0.002 mm</b> ( <i>the smallest particles - highest water-holding capacity</i> ) |

### > Local Names of Shifting (Jhum) Cultivation in the World

| Name of the Cultivation | Related Country / Geographical Region (UPPCS Direct Matching Special) |
|-------------------------|-----------------------------------------------------------------------|
| Ray                     | Vietnam                                                               |
| Masole                  | Congo (Zaire river valley)                                            |
| Fang (Fagang)           | Equatorial African countries                                          |
| Logan                   | Western Africa                                                        |
| Milpa                   | Mexico and Central America                                            |
| Conuco (Koko)           | Venezuela                                                             |
| Roca                    | Brazil                                                                |
| Chitemene               | Uganda, Zambia and Zimbabwe                                           |
| Ladang                  | Java and Indonesia                                                    |

### > Various Regional Names of Cyclones

| Regional Name of the Cyclone | Related Sea / Country / Geographical Region (UPPCS Direct Matching Special) |
|------------------------------|-----------------------------------------------------------------------------|
| Cyclone                      | Indian Ocean                                                                |
| Hurricane                    | Caribbean Sea / Atlantic Ocean                                              |
| Typhoon                      | China Sea                                                                   |
| Taifu                        | Japan                                                                       |
| Willy Willy                  | Northern Australia                                                          |
| Baguio                       | Philippines                                                                 |
| Tornado                      | America (USA)                                                               |

### > Important Ports of the World (Major Global Ports)

| Name of the Port    | Location / Related Country                              | Name of the Port  | Location / Related Country                             |
|---------------------|---------------------------------------------------------|-------------------|--------------------------------------------------------|
| Vancouver           | Canada (Pacific coast)                                  | Yokohama          | Japan                                                  |
| New Orleans         | United States of America (USA)                          | Manila            | Philippines                                            |
| Santos              | <b>Brazil</b> ( <i>the world-famous 'coffee port'</i> ) | Jakarta           | Indonesia                                              |
| Cape Town / Nolloth | South Africa ( <i>Nolloth: Port Diamond</i> )           | Auckland          | New Zealand                                            |
| Cairo / Alexandria  | Egypt                                                   | Fremantle         | Australia (western coast)                              |
| Copenhagen          | Denmark                                                 | Colombo           | Sri Lanka                                              |
| Lisbon              | Portugal                                                | Gwadar            | <b>Pakistan</b> (Balochistan province)                 |
| La Skhira / Tripoli | Tunisia (La Skhira), Libya (Tripoli)                    | Murmansk / Igarka | Russia ( <i>Murmansk remains open even in winter</i> ) |
| St. Petersburg      | Russia (Baltic coast)                                   |                   |                                                        |

### > Special Types of Culture (Special Culture Types)

| Name of the Culture / Production | Subject Matter (Commercial production of what?)                           |
|----------------------------------|---------------------------------------------------------------------------|
| Viticulture                      | Commercial-scale production of <b>grapes</b> .                            |
| Sericulture                      | Rearing of silkworms for <b>silk production</b> .                         |
| Pisciculture                     | The practice of <b>fish farming</b> on a commercial scale.                |
| Apiculture                       | <b>Beekeeping</b> for the production of honey and wax.                    |
| Oliviculture                     | Commercial-scale cultivation of the <b>olive</b> .                        |
| Horticulture                     | Commercial-scale <b>gardening</b> (fruits, flowers, vegetables).          |
| Arboriculture                    | Cultivation and propagation of special kinds of <b>trees and shrubs</b> . |

### > Transformation of Rocks (Metamorphic Rocks Classification)

| Igneous Rock (Original) | Metamorphosed Form | Sedimentary Rock (Original) | Metamorphosed Form |
|-------------------------|--------------------|-----------------------------|--------------------|
| Granite                 | Gneiss             | Shale                       | Shale schist       |
| Syenite                 | Syenite gneiss     | Sandstone                   | Quartzite          |
|                         |                    |                             |                    |

| Igneous Rock (Original) | Metamorphosed Form | Sedimentary Rock (Original) | Metamorphosed Form |
|-------------------------|--------------------|-----------------------------|--------------------|
| Gabbro                  | Serpentine         | Limestone                   | Marble             |
| Basalt                  | Schist             | Lignite coal                | Anthracite coal    |

JAI HO STUDY

03

# Indian Geography

GEOGRAPHY OF INDIA

## > India's Neighbouring Countries and Border States/UTs

| Neighbouring Country | Length of the Border | Associated Indian States / Union Territories (UT)                            |
|----------------------|----------------------|------------------------------------------------------------------------------|
| Bangladesh           | 4096.07 km           | West Bengal, Assam, Meghalaya, Tripura, Mizoram. <i>(the longest border)</i> |
| China                | 3488.00 km           | Ladakh (UT), Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh.       |
| Pakistan             | 3323.00 km           | Ladakh (UT), Jammu and Kashmir (UT), Punjab, Rajasthan, Gujarat.             |
| Nepal                | 1751.00 km           | Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Sikkim.                      |
| Myanmar              | 1643.00 km           | Arunachal Pradesh, Nagaland, Manipur, Mizoram.                               |
| Bhutan               | 699.00 km            | Sikkim, West Bengal, Assam, Arunachal Pradesh.                               |
| Afghanistan          | 106.00 km            | Ladakh (UT). <i>(the shortest border)</i>                                    |

## > Important Channels and Straits of India (Water Channels)

| Channel / Gulf / Passage | Geographical Location (between which?)                              |
|--------------------------|---------------------------------------------------------------------|
| 8 Degree Channel         | Between the Maldives and Minicoy                                    |
| 9 Degree Channel         | Between Lakshadweep and Minicoy                                     |
| 10 Degree Channel        | <b>Between Andaman and Nicobar</b> <i>(repeated often in UPPCS)</i> |
| Duncan Passage           | Between South Andaman and Little Andaman                            |
| Great Channel            | Between Sumatra of Indonesia and Nicobar                            |
| Coco Strait              | Between the Coco Islands of Myanmar and North Andaman               |
| Gulf of Mannar           | Between south-eastern Tamil Nadu and Sri Lanka                      |



### > Regional Classification of the Himalaya according to Sidney Burrard

| Part of the Himalaya | Between which Rivers is it Located?             | Total Length                                      |
|----------------------|-------------------------------------------------|---------------------------------------------------|
| Punjab Himalaya      | Between the <b>Indus</b> and the <b>Sutlej</b>  | 560 kilometres                                    |
| Kumaon Himalaya      | Between the <b>Sutlej</b> and the <b>Kali</b>   | 320 kilometres                                    |
| Nepal Himalaya       | Between the <b>Kali</b> and the <b>Teesta</b>   | <b>800 kilometres</b> ( <i>the longest part</i> ) |
| Assam Himalaya       | Between the <b>Teesta</b> and the <b>Dihang</b> | 750 kilometres                                    |

### > Mountains and Highest Peaks of the Himalaya / Central and South India

### > Classification of the Himalaya and Mountain Ranges (Chronology)

#### A. Order of Physiographic Division from North to South

Trans-Himalaya → Great Himalaya (Himadri) → Himachal (Lesser) Himalaya → Outer / Shiwalik Himalaya → Indo-Gangetic Plain

#### B. Correct Order of the Mountain Ranges from North to South

| Region                  | Correct Order of the Ranges from North to South (UPPCS Chronology Special)                                         |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| Trans-Himalayan ranges  | <b>Karakoram → Ladakh → Zaskar → Kailash</b>                                                                       |
| Lesser Himalayan ranges | <b>Pir Panjal</b> (J&K - the longest) → <b>Dhauladhar</b> (HP) → <b>Mussoorie</b> (UK) → <b>Mahabharat</b> (Nepal) |

### > Principal Mountain Peaks of the Great Himalaya

| Mountain Peak | Official Height (metres) |
|---------------|--------------------------|
| Mount Everest | <b>8,848.86 m</b>        |
| Kanchenjunga  | 8,598 m                  |
| Makalu        | 8,481 m                  |
| Dhaulagiri    | 8,172 m                  |
| Nanga Parbat  | 8,126 m                  |
| Annapurna     | 8,078 m                  |
| Nanda Devi    | 7,817 m                  |

### > The Five Highest Mountain Peaks of India

| Name of the Mountain Peak | Location        | Official Height (metres) |
|---------------------------|-----------------|--------------------------|
| 1. K2 (Godwin Austen)     | Karakoram Range | <b>8,611 m</b>           |
| 2. Kanchenjunga           | Sikkim          | <b>8,586 m</b>           |
| 3. Nanda Devi             | Uttarakhand     | 7,816 m                  |
| 4. Kamet                  | Uttarakhand     | 7,756 m                  |
| 5. Saltoro Kangri         | Ladakh          | 7,742 m                  |

### > Mountains and Highest Peaks of Central and South India

| Mountain Range / Hill | Name of the Highest Peak | Height (metres) | Special Details (UPPCS Facts)                           |
|-----------------------|--------------------------|-----------------|---------------------------------------------------------|
| Aravalli Range        | Guru Shikhar             | 1,722 m         | One of the oldest ranges in the world.                  |
| Satpura Range         | Dhupgarh                 | 1,350 m         | Located on the Mahadeo hills.                           |
| Maikal Range          | Amarkantak               | 1,065 m         | It is the <b>source of the Narmada and Son rivers</b> . |
| Chhota Nagpur Plateau | Parasnath Hill           | —               | A famous pilgrimage centre of Jainism.                  |
| Nilgiri Hills         | Doddabetta               | 2,637 m         | The highest peak of the Nilgiris.                       |
| Anamalai Hills        | Anamudi                  | <b>2,695 m</b>  | <b>The highest peak of South India.</b>                 |
| Cardamom (Elaichi)    | —                        | —               | The southernmost mountain region of peninsular India.   |

**Correct order of the South Indian hills from north to south:** Nilgiri → Anamalai → Cardamom.

### > Eastern Coastal Hills (Geographical Order from North to South)

| Name of the Hill                            | Location (State) | Geographical Order (North to South) |
|---------------------------------------------|------------------|-------------------------------------|
| Mahendragiri                                | Odisha           | 1 (northernmost)                    |
| Nallamalai / Velikonda / Palkonda / Nagari  | Andhra Pradesh   | 2                                   |
| Javadi / Shevaroy / Panchamalai / Sirumalai | Tamil Nadu       | 3 (southernmost)                    |

### > Principal Hills of North-East India

| Name of the State  | Principal Hills Located (UPPCS Matching Special)                                               |
|--------------------|------------------------------------------------------------------------------------------------|
| Arunachal Pradesh  | <b>Dafla → Miri → Abor → Mishmi → Patkai Bum</b> ( <i>order from west towards east/north</i> ) |
| Meghalaya          | <b>Garo → Khasi → Jaintia hills</b> ( <i>exact west-to-east order</i> )                        |
| Assam              | Mikir hills, North Rengma, Barail hills                                                        |
| Nagaland / Manipur | Naga hills (Nagaland), Barail/Bal hills (Manipur)                                              |
| Mizoram            | Mizo hills, Lushai hills                                                                       |

### > Short Quick Points on the Geographical Order of Mountain Ranges and Hills:

- **Correct order from north to south (N to S):** Karakoram → Ladakh → Zaskar → Pir Panjal.
- **Correct order from west to east (W to E):** Garo hills → Khasi hills → Jaintia hills → Mikir hills.

### > Geographical Order of Mountain Ranges and Hills (Pointers)

#### A. Correct Order from North to South (Geographical Order - N to S)

Karakoram → Ladakh → Zaskar → Pir Panjal

#### B. Correct Order from West to East (Geographical Order - W to E)

Garo → Khasi → Jaintia → Mikir

### > Exact North-to-South Order of the Principal Hills of the Eastern Ghats

| State          | Correct Geographical Order from North to South (N to S Chronology)         |
|----------------|----------------------------------------------------------------------------|
| Andhra Pradesh | <b>Nallamalai hills → Velikonda hills → Palkonda hills → Nagari hills</b>  |
| Tamil Nadu     | <b>Javadi hills → Shevaroy hills → Panchamalai hills → Sirumalai hills</b> |

### > Coastal Plains and Delta Regions of India

| Western Coastal Plain   | Geographical Extent                   | Eastern Coastal Plain and Delta                            | Geographical Location         |
|-------------------------|---------------------------------------|------------------------------------------------------------|-------------------------------|
| Konkan coast            | Between Daman (Maharashtra) and Goa   | <b>Utkal coast / Mahanadi delta</b>                        | Odisha                        |
| Kannada or Kanara coast | Between Goa and Mangaluru (Karnataka) | <b>Northern Circars coast / Godavari and Krishna delta</b> | Odisha and Andhra Pradesh     |
| Malabar coast           | From Mangaluru to Kanyakumari         | <b>Coromandel coast / Kaveri delta</b>                     | Andhra Pradesh and Tamil Nadu |

### > Principal Valleys of India

| State / UT        | Names of the Principal Valleys (UPPCS Matching Special)                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ladakh            | • Markha Valley • Suru Valley                                                                                                                                                   |
| Jammu and Kashmir | • Kashmir Valley ( <i>between the Great Himalaya and the Pir Panjal</i> )                                                                                                       |
| Himachal Pradesh  | • Kinnaur Valley • Parvati Valley ( <i>Kullu</i> ) • Sangla Valley • Malana Valley • Kullu Valley • Pangi Valley • Spiti Valley • Lahaul Valley • Kangra Valley • Chamba Valley |
| Uttarakhand       | • Doon Valley • Johar Valley • <b>Valley of Flowers (Chamoli)</b> • Dharma Valley • Sor Valley ( <i>Pithoragarh</i> ) • Nelang Valley ( <i>Uttarkashi</i> )                     |
| West Bengal       | • Neora Valley ( <i>Kalimpong, Darjeeling</i> )                                                                                                                                 |
| Sikkim            | • Chumbi Valley • Yumthang Valley                                                                                                                                               |
| Nagaland          | • Dzukou Valley                                                                                                                                                                 |
| Andhra Pradesh    | • Araku Valley                                                                                                                                                                  |
| Tamil Nadu        | • Cumbum Valley                                                                                                                                                                 |
| Kerala            | • <b>Silent Valley</b>                                                                                                                                                          |

### > Principal Mountain Passes of India

| State / UT        | Names of the Principal Passes (UPPCS Matching Special) |
|-------------------|--------------------------------------------------------|
| Ladakh            | Chang La pass, Khardung La pass                        |
| Uttarakhand       | Lipulekh pass, Mana pass, Niti pass                    |
| Sikkim            | Nathu La pass, Jelep La pass                           |
| Arunachal Pradesh | Bomdi La pass, Yang Yap pass, Diphu pass, Chankan pass |
| Manipur           | Tuju pass                                              |

### > Passes of the Western Ghats (correct chronological order from north to south):

Thal Ghat (Nashik-Mumbai) → Bhor Ghat (Mumbai-Pune) → Palghat (Coimbatore-Cochin) → Shencottah (Thiruvananthapuram-Madurai)

## > Principal Rivers of India and their Sources

| Name of the River | Source (Where does it rise?)                     | Location (State / Country) |
|-------------------|--------------------------------------------------|----------------------------|
| Indus             | Mansarovar lake (near the Bokhar Chu glacier)    | Tibet (China)              |
| Ganga             | Gangotri glacier (as the Bhagirathi)             | Uttarakhand                |
| Yamuna            | Yamunotri glacier (Bandarpunch peak)             | Uttarakhand                |
| Jhelum            | Verinag spring (Sheshnag lake, Pir Panjal range) | Jammu and Kashmir          |
| Ravi              | Rohtang pass (Kullu hills)                       | Himachal Pradesh           |
| Chambal           | Janapav hill (near Mhow)                         | Madhya Pradesh             |
| Kosi              | Saptakoshi region                                | Nepal                      |
| Narmada           | <b>Amarkantak hills</b>                          | Madhya Pradesh             |
| Godavari          | <b>Trimbakeshwar hills</b> (Nashik district)     | Maharashtra                |
| Krishna           | From near <b>Mahabaleshwar</b>                   | Maharashtra                |
| Kaveri            | <b>Brahmagiri hills</b> (Coorg district)         | Karnataka                  |

## > Ganga River System: Tributaries Joining from the Left and Right Banks

| Rivers Joining the Ganga from the Left Bank                                                                               | Rivers Joining the Ganga from the Right Bank |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| • Ramganga • Gomti • Ghaghara • Gandak • Burhi Gandak • Kosi • Mahananda • Brahmaputra (joins as the Padma in Bangladesh) | • Yamuna • Tons • Son                        |

**Correct order of the tributaries from west to east:**

Yamuna → Tons → Gomti → Ghaghara → Son → Gandak → Burhi Gandak → Kosi → Mahananda → Hooghly → Brahmaputra

## > Principal Rivers and Tributaries of South India (Peninsular India)

### A. Decreasing Order of Length of the Rivers:

Godavari → Krishna → Narmada → Mahanadi → Kaveri

### B. Tributary Systems of the Principal Rivers:

| Name of the Main River                  | Its Principal Tributaries (Extremely Important for UPPCS Matching) |
|-----------------------------------------|--------------------------------------------------------------------|
| <b>Godavari</b> ( <i>Vridha Ganga</i> ) | Purna, Pranhita, Wainganga, Penganga, Wardha, Manjira, Indravati.  |

| Name of the Main River | Its Principal Tributaries (Extremely Important for UPPCS Matching)             |
|------------------------|--------------------------------------------------------------------------------|
| Krishna                | Tungabhadra, Koyna, Ghatprabha, Malprabha, Musi, Bhima, Panchganga, Dudhganga. |
| Kaveri                 | Amaravati, Bhavani, Hemavati, Kabini.                                          |
| Mahanadi               | Shivnath, Hasdeo, Ib, Tel.                                                     |

### > Regional (River Valley) Division of the Himalaya

| Part of the Himalaya | Between which Rivers is it Located?                           | Total Length                                      |
|----------------------|---------------------------------------------------------------|---------------------------------------------------|
| Punjab Himalaya      | Between the <b>Indus</b> and the <b>Sutlej</b>                | 560 kilometres                                    |
| Kumaon Himalaya      | Between the <b>Sutlej</b> and the <b>Kali</b>                 | 320 kilometres                                    |
| Nepal Himalaya       | Between the <b>Kali</b> and the <b>Teesta</b>                 | <b>800 kilometres</b> ( <i>the longest part</i> ) |
| Assam Himalaya       | Between the <b>Teesta</b> and the <b>Dihang (Brahmaputra)</b> | 750 kilometres                                    |

### > Principal Waterfalls of India

| Waterfall                       | Related River     | Location / State          | Special Exam Fact                       |
|---------------------------------|-------------------|---------------------------|-----------------------------------------|
| Kunchikal                       | Varahi river      | Karnataka                 | The highest waterfall in India.         |
| Jog / Gersoppa / Mahatma Gandhi | Sharavati river   | Karnataka                 | —                                       |
| Shivasamudram                   | Kaveri river      | Karnataka                 | —                                       |
| Gokak                           | Ghataprabha river | Belagavi (Karnataka)      | —                                       |
| Dudhsagar                       | Mandovi river     | Goa and Karnataka border  | —                                       |
| Dhuandhar                       | Narmada river     | Jabalpur (Madhya Pradesh) | —                                       |
| Dugdhdhara / Kapildhara         | Narmada river     | Madhya Pradesh            | —                                       |
| Chachai                         | Bihad river       | Madhya Pradesh            | —                                       |
| Bhalkund / Rahatgarh            | Bina river        | Sagar (Madhya Pradesh)    | —                                       |
| Chitrakote                      | Indravati river   | Bastar (Chhattisgarh)     | Called " <b>the Niagara of India</b> ". |

| Waterfall             | Related River  | Location / State    | Special Exam Fact      |
|-----------------------|----------------|---------------------|------------------------|
| Tirathgarh            | Kanger river   | Chhattisgarh        | —                      |
| Amritdhara / Devdhara | Hasdeo river   | Chhattisgarh        | —                      |
| Kendai                | —              | Chhattisgarh        | —                      |
| Lodh / Burha Ghagh    | Burha river    | Latehar (Jharkhand) | —                      |
| Jonha / Gautamdharma  | Rarhu river    | Ranchi (Jharkhand)  | —                      |
| Dassam / Dasong       | Kanchi river   | Ranchi (Jharkhand)  | —                      |
| Sadni                 | Shankh river   | Jharkhand           | —                      |
| Chuliya               | Chambal river  | Rajasthan           | —                      |
| Duduma                | Machkund river | Koraput (Odisha)    | —                      |
| Pykara                | Pykara river   | Tamil Nadu          | In the Nilgiri region. |

### > Principal Lakes of India, State-wise

| State / UT        | Names of the Principal Lakes (Quick Pointers)                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| Jammu and Kashmir | Wular lake, Dal lake, Zanskar lake, Anantnag, Nagin, Verinag, Sheshnag, Tso Moriri, Manasbal.                             |
| Uttarakhand       | Nainital, Bhimtal, Naukuchiatal, Khurpatal, Roopkund ( <i>the mysterious lake</i> ), Sattal, Naini, Malatal.              |
| Rajasthan         | Pichola, Sambhar, Pachpadra, Rajsamand, Jaisamand, Didwana, Pushkar, Fateh Sagar, Udai Sagar, Nakki, Dhebar, Lunkaransar. |
| Sikkim            | <b>Gurudongmar</b> ( <i>at the highest altitude</i> ), Cholamu lake, Tsomgo lake.                                         |
| Kerala            | Sasthamkotta, Ashtamudi, Vembanad lake.                                                                                   |
| Tamil Nadu        | Berijam, Ooty, Kodaikanal, Chembarambakkam, Veeranam lake.                                                                |
| Assam             | Deepor Beel, Son Beel, Chapanal, Chandubi.                                                                                |
| Maharashtra       | Lonar ( <i>crater lake</i> ), Powai, Gorewada, Salim Ali Sarovar.                                                         |
| Telangana         | Hussain Sagar, Osman Sagar, Himayat Sagar.                                                                                |
| Gujarat           | Nal Sarovar, Narayan Sarovar, Hamirsar.                                                                                   |
| Andhra Pradesh    | Kolleru lake.                                                                                                             |
| Manipur           | <b>Loktak lake</b> ( <i>the largest freshwater lake of the North-East</i> )                                               |
|                   |                                                                                                                           |



| State / UT          | Names of the Principal Lakes (Quick Pointers) |
|---------------------|-----------------------------------------------|
| Meghalaya / Mizoram | Umiam (Meghalaya), Palak (Mizoram).           |
| Haryana / Punjab    | Surajkund (Haryana), Kanjli, Harike (Punjab). |

### › The Peninsular Plateau of India: At a Glance

| Physiographic Division                               | Principal Features and Divisions (UPPCS Core Points)                                                                                                                                                                                                                                                                                                                        | Highest Peaks / Height                                                                                                                              |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| General introduction                                 | <ul style="list-style-type: none"> <li>The <b>oldest and most stable landmass</b> of India (ancient igneous and metamorphic rocks).</li> <li>Area: <b>~16 lakh sq km (~48% of the country - the largest physical region)</b>.</li> <li>Shape: <b>triangular</b>; general slope: <b>from west to east</b>.</li> </ul>                                                        | —                                                                                                                                                   |
| 1. Central Highlands<br>(north of the Narmada river) | <ul style="list-style-type: none"> <li>Main parts: the Malwa, Bundelkhand, Baghelkhand and Chhota Nagpur plateaus.</li> <li><b>Malwa Plateau</b>: mainly in Madhya Pradesh, average height 300–600 metres.</li> <li><b>Chhota Nagpur Plateau</b>: mainly in Jharkhand (average height 700 metres). Extremely rich in mineral wealth (iron, coal, mica, bauxite).</li> </ul> | <ul style="list-style-type: none"> <li>Principal high peak of Chhota Nagpur: <b>Parasnath (1,365 metres)</b> (a Jain pilgrimage centre).</li> </ul> |
| 2. Deccan Plateau<br>(south of the Narmada river)    | <ul style="list-style-type: none"> <li>Main extent: Maharashtra, Karnataka, Telangana and Andhra Pradesh.</li> <li>Formation: mainly of <b>basalt rocks</b>, because of which <b>black soil (regur)</b> is widespread here.</li> </ul>                                                                                                                                      | —                                                                                                                                                   |
| 3. Aravalli and Vindhya-Satpura                      | <ul style="list-style-type: none"> <li><b>Aravalli</b>: one of the oldest mountain systems in the world.</li> <li><b>Vindhya and Satpura</b>: between them the Narmada flows westward in a rift valley.</li> </ul>                                                                                                                                                          | <ul style="list-style-type: none"> <li>Highest peak of the Aravalli: <b>Guru Shikhar (1,722 metres)</b>.</li> </ul>                                 |
| Drainage system (river slope)                        | <ul style="list-style-type: none"> <li><b>West-flowing rivers (rift valley)</b>: Narmada and Tapi (principal rivers).</li> <li><b>East-flowing rivers (according to the slope)</b>: Mahanadi, Godavari, Krishna and Kaveri.</li> </ul>                                                                                                                                      | —                                                                                                                                                   |

### > Western Ghats versus Eastern Ghats: Comparative Table

| Criterion / Feature                        | Western Ghats                                                                                                                                         | Eastern Ghats                                                                                                                                                                    |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Geographical extent</b>                 | • Parallel to the western coast of India, <b>from the Tapti valley to Kanyakumari</b> (~1,600 km).                                                    | • Parallel to the eastern coast of India, <b>from Odisha to Tamil Nadu</b> (~1,500 km).                                                                                          |
| <b>States covered</b>                      | • Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu.                                                                                                 | • Parts of Odisha, Andhra Pradesh, Telangana, Karnataka and Tamil Nadu.                                                                                                          |
| <b>Continuity and height</b>               | • <b>Higher (average 1,200 metres) and continuous.</b><br>Can be crossed only through passes.                                                         | • <b>Lower and discontinuous/fragmented</b> , because the east-flowing rivers have badly dissected them.                                                                         |
| <b>Highest peak</b>                        | • <b>Anamudi / Anaimudi (2,695 metres)</b> - located in Kerala ( <i>it is also the highest peak of South India</i> ).                                 | • <b>Jindhagada peak (1,690 metres)</b> - located in Andhra Pradesh. ( <i>Note: your data also mentions Mahendragiri, 1,501 m, Odisha, which is the second principal peak</i> ). |
| <b>Monsoon and rainfall</b>                | • They block the south-west monsoon and cause heavy rain on the western slope, while the interior of the Deccan becomes a <b>rain-shadow region</b> . | • Being parallel to the monsoon, such orographic rainfall is low here.                                                                                                           |
| <b>Constitutional/international status</b> | • Recognised as a <b>UNESCO World Natural Heritage Site</b> .                                                                                         | —                                                                                                                                                                                |
| <b>Meeting point</b>                       | The Western Ghats and Eastern Ghats meet at the Nilgiri hills; the highest peak of the Nilgiris is Doddabetta (2,637 metres).                         |                                                                                                                                                                                  |

### > Coastal Plains of India

| Western Coastal Plain     | States Covered      | Eastern Coastal Plain | States Covered |
|---------------------------|---------------------|-----------------------|----------------|
| 1. Gujarat coast          | Gujarat             | 1. Utkal coast        | Odisha         |
| 2. Konkan coast           | Maharashtra and Goa | 2. Andhra coast       | Andhra Pradesh |
| 3. Kannada / Kanara coast | Karnataka           | 3. Coromandel coast   | Tamil Nadu     |

| Western Coastal Plain | States Covered | Eastern Coastal Plain | States Covered |
|-----------------------|----------------|-----------------------|----------------|
| 4. Malabar coast      | Kerala         |                       |                |

### > Major Soils of India

| Type of Soil                                    | Main Characteristics                                                                                                                                                                                                   | Abundance and Deficiency of Minerals / Elements                                                                                                                                                          | Main Area of Extent                                                                   |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Red and yellow soil                             | <ul style="list-style-type: none"> <li>Developed on crystalline igneous rocks in areas of low rainfall.</li> <li>In the eastern and southern parts of the Deccan Plateau (~10.6% of the geographical area).</li> </ul> | <ul style="list-style-type: none"> <li><b>Abundance:</b> rich in potash.</li> <li><b>Deficiency:</b> phosphorus, humus, nitrogen and magnesia.</li> </ul>                                                | Eastern and southern parts of the Deccan Plateau, Chhattisgarh, Odisha.               |
| Black soil<br>(Regur / cotton soil / Chernozem) | <ul style="list-style-type: none"> <li>Has the property of <b>self-ploughing</b> (high moisture-retention capacity).</li> <li>Most suitable for cotton cultivation.</li> </ul>                                         | <ul style="list-style-type: none"> <li><b>Abundance:</b> lime, iron, magnesia, aluminium in abundance.</li> <li><b>Deficiency:</b> phosphate, nitrogen and humus.</li> </ul>                             | Maharashtra, Gujarat, Madhya Pradesh, Rajasthan, Chhattisgarh, Odisha, Tamil Nadu.    |
| Laterite soil                                   | <ul style="list-style-type: none"> <li>Formed by the <b>leaching of lime and silica</b> in areas with rainfall of more than 200 cm.</li> </ul>                                                                         | <ul style="list-style-type: none"> <li><b>Abundance:</b> iron oxide and aluminium oxide in abundance.</li> <li><b>Deficiency:</b> nitrogen, phosphoric acid, potash, lime and organic matter.</li> </ul> | Eastern Ghats, Rajmahal hills, Satpura, the hills of Meghalaya and the Western Ghats. |

### > Principal Nuclear Power Plants of India

| Nuclear Power Plant | Location (State) | Special Exam Fact                                              |
|---------------------|------------------|----------------------------------------------------------------|
| Narora              | Uttar Pradesh    | Located in Bulandshahr district of Uttar Pradesh (UP Special). |
| Tarapur             | Maharashtra      | India's very first nuclear power plant (1969).                 |
| Rawatbhata          | Rajasthan        | Established with the cooperation of Canada.                    |
| Kalpakkam           | Tamil Nadu       | Also called the Indira Gandhi Centre for Atomic Research.      |
| Kakrapar            | Gujarat          | Located near the Tapi river.                                   |
| Kaiga               | Karnataka        | In the Kali river region.                                      |

## > Principal Hydroelectric Stations and Projects of India

| State / UT                   | Name of the Hydroelectric Project / Plant                                                                                                                                                                                                     | Related River                                                                                    | Special Facts for the UPPCS Exam (Exam Insights)                                                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Andaman and Nicobar          | Kalpong project                                                                                                                                                                                                                               | Kalpong river                                                                                    | • The <b>first hydroelectric project</b> of this Union Territory (UT).                                                                                                                                                               |
| Uttarakhand                  | <ul style="list-style-type: none"> <li>• <b>Tehri project</b></li> <li>• Koteswar</li> <li>• Tanakpur</li> <li>• Dhauliganga</li> </ul>                                                                                                       | Bhagirathi + Bhilangana<br>Bhagirathi<br>Sharda (Mahakali)<br>Dhauliganga                        | <ul style="list-style-type: none"> <li>• The Tehri dam is the <b>highest dam</b> of India (260.5 metres).</li> <li>• Tanakpur is located near the India-Nepal border.</li> </ul>                                                     |
| Himachal Pradesh             | <ul style="list-style-type: none"> <li>• <b>Bhakra Nangal</b></li> <li>• Nathpa Jhakri</li> <li>• Karcham-Wangtoo</li> <li>• Kol Dam</li> <li>• Dehar</li> <li>• Chamera (I, II, III)</li> <li>• Baira Siul</li> <li>• Parbati-III</li> </ul> | Sutlej<br>Sutlej<br>Sutlej<br>Sutlej<br>Beas<br>Ravi<br>Baira (tributary of the Ravi)<br>Parbati | <ul style="list-style-type: none"> <li>• Bhakra Nangal is one of the largest multipurpose projects of India (Himachal-Punjab border).</li> <li>• Nathpa Jhakri is one of the largest underground projects of the country.</li> </ul> |
| Jammu-Kashmir (UT)           | <ul style="list-style-type: none"> <li>• Salal</li> <li>• Uri (I, II)</li> <li>• Dulhasti</li> <li>• Kishanganga</li> <li>• Sewa-II</li> </ul>                                                                                                | Chenab<br>Jhelum<br>Chenab<br>Kishanganga (tributary of the Jhelum)<br>Sewa river                | <ul style="list-style-type: none"> <li>• The Kishanganga project has been in dispute between India and Pakistan over the Indus Waters Treaty.</li> <li>• Salal and Dulhasti are '<b>run-of-the-river</b>' projects.</li> </ul>       |
| Karnataka                    | <ul style="list-style-type: none"> <li>• <b>Shivasamudram</b></li> <li>• Sharavati</li> <li>• Kali Nadi</li> <li>• Varahi</li> </ul>                                                                                                          | Kaveri<br>Sharavati<br>Kali river<br>Varahi                                                      | <ul style="list-style-type: none"> <li>• Shivasamudram (1902) is one of the <b>earliest hydroelectric plants of Asia</b>.</li> <li>• The Sharavati project is located near the famous 'Jog Falls'.</li> </ul>                        |
| Kerala                       | <ul style="list-style-type: none"> <li>• <b>Idukki project</b></li> <li>• Sabarigiri</li> </ul>                                                                                                                                               | Periyar<br>Pamba                                                                                 | • Idukki is a distinctive <b>arch dam (double curvature arch dam)</b> on the Periyar river.                                                                                                                                          |
| Gujarat                      | • <b>Sardar Sarovar</b>                                                                                                                                                                                                                       | Narmada                                                                                          | • It is a joint project of four states (Gujarat, Madhya Pradesh, Maharashtra and Rajasthan).                                                                                                                                         |
| Andhra Pradesh and Telangana | <ul style="list-style-type: none"> <li>• <b>Srisaillam project</b></li> <li>• <b>Nagarjuna Sagar</b></li> <li>• Pulichintala</li> <li>• Machkund</li> </ul>                                                                                   | Krishna<br>Krishna<br>Krishna<br>Machkund                                                        | <ul style="list-style-type: none"> <li>• Nagarjuna Sagar is a giant masonry dam located on the border of Andhra Pradesh and Telangana.</li> <li>• Machkund is a joint project of Andhra Pradesh and Odisha.</li> </ul>               |

| State / UT           | Name of the Hydroelectric Project / Plant                                                                                                                 | Related River                                               | Special Facts for the UPPCS Exam (Exam Insights)                                                                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Madhya Pradesh       | <ul style="list-style-type: none"> <li>Indira Sagar</li> <li>Omkareshwar</li> <li>Gandhi Sagar</li> </ul>                                                 | Narmada<br>Narmada<br>Chambal                               | <ul style="list-style-type: none"> <li>Gandhi Sagar is the first of the three principal dams built on the Chambal river.</li> </ul>                                                                                                                       |
| Sikkim               | <ul style="list-style-type: none"> <li>Teesta-III and Teesta-V</li> <li>Rangit</li> </ul>                                                                 | Teesta<br>Rangit (tributary of the Teesta)                  | <ul style="list-style-type: none"> <li>The Teesta project is one of the principal hydroelectric projects of north-eastern India.</li> </ul>                                                                                                               |
| North-East India     | <ul style="list-style-type: none"> <li>Loktak (<b>Manipur</b>)</li> <li>Tuirial (<b>Mizoram</b>)</li> <li>Kameng, Subansiri (<b>Arunachal</b>)</li> </ul> | Leimatak/Loktak lake<br>Tuirial river<br>Kameng / Subansiri | <ul style="list-style-type: none"> <li>The Loktak project is linked with Loktak, the largest natural freshwater lake of the North-East.</li> <li>The Subansiri Lower Project is the largest hydroelectric project under construction in India.</li> </ul> |
| Maharashtra          | <ul style="list-style-type: none"> <li>Koyna</li> <li>Bhira</li> </ul>                                                                                    | Koyna (tributary of the Krishna)<br>Kundalika               | <ul style="list-style-type: none"> <li>The Koyna project is also called the 'lifeline' of Maharashtra.</li> </ul>                                                                                                                                         |
| Odisha and Jharkhand | <ul style="list-style-type: none"> <li>Hirakud, Balimela (<b>Odisha</b>)</li> <li>Subarnarekha-I and II (<b>Jharkhand</b>)</li> </ul>                     | Mahanadi / Sileru<br>Subarnarekha                           | <ul style="list-style-type: none"> <li>The Hirakud dam is the <b>longest dam</b> of India.</li> </ul>                                                                                                                                                     |
| West Bengal          | <ul style="list-style-type: none"> <li>Teesta Low Dam-III and IV</li> </ul>                                                                               | Teesta                                                      | <ul style="list-style-type: none"> <li>Located on the lower course of the Teesta river in the Darjeeling region.</li> </ul>                                                                                                                               |

### > Principal Thermal Power Plants of India

| Name of the State           | Names of the Thermal Power Plants (UPPCS Matching Special)                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Uttar Pradesh               | <ul style="list-style-type: none"> <li>Singrauli • Rihand • Anpara • Obra • NTPC Dadri (<i>extremely important for UP Special</i>)</li> </ul> |
| Chhattisgarh / Odisha       | <ul style="list-style-type: none"> <li>Korba (Chhattisgarh) • Talcher (Odisha)</li> </ul>                                                     |
| Madhya Pradesh              | <ul style="list-style-type: none"> <li>Vindhyachal • Satpura • Sanjay Gandhi</li> </ul>                                                       |
| Maharashtra / Gujarat       | <ul style="list-style-type: none"> <li>Khaperkheda, Chandrapur Super, Koradi (MH) • Ukai, Mundra, Surat (GJ)</li> </ul>                       |
| Jharkhand / West Bengal     | <ul style="list-style-type: none"> <li>Bokaro, Patratu (Jharkhand) • Farakka (West Bengal)</li> </ul>                                         |
| Delhi / Haryana / Rajasthan | <ul style="list-style-type: none"> <li>Badarpur (Delhi) • Rajiv Gandhi (Haryana) • Kota, Suratgarh, Chhabra (Rajasthan)</li> </ul>            |
| Tamil Nadu                  | <ul style="list-style-type: none"> <li>Neyveli (<i>lignite based</i>) • Ennore • Tuticorin • North Chennai</li> </ul>                         |

| Name of the State               | Names of the Thermal Power Plants (UPPCS Matching Special)        |
|---------------------------------|-------------------------------------------------------------------|
| Telangana / Andhra Pradesh / UK | • Ramagundam (Telangana) • Simhadri (AP) • Kashipur (Uttarakhand) |

### > Minerals and their Principal Producing Areas in India (Mines - UPPCS Special)

| Name of the Mineral     | State                                                               | Principal Producing Areas / Famous Mines (Extremely Important for UPPCS Matching Questions)                                                                                                                                                                                                                            |
|-------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iron ore                | Odisha<br>Chhattisgarh<br>Karnataka<br>Jharkhand / Goa              | <ul style="list-style-type: none"> <li>• Keonjhar, Sundargarh, Mayurbhanj</li> <li>• <b>Bailadila</b>, Dantewada (<i>high-quality haematite</i>)</li> <li>• Ballari, Chitradurga, Tumakuru</li> <li>• Singhbhum (Jharkhand), North Goa (Goa)</li> </ul>                                                                |
| Coal                    | Jharkhand<br>West Bengal<br>Odisha / Chhattisgarh<br>MP / Telangana | <ul style="list-style-type: none"> <li>• <b>Jharia</b> (<i>the largest</i>), Bokaro, Giridih, Karanpura</li> <li>• <b>Raniganj</b> (<i>the oldest coalfield of India</i>)</li> <li>• Talcher, Ib Valley (Odisha), Korba (Chhattisgarh)</li> <li>• Singrauli, Umaria (Madhya Pradesh), Singareni (Telangana)</li> </ul> |
| Copper                  | Rajasthan<br>Madhya Pradesh / Jharkhand                             | <ul style="list-style-type: none"> <li>• <b>Khetri</b>, Jhunjhunu</li> <li>• <b>Malanjkhand</b>, Balaghat (MP), Singhbhum (Jharkhand)</li> </ul>                                                                                                                                                                       |
| Bauxite (aluminium ore) | Odisha<br>Jharkhand / Gujarat / Chhattisgarh                        | <ul style="list-style-type: none"> <li>• Koraput, Kalahandi, Rayagada</li> <li>• Lohardaga (Jharkhand), Kutch, Jamnagar (Gujarat), Surguja, Kabirdham (CG)</li> </ul>                                                                                                                                                  |
| Manganese               | Madhya Pradesh<br>Maharashtra / Odisha / Karnataka                  | <ul style="list-style-type: none"> <li>• Balaghat</li> <li>• Nagpur, Bhandara, Sindhudurg (MH), Sundargarh, Keonjhar (OD), Ballari (KN)</li> </ul>                                                                                                                                                                     |
| Uranium                 | Jharkhand<br>Andhra Pradesh / Telangana                             | <ul style="list-style-type: none"> <li>• <b>Jaduguda</b>, Bhatin, Narwapahar</li> <li>• Tummalapalle, Kadapa (AP), Lambapur-Peddagattu (Telangana)</li> </ul>                                                                                                                                                          |
| Mica                    | Jharkhand / Andhra Pradesh / Raj.                                   | <ul style="list-style-type: none"> <li>• Koderma (<i>the mica capital</i>), Giridih, Hazaribagh (JH), Nellore (AP), Bhilwara, Ajmer (RJ)</li> </ul>                                                                                                                                                                    |
| Gold                    | Karnataka / Jharkhand                                               | <ul style="list-style-type: none"> <li>• <b>Hutti, Kolar (KGF)</b> (Karnataka), Singhbhum region (Jharkhand)</li> </ul>                                                                                                                                                                                                |
| Lead and zinc           | Rajasthan                                                           | <ul style="list-style-type: none"> <li>• <b>Zawar</b>, Rampura-Agucha, Rajpura-Dariba (<i>mixed ore</i>)</li> </ul>                                                                                                                                                                                                    |
| Silver                  | Rajasthan / Karnataka                                               | <ul style="list-style-type: none"> <li>• Zawar, Rampura-Agucha (RJ), Hutti region (KN)</li> </ul>                                                                                                                                                                                                                      |
| Chromite / nickel       | Odisha                                                              | <ul style="list-style-type: none"> <li>• <b>Sukinda valley</b> (Jajpur) (<i>India's top region for both chromite and nickel</i>)</li> </ul>                                                                                                                                                                            |

| Name of the Mineral | State                      | Principal Producing Areas / Famous Mines (Extremely Important for UPPCS Matching Questions) |
|---------------------|----------------------------|---------------------------------------------------------------------------------------------|
| Rock phosphate      | Rajasthan / Madhya Pradesh | • Jhamarkotra, Udaipur (RJ), Jhabua (MP)                                                    |
| Diamond             | Madhya Pradesh             | • <b>Panna</b> ( <i>the only active diamond mine in India</i> )                             |
| Tin                 | Chhattisgarh               | • <b>Bastar, Dantewada</b> ( <i>Chhattisgarh is the only tin-producing state in India</i> ) |
| Gypsum              | Rajasthan / Jammu-Kashmir  | • Bikaner, Nagaur, Jaisalmer, Barmer (RJ), Kishtwar, Udhampur (J&K)                         |
| Limestone           | Raj. / MP / CG / AP        | • Chittorgarh (RJ), Satna, Katni (MP), Raipur (CG), Kadapa, Kurnool (AP)                    |
| Dolomite            | Chhattisgarh / Odisha / MP | • Bastar, Bilaspur (CG), Sundargarh (OD), Katni, Jabalpur (MP)                              |
| Magnesite           | Tamil Nadu / Uttarakhand   | • Salem, Dharmapuri (TN), Bageshwar, Pithoragarh (UK)                                       |

### > Types of Coal in India and their Features

| Type of Coal  | Carbon Content | Quality and Main Features (UPPCS Core Pointers)                                                                                                                                                                                            | Main Areas of Occurrence                    |
|---------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1. Anthracite | 90% to 95%     | <ul style="list-style-type: none"> <li>It is a hard coal of the <b>best / highest grade</b>.</li> <li>Gives no smoke while burning and provides the most heat.</li> </ul>                                                                  | From <b>Jammu-Kashmir</b> .                 |
| 2. Bituminous | 55% to 80%     | <ul style="list-style-type: none"> <li>It is second-grade coal. Most of India's coal is of this grade.</li> <li>The coal of the <b>Gondwana period</b> is of this type. It represents 80 of the country's total 113 coalfields.</li> </ul> | Jharkhand, Odisha, Chhattisgarh, W. Bengal. |
| 3. Lignite    | 40% to 55%     | <ul style="list-style-type: none"> <li>It is also called "<b>brown coal</b>" and is of inferior quality.</li> <li>Because of its high sulphur content (2 to 7%), it is unsuitable for some industries.</li> </ul>                          | Tamil Nadu (Neyveli).                       |
| 4. Peat       | Less than 40%  | <ul style="list-style-type: none"> <li>It is coal of the <b>lowest grade (the poorest)</b>, with very high moisture.</li> <li>On burning it gives off a great deal of ash and smoke.</li> </ul>                                            | Marshy areas.                               |



### > Principal Petroleum-Producing Areas of India (Oil Fields)

| State / Offshore Area       | Principal Oil-Producing Wells / Basins (Core Field Pointers)                                                     |
|-----------------------------|------------------------------------------------------------------------------------------------------------------|
| Assam                       | • <b>Digboi</b> ( <i>India's very first oil well</i> ) • Naharkatiya • Moran • Rudrasagar                        |
| Gujarat                     | • <b>Ankleshwar</b> • Kalol • Mehsana • Navagam • Kosamba                                                        |
| Mumbai offshore             | • <b>Mumbai High</b> • Bassein and satellite fields ( <i>India's principal producer</i> )                        |
| Rajasthan                   | • Barmer • <b>Mangala</b> • <b>Bhagyam</b> • <b>Aishwarya</b> ( <i>names have been asked directly in UPPCS</i> ) |
| Andhra Pradesh / Tamil Nadu | • Krishna-Godavari basin (AP) • Kaveri basin, Nagapattinam area (TN)                                             |
| Arunachal Pradesh / Tripura | • Kharsang, Namchik-Namphuk (AR), Baramura, Trishna, Rokhia (TR)                                                 |

### > Additional UPPCS Trending Facts Related to Minerals and Energy (Revision)

- **Largest producer of bauxite:** Odisha (Koraput-Kalahandi belt).
- **Monopoly of chromite:** The Sukinda valley of Odisha produces more than about 90% of the country's chromite.
- **Famous copper refinery:** Khetri (Rajasthan), operated by Hindustan Copper Limited.
- **Largest reserve of lignite coal:** Neyveli (Tamil Nadu).

### > Miscellaneous

### > Principal Tribes of India and their Geographical Distribution

| Name of the Tribe             | Main States / Union Territories of Distribution (UPPCS Matching Special)                             |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| Bhil                          | Rajasthan, Gujarat, Madhya Pradesh, Maharashtra ( <i>one of the largest tribal groups of India</i> ) |
| Gond                          | Madhya Pradesh, Chhattisgarh, Maharashtra, Telangana, Andhra Pradesh, Odisha, Uttar Pradesh          |
| Santhal                       | Jharkhand, West Bengal, Bihar, Odisha                                                                |
| Munda / Oraon                 | Jharkhand, Odisha, West Bengal, Chhattisgarh                                                         |
| Ho / Kharia / Birhor / Pahari | Jharkhand, Odisha, Chhattisgarh                                                                      |
| Garos, Khasi, Jaintia         | Meghalaya ( <i>the tribes are named after these very hills</i> )                                     |
| Bodo                          | Assam                                                                                                |

| Name of the Tribe        | Main States / Union Territories of Distribution (UPPCS Matching Special)                    |
|--------------------------|---------------------------------------------------------------------------------------------|
| Naga                     | Nagaland, Manipur, Arunachal Pradesh                                                        |
| Lepcha / Bhutia          | Sikkim, West Bengal                                                                         |
| Toda / Kota              | Tamil Nadu ( <i>the Toda tribe lives on the Nilgiri hills and is famous for polyandry</i> ) |
| Irula / Kurumba          | Tamil Nadu, Kerala                                                                          |
| Chenchu / Koya           | Andhra Pradesh, Telangana, Chhattisgarh, Odisha                                             |
| Warli / Katkari          | Maharashtra, Gujarat                                                                        |
| Sahariya / Baiga / Korku | Madhya Pradesh, Chhattisgarh, Rajasthan, Uttar Pradesh, Maharashtra                         |
| Jarawa / Sentinelese     | Andaman and Nicobar Islands ( <i>extremely vulnerable PVTG groups</i> )                     |

### > Principal International Airports of India

| Name of the Airport        | Location (City, State)   | Name of the Airport         | Location (City, State)      |
|----------------------------|--------------------------|-----------------------------|-----------------------------|
| Chaudhary Charan Singh     | Lucknow (Uttar Pradesh)  | Indira Gandhi               | New Delhi                   |
| Lal Bahadur Shastri        | Varanasi (Uttar Pradesh) | Chhatrapati Shivaji Maharaj | Mumbai (Maharashtra)        |
| Netaji Subhas Chandra Bose | Kolkata (West Bengal)    | Kempegowda                  | Bengaluru (Karnataka)       |
| Sardar Vallabhbhai Patel   | Ahmedabad (Gujarat)      | Rajiv Gandhi                | Hyderabad (Telangana)       |
| Jai Prakash Narayan        | Patna (Bihar)            | Chennai International       | Chennai (Tamil Nadu)        |
| Lokpriya Gopinath Bordoloi | Guwahati (Assam)         | Cochin International        | Kochi (Kerala)              |
| Birsa Munda                | Ranchi (Jharkhand)       | Thiruvananthapuram          | Thiruvananthapuram (Kerala) |
| Biju Patnaik               | Bhubaneswar (Odisha)     | Goa International           | Mopa (Goa)                  |
| Jaipur International       | Jaipur (Rajasthan)       | Dr. Babasaheb Ambedkar      | Nagpur (Maharashtra)        |
| Srinagar International     | Srinagar (Jammu-Kashmir) | Sri Guru Ram Dass Ji        | Amritsar (Punjab)           |

### > Railway Zones of India and their Headquarters

| Railway Zone           | Headquarters                                  | Railway Zone               | Headquarters                   |
|------------------------|-----------------------------------------------|----------------------------|--------------------------------|
| Central Railway        | Mumbai (Chhatrapati Shivaji Maharaj Terminus) | South Western Railway      | Hubballi (Karnataka)           |
| Western Railway        | Mumbai Central                                | West Central Railway       | Jabalpur (Madhya Pradesh)      |
| Northern Railway       | New Delhi                                     | North Western Railway      | Jaipur (Rajasthan)             |
| North Central Railway  | Prayagraj (Uttar Pradesh)                     | North Eastern Railway      | Gorakhpur (Uttar Pradesh)      |
| Eastern Railway        | Kolkata                                       | East Central Railway       | Hajipur (Bihar)                |
| South Eastern Railway  | Kolkata                                       | East Coast Railway         | Bhubaneswar (Odisha)           |
| Metro Railway, Kolkata | Kolkata                                       | South Coast Railway        | Visakhapatnam (Andhra Pradesh) |
| Southern Railway       | Chennai (Tamil Nadu)                          | Northeast Frontier Railway | Maligaon, Guwahati (Assam)     |
| South Central Railway  | Secunderabad (Telangana)                      | South East Central Railway | Bilaspur (Chhattisgarh)        |

### > Water Transport in India: Principal National Waterways

- **Inland water transport:** Transport of goods and passengers within the country through rivers, canals, backwaters (kayals) and lakes.
- **Total number declared:** At present **111 National Waterways** have been declared in India.

| National Waterway | River System / Area of Extent  | Exact Extent (from where to where)      | Total Length | Special Fact                                   |
|-------------------|--------------------------------|-----------------------------------------|--------------|------------------------------------------------|
| NW - 1            | Ganga-Bhagirathi-Hooghly       | Prayagraj to Haldia                     | 1,620 km     | The longest National Waterway of India.        |
| NW - 2            | Brahmaputra river              | Sadiya to Dhubri (Assam)                | 891 km       | The main waterway of north-eastern India.      |
| NW - 3            | West Coast Canal               | Kottapuram to Kollam (Kerala)           | 205 km       | India's first waterway navigable for 24 hours. |
| NW - 4            | Godavari-Krishna rivers        | Kakinada to the Puducherry canal system | —            | South-eastern coastal network.                 |
| NW - 5            | Brahmani-Mahanadi river system | Along with the East Coast Canal         | —            | Odisha/West Bengal region.                     |

| National Waterway | River System / Area of Extent | Exact Extent (from where to where) | Total Length | Special Fact                            |
|-------------------|-------------------------------|------------------------------------|--------------|-----------------------------------------|
| NW - 16           | Barak river                   | Bhanga to Lakhimpur (Assam)        | —            | A strategic waterway of the North-East. |

### > Principal Agricultural Commodity Boards of India

| Name of the Board      | Headquarters (Location) |
|------------------------|-------------------------|
| Coffee Board           | Bengaluru (Karnataka)   |
| Rubber Board           | Kottayam (Kerala)       |
| Tea Board              | Kolkata (West Bengal)   |
| National Jute Board    | Kolkata (West Bengal)   |
| Tobacco Board          | Guntur (Andhra Pradesh) |
| Grape Processing Board | Pune (Maharashtra)      |

### > Principal Oil Refineries of India

| Name of the Refinery | Year of Establishment | Location (State)                              |
|----------------------|-----------------------|-----------------------------------------------|
| Digboi               | 1901                  | Assam ( <i>the oldest refinery of India</i> ) |
| Mumbai (Trombay)     | 1955                  | Maharashtra                                   |
| Visakhapatnam        | 1957                  | Andhra Pradesh                                |
| Noonmati (Guwahati)  | 1962                  | Assam                                         |
| Barauni              | 1964                  | Bihar                                         |
| Koyali               | 1965                  | Gujarat                                       |
| Kochi                | 1966                  | Kerala                                        |
| Haldia               | 1975                  | West Bengal                                   |
| Mumbai (BP)          | 1979                  | Maharashtra                                   |
| Mathura              | 1982                  | Uttar Pradesh                                 |
| Bangalore            | 1988                  | Karnataka                                     |
| Nagapattinam         | 1993                  | Tamil Nadu                                    |
| Panipat              | 1998                  | Haryana                                       |

| Name of the Refinery | Year of Establishment | Location (State) |
|----------------------|-----------------------|------------------|
| Numaligarh           | 1999                  | Assam            |
| Tatipaka             | 2001                  | Andhra Pradesh   |
| Paradip              | 2016                  | Odisha           |

### > Principal Steel Plants and their Establishment

| Steel Plant | Year of Establishment | Collaborating Country (with whose help?) | Location (State) |
|-------------|-----------------------|------------------------------------------|------------------|
| Bhilai      | 1955                  | Russia (former Soviet Union)             | Chhattisgarh     |
| Rourkela    | 1959                  | Germany                                  | Odisha           |
| Durgapur    | 1959                  | Britain                                  | West Bengal      |
| Bokaro      | 1964                  | Russia (former Soviet Union)             | Jharkhand        |

### > Principal Aluminium Companies of India

| Name of the Company | Year of Establishment | Collaborating Country          | Principal Industrial Centres                                |
|---------------------|-----------------------|--------------------------------|-------------------------------------------------------------|
| INDALCO             | 1938                  | Canada                         | J.K. Nagar (West Bengal), Muri (Jharkhand), Alwaye (Kerala) |
| HINDALCO            | 1958                  | United States of America (USA) | Renukoot (Uttar Pradesh)                                    |
| BALCO               | 1965                  | Soviet Union (USSR)            | Korba (Chhattisgarh), Koyna (Maharashtra)                   |
| MALCO               | 1965                  | Italy                          | Mettur, Salem (Tamil Nadu)                                  |
| VEDANTA             | 1976                  | Germany                        | Jharsuguda (Odisha)                                         |
| NALCO               | 1981                  | France                         | Damanjodi (Odisha), Angul (Odisha)                          |

### > Principal Mineral Institutions

| Name of the Mineral Institution                | Headquarters / Location (Place) |
|------------------------------------------------|---------------------------------|
| Hindustan Copper Limited (HCL)                 | Khetri (Rajasthan)              |
| Hindustan Zinc Limited (HZL)                   | Udaipur (Rajasthan)             |
| Indian Bureau of Mines (IBM)                   | Nagpur (Maharashtra)            |
| Mineral Exploration Corporation Limited (MECL) | Nagpur (Maharashtra)            |

| Name of the Mineral Institution               | Headquarters / Location (Place)           |
|-----------------------------------------------|-------------------------------------------|
| National Metallurgical Laboratory             | Jamshedpur (Jharkhand - Burmahmines area) |
| Central Institute of Mining and Fuel Research | Dhanbad (Jharkhand)                       |

### > Principal River Valley Projects of India

| Project             | Related River           | Beneficiary States / UT (UPPCS Matching Special)    |
|---------------------|-------------------------|-----------------------------------------------------|
| Sabarigiri          | Kakki / Pamba           | Kerala                                              |
| Nagarjuna Sagar     | Krishna                 | Andhra Pradesh and Telangana ( <i>masonry dam</i> ) |
| Pochampad           | Godavari                | Telangana                                           |
| Idukki              | Periyar                 | Kerala                                              |
| Pavana              | Pavana                  | Maharashtra                                         |
| Nathpa Jhakri       | Sutlej                  | Himachal Pradesh                                    |
| Suil                | Suil                    | Himachal Pradesh                                    |
| Bansagar            | Son                     | Bihar, Uttar Pradesh, Madhya Pradesh                |
| Ramganga            | Ramganga                | Uttarakhand                                         |
| Nimoo Bazgo         | Indus                   | Ladakh (UT)                                         |
| Uri / Tulbul        | Jhelum                  | Jammu Kashmir (UT)                                  |
| Chutak              | Suru                    | Kargil (Ladakh)                                     |
| Baglihar / Dulhasti | Chenab                  | Jammu Kashmir (UT)                                  |
| Almatti             | Krishna                 | Andhra Pradesh, Karnataka, Maharashtra              |
| Chambal             | Chambal                 | Madhya Pradesh, Rajasthan                           |
| Damodar Valley      | Damodar and Barakar     | Jharkhand and West Bengal                           |
| Bhakra Nangal       | Sutlej                  | Delhi, Himachal Pradesh, Punjab, Haryana, Rajasthan |
| Hirakud             | Mahanadi                | Odisha                                              |
| Tehri               | Bhagirathi + Bhilangana | Uttarakhand                                         |
| Narmada Valley      | Narmada                 | Gujarat, Rajasthan, Madhya Pradesh, Maharashtra     |
| Rihand              | Rihand                  | Uttar Pradesh                                       |

| Project                  | Related River          | Beneficiary States / UT (UPPCS Matching Special) |
|--------------------------|------------------------|--------------------------------------------------|
| Kishanganga              | Kishanganga            | Jammu Kashmir (UT)                               |
| Gandak                   | Gandak                 | Uttar Pradesh, Bihar                             |
| Srisaïlam                | Krishna                | Andhra Pradesh and Telangana                     |
| Mettur                   | Kaveri                 | Tamil Nadu                                       |
| Sharavati                | Sharavati              | Karnataka, Goa and Tamil Nadu                    |
| Shivasamudram            | Kaveri                 | Karnataka                                        |
| Hidkal                   | Ghataprabha            | Karnataka                                        |
| Purna                    | Purna                  | Maharashtra                                      |
| Kakrapar                 | Tapti                  | Gujarat                                          |
| Ranjit Sagar (Thein)     | Ravi                   | Punjab                                           |
| Pong Dam                 | Beas                   | Himachal Pradesh                                 |
| Sanjay Sarovar           | Wainganga              | Madhya Pradesh                                   |
| Koyna                    | Koyna                  | Maharashtra                                      |
| Matatila / Rani Laxmibai | Betwa                  | Uttar Pradesh                                    |
| Tungabhadra              | Pennar and Tungabhadra | Andhra Pradesh and Karnataka                     |

### > States and their Principal Hill Stations

| State / UT         | Principal Hill Stations                                               |
|--------------------|-----------------------------------------------------------------------|
| Uttarakhand        | Ranikhet, Lansdowne, Mussoorie, Mukteshwar, Nainital, Almora, Bhowali |
| Himachal Pradesh   | Dalhousie, Kasauli, Manali, Mandi                                     |
| Jammu Kashmir (UT) | Gulmarg, Pahalgam, Srinagar                                           |
| Tamil Nadu         | Coonoor, Ooty (Udhagamandalam), Kodaikanal, Yercaud                   |
| Karnataka          | Nandi Hills, Kemmangundi                                              |
| Kerala             | Munnar                                                                |
| Madhya Pradesh     | Pachmarhi                                                             |
|                    |                                                                       |



| State / UT              | Principal Hill Stations                                   |
|-------------------------|-----------------------------------------------------------|
| Meghalaya / West Bengal | Shillong (Meghalaya), Kalimpong, Darjeeling (West Bengal) |

### > Regional Names of Jhuming / Shifting Cultivation in India

| Local Name                                         | Related Region / State (India)                         |
|----------------------------------------------------|--------------------------------------------------------|
| Jhum                                               | North-East India (Assam, Meghalaya, Mizoram, Nagaland) |
| Pamlou                                             | Manipur                                                |
| Deepa                                              | Chhattisgarh (Bastar), Andaman and Nicobar Islands     |
| Bewar or Dahiya (Dahiyar)                          | Madhya Pradesh (Bundelkhand region)                    |
| Podu or Penda                                      | Andhra Pradesh                                         |
| Pama Dabi / Koman / Bringa / Kaman / Vinga / Dhavi | Odisha                                                 |
| Kumari                                             | Hilly areas of the Western Ghats of Kerala             |
| Valre or Waltre / Batra                            | South-eastern Rajasthan                                |
| Khil                                               | Himalayan region                                       |
| Kuruwa                                             | Jharkhand                                              |
| Jara and Erka                                      | Southern states                                        |

### > Main Facts Related to Indian Standard Time (IST)

| Criterion / Component | Exam-Useful Core Points (IST Facts)                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Meridian line         | India's standard time is measured from the <b>82°30' East longitude</b> .                                                                                                                                                                                                 |
| Exact location        | This line passes through <b>Mirzapur</b> in Uttar Pradesh (near Allahabad/Prayagraj).                                                                                                                                                                                     |
| Difference from GMT   | It is <b>5 hours 30 minutes ahead (+5:30)</b> of Greenwich Mean Time (GMT).                                                                                                                                                                                               |
| Longitudinal rule     | <ul style="list-style-type: none"> <li>A difference of <b>15 degrees of longitude</b> causes a difference of <b>1 hour</b> in local time.</li> <li>By mutual global understanding, the standard meridian is chosen at a <b>multiple of 7°30' of longitude</b>.</li> </ul> |
|                       |                                                                                                                                                                                                                                                                           |

| Criterion / Component  | Exam-Useful Core Points (IST Facts)                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other historical facts | Before independence Mumbai and Kolkata had their own local times; these were abolished after independence and <b>only one standard time for the whole of India</b> was adopted. |

### > Principal Mountain Peaks of Andaman and Nicobar

| Name of the Mountain Peak | Geographical Location (exact island)                                    |
|---------------------------|-------------------------------------------------------------------------|
| 1. Saddle Peak            | <b>North Andaman</b> ( <i>the highest peak of the Andaman Islands</i> ) |
| 2. Mount Diavolo          | Middle Andaman                                                          |
| 3. Mount Koyob            | South Andaman                                                           |
| 4. Mount Harriet          | South Andaman                                                           |
| 5. Mount Thullier         | <b>Great Nicobar</b> ( <i>the highest peak of the Nicobar Islands</i> ) |

### > Correct North-to-South Order of the Islands of Andaman and Nicobar

| Order (North to South) | Name of the Island (Andaman and Nicobar chain)                |
|------------------------|---------------------------------------------------------------|
| 1 (northernmost)       | <b>North Andaman</b>                                          |
| 2                      | <b>Middle Andaman</b>                                         |
| 3                      | <b>South Andaman</b>                                          |
| 4                      | <b>Little Andaman</b>                                         |
| 5                      | <b>Car Nicobar</b>                                            |
| 6                      | <b>Little Nicobar</b>                                         |
| 7 (southernmost)       | <b>Great Nicobar</b> ( <i>Indira Point is located on it</i> ) |

### > The Peninsular Plateau of India: Delimitation of Boundaries and Average Height

| Boundary Area (Direction) | Natural / Geographical Boundary                    |
|---------------------------|----------------------------------------------------|
| North-western boundary    | Aravalli range and Delhi                           |
| Eastern boundary          | Rajmahal hills                                     |
| Western boundary          | Gir hills (Gujarat)                                |
| North-eastern boundary    | Shillong (Meghalaya) and the Karbi-Anglong plateau |

| Boundary Area (Direction) | Natural / Geographical Boundary                                             |
|---------------------------|-----------------------------------------------------------------------------|
| Average height            | The average height of this entire plateau region is <b>600-900 metres</b> . |

### > Principal Nuclear Power Plants of India

| Name of the Nuclear Plant | Location (State) | Present Status / Special Fact                                                               |
|---------------------------|------------------|---------------------------------------------------------------------------------------------|
| Tarapur                   | Maharashtra      | <b>India's first nuclear power station (1969).</b>                                          |
| Narora                    | Uttar Pradesh    | <b>Located in Bulandshahr district</b> (UPPCS special).                                     |
| Rawatbhata                | Rajasthan        | Established <b>with Canadian cooperation</b> .                                              |
| Kakrapar / Mithi Virdi    | Gujarat          | Mithi Virdi is proposed <b>with the cooperation of the United States of America (USA)</b> . |
| Kaiga                     | Karnataka        | —                                                                                           |
| Kalpakkam / Kudankulam    | Tamil Nadu       | Kudankulam built <b>with Russian cooperation</b> .                                          |
| Jaitapur                  | Maharashtra      | —                                                                                           |
| Kovvada                   | Andhra Pradesh   | Proposed plant.                                                                             |
| Haripur                   | West Bengal      | Proposed plant.                                                                             |
| Fatehabad                 | Haryana          | Proposed plant.                                                                             |
| Chutka                    | Madhya Pradesh   | Proposed plant.                                                                             |

04

# Indian Polity

INDIAN POLITY AND CONSTITUTION

## > Constitutional Development of India

| Name of the Act          | Principal Provisions and Main Points (UPPCS Prelims Special)                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regulating Act of 1773   | <ul style="list-style-type: none"> <li>The <b>first step</b> to control the affairs of the East India Company in India.</li> <li>The Governor of Bengal was designated '<b>Governor-General of Bengal</b>' (<b>Warren Hastings</b> became the first Governor-General).</li> <li><b>A Supreme Court was established at Calcutta (1774)</b> (1 Chief Justice + 3 other judges).</li> </ul>                                                                              |
| Pitt's India Act of 1784 | <ul style="list-style-type: none"> <li>The <b>political and commercial functions</b> of the Company were <b>separated</b>.</li> <li>A new body named the '<b>Board of Control</b>' was created (beginning of dual government).</li> <li>The British government was given <b>complete control</b> over the affairs and administration of the Company in India.</li> </ul>                                                                                              |
| Charter Act of 1793      | <ul style="list-style-type: none"> <li>The Company's trade monopoly was <b>extended for another 20 years</b>.</li> <li>Members and staff of the Board of Control began to be <b>paid out of Indian revenues</b>.</li> </ul>                                                                                                                                                                                                                                           |
| Charter Act of 1813      | <ul style="list-style-type: none"> <li>The Company's trade monopoly was ended, <b>except for trade in tea and with China</b>.</li> <li><b>Christian missionaries</b> were permitted to come to India to preach/enlighten the people.</li> <li>Local governments got the power to <b>levy taxes</b> on persons and to punish those who did not pay.</li> </ul>                                                                                                         |
| Charter Act of 1833      | <ul style="list-style-type: none"> <li>The Governor-General of Bengal was made the '<b>Governor-General of India</b>' (<b>Lord William Bentinck</b> became the first Governor-General of India).</li> <li>The Governors of Madras and Bombay were <b>deprived of legislative powers</b> (complete centralisation of administration).</li> <li>An (unsuccessful) attempt was made to introduce <b>open competition</b> for the selection of civil servants.</li> </ul> |
| Charter Act of 1853      | <ul style="list-style-type: none"> <li>For the first time the <b>legislative and executive functions</b> of the Governor-General's Council were <b>separated</b>.</li> <li>The <b>Macaulay Committee was appointed in 1854</b> for recruitment to the civil service.</li> </ul>                                                                                                                                                                                       |

| Name of the Act                                                              | Principal Provisions and Main Points (UPPCS Prelims Special)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Government of India Act of 1858</b>                                       | <ul style="list-style-type: none"> <li>The governance of India passed directly under the British Crown; the designation of Governor-General was changed to <b>'Viceroy of India'</b> (Lord Canning became the first Viceroy of India).</li> <li>A new post of <b>'Secretary of State for India'</b> was created.</li> <li>A <b>15-member council</b> (advisory committee) was formed to assist the Secretary of State.</li> </ul>                                                                                                                                                                                                                                                            |
| <b>Indian Councils Act of 1861</b>                                           | <ul style="list-style-type: none"> <li>A beginning was made of <b>associating Indian representatives</b> with the law-making process.</li> <li>In 1862 Lord Canning nominated three Indians (<b>the Raja of Banaras, the Maharaja of Patiala and Sir Dinkar Rao</b>) to the legislative council.</li> <li>The Viceroy was empowered to <b>issue ordinances</b> in an emergency.</li> <li><b>Legislative councils were set up</b> in Bengal (1862), the North-Western Frontier Province (1866) and Punjab (1897).</li> </ul>                                                                                                                                                                  |
| <b>Indian Councils Act of 1892</b>                                           | <ul style="list-style-type: none"> <li>The functions of the legislative councils were increased; members got the right to <b>discuss the budget</b> and to ask questions of the executive.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Indian Councils Act of 1909</b><br>( <i>Morley-Minto Reforms</i> )        | <ul style="list-style-type: none"> <li>The size and scope of functions of the central and provincial legislative councils were enlarged (<b>asking supplementary questions</b>, moving resolutions on the budget).</li> <li>Provision was made for <b>separate electorates / communal representation for Muslims</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                             |
| <b>Government of India Act 1919</b><br>( <i>Montagu-Chelmsford Reforms</i> ) | <ul style="list-style-type: none"> <li>The administrative lists of the centre and the provinces were separated and the provinces were given the right to legislate on their own list.</li> <li><b>Dyarchy was introduced in the provinces</b>; the subjects were divided into two parts—'Transferred' and 'Reserved'.</li> <li><b>Bicameralism</b> and <b>direct elections</b> were introduced in the country for the first time.</li> <li>In 1926 a <b>Central Public Service Commission (on the recommendation of the Lee Commission)</b> was set up for the recruitment of civil servants.</li> <li><b>The central budget was separated from the budgets of the provinces.</b></li> </ul> |
| <b>Government of India Act 1935</b>                                          | <ul style="list-style-type: none"> <li><b>Dyarchy in the provinces was abolished</b> and they were given autonomy, and a <b>system of dyarchy at the centre</b> was introduced.</li> <li><b>Residuary powers were given to the Viceroy.</b></li> <li>The 'Council of India' established by the Act of 1858 was abolished.</li> <li>Along with a <b>Federal Public Service Commission (FPSC)</b>, Provincial Service Commissions and a <b>Joint Service Commission (JPSC)</b> for two or more provinces were established.</li> <li>A <b>Federal Court</b> was established in 1937.</li> </ul>                                                                                                 |

## > The Constituent Assembly

### > Constituent Assembly: Important Facts

| Subject / Event                   | Principal Facts                                                                                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Basis of formation                | On the recommendation of the <b>Cabinet Mission Plan, 1946</b> .                                                                       |
| Allocation of seats               | <b>1 representative</b> for a population of <b>every 10 lakh (1 million)</b> persons.                                                  |
| Time of election                  | Elections were completed in <b>July–August 1946</b> .                                                                                  |
| Original strength                 | <b>A total of 389 fixed</b> (296 from British India + 93 from the princely states).                                                    |
| Strength after Partition          | After the Muslim League separated, <b>299 members in all</b> remained (229 from the Indian provinces + 70 from the princely states).   |
| First meeting                     | On <b>9 December 1946</b> (only <b>207 members</b> attended).                                                                          |
| Temporary (provisional) President | <b>Dr. Sachchidananda Sinha</b> (on 9 December 1946; being the eldest member).                                                         |
| Permanent President               | <b>Dr. Rajendra Prasad</b> (elected unopposed on 11 December 1946).                                                                    |
| Vice-Presidents                   | <b>H.C. Mukherjee</b> and <b>V.T. Krishnamachari</b> (there were two Vice-Presidents).                                                 |
| Secretariat officers              | <ul style="list-style-type: none"> <li>• <b>Secretary:</b> H.V.R. Iyengar</li> <li>• <b>Chief Draftsman:</b> S.N. Mukherjee</li> </ul> |

### > The Drafting Committee and its Members

| Name of the Committee | Date of Formation     | Total Number of Members                   | Members and their Special Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|-----------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drafting Committee    | <b>29 August 1947</b> | <b>7 members</b><br>(Chairman + 6 others) | <ol style="list-style-type: none"> <li>1. <b>Dr. Bhimrao Ambedkar</b> (Chairman)</li> <li>2. <b>N. Gopalaswamy Ayyangar</b></li> <li>3. <b>Alladi Krishnaswamy Ayyar</b></li> <li>4. <b>Dr. K.M. Munshi</b> (the only member who was originally a Congressman)</li> <li>5. <b>Syed Mohammad Saadulla</b></li> <li>6. <b>N. Madhava Rau</b> (he took the place of B.L. Mitter, who withdrew for reasons of health)</li> <li>7. <b>T.T. Krishnamachari</b> (became a member in 1948 after the death of D.P. Khaitan)</li> </ol> |

- **Publication of the draft (Quick Revision):** The first draft was published in **February 1948**, the second draft in **October 1948**, and the final draft was presented to the Assembly by Dr. Ambedkar on **4 November 1948**.

## > Principal Committees of the Constituent Assembly and their Chairmen

| Name of the Committee                                                                               | Name of the Chairman     |
|-----------------------------------------------------------------------------------------------------|--------------------------|
| Union Powers Committee / Union Constitution Committee / Committee for (Negotiating with) the States | Jawaharlal Nehru         |
| Provincial Constitution Committee                                                                   | Sardar Vallabhbhai Patel |
| Drafting Committee                                                                                  | Dr. B.R. Ambedkar        |
| Advisory Committee on Fundamental Rights and Minorities                                             | Sardar Vallabhbhai Patel |
| • <i>Fundamental Rights Sub-Committee</i>                                                           | • J.B. Kripalani         |
| • <i>Minorities Sub-Committee</i>                                                                   | • H.C. Mukherjee         |
| Rules of Procedure Committee / Steering Committee                                                   | Dr. Rajendra Prasad      |

## > Parts of the Indian Constitution/ Schedules

## > Parts of the Indian Constitution and the Related Articles

| Part | Subject                                                                             | Related Articles |
|------|-------------------------------------------------------------------------------------|------------------|
| I    | The Union and its Territory                                                         | 1 – 4            |
| II   | Citizenship                                                                         | 5 – 11           |
| III  | Fundamental Rights                                                                  | 12 – 35          |
| IV   | Directive Principles of State Policy (DPSP)                                         | 36 – 51          |
| IV A | Fundamental Duties <i>(added by the 42nd Amendment)</i>                             | 51A              |
| V    | The Union                                                                           | 52 – 151         |
| VI   | The States                                                                          | 152 – 237        |
| VII  | States in Part B of the First Schedule <i>(repealed by the 7th Amendment, 1956)</i> | 238 (repealed)   |
| VIII | Union Territories                                                                   | 239 – 242        |
| IX   | Panchayats <i>(by the 73rd Constitutional Amendment)</i>                            | 243 – 243O       |
| IX A | Municipalities <i>(by the 74th Constitutional Amendment)</i>                        | 243P – 243ZG     |
| IX B | Co-operative Societies <i>(by the 97th Amendment)</i>                               | 243ZH – 243ZT    |



| Part  | Subject                                                                             | Related Articles |
|-------|-------------------------------------------------------------------------------------|------------------|
| X     | Scheduled and Tribal Areas                                                          | 244 – 244A       |
| XI    | Relations between the Union and the States (legislative and administrative)         | 245 – 263        |
| XII   | Finance, Property, Contracts and Suits (Art. 300A, the right to property, is in it) | 264 – 300A       |
| XIII  | Trade, Commerce and Intercourse within the Territory of India                       | 301 – 307        |
| XIV   | Services under the Union and the States (UPSC and UPPSC, etc.)                      | 308 – 323        |
| XIV A | Tribunals <i>(added by the 42nd Amendment)</i>                                      | 323A – 323B      |
| XV    | Elections                                                                           | 324 – 329A       |
| XVI   | Special Provisions Relating to Certain Classes (SC, ST, OBC, Anglo-Indians)         | 330 – 342        |
| XVII  | Official Language                                                                   | 343 – 351        |
| XVIII | Emergency Provisions                                                                | 352 – 360        |
| XIX   | Miscellaneous                                                                       | 361 – 367        |
| XX    | Amendment of the Constitution                                                       | 368              |
| XXI   | Temporary, Transitional and Special Provisions (such as Art. 370 earlier)           | 369 – 392        |
| XXII  | Short Title, Commencement, Authoritative Text in Hindi and Repeals                  | 393 – 395        |

## > Schedules of the Constitution

| Schedule | Main Subject and Provisions                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First    | • <b>Names of the States and Union Territories (UTs)</b> and their territorial jurisdiction/boundaries.                                                                                                                                                  |
| Second   | • <b>Salaries, allowances and privileges</b> of various constitutional functionaries (such as the President, Governors, Speakers/Deputy Speakers of the Lok Sabha/Legislative Assemblies, judges of the Supreme Court/High Courts, and the <b>CAG</b> ). |
| Third    | • <b>Forms of oaths and affirmations</b> taken by ministers, members of Parliament/legislatures, candidates for election, judges of the Supreme Court/High Courts and the <b>CAG</b> .                                                                   |
| Fourth   | • <b>Allocation of seats in the Rajya Sabha</b> to the States and Union Territories.                                                                                                                                                                     |
|          |                                                                                                                                                                                                                                                          |

| Schedule        | Main Subject and Provisions                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Fifth</b>    | <ul style="list-style-type: none"> <li>Special provisions relating to the administration and control of <b>Scheduled Areas and Scheduled Tribes (ST)</b>.</li> </ul>                                                                                                                                                                                                                          |                                                                                                 |
| <b>Sixth</b>    | <ul style="list-style-type: none"> <li>Provision of <b>autonomous district councils</b> for the administration of the tribal areas of four special states—<b>Assam, Meghalaya, Tripura and Mizoram (AMTM)</b>.</li> </ul>                                                                                                                                                                     |                                                                                                 |
| <b>Seventh</b>  | <ul style="list-style-type: none"> <li><b>Division of legislative powers</b> (between the Centre and the States).</li> <li><b>Union List:</b> laws by the Centre (present subjects: <b>100</b>)</li> <li><b>State List:</b> laws by the States (present subjects: <b>61</b>)</li> <li><b>Concurrent List:</b> laws by both the Centre and the States (present subjects: <b>52</b>)</li> </ul> | The original Constitution had 97 (Union), 66 (State) and 47 (Concurrent) subjects respectively. |
| <b>Eighth</b>   | <ul style="list-style-type: none"> <li><b>Languages recognised by the Constitution.</b></li> <li>The original Constitution had <b>14 languages</b>, which have now become <b>22</b>.</li> </ul>                                                                                                                                                                                               | Languages added later: Sindhi, Konkani, Manipuri, Nepali, Bodo, Dogri, Maithili and Santhali.   |
| <b>Ninth</b>    | <ul style="list-style-type: none"> <li>Giving <b>special constitutional protection</b> to certain central and state laws (land reforms and abolition of zamindari).</li> </ul>                                                                                                                                                                                                                | Added by the <b>First Constitutional Amendment, 1951</b> .                                      |
| <b>Tenth</b>    | <ul style="list-style-type: none"> <li><b>Anti-defection provisions</b> (relating to the disqualification of members of Parliament and legislatures).</li> </ul>                                                                                                                                                                                                                              | Added by the <b>52nd Constitutional Amendment, 1985</b> .                                       |
| <b>Eleventh</b> | <ul style="list-style-type: none"> <li><b>Subjects of the Panchayats</b> (functions of Panchayati Raj institutions).</li> <li>It includes agriculture, drinking water, rural housing, roads, health, education, etc.</li> </ul>                                                                                                                                                               | Added by the <b>73rd Constitutional Amendment, 1992</b> . There are <b>29 subjects</b> in all.  |
| <b>Twelfth</b>  | <ul style="list-style-type: none"> <li><b>Subjects of the Municipalities</b> (functions of urban local bodies).</li> <li>It includes urban planning, land use, water supply, sanitation, fire services, etc.</li> </ul>                                                                                                                                                                       | Added by the <b>74th Constitutional Amendment, 1992</b> . There are <b>18 subjects</b> in all.  |

### › Eighth Schedule: Important Constitutional Amendments Relating to Languages

| Amendment Act         | Year | Languages Added (trick to remember)                                         |
|-----------------------|------|-----------------------------------------------------------------------------|
| <b>21st Amendment</b> | 1967 | • <b>Sindhi</b>                                                             |
| <b>71st Amendment</b> | 1992 | • <b>Nepali, Manipuri, Konkani</b> ( <i>trick: NaMaK - Na, Ma, Ka</i> )     |
| <b>92nd Amendment</b> | 2003 | • <b>Santhali, Bodo, Dogri, Maithili</b> ( <i>trick: BDSM or Bodo-Mas</i> ) |

## > Seventh Schedule: Principal Subjects of the Three Lists

| Name of the List | Present Subjects (Original Subjects) | Principal Law-maker      | Principal Subjects Asked in the Exam (Core Examples)                                                    |
|------------------|--------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------|
| Union List       | <b>100</b> (97)                      | Parliament only          | • Defence • Foreign affairs • Posts and telegraphs • Telephones • Railways • Currency • Banking.        |
| State List       | <b>61</b> (66)                       | State legislature        | • Police • Prisons • Agriculture • Irrigation • Public health • Local government.                       |
| Concurrent List  | <b>52</b> (47)                       | Both Parliament + States | • Education • Forests • Marriage and divorce • Factories • Electricity • Economic planning • Waterways. |

## > Foreign Sources of the Indian Constitution

| Source Country / Act           | Features Incorporated in the Constitution                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Government of India Act, 1935  | • Federal scheme • Office of the Governor • Structure of the judiciary • Public Service Commissions • Emergency provisions • Administrative details.                                                     |
| Constitution of Britain        | • <b>Rule of Law</b> • Parliamentary system of government • Legislative procedure • <b>Single citizenship</b> • Cabinet system • Prerogative writs • Parliamentary privileges • Bicameralism.            |
| United States of America (USA) | • <b>Fundamental Rights</b> • Independence of the judiciary • <b>Judicial review</b> • Office of the President/impeachment • Removal of judges of the Supreme Court and High Courts • Federal supremacy. |
| Constitution of Ireland        | • <b>Directive Principles of State Policy (DPSP)</b> • Method of election of the President • Nomination of members to the Rajya Sabha.                                                                   |
| Constitution of Canada         | • <b>Federation with a strong Centre</b> • Vesting of residuary powers in the Centre • Appointment of state Governors by the Centre • Advisory jurisdiction of the Supreme Court (Art. 143).             |
| Constitution of Australia      | • <b>Concurrent List</b> • Freedom of trade, commerce and intercourse • <b>Joint sitting of the two Houses of Parliament.</b>                                                                            |
| Constitution of France         | • Republican structure • The ideals of <b>liberty, equality and fraternity</b> in the Preamble.                                                                                                          |
| Soviet Union (USSR / Russia)   | • <b>Fundamental Duties</b> • The ideal of <b>justice (social, economic and political)</b> in the Preamble.                                                                                              |
| South Africa                   | • <b>Procedure for amendment of the Constitution (Art. 368)</b> • Election of members of the Rajya Sabha.                                                                                                |
|                                |                                                                                                                                                                                                          |

| Source Country / Act           | Features Incorporated in the Constitution                      |
|--------------------------------|----------------------------------------------------------------|
| Weimar Constitution of Germany | • <b>Suspension of Fundamental Rights during an Emergency.</b> |
| Constitution of Japan          | • <b>Procedure established by law.</b>                         |

## > The Preamble

### > The Preamble: Important Components and Nature

| Component / Dimension   | Important Facts Related to the Preamble (Quick Revision)                                                                                                                                                                                          |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historical background   | <ul style="list-style-type: none"> <li>• The 'Objectives Resolution' was moved by Nehru on <b>13 December 1946</b>.</li> <li>• It was adopted by the Constituent Assembly on <b>22 January 1947</b>, and it later became the Preamble.</li> </ul> |
| The only amendment      | • Three new words were added by the <b>42nd Constitutional Amendment, 1976</b> : <b>Socialist, Secular and Integrity</b> .                                                                                                                        |
| Justice (3 kinds)       | • <b>Social, economic and political justice.</b>                                                                                                                                                                                                  |
| Liberty (5 kinds)       | • <b>Liberty of thought, expression, belief, faith and worship.</b>                                                                                                                                                                               |
| Equality (2 kinds)      | • <b>Equality of status and of opportunity.</b>                                                                                                                                                                                                   |
| Source (foreign)        | • The idea/draft of the Preamble was taken from <b>America</b> and its distinctive language from <b>Australia</b> .                                                                                                                               |
| Nature / justiciability | • The Preamble is <b>not justiciable (non-justiciable)</b> (it cannot be enforced in a court).                                                                                                                                                    |

### > Facts Related to the Preamble and the 42nd Amendment

| Subject                                                  | Words in Order / New Provisions Added                                                                                         |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Correct order of the key words of the Preamble           | <b>Sovereign → Socialist → Secular → Democratic → Republic</b><br><i>(this order has been asked several times in UPPCS)</i>   |
| Words added to the Preamble by the 42nd Amendment (1976) | <ol style="list-style-type: none"> <li>1. <b>Socialist</b></li> <li>2. <b>Secular</b></li> <li>3. <b>Integrity</b></li> </ol> |
|                                                          |                                                                                                                               |

| Subject                                                               | Words in Order / New Provisions Added                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New Articles added to the DPSP (Part IV) by the 42nd Amendment (1976) | <ul style="list-style-type: none"> <li>• <b>Article 39(A)</b>: equal justice and free legal aid to the poor.</li> <li>• <b>Article 39(f)</b>: protection of children and youth against exploitation and their development in conditions of dignity.</li> <li>• <b>Article 43(A)</b>: participation of workers in the management of industries.</li> <li>• <b>Article 48(A)</b>: protection and improvement of the environment and safeguarding of wildlife.</li> </ul> |

### > Historic Judgments of the Supreme Court (SC) on the Preamble

| Case                     | Year | Main Verdict of the Supreme Court (Core Verdict)                                                                                         |
|--------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------|
| Berubari Union case      | 1960 | • The Preamble <b>is not a part of the Constitution</b> , hence Parliament cannot amend it.                                              |
| Kesavananda Bharati case | 1973 | • The Preamble <b>is an integral part of the Constitution</b> and Parliament <b>can amend it</b> without damaging its 'basic structure'. |
| LIC of India case        | 1995 | • The Preamble is an <b>integral part</b> of the Constitution.                                                                           |

**Historic amendment:** By the **42nd Constitutional Amendment (1976)** the Preamble was amended only once and three words, '**Socialist**', '**Secular**' and '**Integrity**', were added.

### > The Preamble: Important Statements and Influences

| Political Expert / Scholar           | Important Statement Made about the Preamble                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Thakurdas Bhargava / M. Hidayatullah | The Preamble is " <b>the soul of the Indian Constitution</b> ".                                                |
| N.A. Palkhivala                      | The Preamble is the " <b>identity card</b> " of the Constitution.                                              |
| K.M. Munshi                          | The Preamble is " <b>the horoscope of the Constitution</b> ".                                                  |
| Historical Event / Country           | Main Influence on the Preamble / Element Borrowed                                                              |
| Objectives Resolution                | • Moved by Nehru on <b>13 December 1946</b> ; passed on <b>22 January 1947</b> . It later became the Preamble. |
| Constitution of America              | • The concept of " <b>We the People of India</b> " and the idea of a Preamble.                                 |
| Constitution of Australia            | • <b>The language and distinctive style of the Preamble.</b>                                                   |
| French Revolution                    | • The ideal words <b>liberty, equality and fraternity</b> included in the Preamble.                            |
| Russian Revolution                   | • The ideal of three kinds of justice in the Preamble: <b>social, economic and political justice.</b>          |

## > Directive Principles of State Policy

### > Directive Principles of State Policy (DPSP - Part IV)

| Article | Main Provision and Directive Principle                                                                                             |
|---------|------------------------------------------------------------------------------------------------------------------------------------|
| 39A     | Equal justice and <b>free legal aid to the poor</b> <i>(by the 42nd Amendment)</i>                                                 |
| 41      | <b>Right to work, to education and to public assistance</b> in certain cases (unemployment, old age, sickness and disablement).    |
| 42      | Provision for just and humane conditions of work and <b>maternity relief</b> .                                                     |
| 43      | <b>Living wage</b> and a decent standard of life for all workers.                                                                  |
| 43A     | Ensuring <b>participation of workers in the management of industries</b> <i>(by the 42nd Amendment)</i> .                          |
| 43B     | Promotion of the voluntary formation and professional management of <b>co-operative societies</b> <i>(by the 97th Amendment)</i> . |
| 48      | Organising <b>agriculture and animal husbandry</b> on modern and scientific lines (prohibition of the slaughter of milch cattle).  |
| 48A     | <b>Protection and improvement of the environment</b> and safeguarding of forests and wildlife <i>(by the 42nd Amendment)</i> .     |
| 49      | Protection of <b>monuments</b> , places and objects <b>of national importance</b> .                                                |

### > Famous Statements of Thinkers on the Directive Principles of State Policy (DPSP)

| Name of the Scholar / Thinker | Famous Statement Made about the DPSP (UPPCS Special)                                                                    |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| K.T. Shah                     | "The Directive Principles of State Policy are <b>like a cheque on a bank, payable at the convenience of the bank.</b> " |
| T.T. Krishnamachari           | Called the DPSP " <b>a veritable dustbin of sentiments</b> ".                                                           |
| K.C. Wheare                   | Regarded them as " <b>a manifesto of aims and aspirations</b> ".                                                        |
| Sir Ivor Jennings             | "These principles have no regular pattern of philosophy; they are <b>Fabian socialism without the socialism.</b> "      |
| Granville Austin              | "The Directive Principles and the Fundamental Rights together are <b>the conscience of the Constitution.</b> "          |
| Dr. B.R. Ambedkar             | "The Directive Principles are like <b>instruments of instructions</b> to the legislature and the executive."            |
| B.N. Rau                      | "These principles have <b>moral precept and educative value</b> for the authorities of the State."                      |

| Name of the Scholar / Thinker | Famous Statement Made about the DPSP (UPPCS Special)                                             |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| K. Santhanam                  | "These principles will cause <b>constitutional conflict between the Centre and the States.</b> " |

## > Fundamental Duties

### > Fundamental Duties: Part IVA, Article 51A

| Subject / Criterion      | Important Facts and Features (UPPCS Special Points)                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historical background    | <ul style="list-style-type: none"> <li>Introduced on the recommendations of the <b>Sardar Swaran Singh Committee (1976)</b>.</li> <li>Borrowed from the Constitution of the <b>Soviet Union (USSR / Russia)</b>.</li> </ul>                                                                                                                                                                        |
| Constitutional amendment | <ul style="list-style-type: none"> <li>Initially <b>10 Fundamental Duties</b> were added by the <b>42nd Constitutional Amendment, 1976</b>.</li> <li>The <b>11th Fundamental Duty</b> (education of children aged 6-14) was added by the <b>86th Constitutional Amendment, 2002</b>.</li> </ul>                                                                                                    |
| Principal features       | <ul style="list-style-type: none"> <li><b>Non-justiciable:</b> the Fundamental Duties are non-justiciable (no punishment can be given directly by a court for their violation).</li> <li>In nature some are moral duties and some are civic duties.</li> <li>They are <b>only for Indian citizens</b>, not for foreigners.</li> <li>Rights and duties are interrelated and inseparable.</li> </ul> |

### > List of the 11 Fundamental Duties (in order)

1. To **abide by the Constitution** and respect its ideals and institutions, the National Flag and the National Anthem.
2. To cherish and follow the noble ideals **which inspired the national struggle** for freedom.
3. To uphold and protect the **sovereignty, unity and integrity** of India.
4. To **defend the country** and render national service when called upon to do so.
5. To promote **harmony and the spirit of common brotherhood** among all the people.
6. To value and preserve the rich heritage of our **composite culture**.
7. To protect and improve the **natural environment** (forests, lakes, rivers, wildlife).
8. To develop the **scientific temper**, humanism and the spirit of inquiry.
9. To safeguard **public property** and to abjure violence.
10. To strive continuously **towards excellence** in all spheres of individual and collective activity.
11. Provision of **primary education for children aged 6 to 14 years** by the parent or guardian (*86th Amendment, 2002*).



## > Principal Articles

### > The Union and its Territory (Articles 1-4) and the Integration of the Princely States

| Article   | Provision (Federal System)                                                                                                                                  | Integration of the Princely States into India (Special Facts)                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article 1 | <b>India is a Union of States.</b> (States cannot secede from the Union; single citizenship).                                                               | • At the time of independence 13 princely states were in Pakistan and <b>552 princely states</b> were within the geographical boundary of India.                                                                                                |
| Article 2 | <b>Admission/establishment of new (foreign) states:</b> Parliament may add foreign territory to India on the prior notice of the President.                 | • The responsibility of integrating the princely states into the Indian Union was entrusted to <b>Sardar Patel, V.P. Menon and Lord Mountbatten</b> .                                                                                           |
| Article 3 | <b>Formation of new states and alteration of names/boundaries:</b> Parliament may alter the names, areas or boundaries of the existing states within India. | • <b>Integration of the 3 rebellious princely states:</b><br>1. <b>Hyderabad:</b> by 'Operation Polo' (military action).<br>2. <b>Junagadh:</b> by a 'plebiscite'.<br>3. <b>Jammu-Kashmir:</b> by the signing of the 'Instrument of Accession'. |
| Article 4 | Amendments made under Articles 2 and 3 will be by simple majority (they will not be deemed constitutional amendments under Art. 368).                       | • <b>Fact about Sikkim:</b> became an 'associate state' by the 35th Amendment, then a full state (the 22nd) by the <b>36th Amendment (1975)</b> .                                                                                               |

### > Important Articles Related to Citizenship (Part II)

| Article    | Main Provision (Quick Revision Line)                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Article 5  | Citizenship at the commencement of the Constitution (26 January 1950).                                                                       |
| Article 6  | Rights of citizenship of persons <b>who migrated to India from Pakistan</b> .                                                                |
| Article 7  | Rights of citizenship of persons <b>who migrated from India to Pakistan</b> .                                                                |
| Article 8  | Citizenship of <b>persons of Indian origin (PIO)</b> residing outside India (abroad).                                                        |
| Article 9  | <b>Automatic termination of Indian citizenship</b> on voluntarily acquiring the citizenship of a foreign state (rule of single citizenship). |
| Article 10 | Continuance of the rights of citizenship (citizenship cannot be taken away except by a law made by Parliament).                              |
| Article 11 | <b>Full power of Parliament to make law/regulate citizenship.</b>                                                                            |

## > Vice-President: Constitutional Articles and Vacancy of Office (Part V)

| Article    | Main Provision                                                                                                                | Causes of Vacancy of Office (Art. 67)                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Article 63 | • There shall be a Vice-President of India.                                                                                   | 1. On <b>expiry of the 5-year term</b> .                                          |
| Article 64 | • The Vice-President shall be the <b>ex-officio Chairman of the Rajya Sabha</b> .                                             | 2. By <b>resignation</b> addressed to the President.                              |
| Article 65 | • To act as President when the office of President falls vacant.                                                              | 3. By <b>removal</b> through a resolution passed by the Rajya Sabha.              |
| Article 66 | • Procedure and qualifications for the election of the Vice-President.                                                        | 4. On <b>death</b> during the term.                                               |
| Article 67 | • Term of office of the Vice-President ( <b>5 years</b> from entering upon office).                                           | 5. When the Supreme Court declares the <b>election void or him disqualified</b> . |
| Article 69 | • <b>Oath</b> by the Vice-President (before the President or a person appointed by him).                                      |                                                                                   |
| Article 71 | • Disputes connected with the election of the President or Vice-President shall be decided <b>only by the Supreme Court</b> . |                                                                                   |

### > Quick Pointers:

- The Vice-President is not a member of the Rajya Sabha, so he **has no ordinary right to vote**, but as Chairman he can give a **casting vote**.
- When the Vice-President acts as 'Acting President', he does not function as Chairman of the Rajya Sabha. During that period the Deputy Chairman conducts the House.
- **Vice-Presidents who later became President:** Dr. Sarvepalli Radhakrishnan, Dr. Zakir Husain, V.V. Giri, R. Venkataraman, Dr. Shankar Dayal Sharma.

## > The Executive: Articles Relating to the President and the Prime Minister

| Article    | Subject / Name                         | Main Provision on the Relationship between the President and the Prime Minister                                                                                    |
|------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article 74 | Council of Ministers and the President | • There shall be a Council of Ministers, with the Prime Minister at its head, to aid and advise the President. The President is bound to act on its advice.        |
| Article 75 | Other provisions as to ministers       | • <b>The Prime Minister shall be appointed by the President</b> , and the other ministers shall be appointed by the President on the advice of the Prime Minister. |

| Article           | Subject / Name                                 | Main Provision on the Relationship between the President and the Prime Minister                                                                                            |
|-------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Article 77</b> | Conduct of business of the Government of India | • All executive action of the Government of India shall be expressed to be taken <b>in the name of the President</b> .                                                     |
| <b>Article 78</b> | Duties of the Prime Minister                   | • It shall be the duty of the Prime Minister to communicate to the President all decisions of the Council of Ministers and matters relating to administration/legislation. |

### > Constitution and Internal Structure of Parliament (Part V: Articles 79 - 95)

| Article   | Subject Matter / Provision                                                                                              | Article   | Subject Matter / Provision                                                                        |
|-----------|-------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------|
| <b>79</b> | <b>Constitution of Parliament</b> (President + Rajya Sabha + Lok Sabha)                                                 | <b>88</b> | <b>Rights of ministers and the Attorney General</b> as respects the Houses                        |
| <b>80</b> | Composition/organisation of the <b>Rajya Sabha</b>                                                                      | <b>89</b> | Chairman (Vice-President) and Deputy Chairman of the Rajya Sabha                                  |
| <b>81</b> | Composition/organisation of the <b>Lok Sabha</b>                                                                        | <b>90</b> | Vacation, resignation and removal of the Deputy Chairman of the Rajya Sabha                       |
| <b>82</b> | Readjustment of seats after each census                                                                                 | <b>92</b> | Chairman/Deputy Chairman not to preside while a resolution for his removal is under consideration |
| <b>83</b> | Duration (term) of the two Houses of Parliament                                                                         | <b>93</b> | <b>Speaker</b> and Deputy Speaker of the Lok Sabha                                                |
| <b>84</b> | Essential qualifications for membership of Parliament                                                                   | <b>94</b> | Vacation, resignation and removal of the Speaker and Deputy Speaker of the Lok Sabha              |
| <b>85</b> | <b>Sessions, prorogation and dissolution</b> of Parliament                                                              | <b>95</b> | Power of the Deputy Speaker or another person to perform the duties of the Speaker                |
| <b>86</b> | Right of the President to address and send messages to the Houses                                                       |           |                                                                                                   |
| <b>87</b> | <b>Special address by the President</b> (first session after the Lok Sabha election and the first session of each year) |           |                                                                                                   |

### > Principal Articles Relating to the Executive and the Legislature

| Article   | Office / Subject                 | Main Constitutional Provision                                                                                            |
|-----------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>76</b> | <b>Attorney General of India</b> | • The chief legal adviser of the Government of India. Has the right to speak in both Houses of Parliament (not to vote). |

| Article | Office / Subject                                      | Main Constitutional Provision                                                                                                           |
|---------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 110     | <b>Money Bill</b>                                     | • Definition of a Money Bill. Whether a bill is a Money Bill or not is decided by the <b>Speaker of the Lok Sabha</b> .                 |
| 112     | <b>Annual Financial Statement</b>                     | • The word 'budget' is not mentioned in the Constitution; in its place is the term 'Annual Financial Statement'.                        |
| 123     | <b>Ordinance-making power of the President</b>        | • Power of the President to promulgate ordinances while Parliament is not in session (recess).                                          |
| 143     | <b>Consultation with the Supreme Court</b>            | • Power of the President to seek the advice of the Supreme Court on a question of law of public importance (the advice is not binding). |
| 148     | <b>Comptroller and Auditor-General of India (CAG)</b> | • Provision for the office of the Comptroller and Auditor-General of India (guardian of the public purse of India).                     |
| 149     | <b>Duties and powers of the CAG</b>                   | • Determination of the functions, duties and powers of the Comptroller and Auditor-General of India.                                    |

### > Constitutional Provisions Relating to the Official Language (Part XVII)

| Article | Main Provision and Direction                                                                                                                 |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 343     | <b>Official language of the Union:</b> the official language of the Union shall be <b>Hindi</b> in the <b>Devanagari</b> script.             |
| 344     | Provision for the constitution of a <b>Commission (Official Language Commission)</b> and a Committee of Parliament on the official language. |
| 350A    | Duty of the states to provide adequate facilities for <b>instruction in the mother tongue</b> at the primary stage.                          |
| 350B    | Provision for the appointment of a Special Officer for <b>linguistic minorities</b> .                                                        |
| 351     | <b>Directive for the development of the Hindi language</b> (the Union government shall work for the spread and development of Hindi).        |

#### Note

**Note:** The **Eighth Schedule of the Indian Constitution** at present contains **22 official languages**, including Hindi.

## > Principal Articles on the Judiciary (Supreme Court and High Courts)

| Supreme Court      | Main Provision                                                                              | High Court and Subordinate Courts | Main Provision                                                                            |
|--------------------|---------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------|
| <b>Article 124</b> | Establishment and constitution of the Supreme Court.                                        | <b>Article 214</b>                | Provision of High Courts for the states.                                                  |
| <b>Article 125</b> | Salaries, allowances and facilities of judges.                                              | <b>Article 215</b>                | High Courts to be <b>courts of record</b> .                                               |
| <b>Article 126</b> | Appointment of an <b>Acting Chief Justice</b> .                                             | <b>Article 226</b>                | <b>Power of the High Courts to issue writs</b> for Fundamental Rights and other purposes. |
| <b>Article 127</b> | Appointment of <b>ad hoc judges</b> (only in the SC).                                       | <b>Article 233</b>                | Appointment of <b>District Judges</b> by the Governor.                                    |
| <b>Article 128</b> | Attendance of retired judges at sittings of the Supreme Court.                              | <b>Article 235</b>                | <b>Control over subordinate courts</b> (administrative control of the High Court).        |
| <b>Article 129</b> | The Supreme Court to be a <b>court of record</b> and its power to punish for contempt.      |                                   |                                                                                           |
| <b>Article 136</b> | <b>Special Leave to Appeal (Special Leave Petition - SLP)</b> by the Supreme Court.         |                                   |                                                                                           |
| <b>Article 137</b> | <b>Review</b> by the Supreme Court of its own judgments or orders (review/reconsideration). |                                   |                                                                                           |
| <b>Article 141</b> | Law declared by the Supreme Court shall be <b>binding</b> on all courts within India.       |                                   |                                                                                           |

## > Principal Articles Relating to the Governor (Part VI)

| Article    | Main Provision Relating to the Governor (UPPCS Matching Special)                                                                                     |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>153</b> | <b>Governors of States:</b> there shall be a Governor for each state (the same person may be Governor of two or more states).                        |
| <b>154</b> | <b>The executive power of the state</b> shall be vested in the Governor and exercised by him either directly or through officers subordinate to him. |
| <b>155</b> | <b>Appointment of the Governor:</b> the Governor is appointed by the President.                                                                      |
| <b>156</b> | <b>Term of office of the Governor:</b> during the pleasure of the President (normally 5 years).                                                      |
| <b>158</b> | <b>Conditions</b> of the Governor's office.                                                                                                          |

| Article | Main Provision Relating to the Governor (UPPCS Matching Special)                                                                                               |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 159     | <b>Oath or affirmation</b> by the Governor (administered by the Chief Justice of the High Court of the state concerned).                                       |
| 160     | Discharge of the functions of the Governor in certain contingencies.                                                                                           |
| 161     | <b>Pardoning power of the Governor:</b> power to grant pardons, reprieves, respites or remissions of punishment (except the death sentence and court-martial). |
| 162     | <b>Extent of the executive power of the state.</b>                                                                                                             |
| 163     | There shall be a <b>Council of Ministers</b> , with the Chief Minister at its head, to aid and advise the Governor.                                            |
| 164     | Appointment of ministers (by the Governor on the advice of the Chief Minister), their tenure, salaries and other provisions.                                   |

### > Power of the President to Make Regulations for UTs (Article 240)

| Article   | Related Union Territory     |
|-----------|-----------------------------|
| 240(1)(A) | Andaman and Nicobar Islands |
| 240(1)(B) | Lakshadweep                 |
| 240(1)(C) | Dadra and Nagar Haveli      |
| 240(1)(D) | Daman and Diu               |
| 240(1)(E) | Puducherry                  |

### > Centre-State Legislative and Administrative Relations

| Article            | Main Legislative / Administrative Provision (UPPCS Matching Special)                                                                                                                   |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Article 245</b> | Parliament may make laws for the whole or any part of the country, and a state legislature for its own state.                                                                          |
| <b>Article 246</b> | Classification of the powers to make laws on the subjects of the <b>three lists (Union, State and Concurrent)</b> of the Seventh Schedule.                                             |
| <b>Article 247</b> | Power of Parliament to establish certain additional courts.                                                                                                                            |
| <b>Article 248</b> | <b>Residuary powers:</b> the right to make laws on subjects not in any of the three lists (such as cyber crime, ISRO, DRDO) belongs <b>only to the Union government (Parliament)</b> . |
| <b>Article 252</b> | Power of Parliament to legislate on a State List subject <b>with the consent/at the request of two or more states</b> .                                                                |

| Article     | Main Legislative / Administrative Provision (UPPCS Matching Special)                                                                    |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Article 253 | Parliament has full power to legislate even on State List subjects <b>for giving effect to international agreements/treaties.</b>       |
| Article 254 | If a state law is repugnant (inconsistent) to a law of Parliament, the law of Parliament shall prevail and the state law shall be void. |
| Article 263 | Constitution of an <b>Inter-State Council</b> (by the President); meetings with Chief Ministers for Centre-State coordination.          |

### > Centre-State Financial Relations (Part XII: Articles 265 - 281)

| Article | Main Financial Provision (UPPCS Fact Pointer)                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------|
| 265     | • <b>Taxes not to be imposed save by authority of law:</b> no tax can be levied without a legal basis.                  |
| 266     | • Details of the <b>Consolidated Funds</b> and Public Accounts of India and of the states.                              |
| 267     | • <b>Contingency Fund:</b> a fund under the President/Governor for unforeseen expenditure.                              |
| 268     | • Duties levied by the Union but <b>collected and appropriated by the states.</b>                                       |
| 270     | • Taxes levied and collected by the Union but <b>distributed between the Union and the states.</b>                      |
| 274     | • <b>Prior recommendation of the President is required</b> for bills affecting taxation in which states are interested. |
| 275     | • <b>Grants-in-aid</b> from the Union to certain states.                                                                |
| 280     | • <b>Constitution of the Finance Commission:</b> by the President at the expiration of every fifth year.                |
| 281     | • The recommendations of the Finance Commission to be <b>laid before both Houses of Parliament</b> by the President.    |

### > Public Service Commissions (UPSC and UPPSC: Part XIV, Articles 315 - 323)

| Article | Main Constitutional Provision (Quick Pointers)                                                                              |
|---------|-----------------------------------------------------------------------------------------------------------------------------|
| 315     | • Provision for the constitution of <b>Public Service Commissions</b> for the Union (UPSC) and for the states (SPSC/UPPSC). |
| 316     | • <b>Appointment and term of office</b> of the Chairman and members of the Commission.                                      |
| 317     | • <b>Removal and suspension</b> of members of a Public Service Commission.                                                  |
| 318     | • Power to make regulations regarding the conditions of service of members and staff of the Commission.                     |
|         |                                                                                                                             |



| Article | Main Constitutional Provision (Quick Pointers)                                                                                                                                                 |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 319     | • Prohibition on members of the Commission holding any other government office on ceasing to be members.                                                                                       |
| 320     | • <b>Functions of the Union and State Public Service Commissions</b> (conducting examinations, etc.).                                                                                          |
| 321     | • <b>Power of Parliament</b> to extend the functions of the Public Service Commissions.                                                                                                        |
| 322     | • <b>Expenses:</b> the expenses of the UPSC are charged on the Consolidated Fund of India, while those of an SPSC (UPPSC) are charged on the <b>Consolidated Fund of the state concerned</b> . |
| 323     | • <b>Reports (annual reports)</b> of the Public Service Commissions: the UPSC submits its report to the President and the UPPSC to the Governor.                                               |

### ➤ Election Commission of India (ECI): Constitutional Provisions (Part XV)

| Article                  | Main Provision and Jurisdiction (Quick Pointers)                                                                                                                                                                                                                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Establishment and nature | • An <b>autonomous constitutional body</b> ; established on <b>25 January 1950</b> .                                                                                                                                                                                                                                        |
| Electoral responsibility | • Conducting the elections to the <b>Lok Sabha, Rajya Sabha, state Legislative Assemblies, and of the President and Vice-President</b> in the country.<br>• <b>Note:</b> it has <b>nothing to do with panchayat and municipal elections</b> in the states (for these there is provision for a 'State Election Commission'). |
| Article 324              | • Superintendence, direction and control of elections shall be vested in the Election Commission.                                                                                                                                                                                                                           |
| Article 325              | • Provision that no person shall be excluded from the electoral roll or declared ineligible on grounds of religion, race, caste or sex.                                                                                                                                                                                     |
| Article 326              | • Elections to the Lok Sabha and state Legislative Assemblies shall be on the basis of <b>universal adult suffrage</b> .                                                                                                                                                                                                    |
| Article 327              | • <b>Power of Parliament</b> to make provision with respect to elections to legislatures.                                                                                                                                                                                                                                   |
| Article 328              | • <b>Power of the state legislature</b> to make laws for elections to the legislature of that state.                                                                                                                                                                                                                        |
| Article 329              | • <b>Bar to interference by courts</b> in electoral matters/delimitation of constituencies.                                                                                                                                                                                                                                 |

### ➤ Special Provisions for States/Regions (Articles 370 and 371), Part 21

| Article     | Related State (Extremely Important for UPPCS Prelims Matching)                                                                         |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Article 370 | <b>Jammu Kashmir (rendered completely inoperative from 31 October 2019;</b> Jammu-Kashmir is now an ordinary Union Territory of India) |

| Article            | Related State (Extremely Important for UPPCS Prelims Matching) |
|--------------------|----------------------------------------------------------------|
| Article 371        | Maharashtra and Gujarat                                        |
| Article 371A       | Nagaland                                                       |
| Article 371B       | Assam                                                          |
| Article 371C       | Manipur                                                        |
| Article 371D and E | Andhra Pradesh (and Telangana)                                 |
| Article 371F       | Sikkim                                                         |
| Article 371G       | Mizoram                                                        |
| Article 371H       | Arunachal Pradesh                                              |
| Article 371I       | Goa                                                            |
| Article 371J       | Karnataka (Hyderabad-Karnataka region)                         |

### > Important Constitutional Amendment Acts

| Amendment | Year | Main Provisions and UPPCS Special Points                                                                                                             |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1st       | 1951 | • <b>The Ninth Schedule was added</b> ; an attempt to keep land reform laws outside judicial review.                                                 |
| 10th      | 1961 | • <b>Dadra and Nagar Haveli</b> was incorporated into the Indian Union (UT).                                                                         |
| 11th      | 1961 | • The election of the President/Vice-President cannot be challenged on the ground of <b>any vacancy in the electoral college</b> .                   |
| 12th      | 1962 | • <b>Goa, Daman and Diu</b> were incorporated into the Indian Union (UT).                                                                            |
| 31st      | 1973 | • The number of Lok Sabha seats was increased <b>from 525 to 545</b> .                                                                               |
| 36th      | 1975 | • <b>Sikkim was made a full (22nd) state of India</b> .                                                                                              |
| 80th      | 2000 | • An alternative scheme of devolution of taxes between the Union and the states was made (on the recommendation of the 10th Finance Commission).     |
| 85th      | 2001 | • Provision of <b>consequential seniority in promotion</b> for government employees of the SC/ST categories.                                         |
| 86th      | 2002 | • By adding <b>Article 21A</b> , <b>education was made a Fundamental Right</b> for children aged 6 to 14 (the 11th Fundamental Duty was also added). |
|           |      |                                                                                                                                                      |

| Amendment | Year | Main Provisions and UPPCS Special Points                                                                                                                                                   |
|-----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 97th      | 2012 | • Constitutional status to <b>co-operative societies</b> . <b>Part IX-B</b> and <b>Article 43B (DPSP)</b> were added. Forming a society became a Fundamental Right under Article 19(1)(c). |
| 99th      | 2014 | • Creation of the <b>National Judicial Appointments Commission (NJAC)</b> in place of the collegium. ( <i>The Supreme Court declared it unconstitutional and struck it down</i> ).         |
| 100th     | 2015 | • <b>India-Bangladesh Land Boundary Agreement (LBA)</b> ; amendment of the First Schedule for the exchange of territories.                                                                 |
| 101st     | 2016 | • The <b>Goods and Services Tax (GST)</b> was introduced.                                                                                                                                  |
| 102nd     | 2018 | • Constitutional status was given to the <b>National Commission for Backward Classes (NCBC)</b> (Article 338B added).                                                                      |
| 103rd     | 2019 | • Provision of <b>10% reservation</b> for the economically weaker sections ( <b>EWS</b> ) (Articles 15(6) and 16(6) added).                                                                |
| 104th     | 2020 | • Reservation of SC/ST seats in the Lok Sabha/Assemblies extended by 10 years (till 2030). <b>The Anglo-Indian quota was abolished</b> .                                                   |
| 105th     | 2021 | • <b>The power of the states</b> to prepare the list of socially and educationally backward classes ( <b>SEBC</b> ) <b>was restored</b> .                                                  |
| 106th     | 2023 | • <b>Women's Reservation Act (Nari Shakti Vandan Adhiniyam)</b> ; provision for <b>reserving 33% of seats</b> for women in the Lok Sabha and state Legislative Assemblies.                 |

## > Reorganisation of States

## > Commissions and Committees Related to the Reorganisation of States

| Name of the Commission / Committee | Year    | Principal Members                                         | Main Recommendations / UPPCS Special Facts                                                                                                                            |
|------------------------------------|---------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dhar Commission                    | 1947-48 | S.K. Dhar (Chairman)                                      | • The basis of reorganisation of states should be <b>administrative convenience, not language</b> (opposed the linguistic basis).                                     |
| JVP Committee                      | 1948    | Jawaharlal Nehru, Vallabhbhai Patel, Pattabhi Sitaramayya | • Regarded <b>the security, unity and economic prosperity of the country</b> as the main basis of reorganisation of states (rejected the immediate linguistic basis). |
|                                    |         |                                                           |                                                                                                                                                                       |

| Name of the Commission / Committee | Year | Principal Members                                           | Main Recommendations / UPPCS Special Facts                                                                                                                                      |
|------------------------------------|------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| States Reorganisation Commission   | 1953 | Syed Fazl Ali (Chairman), Hriday Nath Kunzru, K.M. Panikkar | <ul style="list-style-type: none"> <li>Accepted language as a basis for the reorganisation of states. The States Reorganisation Act, 1956 was enacted on its report.</li> </ul> |

**Important fact:** The first state formed on a linguistic basis in the country was **Andhra Pradesh (1 October 1953)**.

### > Reorganisation of States (Fazl Ali Commission) and the Chronological Years of New States

| Main Facts of the Fazl Ali Commission (1953-1956)                                                                                                                                      | Name of the State           | Year / Date of Formation (UPPCS Chronology Special) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------|
| • Constituted: 1953; report: 1956                                                                                                                                                      | Andhra Pradesh              | 1 October 1953                                      |
| • Members: Syed Fazl Ali (Chairman), Hriday Nath Kunzru, K.M. Panikkar.                                                                                                                | Gujarat and Maharashtra     | 1 May 1960                                          |
| • Recommendation: accepted language as a basis but rejected the principle of 'one state-one language'.                                                                                 | Nagaland                    | 1 December 1963                                     |
| • Result: the 7th Constitutional Amendment, 1956 was made. The classification of states in the original Constitution was abolished and 14 states and 6 Union Territories were created. | Haryana                     | 1 November 1966                                     |
|                                                                                                                                                                                        | Himachal Pradesh            | 25 January 1971                                     |
|                                                                                                                                                                                        | Meghalaya, Manipur, Tripura | 21 January 1972<br>(formed together)                |
|                                                                                                                                                                                        | Sikkim                      | 16 May 1975 (full state)                            |
|                                                                                                                                                                                        | Mizoram and Arunachal       | 20 February 1987                                    |
|                                                                                                                                                                                        | Goa                         | 30 May 1987                                         |
|                                                                                                                                                                                        | Chhattisgarh                | 1 November 2000                                     |
|                                                                                                                                                                                        | Uttarakhand                 | 9 November 2000                                     |
|                                                                                                                                                                                        | Jharkhand                   | 15 November 2000                                    |
|                                                                                                                                                                                        |                             |                                                     |

| Main Facts of the Fazl Ali Commission (1953-1956) | Name of the State | Year / Date of Formation (UPPCS Chronology Special) |
|---------------------------------------------------|-------------------|-----------------------------------------------------|
|                                                   | Telangana         | 2 June 2014                                         |

## > Panchayati Raj

### > Principal Committees and Development Related to Panchayati Raj

| Name of the Committee       | Year of Formation | Main Recommendations / UPPCS Special Facts                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Balwant Rai Mehta Committee | 1957              | <ul style="list-style-type: none"> <li>Recommended a <b>three-tier Panchayati Raj system</b> (village, block, district) and 'democratic decentralisation'.</li> <li><b>Note:</b> on this basis Pt. Nehru inaugurated Panchayati Raj for the first time in the country on <b>2 October 1959</b> in <b>Nagaur</b> district of Rajasthan.</li> </ul> |
| Ashok Mehta Committee       | 1977              | <ul style="list-style-type: none"> <li>Suggested a <b>two-tier panchayat system</b> (Zila Parishad and Mandal Panchayat).</li> </ul>                                                                                                                                                                                                              |
| L.M. Singhvi Committee      | 1986              | <ul style="list-style-type: none"> <li>Strongly recommended giving <b>constitutional status</b> to Panchayati Raj institutions.</li> </ul>                                                                                                                                                                                                        |
| P.K. Thungon Committee      | 1988              | <ul style="list-style-type: none"> <li>Recommended constitutional recognition of Panchayati Raj and regular elections.</li> </ul>                                                                                                                                                                                                                 |

**Constitutional status (Quick Fact):** During the tenure of Prime Minister P.V. Narasimha Rao, the panchayats got constitutional status through the **73rd Constitutional Amendment Act, 1992**, which came into effect from **24 April 1993** (that is why 24 April is celebrated as National Panchayati Raj Day).

## > Federal versus Unitary Features of the Constitution

| Federal Features of the Constitution                        | Unitary Features of the Constitution                                   |
|-------------------------------------------------------------|------------------------------------------------------------------------|
| • Dual polity (governments at the central and state levels) | • <b>A strong Centre</b> (the Centre being more powerful)              |
| • Written and rigid Constitution                            | • <b>Single Constitution</b> (states having no separate constitutions) |
| • Clear division of powers (Seventh Schedule)               | • <b>Single citizenship</b>                                            |
| • Supremacy of the Constitution                             | • <b>Flexibility of the Constitution</b> (in special circumstances)    |
| • Independent judiciary                                     | • <b>Integrated judiciary</b>                                          |

| Federal Features of the Constitution                       | Unitary Features of the Constitution            |
|------------------------------------------------------------|-------------------------------------------------|
| • <b>Bicameral legislature</b> (Lok Sabha and Rajya Sabha) | • <b>Appointment of Governors by the Centre</b> |
|                                                            | • <b>All-India Services</b> (IAS, IPS, IFS)     |
|                                                            | • <b>Emergency provisions</b> (Part XVIII)      |

## > The Four Main Instruments of Direct Democracy

| Type                 | Simple Definition / Meaning (Concept Pointer)                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. <b>Referendum</b> | Under it a proposed law or policy is sent directly to the voters for their vote (yes or no).                                                   |
| 2. <b>Initiative</b> | Through it the people themselves can prepare the draft of a law and ask the legislature (Parliament) to make the law.                          |
| 3. <b>Recall</b>     | Through it the voters can remove any of their representatives or officials from office even before the end of his term.                        |
| 4. <b>Plebiscite</b> | It is generally the process of ascertaining the opinion of the general public on an issue of great public importance or a territorial dispute. |

## 1 Constitutional Writs: Types and their Jurisdiction

| Name of the Writ     | Literal Meaning                | Against whom can it be Issued?                                                                                                                   | Against whom can it not be Issued?                                                                                                   | Special UPPCS Prelims Fact                                                                                                       |
|----------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Habeas Corpus</b> | "Let the body be produced"     | <ul style="list-style-type: none"> <li>• Against both public officials</li> <li>• and private individuals</li> </ul>                             | —                                                                                                                                    | <ul style="list-style-type: none"> <li>• It is the greatest protector of personal liberty.</li> </ul>                            |
| <b>Mandamus</b>      | "We command"                   | <ul style="list-style-type: none"> <li>• Public officials, bodies, corporations</li> <li>• Lower courts, tribunals and the government</li> </ul> | <ul style="list-style-type: none"> <li>• Private individuals or private bodies</li> <li>• President / Governor</li> </ul>            | <ul style="list-style-type: none"> <li>• Issued only to enforce public duties.</li> </ul>                                        |
| <b>Quo-Warranto</b>  | "By what authority or warrant" | <ul style="list-style-type: none"> <li>• Against a person who has illegally usurped a public (government) office</li> </ul>                      | <ul style="list-style-type: none"> <li>• Private offices</li> <li>• Ministerial offices</li> </ul>                                   | <ul style="list-style-type: none"> <li>• It can be filed by any interested person (need not be the aggrieved person).</li> </ul> |
| <b>Prohibition</b>   | "To forbid or prohibit"        | <ul style="list-style-type: none"> <li>• Judicial authorities (lower courts)</li> <li>• Quasi-judicial authorities</li> </ul>                    | <ul style="list-style-type: none"> <li>• Administrative authorities</li> <li>• Legislative bodies and private individuals</li> </ul> | <ul style="list-style-type: none"> <li>• It is issued only to stop proceedings while they are pending.</li> </ul>                |

| Name of the Writ  | Literal Meaning                    | Against whom can it be Issued?                                                                                                                    | Against whom can it not be Issued?                                                                          | Special UPPCS Prelims Fact                                                                                                                              |
|-------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Certiorari</b> | "To be informed / to be certified" | <ul style="list-style-type: none"> <li>Judicial and quasi-judicial authorities</li> <li>Administrative authorities (<i>since 1991</i>)</li> </ul> | <ul style="list-style-type: none"> <li>Legislative bodies</li> <li>Private individuals or bodies</li> </ul> | <ul style="list-style-type: none"> <li>Also called the "<b>post-mortem</b>" writ.</li> <li>To call up a pending case or to quash a decision.</li> </ul> |

## > Emergency Provisions in the Indian Constitution

| Type of Emergency                                                         | Related Article    | Source / Borrowed from Where? | Special Facts and Amendments (UPPCS Special)                                                                                                                                           |
|---------------------------------------------------------------------------|--------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>National Emergency</b>                                                 | <b>Article 352</b> | Government of India Act, 1935 | <ul style="list-style-type: none"> <li>By the <b>44th Amendment (1978)</b> the words '<b>internal disturbance</b>' were removed and replaced by '<b>armed rebellion</b>'.</li> </ul>   |
| <b>President's Rule</b><br>( <i>failure of constitutional machinery</i> ) | <b>Article 356</b> | Government of India Act, 1935 | <ul style="list-style-type: none"> <li>Imposed in the states on the report of the Governor or on failure to comply with the directions of the Centre (also under Art. 365).</li> </ul> |
| <b>Financial Emergency</b>                                                | <b>Article 360</b> | Government of India Act, 1935 | <ul style="list-style-type: none"> <li><b>A financial emergency has not been imposed even once</b> in India so far.</li> </ul>                                                         |

## > Emergency Provisions of the Indian Constitution (Comparative Table)

| Criterion / Feature                            | National Emergency (Article 352)                                                                                                              | President's Rule (Article 356)                                                                                                                               | Financial Emergency (Article 360)                                                                                                              |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main ground of proclamation</b>             | <ul style="list-style-type: none"> <li>War, external aggression or <b>armed rebellion</b>.</li> </ul>                                         | <ul style="list-style-type: none"> <li>Failure of the constitutional machinery in the states (on the report of the Governor or without a report).</li> </ul> | <ul style="list-style-type: none"> <li>Threat to the financial stability or credit of India or of any part of it.</li> </ul>                   |
| <b>Role of the cabinet</b>                     | <ul style="list-style-type: none"> <li>Proclaimed by the President only on the <b>written recommendation</b> of the Union Cabinet.</li> </ul> | <ul style="list-style-type: none"> <li>On the ordinary advice of the Union Cabinet.</li> </ul>                                                               | <ul style="list-style-type: none"> <li>On the ordinary advice of the Union Cabinet.</li> </ul>                                                 |
| <b>Period for first approval by Parliament</b> | <ul style="list-style-type: none"> <li>Must be approved by Parliament within <b>30 days (1 month)</b> of the date of proclamation.</li> </ul> | <ul style="list-style-type: none"> <li>Must be approved by Parliament within <b>2 months (60 days)</b> of the date of proclamation.</li> </ul>               | <ul style="list-style-type: none"> <li>Must be approved by Parliament within <b>2 months (60 days)</b> of the date of proclamation.</li> </ul> |
| <b>Majority for parliamentary approval</b>     | <ul style="list-style-type: none"> <li>Must be passed by both Houses of Parliament by a <b>special majority</b>.</li> </ul>                   | <ul style="list-style-type: none"> <li>Passed by both Houses of Parliament by a <b>simple majority</b>.</li> </ul>                                           | <ul style="list-style-type: none"> <li>Passed by both Houses of Parliament by a <b>simple majority</b>.</li> </ul>                             |



| Criterion / Feature             | National Emergency (Article 352)                                                                                                     | President's Rule (Article 356)                                                                    | Financial Emergency (Article 360)                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Duration of operation           | • 6 months at first; with Parliament's approval it can continue <b>indefinitely, being extended every 6 months</b> .                 | • 6 months at first; it can remain in force only for a maximum <b>total period of 3 years</b> .   | • Once approved, <b>its duration is not fixed</b> (it remains in force until Parliament revokes it). |
| Effect on Fundamental Rights    | • <b>Affected:</b> Fundamental Rights are suspended under Articles 358 and 359 (but <b>Articles 20 and 21 are never suspended</b> ). | • <b>No effect on the Fundamental Rights</b> of citizens.                                         | • <b>No effect on the Fundamental Rights</b> of citizens.                                            |
| Termination of the proclamation | • By the Lok Sabha passing a resolution by a <b>simple majority</b> , or by the President at any time.                               | • The President can himself withdraw the proclamation at any time without parliamentary approval. | • The President can revoke it at any time by a subsequent proclamation.                              |
| Financial effect                | —                                                                                                                                    | —                                                                                                 | • The financial powers of the states pass to the Centre; reduction of salaries possible.             |

### > Types of Bills and their Comparative Features

| Feature / Criterion                          | Government Bill                                        | Private Member's Bill                                            | Ordinary Bill                                                                                | Constitution Amendment Bill (Art. 368)                                      |
|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Introduced by                                | • A <b>minister</b> of the government.                 | • <b>Any member of Parliament</b> other than a minister.         | • Any member of Parliament (minister or non-minister).                                       | • Any member of Parliament.                                                 |
| House                                        | • Introduced in either House.                          | • Introduced in either House.                                    | • In <b>either House</b> (Lok Sabha or Rajya Sabha).                                         | • Can be introduced in <b>either House</b> .                                |
| Prior approval of the President              | —                                                      | —                                                                | • <b>Prior permission</b> of the President is <b>not required</b> .                          | • <b>Prior permission</b> of the President is <b>not required</b> .         |
| Chances of passing                           | • <b>High</b> (because the government has a majority). | • <b>Negligible / very low</b> (only 14 have become law so far). | • Passed by a simple majority of Parliament.                                                 | • Passed separately by each House by a <b>special majority</b> .            |
| Joint sitting in case of deadlock (Art. 108) | —                                                      | —                                                                | • <b>There is provision for a joint sitting</b> if the other House holds it up for 6 months. | • <b>There is no provision for a joint sitting</b> in case of disagreement. |

| Feature / Criterion    | Government Bill | Private Member's Bill | Ordinary Bill                                                                                                                         | Constitution Amendment Bill (Art. 368)                                                                   |
|------------------------|-----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Power of the President | —               | —                     | <ul style="list-style-type: none"> <li>The President may give assent, withhold it or <b>return it for reconsideration.</b></li> </ul> | <ul style="list-style-type: none"> <li>The President <b>must give assent</b> (it is binding).</li> </ul> |

## > Delimitation Commission

| Criterion / Component | Important Facts (Core Pointers)                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appointment           | Made by the <b>President</b> of India.                                                                                                                                                                           |
| Main function         | To determine the boundaries of constituencies throughout the country on the basis of the census.                                                                                                                 |
| Cooperation           | This commission works together with the <b>Election Commission of India (ECI)</b> .                                                                                                                              |
| Composition (members) | <ol style="list-style-type: none"> <li><b>Chairman:</b> a retired judge of the Supreme Court</li> <li>Chief Election Commissioner (CEC)</li> <li>State Election Commissioners of the states concerned</li> </ol> |
| Judicial review       | The decisions of the Delimitation Commission <b>cannot be challenged in any court.</b>                                                                                                                           |
| Constituted so far    | It has been constituted <b>4 times</b> in India so far: <b>1952, 1963, 1972 and 2002.</b>                                                                                                                        |

## > Composition of the Houses of Parliament

| Name of the House                                | Maximum Strength                       | Elected / Representation of Union Territories                                                                                        | Number of Nominated Members                                                             |
|--------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Lok Sabha<br>(Lower House / House of the People) | <b>550</b> (after the 104th Amendment) | <ul style="list-style-type: none"> <li><b>530</b> (from the states)</li> <li><b>20</b> (from the Union Territories - UTs)</li> </ul> | <b>0</b> (Anglo-Indian nomination abolished)                                            |
| Rajya Sabha<br>(Upper House / Council of States) | <b>250</b>                             | <ul style="list-style-type: none"> <li><b>238</b> (representatives of the states and Union Territories)</li> </ul>                   | <b>12</b> (nominated by the President from literature, science, art and social service) |

## > Public Accounts Committee (PAC)

| Component               | Principal Feature / Rule                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| History / establishment | The oldest parliamentary committee; established <b>in 1921 (under the Government of India Act 1919).</b> |

| Component                   | Principal Feature / Rule                                                                                                                      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Total membership            | <b>22 members [ 15 from the Lok Sabha + 7 from the Rajya Sabha ].</b>                                                                         |
| Term                        | The term of members is <b>1 year</b> (elected by the system of proportional representation).                                                  |
| Membership of ministers     | <b>No minister can become a member of this committee.</b>                                                                                     |
| Appointment of the Chairman | Made by the <b>Speaker of the Lok Sabha</b> from among its members (by convention a member of the opposition becomes Chairman).               |
| Main function               | To examine the appropriation accounts of the Government of India and the <b>reports of the CAG (Comptroller and Auditor-General)</b> on them. |

### > Attorney General of India

| Criterion / Component | Constitutional Provision (Article 76)                                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nature of the office  | The <b>highest law officer</b> of the country (chief legal adviser to the Government of India).                                                                                                                                     |
| Appointment           | <b>By the President</b> on the advice of the Union Cabinet.                                                                                                                                                                         |
| Term                  | <b>Not fixed by the Constitution.</b>                                                                                                                                                                                               |
| Removal               | The procedure/grounds of removal are not mentioned in the Constitution. He holds office <b>during the pleasure of the President.</b>                                                                                                |
| Principal functions   | <ul style="list-style-type: none"> <li>• To advise the Government of India on legal matters.</li> <li>• To appear on behalf of the government in the Supreme Court or any High Court in cases concerning the government.</li> </ul> |

### > Allocation of Lok Sabha Seats to the States and Union Territories (UTs)

| Name of the State / UT | Lok Sabha Seats | Name of the State | Lok Sabha Seats | Name of the State | Lok Sabha Seats |
|------------------------|-----------------|-------------------|-----------------|-------------------|-----------------|
| Uttar Pradesh          | 80              | Rajasthan         | 25              | Tripura           | 2               |
| Maharashtra            | 48              | Odisha            | 21              | Manipur           | 2               |
| Bihar                  | 40              | Telangana         | 17              | Goa               | 2               |
| Tamil Nadu             | 39              | Jharkhand         | 14              | Arunachal Pradesh | 2               |
| Madhya Pradesh         | 29              | Punjab            | 13              | Meghalaya         | 2               |

| Name of the State / UT | Lok Sabha Seats | Name of the State       | Lok Sabha Seats                                                                          | Name of the State                | Lok Sabha Seats |
|------------------------|-----------------|-------------------------|------------------------------------------------------------------------------------------|----------------------------------|-----------------|
| Karnataka              | 28              | Chhattisgarh            | 11                                                                                       | Dadra-Nagar Haveli and Daman-Diu | 2               |
| Gujarat                | 26              | Haryana                 | 10                                                                                       | Delhi (UT)                       | 7               |
| Andhra Pradesh         | 25              | Jammu-Kashmir (UT)      | 5                                                                                        | Uttarakhand                      | 5               |
| Himachal Pradesh       | 4               | States/UTs with 1 seat: | Ladakh, Sikkim, Chandigarh, Lakshadweep, Puducherry, Nagaland, Mizoram, Andaman-Nicobar. | 1 (each)                         |                 |

### > Recognised National Political Parties of India (Current 2026 Updated)

| Name of the Political Party                   | Year of Founding |
|-----------------------------------------------|------------------|
| 1. Indian National Congress (INC)             | 1885             |
| 2. Communist Party of India - Marxist (CPI-M) | 1964             |
| 3. Bharatiya Janata Party (BJP)               | 1980             |
| 4. Bahujan Samaj Party (BSP)                  | 1984             |
| 5. Aam Aadmi Party (AAP)                      | 2012             |
| 6. National People's Party (NPP)              | 2013             |

At present, according to the rules of the Election Commission of India, only **6 national political parties in all** are recognised in the country.

### > Fundamental Rights Available Only to Indian Citizens (Not to Foreigners)

| Article    | Subject Matter of the Fundamental Right (Core Concept)                                                                                                                              |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article 15 | Prohibition of discrimination on grounds only of religion, race, caste, sex or place of birth.                                                                                      |
| Article 16 | Equality of opportunity in matters of public employment.                                                                                                                            |
| Article 19 | 6 democratic freedoms: speech and expression, peaceful assembly, forming associations/unions, movement throughout the country, residence, and practising any profession/occupation. |

| Article    | Subject Matter of the Fundamental Right (Core Concept)                    |
|------------|---------------------------------------------------------------------------|
| Article 29 | Protection of the interests of minorities (language, script and culture). |
| Article 30 | Right of minorities to establish and administer educational institutions. |

### > Right to Equality (Part III: Articles 14 - 18) and its Legal Concept

| Article | Fundamental Right (Right to Equality)                                                          | 'Equality before law' (Art. 14)<br>(inspired by Britain / negative concept) | 'Equal protection of laws' (Art. 14)<br>(inspired by America / positive concept)                                          |
|---------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 14      | Equality before law and equal protection of laws.                                              | • Absence of special privileges in favour of any person.                    | • Equal treatment in both the rights and the obligations conferred by laws.                                               |
| 15      | Prohibition of discrimination on grounds only of religion, race, caste, sex or place of birth. | • The same uniform system for all persons under the ordinary courts.        | • The same law applied without discrimination to persons similarly situated.                                              |
| 16      | Equality of opportunity in matters of public employment (government jobs).                     | • No person (rich or poor, official or ordinary) is above the law.          | • <b>Rule of Law:</b> the Supreme Court has regarded this principle of A.V. Dicey as part of the <b>Basic Structure</b> . |
| 17      | <b>Abolition of untouchability</b> , and its practice in any form is forbidden.                |                                                                             |                                                                                                                           |
| 18      | <b>Abolition of titles</b> (except military and academic distinctions).                        |                                                                             |                                                                                                                           |

### > Principal Financial Committees of Parliament (Parliamentary Committees)

| Name of the Committee | Membership and Composition                                                                            | Term   | Nomination / Selection of the Chairman                            | Main Functions and Other Exam Facts                                                                                                                                                                                            |
|-----------------------|-------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Estimates Committee   | <b>30 members</b> (all of them <b>only from the Lok Sabha</b> ; no representation of the Rajya Sabha) | 1 year | Nominated by the Speaker of the Lok Sabha from among the members. | <ul style="list-style-type: none"> <li>• To bring about <b>economy</b> in government expenditure, organisational efficiency and administrative reform.</li> <li>• It is the <b>largest parliamentary committee</b>.</li> </ul> |

| Name of the Committee                   | Membership and Composition                                         | Term   | Nomination / Selection of the Chairman                                                             | Main Functions and Other Exam Facts                                                                                                                                                                                                                                       |
|-----------------------------------------|--------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Public Accounts Committee (PAC)</b>  | <b>22 members (15 from the Lok Sabha + 7 from the Rajya Sabha)</b> | 1 year | By convention the Speaker of the Lok Sabha appoints a <b>leader of the opposition</b> as Chairman. | <ul style="list-style-type: none"> <li>To examine the <b>annual reports of the CAG</b>, examine the appropriation accounts and give its report to the Lok Sabha.</li> <li>Nicknames: '<b>mini Lok Sabha</b>', '<b>twin sister of the Estimates Committee</b>'.</li> </ul> |
| <b>Committee on Public Undertakings</b> | <b>22 members (15 from the Lok Sabha + 7 from the Rajya Sabha)</b> | 1 year | Chosen by the Speaker of the Lok Sabha only from among the Lok Sabha members.                      | <ul style="list-style-type: none"> <li>To examine the reports and accounts of government/public undertakings (PSUs) and the CAG's reports on them.</li> <li><b>Special rule:</b> No minister can become a member of this committee.</li> </ul>                            |

05

# General Science

GENERAL SCIENCE

## > Important Discoveries Related to Science

## > Important Discoveries and Scientists Related to Chemistry

| Name of the Discovery / Theory                     | Related Scientist / Discoverer | UPPCS Exam Key Point                                                                                         |
|----------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------|
| Periodic Table                                     | Mendeleev                      | The first systematic table based on the atomic weights of elements.                                          |
| Modern Periodic Table                              | Moseley                        | A table based on the <b>atomic number</b> of elements.                                                       |
| Discovery of the electron                          | J.J. Thomson                   | The negatively charged fundamental particle of the atom.                                                     |
| Charge of the electron                             | Millikan                       | Determination of the charge by the <b>oil drop experiment</b> .                                              |
| Wave nature of the electron                        | De Broglie                     | Theory of the dual nature of matter.                                                                         |
| Discovery of the proton                            | Rutherford                     | The positively charged fundamental particle of the atom (note: Goldstein made the naming/initial discovery). |
| Discovery of the neutron                           | Chadwick                       | The neutral (charge-less) fundamental particle of the atom.                                                  |
| Atomic model / detailed form of the periodic table | Niels Bohr                     | A stable model of the distribution of electrons in orbits (shells).                                          |
| Wave theory of light / quantum theory              | Max Planck                     | Light energy travels in the form of small packets (quanta/photons).                                          |
| Discovery of X-rays                                | Roentgen                       | High-energy waves used in medicine and security.                                                             |
| Theory of the catalyst                             | Berzelius                      | Discovery of substances that change the rate of a chemical reaction.                                         |
| Radioactivity                                      | Henri Becquerel                | Discovery of substances that disintegrate spontaneously (SI unit: becquerel).                                |
| Artificial radioactivity                           | Madame Curie and F. Joliot     | Making elements artificially radioactive in the laboratory.                                                  |



| Name of the Discovery / Theory                | Related Scientist / Discoverer | UPPCS Exam Key Point                                                                  |
|-----------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------|
| Radioactive age determination (carbon dating) | W.F. Libby                     | Method of finding the age of fossils and old objects.                                 |
| Principle of electrolysis                     | Michael Faraday                | Laws of decomposition of chemical compounds by electric current.                      |
| Cyclotron                                     | Lawrence                       | A device that accelerates charged particles.                                          |
| Scattering of light by solid particles        | John Tyndall                   | The path of light becoming visible in a colloidal solution ( <b>Tyndall effect</b> ). |
| Discovery of the meson                        | Hideki Yukawa                  | Sub-atomic particles that explain nuclear forces.                                     |
| Discovery of argon (Ar)                       | Rayleigh and Ramsay            | The inert gas present in the largest amount in the atmosphere.                        |
| Discovery of neon, krypton and xenon          | Ramsay and Travers             | Discovery of other important noble/inert gases.                                       |
| pH value (pH scale)                           | Sorensen                       | A scale for measuring acidity and alkalinity (0 to 14).                               |

### › Order of Waves, their Discoverers and Principal Applications

| Name of the Wave               | Discoverer     | Main Applications Useful for the Exam (Exam Notes)                                                             |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------------------------|
| Gamma rays                     | Becquerel      | In nuclear reactions and for destroying cancer cells. ( <i>highest energy/frequency</i> )                      |
| X-rays                         | Roentgen       | In medicine (to see bone fractures, etc.) and in security checks.                                              |
| Ultraviolet rays               | Ritter         | In disinfection, water purification and detecting forged documents.                                            |
| Visible radiation              | Newton         | For seeing objects with human eyes and in photography.                                                         |
| Infrared radiation             | Herschel       | In remote controls and for heat treatment of patients' muscles.                                                |
| Short radio waves / microwaves | Heinrich Hertz | In radar systems, satellite communication and microwave ovens.                                                 |
| Long radio waves               | Marconi        | In cellular phones and in the transmission (broadcasting) of radio and TV waves. ( <i>longest wavelength</i> ) |

## > Medical Equipment and Diagnostic Tools and their Applications

| Medical Instrument                      | Main Function and Practical Use (Exam Facts)                                                                                 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Stethoscope</b>                      | Used to hear the rate and sound of the heartbeat.                                                                            |
| <b>Pacemaker</b>                        | To control the heart rate and bring it back to normal when it becomes too fast or too slow.                                  |
| <b>Sphygmomanometer</b>                 | Used to measure the blood pressure of the human body.                                                                        |
| <b>CT Scan</b>                          | To view bones, muscles, joints (knee, shoulder), internal organs (heart, stomach, intestines), fat tissue and blood vessels. |
| <b>Electrocardiography (ECG)</b>        | Used to detect abnormalities and diseases of the heart.                                                                      |
| <b>Electroencephalogram (EEG)</b>       | To measure/examine disorders of the brain and its electrical activity.                                                       |
| <b>Auto Analyser</b>                    | Used to test biological constituents such as glucose, urea and cholesterol in the blood.                                     |
| <b>Magnetic Resonance Imaging (MRI)</b> | To detect accurately any abnormality or internal disorder in the whole body or a particular organ.                           |
| <b>Endoscope</b>                        | An instrument for direct inspection of the internal organs of the body, working on the principle of optical fibres.          |

## > Principal Measuring Instruments and their Uses (Scientific Instruments Master Matrix)

| Measuring Instrument             | Field of Measurement / Basic Use (Core Function)                  |
|----------------------------------|-------------------------------------------------------------------|
| <b>Audiometer / Phonometer</b>   | Used to measure the intensity of sound.                           |
| <b>Odometer</b>                  | To measure the total distance covered by a wheeled vehicle.       |
| <b>Altimeter</b>                 | To measure the altitude of aircraft or mountains.                 |
| <b>Auxanometer / Crescograph</b> | An instrument that measures the growth of plants.                 |
| <b>Luxmeter / Photometer</b>     | An instrument to measure luminous intensity (intensity of light). |
| <b>Lactometer</b>                | To measure the purity of milk and its relative density.           |
| <b>Hydrometer</b>                | To measure the relative density of liquids.                       |
| <b>Hygrometer</b>                | To measure the humidity of the air or atmosphere.                 |
| <b>Manometer</b>                 | To measure the pressure of liquids or gases.                      |
| <b>Galvanometer</b>              | To detect the presence and direction of an electric current.      |

| Measuring Instrument            | Field of Measurement / Basic Use (Core Function)                                         |
|---------------------------------|------------------------------------------------------------------------------------------|
| <b>Ammeter</b>                  | To measure the actual magnitude of an electric current.                                  |
| <b>Anemometer</b>               | To measure the speed and velocity of wind.                                               |
| <b>Voltmeter / Electrometer</b> | Measures electric potential difference or electric charge.                               |
| <b>Seismograph</b>              | To measure earthquake waves and their intensity.                                         |
| <b>Thermometer</b>              | To measure ordinary human or industrial temperature.                                     |
| <b>Pyrometer / Actinometer</b>  | To measure high temperature and solar radiation.                                         |
| <b>Pyrheliometer</b>            | A special instrument to measure very intense solar radiation.                            |
| <b>Spectroheliograph</b>        | A special instrument used for photographing the sun.                                     |
| <b>Caratmeter</b>               | For exact determination of the purity of gold.                                           |
| <b>Fathometer</b>               | Used to measure the depth of the ocean.                                                  |
| <b>Tachometer</b>               | An instrument to measure rotational speed or RPM (revolutions per minute).               |
| <b>Polygraph</b>                | An instrument for detecting lies in criminal investigation (lie detector).               |
| <b>Bolometer</b>                | Measures thermal or electromagnetic radiation.                                           |
| <b>Evaporimeter</b>             | Measures the rate of evaporation of water.                                               |
| <b>Calorimeter</b>              | Measures the quantity of heat absorbed or released in a chemical or physical process.    |
| <b>Cryometer</b>                | A thermometer that measures very low (cryogenic) temperatures.                           |
| <b>Dynamometer</b>              | An instrument that measures the force, torque or power produced by an engine.            |
| <b>Electroscope</b>             | Detects the presence of electric charge in a body or object.                             |
| <b>Gravimeter</b>               | An instrument for the precise measurement of the Earth's gravitational acceleration (g). |
| <b>Gyroscope</b>                | An instrument measuring angular velocity and the direction of rotating bodies.           |
| <b>Hydrophone</b>               | An instrument that records sound waves under water.                                      |
| <b>Nephoscope</b>               | Measures the height of clouds, their velocity and direction of movement.                 |
| <b>Microphone</b>               | Converts sound waves into electrical signals.                                            |

## > Important Laws and Principles of Science (Scientific Laws)

| Law / Principle              | Basic Definition and Scientific Concept                                                                                                 | Practical Applications and Instruments                                                                        |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Pascal's law</b>          | Pressure applied at any one point of a confined fluid in equilibrium is transmitted equally in all directions.                          | Hydraulic lift, hydraulic press, hydraulic brakes.                                                            |
| <b>Hooke's law</b>           | Within the elastic limit, the stress on a body is always proportional to its strain (stress $\propto$ strain).                          | <b>Young's modulus</b> , which indicates the stiffness of skin or a material / its likelihood of deformation. |
| <b>Archimedes' principle</b> | The upthrust (buoyant force) on a solid immersed in a liquid is equal to the total weight of the liquid displaced by that solid.        | In the hydrometer and in the construction of ships and submarines.                                            |
| <b>Boyle's law</b>           | At a fixed/constant temperature, the volume of a given quantity of a gas is inversely proportional to its pressure ( $V \propto 1/P$ ). | Refilling system of gas cylinders and the human breathing process.                                            |

## > Newton's Laws of Motion

Note: The physicist and astronomer Sir Isaac Newton presented these laws in **1687** in his fundamental work "**Philosophiae Naturalis Principia Mathematica**" (Mathematical Principles of Natural Philosophy).

| Order of the Law  | Name of the Law                   | Basic Scientific Concept                                                                                                                                                                                 | Practical Examples Useful for the Exam (Exam Hits)                                                                                                                                                                                                                                          |
|-------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First law</b>  | <b>Law of inertia</b>             | A body continues in its state of motion or rest until an external unbalanced force is applied to it.                                                                                                     | Passengers leaning forward when a moving bus stops suddenly, or falling backward when a stationary vehicle starts.                                                                                                                                                                          |
| <b>Second law</b> | <b>Law of momentum</b>            | The rate of change of momentum is proportional to the applied force, and the change takes place in the direction of the force. Formula: <b><math>F = ma</math></b> (force = mass $\times$ acceleration). | A cricketer pulling his hands back while catching a ball so as to reduce injury.                                                                                                                                                                                                            |
| <b>Third law</b>  | <b>Law of action and reaction</b> | To every action there is an equal and opposite reaction.                                                                                                                                                 | <ul style="list-style-type: none"> <li>* <b>The backward jerk (recoil) when a bullet is fired from a gun.</b></li> <li>* <b>A rocket moving forward by the pressure of fast-moving gases.</b></li> <li>* Falling backwards when the rope breaks while drawing water from a well.</li> </ul> |

## > The Electromagnetic Spectrum in Increasing Order of Wavelength

| Electromagnetic Wave  | Wavelength Range                                    |
|-----------------------|-----------------------------------------------------|
| Gamma rays            | Less than 10 picometres                             |
| X-rays                | Between 0.01 nanometre (nm) and 10 nanometres (nm)  |
| Ultraviolet (UV) rays | Between 10 nanometres (nm) and 400 nanometres (nm)  |
| Visible light         | Between 400 nanometres (nm) and 700 nanometres (nm) |
| Infrared rays         | Between 700 nanometres (nm) and 1 millimetre (mm)   |
| Microwaves            | Between 1 millimetre (mm) and 1 metre (m)           |
| Radio waves           | Longer than 1 metre (m)                             |

**Decreasing order of wavelength:** radio waves > microwaves > infrared waves > visible waves > ultraviolet waves > X-rays > gamma rays.

- **Decreasing order of energy / frequency:** gamma rays > X-rays > ultraviolet waves > visible waves > infrared waves > microwaves > radio waves.

## > Spectrum Order

- **Order of increasing wavelength:** violet → blue → green → yellow → orange → red (*red has the longest wavelength*).
- **Order of increasing frequency / energy:** red → orange → yellow → green → blue → violet (*violet has the highest frequency/energy*).

## > Electromagnetic Waves and their Specific Applications

| Name of the Wave | Main Practical Uses Useful for the Exam (Exam Target)                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------|
| Gamma rays       | In medical science, to destroy harmful cancer cells.                                                              |
| X-rays           | In diagnostic work, for seeing the structure of bone fractures and for examining the internal organs of the body. |
| Ultraviolet rays | In water purifiers, to destroy harmful germs and bacteria.                                                        |
| Infrared rays    | In physiotherapy (treatment), TV remote controls and AC operating systems.                                        |
| Microwaves       | In radar systems, weather forecasting, microwave ovens, mobile phone networks and Wi-Fi.                          |
| Radio waves      | In radio broadcasting, television (TV) telecommunication systems and for long-distance commands/signals.          |

## > Classification of Sound Waves on the Basis of Frequency

| Type of Wave / Motion | Frequency Range / Mach Number (Core Concept)                                                                                   |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Infrasonic            | The frequency of the wave is <b>less than 20 hertz (Hz)</b> , below the range of human hearing (e.g., earthquake waves).       |
| Ultrasonic            | The frequency of the wave is <b>more than 20,000 hertz</b> , beyond the range of human hearing (used by bats and in sonar).    |
| Subsonic              | When a body or wave travels <b>slower than the speed of sound</b> (Mach number < 1).                                           |
| Supersonic            | When a body moves faster than the speed of sound ( <b>more than Mach 1</b> ) (e.g., fighter jets).                             |
| Hypersonic            | When a body travels at a speed far greater than the speed of sound ( <b>between Mach 5.0 and Mach 10.0</b> ) (e.g., missiles). |

## > Various Uses of Ultrasonic Waves (>20,000 Hz)

- **In SONAR equipment:** for locating hidden objects or submarines in the sea.
- **In cleaning work:** for removing dust from costly clothes, aircraft parts and the parts of expensive watches.
- **Industrial use:** for removing soot from factory chimneys and for heating liquids.
- **Medical field:** in the ultrasound machine, ECG, sonography and ultrasonography, etc.
- **Security and control:** for controlling automatic doors and for destroying harmful insects.

## > Speed of Sound - Important Rules at 25°C

- **Rule of the medium:** the speed of sound is always in the order **solid > liquid > gas** (sound cannot travel in a vacuum).
- **Correct order in solids:** aluminium (highest: **6420 m/s**) > nickel > steel > iron > brass.
- **Descending order in various media (Exam Note):** glass > water > hydrogen > helium > air.

## > Principal Fuel Gases and their Main Components

| Name of the Fuel Gas          | Principal Components / Constituents                 |
|-------------------------------|-----------------------------------------------------|
| CNG (Compressed Natural Gas)  | Methane, ethane                                     |
| LPG (Liquefied Petroleum Gas) | Butane, propane                                     |
| Coal gas                      | Hydrogen, methane, carbon monoxide                  |
| Water gas                     | Carbon monoxide and hydrogen (CO + H <sub>2</sub> ) |
| Biogas                        | Methane (CH <sub>4</sub> )                          |

| Name of the Fuel Gas | Principal Components / Constituents                |
|----------------------|----------------------------------------------------|
| Natural gas          | Methane (with small amounts of ethane and propane) |

### > Types of Glass and their Principal Uses

| Type of Glass | Main Practical Uses (Core Uses)                                    |
|---------------|--------------------------------------------------------------------|
| Flint glass   | For making cameras, telescopes and lenses.                         |
| Pyrex glass   | For making laboratory apparatus that withstands high temperatures. |
| Soda glass    | In everyday tube lights, glass bottles and window panes.           |
| Crown glass   | For making electric bulbs.                                         |
| Crookes glass | In the lenses of sunglasses (it blocks ultraviolet rays).          |
| Potash glass  | For making strong commercial glassware and beakers.                |
| Soft glass    | For making ordinary laboratory test tubes.                         |

### > Phenomena Caused by Surface Tension (Surface Tension Examples)

| Serial No. | Practical Examples of Surface Tension Asked Repeatedly in the Exam                                          |
|------------|-------------------------------------------------------------------------------------------------------------|
| 1          | A thin steel needle or blade floating on the surface of water/mercury.                                      |
| 2          | Adding soap or detergent lowers the surface tension of water, making washing of clothes easier.             |
| 3          | Formation of large bubbles of soap solution.                                                                |
| 4          | Spraying kerosene on the surface of water lowers the surface tension, so that mosquito larvae sink and die. |
| 5          | Hot soup tasting delicious (because, owing to lower surface tension, it spreads fully over the tongue).     |
| 6          | The bristles of a shaving brush cling together when taken out of water.                                     |
| 7          | Oil being sprayed to calm the stormy waves of the sea.                                                      |
| 8          | A drop of oil spreading very quickly over the surface of water.                                             |
| 9          | Falling raindrops being spherical.                                                                          |
| 10         | Cooling produced by surface tension when water is broken into fine droplets by a spray.                     |



## > Important Epidemiological Terminology

| Term            | Simple Medical Definition                                                                                                    | UPPCS Exam Point                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Endemic</b>  | When a disease or infection is always present permanently (regularly) in a particular geographical area or population.       | A disease confined to a definite area.      |
| <b>Epidemic</b> | An infectious disease spreading very rapidly in a short time in a definite population or large area.                         | A sudden local outbreak.                    |
| <b>Pandemic</b> | When an epidemic takes a severe form and spreads over a very large part of the world, several continents or the whole world. | A global epidemic (e.g., <b>COVID-19</b> ). |

## > Principal Isotopes and their Medical/Scientific Use

| Isotope                | Main Function / Practical Use Useful for the Exam                                  |
|------------------------|------------------------------------------------------------------------------------|
| <b>Uranium - 235</b>   | For finding the age of rocks, minerals and the ancient Earth (radiometric dating). |
| <b>Sodium - 24</b>     | For accurately detecting blood clots inside the human body.                        |
| <b>Cobalt - 60</b>     | For destroying cancer cells and controlling blood cancer (leukaemia).              |
| <b>Arsenic - 74</b>    | For detecting the position of tumours hidden inside the human body.                |
| <b>Phosphorus - 32</b> | In the treatment of leukaemia (blood cancer) and some bone disorders.              |
| <b>Iodine - 131</b>    | For testing the activity of the thyroid gland (goitre) and its treatment.          |

## > Important Natural Acids and their Sources

| Name of the Acid                 | Natural Source / Place of Occurrence                             | Specific Practical Use (Exam Facts)                                                  |
|----------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Acetic acid</b>               | Vinegar                                                          | In the preservation of pickles and foodstuffs.                                       |
| <b>Citric acid</b>               | Fruits of the citrus family such as orange, lemon, mosambi, etc. | For giving a sour taste to foodstuffs.                                               |
| <b>Lactic acid</b>               | Sour milk, curd                                                  | Produced by Lactobacillus in the process of milk setting into curd.                  |
| <b>Oxalic acid</b>               | Spinach, tomato, sorrel plants                                   | It is especially used <b>for removing rust or ink (colour) stains from clothes</b> . |
| <b>Ascorbic acid (Vitamin C)</b> | Amla, sour citrus fruits, guava                                  | Helps increase the body's immunity.                                                  |
| <b>Tartaric acid</b>             | Tamarind, grapes, raw mango                                      | Helpful in the manufacture of baking powder.                                         |

| Name of the Acid                    | Natural Source / Place of Occurrence          | Specific Practical Use (Exam Facts)                                                                            |
|-------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Formic acid (methanoic acid)</b> | Sting of red ants, bee sting, stinging nettle | Its sting causes burning and swelling on the skin.                                                             |
| <b>Hydrochloric acid (HCl)</b>      | Gastric juice of animals/humans               | <b>Performs the main function of digesting the food taken by animals in the internal process of digestion.</b> |

### > Principal Alkaline Compounds and their Natural Sources

| Name of the Base                              | Commercial/Natural Source                                   | Principal Practical Uses (Exam Points)                                                                               |
|-----------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Calcium hydroxide</b>                      | Lime water                                                  | In whitewashing, making bleaching powder and treating acidic soil.                                                   |
| <b>Ammonium hydroxide [NH<sub>4</sub>OH]</b>  | Detergents used for cleaning window glass (window cleaners) | In household cleaning and in chemicals that remove grease stains.                                                    |
| <b>Sodium hydroxide / potassium hydroxide</b> | Everyday soap                                               | Caustic soda (NaOH) is used in hard soap (for washing clothes) and potassium (hydroxide) in soft soap (for bathing). |
| <b>Magnesium hydroxide</b>                    | Milk of Magnesia                                            | As an excellent <b>antacid</b> to remove excess acidity of the stomach.                                              |

### > International Standard Units of Measurement (SI Units)

| Physical Quantity          | Standard SI Unit (Symbol) |
|----------------------------|---------------------------|
| <b>Length</b>              | Metre (m)                 |
| <b>Mass</b>                | Kilogram (kg)             |
| <b>Time</b>                | Second (s)                |
| <b>Temperature</b>         | Kelvin (K)                |
| <b>Amount of substance</b> | Mole (mol)                |
| <b>Luminous intensity</b>  | Candela (cd)              |
| <b>Power of a lens</b>     | Dioptre (D)               |
| <b>Heat / energy</b>       | Joule (J)                 |
| <b>Pressure</b>            | Pascal (Pa)               |
| <b>Inductance</b>          | Henry (H)                 |

| Physical Quantity                  | Standard SI Unit (Symbol)      |
|------------------------------------|--------------------------------|
| Resistivity or specific resistance | Ohm-metre ( $\Omega \cdot m$ ) |
| Radioactivity                      | Becquerel (Bq)                 |

## > Various Phenomena of Light

| Phenomenon of Light              | Real Examples and Applications Useful for the Exam (Exam Hotspots)                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Refraction of light</b>       | <ul style="list-style-type: none"> <li>* An object or fish lying in water appearing less deep than its real depth.</li> <li>* The bottom of a vessel or pond filled with water appearing raised.</li> <li>* The sun being visible a few minutes before actual sunrise and after sunset.</li> <li>* The sun appearing flattened or oval near the horizon at sunrise and sunset.</li> <li>* Twinkling of stars in a clear sky (because light passes through atmospheric layers).</li> </ul> |
| <b>Reflection of light</b>       | <ul style="list-style-type: none"> <li>* Seeing one's face/image in an ordinary mirror.</li> <li>* Images (reflections) of outside scenes appearing on the surface of calm or still water.</li> <li>* Light returning after striking any shiny, smooth surface.</li> </ul>                                                                                                                                                                                                                |
| <b>Total internal reflection</b> | <ul style="list-style-type: none"> <li>* A diamond sparkling brilliantly because of its special cut.</li> <li>* Rapid communication without energy loss in optical fibres.</li> <li>* Formation of a 'mirage' in deserts or on roads in summer.</li> <li>* In the 'endoscope', an instrument used in medicine to examine the internal organs of the body.</li> </ul>                                                                                                                      |
| <b>Interference of light</b>     | <ul style="list-style-type: none"> <li>* A thin film of soap foam or bubbles appearing coloured in sunlight.</li> <li>* A film of kerosene or diesel oil spread on the surface of water appearing coloured.</li> <li>* The surface of a CD (compact disc) appearing rainbow-coloured when light falls on it.</li> </ul>                                                                                                                                                                   |
| <b>Scattering of light</b>       | <ul style="list-style-type: none"> <li>* The clear sky appearing blue.</li> <li>* The water of the deep sea appearing blue.</li> <li>* The sun appearing red at sunrise and sunset.</li> <li>* <b>Danger signals being made red</b> on roads and railways (because red has the longest wavelength, it is scattered least and is visible from a distance).</li> </ul>                                                                                                                      |

## > Scalar and Vector Quantities

| Type of Quantity         | Basic Definition and Feature                               | Exact Examples Useful for the Exam (Exam Examples)                                                               |
|--------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Scalar quantities</b> | They have only <b>magnitude</b> and no definite direction. | Distance, volume, mass, density, work, energy, time, temperature, power, charge, speed, electric potential, etc. |
|                          |                                                            |                                                                                                                  |

| Type of Quantity         | Basic Definition and Feature                         | Exact Examples Useful for the Exam (Exam Examples)                                                   |
|--------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Vector quantities</b> | They must have both <b>magnitude and direction</b> . | Displacement, velocity, acceleration, force, momentum, impulse, electric field, magnetic field, etc. |

### > Principal Characteristics of Sound Waves

| Characteristic     | Simple Physical Meaning and its Effect                                                                            |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Wavelength</b>  | The total distance travelled by a sound wave before it repeats itself (completes one cycle).                      |
| <b>Amplitude</b>   | Indicates the maximum displacement of the particles of the medium. <b>High amplitude = loud sound</b> .           |
| <b>Frequency</b>   | The number of wave cycles produced per second. <b>High frequency = thin/sharp sound (high pitch/shrillness)</b> . |
| <b>Time period</b> | The total time required to complete one full wave cycle.                                                          |
| <b>Velocity</b>    | The speed with which a sound wave travels through a medium.                                                       |

### > Important Facts about Convex and Concave Lenses (Lenses Matrix)

| Feature / Point                     | Convex Lens (converging)                                                                                                                                                                                                                                                                                                                | Concave Lens (diverging)                                                                                                                                                                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Action on light</b>              | <b>Converges</b> parallel rays of light to a point.                                                                                                                                                                                                                                                                                     | <b>Diverges</b> parallel rays of light outward.                                                                                                                                                                                                                              |
| <b>Physical structure</b>           | This lens is <b>thick in the middle</b> and thin at the edges.                                                                                                                                                                                                                                                                          | This lens is <b>thin in the middle</b> and thick at the edges.                                                                                                                                                                                                               |
| <b>Correction of vision defects</b> | Spectacles of this lens are made to correct <b>hypermetropia or long-sightedness</b> .                                                                                                                                                                                                                                                  | Spectacles of this lens are made to correct <b>myopia or short-sightedness</b> .                                                                                                                                                                                             |
| <b>Nature of the image</b>          | Depending on the position of the object, it can form both <b>real and virtual</b> images.                                                                                                                                                                                                                                               | Always forms an image that is <b>smaller than the object, erect and virtual</b> .                                                                                                                                                                                            |
| <b>Main practical applications</b>  | <ul style="list-style-type: none"> <li>* Side mirrors (rear-view) of vehicles.</li> <li>* The natural lens inside the human eye.</li> <li>* In movie projectors, cameras and magnifying glasses.</li> <li>* In solar cookers for concentrating sunlight at one place.</li> <li>* In astronomical telescopes and microscopes.</li> </ul> | <ul style="list-style-type: none"> <li>* The mirror used by dentists to look inside the mouth (dentist mirror).</li> <li>* In spectacles for correcting short-sightedness.</li> <li>* In flashlights, peepholes (the small hole in a door) and laser instruments.</li> </ul> |

## > Transformation of Energy

| Device                                | Initial Form of Energy | Transformed Form of Energy (Final Output) |
|---------------------------------------|------------------------|-------------------------------------------|
| Electric motor                        | Electrical energy      | Into mechanical energy                    |
| Engine (heat engine)                  | Heat energy            | Into mechanical energy                    |
| Loudspeaker                           | Electrical energy      | Into sound energy                         |
| Transmitter                           | Electrical energy      | Into electromagnetic energy               |
| Candle                                | Chemical energy        | Into light and heat energy                |
| Photoelectric cell                    | Light energy           | Into electrical energy                    |
| Sitar                                 | Mechanical energy      | Into sound energy                         |
| Electric generator (dynamo/generator) | Mechanical energy      | Into electrical energy                    |
| Electric bulb                         | Electrical energy      | Into light and heat energy                |
| Electric cell (battery)               | Chemical energy        | Into electrical energy                    |
| Tubelight                             | Electrical energy      | Into light energy                         |

## > Practical Applications of Mirrors

| Type of Mirror        | Main Practical Uses and Examples (Exam Hotspot)                                                                                                                                                                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concave mirror</b> | <ul style="list-style-type: none"> <li>* In torches, searchlights and vehicle headlights (to obtain a powerful parallel beam of light).</li> <li>* As a shaving mirror for seeing an enlarged image of the face.</li> <li>* Used by dentists to see enlarged images of patients' teeth.</li> <li>* As a reflector in solar furnaces for concentrating sunlight.</li> </ul> |
| <b>Convex mirror</b>  | <ul style="list-style-type: none"> <li>* In the side-view (wing) mirrors of vehicles for a wide view of the rear (because it always forms an erect, diminished image and has a wide field of view).</li> <li>* In street lamps for spreading light over a large area.</li> <li>* In sunglasses.</li> </ul>                                                                 |

## > Principal Gases, Chemical Compounds and their Industrial Uses

| Name of the Gas / Chemical  | Main Industrial and Practical Uses Useful for the Exam (Exam Notes)                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hydrogen peroxide</b>    | <ul style="list-style-type: none"> <li>* In medicine as a disinfectant and as a wash for cleaning wounds, gargling and cleaning ears.</li> <li>* In bleaching delicate materials such as hair, cotton, silk, leather and feathers.</li> <li>* As a tooth and mouth wash and as a rocket fuel (oxidiser).</li> <li>* In the food industry as a means of preserving milk, fish, meat and other foods from spoiling.</li> </ul>                                                 |
| <b>Helium (He)</b>          | <ul style="list-style-type: none"> <li>* Being extremely light and non-inflammable, for filling the tyres of aircraft and balloons.</li> <li>* Used in place of pure air in delicate eye surgeries (medical surgeries).</li> </ul>                                                                                                                                                                                                                                           |
| <b>Neon (Ne)</b>            | <ul style="list-style-type: none"> <li>* Used in television screens, advertisement boards (neon signs) and plasma display tubes.</li> <li>* For bright light in high-voltage discharge tubes.</li> </ul>                                                                                                                                                                                                                                                                     |
| <b>Xenon (Xe)</b>           | <ul style="list-style-type: none"> <li>* It is also called the '<b>stranger gas</b>'. It is used in high-speed photography.</li> <li>* In medical science as an anaesthetic, and in smartphone flashes and high-intensity discharge lamps.</li> </ul>                                                                                                                                                                                                                        |
| <b>Sulphur hexafluoride</b> | <ul style="list-style-type: none"> <li>* As an excellent gaseous electrical insulator (dielectric) in high-voltage electrical equipment.</li> <li>* In the construction of special modern windows and doors for sound-proofing and thermal insulation.</li> </ul>                                                                                                                                                                                                            |
| <b>Sulphuric acid</b>       | <ul style="list-style-type: none"> <li>* In the production of chemical fertilisers (such as superphosphate). It is called the '<b>king of acids</b>'.</li> <li>* In the manufacture of industrial explosives and in the nylon/rayon synthetic textile industries.</li> <li>* In petroleum refining industries, in the manufacture of medicines and in the storage batteries (lead-acid batteries) of automobiles.</li> </ul>                                                 |
| <b>Nitrogen</b>             | <ul style="list-style-type: none"> <li>* In the industrial refining of sulphur and the highly inflammable acetylene gas.</li> <li>* For creating an inert atmosphere to protect (preserve) canned grains and fruit juices from oxidation.</li> <li>* In the production of ammonia gas and of sodium cyanamide, sulphocyanide and calcium cyanamide fertilisers.</li> <li>* In military and industrial fertilisers, explosives and as high-security preservatives.</li> </ul> |
| <b>Ammonia</b>              | <ul style="list-style-type: none"> <li>* In the large-scale production of nitrogenous chemical fertilisers (such as ammonium sulphate, ammonium nitrate, urea, ammonium phosphate and ammonium carbonate).</li> <li>* In the synthesis of nitric acid and other important industrial chemicals.</li> </ul>                                                                                                                                                                   |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Name of the Gas / Chemical | Main Industrial and Practical Uses Useful for the Exam (Exam Notes)                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chlorine</b>            | <ul style="list-style-type: none"> <li>* In the production of wood pulp in paper mills and in the manufacture of bleaching powder.</li> <li>* In the synthesis of various products of the chemical industry and of pesticides (such as PVC).</li> <li>* As a germicide, for making drinking water (water supply) completely pure and bacteria-free.</li> <li>* In the production of special industrial gases for photography, pharmaceuticals and the match industry.</li> </ul> |

### ➤ Principal Chemical Compounds: Names, Formulae and Specific Uses

| Trade / Common Name           | Chemical Name                | Specific Uses and Key Points for the UPPCS Exam                                                                                                                                                    |
|-------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Common salt</b>            | Sodium chloride              | <ul style="list-style-type: none"> <li>• For taste and preservation of food.</li> <li>• For making ORS solution in dehydration.</li> </ul>                                                         |
| <b>Baking soda</b>            | Sodium bicarbonate           | <ul style="list-style-type: none"> <li>• For leavening in the bakery industry (bread, cake).</li> <li>• In <b>fire extinguishers</b> and for removing acidity of the stomach (antacid).</li> </ul> |
| <b>Washing soda</b>           | Sodium carbonate             | <ul style="list-style-type: none"> <li>• In washing clothes and in the glass, soap and paper industries.</li> <li>• For removing the <b>permanent hardness of water</b>.</li> </ul>                |
| <b>Bleaching powder</b>       | Calcium oxychloride          | <ul style="list-style-type: none"> <li>• For disinfecting drinking water.</li> <li>• As a bleaching agent in the textile and paper industries.</li> </ul>                                          |
| <b>Plaster of Paris (POP)</b> | Calcium sulphate hemihydrate | <ul style="list-style-type: none"> <li>• For plastering broken bones (to set them).</li> <li>• For making statues, toys and ceiling designs (POP) in houses.</li> </ul>                            |
| <b>Gypsum</b>                 | Calcium sulphate dihydrate   | <ul style="list-style-type: none"> <li>• For slowing down the setting rate of cement (retarder).</li> <li>• For the improvement/treatment of <b>alkaline soil</b>.</li> </ul>                      |
| <b>Quick lime</b>             | Calcium oxide                | <ul style="list-style-type: none"> <li>• In the manufacture of cement and glass.</li> <li>• For drying gases in the laboratory (drying agent).</li> </ul>                                          |
| <b>Slaked lime</b>            | Calcium hydroxide            | <ul style="list-style-type: none"> <li>• For whitewashing houses.</li> <li>• For neutralising <b>acidic soil</b>.</li> </ul>                                                                       |
| <b>Blue vitriol</b>           | Copper sulphate              | <ul style="list-style-type: none"> <li>• As a <b>fungicide</b> (e.g., Bordeaux mixture).</li> <li>• In electroplating and the dyeing industry.</li> </ul>                                          |
| <b>Green vitriol</b>          | Ferrous sulphate             | <ul style="list-style-type: none"> <li>• In making medicine for anaemia and in making ink.</li> </ul>                                                                                              |



| Trade / Common Name       | Chemical Name                | Specific Uses and Key Points for the UPPCS Exam                                         |
|---------------------------|------------------------------|-----------------------------------------------------------------------------------------|
| White vitriol             | Zinc sulphate                | • In eye medicines (eye drops) and the leather industry.                                |
| Caustic soda              | Sodium hydroxide             | • In the manufacture of soap, detergents and artificial silk (rayon).                   |
| Caustic potash            | Potassium hydroxide          | • In making <b>soft (shaving) soap</b> and liquid soap.                                 |
| Laughing gas              | Nitrous oxide                | • As an <b>anaesthetic</b> in surgery and dentistry.                                    |
| Chloroform                | Trichloromethane             | • As a solvent and formerly as an anaesthetic (for unconsciousness).                    |
| Heavy water               | Deuterium oxide              | • As a <b>moderator</b> and coolant in nuclear reactors.                                |
| Alum                      | Potassium aluminium sulphate | • For purification of water (by coagulation).<br>• As an antiseptic (to stop bleeding). |
| Main constituent of glass | Silica (silicon dioxide)     | • In the manufacture of glass and quartz.                                               |
| Vinegar                   | Acetic acid (dilute)         | • For preservation of foodstuffs and for giving sourness.                               |
| Shora / saltpetre         | Potassium nitrate            | • In making <b>gunpowder</b> , fireworks and fertilisers.                               |
| Lunar caustic             | Silver nitrate               | • In making <b>voting ink (indelible ink)</b> .<br>• For silvering the back of mirrors. |
| Agent of artificial rain  | Silver iodide                | • For causing <b>artificial rain</b> through cloud seeding.                             |

### > Principal Organic Compounds and their Applications (Organic Chemistry Matrix)

| Organic Compound              | Main Practical Applications / Special Facts Useful for the Exam (Exam Notes)           |
|-------------------------------|----------------------------------------------------------------------------------------|
| Citric acid                   | For the long-term preservation of pickles, murabbas and fruit juices.                  |
| Naphthalene                   | Its white balls kept among clothes prevent insects (moths) from damaging the clothes.  |
| Tear gas (chloroacetophenone) | Used by the police to disperse riots or uncontrolled crowds.                           |
| Carotene                      | The natural pigment <b>that gives carrots their characteristic red/orange colour</b> . |
|                               |                                                                                        |

| Organic Compound                                    | Main Practical Applications / Special Facts Useful for the Exam (Exam Notes)                                                      |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Gammexane (BHC) and DDT                             | As a powerful insecticide in agricultural and domestic use.                                                                       |
| Mercaptan (ethyl mercaptan)                         | A strongly smelling substance added <b>to detect leakage of LPG gas immediately</b> .                                             |
| Potassium permanganate (KMnO <sub>4</sub> )         | For cleaning well water and in medicine as an antibacterial substance (red medicine).                                             |
| Ethylene / calcium carbide                          | Chemical substances used for artificially (quickly) ripening raw green fruits.                                                    |
| Polycarbonates                                      | Polymers used in the manufacture of very strong <b>bullet-proof glass</b> .                                                       |
| Oxalic acid                                         | Used for removing ink stains from clothes and in the manufacturing process of photography.                                        |
| Allethrin and pyrethrin                             | The main constituents used <b>in making mosquito-repellent liquids and coils</b> for homes.                                       |
| Sodium benzoate / benzoic acid                      | <b>The most widely used in the preservation of foodstuffs such as pickles, jams, sauces and fruit juices (food preservation).</b> |
| Ethylene glycol                                     | In cold countries as an <b>anti-freeze in car radiators</b> (prevents water from freezing).                                       |
| Carbon tetrachloride (CCl <sub>4</sub> ) / paraffin | In liquid fire extinguishers (Pyrene) and in industry as a solvent/coolant.                                                       |
| Bakelite                                            | A thermosetting plastic; used <b>in making handles, switches and plugs of electrical equipment</b> .                              |

### > Principal Human Diseases and their Responsible Agents (Environmental Diseases)

| Name of the Disease      | Responsible Pollutant / Metallic Agent (Core Cause)                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| Blood cancer (leukaemia) | Due to industrial exposure to <b>benzene</b> .                                                                |
| Itai-Itai                | Due to pollution caused by the metal <b>cadmium</b> (a disease of the bones).                                 |
| Minamata                 | Due to consuming fish/water contaminated with <b>mercury (Hg)</b> (a disease of the nervous system).          |
| Blue Baby Syndrome       | Due to an excess of <b>nitrate (–NO<sub>3</sub>)</b> in drinking water (methaemoglobinaemia).                 |
| Blackfoot disease        | Due to drinking water contaminated with <b>arsenic (As)</b> for a long time (a disease of the skin and feet). |

| Name of the Disease | Responsible Pollutant / Metallic Agent (Core Cause)                                               |
|---------------------|---------------------------------------------------------------------------------------------------|
| Wilson's disease    | Due to excessive accumulation/toxicity of the metal <b>copper (Cu)</b> in the body.               |
| Siderosis           | Due to the deposition of fine particles of <b>iron (Fe)</b> in the lungs (a respiratory disease). |

### > Gases and their Characteristics Useful for the Exam

| Name of the Gas (Compound) | Main Characteristic Useful for the Exam (Exam Facts)                                                                                                                                                                          |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carbon monoxide (CO)       | <b>It combines with haemoglobin in the blood to form 'carboxyhaemoglobin',</b> which completely blocks the oxygen-carrying capacity of the blood and the person dies of suffocation. It is also called the 'suffocating gas'. |
| Nitrous oxide              | It is called <b>laughing gas</b> ; used as an anaesthetic in minor surgery.                                                                                                                                                   |
| Carbon dioxide             | The main gas responsible for the <b>greenhouse effect</b> . Its amount in the atmosphere is about <b>0.03%</b> .                                                                                                              |
| Methane                    | A gas emitted from marshy areas and paddy fields; <b>the main constituent of biogas and CNG</b> .                                                                                                                             |
| Radon                      | The only radioactive <b>noble/inert gas that is not normally found in the atmosphere</b> .                                                                                                                                    |
| Chlorine                   | The gas used by municipal corporations to make <b>drinking water pure and bacteria-free</b> .                                                                                                                                 |
| Butane                     | <b>The main constituent of the domestic LPG cylinder</b> (along with propane).                                                                                                                                                |

### > Carbon and its Allotropes

- **The 3 main allotropes of carbon:** (1) diamond, (2) graphite, (3) buckminsterfullerene (C<sub>60</sub>).
- **Graphene:** a layer of pure carbon **only one atom thick** (2D structure), which is extremely strong and a good conductor of electricity.
- **Diamond:** the **hardest allotrope** of carbon found naturally (it sparkles because of total internal reflection).
- **Graphite:** thermodynamically **the most stable form of carbon**; it is soft and a good conductor of electricity.

### > Prelims Facts Related to Metals and Non-metals

- **Metal found in blood:** iron (at the centre of haemoglobin).
- **Lightest metal:** lithium (used in batteries).
- **Heaviest metal:** osmium.
- **Hardest metal:** platinum (also called white gold).
- **Filament of the electric bulb:** tungsten (because of its high melting point).

- **Best conductor of heat and electricity among metals:** silver.
- **Soft metals that can be easily cut with a knife:** sodium (Na) and potassium (K).
- **The only non-metal that remains liquid at room temperature (25°C):** bromine (a dark red liquid).

## > Difference between Physical Change and Chemical Change

| Point of Comparison                | Physical Change                                                                                                                                                                            | Chemical Change                                                                                                                                                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New substance                      | No new substance is formed in it.                                                                                                                                                          | A new substance is necessarily formed, whose properties differ from those of the original substance.                                                                                                                                                                                                 |
| Permanence and reversibility       | It is a <b>temporary change</b> which is easily <b>reversible</b> .                                                                                                                        | It is a <b>permanent change</b> , which is usually <b>irreversible</b> .                                                                                                                                                                                                                             |
| Exchange of energy                 | Usually a very small amount of heat or light energy is absorbed or released.                                                                                                               | A large amount of heat or light energy is absorbed or released.                                                                                                                                                                                                                                      |
| Basic nature of the substance      | The chemical composition/nature of the substance does not change.                                                                                                                          | The basic chemical nature of the substance changes completely.                                                                                                                                                                                                                                       |
| Principal examples (Exam Hotspots) | <ul style="list-style-type: none"> <li>* Water freezing into ice.</li> <li>* Water boiling into steam.</li> <li>* Melting of wax.</li> <li>* Sugar or salt dissolving in water.</li> </ul> | <ul style="list-style-type: none"> <li>* Digestion of food.</li> <li>* Burning of paper or wood.</li> <li>* Rusting of metals.</li> <li>* Silver utensils turning black/tarnishing.</li> <li>* Burning of fuel (LPG, petrol).</li> <li>* Curdling of milk or formation of curd from milk.</li> </ul> |

## 2 Energy-Transforming Devices (Energy Transducers)

| Name of the Device | Initial Form of Energy | Transformed Form of Energy (Final Output) |
|--------------------|------------------------|-------------------------------------------|
| Dynamo             | Mechanical energy      | Into electrical energy                    |
| Microphone         | Sound energy           | Into electrical energy                    |
| Loudspeaker        | Electrical energy      | Into sound energy                         |
| Solar cell         | Solar energy           | Into electrical energy                    |
| Tubelight          | Electrical energy      | Into light energy                         |
| Electric motor     | Electrical energy      | Into mechanical energy                    |
| Electric bulb      | Electrical energy      | Into light and heat energy                |

| Name of the Device      | Initial Form of Energy | Transformed Form of Energy (Final Output) |
|-------------------------|------------------------|-------------------------------------------|
| Electric cell (battery) | Chemical energy        | Into electrical energy                    |
| Sitar                   | Mechanical energy      | Into sound energy                         |

### > Principal Common Alloys: Chemical Composition and Uses (Master Alloys Matrix)

| Alloy                        | Chemical Composition                                                       | Main Practical Uses (Core Uses)                                                               |
|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Bronze</b>                | Copper (88%) + tin (12%)                                                   | In making statues, coins, medals and strong utensils.                                         |
| <b>Brass</b>                 | Copper (70%) + zinc (30%)                                                  | In band instruments, radio parts, wires, mridangams and machine tools.                        |
| <b>Gun metal</b>             | Copper (88%) + tin (8-10%) + zinc (2-4%)                                   | In making statues, gears, bearings, shutters, gutters and weapons.                            |
| <b>Solder</b>                | Tin (50-60%) + lead (40-50%)                                               | For <b>soldering (joining) metals</b> in electronic circuits.                                 |
| <b>Bell metal</b>            | Copper (80%) + tin (20%)                                                   | In making temple bells, household utensils and statues.                                       |
| <b>Rolled gold</b>           | Copper (90%) + aluminium (10%)                                             | In making cheap ornaments <b>that look like gold</b> .                                        |
| <b>Dutch metal</b>           | Copper (80-85%) + zinc (15-20%)                                            | In making very thin foils or leaves resembling copper or gold.                                |
| <b>Stainless steel</b>       | Chromium (15-20%) + nickel (8-10%) + iron                                  | Rust-free household utensils, <b>surgical instruments</b> , blades, etc.                      |
| <b>Chrome steel</b>          | Chromium (2-3%) + iron (95%) + carbon (0.5%) + manganese (0.5-1%)          | In making parts of heavy machinery, cutting tools and strong safes.                           |
| <b>Magnalium</b>             | Aluminium (95-96%) + magnesium (2%) + iron (2.3%)                          | In making light metallic mirrors and scientific instruments.                                  |
| <b>Duralumin</b>             | Aluminium (95%) + copper (4%) + magnesium (0.5%) + manganese (0.5%)        | Being very light and strong, in making the <b>frames of aircraft, buses and automobiles</b> . |
| <b>German silver</b>         | Copper (50%) + zinc (35%) + nickel (15%)                                   | <b>(Note: it contains 0% silver)</b> ; in heating equipment, utensils, fancy belts, etc.      |
| <b>Chrome-vanadium steel</b> | Iron (97-98%) + chromium (0.8-1.1%) + manganese (0.7-0.9%) + nickel (0.5%) | In making high-stress automobile springs, axles and strong tools.                             |

| Alloy            | Chemical Composition                             | Main Practical Uses (Core Uses)                                                                              |
|------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Tungsten steel   | Iron + tungsten + carbon                         | In rocket-engine nozzles that withstand very high temperatures and in heavy machine tools.                   |
| Aluminium bronze | Copper (90%) + aluminium (10%)                   | In making corrosion-free (rust-free) utensils, jewellery and shiny paints.                                   |
| Nichrome         | Nickel (55-60%) + chromium (20%) + iron (20-25%) | Used <b>in making the filaments of electric heaters and heating coils</b> (because of its high resistivity). |

## > Branches of Biology

| Branches of Biology |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| Ornithology         | ■ Study of birds                                                             |
| Ecology             | ■ Study of the effect of the surrounding environment on living beings        |
| Palaeobiology       | ■ Study of fossils                                                           |
| Phrenology          | ■ Study of the functioning and derangement of the various parts of the brain |
| Anthology           | ■ Study of flowers                                                           |
| Agrostology         | ■ Study related to grasses                                                   |
| Entomology          | ■ Scientific study of insects                                                |
| Parasitology        | ■ Study of parasitic organisms                                               |
| Malacology          | ■ Study of molluscs and their shells                                         |
| Genetics            | ■ Study of the hereditary characters of organisms and their inheritance      |
| Demography          | ■ Study of human population and of the vital statistics of mankind           |
| Virology            | ■ Study of viruses                                                           |
| Ethnology           | ■ Comparative study of the scientific descriptions of various cultures       |
| Ethnography         | ■ Study of a particular culture                                              |
| Ethology            | ■ Study of the behaviour of all animals including humans                     |

| Branches of Biology |                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Ethics</b>       | ■ Psychological study of moral conduct and duty                                                                                |
| <b>Bionics</b>      | ■ Study of the functions, properties and methods occurring in the biological world and their application to non-living systems |
| <b>Bionomics</b>    | ■ Study of the relationship of organisms with their environment                                                                |
| <b>Bionomy</b>      | ■ Study of the laws of life                                                                                                    |
| <b>Palaeobotany</b> | ■ Study of plant fossils                                                                                                       |
| <b>Agronomy</b>     | ■ Cultivation of crops                                                                                                         |
| <b>Gerontology</b>  | ■ A branch of biology related to old age and ageing in humans                                                                  |
| <b>Oncology</b>     | ■ A branch of medical science for the study of tumours related to cancer                                                       |
| <b>Teratology</b>   | ■ Study of abnormalities in physical development                                                                               |
| <b>Pedology</b>     | ■ Scientific study of soil formation and types                                                                                 |
| <b>Osteology</b>    | ■ Study of bones                                                                                                               |
| <b>Orology</b>      | ■ Study of mountains                                                                                                           |
| <b>Serpentology</b> | ■ Study of snakes                                                                                                              |
| <b>Herpetology</b>  | ■ Study of amphibians and reptiles                                                                                             |
| <b>Ichthyology</b>  | ■ Study of fishes                                                                                                              |
| <b>Dermatology</b>  | ■ Study of the skin and its diseases                                                                                           |
| <b>Trophology</b>   | ■ Study of nutrition and methods of nutrition                                                                                  |
| <b>Tetology</b>     | ■ Study of acquired characters                                                                                                 |
| <b>Bacteriology</b> | ■ Study of bacteria                                                                                                            |
| <b>Entomology</b>   | ■ Study of insects                                                                                                             |
| <b>Myrmecology</b>  | ■ Study of ants                                                                                                                |
| <b>Pathology</b>    | ■ Study of the nature, symptoms and causes of diseases                                                                         |



### Branches of Biology

|                    |                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lithotripsy</b> | <ul style="list-style-type: none"> <li>■ A medical procedure in which the patient is treated by breaking up stones in the kidney, gall bladder or urinary bladder with the help of rays.</li> </ul> |
| <b>Neurology</b>   | <ul style="list-style-type: none"> <li>■ Study related to the nervous system</li> </ul>                                                                                                             |
| <b>Haematology</b> | <ul style="list-style-type: none"> <li>■ Study of blood</li> </ul>                                                                                                                                  |
| <b>Serology</b>    | <ul style="list-style-type: none"> <li>■ Scientific study of blood serum</li> </ul>                                                                                                                 |
| <b>Chondrology</b> | <ul style="list-style-type: none"> <li>■ Study related to cartilage</li> </ul>                                                                                                                      |
| <b>Mammalogy</b>   | <ul style="list-style-type: none"> <li>■ Study of mammals</li> </ul>                                                                                                                                |
| <b>Myology</b>     | <ul style="list-style-type: none"> <li>■ Study of human muscles</li> </ul>                                                                                                                          |
| <b>Physiology</b>  | <ul style="list-style-type: none"> <li>■ Study of the functions and working of the various parts of the body</li> </ul>                                                                             |

### Biofertilisers

|                                |                                                                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nitrogen-fixing</b>         | <ul style="list-style-type: none"> <li>■ Rhizobium</li> <li>■ Azotobacter</li> <li>■ Azospirillum</li> <li>■ Acetobacter</li> <li>■ Blue-green algae</li> <li>■ Azolla</li> </ul>                                                      |
| <b>Phosphorus-solubilising</b> | <ul style="list-style-type: none"> <li>■ Bacillus</li> <li>■ Pseudomonas</li> <li>■ Aspergillus</li> </ul>                                                                                                                             |
| <b>Plant-growth promoting</b>  | <ul style="list-style-type: none"> <li>■ Rhizobacteria</li> <li>■ Pseudomonas</li> </ul>                                                                                                                                               |
| <b>Blue-green algae</b>        | <ul style="list-style-type: none"> <li>■ Other name:- Cyanobacteria</li> <li>■ Prokaryotic algae</li> <li>■ Symbiont of Azolla</li> <li>■ Helpful in nitrogen fixation</li> <li>■ Used as a biofertiliser in the paddy crop</li> </ul> |
| <b>Azolla</b>                  | <ul style="list-style-type: none"> <li>■ Aquatic fern</li> <li>■ Feeding it to milch cattle increases the quantity of milk</li> </ul>                                                                                                  |

| Fruits and their Edible Parts  |                           |
|--------------------------------|---------------------------|
| Edible part                    | ■ Fruit                   |
| Thalamus                       | ■ Apple and pear          |
| Pericarp, placenta             | ■ Guava, grape and tomato |
| Mesocarp                       | ■ Papaya and mango        |
| Endosperm                      | ■ Coconut                 |
| Mesocarp and endocarp          | ■ Banana                  |
| Endosperm and embryo           | ■ Wheat                   |
| Peduncle (pedicel), cotyledons | ■ Cashew                  |
| Aril                           | ■ Litchi                  |
| Cotyledons and embryo          | ■ Gram, groundnut         |
| Succulent perianth             | ■ Mulberry                |
| Perianth and seeds             | ■ Jackfruit               |
| Perianth                       | ■ Pineapple               |
| Juicy hairs                    | ■ Orange                  |

## > Human Diseases

| Human Diseases         |                                                                                                                                                                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two classes            | <ul style="list-style-type: none"> <li>■ Congenital diseases</li> <li>■ Acquired diseases:- infectious and non-infectious</li> </ul>                                                                                                                                |
| Germ Theory of Disease |                                                                                                                                                                                                                                                                     |
| Germ theory of disease | <ul style="list-style-type: none"> <li>■ Propounded by Robert Koch</li> <li>■ Diseases are caused by various micro-organisms and harmful substances</li> <li>■ Robert Koch proved that the anthrax disease of animals is caused by a microbial bacterium</li> </ul> |
| Infectious Diseases    |                                                                                                                                                                                                                                                                     |

| Human Diseases        |                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>   | <ul style="list-style-type: none"> <li>Such diseases spread when organisms come into contact with one another</li> </ul>                                                                                                              |
| <b>Agents</b>         | <ul style="list-style-type: none"> <li>Bacterial diseases</li> <li>Viral diseases</li> <li>Protozoan diseases</li> <li>Fungal diseases</li> </ul>                                                                                     |
| <b>Viral Diseases</b> |                                                                                                                                                                                                                                       |
| <b>Virus</b>          | <ul style="list-style-type: none"> <li>Origin of the word :- from the Latin word 'virus'</li> <li>Chemical composition :- a segment of nucleic acid surrounded by a protein coat</li> <li>Multiplies only in a living cell</li> </ul> |
| <b>Virology</b>       | <ul style="list-style-type: none"> <li>Science of viruses (branch of study)</li> </ul>                                                                                                                                                |
| <b>Smallpox</b>       |                                                                                                                                                                                                                                       |
| <b>Virus</b>          | <ul style="list-style-type: none"> <li>Variola virus</li> </ul>                                                                                                                                                                       |
| <b>Transmission</b>   | <ul style="list-style-type: none"> <li>Through the patient's skin, sputum, air, clothes, etc.</li> </ul>                                                                                                                              |
| <b>Vaccine</b>        | <ul style="list-style-type: none"> <li>Edward Jenner (father of immunology)</li> </ul>                                                                                                                                                |
| <b>Mumps</b>          |                                                                                                                                                                                                                                       |
| <b>Other name</b>     | <ul style="list-style-type: none"> <li>Galsua</li> </ul>                                                                                                                                                                              |
| <b>Virus</b>          | <ul style="list-style-type: none"> <li>Mumps virus</li> </ul>                                                                                                                                                                         |
| <b>Affected organ</b> | <ul style="list-style-type: none"> <li>Parotid gland</li> </ul>                                                                                                                                                                       |
| <b>Polio</b>          |                                                                                                                                                                                                                                       |
| <b>Agent</b>          | <ul style="list-style-type: none"> <li>Poliomyelitis virus</li> </ul>                                                                                                                                                                 |
| <b>Transmission</b>   | <ul style="list-style-type: none"> <li>Through water and food</li> </ul>                                                                                                                                                              |
| <b>Affected organ</b> | <ul style="list-style-type: none"> <li>Due to destruction of nerve tissues, messages stop reaching the muscles (paralysis)</li> </ul>                                                                                                 |
| <b>Programme</b>      | <ul style="list-style-type: none"> <li>Polio eradication in India:- since 1995</li> </ul>                                                                                                                                             |
| <b>Jonas Salk</b>     | <ul style="list-style-type: none"> <li>Made the first polio vaccine in 1952</li> </ul>                                                                                                                                                |

| Human Diseases         |                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Albert Sabin</b>    | <ul style="list-style-type: none"> <li>■ Made the first oral polio vaccine</li> <li>■ Taken by mouth</li> </ul>                     |
| <b>India</b>           | <ul style="list-style-type: none"> <li>■ Polio-free since 2014</li> </ul>                                                           |
| <b>Hepatitis – B</b>   |                                                                                                                                     |
| <b>Cause</b>           | <ul style="list-style-type: none"> <li>■ Hepadnavirus family</li> </ul>                                                             |
| <b>Affected organ</b>  | <ul style="list-style-type: none"> <li>■ Swelling and inflammation in the liver</li> </ul>                                          |
| <b>Transmission</b>    | <ul style="list-style-type: none"> <li>■ Infected body fluids</li> </ul>                                                            |
| <b>Dengue</b>          |                                                                                                                                     |
| <b>Other name</b>      | <ul style="list-style-type: none"> <li>■ Break-bone fever</li> </ul>                                                                |
| <b>Virus</b>           | <ul style="list-style-type: none"> <li>■ Dengue virus (Flaviviridae family)</li> </ul>                                              |
| <b>Vector mosquito</b> | <ul style="list-style-type: none"> <li>■ Female Aedes aegypti</li> <li>■ Aedes albopictus</li> <li>■ Culex fatigans</li> </ul>      |
| <b>Affected organ</b>  | <ul style="list-style-type: none"> <li>■ Rapid fall in the number of platelets in the blood</li> <li>■ Internal bleeding</li> </ul> |
| <b>Rabies</b>          |                                                                                                                                     |
| <b>Other names</b>     | <ul style="list-style-type: none"> <li>■ Jalantak</li> <li>■ Hydrophobia</li> </ul>                                                 |
| <b>Affected organ</b>  | <ul style="list-style-type: none"> <li>■ Central nervous system</li> </ul>                                                          |
| <b>Agent</b>           | <ul style="list-style-type: none"> <li>■ Rabies virus</li> </ul>                                                                    |
| <b>Transmission</b>    | <ul style="list-style-type: none"> <li>■ From dogs, cats, bats, foxes, raccoons, etc.</li> </ul>                                    |
| <b>Vaccine</b>         | <ul style="list-style-type: none"> <li>■ Discovery :- Louis Pasteur</li> <li>■ Year :- 1885 CE</li> </ul>                           |
| <b>Main symptoms</b>   | <ul style="list-style-type: none"> <li>■ Fear of water</li> <li>■ High fever</li> </ul>                                             |
| <b>Yellow Fever</b>    |                                                                                                                                     |

| Human Diseases               |                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>                 | <ul style="list-style-type: none"> <li>Yellow fever virus</li> </ul>                                                                                 |
| <b>Vector mosquito</b>       | <ul style="list-style-type: none"> <li>Aedes aegypti</li> </ul>                                                                                      |
| <b>Aedes Mosquito</b>        |                                                                                                                                                      |
| <b>Diseases carried</b>      | <ul style="list-style-type: none"> <li>Yellow fever, dengue fever, chikungunya, Zika fever, Mayaro fever</li> </ul>                                  |
| <b>Other name</b>            | <ul style="list-style-type: none"> <li>Tiger mosquito</li> </ul>                                                                                     |
| <b>Identification</b>        | <ul style="list-style-type: none"> <li>White stripes on the legs</li> </ul>                                                                          |
| <b>Japanese Encephalitis</b> |                                                                                                                                                      |
| <b>Cause</b>                 | <ul style="list-style-type: none"> <li>A virus of the Flaviviridae family</li> </ul>                                                                 |
| <b>Vector</b>                | <ul style="list-style-type: none"> <li>Culex mosquito</li> </ul>                                                                                     |
| <b>Purvanchal (UP)</b>       | <ul style="list-style-type: none"> <li>Endemic (epidemic) affected region</li> </ul>                                                                 |
| <b>AIDS</b>                  |                                                                                                                                                      |
| <b>Cause</b>                 | <ul style="list-style-type: none"> <li>Human Immunodeficiency Virus</li> </ul>                                                                       |
| <b>Transmission</b>          | <ul style="list-style-type: none"> <li>Unsafe sexual contact</li> <li>Contaminated blood transfusion</li> </ul>                                      |
| <b>Affected organ</b>        | <ul style="list-style-type: none"> <li>The body's immune system</li> <li>Destruction of the immune white blood cells called T-lymphocytes</li> </ul> |
| <b>1 December</b>            | <ul style="list-style-type: none"> <li>World AIDS Day</li> </ul>                                                                                     |
| <b>Diagnostic tests</b>      | <ul style="list-style-type: none"> <li>ELISA</li> <li>Western blot</li> </ul>                                                                        |
| <b>Influenza</b>             |                                                                                                                                                      |
| <b>Agent</b>                 | <ul style="list-style-type: none"> <li>Influenza virus</li> </ul>                                                                                    |
| <b>Infection</b>             | <ul style="list-style-type: none"> <li>Air-borne disease</li> </ul>                                                                                  |
| <b>Symptoms</b>              | <ul style="list-style-type: none"> <li>Cough, phlegm, sneezing, fever, etc.</li> </ul>                                                               |
| <b>Measles</b>               |                                                                                                                                                      |
| <b>Agent</b>                 | <ul style="list-style-type: none"> <li>Morbilli virus</li> </ul>                                                                                     |

| Human Diseases                    |                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Infection                         | <ul style="list-style-type: none"> <li>Air-borne disease</li> </ul>                                                             |
| Chickenpox                        |                                                                                                                                 |
| Agent                             | <ul style="list-style-type: none"> <li>Varicella zoster virus</li> </ul>                                                        |
| Infection                         | <ul style="list-style-type: none"> <li>Through breath or sneezes</li> </ul>                                                     |
| Symptoms                          | <ul style="list-style-type: none"> <li>Mild fever and pain in the joints</li> </ul>                                             |
| Herpes                            |                                                                                                                                 |
| Agent                             | <ul style="list-style-type: none"> <li>Herpes virus</li> </ul>                                                                  |
| Affected organ                    | <ul style="list-style-type: none"> <li>Skin</li> </ul>                                                                          |
| Other Diseases                    |                                                                                                                                 |
| Ebola                             | <ul style="list-style-type: none"> <li>Ebola virus</li> </ul>                                                                   |
| Diarrhoea                         | <ul style="list-style-type: none"> <li>Rotavirus</li> </ul>                                                                     |
| Rubella                           | <ul style="list-style-type: none"> <li>A viral disease</li> </ul>                                                               |
| Nipah virus disease               | <ul style="list-style-type: none"> <li>Agent :- Nipah virus</li> <li>Zoonotic disease</li> </ul>                                |
| Bacterial Diseases                |                                                                                                                                 |
| Tuberculosis                      |                                                                                                                                 |
| Other names                       | <ul style="list-style-type: none"> <li>Yakshma, tuberculosis (TB), tapedik</li> </ul>                                           |
| Discoverer                        | <ul style="list-style-type: none"> <li>Robert Koch</li> </ul>                                                                   |
| Affected organ                    | <ul style="list-style-type: none"> <li>Lungs</li> </ul>                                                                         |
| Agent                             | <ul style="list-style-type: none"> <li>Mycobacterium tuberculosis</li> </ul>                                                    |
| Treatment method                  | <ul style="list-style-type: none"> <li>DOTS :- Directly Observed Treatment</li> <li>MDR TB:- Multi-Drug Resistant TB</li> </ul> |
| On not completing the DOTS course | <ul style="list-style-type: none"> <li>Antibiotic drugs stop having any effect on the tuberculosis bacteria</li> </ul>          |
| Vaccine                           | <ul style="list-style-type: none"> <li>BCG :- Bacillus Calmette–Guérin</li> </ul>                                               |

| Human Diseases                    |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Plague</b>                     |                                                                                                                                                        |
| <b>Cause</b>                      | <ul style="list-style-type: none"> <li>Caused by the bacterium <i>Pasteurella pestis</i> or <i>Yersinia pestis</i></li> </ul>                          |
| <b>Affected organ</b>             | <ul style="list-style-type: none"> <li>Lungs and lymph glands</li> </ul>                                                                               |
| <b>Vector</b>                     | <ul style="list-style-type: none"> <li>Fleas carrying the germs of infected rats</li> </ul>                                                            |
| <b>Salmonella Bacilli</b>         |                                                                                                                                                        |
| <b>Structure</b>                  | <ul style="list-style-type: none"> <li>Rod-shaped</li> </ul>                                                                                           |
| <b>Disease</b>                    | <ul style="list-style-type: none"> <li>Food poisoning</li> </ul>                                                                                       |
| <b>Spread</b>                     | <ul style="list-style-type: none"> <li>Infected water</li> <li>Contaminated food items</li> </ul>                                                      |
| <b>Cholera</b>                    |                                                                                                                                                        |
| <b>Bacterium</b>                  | <ul style="list-style-type: none"> <li><i>Vibrio cholerae</i> (discovery:- Robert Koch)</li> </ul>                                                     |
| <b>Spread</b>                     | <ul style="list-style-type: none"> <li>Contaminated water and food</li> </ul>                                                                          |
| <b>Affected organ</b>             | <ul style="list-style-type: none"> <li>Intestine</li> </ul>                                                                                            |
| <b>Symptoms</b>                   | <ul style="list-style-type: none"> <li>Continuous diarrhoea and vomiting</li> </ul>                                                                    |
| <b>Diphtheria</b>                 |                                                                                                                                                        |
| <b>Bacterium</b>                  | <ul style="list-style-type: none"> <li><i>Corynebacterium diphtheriae</i></li> </ul>                                                                   |
| <b>Other names</b>                | <ul style="list-style-type: none"> <li>Rohini</li> <li>Galaghontu</li> </ul>                                                                           |
| <b>Affected organ</b>             | <ul style="list-style-type: none"> <li>Windpipe (respiratory tract)</li> </ul>                                                                         |
| <b>Vaccine</b>                    | <ul style="list-style-type: none"> <li>DPT (combined vaccine)</li> <li>This vaccine protects against diphtheria, tetanus and whooping cough</li> </ul> |
| <b>Whooping Cough (Pertussis)</b> |                                                                                                                                                        |
| <b>Bacterium</b>                  | <ul style="list-style-type: none"> <li><i>Haemophilus pertussis</i> or <i>Bordetella pertussis</i></li> </ul>                                          |
| <b>Other name</b>                 | <ul style="list-style-type: none"> <li>Kukur khansi</li> </ul>                                                                                         |



| Human Diseases     |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Affected organ     | <ul style="list-style-type: none"> <li>Respiratory system</li> </ul>                                            |
| Vaccine            | <ul style="list-style-type: none"> <li>DPT</li> </ul>                                                           |
| Pneumonia          |                                                                                                                 |
| Other names        | <ul style="list-style-type: none"> <li>Phupphus shoth or phupphus pradah (inflammation of the lungs)</li> </ul> |
| Bacterium          | <ul style="list-style-type: none"> <li>Streptococcus pneumoniae</li> </ul>                                      |
| Symptoms           | <ul style="list-style-type: none"> <li>High fever</li> <li>Water in the lungs</li> </ul>                        |
| Affected organ     | <ul style="list-style-type: none"> <li>Lungs</li> </ul>                                                         |
| Vaccine            | <ul style="list-style-type: none"> <li>Pneumococcal vaccine</li> </ul>                                          |
| Typhoid            |                                                                                                                 |
| Other names        | <ul style="list-style-type: none"> <li>Antra jwar (enteric fever)</li> <li>Miyadi bukhar</li> </ul>             |
| Bacterium          | <ul style="list-style-type: none"> <li>Salmonella typhi (discovery:- Dr. Daniel Salmon, 1885 CE)</li> </ul>     |
| Affected organ     | <ul style="list-style-type: none"> <li>Intestine</li> </ul>                                                     |
| Vaccine            | <ul style="list-style-type: none"> <li>TAB vaccine</li> </ul>                                                   |
| Transmission       | <ul style="list-style-type: none"> <li>Dirty drinking water</li> </ul>                                          |
| Symptoms           | <ul style="list-style-type: none"> <li>High fever</li> <li>Headache</li> </ul>                                  |
| Tetanus            |                                                                                                                 |
| Bacterium          | <ul style="list-style-type: none"> <li>Clostridium tetani</li> </ul>                                            |
| Other names        | <ul style="list-style-type: none"> <li>Dhanushtankar, lockjaw</li> </ul>                                        |
| Vaccine            | <ul style="list-style-type: none"> <li>DPT vaccine</li> </ul>                                                   |
| Affected organ     | <ul style="list-style-type: none"> <li>Nervous system</li> </ul>                                                |
| Sujak / Gonorrhoea |                                                                                                                 |
|                    |                                                                                                                 |

| Human Diseases          |                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bacterium</b>        | <ul style="list-style-type: none"> <li>Neisseria gonorrhoeae</li> </ul>                                                                                                       |
| <b>Affected organ</b>   | <ul style="list-style-type: none"> <li>Urinary tract</li> </ul>                                                                                                               |
| <b>Syphilis</b>         |                                                                                                                                                                               |
| <b>Bacterium</b>        | <ul style="list-style-type: none"> <li>Treponema pallidum</li> </ul>                                                                                                          |
| <b>Affected organ</b>   | <ul style="list-style-type: none"> <li>Penis</li> </ul>                                                                                                                       |
| <b>Leprosy</b>          |                                                                                                                                                                               |
| <b>Transmission</b>     | <ul style="list-style-type: none"> <li>Through physical contact</li> </ul>                                                                                                    |
| <b>Bacterium</b>        | <ul style="list-style-type: none"> <li>Mycobacterium leprae or lepromatosis</li> </ul>                                                                                        |
| <b>Symptoms</b>         | <ul style="list-style-type: none"> <li>Patches on the body and loss of sensation at those places</li> <li>Degeneration of tissues</li> </ul>                                  |
| <b>Eradication</b>      | <ul style="list-style-type: none"> <li>National Leprosy Eradication Programme since 1983</li> <li>India declared leprosy-free in 2005</li> </ul>                              |
| <b>Treatment drugs</b>  | <ul style="list-style-type: none"> <li>Dapsone</li> <li>Rifampicin</li> <li>Clofazimine</li> </ul>                                                                            |
| <b>Antibiotic Drugs</b> |                                                                                                                                                                               |
| <b>Effective</b>        | <ul style="list-style-type: none"> <li>Against various bacterial diseases</li> </ul>                                                                                          |
| <b>Examples</b>         | <ul style="list-style-type: none"> <li>Chloromycetin, erythromycin</li> </ul>                                                                                                 |
| <b>Penicillin</b>       | <ul style="list-style-type: none"> <li>Discoverer:- Alexander Fleming (1928)</li> <li>The first antibiotic</li> <li>Obtained:- from the fungus Penicillium notatum</li> </ul> |
| <b>Sulpha Drugs</b>     |                                                                                                                                                                               |
| <b>What it is</b>       | <ul style="list-style-type: none"> <li>A group of synthetic drugs</li> </ul>                                                                                                  |
| <b>Examples</b>         | <ul style="list-style-type: none"> <li>Sulphonamide, sulphamerazine, sulphadiazine, etc.</li> </ul>                                                                           |
| <b>Prontosil</b>        | <ul style="list-style-type: none"> <li>The first sulpha drug</li> <li>Year 1932</li> </ul>                                                                                    |

| Human Diseases               |                                                                         |
|------------------------------|-------------------------------------------------------------------------|
| <b>Trachoma or Roha</b>      |                                                                         |
| <b>Bacterium</b>             | ■ The bacterium Chlamydia trachomatis                                   |
| <b>Infection</b>             | ■ By using articles infected with the above bacterium                   |
| <b>Symptoms</b>              | ■ The eyes become red and there is growth in the cornea                 |
| <b>Fungal Diseases</b>       |                                                                         |
| <b>Athlete's Foot</b>        |                                                                         |
| <b>Fungus</b>                | ■ The fungi Trichophyton and Tinea pedis                                |
| <b>Transmission</b>          | ■ Infected soil                                                         |
| <b>Affected organ</b>        | ■ Soft parts of the skin, especially the parts between the toes/fingers |
| <b>Aspergillosis</b>         |                                                                         |
| <b>Fungus</b>                | ■ Fungi of the Aspergillus species                                      |
| <b>Affected organ</b>        | ■ Lungs                                                                 |
| <b>Symptoms</b>              | ■ Suffocation, loss of smell                                            |
| <b>Ergotism</b>              |                                                                         |
| <b>Fungus</b>                | ■ Claviceps purpurea                                                    |
| <b>Transmission</b>          | ■ Consumption of contaminated grain                                     |
| <b>Affected organ</b>        | ■ Digestive system, nervous system                                      |
| <b>Other Fungal Diseases</b> |                                                                         |
| <b>Asthma</b>                | ■ The fungus Aspergillus fumigatus                                      |
| <b>Scabies (itch)</b>        | ■ Acarus scabiei fungus                                                 |
| <b>Baldness</b>              | ■ Tinea capitis fungus                                                  |
| <b>Ringworm</b>              | ■ Trichophyton verrucosum fungus                                        |
| <b>Protozoan Diseases</b>    |                                                                         |

| Human Diseases                  |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Kala-azar</b>                |                                                                                                                                     |
| <b>Other name</b>               | ■ Leishmaniasis                                                                                                                     |
| <b>Cause</b>                    | ■ The protozoan <i>Leishmania donovani</i>                                                                                          |
| <b>Affected organ</b>           | ■ Skin and the entire body                                                                                                          |
| <b>Vector</b>                   | ■ Sand fly                                                                                                                          |
| <b>Sleeping Sickness</b>        |                                                                                                                                     |
| <b>Cause</b>                    | ■ The protozoan <i>Trypanosoma</i>                                                                                                  |
| <b>Vector</b>                   | ■ Tsetse fly                                                                                                                        |
| <b>Affected organ</b>           | <ul style="list-style-type: none"> <li>■ Blood</li> <li>■ Cerebrospinal fluid</li> <li>■ Heart muscles</li> <li>■ Gonads</li> </ul> |
| <b>Malaria</b>                  |                                                                                                                                     |
| <b>Discovery</b>                | ■ By Sir Ronald Ross                                                                                                                |
| <b>Affected organ</b>           | ■ RBCs and spleen                                                                                                                   |
| <b>Cause</b>                    | ■ The protozoan <i>Plasmodium</i>                                                                                                   |
| <b>Vector</b>                   | ■ Female <i>Anopheles</i> mosquito                                                                                                  |
| <b>25 April</b>                 | ■ World Malaria Day                                                                                                                 |
| <b>Quinine</b>                  | ■ A natural drug obtained from the bark of the cinchona tree                                                                        |
| <b>Synthetic drugs</b>          | ■ Camoquin, chloroquine, atebirin                                                                                                   |
| <b>Gambusia</b>                 | ■ A fish released into water bodies to eat mosquito larvae                                                                          |
| <b>Dysentery</b>                |                                                                                                                                     |
| <b>Cause</b>                    | ■ <i>Entamoeba histolytica</i>                                                                                                      |
| <b>Infection</b>                | ■ Through impure water                                                                                                              |
| <b>Other Parasitic Diseases</b> |                                                                                                                                     |

| Human Diseases      |                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Diarrhoea           | <ul style="list-style-type: none"> <li>Giardia intestinalis - small intestine</li> </ul>                                                    |
| Pyorrhoea           | <ul style="list-style-type: none"> <li>Entamoeba gingivalis - gums affected</li> </ul>                                                      |
| Schistosomiasis     |                                                                                                                                             |
| Other name          | <ul style="list-style-type: none"> <li>Bilharzia</li> </ul>                                                                                 |
| Agent               | <ul style="list-style-type: none"> <li>The fluke (flatworm) Schistosoma</li> </ul>                                                          |
| Affected organ      | <ul style="list-style-type: none"> <li>Liver</li> <li>Intestine and other organs</li> </ul>                                                 |
| Symptoms            | <ul style="list-style-type: none"> <li>Itching of the skin, fever, headache, stomach ache</li> </ul>                                        |
| Medium              | <ul style="list-style-type: none"> <li>Unsafe drinking water, lack of sanitation</li> </ul>                                                 |
| Worm-borne Diseases |                                                                                                                                             |
| Filaria             |                                                                                                                                             |
| Other names         | <ul style="list-style-type: none"> <li>Shleepad, elephantiasis (hathi paon), feel paon</li> </ul>                                           |
| Cause               | <ul style="list-style-type: none"> <li>From the roundworm Wuchereria bancrofti of the phylum Nematoda</li> </ul>                            |
| Vector              | <ul style="list-style-type: none"> <li>Female Culex mosquito</li> </ul>                                                                     |
| Symptoms            | <ul style="list-style-type: none"> <li>Swelling and deformity of the hands, legs, genitals and other organs</li> </ul>                      |
| Affected organ      | <ul style="list-style-type: none"> <li>Lymph glands</li> </ul>                                                                              |
| Ascariasis          |                                                                                                                                             |
| Agent               | <ul style="list-style-type: none"> <li>The roundworm Ascaris</li> </ul>                                                                     |
| Effect              | <ul style="list-style-type: none"> <li>Improper nutrition of children</li> <li>Adverse effect on physical and mental development</li> </ul> |
| Affected organ      | <ul style="list-style-type: none"> <li>Intestine</li> </ul>                                                                                 |
| Drugs               | <ul style="list-style-type: none"> <li>Alcopar</li> <li>Oil of chenopodium</li> </ul>                                                       |
| Fascioliasis        |                                                                                                                                             |

| Human Diseases             |                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>               | <ul style="list-style-type: none"> <li>The flat liver fluke of the phylum Platyhelminthes</li> </ul>                          |
| <b>Location</b>            | <ul style="list-style-type: none"> <li>In the large bile ducts of the liver of sheep, goats, pigs, buffaloes, etc.</li> </ul> |
| <b>Neurocysticercosis</b>  |                                                                                                                               |
| <b>Cause</b>               | <ul style="list-style-type: none"> <li>The tapeworm <i>Taenia solium</i></li> </ul>                                           |
| <b>Affected organ</b>      | <ul style="list-style-type: none"> <li>Nervous system</li> </ul>                                                              |
| <b>Spread</b>              | <ul style="list-style-type: none"> <li>Eating pork</li> </ul>                                                                 |
| <b>Hereditary Diseases</b> |                                                                                                                               |
| <b>Colour Blindness</b>    |                                                                                                                               |
| <b>Other names</b>         | <ul style="list-style-type: none"> <li>Red-green blindness</li> <li>Daltonism</li> </ul>                                      |
| <b>Effect</b>              | <ul style="list-style-type: none"> <li>Inability to distinguish red and green colours</li> </ul>                              |
| <b>Carriers</b>            | <ul style="list-style-type: none"> <li>Women</li> </ul>                                                                       |
| <b>Sufferers</b>           | <ul style="list-style-type: none"> <li>Men</li> </ul>                                                                         |
| <b>Haemophilia</b>         |                                                                                                                               |
| <b>Other names</b>         | <ul style="list-style-type: none"> <li>Bleeder's disease</li> <li>Royal disease</li> </ul>                                    |
| <b>Beginning</b>           | <ul style="list-style-type: none"> <li>From Queen Victoria</li> </ul>                                                         |
| <b>Type</b>                | <ul style="list-style-type: none"> <li>Sex-linked hereditary disease</li> </ul>                                               |
| <b>Carriers</b>            | <ul style="list-style-type: none"> <li>Women are carriers while men get the disease</li> </ul>                                |
| <b>Effect</b>              | <ul style="list-style-type: none"> <li>Blood does not clot</li> </ul>                                                         |
| <b>Down Syndrome</b>       |                                                                                                                               |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>Mongolian idiocy</li> </ul>                                                            |
| <b>Cause</b>               | <ul style="list-style-type: none"> <li>Three chromosomes instead of two in the 21st pair of chromosomes (trisomic)</li> </ul> |

| Human Diseases       |                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effect               | <ul style="list-style-type: none"> <li>■ Impotence in affected males</li> </ul>                                                                                                           |
| Symptoms             | <ul style="list-style-type: none"> <li>■ The person has a round head, thick neck, open mouth and slanting eyes</li> </ul>                                                                 |
| Turner Syndrome      |                                                                                                                                                                                           |
| Cause                | <ul style="list-style-type: none"> <li>■ Only one chromosome present among the sex chromosomes (monosomic)</li> </ul>                                                                     |
| Effect               | <ul style="list-style-type: none"> <li>■ Female sterility</li> </ul>                                                                                                                      |
| Klinefelter Syndrome |                                                                                                                                                                                           |
| Cause                | <ul style="list-style-type: none"> <li>■ Three sex chromosomes instead of two (trisomic)</li> </ul>                                                                                       |
| Effect               | <ul style="list-style-type: none"> <li>■ Male sterility</li> </ul>                                                                                                                        |
| Albinism             |                                                                                                                                                                                           |
| Other name           | <ul style="list-style-type: none"> <li>■ Ranjak heenta (lack of pigment)</li> </ul>                                                                                                       |
| Effect               | <ul style="list-style-type: none"> <li>■ Absence of colour in the skin, hair and eyes</li> </ul>                                                                                          |
| Other Diseases       |                                                                                                                                                                                           |
| Cancer               |                                                                                                                                                                                           |
| Oncogene             | <ul style="list-style-type: none"> <li>■ Responsible for cancer</li> <li>■ First identified as SRC</li> <li>■ Year of discovery:- 1970</li> <li>■ First discovered in chickens</li> </ul> |
| Cause                | <ul style="list-style-type: none"> <li>■ Disorganised tissue growth due to uncontrolled division and growth of cells</li> </ul>                                                           |
| Metastasis           | <ul style="list-style-type: none"> <li>■ Spread of cancer cells from one organ of the body to another nearby organ</li> </ul>                                                             |
| Treatment            | <ul style="list-style-type: none"> <li>■ Radio-phosphorus (P-32)</li> <li>■ Cobalt-60</li> <li>■ Arsenic</li> <li>■ Gold-198</li> </ul>                                                   |



## Human Diseases

### Cancer stages

- First:- cancer cells confined to one part of the body
- Second:- cancer cells reach the lymph nodes of the infected organ
- Third:- lymph nodes carry the cancer cells to other organs of the body (metastasis)

### Gout (Arthritis)

#### Cause

- Deposition of uric acid in the joints of the body

#### Effect

- Unbearable pain in the joints

### Dropsy

#### Cause

- Adulteration of mustard oil with argemone oil

#### Effect

- Swelling in the body

### Hysteria (Vatonmad)

#### Cause

- A nervous disorder

#### Affected

- Generally young and unmarried women

### Alzheimer's Disease

#### Affected organ

- Brain

#### Discovery

- German psychiatrist Dr. Alzheimer

#### Effect

- Loss of memory

### Parkinson's Disease

#### Cause

- Decrease in the level of dopamine due to damage to nerve cells in the brain

#### Affected organ

- Nervous system

### Astigmatism

#### Symptoms

- Reduction in the overall vision of the person

#### Treatment

- Cylindrical lens

### Cataract

| Human Diseases             |                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms</b>            | <ul style="list-style-type: none"> <li>The lens of the eye becomes opaque</li> </ul>                                                                           |
| <b>Treatment</b>           | <ul style="list-style-type: none"> <li>Operation and fitting of a new artificial lens</li> </ul>                                                               |
| <b>Myopia</b>              |                                                                                                                                                                |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>Short-sightedness (near-sightedness)</li> </ul>                                                                         |
| <b>Symptoms</b>            | <ul style="list-style-type: none"> <li>Distant objects are not seen</li> </ul>                                                                                 |
| <b>Treatment</b>           | <ul style="list-style-type: none"> <li>Concave lens</li> </ul>                                                                                                 |
| <b>Hypermetropia</b>       |                                                                                                                                                                |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>Long-sightedness (far-sightedness)</li> </ul>                                                                           |
| <b>Symptoms</b>            | <ul style="list-style-type: none"> <li>Nearby objects are not seen</li> </ul>                                                                                  |
| <b>Treatment</b>           | <ul style="list-style-type: none"> <li>Convex lens</li> </ul>                                                                                                  |
| <b>Diabetes</b>            |                                                                                                                                                                |
| <b>Cause</b>               | <ul style="list-style-type: none"> <li>Reduced secretion of insulin</li> <li>Increase of sugar in the blood</li> </ul>                                         |
| <b>Ayurvedic treatment</b> | <ul style="list-style-type: none"> <li>Fenugreek seeds</li> <li>Curry leaves</li> </ul>                                                                        |
| <b>Insulin</b>             | <ul style="list-style-type: none"> <li>It was discovered by Banting and Macleod</li> </ul>                                                                     |
| <b>Heart Attack</b>        |                                                                                                                                                                |
| <b>Cause</b>               | <ul style="list-style-type: none"> <li>Blood clot in the coronary arteries due to deposition of cholesterol and lipid substances in fibrous tissues</li> </ul> |
| <b>Affected organ</b>      | <ul style="list-style-type: none"> <li>Heart</li> </ul>                                                                                                        |
| <b>Treatment</b>           | <ul style="list-style-type: none"> <li>Anticoagulant drug:- aspirin</li> <li>Surgery:- bypass surgery</li> </ul>                                               |

### Other Miscellaneous Facts about Plants

#### Dichogamy

- Maturation of the androecium and gynoecium at different times in a bisexual flower
- Self-pollination is not possible in these flowers
- Two types:-
- 1. Protandry
- 2. Protogyny

#### Stem cutting

- Developing a new plant from a part of the stem of the parent plant
- Examples:- cocoa and sugarcane

#### Layering technique

- Pressing a stem branch of a plant so that it touches the ground, then cutting off the roots that grow from the stem and raising them as new plants
- Example:- jasmine

#### Sunflower

- Botanical name:- *Helianthus annuus*
- Very low amount of fat:- beneficial for heart patients
- An angiosperm plant of the family Asteraceae

#### Aromatic oil is obtained

- From sandalwood, camphor, peppermint and clove

#### Neem

- Azadirachtin chemical (insecticide and fungicide)
- Production of mosquito repellents
- Uses:- helpful in checking pest growth in crops, biodiesel production, used in coating urea, useful in many diseases

#### Biodiesel

- Chemical composition:- abundant hydrocarbons and lipids
- Feature:- produced from plants such as jatropha, karanj, nagchampa and neem

#### Karanj or Karanja

- Example of a biodiesel plant
- Botanical name:- *Pongamia pinnata*

#### Jatropha

- Other name:- Ratanjot
- Botanical name:- *Jatropha curcas*
- Angiosperm plant
- Requires less water compared to other cash crops

#### Sadabahar (periwinkle)

- A flowering plant that blooms all 12 months
- Helpful in the treatment of diabetes

### Other Miscellaneous Facts about Plants

|                          |                                                                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tulsi</b>             | <ul style="list-style-type: none"> <li>■ Dicotyledonous and herbaceous plant</li> <li>■ Antibiotic properties, expectorant</li> <li>■ Flavonoids, saponins, glycosides, phenols, etc. present in the extract of tulsi leaves</li> </ul> |
| <b>Clove</b>             | <ul style="list-style-type: none"> <li>■ Place of origin:- Indonesia</li> <li>■ Family:- Myrtaceae</li> <li>■ Part used:- flower bud</li> </ul>                                                                                         |
| <b>Cricket bat</b>       | <ul style="list-style-type: none"> <li>■ Made:- from willow wood</li> </ul>                                                                                                                                                             |
| <b>Hockey stick</b>      | <ul style="list-style-type: none"> <li>■ Made:- from mulberry wood</li> </ul>                                                                                                                                                           |
| <b>Sugarcane</b>         | <ul style="list-style-type: none"> <li>■ Biofuel (ethanol) is obtained from it</li> </ul>                                                                                                                                               |
| <b>Chilgoza</b>          | <ul style="list-style-type: none"> <li>■ Botanical name:- Pinus gerardiana</li> <li>■ Abundant carbohydrates and proteins</li> <li>■ Obtained from the seed of the pine plant</li> </ul>                                                |
| <b>Brahmi</b>            | <ul style="list-style-type: none"> <li>■ Grows in marshy land</li> <li>■ Use:- nourishment of the brain, invigorating</li> </ul>                                                                                                        |
| <b>Mint (pudina)</b>     | <ul style="list-style-type: none"> <li>■ Botanical name:- Mentha arvensis</li> <li>■ Use:- production of menthol oil</li> </ul>                                                                                                         |
| <b>Caffeine</b>          | <ul style="list-style-type: none"> <li>■ Chemical nature:- alkaloid substance</li> <li>■ Main sources:- tea, coffee, cocoa</li> </ul>                                                                                                   |
| <b>Seaweeds</b>          | <ul style="list-style-type: none"> <li>■ Are sources of iodine</li> </ul>                                                                                                                                                               |
| <b>Asafoetida (hing)</b> | <ul style="list-style-type: none"> <li>■ Produced from the secretion of the stem and root</li> </ul>                                                                                                                                    |
| <b>Camphor</b>           | <ul style="list-style-type: none"> <li>■ From the wood of the evergreen tree Cinnamomum camphora</li> </ul>                                                                                                                             |
| <b>Vanilla</b>           | <ul style="list-style-type: none"> <li>■ Obtained from plants of the orchid family</li> </ul>                                                                                                                                           |

### Various Plant Organs

#### Root

|                 |                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function</b> | <ul style="list-style-type: none"> <li>■ Absorption of water and nutrients from the soil and providing support to plants</li> </ul> |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|

## Various Plant Organs

### Types

Three types

- Tap root:- mustard plant
- Fibrous root:- wheat plant
- Adventitious root:- grass, banyan

### Modification of Roots

#### Modification of tap roots

For storage of food materials

- Fusiform:- radish
- Napiform:- turnip
- Conical:- carrot
- Tuberous- sweet potato

#### Modification of adventitious roots

For storage of food materials

- Tuberous roots:- sweet potato
- Fasciculated roots:- dahlia, shatavar (asparagus)
- Nodulose roots:- amba haldi (mango ginger)
- Moniliform (beaded) roots:- grape, bitter gourd

### Stem

#### Introduction

- The aerial part of the plant axis which is normally situated above the ground and bears branches, leaves and flowers.

#### Functions

- Spreading the branches
- Holding up the leaves, flowers and fruits
- Conduction of water, mineral salts and photosynthetic products

### Modification of Stems

#### In three ways

- Underground
- Aerial
- Sub-aerial

#### Underground stem

- Rhizome:- turmeric, ginger
- Tubers:- potato
- Bulb :- onion, garlic
- Corm :- jimikand (elephant-foot yam), arbi (colocasia)

### Leaf

## Various Plant Organs

### Functions

- Preparing food by photosynthesis
- Controlling transpiration
- Some leaves store food, such as onion and garlic

### Flower

#### Introduction

- The reproductive organ of the plant

#### Whorls of a flower

- Calyx
- Corolla
- Androecium (male reproductive organ)
- Gynoecium (female reproductive organ)

#### Complete flower

- In which all four whorls are found

#### Incomplete flower

- In which all four whorls are not found

### Fruit

#### Introduction

- A mature or fully formed ovary

#### Formation

- After fertilisation

#### False fruits

- When a floral part of the plant changes into a part of the fruit
- Examples:- apple and cashew

#### True fruits

- All other fruits formed from the ovary

## Plant Hormones

### Growth promoters

- Those which increase the rate of growth
- Such as:- auxin, gibberellin, cytokinin

### Growth inhibitors

- Those which decrease the rate of growth
- Such as:- abscisic acid, ethylene

### Auxin

#### Introduction

- The most important plant hormone
- Formed at the upper tips of plants

## Plant Hormones

### Uses

- A high amount promotes stem growth while a low amount promotes root growth
- Important role in vegetative propagation
- In destroying weeds by spraying in fields
- Prevents plants from falling
- Development of fruits even without pollination and fertilisation in orange, lemon, grape, banana, tomato, etc.

### Gibberellin

#### Introduction

- Formed at the upper tips of plants and in the embryo
- Promotes growth in all parts of plants

#### Uses

- Increase the height of plants
- Help in the germination of seeds
- Flowering and parthenocarpy (development of fruits without fertilisation)

### Cytokinin

#### Other name

- Phytokinin

#### Uses

- Controls organ formation in plants
- Increases the rate of cell division
- Reduces the dormancy period and prevents senescence
- Protects chlorophyll from being destroyed

### Ethylene

#### Other name

- Fruit-ripening hormone

#### Introduction

- Exists in the gaseous state
- Formed in the apical part of the stem of the plant
- It is a growth-inhibiting hormone

#### Functions

- Ripening of fruits
- Accelerating the abscission of leaves, fruits and flowers

### Absciscic Acid

#### Functions

- Inhibitor of plant growth and plant metabolism
- Growth-inhibiting hormone:- stops cell division

### Florigen

### Plant Hormones

|                  |                                                                                 |
|------------------|---------------------------------------------------------------------------------|
| <b>Formation</b> | <ul style="list-style-type: none"> <li>■ Takes place in the leaves</li> </ul>   |
| <b>Function</b>  | <ul style="list-style-type: none"> <li>■ Induces flowering in plants</li> </ul> |

### Essential Nutrients for Plants

|                           |                                                                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function</b>           | <ul style="list-style-type: none"> <li>■ Necessary for the regular, complete and disorder-free life cycle and proper development of plants</li> </ul>                                                       |
| <b>Essential elements</b> | <ul style="list-style-type: none"> <li>■ 17</li> </ul>                                                                                                                                                      |
| <b>Macronutrients</b>     | <ul style="list-style-type: none"> <li>■ Total number:- 09</li> <li>■ Elements:- carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, magnesium</li> </ul>                          |
| <b>Micronutrients</b>     | <ul style="list-style-type: none"> <li>■ Total number:- 08</li> <li>■ Requirement:- in small amounts</li> <li>■ Elements:- boron, chlorine, copper, iron, manganese, molybdenum, nickel and zinc</li> </ul> |
| <b>Boron</b>              | <ul style="list-style-type: none"> <li>■ Necessary for nodule formation in leguminous (pulse) crops</li> </ul>                                                                                              |

### Plant Physiology

#### Photosynthesis

|                            |                                                                                                                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>          | <ul style="list-style-type: none"> <li>■ Production of carbohydrates and oxygen by plants using the energy of sunlight, carbon dioxide and water</li> </ul>                            |
| <b>Reaction</b>            | <ul style="list-style-type: none"> <li>■ <math>6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2</math></li> </ul> |
| <b>Energy</b>              | <ul style="list-style-type: none"> <li>■ Radiant energy is converted into chemical energy and stored in food materials</li> </ul>                                                      |
| <b>From CO<sub>2</sub></b> | <ul style="list-style-type: none"> <li>■ Carbon is taken</li> </ul>                                                                                                                    |
| <b>From water</b>          | <ul style="list-style-type: none"> <li>■ Hydrogen is taken</li> </ul>                                                                                                                  |
| <b>Oxygen</b>              | <ul style="list-style-type: none"> <li>■ Released into the atmosphere as a by-product of the reaction</li> </ul>                                                                       |
| <b>Glucose</b>             | <ul style="list-style-type: none"> <li>■ The final product of photosynthesis</li> <li>■ Soon changes into starch</li> </ul>                                                            |



## Plant Physiology

### Red colour

- Most effective in photosynthesis

### Photosynthetic pigments

- Location:- in the grana of the chloroplast
- Colours absorbed:- orange, red, blue, violet

### Chlorophyll

- Pigment essential in photosynthesis
- Main element:- magnesium

### Carotenoids

- Pigment that protects plants from ultraviolet damage

## Respiration in Plants

### Process of respiration

- A continuous process
- In it the plant takes in oxygen and releases carbon dioxide

### Respiring parts

- All parts of plants
- Such as:- root, stem, leaf

## Reproduction in Plants

### In two ways

- Sexual and asexual

## Asexual Reproduction

### Vegetative propagation

- By stem :- ginger, garlic, etc.
- By leaf:- Bryophyllum, Bignonia
- By root:- sweet potato, dahlia, etc.

### Budding

- Reproduction in yeast

### Fragmentation

- Reproduction in algae

### Spore formation

- Reproduction in fungi

## Sexual Reproduction

### Two methods

- Pollination and fertilisation

### Flower

- The reproductive part of plants

### Parts of a flower

- Sepals
- Petals
- Stamens
- Carpels

## Plant Physiology

### Reproductive organs in a flower

- Stamens and carpels

### Stamen

- Filament:- a stalk
- Anther:- a flattened top part

### Anther

- Pollen grains are produced here

### Carpel

Three main parts

- Ovary
- Style
- Stigma

## Pollination

### What is it?

- Transfer of pollen grains from the anther to the stigma

### Two types

- Self-pollination
- Cross-pollination

### Self-pollination

- When pollen grains are transferred from the anther of a flower to the stigma of the same flower or of another flower of the same plant

### Cross-pollination

- When pollen grains are transferred from the anther of one flower to the stigma of a flower of another plant of the same species

### Agents

- Abiotic agents (wind and water)
- Biotic agents (animals)

### Honeybee

- The most important biotic pollinator

## > Vitamins and Other Nutrients

## Nutrients

### Vitamins

#### Introduction

- They are complex organic compounds

#### Discoverer

- Casimir Funk (1911-12)

#### Other name

- Accessory food factors
- Propounded by Frederick Gowland Hopkins

| Nutrients           |                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Functions           | <ul style="list-style-type: none"> <li>Control and regulation of metabolic activities</li> <li>Protection of the body from diseases</li> </ul> |
| Water-soluble       | <ul style="list-style-type: none"> <li>Vitamin B</li> <li>Vitamin C</li> </ul>                                                                 |
| Fat-soluble         | <ul style="list-style-type: none"> <li>Vitamin A</li> <li>Vitamin D</li> <li>Vitamin E</li> <li>Vitamin K</li> </ul>                           |
| Vitamin A           |                                                                                                                                                |
| Other name          | <ul style="list-style-type: none"> <li>Retinol</li> </ul>                                                                                      |
| Deficiency diseases | <ul style="list-style-type: none"> <li>Night blindness, xerophthalmia, changes in skin cells, keratomalacia, stunted growth</li> </ul>         |
| Sources             | <ul style="list-style-type: none"> <li>Carrot, green vegetables, milk, liver, fish oil</li> </ul>                                              |
| Vitamin B1          |                                                                                                                                                |
| Other name          | <ul style="list-style-type: none"> <li>Thiamine</li> </ul>                                                                                     |
| Deficiency diseases | <ul style="list-style-type: none"> <li>Beri-beri, stoppage of growth</li> </ul>                                                                |
| Sources             | <ul style="list-style-type: none"> <li>Cereals, pulses, soyabean, milk, etc.</li> </ul>                                                        |
| Vitamin B2          |                                                                                                                                                |
| Other name          | <ul style="list-style-type: none"> <li>Riboflavin</li> </ul>                                                                                   |
| Deficiency diseases | <ul style="list-style-type: none"> <li>Cheilosis</li> </ul>                                                                                    |
| Sources             | <ul style="list-style-type: none"> <li>Cheese (paneer), eggs, meat, leafy vegetables, yeast</li> </ul>                                         |
| Vitamin B3          |                                                                                                                                                |
| Other name          | <ul style="list-style-type: none"> <li>Niacin</li> </ul>                                                                                       |
| Deficiency diseases | <ul style="list-style-type: none"> <li>Pellagra or skin dermatitis</li> </ul>                                                                  |
| Sources             | <ul style="list-style-type: none"> <li>Yeast, meat, fish, eggs, milk, pulses, etc.</li> </ul>                                                  |

| Nutrients           |                                                                                     |
|---------------------|-------------------------------------------------------------------------------------|
| <b>Vitamin B5</b>   |                                                                                     |
| Other name          | ■ Pantothenic acid                                                                  |
| Deficiency diseases | ■ Reduced growth, skin diseases, greying of hair, reduced fertility, asthma, stress |
| Sources             | ■ Eggs, milk, meat and groundnut                                                    |
| <b>Vitamin B6</b>   |                                                                                     |
| Other name          | ■ Pyridoxine                                                                        |
| Deficiency diseases | ■ Anaemia, skin diseases, muscular cramps                                           |
| Sources             | ■ Milk, yeast, cereals and meat                                                     |
| <b>Vitamin B7</b>   |                                                                                     |
| Other name          | ■ Biotin                                                                            |
| Deficiency diseases | ■ Skin diseases, hair fall                                                          |
| Sources             | ■ Meat, wheat, egg, vegetables, fruits, etc.                                        |
| <b>Vitamin B9</b>   |                                                                                     |
| Other name          | ■ Folic acid                                                                        |
| Deficiency diseases | ■ Anaemia, stunted growth                                                           |
| Sources             | ■ Green vegetables, pulses, yeast, meat, eggs                                       |
| <b>Vitamin B12</b>  |                                                                                     |
| Other name          | ■ Cyanocobalamin                                                                    |
| Deficiency diseases | ■ Anaemia, disorders of the nervous system                                          |
| Sources             | ■ Meat, fish, egg, milk, intestinal bacteria                                        |
| <b>Vitamin C</b>    |                                                                                     |
| Other name          | ■ Ascorbic acid                                                                     |

| Nutrients                  |                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Deficiency diseases</b> | <ul style="list-style-type: none"> <li>■ Scurvy</li> </ul>                                                                  |
| <b>Sources</b>             | <ul style="list-style-type: none"> <li>■ Amla, fruits of the lemon family (citrus fruits), tomato and vegetables</li> </ul> |
| <b>Vitamin D</b>           |                                                                                                                             |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>■ Calciferol</li> </ul>                                                              |
| <b>Deficiency diseases</b> | <ul style="list-style-type: none"> <li>■ Rickets (rickets, osteomalacia)</li> </ul>                                         |
| <b>Sources</b>             | <ul style="list-style-type: none"> <li>■ Butter, liver, eggs, sunlight, fish oil, milk</li> </ul>                           |
| <b>Vitamin E</b>           |                                                                                                                             |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>■ Tocopherol</li> </ul>                                                              |
| <b>Deficiency diseases</b> | <ul style="list-style-type: none"> <li>■ Reduced fertility, weakness of reproductive organs and muscles</li> </ul>          |
| <b>Sources</b>             | <ul style="list-style-type: none"> <li>■ Oils, wheat, egg yolk, soyabean</li> </ul>                                         |
| <b>Vitamin K</b>           |                                                                                                                             |
| <b>Other name</b>          | <ul style="list-style-type: none"> <li>■ Naphthoquinone</li> </ul>                                                          |
| <b>Deficiency diseases</b> | <ul style="list-style-type: none"> <li>■ Blood does not clot on injury</li> </ul>                                           |
| <b>Sources</b>             | <ul style="list-style-type: none"> <li>■ Intestinal bacteria, green vegetables, egg and soyabean</li> </ul>                 |

| Carbohydrates          |                                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>    | <ul style="list-style-type: none"> <li>■ Organic substances made of carbon, hydrogen and oxygen in the ratio 1:2:1</li> </ul>                                             |
| <b>Functions</b>       | <ul style="list-style-type: none"> <li>■ Providing energy to the body</li> <li>■ Acting as an energy store</li> <li>■ Formation of nucleic acids (RNA and DNA)</li> </ul> |
| <b>Monosaccharides</b> | <ul style="list-style-type: none"> <li>■ The simple form of carbohydrates</li> <li>■ Glucose, galactose, fructose, triose, etc.</li> </ul>                                |
| <b>Disaccharides</b>   | <ul style="list-style-type: none"> <li>■ Made of two molecules of monosaccharides</li> <li>■ Sucrose, maltose, lactose, fructose</li> </ul>                               |

| Carbohydrates                 |                                                                                                                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polysaccharides               | <ul style="list-style-type: none"> <li>Made of many molecules of monosaccharides</li> <li>Starch, chitin, glycogen</li> </ul>                                                                                             |
| Proteins                      |                                                                                                                                                                                                                           |
| Naming                        | <ul style="list-style-type: none"> <li>G. J. Mulder</li> </ul>                                                                                                                                                            |
| Discoverer                    | <ul style="list-style-type: none"> <li>J. J. Berzelius</li> </ul>                                                                                                                                                         |
| Amino acids                   | <ul style="list-style-type: none"> <li>Made of 20 types of amino acids</li> <li>Essential amino acids:- 9 (from food)</li> <li>Non-essential amino acids:- 11 (in the body)</li> </ul>                                    |
| Essential amino acids         | <ul style="list-style-type: none"> <li>Obtained only from food</li> <li>Histidine, isoleucine, leucine, lysine, methionine, phenylalanine, tryptophan, valine, threonine</li> </ul>                                       |
| Non-essential amino acids     | <ul style="list-style-type: none"> <li>Also synthesised in the cells of the body</li> <li>Glycine, serine, tyrosine, alanine, glutamic acid, glutamine, proline, aspartic acid, asparagine, arginine, cysteine</li> </ul> |
| Functions of protein          | <ul style="list-style-type: none"> <li>Growth and repair of the body</li> <li>Formation of cells, protoplasm and tissues</li> <li>Also provides energy to the body</li> </ul>                                             |
| Elements contained in protein | <ul style="list-style-type: none"> <li>Carbon, hydrogen, oxygen, nitrogen</li> <li>Sulphur, phosphorus, iodine, iron</li> </ul>                                                                                           |
| Simple proteins               | <ul style="list-style-type: none"> <li>Those made only of amino acids</li> <li>Such as:- globulin, histone, etc.</li> </ul>                                                                                               |
| Conjugated proteins           | <ul style="list-style-type: none"> <li>Proteins whose molecules are joined with other groups</li> <li>Such as:- glycoprotein, chromoprotein, etc.</li> </ul>                                                              |
| Derived proteins              | <ul style="list-style-type: none"> <li>Proteins formed by hydrolysis</li> <li>Such as:- peptone, proteoses, peptides, etc.</li> </ul>                                                                                     |
| Hair                          | <ul style="list-style-type: none"> <li>Made of the protein keratin</li> </ul>                                                                                                                                             |
| Milk                          | <ul style="list-style-type: none"> <li>White colour due to the protein casein</li> </ul>                                                                                                                                  |
| Beta carotene                 | <ul style="list-style-type: none"> <li>The light yellow colour of cow's milk is due to the presence of this protein</li> </ul>                                                                                            |

| Carbohydrates                 |                                                                                                                                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Soyabean</b>               | <ul style="list-style-type: none"> <li>The only non-animal protein in which all the essential amino acids are found</li> </ul>                                                                                                    |
| <b>Other facts</b>            | <ul style="list-style-type: none"> <li>Proteins are denatured by heat, X-rays, heavy metal salts, etc., but not by infrared radiation</li> </ul>                                                                                  |
| <b>Fats</b>                   |                                                                                                                                                                                                                                   |
| <b>Formation</b>              | <ul style="list-style-type: none"> <li>From a combination of carbon and hydrogen</li> <li>Made of glycerol and fatty acids</li> </ul>                                                                                             |
| <b>Functions</b>              | <ul style="list-style-type: none"> <li>Providing energy to the body</li> <li>Getting deposited under the skin and preventing body heat from escaping</li> </ul>                                                                   |
| <b>Saturated fats</b>         | <ul style="list-style-type: none"> <li>Less reactive</li> <li>Single bonds</li> <li>Harmful to health</li> <li>Examples:- palm and coconut oil</li> <li>Called saturated fats because they are saturated with hydrogen</li> </ul> |
| <b>Unsaturated fats</b>       | <ul style="list-style-type: none"> <li>More reactive</li> <li>Double bonds between carbon atoms</li> <li>Rapid rate of decomposition</li> <li>Beneficial to health</li> <li>Example:- fish oil</li> </ul>                         |
| <b>Trans fats</b>             | <ul style="list-style-type: none"> <li>Increase in the level of bad LDL cholesterol</li> <li>Damage to the cardiovascular system</li> <li>Examples:- animal fats and hydrogenated oils</li> </ul>                                 |
| <b>Mineral Salts</b>          |                                                                                                                                                                                                                                   |
| <b>Minor (Macro) Minerals</b> |                                                                                                                                                                                                                                   |
| <b>Calcium</b>                | <ul style="list-style-type: none"> <li>Deficiency:- weak teeth and bones, stunted body growth</li> <li>Importance:- formation of teeth and bones, blood clotting, functioning of nerves and muscles</li> </ul>                    |
| <b>Phosphorus</b>             | <ul style="list-style-type: none"> <li>Deficiency:- weak teeth and bones</li> <li>Importance:- providing strength to teeth and bones and acid-base balance</li> </ul>                                                             |

| Carbohydrates                                         |                                                                                                                                                                                                                                                 |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sulphur</b>                                        | <ul style="list-style-type: none"> <li>■ Deficiency:- protein deficiency</li> <li>■ Importance:- constituent of many amino acids and some hormones and vitamins, and necessary for the synthesis of carotene</li> </ul>                         |
| <b>Chlorine</b>                                       | <ul style="list-style-type: none"> <li>■ Deficiency:- loss of appetite, fatigue</li> <li>■ Importance:- acid-base balance, water balance</li> </ul>                                                                                             |
| <b>Sodium</b>                                         | <ul style="list-style-type: none"> <li>■ Deficiency:- low blood pressure, fatigue, muscular cramps</li> <li>■ Importance:- acid-base balance, heartbeat</li> </ul>                                                                              |
| <b>Magnesium</b>                                      | <ul style="list-style-type: none"> <li>■ Deficiency:- the physiology of the nervous system is affected</li> <li>■ Necessary for the functioning of the nervous and muscular systems</li> </ul>                                                  |
| <b>Trace Minerals</b>                                 |                                                                                                                                                                                                                                                 |
| <b>Iodine</b>                                         | <ul style="list-style-type: none"> <li>■ Deficiency:- goitre and slow development</li> <li>■ Importance:- an important constituent of the hormone thyroxine</li> </ul>                                                                          |
| <b>Iron</b>                                           | <ul style="list-style-type: none"> <li>■ Deficiency:- haemoglobin deficiency &gt; anaemia</li> <li>■ Importance:- constituent of haemoglobin, myoglobin and the cytochrome enzymes</li> </ul>                                                   |
| <b>Zinc</b>                                           | <ul style="list-style-type: none"> <li>■ Deficiency:- rough skin, loss of fertility, loss of appetite, stunted growth</li> <li>■ Importance:- necessary for normal growth, development of reproductive organs and insulin physiology</li> </ul> |
| <b>Copper</b>                                         | <ul style="list-style-type: none"> <li>■ Deficiency:- anaemia, dryness of hair</li> <li>■ Importance:- iron metabolism, development of connective tissue and blood vessels</li> </ul>                                                           |
| <b>Manganese</b>                                      | <ul style="list-style-type: none"> <li>■ Deficiency:- irregularity in the growth of cartilage, bone and connective tissues</li> </ul>                                                                                                           |
| <b>Cobalt</b>                                         | <ul style="list-style-type: none"> <li>■ Deficiency:- anaemia</li> <li>■ Importance:- an important constituent of vitamin B12, formation of RBCs</li> </ul>                                                                                     |
| <b>Other Miscellaneous Facts Related to Nutrients</b> |                                                                                                                                                                                                                                                 |
|                                                       |                                                                                                                                                                                                                                                 |



| Carbohydrates            |                                                                                                                                                                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water                    | <ul style="list-style-type: none"> <li>Calorie-free diet</li> <li>Helps in absorbing the nutrients present in food</li> </ul>                                                                                                                                                               |
| Milk                     | <ul style="list-style-type: none"> <li>Other name:- complete food</li> <li>Rennin:- the enzyme that digests milk protein</li> <li>The sourness of milk is due to Lactobacillus bacteria</li> <li>Composition:- water (about 87%), fat, protein, carbohydrate, calcium, potassium</li> </ul> |
| Starch                   | <ul style="list-style-type: none"> <li>Obtained from plants</li> <li>Reacts with iodine solution to give a blue-black colour</li> <li>Present in plant seeds in the form of amylose or amylopectin</li> </ul>                                                                               |
| Cellulose                | <ul style="list-style-type: none"> <li>Obtained from plants</li> <li>Does not react with iodine solution</li> <li>Forms the cell wall</li> </ul>                                                                                                                                            |
| Chilli                   | <ul style="list-style-type: none"> <li>Vitamin C is found in it</li> </ul>                                                                                                                                                                                                                  |
| Liver                    | <ul style="list-style-type: none"> <li>Vitamin A is stored here</li> </ul>                                                                                                                                                                                                                  |
| Vit. B6                  | <ul style="list-style-type: none"> <li>The vitamin that helps in remembering dreams</li> </ul>                                                                                                                                                                                              |
| Vit. E                   | <ul style="list-style-type: none"> <li>Important source:- wheat germ oil</li> </ul>                                                                                                                                                                                                         |
| Vit. D                   | <ul style="list-style-type: none"> <li>It is also called a hormone</li> </ul>                                                                                                                                                                                                               |
| Essential macronutrients | <ul style="list-style-type: none"> <li>Potassium, sodium, magnesium, calcium, chlorine, sulphur</li> </ul>                                                                                                                                                                                  |
| Alpha-keratin            | <ul style="list-style-type: none"> <li>Protein present in horns, hair, nails, hooves and skin</li> </ul>                                                                                                                                                                                    |
| Iron                     | <ul style="list-style-type: none"> <li>Highest content in green leafy vegetables</li> </ul>                                                                                                                                                                                                 |
| Papaya                   | <ul style="list-style-type: none"> <li>Mainly vitamin C is found in it</li> </ul>                                                                                                                                                                                                           |
| Water                    | <ul style="list-style-type: none"> <li>About 60 to 70% of the body weight</li> <li>Controls body temperature</li> <li>Helps in the excretion of waste materials</li> </ul>                                                                                                                  |
| Tyrosine                 | <ul style="list-style-type: none"> <li>Due to deficiency of this protein a person feels sad and sluggish.</li> </ul>                                                                                                                                                                        |

## Organic Evolution

### Origin of Life

|                                 |                                                                                                                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Materialism</b>              | <ul style="list-style-type: none"> <li>Hypothesis presented by a Russian biochemist</li> <li>In the book 'The Origin of Life'</li> </ul>                                                                                                                                         |
| <b>Lamarckism</b>               | <ul style="list-style-type: none"> <li>In the book 'Philosophie Zoologique' in 1809 CE</li> <li>The first rational theory on the hypothesis of organic evolution</li> <li>By the French biologist Jean-Baptiste Lamarck</li> </ul>                                               |
| <b>Principles of Lamarckism</b> | <ul style="list-style-type: none"> <li>Tendency to increase in size</li> <li>Direct effect of the environment</li> <li>Effect of use or disuse of organs</li> <li>Inheritance of acquired characters</li> </ul>                                                                  |
| <b>Darwinism</b>                | <ul style="list-style-type: none"> <li>By Charles Darwin of England</li> <li>Other names:- evolutionism or the theory of natural selection</li> <li>Book:- On the Origin of Species by Means of Natural Selection (1859)</li> </ul>                                              |
| <b>Principles of Darwinism</b>  | <ul style="list-style-type: none"> <li>Capacity for prolific reproduction in organisms</li> <li>Struggle for existence</li> <li>Variation and its inheritance</li> <li>Survival of the fittest or natural selection</li> </ul>                                                   |
| <b>Mutation theory</b>          | <ul style="list-style-type: none"> <li>By Hugo de Vries of the Netherlands</li> <li>Experiment:- on the plant <i>Oenothera lamarckiana</i></li> <li>According to him, a new species originates in a single step as a result of clear, permanent and sudden mutations.</li> </ul> |

### Other Facts

|                      |                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Archaeopteryx</b> | <ul style="list-style-type: none"> <li>The most ancient bird of the Jurassic period</li> <li>It was the connecting link between reptiles and birds.</li> </ul> |
| <b>Euglena</b>       | <ul style="list-style-type: none"> <li>Considered the connecting link between plants and animals.</li> </ul>                                                   |
| <b>Neopilina</b>     | <ul style="list-style-type: none"> <li>Considered the connecting link between Annelida and Mollusca.</li> </ul>                                                |
| <b>Dinosaur</b>      | <ul style="list-style-type: none"> <li>Gigantic reptiles of the Mesozoic era</li> </ul>                                                                        |
| <b>Mammoth</b>       | <ul style="list-style-type: none"> <li>It is considered the ancestor of the elephant.</li> </ul>                                                               |
| <b>Cro-Magnon</b>    | <ul style="list-style-type: none"> <li>This man is considered the last direct ancestor of modern man, Homo sapiens.</li> </ul>                                 |

| Biology and Classification              |                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The word 'Biology'</b>               | <ul style="list-style-type: none"> <li>First used:- by Lamarck (France) and Treviranus (Germany) in 1801</li> </ul>                                                                                                                                                                                                            |
| <b>Aristotle</b>                        | <ul style="list-style-type: none"> <li>Father of biology</li> <li>Father of zoology</li> <li>Book:- Historia Animalium</li> </ul>                                                                                                                                                                                              |
| <b>Theophrastus</b>                     | <ul style="list-style-type: none"> <li>Father of botany</li> <li>Book:- Historia Plantarum</li> </ul>                                                                                                                                                                                                                          |
| <b>Characteristics of Living Beings</b> |                                                                                                                                                                                                                                                                                                                                |
| <b>Cellular organisation</b>            | <ul style="list-style-type: none"> <li>The structural and functional unit of all living beings</li> <li>A defining characteristic of organisms</li> </ul>                                                                                                                                                                      |
| <b>Metabolism</b>                       | <ul style="list-style-type: none"> <li>All the biochemical reactions taking place in a living being to sustain life</li> <li>Two types</li> <li>Anabolism :- formation of complex molecules from simple molecules (growth)</li> <li>Catabolism:- formation of simple molecules from complex molecules (respiration)</li> </ul> |
| <b>Growth</b>                           | <ul style="list-style-type: none"> <li>Formation of new cells in a living being by using food</li> </ul>                                                                                                                                                                                                                       |
| <b>Reproduction</b>                     | <ul style="list-style-type: none"> <li>The ability to give birth to organisms similar to oneself</li> </ul>                                                                                                                                                                                                                    |
| <b>Binomial Nomenclature</b>            |                                                                                                                                                                                                                                                                                                                                |
| <b>Beginning</b>                        | <ul style="list-style-type: none"> <li>By Carolus Linnaeus (father of taxonomy)</li> <li>From the two-kingdom theory (since 1753)</li> </ul>                                                                                                                                                                                   |
| <b>Linnaeus's classification</b>        | <ul style="list-style-type: none"> <li>Division into the animal kingdom and the plant kingdom</li> <li>The first word is the genus name, the second word the species name</li> <li>In Homo sapiens, Homo is the genus and sapiens the species</li> </ul>                                                                       |
| <b>Five-Kingdom Classification</b>      |                                                                                                                                                                                                                                                                                                                                |
|                                         |                                                                                                                                                                                                                                                                                                                                |

| Biology and Classification |                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Propounder</b>          | <ul style="list-style-type: none"> <li>■ R. H. Whittaker</li> <li>■ The most widely accepted classification in modern biology</li> </ul>                     |
| <b>Five kingdoms</b>       | <ul style="list-style-type: none"> <li>■ Monera, Protista, Fungi, Plantae and Animalia</li> </ul>                                                            |
| <b>Monera</b>              |                                                                                                                                                              |
| <b>Organisms</b>           | <ul style="list-style-type: none"> <li>■ Unicellular prokaryotic</li> </ul>                                                                                  |
| <b>Cell organelles</b>     | <ul style="list-style-type: none"> <li>■ Absence of an organised nucleus</li> <li>■ Undeveloped organelles</li> <li>■ Absence of nuclear membrane</li> </ul> |
| <b>Mode of nutrition</b>   | <ul style="list-style-type: none"> <li>■ Autotrophic</li> </ul>                                                                                              |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Bacteria</li> <li>■ Blue-green algae / cyanobacteria</li> <li>■ Mycoplasma (capable of reproduction)</li> </ul>     |
| <b>Protista</b>            |                                                                                                                                                              |
| <b>Organisms</b>           | <ul style="list-style-type: none"> <li>■ Unicellular eukaryotic</li> <li>■ Exception:- sea-weed (multicellular)</li> </ul>                                   |
| <b>Link</b>                | <ul style="list-style-type: none"> <li>■ Between the animal and fungal kingdoms</li> </ul>                                                                   |
| <b>Mode of nutrition</b>   | <ul style="list-style-type: none"> <li>■ Both autotrophic and heterotrophic</li> </ul>                                                                       |
| <b>Cell organelles</b>     | <ul style="list-style-type: none"> <li>■ Cell wall in some</li> <li>■ Nuclear membrane is found</li> </ul>                                                   |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Unicellular algae, diatoms, protozoa, Euglena, Paramecium, Chlorella, Amoeba</li> </ul>                             |
| <b>Fungi</b>               |                                                                                                                                                              |
| <b>Organisms</b>           | <ul style="list-style-type: none"> <li>■ Multicellular eukaryotic</li> <li>■ Without chlorophyll</li> <li>■ Thallophytes</li> </ul>                          |
| <b>Nutrition</b>           | <ul style="list-style-type: none"> <li>■ Dependent on decaying organic matter</li> <li>■ Heterotrophic, saprophytic, parasitic</li> </ul>                    |

| Biology and Classification         |                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cell wall</b>                   | <ul style="list-style-type: none"> <li>■ Made of a complex sugar called chitin</li> <li>■ Cellulose absent</li> </ul>                                                                                                                                                                                                                                  |
| <b>Examples</b>                    | <ul style="list-style-type: none"> <li>■ Yeast, Penicillium, mushroom</li> </ul>                                                                                                                                                                                                                                                                       |
| <b>Mycology</b>                    | <ul style="list-style-type: none"> <li>■ Study of fungi</li> </ul>                                                                                                                                                                                                                                                                                     |
| <b>Lichen</b>                      | <ul style="list-style-type: none"> <li>■ Fungus + alga</li> </ul>                                                                                                                                                                                                                                                                                      |
| <b>Micro-organisms</b>             |                                                                                                                                                                                                                                                                                                                                                        |
| <b>Bacteria</b>                    | <ul style="list-style-type: none"> <li>■ Simple and unicellular prokaryotic micro-organisms (discovery:- Leeuwenhoek)</li> <li>■ Two types:- heterotrophic and autotrophic</li> </ul>                                                                                                                                                                  |
| <b>Virus</b>                       | <ul style="list-style-type: none"> <li>■ Subcellular structure:- DNA or RNA surrounded by a protein coat</li> <li>■ Nutrition in a host, not on its own</li> <li>■ The link between non-living and living</li> <li>■ Discovery:- Dmitri Ivanovsky (1892 CE)</li> <li>■ It was discovered through the study of the mosaic disease of tobacco</li> </ul> |
| <b>Plants and their Sub-groups</b> |                                                                                                                                                                                                                                                                                                                                                        |
| <b>Organisms</b>                   | <ul style="list-style-type: none"> <li>■ Cell wall made of cellulose</li> <li>■ Multicellular eukaryotic organisms</li> </ul>                                                                                                                                                                                                                          |
| <b>Mode of nutrition</b>           | <ul style="list-style-type: none"> <li>■ Autotrophic</li> <li>■ Food is made by the process of photosynthesis</li> </ul>                                                                                                                                                                                                                               |
| <b>Thallophyta</b>                 |                                                                                                                                                                                                                                                                                                                                                        |
| <b>Structure</b>                   | <ul style="list-style-type: none"> <li>■ Not differentiated into root, stem and leaves</li> </ul>                                                                                                                                                                                                                                                      |
| <b>Examples</b>                    | <ul style="list-style-type: none"> <li>■ Mainly aquatic plants (algae)</li> <li>■ Ulothrix, Spirogyra, Chara</li> </ul>                                                                                                                                                                                                                                |
| <b>Bryophyta</b>                   |                                                                                                                                                                                                                                                                                                                                                        |
| <b>Structure</b>                   | <ul style="list-style-type: none"> <li>■ Differentiated into stem- and leaf-like structures</li> <li>■ Absence of vascular tissues</li> </ul>                                                                                                                                                                                                          |
| <b>Nickname</b>                    | <ul style="list-style-type: none"> <li>■ Amphibians of the plant kingdom</li> </ul>                                                                                                                                                                                                                                                                    |

| Biology and Classification |                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples                   | <ul style="list-style-type: none"> <li>■ Moss (Funaria), Marchantia</li> </ul>                                                                                               |
| Pteridophyta               |                                                                                                                                                                              |
| Structure                  | <ul style="list-style-type: none"> <li>■ Body differentiated into root, stem and leaves</li> <li>■ Vascular tissues also present</li> </ul>                                  |
| Examples                   | <ul style="list-style-type: none"> <li>■ Fern, horsetail, Marsilea</li> </ul>                                                                                                |
| Gymnosperms                |                                                                                                                                                                              |
| Structure                  | <ul style="list-style-type: none"> <li>■ Naked-seeded</li> <li>■ Perennial, evergreen and woody</li> </ul>                                                                   |
| Examples                   | <ul style="list-style-type: none"> <li>■ Pinus and Cycas</li> </ul>                                                                                                          |
| Angiosperms                |                                                                                                                                                                              |
| Structure                  | <ul style="list-style-type: none"> <li>■ The seeds of the plants develop inside fruits, in the ovary</li> </ul>                                                              |
| Other name                 | <ul style="list-style-type: none"> <li>■ Flowering plants</li> </ul>                                                                                                         |
| Two parts                  | <ul style="list-style-type: none"> <li>■ Monocotyledons</li> <li>■ Dicotyledons</li> </ul>                                                                                   |
| Examples                   | <ul style="list-style-type: none"> <li>■ Fruit-bearing plants</li> <li>■ Mango, guava, papaya</li> </ul>                                                                     |
| Plants: Other Facts        |                                                                                                                                                                              |
| Cryptogams                 | <ul style="list-style-type: none"> <li>■ Hidden reproductive organs</li> <li>■ Unable to produce seeds</li> <li>■ Examples:- Thallophyta, Bryophyta, Pteridophyta</li> </ul> |
| Phanerogams                | <ul style="list-style-type: none"> <li>■ Fully developed reproductive organs</li> <li>■ Able to produce seeds</li> <li>■ Examples:- gymnosperms, angiosperms</li> </ul>      |
| Animal Kingdom             |                                                                                                                                                                              |
| Organisms                  | <ul style="list-style-type: none"> <li>■ Eukaryotic, multicellular, heterotrophic</li> </ul>                                                                                 |
|                            |                                                                                                                                                                              |

| Biology and Classification      |                                                                                                                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Classification</b>           | <ul style="list-style-type: none"> <li>■ On many bases</li> <li>■ Mainly on the basis of the notochord:- Non-Chordata and Chordata</li> </ul>                                                                                               |
| <b>Porifera</b>                 |                                                                                                                                                                                                                                             |
| <b>Sponges</b>                  | <ul style="list-style-type: none"> <li>■ Members of this phylum are known by this name</li> </ul>                                                                                                                                           |
| <b>Structure</b>                | <ul style="list-style-type: none"> <li>■ Body with innumerable pores</li> <li>■ Water system or canal system in the body</li> </ul>                                                                                                         |
| <b>Other facts</b>              | <ul style="list-style-type: none"> <li>■ Intake of food, respiration, removal of wastes, etc. through the route of water transport</li> <li>■ Water enters through fine pores (ostia) and goes out through large pores (osculum)</li> </ul> |
| <b>Examples</b>                 | <ul style="list-style-type: none"> <li>■ Sycon, Spongilla, Euspongia</li> </ul>                                                                                                                                                             |
| <b>Coelenterata or Cnidaria</b> |                                                                                                                                                                                                                                             |
| <b>Organisms / Structure</b>    | <ul style="list-style-type: none"> <li>■ Tissue-level body organisation (diploblastic)</li> <li>■ Aquatic, marine</li> <li>■ Thread-like structures (tentacles) around the mouth for catching food</li> </ul>                               |
| <b>Examples</b>                 | <ul style="list-style-type: none"> <li>■ Sea anemone, Hydra, coral, jellyfish</li> </ul>                                                                                                                                                    |
| <b>Hydra</b>                    | <ul style="list-style-type: none"> <li>■ Respiratory organs and blood absent</li> <li>■ Respiration takes place through diffusion</li> </ul>                                                                                                |
| <b>Ctenophora</b>               |                                                                                                                                                                                                                                             |
| <b>Related facts</b>            | <ul style="list-style-type: none"> <li>■ Tissue-grade body organisation</li> <li>■ Emission of light by bioluminescence</li> </ul>                                                                                                          |
| <b>Other names</b>              | <ul style="list-style-type: none"> <li>■ Sea walnuts, comb jellies</li> </ul>                                                                                                                                                               |
| <b>Examples</b>                 | <ul style="list-style-type: none"> <li>■ Pleurobrachia, Ctenoplana</li> </ul>                                                                                                                                                               |
| <b>Platyhelminthes</b>          |                                                                                                                                                                                                                                             |
| <b>Other name</b>               | <ul style="list-style-type: none"> <li>■ Flatworms</li> </ul>                                                                                                                                                                               |
|                                 |                                                                                                                                                                                                                                             |

| Biology and Classification       |                                                                                                                                                                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structure</b>                 | <ul style="list-style-type: none"> <li>■ Triploblastic, bilaterally symmetrical, hermaphrodite</li> <li>■ Organ-level body organisation</li> <li>■ True body cavity (coelom) absent</li> <li>■ Undeveloped digestive system</li> <li>■ Flame cells for excretion</li> </ul> |
| <b>Examples</b>                  | <ul style="list-style-type: none"> <li>■ Taenia solium (tapeworm), Fasciola</li> </ul>                                                                                                                                                                                      |
| <b>Aschelminthes or Nematoda</b> |                                                                                                                                                                                                                                                                             |
| <b>Other name</b>                | <ul style="list-style-type: none"> <li>■ Roundworms</li> </ul>                                                                                                                                                                                                              |
| <b>Structure</b>                 | <ul style="list-style-type: none"> <li>■ Triploblastic, bilaterally symmetrical, true body cavity</li> <li>■ Tissues are found but not true organs</li> <li>■ Complete alimentary canal</li> </ul>                                                                          |
| <b>Other facts</b>               | <ul style="list-style-type: none"> <li>■ Male small, female large</li> <li>■ Unisexual and internal fertilisation</li> </ul>                                                                                                                                                |
| <b>Examples</b>                  | <ul style="list-style-type: none"> <li>■ Ascaris (roundworm), threadworm, Wuchereria (the filaria worm)</li> </ul>                                                                                                                                                          |
| <b>Annelida</b>                  |                                                                                                                                                                                                                                                                             |
| <b>Structure</b>                 | <ul style="list-style-type: none"> <li>■ Bilaterally symmetrical, triploblastic and organ-level organisation (true body cavity)</li> <li>■ Body divided into many segments</li> <li>■ Both unisexual and hermaphrodite</li> </ul>                                           |
| <b>Examples</b>                  | <ul style="list-style-type: none"> <li>■ Nereis, Pheretima (earthworm), Hirudinaria (blood-sucking leech)</li> </ul>                                                                                                                                                        |
| <b>Arthropoda</b>                |                                                                                                                                                                                                                                                                             |
| <b>Structure</b>                 | <ul style="list-style-type: none"> <li>■ Bilaterally symmetrical, triploblastic</li> <li>■ Open circulatory system (blood vessels absent)</li> </ul>                                                                                                                        |
| <b>Blood</b>                     | <ul style="list-style-type: none"> <li>■ Colourless, absence of haemoglobin (haemolymph)</li> </ul>                                                                                                                                                                         |
| <b>Body</b>                      | <ul style="list-style-type: none"> <li>■ Covered by an exoskeleton (chitin)</li> <li>■ Divided into head, thorax and abdomen</li> <li>■ Jointed legs</li> <li>■ Sense organs are found</li> </ul>                                                                           |



| Biology and Classification                                                                                                                                                          |                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other facts</b>                                                                                                                                                                  | <ul style="list-style-type: none"> <li>■ Mostly egg-laying</li> <li>■ Male and female are separate</li> </ul>                                                                                                                                                                            |
| <b>Crustacea</b>                                                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Mainly aquatic</li> <li>■ 5 pairs of legs</li> <li>■ Respiration by gills</li> <li>■ Examples:- crab, prawn, crayfish</li> </ul>                                                                                                                |
| <b>Myriapoda</b>                                                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Terrestrial</li> <li>■ Many pairs of legs</li> <li>■ Respiration by tracheae</li> <li>■ Examples:- centipede (kankhajura), millipede</li> </ul>                                                                                                 |
| <b>Arachnida</b>                                                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Terrestrial</li> <li>■ 4 pairs of legs</li> <li>■ Respiration by tracheae and book lungs</li> <li>■ Examples:- spider, scorpion, mite, tick</li> </ul>                                                                                          |
| <b>Insecta</b>                                                                                                                                                                      | <ul style="list-style-type: none"> <li>■ Generally terrestrial and rarely aquatic</li> <li>■ 3 pairs of legs</li> <li>■ Respiration by tracheae</li> <li>■ Examples:- butterfly, housefly, grasshopper, lice, cockroach, termite, bed bug, moth</li> </ul>                               |
| <ul style="list-style-type: none"> <li>■ Arthropoda is the largest phylum of the animal kingdom.</li> <li>■ About two-thirds of the species on the Earth are arthropods.</li> </ul> |                                                                                                                                                                                                                                                                                          |
| <b>Mollusca</b>                                                                                                                                                                     |                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>■ The second largest phylum of the animal kingdom</li> <li>■ These animals are terrestrial or aquatic</li> </ul>                             |                                                                                                                                                                                                                                                                                          |
| <b>Related facts</b>                                                                                                                                                                | <ul style="list-style-type: none"> <li>■ Bilaterally symmetrical, triploblastic, coelomate</li> <li>■ Body soft and covered by a hard calcium shell</li> <li>■ Open circulatory system</li> <li>■ Kidneys for excretion, gills for respiration</li> <li>■ Blood is colourless</li> </ul> |

| Biology and Classification |                                                                                                                                                                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Male / female              | <ul style="list-style-type: none"> <li>Male and female separate and egg-laying</li> </ul>                                                                                                                                                                                |
| Examples                   | <ul style="list-style-type: none"> <li>Snail (Pila), Sepia (cuttlefish), Loligo (squid), Octopus (devilfish)</li> <li>Aplysia (sea hare), Dentalium (tusk shell), Chaetopleura (chiton)</li> </ul>                                                                       |
| Echinodermata              |                                                                                                                                                                                                                                                                          |
| Structure                  | <ul style="list-style-type: none"> <li>Endoskeleton made of calcareous ossicles</li> <li>Triploblastic and coelomate animals</li> <li>Complete digestive system</li> <li>Distinct excretory system</li> </ul>                                                            |
| Other facts                | <ul style="list-style-type: none"> <li>Spiny skin</li> <li>Marine animals</li> <li>Capacity for regeneration</li> </ul>                                                                                                                                                  |
| Examples                   | <ul style="list-style-type: none"> <li>Asterias or starfish, Ophiura or brittle star, Echinus or sea urchin, sea cucumber, hagfish</li> </ul>                                                                                                                            |
| Hemichordata               |                                                                                                                                                                                                                                                                          |
| Organisms                  | <ul style="list-style-type: none"> <li>Worm-like marine organisms</li> </ul>                                                                                                                                                                                             |
| Structure                  | <ul style="list-style-type: none"> <li>Organ-system level of organisation</li> <li>Bilaterally symmetrical, triploblastic and coelomate</li> <li>Closed circulatory system</li> </ul>                                                                                    |
| Examples                   | <ul style="list-style-type: none"> <li>Balanoglossus, Saccoglossus</li> </ul>                                                                                                                                                                                            |
| Chordata                   |                                                                                                                                                                                                                                                                          |
| Three sub-phyla            | <ul style="list-style-type: none"> <li>Urochordata:- notochord present only in the tail</li> <li>Cephalochordata:- notochord from head to tail</li> <li>Vertebrata:- notochord present in the embryonic stage, changed into the vertebral column in the adult</li> </ul> |
| Class Pisces (Fishes)      |                                                                                                                                                                                                                                                                          |
| Chondrichthyes             |                                                                                                                                                                                                                                                                          |
|                            |                                                                                                                                                                                                                                                                          |

| Biology and Classification |                                                                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structure</b>           | <ul style="list-style-type: none"> <li>■ Powerful jaws</li> <li>■ Two-chambered heart</li> <li>■ Skeleton of cartilage</li> </ul>                                                                                                  |
| <b>Other facts</b>         | <ul style="list-style-type: none"> <li>■ Viviparous, i.e., give birth to young ones</li> <li>■ Cold-blooded (poikilothermic)</li> </ul>                                                                                            |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Scoliodon (dogfish), Pristis (sawfish), blue shark, whale shark, Carcharodon</li> </ul>                                                                                                   |
| <b>Osteichthyes</b>        |                                                                                                                                                                                                                                    |
| <b>Structure</b>           | <ul style="list-style-type: none"> <li>■ The heart is two-chambered</li> <li>■ Endoskeleton bony</li> <li>■ Air bladder is present</li> </ul>                                                                                      |
| <b>Other facts</b>         | <ul style="list-style-type: none"> <li>■ Cold-blooded</li> <li>■ Mostly egg-laying</li> </ul>                                                                                                                                      |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Marine :- Exocoetus (flying fish), Hippocampus (sea horse)</li> <li>■ Freshwater :- rohu, catla, magur</li> <li>■ Aquarium :- Pterophyllum (angel fish), Betta (fighting fish)</li> </ul> |
| <b>Class Tetrapoda</b>     |                                                                                                                                                                                                                                    |
| <b>Amphibia</b>            |                                                                                                                                                                                                                                    |
| <b>Related facts</b>       | <ul style="list-style-type: none"> <li>■ Heart made of three chambers</li> <li>■ Respiration through gills, lungs and skin</li> <li>■ Cold-blooded animals and lay eggs</li> <li>■ Live both in water and on land</li> </ul>       |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Bufo (toad), frog, Hyla (tree frog), salamander, Ichthyophis</li> </ul>                                                                                                                   |
| <b>Lung fish</b>           | <ul style="list-style-type: none"> <li>■ The link between fishes and amphibians</li> </ul>                                                                                                                                         |
| <b>Frog</b>                | <ul style="list-style-type: none"> <li>■ Being highly permeable, it breathes through its skin itself.</li> <li>■ On land the adult frog breathes through its lungs</li> </ul>                                                      |

| Biology and Classification |                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reptilia</b>            |                                                                                                                                                                                      |
| <b>Related facts</b>       | <ul style="list-style-type: none"> <li>■ Body covered with scales made of keratin</li> <li>■ Heart generally three-chambered</li> <li>■ Cold-blooded and egg-laying</li> </ul>       |
| <b>Examples</b>            | <ul style="list-style-type: none"> <li>■ Tortoise, tree lizard, house lizard, chameleon, gharial, cobra, krait, viper, alligator</li> </ul>                                          |
| <b>Crocodile</b>           | <ul style="list-style-type: none"> <li>■ It has a four-chambered heart</li> </ul>                                                                                                    |
| <b>King cobra</b>          | <ul style="list-style-type: none"> <li>■ A nest-building snake</li> </ul>                                                                                                            |
| <b>Mesozoic</b>            | <ul style="list-style-type: none"> <li>■ Is called the age of reptiles</li> </ul>                                                                                                    |
| <b>Poisonous snakes</b>    | <ul style="list-style-type: none"> <li>■ Their fangs are modified forms of the maxillary teeth</li> </ul>                                                                            |
| <b>Birds (Aves)</b>        |                                                                                                                                                                                      |
| <b>Related facts</b>       | <ul style="list-style-type: none"> <li>■ The heart is made of 4 chambers</li> <li>■ Are warm-blooded (homeothermic)</li> <li>■ Breathe through lungs</li> </ul>                      |
| <b>Ostrich</b>             | <ul style="list-style-type: none"> <li>■ The tallest and largest living bird</li> </ul>                                                                                              |
| <b>Emu</b>                 | <ul style="list-style-type: none"> <li>■ The second largest bird in the world</li> <li>■ Found in Australia.</li> </ul>                                                              |
| <b>Penguin</b>             | <ul style="list-style-type: none"> <li>■ A flightless aquatic bird</li> </ul>                                                                                                        |
| <b>Hummingbird</b>         | <ul style="list-style-type: none"> <li>■ The smallest bird in the world</li> <li>■ The only bird that can fly both forwards and backwards</li> </ul>                                 |
| <b>Flightless birds</b>    | <ul style="list-style-type: none"> <li>■ Ostrich, emu, kiwi, penguin, rhea, cassowary</li> </ul>                                                                                     |
| <b>Mammals</b>             |                                                                                                                                                                                      |
| <b>Related facts</b>       | <ul style="list-style-type: none"> <li>■ Most important characteristic :- presence of mammary glands</li> <li>■ Two pairs of limbs</li> <li>■ The heart is four-chambered</li> </ul> |

| Biology and Classification                           |                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototheria</b>                                   | <ul style="list-style-type: none"> <li>■ The link between mammals and reptiles</li> <li>■ Egg-laying mammals</li> <li>■ Examples:- platypus, echidna</li> </ul>                                                                                                                                                                                                           |
| <b>Metatheria</b>                                    | <ul style="list-style-type: none"> <li>■ Mammals giving birth to immature young ones</li> <li>■ Examples:- kangaroo or Macropus, koala</li> </ul>                                                                                                                                                                                                                         |
| <b>Eutheria</b>                                      | <ul style="list-style-type: none"> <li>■ Birth of fully developed young ones</li> <li>■ Examples:- dog, man, elephant</li> </ul>                                                                                                                                                                                                                                          |
| <b>Blue whale</b>                                    | <ul style="list-style-type: none"> <li>■ The largest mammal in the world</li> </ul>                                                                                                                                                                                                                                                                                       |
| <b>African elephant</b>                              | <ul style="list-style-type: none"> <li>■ The largest wild land animal on the Earth</li> </ul>                                                                                                                                                                                                                                                                             |
| <b>Primates</b>                                      | <ul style="list-style-type: none"> <li>■ Man, great apes, gorilla, chimpanzee, orang-utan</li> </ul>                                                                                                                                                                                                                                                                      |
| <b>Miscellaneous</b>                                 |                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Different organisms resembling fishes in name</b> | <ul style="list-style-type: none"> <li>■ Jellyfish :- phylum Cnidaria</li> <li>■ Starfish :- phylum Echinodermata</li> <li>■ Silverfish :- phylum Arthropoda</li> <li>■ Cuttlefish :- phylum Mollusca</li> <li>■ Hagfish :- Cyclostomata</li> <li>■ Devilfish:- phylum Mollusca</li> <li>■ Crayfish:- phylum Arthropoda</li> <li>■ Razorfish:- phylum Mollusca</li> </ul> |
| <b>True fishes</b>                                   | <ul style="list-style-type: none"> <li>■ Flying fish, pipefish, paddlefish, goldfish, globefish, catfish</li> </ul>                                                                                                                                                                                                                                                       |
| <b>Aquatic mammals</b>                               | <ul style="list-style-type: none"> <li>■ Sea cow, dolphin, seal or Phoca, sea lion, walrus, sperm whale</li> <li>■ Blue whale:- the largest mammal on the Earth</li> </ul>                                                                                                                                                                                                |
| <b>Nilgai</b>                                        | <ul style="list-style-type: none"> <li>■ The largest antelope species</li> </ul>                                                                                                                                                                                                                                                                                          |
| <b>Bat</b>                                           | <ul style="list-style-type: none"> <li>■ A flying mammal</li> </ul>                                                                                                                                                                                                                                                                                                       |
| <b>Cockroach</b>                                     | <ul style="list-style-type: none"> <li>■ An organism with white (colourless) blood</li> </ul>                                                                                                                                                                                                                                                                             |
|                                                      |                                                                                                                                                                                                                                                                                                                                                                           |

## Biology and Classification

### Kangaroo

- An animal that excretes solid urine
- Keeps its young in the marsupial pouch until they develop

### Class Insecta

- Six-legged organisms

### Scientific Names of Important Organisms

#### Man

- Homo sapiens

#### Cow

- Bos taurus

#### Cat

- Felis catus

#### Dog

- Canis lupus familiaris

#### Housefly

- Musca domestica

#### Lion

- Panthera leo

#### Mango

- Mangifera indica

#### Paddy (rice)

- Oryza sativa

#### Wheat

- Triticum aestivum

#### Gram

- Cicer arietinum

#### Pea

- Pisum sativum

## > Cell and Tissue

### Cell

#### What it is

- The basic structural and functional unit of every living being
- Has the capacity of self-reproduction

#### Discovery

- First by Robert Hooke in 1665 CE
- Living and free cells were discovered by Leeuwenhoek

#### Study

- Under cell biology or cytology

#### Smallest cell

- Mycoplasma gallisepticum

| Cell                |                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Largest cell</b> | <ul style="list-style-type: none"> <li>That of the ostrich egg</li> </ul>                                                                                                                                     |
| <b>Human body</b>   | <ul style="list-style-type: none"> <li>The largest cell is the ovum</li> <li>The longest cell is the neuron of the nervous system</li> <li>The smallest cell is the granule cell of the cerebellum</li> </ul> |
| <b>Types</b>        | <p>Two types on the basis of the presence of a nucleus</p> <ul style="list-style-type: none"> <li>Prokaryotic</li> <li>Eukaryotic</li> </ul>                                                                  |

| Types of Cell                 |                                                   |                                                    |
|-------------------------------|---------------------------------------------------|----------------------------------------------------|
|                               | Prokaryotic                                       | Eukaryotic                                         |
| <b>Nucleus</b>                | Absence of a distinct nucleus                     | Fully developed nucleus                            |
| <b>Nucleolus and membrane</b> | Nuclear membrane and nucleolus absent             | Nuclear membrane and nucleolus present             |
| <b>Cytoplasm</b>              | Membrane-bound organelles absent in the cytoplasm | All membrane-bound organelles present              |
| <b>Ribosomes</b>              | Only of the 70S type                              | Both 70S and 80S types                             |
| <b>Chromosomes</b>            | Only one                                          | More than one                                      |
| <b>DNA</b>                    | Separate from protein, histone protein absent     | Attached to protein and histone protein present    |
| <b>Found in</b>               | Bacteria and blue-green algae                     | All plants and animals except viruses and bacteria |

| Principal Cell Organelles |                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mitochondria</b>       |                                                                                                                                                                                         |
| <b>Discovery</b>          | <ul style="list-style-type: none"> <li>By Altmann in 1886 CE</li> </ul>                                                                                                                 |
| <b>Naming</b>             | <ul style="list-style-type: none"> <li>By Benda</li> </ul>                                                                                                                              |
| <b>Main function</b>      | <ul style="list-style-type: none"> <li>Carrying out respiration</li> </ul>                                                                                                              |
| <b>Other name</b>         | <ul style="list-style-type: none"> <li>Power house of the cell</li> <li>Energy released by the breakdown of food in the presence of oxygen is stored here in the form of ATP</li> </ul> |

| Principal Cell Organelles        |                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other facts                      | <ul style="list-style-type: none"> <li>■ The Krebs cycle takes place in it</li> <li>■ Circular DNA and 70S ribosomes are found</li> </ul>                                                                                                                                                                                                                                                            |
| Plastids                         |                                                                                                                                                                                                                                                                                                                                                                                                      |
| Found in                         | <ul style="list-style-type: none"> <li>■ Plant cells and Euglena</li> </ul>                                                                                                                                                                                                                                                                                                                          |
| Chloroplasts                     | <ul style="list-style-type: none"> <li>■ Green colour due to the presence of chlorophyll</li> <li>■ Centres of photosynthesis</li> <li>■ Location:- in leaves</li> </ul>                                                                                                                                                                                                                             |
| Chromoplasts                     | <ul style="list-style-type: none"> <li>■ Responsible for the colours other than green found in plants</li> <li>■ Location:- in flowers, fruit walls, seeds, etc.</li> </ul>                                                                                                                                                                                                                          |
| Leucoplasts                      | <ul style="list-style-type: none"> <li>■ In plant parts deprived of sunlight</li> <li>■ Colourless plastids, storage tissue</li> <li>■ Location:- in roots and underground stems</li> </ul>                                                                                                                                                                                                          |
| Examples of chromoplast pigments | <ul style="list-style-type: none"> <li>■ Lycopene:- red colour of tomato</li> <li>■ Carotene:- colour of carrot</li> <li>■ Betanin:- colour of beetroot</li> <li>■ Carotene:- yellow colour of leaves</li> <li>■ Caricaxanthin:- yellow colour of papaya</li> <li>■ Anthocyanin:- red colour of apple</li> <li>■ Phycocyanin:- light blue coloured; protects plants from ultraviolet rays</li> </ul> |
| Endoplasmic Reticulum            |                                                                                                                                                                                                                                                                                                                                                                                                      |
| Location                         | <ul style="list-style-type: none"> <li>■ Spread from the nucleus to the cell membrane</li> </ul>                                                                                                                                                                                                                                                                                                     |
| Types                            | <p>Two types</p> <ul style="list-style-type: none"> <li>■ Smooth endoplasmic reticulum</li> <li>■ Rough endoplasmic reticulum</li> </ul>                                                                                                                                                                                                                                                             |
| Other facts                      | <ul style="list-style-type: none"> <li>■ Ribosomes are attached to it</li> <li>■ The main site of lipid synthesis</li> </ul>                                                                                                                                                                                                                                                                         |
| Golgi Body                       |                                                                                                                                                                                                                                                                                                                                                                                                      |
| Discoverer                       | <ul style="list-style-type: none"> <li>■ Camillo Golgi in 1898 CE</li> </ul>                                                                                                                                                                                                                                                                                                                         |



| Principal Cell Organelles        |                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cisternae</b>                 | <ul style="list-style-type: none"> <li>Functional unit</li> </ul>                                                                                     |
| <b>Main function</b>             | <ul style="list-style-type: none"> <li>Packaging, storage and secretion of macromolecules:- carbohydrates, lipids, proteins, nucleic acids</li> </ul> |
| <b>Other facts</b>               | <ul style="list-style-type: none"> <li>The main site of formation of glycolipids and glycoproteins</li> </ul>                                         |
| <b>Other name</b>                | <ul style="list-style-type: none"> <li>Traffic manager of the molecules of the cell</li> </ul>                                                        |
| <b>Ribosomes</b>                 |                                                                                                                                                       |
| <b>Structure</b>                 | <ul style="list-style-type: none"> <li>A structure made of RNA and protein</li> </ul>                                                                 |
| <b>Discovery</b>                 | <ul style="list-style-type: none"> <li>First in the animal cell by George Palade</li> </ul>                                                           |
| <b>Naming</b>                    | <ul style="list-style-type: none"> <li>By Roberts</li> </ul>                                                                                          |
| <b>Other name</b>                | <ul style="list-style-type: none"> <li>Protein factory</li> </ul>                                                                                     |
| <b>Other facts</b>               | <ul style="list-style-type: none"> <li>Help in protein synthesis</li> <li>Formation:- by the nucleolus</li> </ul>                                     |
| <b>Lysosome</b>                  |                                                                                                                                                       |
| <b>Discoverer</b>                | <ul style="list-style-type: none"> <li>de Duve (1958)</li> </ul>                                                                                      |
| <b>Formation</b>                 | <ul style="list-style-type: none"> <li>In the Golgi body</li> </ul>                                                                                   |
| <b>Other names</b>               | <ul style="list-style-type: none"> <li>Digestive bag or suicide bag</li> <li>They destroy their own cell by digesting it</li> </ul>                   |
| <b>Other facts</b>               | <ul style="list-style-type: none"> <li>Absent in the red blood corpuscles of mammals</li> </ul>                                                       |
| <b>Spherosomes</b>               |                                                                                                                                                       |
|                                  | <ul style="list-style-type: none"> <li>Are called the lysosomes of the plant cell</li> <li>Synthesise and store fat</li> </ul>                        |
| <b>Centrosome and Centrioles</b> |                                                                                                                                                       |
|                                  |                                                                                                                                                       |

## Principal Cell Organelles

- In the cells of all animals and lower plants
- The centrosome is made of two cylindrical structures — the centrioles

### Vacuole

#### What is it?

- A structure in the cytoplasm that encloses water, sap and excretory substances

#### Other facts

- In the plant cell it occupies 90% of the space of the cell

### Nucleus

#### Discovery

- By Robert Brown in 1831 CE

- It is the control centre of the cell.
- Chromosomes and genes are present in the nucleus.

#### Parts

Made of four parts

- Nuclear membrane
- Nucleoplasm
- Nucleolus
- Chromatin network

#### Nuclear membrane

- A double-layered membrane surrounding the nucleus
- It separates the nucleus from the cytoplasm

#### Nucleolus

- Of special importance in cell division
- Site of synthesis of ribosomal RNA
- Storehouse of RNA

#### Chromatin network

- Chromatin is a structure made of DNA and protein
- Fine and extensive thread-like structures
- At the time of division they become organised into chromosomes

#### Chromosome

### Principal Cell Organelles

- In the form of fine, long and coiled threads
- Telomere:- the end part of the chromosome
- DNA about 40%
- Histone protein about 60%
- In humans 23 pairs = 46 chromosomes
- 22 pairs are the same in women and men
- The 23rd pair is XX in women and XY in men
- Sex chromosomes = the 23rd pair
- Carries hereditary information from one generation to the next
- Naming:- Waldeyer (1888 CE);  
discovery:- Wilhelm Nägeli (1842)

### Deoxyribonucleic Acid (DNA)

#### What is it?

- A nucleic acid which, together with protein, forms the structure of the chromosome
- Basically the hereditary material that transmits characters from parents to offspring

#### Location

- Besides the nucleus of the cell, also in mitochondria and chloroplasts

#### Constituent substances

- Nitrogenous bases (adenine, guanine, cytosine, thymine)
- Sugar
- Phosphoric acid

#### Watson and Crick

- Presented the double-helix structure of DNA in 1953
- Awarded the Nobel Prize in 1962

#### Friedrich Miescher

- Discoverer:- Swiss chemist
- In 1869

#### Dr. Har Gobind Khorana

- First synthesised it in the laboratory

| Principal Cell Organelles |                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNA                       | <ul style="list-style-type: none"> <li>Is present in the nucleus, mitochondria and chloroplasts.</li> </ul>                                                         |
| Ribonucleic Acid (RNA)    |                                                                                                                                                                     |
| Structure                 | <ul style="list-style-type: none"> <li>Single-stranded</li> </ul>                                                                                                   |
| Other facts               | <ul style="list-style-type: none"> <li>Takes part in the process of protein formation</li> <li>Is a non-genetic material</li> </ul>                                 |
| Location                  | <ul style="list-style-type: none"> <li>In both the nucleus and the cytoplasm</li> </ul>                                                                             |
| t-RNA                     | <ul style="list-style-type: none"> <li>The smallest of all RNAs</li> <li>Main function:- bringing amino acids to the site of protein synthesis</li> </ul>           |
| r-RNA                     | <ul style="list-style-type: none"> <li>Formation :- in the nucleolus</li> <li>About 80% of the total RNA of the cell</li> <li>Helps in protein synthesis</li> </ul> |
| m-RNA                     | <ul style="list-style-type: none"> <li>Messenger RNA</li> <li>Brings the information recorded in DNA to the site of protein synthesis.</li> </ul>                   |
| RNA                       | <ul style="list-style-type: none"> <li>Is present in the cytoplasm and the nucleoplasm.</li> </ul>                                                                  |
| Gene                      |                                                                                                                                                                     |
| Introduction              | <ul style="list-style-type: none"> <li>The unit of heredity</li> <li>The part of DNA that performs a specific function</li> </ul>                                   |
| In viruses                | <ul style="list-style-type: none"> <li>Made of DNA and RNA</li> </ul>                                                                                               |
| In eukaryotic organisms   | <ul style="list-style-type: none"> <li>Made only of DNA</li> </ul>                                                                                                  |

| Plant Cell and Animal Cell |                                          |                       |
|----------------------------|------------------------------------------|-----------------------|
|                            | Animal Cell                              | Plant Cell            |
| Size                       | Usually small                            | Comparatively large   |
| Shape                      | Round or irregular                       | Square or cuboidal    |
| Energy storage             | In the form of carbohydrate and glycogen | In the form of starch |

**Plant Cell and Animal Cell**

|                  |                                    |                                  |
|------------------|------------------------------------|----------------------------------|
| <b>Growth</b>    | By increase in the number of cells | By increase in the size of cells |
| <b>Cell wall</b> | Not found                          | Cell wall made of cellulose      |
| <b>Cilia</b>     | Provide movement to the cell       | Not found in the plant cell      |
| <b>Plastids</b>  | Not found                          | Found                            |
| <b>Vacuole</b>   | Many small vacuoles                | One large central vacuole        |

**Cell Division****Mitosis****Meiosis**

- |                                                         |                                                                                              |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ■ Takes place in somatic cells                          | ■ Takes place in reproductive (germ) cells                                                   |
| ■ Completed in only one stage                           | ■ Completed in two stages                                                                    |
| ■ Only one division of the nucleus                      | ■ The nucleus divides twice                                                                  |
| ■ Two daughter cells are formed as a result of division | ■ Four daughter cells are formed as a result of division                                     |
| ■ Mitosis is necessary for growth and repair            | ■ Necessary for maintaining the same number of chromosomes in sexually reproducing organisms |

**Miscellaneous Facts Related to the Cell**

- |                             |                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory function</b> | <ul style="list-style-type: none"> <li>■ Mesosome in the prokaryotic cell</li> <li>■ Mitochondria in the eukaryotic cell</li> </ul> |
| <b>Chloroplast</b>          | ■ Kitchen of the cell                                                                                                               |
| <b>Golgi body</b>           | ■ Traffic controller of the molecules of the cell                                                                                   |
| <b>Cell membrane</b>        | ■ Semi-permeable membrane                                                                                                           |
| <b>Ageing gene</b>          | ■ Is responsible for old age                                                                                                        |
| <b>Tissue</b>               |                                                                                                                                     |
| <b>The word</b>             | ■ First used by Bichat                                                                                                              |

## Miscellaneous Facts Related to the Cell

### Introduction

- A group of cells in which the origin, structure and function of the cells are the same

### Plant Tissue

#### Two classes

- Meristematic tissue
- Permanent tissue

### Meristematic Tissue

#### Function

- Continuous formation of new cells by cell division

#### Apical

- Increases length at the tip

#### Intercalary

- Increases length between the nodes

#### Lateral meristem

- Increases the width (girth) of the plant

### Permanent Tissue

#### Introduction

- Are living or dead cells
- Shape is definite
- Cytoplasm with vacuoles

#### Simple tissue

- Tissues that perform a similar function

#### Parenchyma

- Protects plants and prepares food in the leaves
- Also stores excretory substances:- gum, resin and tannin

#### Collenchyma

- Gives mechanical support to plants

#### Sclerenchyma

- Found in abundance in fibrous plants
- Gives mechanical support to plants

#### Complex tissue

- Made of more than one type of cells
- Two types:- xylem and phloem

#### Xylem

- Carry water/mineral salts from the roots to the other parts of plants (one-way conduction)
- Dead cells

### Miscellaneous Facts Related to the Cell

#### Phloem

- Carry food from the leaves to the other parts of plants (two-way conduction)
- Living cells, in the inner layer of the bark

#### Animal Tissue

##### Types

- Epithelial tissue
- Connective tissue
- Muscular tissue
- Nervous tissue

#### Epithelial Tissue

##### Function

- Providing an outer covering or protection to the body, as well as carrying out secretion or excretion, sensation and absorption

#### Connective Tissue

##### True connective tissue

- Adipose connective tissue :- in the form of fat cells under the skin and around the kidneys
- Tendon :- joins muscles to bones or to other muscles
- Ligament:- does the work of joining bones at the joints

##### Skeletal tissue

- It forms the endoskeleton of the body
- Two types:- cartilage and bone tissue

##### Fluid tissue

- Blood is a fluid connective tissue
- Three types:- red blood cells, white blood cells, platelets

#### Muscular Tissue

##### Function

- Necessary for locomotion and various kinds of movements

##### Types

Are of three types:-

- Unstriated / involuntary
- Striated or voluntary
- Cardiac muscle

##### Special

- The strongest muscles are those of the jaw.

#### Nervous Tissue

##### Introduction

- Made of a special type of cells called nerve cells or neurons

### Miscellaneous Facts Related to the Cell

|                                           |                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Axon</b>                               | <ul style="list-style-type: none"> <li>■ A long process found in the neuron</li> </ul>                                                        |
| <b>Cell and Tissue:<br/>Miscellaneous</b> |                                                                                                                                               |
| <b>Genetics</b>                           | <ul style="list-style-type: none"> <li>■ Father:- Gregor Johann Mendel of Austria</li> <li>■ Through experiments on the garden pea</li> </ul> |
| <b>Mendel's laws</b>                      | <ul style="list-style-type: none"> <li>■ Law of dominance, law of independent assortment, law of segregation</li> </ul>                       |

## > Systems of the Human Body

### Digestive System

#### Alimentary Canal

|                                    |                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                | <ul style="list-style-type: none"> <li>■ 30 feet long, extending from the mouth to the anus</li> </ul>                               |
| <b>Parts</b>                       | <ul style="list-style-type: none"> <li>■ Buccal cavity, pharynx, oesophagus, stomach, small intestine and large intestine</li> </ul> |
| <b>Oral Cavity / Buccal Cavity</b> |                                                                                                                                      |
| <b>Organs included</b>             | <ul style="list-style-type: none"> <li>■ Tongue and teeth</li> </ul>                                                                 |
| <b>Digestion</b>                   | <ul style="list-style-type: none"> <li>■ Begins in the buccal cavity itself with the help of saliva</li> </ul>                       |
| <b>Saliva</b>                      | <ul style="list-style-type: none"> <li>■ Contains 99% water and 1% enzymes</li> <li>■ Enzymes:- ptyalin and lysozyme</li> </ul>      |
| <b>Ptyalin</b>                     | <ul style="list-style-type: none"> <li>■ Breaks starch into the disaccharide maltose</li> </ul>                                      |
| <b>Lysozyme</b>                    | <ul style="list-style-type: none"> <li>■ Destroys micro-organisms and bacteria</li> </ul>                                            |
| <b>Teeth</b>                       |                                                                                                                                      |
| <b>Humans</b>                      | <ul style="list-style-type: none"> <li>■ Heterodont</li> <li>■ Four types of teeth:- incisor, canine, premolar, molar</li> </ul>     |
| <b>Incisors</b>                    | <ul style="list-style-type: none"> <li>■ The front-most teeth that cut food</li> </ul>                                               |
| <b>Canines</b>                     | <ul style="list-style-type: none"> <li>■ Pointed teeth that tear food</li> </ul>                                                     |



| Digestive System                          |                                                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Premolars + molars</b>                 | <ul style="list-style-type: none"> <li>■ Cheek teeth</li> <li>■ Teeth that grind food</li> </ul>                                                                                      |
| <b>Enamel</b>                             | <ul style="list-style-type: none"> <li>■ The outer layer of the tooth</li> <li>■ The hardest part of the human body</li> <li>■ 98% made of calcium phosphate and carbonate</li> </ul> |
| <b>Stomach</b>                            |                                                                                                                                                                                       |
| <b>Digestion</b>                          | <ul style="list-style-type: none"> <li>■ The main site of protein digestion</li> </ul>                                                                                                |
| <b>Gastric glands</b>                     | <ul style="list-style-type: none"> <li>■ Present on the inner wall of the stomach</li> <li>■ Secrete highly acidic gastric juice</li> </ul>                                           |
| <b>Digestive enzymes in gastric juice</b> | <ul style="list-style-type: none"> <li>■ Pepsin, rennin, hydrochloric acid and mucus</li> </ul>                                                                                       |
| <b>Pepsin</b>                             | <ul style="list-style-type: none"> <li>■ Protein into peptones and proteoses</li> </ul>                                                                                               |
| <b>Rennin</b>                             | <ul style="list-style-type: none"> <li>■ Milk protein caseinogen into casein</li> </ul>                                                                                               |
| <b>Tongue</b>                             |                                                                                                                                                                                       |
| <b>Taste buds</b>                         | <ul style="list-style-type: none"> <li>■ Taste Buds</li> <li>■ Provide the sense of the taste of food</li> </ul>                                                                      |
| <b>Sweet taste</b>                        | <ul style="list-style-type: none"> <li>■ Sensed by the front part of the tongue</li> </ul>                                                                                            |
| <b>Bitter (pungent) taste</b>             | <ul style="list-style-type: none"> <li>■ Sensed by the back part of the tongue</li> </ul>                                                                                             |
| <b>Sour taste</b>                         | <ul style="list-style-type: none"> <li>■ Sensed by both side parts of the tongue</li> </ul>                                                                                           |
| <b>Small Intestine</b>                    |                                                                                                                                                                                       |
| <b>Introduction</b>                       | <ul style="list-style-type: none"> <li>■ Highly coiled</li> <li>■ Longer than the large intestine</li> <li>■ The digestion of food is completed here</li> </ul>                       |
| <b>Parts</b>                              | Divided into three parts <ul style="list-style-type: none"> <li>■ Duodenum:- 30 cm</li> <li>■ Jejunum:- 2.5 metres</li> <li>■ Ileum:- 2.5 metres to 3.5 metres</li> </ul>             |

| Digestive System                   |                                                                                                                                                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Digestion</b>                   | <ul style="list-style-type: none"> <li>As soon as food comes into the small intestine, three digestive juices (bile juice, pancreatic juice and intestinal juice) are mixed with it.</li> </ul>                                                                                      |
| <b>Function</b>                    | <ul style="list-style-type: none"> <li>Final digestion and absorption of food</li> </ul>                                                                                                                                                                                             |
| <b>Enzymes in intestinal juice</b> | <ul style="list-style-type: none"> <li>Erepsin:- conversion of peptones into amino acids</li> <li>Maltase:- conversion of maltose into glucose</li> <li>Sucrase:- changes sucrose into glucose and fructose</li> <li>Lactase:- changes lactose into glucose and galactose</li> </ul> |
| <b>Enzymes in pancreatic juice</b> | <ul style="list-style-type: none"> <li>Trypsin:- protein into proteoses and peptones</li> <li>Amylase:- digestion of starch</li> <li>Lipase:- digests emulsified fat and converts it into fatty acids</li> </ul>                                                                     |
| <b>Absorption</b>                  | <ul style="list-style-type: none"> <li>Digested food reaching the blood</li> </ul>                                                                                                                                                                                                   |
| <b>Assimilation</b>                | <ul style="list-style-type: none"> <li>The process by which absorbed food materials are put to use by the body</li> </ul>                                                                                                                                                            |
| <b>Liver</b>                       |                                                                                                                                                                                                                                                                                      |
| <b>Introduction</b>                | <ul style="list-style-type: none"> <li>The largest gland of the human body</li> </ul>                                                                                                                                                                                                |
| <b>Weight</b>                      | <ul style="list-style-type: none"> <li>About 1.5 kg</li> </ul>                                                                                                                                                                                                                       |
| <b>Main functions</b>              | <ul style="list-style-type: none"> <li>Synthesis and production of bile juice</li> <li>Stores glucose in the form of glycogen</li> <li>Secretes bile continuously</li> </ul>                                                                                                         |
| <b>Bile juice</b>                  | <ul style="list-style-type: none"> <li>Is stored in the gall bladder</li> <li>A thick greenish-yellow, mildly alkaline fluid</li> <li>No digestive enzyme is found in it</li> </ul>                                                                                                  |
| <b>Bile pigments in bile juice</b> | <ul style="list-style-type: none"> <li>Bilirubin and biliverdin</li> </ul>                                                                                                                                                                                                           |
| <b>Emulsification of fat</b>       | <ul style="list-style-type: none"> <li>By two salts present in bile juice:- sodium glycocholate and sodium taurocholate</li> </ul>                                                                                                                                                   |
| <b>Pancreas</b>                    |                                                                                                                                                                                                                                                                                      |

| Digestive System             |                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>          | <ul style="list-style-type: none"> <li>■ The second largest gland of the human body</li> <li>■ A mixed gland (endocrine + exocrine)</li> </ul>                                           |
| <b>Pancreatic juice</b>      | <ul style="list-style-type: none"> <li>■ 98% water and 2% enzymes and salts</li> <li>■ Is alkaline</li> <li>■ A complete digestive juice</li> </ul>                                      |
| <b>Islets of Langerhans</b>  | <ul style="list-style-type: none"> <li>■ Discoverer:- Langerhans</li> <li>■ Is a part of the pancreas</li> <li>■ The hormones insulin, glucagon and somatostatin are secreted</li> </ul> |
| <b>Large Intestine</b>       |                                                                                                                                                                                          |
| <b>Introduction</b>          | <ul style="list-style-type: none"> <li>■ Wider than the small intestine but shorter in length</li> </ul>                                                                                 |
| <b>Three parts</b>           | <ul style="list-style-type: none"> <li>■ Caecum</li> <li>■ Colon</li> <li>■ Rectum</li> </ul>                                                                                            |
| <b>Functions</b>             | <ul style="list-style-type: none"> <li>■ Absorption of water</li> <li>■ Storing undigested food for some time</li> </ul>                                                                 |
| <b>Appendix</b>              |                                                                                                                                                                                          |
| <b>Other name</b>            | <ul style="list-style-type: none"> <li>■ Vermiform appendix</li> </ul>                                                                                                                   |
| <b>Example</b>               | <ul style="list-style-type: none"> <li>■ Of a vestigial organ</li> </ul>                                                                                                                 |
| <b>Attached to</b>           | <ul style="list-style-type: none"> <li>■ The caecum of the large intestine</li> </ul>                                                                                                    |
| <b>Other Important Facts</b> |                                                                                                                                                                                          |
| <b>Zymase</b>                | <ul style="list-style-type: none"> <li>■ Converts glucose into ethanol</li> </ul>                                                                                                        |
| <b>Enzymes</b>               | <ul style="list-style-type: none"> <li>■ Are basically biocatalysts</li> </ul>                                                                                                           |
| <b>Respiratory System</b>    |                                                                                                                                                                                          |
| <b>Respiratory organs</b>    | <ul style="list-style-type: none"> <li>■ Nose, pharynx, larynx, trachea, lungs</li> </ul>                                                                                                |
| <b>Respiration</b>           |                                                                                                                                                                                          |
|                              |                                                                                                                                                                                          |

| Digestive System                                                 |                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Process of respiration</b>                                    | <ul style="list-style-type: none"> <li>Energy is released by the combustion of glucose in the presence of oxygen</li> <li>Exchange of oxygen and carbon dioxide</li> </ul>                                                                                                                                                                                                     |
| <b>Respiratory rate</b>                                          | <ul style="list-style-type: none"> <li>12 to 15 times per minute in an adult human</li> <li>About 44 times per minute in infants</li> </ul>                                                                                                                                                                                                                                    |
| <b>Other facts</b>                                               | <ul style="list-style-type: none"> <li>A biochemical process</li> <li>An involuntary process</li> <li>Main stages:- glycolysis, Krebs cycle and electron transport system</li> </ul>                                                                                                                                                                                           |
| <b>Lungs</b>                                                     | <ul style="list-style-type: none"> <li>The right lung is bigger and wider</li> <li>The left lung is smaller but longer</li> <li>Pulmonary respiration:- exchange of gases</li> <li>Protected or enclosed by the pleural membrane</li> <li>The alveoli, the sites of gas exchange, are surrounded by blood capillaries</li> <li>Keeps body temperature under control</li> </ul> |
| <b>Internal respiration</b>                                      | <ul style="list-style-type: none"> <li>Gaseous exchange between tissue fluid and blood</li> </ul>                                                                                                                                                                                                                                                                              |
| <b>External respiration</b>                                      | <ul style="list-style-type: none"> <li>Inspiration and expiration</li> <li>Gaseous exchange in the lungs</li> </ul>                                                                                                                                                                                                                                                            |
| <b>Transport of gases</b>                                        | <ul style="list-style-type: none"> <li>Transport of oxygen and CO<sub>2</sub> from the lungs to the cells and back from the cells to the lungs</li> <li>Oxygen is transported by haemoglobin</li> <li>Transport of carbon dioxide:- by haemoglobin, dissolved in plasma, and in the form of bicarbonate</li> </ul>                                                             |
| <b>Control of respiration</b>                                    | <ul style="list-style-type: none"> <li>By the medulla oblongata of the brain</li> </ul>                                                                                                                                                                                                                                                                                        |
| <b>Breakdown of Glucose by Different Pathways in Respiration</b> |                                                                                                                                                                                                                                                                                                                                                                                |
| <b>In yeast</b>                                                  | <ul style="list-style-type: none"> <li>Absence of oxygen</li> <li>Ethanol + CO<sub>2</sub> + energy</li> </ul>                                                                                                                                                                                                                                                                 |
| <b>In muscle cells</b>                                           | <ul style="list-style-type: none"> <li>Lack of oxygen</li> <li>Lactic acid + energy</li> </ul>                                                                                                                                                                                                                                                                                 |

|                           |                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Digestive System</b>   |                                                                                                                                                                                            |
| <b>In mitochondria</b>    | <ul style="list-style-type: none"> <li>■ Presence of oxygen</li> <li>■ CO<sub>2</sub> + water + energy</li> </ul>                                                                          |
| <b>Circulatory System</b> |                                                                                                                                                                                            |
| <b>Discovery</b>          | <ul style="list-style-type: none"> <li>■ By William Harvey</li> </ul>                                                                                                                      |
| <b>Blood circulation</b>  | <ul style="list-style-type: none"> <li>■ Always in a fixed direction</li> <li>■ Carried out by the heart</li> </ul>                                                                        |
| <b>Heart</b>              |                                                                                                                                                                                            |
| <b>Location</b>           | <ul style="list-style-type: none"> <li>■ In the thoracic cavity between the two lungs</li> <li>■ Enclosed by a double membrane called the pericardium</li> </ul>                           |
| <b>Chambers</b>           | <ul style="list-style-type: none"> <li>■ Four chambers</li> <li>■ Atria (right and left)</li> <li>■ Ventricles (right and left)</li> </ul>                                                 |
| <b>Weight</b>             | <ul style="list-style-type: none"> <li>■ About 300 grams</li> </ul>                                                                                                                        |
| <b>Heartbeat</b>          | <ul style="list-style-type: none"> <li>■ 70 to 80 times per minute</li> <li>■ The heart rests between two beats</li> <li>■ Potassium is necessary for controlling the heartbeat</li> </ul> |
| <b>Blood pressure</b>     | <ul style="list-style-type: none"> <li>■ 120/80 (normal)</li> <li>■ More than atmospheric pressure</li> </ul>                                                                              |
| <b>Tricuspid valve</b>    | <ul style="list-style-type: none"> <li>■ Between the right atrium and the right ventricle</li> </ul>                                                                                       |
| <b>Bicuspid valve</b>     | <ul style="list-style-type: none"> <li>■ Between the left atrium and the left ventricle</li> </ul>                                                                                         |
| <b>Systolic pressure</b>  | <ul style="list-style-type: none"> <li>■ 120 mmHg</li> </ul>                                                                                                                               |
| <b>ECG</b>                | <ul style="list-style-type: none"> <li>■ Electrocardiogram</li> <li>■ A type of medical test that shows the activity of the heart</li> </ul>                                               |
| <b>Blood Vessels</b>      |                                                                                                                                                                                            |
| <b>Arteries</b>           |                                                                                                                                                                                            |
| <b>Function</b>           | <ul style="list-style-type: none"> <li>■ Carrying pure blood from the heart to the various organs</li> </ul>                                                                               |

| Digestive System        |                                                                                                                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aorta</b>            | <ul style="list-style-type: none"> <li>■ The largest artery of the body</li> <li>■ Carries pure blood from the left ventricle into the body</li> </ul>                                                                                                                        |
| <b>Pulmonary artery</b> | <ul style="list-style-type: none"> <li>■ The only artery, going to the lungs, in which impure blood flows</li> <li>■ It carries impure blood from the right ventricle to the lungs</li> </ul>                                                                                 |
| <b>Right atrium</b>     | <ul style="list-style-type: none"> <li>■ It receives impure blood from the body</li> </ul>                                                                                                                                                                                    |
| <b>Left atrium</b>      | <ul style="list-style-type: none"> <li>■ Pure blood from the lungs comes into the left atrium and goes into the left ventricle</li> </ul>                                                                                                                                     |
| <b>Veins</b>            |                                                                                                                                                                                                                                                                               |
| <b>Function</b>         | <ul style="list-style-type: none"> <li>■ Carry impure blood from the various organs to the heart</li> </ul>                                                                                                                                                                   |
| <b>Pulmonary vein</b>   | <ul style="list-style-type: none"> <li>■ The only vein in which pure blood flows</li> <li>■ It carries pure blood from the lungs to the left atrium</li> </ul>                                                                                                                |
| <b>Vena cava</b>        | <ul style="list-style-type: none"> <li>■ Supplies oxygen-deficient blood to the right side of the heart</li> <li>■ Two types:- anterior (superior) and posterior (inferior)</li> </ul>                                                                                        |
| <b>Right ventricle</b>  | <ul style="list-style-type: none"> <li>■ From here impure blood goes into the pulmonary artery</li> </ul>                                                                                                                                                                     |
| <b>Left ventricle</b>   | <ul style="list-style-type: none"> <li>■ From here pure blood reaches all the tissues of the body</li> </ul>                                                                                                                                                                  |
| <b>Blood</b>            |                                                                                                                                                                                                                                                                               |
| <b>What is it?</b>      | <ul style="list-style-type: none"> <li>■ A fluid connective tissue</li> <li>■ Made of plasma and blood cells</li> <li>■ pH value:- 7.4 (alkaline)</li> </ul>                                                                                                                  |
| <b>Plasma</b>           | <ul style="list-style-type: none"> <li>■ About 55% of the total blood volume</li> <li>■ A colourless fluid (about 90% water)</li> <li>■ Carries CO<sub>2</sub>, hormones and wastes</li> <li>■ Plasma proteins formed in the liver:- globulin, albumin, fibrinogen</li> </ul> |
| <b>Blood Cells</b>      |                                                                                                                                                                                                                                                                               |
|                         |                                                                                                                                                                                                                                                                               |

| Digestive System              |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blood cells</b>            | <p>Of three types</p> <ul style="list-style-type: none"> <li>Red blood cells (Erythrocytes)</li> <li>White blood cells (Leukocytes)</li> <li>Platelets</li> </ul>                                                                                                                                                                                                                                                                                    |
| <b>Red blood cells</b>        | <ul style="list-style-type: none"> <li>Round, and nucleus absent</li> <li>Red colour due to haemoglobin</li> <li>Carry oxygen</li> <li>Lifespan:- about 120 days</li> <li>Develop in the bone marrow; old blood corpuscles are destroyed by the spleen</li> </ul>                                                                                                                                                                                    |
| <b>White blood corpuscles</b> | <ul style="list-style-type: none"> <li>Nucleated and of irregular shape</li> <li>Other names:- soldiers or brave soldiers</li> <li>Function:- protection of the body from infection</li> <li>The largest of the blood cells</li> <li>Lifespan:- 10 to 13 days</li> <li>Monocyte:- the largest white blood corpuscle</li> <li>Lymphocyte:- the smallest WBC</li> <li>Formation:- in the lymph glands, spleen, thymus gland and bone marrow</li> </ul> |
| <b>Granular WBCs</b>          | <ul style="list-style-type: none"> <li>Eosinophils</li> <li>Basophils</li> <li>Neutrophils</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
| <b>Agranular WBCs</b>         | <ul style="list-style-type: none"> <li>Lymphocytes</li> <li>Monocytes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Platelets</b>              | <ul style="list-style-type: none"> <li>Minute, colourless and without a nucleus</li> <li>Functions:- blood coagulation, forming blood clots, repair of blood vessels</li> </ul>                                                                                                                                                                                                                                                                      |
| <b>Blood Pressure</b>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Measurement</b>            | <ul style="list-style-type: none"> <li>Blood pressure is measured by the sphygmomanometer</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| <b>Normal blood pressure</b>  | <ul style="list-style-type: none"> <li>Systolic pressure:- 115 to 120 mmHg</li> <li>Diastolic pressure:- 75 to 80 mmHg</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| <b>With increasing age</b>    | <ul style="list-style-type: none"> <li>Blood pressure also increases</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      |

| Digestive System                           |                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other Facts Related to the Heart and Blood |                                                                                                                                                                                                                                                                                   |
| Red bone marrow                            | <ul style="list-style-type: none"> <li>● Blood cells are formed in the red bone marrow</li> </ul>                                                                                                                                                                                 |
| Spleen                                     | <ul style="list-style-type: none"> <li>● Is called the graveyard of RBCs</li> </ul>                                                                                                                                                                                               |
| Adrenal gland                              | <ul style="list-style-type: none"> <li>● Controls blood pressure</li> </ul>                                                                                                                                                                                                       |
| Ischaemia                                  | <ul style="list-style-type: none"> <li>● Insufficient supply of blood in the body</li> </ul>                                                                                                                                                                                      |
| Quantity of blood                          | <ul style="list-style-type: none"> <li>● Blood in a healthy human:- 5 to 6 litres</li> </ul>                                                                                                                                                                                      |
| Heparin                                    | <ul style="list-style-type: none"> <li>● Because this substance is found in the blood, blood does not clot (inside the vessels)</li> </ul>                                                                                                                                        |
| Sugar in the blood                         | <ul style="list-style-type: none"> <li>● Measurement:- glucometer</li> <li>● Normal amount:- 100 milligrams per decilitre</li> <li>● Insulin:- the hormone responsible for its regulation</li> <li>● Excess causes diabetes</li> <li>● Deficiency causes hypoglycaemia</li> </ul> |
| SAN                                        | <ul style="list-style-type: none"> <li>● Situated in the right atrium of the heart, controls the heart rate (natural pacemaker)</li> </ul>                                                                                                                                        |
| Hormones controlling heart rate            | <ul style="list-style-type: none"> <li>● Thyroxine</li> <li>● Adrenaline</li> </ul>                                                                                                                                                                                               |
| Blood Groups                               |                                                                                                                                                                                                                                                                                   |
| Discovery                                  | <ul style="list-style-type: none"> <li>● By Karl Landsteiner in 1900</li> </ul>                                                                                                                                                                                                   |
| Rh factor                                  | <ul style="list-style-type: none"> <li>● In the RBCs of the rhesus monkey (1940)</li> <li>● Discovery:- by Landsteiner and Wiener</li> </ul>                                                                                                                                      |
| Antigens                                   |                                                                                                                                                                                                                                                                                   |
| Introduction                               | <ul style="list-style-type: none"> <li>● Those that stimulate the formation of antibodies</li> <li>● Glycoproteins transferred from generation to generation</li> </ul>                                                                                                           |
| Other name                                 | <ul style="list-style-type: none"> <li>● Agglutinogens</li> </ul>                                                                                                                                                                                                                 |
| Location                                   | <ul style="list-style-type: none"> <li>● Attached to the plasma membrane of the blood corpuscles</li> </ul>                                                                                                                                                                       |



| Digestive System              |                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Types</b>                  | <ul style="list-style-type: none"> <li>Antigen-A</li> <li>Antigen-B</li> </ul>                                                                                                                      |
| <b>Antibodies</b>             |                                                                                                                                                                                                     |
| <b>Other name</b>             | <ul style="list-style-type: none"> <li>Agglutinins</li> </ul>                                                                                                                                       |
| <b>In what form</b>           | <ul style="list-style-type: none"> <li>Exist in the form of pure proteins</li> </ul>                                                                                                                |
| <b>Location</b>               | <ul style="list-style-type: none"> <li>In the blood plasma or serum</li> </ul>                                                                                                                      |
| <b>Types</b>                  | <ul style="list-style-type: none"> <li>Anti A or a</li> <li>Anti B or b</li> </ul>                                                                                                                  |
| <b>Blood Groups in Humans</b> |                                                                                                                                                                                                     |
| <b>A, B, O</b>                | <ul style="list-style-type: none"> <li>These three blood groups were discovered by Karl Landsteiner</li> <li>In 1900 CE</li> </ul>                                                                  |
| <b>AB</b>                     | <ul style="list-style-type: none"> <li>Discovery:- by Decastello and Sturli in 1901</li> </ul>                                                                                                      |
| <b>Blood group A</b>          | <ul style="list-style-type: none"> <li>Antigen:- A; antibody:- b</li> <li>Can donate blood to:- A and AB</li> <li>Can receive from:- O and A</li> </ul>                                             |
| <b>Blood group B</b>          | <ul style="list-style-type: none"> <li>Antigen:- B; antibody:- a</li> <li>Can donate blood to:- B and AB</li> <li>Can receive from:- O and B</li> </ul>                                             |
| <b>Blood group AB</b>         | <ul style="list-style-type: none"> <li>Antigen:- AB; antibody:- x</li> <li>Can donate blood to:- AB</li> <li>Can receive from:- O, A, B and AB</li> <li>Other name:- universal recipient</li> </ul> |
| <b>Blood group O</b>          | <ul style="list-style-type: none"> <li>Antigen:- x ; antibody:- ab</li> <li>Can donate blood to:- A, B, AB, O</li> <li>Can receive from:- only O</li> <li>Other name:- universal donor</li> </ul>   |
| <b>Other Facts</b>            |                                                                                                                                                                                                     |

| Digestive System         |                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lymphocytes</b>       | <ul style="list-style-type: none"> <li>Responsible for the formation of antibodies in the blood plasma</li> </ul>                                                                                                                                                                                               |
| <b>Thrombin enzyme</b>   | <ul style="list-style-type: none"> <li>Converts fibrinogen into fibrin in the formation of a blood clot</li> </ul>                                                                                                                                                                                              |
| <b>Lymph</b>             |                                                                                                                                                                                                                                                                                                                 |
| <b>Introduction</b>      | <ul style="list-style-type: none"> <li>A colourless, clear fluid in the empty spaces between the blood vessels and the tissues</li> </ul>                                                                                                                                                                       |
| <b>Direction of flow</b> | <ul style="list-style-type: none"> <li>From the tissues towards the heart</li> </ul>                                                                                                                                                                                                                            |
| <b>Related facts</b>     | <ul style="list-style-type: none"> <li>RBCs are not found in it</li> <li>WBCs are more in it</li> <li>Centre of formation of lymphocytes</li> </ul>                                                                                                                                                             |
| <b>Functions</b>         | <ul style="list-style-type: none"> <li>Immunity (disease resistance)</li> <li>Filtering out and separating dust particles, cancer cells, bacteria, viruses, etc. by the lymph nodes</li> <li>Helps in healing wounds</li> <li>Helps in the circulation of substances from the tissues into the veins</li> </ul> |
| <b>Excretory System</b>  |                                                                                                                                                                                                                                                                                                                 |
| <b>Excretory organs</b>  | <ul style="list-style-type: none"> <li>Kidney, lungs, skin, liver, large intestine</li> </ul>                                                                                                                                                                                                                   |
| <b>Kidney</b>            |                                                                                                                                                                                                                                                                                                                 |
| <b>Nephron</b>           | <ul style="list-style-type: none"> <li>The structural and functional unit of the kidney</li> <li>The filtration unit of the kidney</li> <li>Bowman's capsule:- a small structure in the nephron that purifies the blood</li> </ul>                                                                              |
| <b>Two parts</b>         | <ul style="list-style-type: none"> <li>Outer part:- cortex</li> <li>Inner part:- medulla</li> </ul>                                                                                                                                                                                                             |
| <b>Functions</b>         | <ul style="list-style-type: none"> <li>Excretion and osmoregulation</li> </ul>                                                                                                                                                                                                                                  |
| <b>Excretion</b>         | <ul style="list-style-type: none"> <li>Urea, uric acid, ammonia, etc.</li> <li>Urochrome:- the yellow pigment found in urine</li> </ul>                                                                                                                                                                         |
| <b>Dialysis</b>          | <ul style="list-style-type: none"> <li>Purification of blood (in place of) the kidney</li> </ul>                                                                                                                                                                                                                |

| Digestive System                                                                                                                                                                                                                                      |                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Osmoregulation                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Control of the amount of salts and water in the body</li> </ul>                                                                |
| Human Urine                                                                                                                                                                                                                                           |                                                                                                                                                                       |
| Composition                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Water:- 95%, urea:- 2.6 per cent</li> <li>Salts:- 2.0%, uric acid:- trace amount</li> <li>Creatinine:- 0.3%</li> </ul>         |
| Quantity                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>About 1.5 litres per day</li> </ul>                                                                                            |
| pH value                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>6 (mildly acidic)</li> </ul>                                                                                                   |
| Formation                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Urea is formed in the liver, then urine is formed in the kidney by ultrafiltration</li> </ul>                                  |
| Liver                                                                                                                                                                                                                                                 |                                                                                                                                                                       |
| Functions                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Synthesis of urea (ammonia &gt; urea)</li> <li>Detoxification of alcohol and other harmful substances from the body</li> </ul> |
| Skin                                                                                                                                                                                                                                                  |                                                                                                                                                                       |
| Function                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Taking water, salts and urea through the sweat glands and expelling them from the body in the form of sweat</li> </ul>         |
| Lungs                                                                                                                                                                                                                                                 |                                                                                                                                                                       |
| Function                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Excretion of carbon dioxide dissolved in the blood and, besides it, water vapour</li> </ul>                                    |
| Excretion in Different Animals                                                                                                                                                                                                                        |                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>Unicellular organisms:- by diffusion</li> <li>Coelenterata:- directly by the cells</li> <li>In flatworms:- by flame cells</li> <li>Phylum Annelida:- by nephridia</li> <li>Vertebrates:- by kidneys</li> </ul> |                                                                                                                                                                       |
| Central Nervous System                                                                                                                                                                                                                                |                                                                                                                                                                       |

| Digestive System |                                                                                                                                                                                                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Function         | <ul style="list-style-type: none"> <li>Regulation and control of all the activities of the body</li> </ul>                                                                                                                                                                                                         |
| Unit             | <ul style="list-style-type: none"> <li>Neuron or nerve cell</li> </ul>                                                                                                                                                                                                                                             |
| Parts            | <ul style="list-style-type: none"> <li>Central nervous system:- brain and spinal cord</li> <li>Peripheral nervous system:- cranial and spinal nerves</li> <li>Autonomic nervous system:- sympathetic nervous system, parasympathetic nervous system</li> </ul>                                                     |
| Brain            |                                                                                                                                                                                                                                                                                                                    |
| Weight           | <ul style="list-style-type: none"> <li>About 1400 grams</li> <li>About 2% of the total (body) weight</li> </ul>                                                                                                                                                                                                    |
| Functions        | <ul style="list-style-type: none"> <li>Thinking, control of heart rate, balance of the body, etc.</li> </ul>                                                                                                                                                                                                       |
| 3 parts          | <ul style="list-style-type: none"> <li>Forebrain, midbrain, hindbrain</li> </ul>                                                                                                                                                                                                                                   |
| Forebrain        |                                                                                                                                                                                                                                                                                                                    |
| Two parts        | <ul style="list-style-type: none"> <li>Cerebrum</li> <li>Diencephalon</li> </ul>                                                                                                                                                                                                                                   |
| Cerebrum         | <ul style="list-style-type: none"> <li>About 80% of the brain (the largest part)</li> <li>It has separate centres for seeing, touching and smelling</li> <li>Voluntary actions are controlled from here</li> <li>It is also the centre of human consciousness, reasoning, memory and intelligence</li> </ul>       |
| Diencephalon     | <ul style="list-style-type: none"> <li>Has 2 parts</li> <li>Thalamus and hypothalamus</li> </ul>                                                                                                                                                                                                                   |
| Thalamus         | <ul style="list-style-type: none"> <li>The upper part of the diencephalon</li> <li>Recognises cold, heat and pain</li> </ul>                                                                                                                                                                                       |
| Hypothalamus     | <ul style="list-style-type: none"> <li>The lower part of the diencephalon</li> <li>Controls emotional activities</li> <li>Controls hormones</li> <li>Regulation of hunger, thirst, love, hatred, etc.</li> <li>Controls body temperature, blood pressure, sleep, reproductive capacity, heartbeat, etc.</li> </ul> |
| Midbrain         |                                                                                                                                                                                                                                                                                                                    |

| Digestive System                          |                                                                                                                                                                                                                                                            |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Two parts</b>                          | <ul style="list-style-type: none"> <li>■ Cerebral peduncle</li> <li>■ Corpora quadrigemina</li> </ul>                                                                                                                                                      |
| <b>Functions</b>                          | <ul style="list-style-type: none"> <li>■ Control over sight, hearing and reflex actions</li> </ul>                                                                                                                                                         |
| <b>Cerebral peduncle</b>                  | <ul style="list-style-type: none"> <li>■ The front part of the midbrain</li> <li>■ Made of fibres</li> </ul>                                                                                                                                               |
| <b>Corpora quadrigemina</b>               | <ul style="list-style-type: none"> <li>■ The rear part of the midbrain</li> <li>■ Related to visual perception and hearing</li> </ul>                                                                                                                      |
| <b>Hindbrain</b>                          |                                                                                                                                                                                                                                                            |
| <b>Parts</b>                              | <ul style="list-style-type: none"> <li>■ Cerebellum</li> <li>■ Medulla oblongata</li> </ul>                                                                                                                                                                |
| <b>Cerebellum</b>                         | <ul style="list-style-type: none"> <li>■ The second largest part of the human brain</li> <li>■ Controls movement and maintains the balance of the body</li> <li>■ Control over the contraction of voluntary muscles</li> </ul>                             |
| <b>Medulla oblongata</b>                  | <ul style="list-style-type: none"> <li>■ The rear-most triangular part of the brain</li> <li>■ Centre of swallowing food and vomiting</li> <li>■ Centre of all involuntary actions</li> <li>■ Respiration, metabolism, coughing, sneezing, etc.</li> </ul> |
| <b>Spinal Cord (Sushumna / Merurajju)</b> |                                                                                                                                                                                                                                                            |
| <b>Discoverer</b>                         | <ul style="list-style-type: none"> <li>■ Marshall Hall</li> </ul>                                                                                                                                                                                          |
| <b>Location</b>                           | <ul style="list-style-type: none"> <li>■ Emerges from the foramen magnum at the medulla oblongata and extends down through the vertebral column to the end</li> </ul>                                                                                      |
| <b>Functions</b>                          | <ul style="list-style-type: none"> <li>■ The main centre of reflex actions</li> <li>■ Balances involuntary actions</li> <li>■ Conduction of impulses coming to and going from the brain</li> </ul>                                                         |
| <b>Peripheral Nervous System</b>          |                                                                                                                                                                                                                                                            |
| <b>Includes</b>                           | <ul style="list-style-type: none"> <li>■ 12 pairs of cranial nerves</li> <li>■ 31 pairs of spinal nerves</li> </ul>                                                                                                                                        |

| Digestive System                      |                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cranial nerves</b>                 | <ul style="list-style-type: none"> <li>■ The first, second and eighth pairs are sensory</li> <li>■ The rest are mixed, i.e., sensory and motor</li> </ul>                                                                                                                                                                                                                                         |
| <b>Spinal nerves</b>                  | <ul style="list-style-type: none"> <li>■ Connect the muscles of the hands and legs, the skin, the sense organs, etc. with the spinal cord</li> </ul>                                                                                                                                                                                                                                              |
| <b>Autonomic Nervous System</b>       |                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Two parts</b>                      | <ul style="list-style-type: none"> <li>■ Sympathetic nervous system</li> <li>■ Parasympathetic nervous system</li> </ul>                                                                                                                                                                                                                                                                          |
| <b>Sympathetic nervous system</b>     | <ul style="list-style-type: none"> <li>■ Speeds up the heartbeat</li> <li>■ Dilates the pupils of the eyes</li> <li>■ Secretion from the sweat glands</li> <li>■ Increases sugar in the blood</li> <li>■ Reduces the secretion of the salivary glands</li> <li>■ Its collective effect gives the feeling of fear, pain and anger</li> </ul>                                                       |
| <b>Parasympathetic nervous system</b> | <ul style="list-style-type: none"> <li>■ Constricts the pupils of the eyes</li> <li>■ Increases the secretion of saliva</li> <li>■ Contraction in the muscles of the urinary bladder</li> <li>■ Contraction in the intestinal wall</li> <li>■ Increase in digestive juices</li> <li>■ Slows down the heartbeat</li> <li>■ Its collective effect gives the feeling of rest and pleasure</li> </ul> |
| <b>Skeletal System</b>                |                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Introduction</b>                   | <ul style="list-style-type: none"> <li>■ A framework of bones and cartilages</li> <li>■ Gives shape to the body</li> </ul>                                                                                                                                                                                                                                                                        |
| <b>Two parts</b>                      | <ul style="list-style-type: none"> <li>■ Exoskeleton :- hair and nails</li> <li>■ Endoskeleton :- the inner skeleton</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>Endoskeleton</b>                   | <ul style="list-style-type: none"> <li>■ Axial skeleton:- 80 bones</li> <li>■ Appendicular skeleton:- 126 bones</li> </ul>                                                                                                                                                                                                                                                                        |
| <b>Bone</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Total bones</b>                    | <ul style="list-style-type: none"> <li>■ 206 in an adult human</li> <li>■ About 300 in a newborn infant</li> </ul>                                                                                                                                                                                                                                                                                |

| Digestive System                   |                                                                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Constituents</b>                | <ul style="list-style-type: none"> <li>■ Important constituent:- calcium phosphate</li> <li>■ Collagen protein:- about 33%</li> </ul>                                                  |
| <b>Marrow cavity</b>               | <ul style="list-style-type: none"> <li>■ A hollow cavity in the middle of long and thick bones</li> <li>■ The fluid bone marrow is in this cavity</li> </ul>                           |
| <b>Vitamin D</b>                   | <ul style="list-style-type: none"> <li>■ Necessary for the normal growth of bones</li> </ul>                                                                                           |
| <b>Femur</b>                       | <ul style="list-style-type: none"> <li>■ The thigh bone</li> <li>■ The largest bone of the human body</li> </ul>                                                                       |
| <b>Stapes</b>                      | <ul style="list-style-type: none"> <li>■ A bone of the ear</li> <li>■ The smallest bone of the human body</li> </ul>                                                                   |
| <b>Ligaments</b>                   | <ul style="list-style-type: none"> <li>■ The joint of bone with bone is called this</li> </ul>                                                                                         |
| <b>Tendon</b>                      | <ul style="list-style-type: none"> <li>■ The joint of muscle and bone is called this</li> </ul>                                                                                        |
| <b>Patella (janu-phalak)</b>       | <ul style="list-style-type: none"> <li>■ The bone that protects the knee joint and helps in bending (knee cap)</li> <li>■ The largest sesamoid bone</li> </ul>                         |
| <b>Axial Skeleton</b>              |                                                                                                                                                                                        |
| <b>Human skull</b>                 | 29 bones in total <ul style="list-style-type: none"> <li>■ Cranium:- 8 bones</li> <li>■ Face:- 14 bones</li> <li>■ In the middle ear:- 6 bones</li> </ul>                              |
| <b>Vertebral column (backbone)</b> | 26 bones in total <ul style="list-style-type: none"> <li>■ Neck:- 7</li> <li>■ Thorax:- 12</li> <li>■ Waist (lumbar):- 5</li> <li>■ Sacrum:- 1</li> <li>■ Coccyx (tail):- 1</li> </ul> |
| <b>Sternum</b>                     | <ul style="list-style-type: none"> <li>■ 1 bone</li> </ul>                                                                                                                             |
| <b>Ribs</b>                        | <ul style="list-style-type: none"> <li>■ 24 bones (12 pairs)</li> </ul>                                                                                                                |
| <b>Appendicular Skeleton</b>       |                                                                                                                                                                                        |
| <b>Girdles</b>                     | <ul style="list-style-type: none"> <li>■ 6 bones</li> </ul>                                                                                                                            |

|                                                   |                                                                                                                                                                                                      |
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| <b>Digestive System</b>                           |                                                                                                                                                                                                      |
| <b>Limbs</b>                                      | <ul style="list-style-type: none"> <li>120 bones</li> </ul>                                                                                                                                          |
| <b>Nails</b>                                      |                                                                                                                                                                                                      |
| <b>Formation</b>                                  | <ul style="list-style-type: none"> <li>From the matter of dead cells</li> <li>From a protein called alpha keratin</li> </ul>                                                                         |
| <b>Exoskeleton</b>                                | <ul style="list-style-type: none"> <li>Absence of blood circulation</li> <li>No pain on cutting</li> </ul>                                                                                           |
| <b>Other Facts Related to the Skeletal System</b> |                                                                                                                                                                                                      |
| <b>Arthroplasty</b>                               | <ul style="list-style-type: none"> <li>Replacement of the hip joint is called this</li> </ul>                                                                                                        |
| <b>Calcium phosphate</b>                          | <ul style="list-style-type: none"> <li>Is the salt found in the largest amount in human bones</li> </ul>                                                                                             |
| <b>Hyoid</b>                                      | <ul style="list-style-type: none"> <li>The bone situated below the tongue between the jaw and the throat</li> </ul>                                                                                  |
| <b>Principal bones</b>                            | <ul style="list-style-type: none"> <li>Clavicle:- jatrak (collar bone)</li> <li>Scapula:- skandh phalak (shoulder blade)</li> <li>Sternum:- urosthi (breast bone)</li> <li>Patella:- knee</li> </ul> |
| <b>Endocrine System</b>                           |                                                                                                                                                                                                      |
| <b>Glands</b>                                     |                                                                                                                                                                                                      |
| <b>Introduction</b>                               | <ul style="list-style-type: none"> <li>Organs of the human body that secrete various kinds of fluids</li> </ul>                                                                                      |
| <b>Function</b>                                   | <ul style="list-style-type: none"> <li>Secretion of substances necessary for the body</li> </ul>                                                                                                     |
| <b>Three types</b>                                | <ul style="list-style-type: none"> <li>Endocrine</li> <li>Exocrine</li> <li>Mixed gland</li> </ul>                                                                                                   |
| <b>Endocrine Glands</b>                           |                                                                                                                                                                                                      |
| <b>Introduction</b>                               | <ul style="list-style-type: none"> <li>Ductless glands</li> <li>Secrete hormones</li> </ul>                                                                                                          |
|                                                   |                                                                                                                                                                                                      |



| Digestive System                                       |                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hormones</b>                                        | <ul style="list-style-type: none"> <li>Reach the target organ through the blood</li> <li>Chemically they are of the nature of proteins, peptones, amino acids, steroids</li> <li>They decompose after taking part in a chemical reaction once</li> </ul>                            |
| <b>Pituitary Gland and the Hormones Secreted by it</b> |                                                                                                                                                                                                                                                                                     |
| <b>Introduction</b>                                    | <ul style="list-style-type: none"> <li>The smallest endocrine gland of the human body</li> <li>In the 'sella turcica' of the sphenoid bone of the skull</li> </ul>                                                                                                                  |
| <b>Master gland</b>                                    | <ul style="list-style-type: none"> <li>It is so called because it controls a person's growth, health, sexual development, etc.</li> </ul>                                                                                                                                           |
| <b>Somatotropic hormone</b>                            | <ul style="list-style-type: none"> <li>Other name:- growth hormone</li> <li>Function:- normal and balanced growth of the body</li> <li>Hyposecretion causes dwarfism in children and Simmonds' disease in adults</li> <li>Hypersecretion causes gigantism in adolescents</li> </ul> |
| <b>TSH hormone</b>                                     | <ul style="list-style-type: none"> <li>Thyroid stimulating hormone</li> <li>Function:- stimulating the thyroid gland to secrete the hormone thyroxine</li> </ul>                                                                                                                    |
| <b>ACTH hormone</b>                                    | <ul style="list-style-type: none"> <li>Adrenocorticotrophic hormone</li> <li>Stimulates the secretion of hormones from the adrenal gland</li> <li>Regulation of the secretion of the hormone cortisol</li> </ul>                                                                    |
| <b>LTH hormone</b>                                     | <ul style="list-style-type: none"> <li>Luteotropic hormone (Prolactin)</li> <li>Other name:- lactogenic hormone</li> <li>Stimulates the secretion of milk from the mammary glands</li> </ul>                                                                                        |
| <b>Vasopressin</b>                                     | <ul style="list-style-type: none"> <li>Anti-diuretic hormone</li> <li>Synthesis:- in the hypothalamus</li> <li>Storage:- in the pituitary gland</li> <li>Function:- water balance in the body and concentration of urine</li> </ul>                                                 |
|                                                        |                                                                                                                                                                                                                                                                                     |

| Digestive System                                         |                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FSH hormone</b>                                       | <ul style="list-style-type: none"> <li>● Follicle stimulating hormone</li> <li>● Stimulates the formation of sperms in the testes</li> <li>● Regulation of the menstrual cycle in women</li> </ul>                                                                                                                       |
| <b>LH hormone</b>                                        | <ul style="list-style-type: none"> <li>● Luteinizing hormone</li> <li>● In women it is called luteotropin</li> <li>● Develops sexual characteristics</li> </ul>                                                                                                                                                          |
| <b>Oxytocin</b>                                          | <ul style="list-style-type: none"> <li>● Birth hormone:- stimulates the contraction of the uterine muscles at the time of delivery</li> <li>● Milk ejection hormone:- stimulates the secretion of milk from the mammary glands after childbirth</li> <li>● Love hormone:- affection between mother and infant</li> </ul> |
| <b>Thyroid Gland and the Hormones Secreted by it</b>     |                                                                                                                                                                                                                                                                                                                          |
| <b>Location</b>                                          | <ul style="list-style-type: none"> <li>● In the human neck, below the larynx</li> </ul>                                                                                                                                                                                                                                  |
| <b>Thyroxine hormone</b>                                 | <ul style="list-style-type: none"> <li>● Structure:- an iodine-containing amino acid</li> <li>● Control of metabolic activities</li> <li>● Stimulant for growth</li> <li>● Regulation of the functions of the reproductive organs</li> <li>● Control of water balance</li> </ul>                                         |
| <b>Related diseases</b>                                  | <ul style="list-style-type: none"> <li>● Excess:- exophthalmic goitre and Plummer's disease</li> <li>● Deficiency :- cretinism in children, myxoedema and goitre in adults</li> </ul>                                                                                                                                    |
| <b>Hashimoto's disease</b>                               | <ul style="list-style-type: none"> <li>● Attack on the thyroid gland by the body's immune system</li> <li>● This disease is also called the suicide of the thyroid</li> </ul>                                                                                                                                            |
| <b>Parathyroid Gland and the Hormones Secreted by it</b> |                                                                                                                                                                                                                                                                                                                          |
| <b>Location</b>                                          | <ul style="list-style-type: none"> <li>● Just behind the thyroid gland situated in the neck</li> </ul>                                                                                                                                                                                                                   |
| <b>Derived</b>                                           | <ul style="list-style-type: none"> <li>● From the endoderm of the embryo</li> </ul>                                                                                                                                                                                                                                      |
|                                                          |                                                                                                                                                                                                                                                                                                                          |

| Digestive System                              |                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PTH hormone</b>                            | <ul style="list-style-type: none"> <li>■ Parathyroid hormone</li> <li>■ Regulates the amount of calcium in the blood</li> <li>■ Its hyposecretion causes cramps and stiffness in the skeletal muscles</li> <li>■ Its hypersecretion causes osteoporosis</li> </ul>                                                      |
| <b>Hormones Secreted by the Adrenal Gland</b> |                                                                                                                                                                                                                                                                                                                         |
| <b>Location</b>                               | <ul style="list-style-type: none"> <li>■ On the upper end of both kidneys</li> </ul>                                                                                                                                                                                                                                    |
| <b>Two parts</b>                              | <ul style="list-style-type: none"> <li>■ Adrenal cortex</li> <li>■ Adrenal medulla</li> </ul>                                                                                                                                                                                                                           |
| <b>Adrenal cortex</b>                         | <ul style="list-style-type: none"> <li>■ Aldosterone hormone</li> <li>■ Cortisol:- secreted at the time of stress; helps in the decision to fight or flee in adverse situations</li> <li>■ Androgen:- responsible for male characteristics and reproduction</li> <li>■ Its disorder causes Addison's disease</li> </ul> |
| <b>Adrenal medulla</b>                        | <ul style="list-style-type: none"> <li>■ Adrenaline:- called the 'fight and flight' hormone and the emergency hormone.</li> <li>■ Noradrenaline hormone</li> </ul>                                                                                                                                                      |
| <b>Hormones Secreted by the Pineal Gland</b>  |                                                                                                                                                                                                                                                                                                                         |
| <b>Melatonin hormone</b>                      | <ul style="list-style-type: none"> <li>■ Its main function is to maintain the sleep-wake cycle and the biological rhythm of the body</li> </ul>                                                                                                                                                                         |
| <b>Melanin</b>                                | <ul style="list-style-type: none"> <li>■ It is a pigment</li> <li>■ Found in the skin, eyes, hair and the central nervous system</li> </ul>                                                                                                                                                                             |
| <b>Serotonin</b>                              | <ul style="list-style-type: none"> <li>■ Exchange of information between nerve cells</li> </ul>                                                                                                                                                                                                                         |
| <b>Other facts</b>                            | <ul style="list-style-type: none"> <li>■ By regulating the biological clock in the body, the pineal gland also determines the sleep-wake cycle, body temperature and the menstrual cycle.</li> </ul>                                                                                                                    |
| <b>Hormones Secreted by the Thymus Gland</b>  |                                                                                                                                                                                                                                                                                                                         |
|                                               |                                                                                                                                                                                                                                                                                                                         |

| Digestive System                               |                                                                                                                                                                                                                                                                                        |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thymosin hormone</b>                        | <ul style="list-style-type: none"> <li>Helps the body's immune system by forming T-lymphocytes in the body</li> </ul>                                                                                                                                                                  |
| <b>Other hormones</b>                          | <ul style="list-style-type: none"> <li>Thymin or thymopoietin</li> </ul>                                                                                                                                                                                                               |
| <b>Exocrine Glands</b>                         |                                                                                                                                                                                                                                                                                        |
| <b>Introduction</b>                            | <ul style="list-style-type: none"> <li>Glands with ducts</li> <li>The fluid secreted by them reaches the concerned organ through ducts</li> </ul>                                                                                                                                      |
| <b>Secretion</b>                               | <ul style="list-style-type: none"> <li>Of enzymes</li> </ul>                                                                                                                                                                                                                           |
| <b>Examples</b>                                | <ul style="list-style-type: none"> <li>Salivary glands, sweat glands, oil glands, etc.</li> </ul>                                                                                                                                                                                      |
| <b>Other Glands</b>                            |                                                                                                                                                                                                                                                                                        |
| <b>Hormones Secreted by the Pancreas Gland</b> |                                                                                                                                                                                                                                                                                        |
| <b>Introduction</b>                            | <ul style="list-style-type: none"> <li>The second largest gland of the human body after the liver</li> <li>A mixed gland (endocrine + exocrine)</li> </ul>                                                                                                                             |
| <b>Pancreatic juice</b>                        | <ul style="list-style-type: none"> <li>Is called the complete digestive juice</li> </ul>                                                                                                                                                                                               |
| <b>Cells of the islets of Langerhans</b>       | <p>The endocrine part of the gland</p> <ul style="list-style-type: none"> <li>Alpha cells:- glucagon</li> <li>Beta cells:- insulin</li> <li>Delta cells:- somatostatin</li> <li>F cells:- pancreatic polypeptide</li> </ul>                                                            |
| <b>Alpha cells</b>                             | <ul style="list-style-type: none"> <li>Secretion of the hormone glucagon</li> <li>Glucagon:- keeps the amount of glucose in the blood normal and synthesises glucose from the glycogen stored in the liver</li> </ul>                                                                  |
| <b>Beta cells</b>                              | <ul style="list-style-type: none"> <li>Secrete the hormone insulin.</li> </ul>                                                                                                                                                                                                         |
| <b>Insulin</b>                                 | <ul style="list-style-type: none"> <li>The metal zinc is present</li> <li>A protein chain or peptide hormone</li> <li>Function:- converting excess sugar of the blood into glycogen</li> <li>Its deficiency causes diabetes</li> <li>Also obtained from the roots of dahlia</li> </ul> |

| Digestive System                       |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frederick Sanger</b>                | <ul style="list-style-type: none"> <li>Received the Nobel Prize for discovering the sequence of amino acids in insulin.</li> <li>He was the first to give the complete chemical structure of bovine (cow) insulin.</li> </ul>                                                                                                                                                                                            |
| Reproductive Hormones                  |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Female Hormones                        |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Estrogen</b>                        | <ul style="list-style-type: none"> <li>Production:- from the ovary</li> <li>Age:- from the onset of puberty up to the age of menopause</li> <li>Estradiol:- the principal estrogen</li> <li>Other name:- female development hormone</li> <li>Secondary sexual characters of the female (proper development of breasts, uterus, mammary glands, etc.)</li> <li>Controlled by the pituitary hormones FSH and LH</li> </ul> |
| <b>Relaxin</b>                         | <ul style="list-style-type: none"> <li>Secretion:- from the corpus luteum and the ovary during pregnancy</li> </ul>                                                                                                                                                                                                                                                                                                      |
| <b>Progesterone</b>                    | <ul style="list-style-type: none"> <li>Production:- by the corpus luteum</li> <li>Function:- stimulating the development of the characteristics necessary for pregnancy.</li> </ul>                                                                                                                                                                                                                                      |
| Male Hormones                          |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Testosterone</b>                    | <ul style="list-style-type: none"> <li>Other name:- male development hormone</li> <li>Function:- development of masculine sexual characteristics</li> </ul>                                                                                                                                                                                                                                                              |
| Human Body Systems:<br>Miscellaneous   |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Fertilisation</b>                   | <ul style="list-style-type: none"> <li>Male gamete (sperm) &gt; female gamete (ovum) &gt; fertilisation &gt; zygote &gt; diploid &gt; formation and development of the embryo</li> <li>The process of fertilisation takes place in the oviduct or Fallopian tube</li> </ul>                                                                                                                                              |
| <b>Development of the human embryo</b> | <ul style="list-style-type: none"> <li>Heartbeat:- from the fourth-fifth week of development</li> <li>Embryonic development:- from the third to the eighth week</li> </ul>                                                                                                                                                                                                                                               |
| <b>Neuron</b>                          | <ul style="list-style-type: none"> <li>Is the unit of nervous tissue</li> </ul>                                                                                                                                                                                                                                                                                                                                          |

| Digestive System             |                                                                                                                                                                                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bernstein</b>             | <ul style="list-style-type: none"><li>■ In 1924 he showed that the blood group is a hereditary character.</li></ul>                                                                                                                                    |
| <b>Heart</b>                 | <ul style="list-style-type: none"><li>■ Main function:- supplying blood to the organs of the body</li></ul>                                                                                                                                            |
| <b>Anaerobic respiration</b> | <ul style="list-style-type: none"><li>■ In the absence of oxygen</li><li>■ Glucose — bacteria / ethyl alcohol in yeast</li><li>■ In the end, pyruvic acid</li></ul>                                                                                    |
| <b>Aerobic respiration</b>   | <ul style="list-style-type: none"><li>■ In the presence of oxygen</li><li>■ <math>\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{energy}</math></li><li>■ Two parts :- glycolysis and the Krebs cycle</li></ul>      |
| <b>Krebs cycle</b>           | <ul style="list-style-type: none"><li>■ First described:- by Hans Krebs</li><li>■ Other names:- citric acid cycle / tricarboxylic acid cycle</li><li>■ Location:- mitochondria</li><li>■ Functions:- production of ATP, production of energy</li></ul> |

06

## Environment and Ecology

ENVIRONMENT AND ECOLOGY

> Ramsar Sites / Biosphere Reserves / National Parks / Leopard Status Report / UNESCO World Heritage Sites - UPDATED 2026

> India's 18 Biosphere Reserves (September 2025)

| S.No. | Name of the Biosphere Reserve  | Year of Establishment | Year of Inclusion in the UNESCO MAB List / Fact                        |
|-------|--------------------------------|-----------------------|------------------------------------------------------------------------|
| 1     | Nilgiri Biosphere Reserve      | 1986                  | India's first biosphere reserve; also included in the UNESCO MAB list. |
| 2     | Nokrek Biosphere Reserve       | 1988                  | (Meghalaya); included in the UNESCO MAB list.                          |
| 3     | Nanda Devi Biosphere Reserve   | 1988                  | (Uttarakhand); included in the UNESCO MAB list.                        |
| 4     | Manas Biosphere Reserve        | 1989                  | (Assam); only in the national list (not included in MAB).              |
| 5     | Gulf of Mannar Biosphere       | 1989                  | (Tamil Nadu); included in the UNESCO MAB list.                         |
| 6     | Great Nicobar Biosphere        | 1989                  | (Andaman and Nicobar); included in the UNESCO MAB list.                |
| 7     | Sundarban Biosphere Reserve    | 1989                  | (West Bengal); included in the UNESCO MAB list.                        |
| 8     | Simlipal Biosphere Reserve     | 1994                  | (Odisha); included in the UNESCO MAB list.                             |
| 9     | Dibru-Saikhowa Biosphere       | 1997                  | India's smallest biosphere reserve by area (Assam).                    |
| 10    | Dihang-Dibang Biosphere        | 1998                  | (Arunachal Pradesh); not included in the MAB list.                     |
| 11    | Pachmarhi Biosphere Reserve    | 1999                  | (Madhya Pradesh); included in the UNESCO MAB list.                     |
| 12    | Kanchenjunga Biosphere Reserve | 2000                  | (Sikkim); included in the UNESCO MAB list (mixed site).                |
| 13    | Agasthyamalai Biosphere        | 2001                  | (Kerala-Tamil Nadu); included in the UNESCO MAB list.                  |
| 14    | Achanakmar-Amarkantak          | 2005                  | (Madhya Pradesh-Chhattisgarh); included in the UNESCO MAB list.        |

| S.No. | Name of the Biosphere Reserve | Year of Establishment | Year of Inclusion in the UNESCO MAB List / Fact                                                |
|-------|-------------------------------|-----------------------|------------------------------------------------------------------------------------------------|
| 15    | Rann of Kutch Biosphere       | 2008                  | India's largest biosphere reserve by area (Gujarat).                                           |
| 16    | Cold Desert                   | 2009                  | (Himachal Pradesh); not included in the MAB list.                                              |
| 17    | Seshachalam Hills Biosphere   | 2010                  | (Andhra Pradesh); not included in the MAB list.                                                |
| 18    | Panna Biosphere Reserve       | 2011                  | (Madhya Pradesh); the <b>latest (12th)</b> site to be included in the UNESCO <b>MAB list</b> . |

### India's total of 13 biosphere reserves included in UNESCO's MAB (Man and Biosphere) Programme:

At present, out of India's total of **18 notified biosphere** reserves, **13 reserves** have become part of UNESCO's global network (MAB), in chronological order: (1) Nilgiri, (2) Gulf of Mannar, (3) Sundarban, (4) Nanda Devi, (5) Nokrek, (6) Pachmarhi, (7) Simlipal, (8) Achanakmar-Amarkantak, (9) Great Nicobar, (10) Agasthyamalai, (11) Kanchenjunga, and (12) Panna, (13) **Cold Desert Biosphere Reserve**

#### > NOTE:

- India's **13th** UNESCO (UNESCO MAB) site is the **Cold Desert Biosphere Reserve**.
- This new status was given in **September 2025** during UNESCO's 5th World Congress of Biosphere Reserves held in Hangzhou, China.
- It is situated in the Lahaul and Spiti district of **Himachal Pradesh**.

#### > A Total of 101 Ramsar Sites in India (as of August 2026)

**NOTE** - At present (as of August 2026) India has a total of **101 Ramsar Sites**, in which **Glaw Lake** of **Arunachal Pradesh** has become India's 101st Ramsar site.

#### > Year-wise and Serial List of India's Ramsar Sites (1981 - 2026)

| S.No. | Name of the Ramsar Site | State / Union Territory | Year Declared |
|-------|-------------------------|-------------------------|---------------|
| 1     | Chilika Lake            | Odisha                  | 1981          |
| 2     | Keoladeo National Park  | Rajasthan               | 1981          |
| 3     | Wular Lake              | Jammu and Kashmir       | 1990          |
| 4     | Harike Wetland          | Punjab                  | 1990          |
| 5     | Loktak Lake             | Manipur                 | 1990          |
| 6     | Sambhar Lake            | Rajasthan               | 1990          |
| 7     | Kolleru Lake            | Andhra Pradesh          | 2002          |
| 8     | Deepor Beel             | Assam                   | 2002          |
| 9     | Pong Dam Lake           | Himachal Pradesh        | 2002          |



| S.No. | Name of the Ramsar Site                        | State / Union Territory | Year Declared |
|-------|------------------------------------------------|-------------------------|---------------|
| 10    | Ashtamudi Wetland                              | Kerala                  | 2002          |
| 11    | Sasthamkotta Lake                              | Kerala                  | 2002          |
| 12    | Vembanad-Kol Wetland                           | Kerala                  | 2002          |
| 13    | Bhoj Wetland                                   | Madhya Pradesh          | 2002          |
| 14    | Bhitarkanika Mangroves                         | Odisha                  | 2002          |
| 15    | Ropar Wetland                                  | Punjab                  | 2002          |
| 16    | Kanjli Wetland                                 | Punjab                  | 2002          |
| 17    | Tsomoriri Lake                                 | Ladakh                  | 2002          |
| 18    | Point Calimere Wildlife and Bird Sanctuary     | Tamil Nadu              | 2002          |
| 19    | East Kolkata Wetlands                          | West Bengal             | 2002          |
| 20    | Chandra Taal                                   | Himachal Pradesh        | 2005          |
| 21    | Renuka Wetland                                 | Himachal Pradesh        | 2005          |
| 22    | Hokersar Wetland                               | Jammu and Kashmir       | 2005          |
| 23    | Surinsar-Mansar Lakes                          | Jammu and Kashmir       | 2005          |
| 24    | Rudrasagar Lake                                | Tripura                 | 2005          |
| 25    | Upper Ganga River (Brijghat to Narora stretch) | Uttar Pradesh           | 2005          |
| 26    | Nalsarovar Bird Sanctuary                      | Gujarat                 | 2012          |
| 27    | Sundarban Wetland                              | West Bengal             | 2019          |
| 28    | Nandur Madhameshwar                            | Maharashtra             | 2019          |
| 29    | Nawabganj Bird Sanctuary                       | Uttar Pradesh           | 2019          |
| 30    | Parvati Arga Bird Sanctuary                    | Uttar Pradesh           | 2019          |
| 31    | Saman Bird Sanctuary                           | Uttar Pradesh           | 2019          |
| 32    | Samaspur Bird Sanctuary                        | Uttar Pradesh           | 2019          |
| 33    | Sandi Bird Sanctuary                           | Uttar Pradesh           | 2019          |
| 34    | Sarsai Nawar Jheel                             | Uttar Pradesh           | 2019          |
| 35    | Beas Conservation Reserve                      | Punjab                  | 2019          |
| 36    | Keshopur-Miani Community Reserve               | Punjab                  | 2019          |
| 37    | Nangal Wildlife Sanctuary                      | Punjab                  | 2019          |
| 38    | Asan Conservation Reserve                      | Uttarakhand             | 2020          |

| S.No. | Name of the Ramsar Site                 | State / Union Territory | Year Declared |
|-------|-----------------------------------------|-------------------------|---------------|
| 39    | Kanwar or Kabar Taal Wetland            | Bihar                   | 2020          |
| 40    | Lonar Lake                              | Maharashtra             | 2020          |
| 41    | Sur Sarovar (Keetham Lake)              | Uttar Pradesh           | 2020          |
| 42    | Tso Kar Wetland Complex                 | Ladakh                  | 2020          |
| 43    | Sultanpur National Park                 | Haryana                 | 2021          |
| 44    | Bhindawas Wildlife Sanctuary            | Haryana                 | 2021          |
| 45    | Thol Lake Bird Sanctuary                | Gujarat                 | 2021          |
| 46    | Wadhvana Wetland                        | Gujarat                 | 2021          |
| 47    | Haiderpur Wetland                       | Uttar Pradesh           | 2021          |
| 48    | Khijadia Wildlife Sanctuary             | Gujarat                 | 2021          |
| 49    | Bakhira Wildlife Sanctuary              | Uttar Pradesh           | 2021          |
| 50    | Karikili Bird Sanctuary                 | Tamil Nadu              | 2022          |
| 51    | Pallikaranai Marsh Reserve Forest       | Tamil Nadu              | 2022          |
| 52    | Pichavaram Mangroves                    | Tamil Nadu              | 2022          |
| 53    | Pala Wetland                            | Mizoram                 | 2022          |
| 54    | Sakhya Sagar                            | Madhya Pradesh          | 2022          |
| 55    | Koonthankulam Bird Sanctuary            | Tamil Nadu              | 2022          |
| 56    | Gulf of Mannar Marine Biosphere Reserve | Tamil Nadu              | 2022          |
| 57    | Vembannur Wetland Complex               | Tamil Nadu              | 2022          |
| 58    | Vellode Bird Sanctuary                  | Tamil Nadu              | 2022          |
| 59    | Vedanthangal Bird Sanctuary             | Tamil Nadu              | 2022          |
| 60    | Udhayamarthandapuram Bird Sanctuary     | Tamil Nadu              | 2022          |
| 61    | Sirpur Wetland                          | Madhya Pradesh          | 2022          |
| 62    | Satkosia Gorge                          | Odisha                  | 2022          |
| 63    | Nanda Lake                              | Goa                     | 2022          |
| 64    | Ranganathittu Bird Sanctuary            | Karnataka               | 2022          |
| 65    | Tampara Lake                            | Odisha                  | 2022          |
| 66    | Hirakud Reservoir                       | Odisha                  | 2022          |
| 67    | Ansupa Lake                             | Odisha                  | 2022          |

| S.No. | Name of the Ramsar Site                | State / Union Territory | Year Declared |
|-------|----------------------------------------|-------------------------|---------------|
| 68    | Yashwant Sagar                         | Madhya Pradesh          | 2022          |
| 69    | Chitrangudi Bird Sanctuary             | Tamil Nadu              | 2022          |
| 70    | Suchindram Theroor Wetland Complex     | Tamil Nadu              | 2022          |
| 71    | Vaduvur Bird Sanctuary                 | Tamil Nadu              | 2022          |
| 72    | Kanjirankulam Bird Sanctuary           | Tamil Nadu              | 2022          |
| 73    | Thane Creek                            | Maharashtra             | 2022          |
| 74    | Hygam Wetland Conservation Reserve     | Jammu and Kashmir       | 2022          |
| 75    | Shallbugh Wetland Conservation Reserve | Jammu & Kashmir         | 2022          |
| 76    | Nagi Bird Sanctuary                    | Bihar                   | 2023          |
| 77    | Nakti Bird Sanctuary                   | Bihar                   | 2023          |
| 78    | Aghanashini Estuary                    | Karnataka               | 2024          |
| 79    | Ankasamudra Bird Conservation Reserve  | Karnataka               | 2024          |
| 80    | Magadi Kere Conservation Reserve       | Karnataka               | 2024          |
| 81    | Karaivetti Bird Sanctuary              | Tamil Nadu              | 2024          |
| 82    | Longwood Shola Reserve Forest          | Tamil Nadu              | 2024          |
| 83    | Tawa Reservoir                         | Madhya Pradesh          | 2024          |
| 84    | Nanjarayan Bird Sanctuary              | Tamil Nadu              | 2024          |
| 85    | Kazhuveli Bird Sanctuary               | Tamil Nadu              | 2024          |
| 86    | Sakkarakottai Bird Sanctuary           | Tamil Nadu              | 2024          |
| 87    | Therthangal Bird Sanctuary             | Tamil Nadu              | 2024          |
| 88    | Khecheopalri Wetland                   | Sikkim                  | 2024          |
| 89    | Udhwa Lake Bird Sanctuary              | Jharkhand               | 2024          |
| 90    | Gogabeel Lake                          | Bihar                   | 2025          |
| 91    | Udaipur Jheel                          | Bihar                   | 2025          |
| 92    | Gokul Jalashay                         | Bihar                   | 2025          |
| 93    | Kopra Reservoir                        | Chhattisgarh            | 2025          |
| 94    | Siliserh Lake                          | Rajasthan               | 2025          |
| 95    | Menar Wetland Complex                  | Rajasthan               | 2025          |
| 96    | Khichan Wetland                        | Rajasthan               | 2025          |

| S.No. | Name of the Ramsar Site                         | State / Union Territory | Year Declared |
|-------|-------------------------------------------------|-------------------------|---------------|
| 97    | Patna Bird Sanctuary                            | Uttar Pradesh           | 2026          |
| 98    | Chhari-Dhand Wetland                            | Gujarat                 | 2026          |
| 99    | Shekha Jheel Bird Sanctuary                     | Uttar Pradesh           | 2026          |
| 100   | Jayaprakash Narayan Bird Sanctuary (Surha Taal) | Uttar Pradesh           | 2026          |
| 101   | Glaw Lake                                       | Arunachal Pradesh       | 2026          |

### > Key Statistics and Facts:

- **State with the most sites:** **Tamil Nadu** is in the lead with the highest number of **20 Ramsar sites**. In second place is **Uttar Pradesh** with **13 sites**.
- **Largest Ramsar site:** the **Sundarban Wetland** of West Bengal (about 4,230 sq km).
- **Smallest Ramsar site:** the **Renuka Wetland** of Himachal Pradesh (about 0.2 sq km / 20 hectares).

### > Status of Leopards in India 2022 Report (Leopard Status Report)

Note: This updated report was released in **February 2024** by the Union Ministry of Environment, Forest and Climate Change. According to it, the total number of leopards in India has increased from 12,852 in 2018 to **13,874 in 2022**.

| Indicator / Parameter                     | Main Findings of the Report and State-wise Data (Exam Trends)                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------|
| State with the most leopards              | <b>Madhya Pradesh (total number: 3,907).</b>                                            |
| Descending order of states                | <b>Madhya Pradesh (3,907) &gt; Maharashtra (1,985) &gt; Karnataka (1,879).</b>          |
| Status of Uttar Pradesh                   | The total number of leopards in Uttar Pradesh is <b>371</b> (important for UP Special). |
| Tiger reserve with the highest population | <b>Nagarjunsagar Srisailem Tiger Reserve</b> (Andhra Pradesh).                          |
| Reserve with the highest density          | <b>Sariska Tiger Reserve</b> (Rajasthan).                                               |

### > Natural World Heritage Sites of India (UNESCO Natural Heritage Sites)

| Name of the Heritage Site | Location (State) | Name of the Heritage Site        | Location (State) |
|---------------------------|------------------|----------------------------------|------------------|
| Kaziranga National Park   | Assam            | Nanda Devi and Valley of Flowers | Uttarakhand      |
| Manas Wildlife Sanctuary  | Assam            | Great Himalayan National Park    | Himachal Pradesh |
|                           |                  |                                  |                  |

| Name of the Heritage Site | Location (State)     | Name of the Heritage Site | Location (State)                                                      |
|---------------------------|----------------------|---------------------------|-----------------------------------------------------------------------|
| Keoladeo National Park    | Bharatpur, Rajasthan | <b>Western Ghats</b>      | Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat (jointly) |
| Sundarbans National Park  | West Bengal          |                           |                                                                       |

## › National Parks of India (State-wise National Parks)

| State / UT             | Names of National Parks (Quick Pointers)                                                                                                                                            |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uttar Pradesh          | • <b>Dudhwa</b> ( <i>Lakhimpur Kheri - extremely important for UP Special</i> )                                                                                                     |
| Uttarakhand            | • <b>Jim Corbett</b> ( <i>India's first national park</i> ) • Rajaji • Nanda Devi • Valley of Flowers • Gangotri • Govind Pashu Vihar                                               |
| Madhya Pradesh         | • Kanha • Bandhavgarh • Madhav • Panna • Pench • Satpura • Sanjay • <b>Kuno</b> ( <i>famous for the rehabilitation of cheetahs</i> ) • Van Vihar • Ghughua Fossil • Dinosaur Fossil |
| Assam                  | • Kaziranga ( <i>for the one-horned rhinoceros</i> ) • Manas • Nameri • Orang (Rajiv Gandhi) • Dibru-Saikhowa • Dehing Patkai • Raimona                                             |
| Bihar / Jharkhand      | • <b>Valmiki</b> (Bihar) • <b>Betla</b> (Jharkhand)                                                                                                                                 |
| Chhattisgarh           | • Guru Ghasidas (Sanjay) • Indravati • Kanger Valley                                                                                                                                |
| Himachal / Ladakh      | • Great Himalayan • Inderkilla • Khirganga • Pin Valley • Simbalbara (HP) • <b>Hemis</b> (Ladakh - <i>largest in area</i> )                                                         |
| Jammu-Kashmir          | • Dachigam ( <i>for the Hangul/Kashmir stag</i> ) • Kishtwar • Kazinag • City Forest (Salim Ali)                                                                                    |
| Rajasthan              | • Ranthambore • Sariska • <b>Keoladeo (Bharatpur)</b> ( <i>for the Siberian crane</i> ) • Desert National Park • Mukundra Hills                                                     |
| Gujarat / Haryana      | • <b>Gir</b> ( <i>for Asiatic lions</i> ) • Velavadar Blackbuck • Marine (Gulf of Kutch) • Vansda (GJ) • Kalesar • Sultanpur (HR)                                                   |
| Karnataka              | • Bandipur • Bannerghatta • Kudremukh • Nagarhole • Anshi (Kali)                                                                                                                    |
| Kerala                 | • Eravikulam • Periyar • <b>Silent Valley</b> • Anamudi Shola • Mathikettan Shola • Pambadum Shola                                                                                  |
| Tamil Nadu / Telangana | • Mudumalai • Mukurthi • Guindy • Indira Gandhi (Anamalai) • Gulf of Mannar Marine (TN) • Kasu Brahmananda Reddy • Mahavir Harina Vanasthali • Mrugavani (TL)                       |
| Odisha / W. Bengal     | • <b>Bhitarkanika</b> • Simlipal (OD) • <b>Sundarban</b> • Buxa • Gorumara • Jaldapara • Neora Valley • Singalila (WB)                                                              |

| State / UT                 | Names of National Parks (Quick Pointers)                                                                                                                                                                                                                           |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Andhra / Arunachal         | • Papikonda • Rajiv Gandhi (Rameswaram) • Sri Venkateswara (AP) • <b>Namdapha</b> • Mouling (AR)                                                                                                                                                                   |
| Other North-Eastern states | • <b>Keibul Lamjao</b> ( <i>the world's only floating park</i> ), Shirui (Manipur) • Nokrek, Balphakram (Meghalaya) • Murlen, Phawngpui/Blue Mountain (Mizoram) • Intanki/Ntangki (Nagaland) • Clouded Leopard, Bison/Rajbari (Tripura) • Khangchendzonga (Sikkim) |
| Goa / Andaman              | • Bhagwan Mahavir/Mollem (Goa) • Campbell Bay • Galathea Bay • Mahatma Gandhi Marine • Mount Manipur (Mount Harriet) • North Button • Saddle Peak (Andaman & Nicobar)                                                                                              |

### › Tiger Reserves of India (State-wise Tiger Reserves)

| State / UT         | Names of Tiger Reserves (UPPCS Matching Special)                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| Uttar Pradesh      | • <b>Dudhwa</b> • <b>Pilibhit</b> • <b>Ranipur</b> ( <i>the first in the Bundelkhand region</i> )             |
| Uttarakhand        | • Corbett • Rajaji                                                                                            |
| Madhya Pradesh     | • Kanha • Bandhavgarh • Panna • Pench • Satpura • Sanjay • <b>Veerangana Durgavati</b> • Ratapani • Madhav    |
| Maharashtra        | • Melghat • Tadoba-Andhari • Pench • Navegaon-Nagzira • Bor ( <i>the smallest</i> ) • Sahyadri                |
| Rajasthan          | • Ranthambore • Sariska • Mukundra Hills • Ramgarh Vishdhari • Dholpur-Karauli                                |
| Karnataka          | • Bandipur • Bhadra • Kali (Anshi) • Nagarhole • Biligiriranganatha (BRT)                                     |
| Tamil Nadu         | • Kalakkad-Mundanthurai • Mudumalai • Anamalai • <b>Sathyamangalam</b> • Srivilliputhur-Megamalai             |
| Chhattisgarh       | • Indravati • Achanakmar • Udanti-Sitanadi • Guru Ghasidas-Tamor Pingla                                       |
| Assam              | • Kaziranga • Manas • Nameri • Orang                                                                          |
| Arunachal Pradesh  | • Namdapha • Pakke (Pakhui) • Kamlang                                                                         |
| Andhra / Telangana | • <b>Nagarjunsagar-Srisailem</b> ( <i>India's largest tiger reserve by area</i> ) (AP) • Kawal • Amrabad (TL) |
| Kerala / Mizoram   | • Periyar • Parambikulam (KL) • <b>Dampa</b> (MZ)                                                             |
| Odisha / W. Bengal | • Simlipal • Satkosia (OD) • Sundarban • Buxa (WB)                                                            |
| Bihar / Jharkhand  | • Valmiki (BR) • Palamau (JH)                                                                                 |

## > Ecology

### > Types of Ecological Species

| Type of Species               | Ecological Definition and Nature                                                                                                                        | Precise Exam-useful Examples (Exam Hits)                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Keystone species</b>       | A species which, even if small in number or biomass in its ecosystem, is of very great importance in controlling the structure of the entire ecosystem. | <b>Tiger</b> , lion, elephant (top carnivores or large animals).                                            |
| <b>Indicator species</b>      | A species which is highly sensitive to any subtle chemical or physical change in the environment and gives immediate information.                       | <b>Lichen - they automatically die out as soon as SO<sub>2</sub> pollution increases</b> in the atmosphere. |
| <b>Endemic species</b>        | A species which is naturally found only in one specific geographical region and is found nowhere else in the world.                                     | <b>The lion-tailed macaque of the Western Ghats.</b>                                                        |
| <b>Exotic / Alien species</b> | Species which are not native to that local ecosystem but are brought from outside and established, unknowingly or deliberately.                         | <b>Tea</b> (brought from China), <b>potato</b> (brought by the Portuguese), water hyacinth.                 |
| <b>Umbrella species</b>       | Species with an extensive range, efforts to conserve which automatically conserve the other smaller species of the entire ecosystem coming under them.  | <b>One-horned rhinoceros</b> , giant panda, Bengal tiger.                                                   |
| <b>Flagship species</b>       | An attractive or distinctive species used as a 'symbol' or identity for the conservation campaign of a particular ecosystem.                            | <b>Sangai deer</b> (for Keibul Lamjao of Manipur), <b>Bengal tiger</b> (for the Sundarbans).                |

## 8 Biological Interactions, Categories of Biodiversity and Conservation Systems

### > Main Types of Biological Interactions

| Type of Interaction          | Effect on Both Species                                                 | Practical Ecological Example (Core Example)                                                           |
|------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Mutualism / Symbiosis</b> | Both species benefit mutually ( + , + ).                               | Honeybee and flower (the honeybee gets nectar and the plants are pollinated), lichen (fungus + alga). |
| <b>Parasitism</b>            | One organism benefits and the other host organism is harmed ( + , - ). | Amarbel (dodder, on plants), parasitic insects or worms in the human stomach.                         |

| Type of Interaction | Effect on Both Species                                                            | Practical Ecological Example (Core Example)                                                                                          |
|---------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Commensalism        | One organism benefits and the other organism is neither harmed nor helped (+, 0). | <b>Epiphytic plants such as orchids</b> , which grow on the branches of large trees only for shelter, without causing them any harm. |

## > Biodiversity

### > Three Main Categories of Biodiversity

- 1. Species diversity:** focused on the total number of different species present within a particular area or ecosystem and their abundance.
- 2. Genetic diversity:** the internal diversity of genes and genetic information present within organisms of the same species (such as the thousands of varieties of paddy).
- 3. Ecosystem diversity:** the diversity of the different types of habitats, biological communities and ecological processes present within a large area.

### > Types and Measurement of Biodiversity

| Type of Biodiversity                         | Basic Scientific Definition and Concept (Core Logic)                                                                                       | Precise Exam-useful Example (Exam Example)                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Alpha (<math>\alpha</math>) diversity</b> | The total number (richness) of species found within <b>one particular place, local community or a single ecosystem</b> .                   | The total bird species found in one particular forest area or bird sanctuary.                 |
| <b>Beta (<math>\beta</math>) diversity</b>   | The amount of difference/change in species diversity found <b>between two different places, habitats or communities</b> .                  | Comparison of the bird communities of two different forests, or of a forest and a grassland.  |
| <b>Gamma (<math>\gamma</math>) diversity</b> | The overall/total diversity of all the ecosystems present in <b>one vast geographical region (such as a state, country or continent)</b> . | The total plant species found in the whole of North India or the entire Western Ghats region. |

### > Two Main Systems of Biodiversity Conservation

| Type of Conservation        | Basic Definition and Scientific Philosophy                                                        | Included Examples (Exam Hotspots)                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>In-situ conservation</b> | The system of conserving organisms, flora or species <b>within their natural habitat itself</b> . | <b>National Parks, Biosphere Reserves, Wildlife Sanctuaries, Sacred Groves.</b> |
|                             |                                                                                                   |                                                                                 |



| Type of Conservation | Basic Definition and Scientific Philosophy                                                                                                                     | Included Examples (Exam Hotspots)                                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Ex-situ conservation | The system of conserving genetic diversity by taking endangered organisms or plants away from their natural habitat to some artificial or man-made safe place. | <b>Seed banks, gene banks, botanical gardens, zoological gardens (zoos), cryopreservation labs.</b> |

### › The 3 Zones of a Biosphere Reserve

- 1. Core zone:** This is the innermost part, which includes legally strictly protected areas. Here **any kind of human activity or entry is completely prohibited.**
- 2. Buffer zone:** This is the area around the core zone. Only controlled scientific research, environmental education, eco-tourism and training activities are permitted here.
- 3. Transition zone:** This is the outermost part. In this area, with the cooperation of local communities, sustainable human settlements, socio-cultural activities, crop agriculture and other sustainable economic activities are fully permitted.

### › India's Four Major Biodiversity Hotspots

- 1. The Himalayas / Eastern Himalayas**
- 2. Indo-Burma region**
- 3. Sundaland** (includes the Nicobar Islands)
- 4. Western Ghats and Sri Lanka**

### › Projects for the Conservation of Fauna in India

| Name of the Project            | Year / Date of Launch | Updated and Specific Exam-useful Facts (Exam Notes)                                                                                                              |
|--------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Musk Deer Project              | 1970                  | Started from Kedarnath Sanctuary with the cooperation of the <b>IUCN</b> .                                                                                       |
| Hangul Project                 | 1970                  | Started in Dachigam National Park for the conservation of the Kashmir stag (Hangul).                                                                             |
| Project Tiger                  | <b>1 April 1973</b>   | <p>* <b>Update (2026):</b> At present India has a total of <b>58 tiger reserves</b>.</p> <p>* <b>Most tiger reserves: 9 in Madhya Pradesh</b> (Tiger State).</p> |
| Crocodile Conservation Project | 1974                  | Started from Odisha with the cooperation of the United Nations Development Programme (UNDP).                                                                     |
| Gharial Project                | 1975                  | For the breeding and conservation of gharials in the Chambal and wildlife sanctuaries.                                                                           |
| Project Elephant               | <b>February 1992</b>  | <p>* <b>Update (2026):</b> At present India has a total of <b>33 elephant reserves</b>.</p>                                                                      |

| Name of the Project         | Year / Date of Launch | Updated and Specific Exam-useful Facts (Exam Notes)                                       |
|-----------------------------|-----------------------|-------------------------------------------------------------------------------------------|
| Project Rhino               | 1987                  | One-horned rhino action plan for the conservation of the one-horned rhinoceros.           |
| Red Panda Project           | 1996                  | In the Padmaja Naidu Himalayan Zoological Park with the cooperation of <b>WWF-India</b> . |
| Sea Turtle Project          | 1999                  | By the <b>Wildlife Institute of India (WII)</b> , Dehradun, for turtles (Olive Ridley).   |
| Project Vulture             | 2006                  | Action plan to stop the deaths of vultures caused by the drug <b>Diclofenac</b> .         |
| Project Snow Leopard        | 2009                  | For conservation in the high Himalayan regions (such as Hemis National Park).             |
| Haathi Mere Saathi Campaign | 2011                  | By the Ministry of Environment and the Wildlife Trust; mascot: ' <b>Gaju</b> '.           |

## > Sustainable Development

### > 17 Sustainable Development Goals: 2030 (17 Sustainable Development Goals - SDGs)

Note: Adopted by the United Nations in 2015, with the aim of ending poverty in all its forms from the entire world and ensuring sustainable development by 2030.

| SDG No. | Sustainable Development Goal (SDG Target) | SDG No. | Sustainable Development Goal (SDG Target) |
|---------|-------------------------------------------|---------|-------------------------------------------|
| Goal 1  | No Poverty                                | Goal 10 | Reduced Inequalities                      |
| Goal 2  | Zero Hunger                               | Goal 11 | Sustainable Cities and Communities        |
| Goal 3  | Good Health and Well-being                | Goal 12 | Responsible Consumption and Production    |
| Goal 4  | Quality Education                         | Goal 13 | Climate Action                            |
| Goal 5  | Gender Equality                           | Goal 14 | Life Below Water                          |
| Goal 6  | Clean Water and Sanitation                | Goal 15 | Life on Land                              |
| Goal 7  | Affordable and Clean Energy               | Goal 16 | Peace, Justice and Strong Institutions    |
| Goal 8  | Decent Work and Economic Growth           | Goal 17 | Partnerships for the Goals                |
| Goal 9  | Industry, Innovation and Infrastructure   |         |                                           |

## > UNESCO World Heritage Sites

## > Environmental Pollution and Pollutants

### > Primary Pollutants versus Secondary Pollutants

| Category of Pollutant       | Chemical Nature of Formation                                                                                                                                    | Precise Exam-useful Examples (Exam Hotspots)                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary pollutants</b>   | Those which are emitted directly from the source (such as vehicle smoke or factories) and remain in the atmosphere in the same form.                            | Carbon monoxide ( <b>CO</b> ), carbon dioxide ( <b>CO<sub>2</sub></b> ), sulphur dioxide ( <b>SO<sub>2</sub></b> ), oxides of nitrogen ( <b>NO</b> , <b>NO<sub>2</sub></b> ), methane ( <b>CH<sub>4</sub></b> ), lead ( <b>Pb</b> ), chlorofluorocarbons ( <b>CFCs</b> ), and particulate matter ( <b>PM<sub>10</sub></b> , <b>PM<sub>2.5</sub></b> ). |
| <b>Secondary pollutants</b> | Those which are not emitted directly but are formed in the atmosphere by chemical reactions among primary pollutants or by the effect of sunlight and moisture. | Ozone ( <b>O<sub>3</sub></b> - tropospheric/ground-level), peroxyacetyl nitrate ( <b>PAN</b> ), <b>photochemical smog</b> , sulphuric acid ( <b>H<sub>2</sub>SO<sub>4</sub></b> ), nitric acid ( <b>HNO<sub>3</sub></b> ), and organic aerosols.                                                                                                       |

- **Biodegradable pollutants:** pollutants that are completely decomposed over time by biological agents (such as bacteria) (e.g., domestic sewage, paper, agricultural waste).

### > Specific Pollutants and the Organs of the Human Body Affected by them

| Pollutant Element / Particle   | Human Organ / System Affected         | Pathological Effect (Impact)                                                |
|--------------------------------|---------------------------------------|-----------------------------------------------------------------------------|
| <b>Asbestos dust</b>           | <b>Lungs</b>                          | Asbestosis (serious difficulty in breathing).                               |
| <b>Lead (Pb)</b>               | <b>Central nervous system (brain)</b> | Damage to the nervous system, obstruction of children's mental development. |
| <b>Carbon monoxide (CO)</b>    | <b>Blood streams</b>                  | Reacts with haemoglobin and stops the oxygen-carrying capacity.             |
| <b>Mercury (Hg)</b>            | <b>Stomach / nervous system</b>       | Minamata disease, serious diseases of the stomach and kidneys.              |
| <b>Fluoride (fluoride ion)</b> | <b>Teeth and bones</b>                | Fluorosis (yellowing of teeth and bending of bones).                        |

### > Ozone-depleting Gases and their Industrial Sources

| Ozone Depleting Substance (ODS)               | Main Industrial / Practical Source                                          | Ozone Depletion Potential (ODP) Effect                              |
|-----------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Chlorofluorocarbons (CFCs)</b>             | Domestic refrigerators (fridges), air conditioners (AC) and aerosol sprays. | Release chlorine in the stratosphere and destroy the ozone layer.   |
| <b>Halons</b>                                 | Fire extinguishers.                                                         | They contain bromine, which is even more destructive than chlorine. |
| <b>Carbon tetrachloride (CCl<sub>4</sub>)</b> | Industrial cleaning solvents, dry-cleaning chemicals.                       | Highly toxic and causes heavy damage to ozone.                      |
| <b>Hydrochlorofluorocarbons (HCFCs)</b>       | Transitional coolants (as temporary substitutes for CFCs).                  | Less harmful than CFCs, but in the category of complete ban.        |
| <b>Methyl bromide</b>                         | Agricultural pesticide, soil fumigation process.                            | Used to kill soil-borne insects; acutely destructive for ozone.     |
| <b>Methyl chloroform</b>                      | Industrial degreasing (cleaning) solvents and coatings.                     | Capable of evaporating and reaching the stratosphere.               |

### > Principal Gases Emitted by Volcanic Eruptions

| Gases Emitted from Volcanoes              | Special Exam-useful Fact (Exam Hit)                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Water vapour</b>                       | <b>The largest amount (60% to 90%) in a volcanic eruption is of water vapour.</b> |
| <b>Carbon dioxide (CO<sub>2</sub>)</b>    | The second main gas, which increases the greenhouse effect.                       |
| <b>Sulphur dioxide (SO<sub>2</sub>)</b>   | Because of it, conditions of acid rain develop in the atmosphere.                 |
| <b>Hydrogen sulphide (H<sub>2</sub>S)</b> | A gas with a strong smell like rotten eggs.                                       |
| <b>Other gases</b>                        | Hydrogen (H), carbon monoxide (CO), hydrogen chloride (HCl), and helium (He).     |

## > The Sequential Process of Soil Erosion

| Type of Erosion                  | Process of Formation and Simple Definition                                                                     | UPPCS Exam Key-point                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>1. Splash erosion</b>         | Scattering and detachment of soil particles due to heavy raindrops falling directly on the soil.               | The initial stage of erosion.                                                         |
| <b>2. Sheet erosion</b>          | When, due to excessive rainfall, the fertile top layer of soil begins to flow with water as a uniform sheet.   | This erosion is not immediately visible to the farmer, but is the most harmful.       |
| <b>3. Rill erosion</b>           | When flowing water, due to the slope, makes <b>small and shallow channels</b> in the soil.                     | Fragmentation of agricultural land begins.                                            |
| <b>4. Gully erosion</b>          | When the small channels of rill erosion become very large, deep and wide due to the rapid flow of water.       | Because of it the <b>"ravines" (badlands) of the Chambal valley</b> have been formed. |
| <b>5. Stream channel erosion</b> | When water, instead of flowing over the land, flows as a very thick, deep and large stream and cuts its banks. | The extreme form of erosion.                                                          |

## > India's Principal Environmental Protection Acts

| Name of the Act / Policy                                      | Year of Passing | Regulatory Area / Special Fact (Exam Context)                                                                |
|---------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------|
| <b>River Boards Act</b>                                       | 1956            | For the regulation and development of inter-state rivers and river valleys.                                  |
| <b>Wild Life (Protection) Act</b>                             | 1972            | Complete ban on hunting and illegal trade of endangered wildlife (stringent amendment in 1991).              |
| <b>Water (Prevention and Control of Pollution) Act</b>        | 1974            | For maintaining water quality and establishing pollution control boards (amended in 1988).                   |
| <b>Water (Prevention and Control of Pollution) Cess Rules</b> | 1977            | For levying a cess on water consumed by local bodies and industries.                                         |
| <b>Forest (Conservation) Act</b>                              | 1980            | Strict control on stopping the destruction of forests and on diverting forest land to non-forestry purposes. |
| <b>Air (Prevention and Control of Pollution) Act</b>          | 1981            | For the prevention of air pollution; <b>'noise pollution'</b> was also added to it by the 1987 amendment.    |
|                                                               |                 |                                                                                                              |

| Name of the Act / Policy                      | Year of Passing | Regulatory Area / Special Fact (Exam Context)                                                                                                            |
|-----------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environment (Protection) Act                  | 1986            | <b>(It is also called Umbrella Legislation)</b> ; passed after the Bhopal gas tragedy (1984) for the comprehensive protection of the entire environment. |
| National Environment Tribunal Act             | 1995            | For giving speedy compensation to the victims of accidents occurring while handling hazardous substances.                                                |
| National Environment Appellate Authority Act  | 1997            | For hearing appeals regarding areas where the setting up of industries is restricted under the Environment Protection Act.                               |
| Ozone Depleting Substances (Regulation) Rules | 2000            |                                                                                                                                                          |
| Biological Diversity Act                      | 2002            | For the conservation of the country's rich biological resources and the sustainable sharing of their benefits.                                           |
| National Environment Policy                   | 2004            | A guiding policy for balancing sustainable development and environmental protection in the country.                                                      |
| Forest Rights Act (FRA)                       | 2006            | Legal recognition of the land rights of Scheduled Tribes (ST) and other traditional forest dwellers living in forests.                                   |

### › India's Upcoming Environmental Target Years

| Target Objective                                                              | Target Year Fixed by the Government of India                  |
|-------------------------------------------------------------------------------|---------------------------------------------------------------|
| Goal of making India a fully 'Developed India' (Viksit Bharat)                | <b>2047</b> (on the completion of 100 years of independence). |
| Goal of making India energy independent / self-reliant                        | <b>2047</b>                                                   |
| Goal of reducing the use of hydrofluorocarbons (HFC) by 80%                   | <b>2047</b> (in line with the Kigali Amendment).              |
| Goal of increasing India's nuclear power capacity to 1 lakh megawatt (100 GW) | <b>2047</b>                                                   |
| Goal of achieving complete self-reliance in shipbuilding                      | <b>2047</b>                                                   |
| Goal of Mumbai city achieving 'zero carbon emission' (Net Zero)               | <b>2050</b>                                                   |
| Goal of 'Net Zero' (Net Zero Carbon Emission) by the entire Indian nation     | <b>2070</b> (announced at Glasgow COP26).                     |

## > India's Upcoming Targets for the Year 2030 (India's Targets for 2030)

| Target Objective              | Specific Numerical Value / Description (Exam Target)                                                |
|-------------------------------|-----------------------------------------------------------------------------------------------------|
| Coal gasification             | Target of achieving <b>100 million tonnes</b> of coal gasification.                                 |
| Renewable energy capacity     | Target of achieving <b>500 gigawatt (GW)</b> of non-fossil/renewable energy capacity.               |
| Solar power capacity          | Target of obtaining <b>280 gigawatt (GW)</b> of the total renewable energy from solar energy alone. |
| Reduction in carbon emissions | Reducing the projected total carbon emissions by <b>1 billion tonnes</b> .                          |
| Electric vehicles (EVs)       | Target of making <b>30% of total vehicles electric</b> .                                            |
| Malaria eradication           | <b>Complete eradication of malaria</b> from India.                                                  |
| HIV AIDS                      | Target of completely ending HIV AIDS infection from the country.                                    |
| Green Railway Mission         | Target of making <b>Indian Railways 'Net Zero Carbon Emission'</b> or a ' <b>Green Railway</b> '.   |

## > Environmental Terminology

## > Important Terms, Concepts and the Experts who Propounded them

| Specific Term / Environmental Concept               | Name of the Propounder / Related Expert            |
|-----------------------------------------------------|----------------------------------------------------|
| Green Development                                   | W. M. Adams                                        |
| Inventor / father of natural farming                | Masanobu Fukuoka (Japanese farmer and philosopher) |
| Ecosystem                                           | A. G. Tansley                                      |
| Ecological niche                                    | Joseph Grinnell                                    |
| Greenhouse gas (discovery of the greenhouse effect) | Joseph Fourier                                     |
| Ecology                                             | Ernst Haeckel                                      |
| Biological Diversity                                | Raymond F. Dasmann                                 |
| Biodiversity (popularising the term)                | Walter G. Rosen                                    |
| Biodiversity hotspot                                | Norman Myers                                       |
|                                                     |                                                    |

| Specific Term / Environmental Concept                 | Name of the Propounder / Related Expert |
|-------------------------------------------------------|-----------------------------------------|
| Biosphere reserve (first use of the term 'biosphere') | Eduard Suess                            |
| Ecotone                                               | F. E. Clements                          |

## > Important Environmental Terminology and Definitions

| Term                           | Basic Scientific Definition and Concept (Core Logic)                                                                           | Special Exam-useful Fact                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sustainable Development</b> | Development that meets the needs of the present generation without compromising the needs of future generations.               | This concept was given by the <b>Brundtland Commission (1987)</b> in its report ' <b>Our Common Future</b> '.                               |
| <b>Ecomark</b>                 | A mark certifying environment-friendly and sustainable household products.                                                     | Started by the <b>Bureau of Indian Standards (BIS)</b> in <b>1991</b> (symbol: earthen pot).                                                |
| <b>AGMARK</b>                  | A mark certifying the quality and purity of agricultural and food products.                                                    | Issued by the Directorate of Marketing and Inspection (DMI).                                                                                |
| <b>Greenwash</b>               | A <b>false or misleading claim</b> made by a company or organisation to present its products as environment-friendly or green. | Mere marketing/show in the name of environmental protection.                                                                                |
| <b>Habitat</b>                 | The specific physical place or environment where a particular organism or species naturally lives.                             | It indicates the 'address' of the organism.                                                                                                 |
| <b>Lentic habitat</b>          | Ecological habitats of <b>still water</b> .                                                                                    | Examples: <b>ponds, lakes, wetlands</b> , pools, etc.                                                                                       |
| <b>Lotic habitat</b>           | Ecological habitats of <b>moving or running water</b> .                                                                        | Examples: <b>rivers, springs, streams</b> , etc.                                                                                            |
| <b>Niche</b>                   | The specific <b>functional role</b> played by a species in its ecosystem and its position.                                     | This concept was first given by " <b>Joseph Grinnell</b> ".                                                                                 |
| <b>Ecosystem</b>               | The community of organisms in a definite area and the non-living environment around them, which interact with one another.     | * This term was first used by <b>A. G. Tansley (1935)</b> .<br>* The limit of environmental support is called " <b>carrying capacity</b> ". |



| Term                    | Basic Scientific Definition and Concept (Core Logic)                                                                                                                               | Special Exam-useful Fact                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Biomagnification</b> | The entry of non-biodegradable harmful chemicals / heavy metals into organisms through the food chain and <b>the increase of their concentration at each higher trophic level.</b> | E.g., accumulation of DDT or heavy metals in top consumers. |

### > Movements Related to Environmental Protection and their Leaders

| Name of the Movement                            | Main Leader / Pioneer                   | Special Exam-useful Geographical Fact (Exam Hits)                                                 |
|-------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Chipko Movement (1973)</b>                   | Sunderlal Bahuguna, Chandi Prasad Bhatt | To stop the felling of trees in Chamoli in Uttarakhand (then UP).                                 |
| <b>Appiko Movement (1983)</b>                   | Pandurang Hegde                         | A movement run on the lines of Chipko in <b>Karnataka (South India).</b>                          |
| <b>Narmada Bachao Andolan</b>                   | Medha Patkar                            | A broad movement against the Sardar Sarovar Dam being built on the Narmada river.                 |
| <b>Opposition to the Narmada Valley Project</b> | Baba Amte                               | For the human rights of dam-displaced people and environmental protection.                        |
| <b>Silent Valley Movement</b>                   | M. K. Prasad                            | To save the Silent Valley in <b>Kerala</b> from a hydroelectric project on the Kunthipuzha river. |
| <b>Bishnoi Movement</b>                         | Amrita Devi Bishnoi                     | Historic supreme sacrifice for the protection of Khejri trees in <b>Rajasthan</b> (1730).         |
| <b>Anti-Tehri Dam Movement</b>                  | Sunderlal Bahuguna                      | Protective opposition to the huge dam built on the Bhagirathi river in Uttarakhand.               |
| <b>Dongri Adivasi Movement</b>                  | Chandi Prasad Bhatt                     | For the protection of the forest rights of local tribal communities.                              |
| <b>Nilgiri Movement</b>                         | N. Gopalakrishnan                       | For the conservation of the Nilgiri ecosystem of the Western Ghats.                               |
| <b>Chilika Lake Movement</b>                    | S. K. Patnaik                           | Against prawn farming and commercial exploitation in the Chilika Lake of <b>Odisha.</b>           |
| <b>Mithun Movement</b>                          | Nagarika Bahini and tribal groups       | For local wildlife and tribal forest rights in the Jharkhand region.                              |

## > Principal Personalities Associated with the Environment

| Name of the Personality    | Famous Epithet / Related Field (Exam Pointer)                                          |
|----------------------------|----------------------------------------------------------------------------------------|
| Ravindra Kumar Sinha       | <b>Dolphin Man of India.</b>                                                           |
| Kailash Sankhala           | <b>Tiger Man of India.</b>                                                             |
| Dr. Ramdeo Misra           | <b>Father of Ecology in India.</b>                                                     |
| E. P. Odum                 | <b>Global father of ecology and environmental science.</b>                             |
| Richard St. Barbe Baker    | He is called the " <b>Man of the Trees</b> ".                                          |
| Dr. Norman Borlaug         | <b>Originator of the Green Revolution in the world</b> (Nobel Prize winner).           |
| Dr. M. S. Swaminathan      | <b>Father of the Green Revolution in India.</b>                                        |
| Dr. Verghese Kurien        | <b>Father of the White Revolution / Milk Revolution in India</b> (Operation Flood).    |
| Dr. Salim Ali              | Famous Indian bird expert (ornithologist).                                             |
| Sunderlal Bahuguna         | Main leader associated with the famous <b>Chipko Movement (1973)</b> of Uttarakhand.   |
| Pandurang Hegde            | Leader of the <b>Appiko Movement (1983)</b> , run on the lines of Chipko in Karnataka. |
| Baba Amte and Medha Patkar | Main leaders of the <b>Narmada Bachao Andolan.</b>                                     |

## > Important Environment Days

| Date       | Name of the Day                                   |
|------------|---------------------------------------------------|
| 2 February | <b>World Wetlands Day</b>                         |
| 3 March    | <b>World Wildlife Day</b>                         |
| 20 March   | <b>World Sparrow Day</b>                          |
| 21 March   | <b>World Forestry Day</b>                         |
| 22 March   | <b>World Water Day</b>                            |
| 23 March   | <b>World Meteorological Day</b>                   |
| 22 April   | <b>World Earth Day</b>                            |
| 22 May     | <b>International Day for Biological Diversity</b> |
|            |                                                   |

| Date           | Name of the Day                           |
|----------------|-------------------------------------------|
| 31 May         | World No Tobacco Day                      |
| 5 June         | World Environment Day                     |
| 8 June         | World Oceans Day                          |
| 29 July        | International Tiger Day                   |
| 10 August      | World Lion Day                            |
| 12 August      | World Elephant Day                        |
| 29 August      | National Sports Day                       |
| 16 September   | World Ozone Day (International Ozone Day) |
| 1 to 7 October | National Wildlife Week                    |
| 3 October      | World Habitat Day                         |
| 4 October      | World Animal Welfare Day                  |
| 13 October     | International Day for Disaster Reduction  |
| 19 November    | World Toilet Day                          |
| 26 November    | National Environment Protection Day       |
| 5 December     | World Soil Day                            |
| 11 December    | International Mountain Day                |
| 14 December    | National Energy Conservation Day          |

### › Short Trick for UPPCS Prelims (Chronological Revision):

- **Sequence of March:** 20 (Sparrow) → 21 (Forest) → 22 (Water) → 23 (Meteorology).
- **The magic figure of 22:** 22 March (Water) → 22 April (Earth) → 22 May (Biodiversity).

### › Energy Resources

#### › Classification of Energy Resources

#### › Non-renewable versus Renewable Energy Resources

| Non-Renewable Resources<br>(limited and polluting) | Renewable Energy Resources<br>(unlimited and environment-friendly) |
|----------------------------------------------------|--------------------------------------------------------------------|
| * Coal                                             | * Wind power                                                       |

| <b>Non-Renewable Resources</b><br>(limited and polluting) | <b>Renewable Energy Resources</b><br>(unlimited and environment-friendly) |
|-----------------------------------------------------------|---------------------------------------------------------------------------|
| * Petroleum / mineral oil                                 | * Solar power                                                             |
| * Natural gas                                             | * Hydro-electric power                                                    |
| * Atomic energy / nuclear energy                          | * Biogas                                                                  |
| * Shale gas                                               | * Geothermal energy                                                       |
|                                                           | * Tidal power                                                             |
|                                                           | * Ocean thermal energy (OTEC)                                             |
|                                                           | * Energy produced from waste (Waste-to-Energy)                            |

### > Conventional versus Non-conventional Energy Sources

| <b>Conventional Sources of Energy</b><br>(historically used for a long time) | <b>Non-Conventional Sources of Energy</b><br>(modern / newly discovered technologies) |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| * Coal, lignite                                                              | * Solar energy                                                                        |
| * Petroleum, natural gas                                                     | * Wind energy                                                                         |
| * Wood / forest waste                                                        | * Biogas                                                                              |
| * Animal waste (cow-dung cakes, etc.)                                        | * Geothermal energy                                                                   |
| * Mineral oil                                                                | * Tidal energy / ocean wave energy                                                    |
|                                                                              | * Hydrogen energy / green hydrogen                                                    |
|                                                                              | * Fuel cells                                                                          |
|                                                                              | * Atomic energy (nuclear energy)                                                      |

- **Note (Exam Point):** As a source, **atomic energy** is 'non-renewable' because it is based on a limited mineral (uranium), but being a new technology it is counted among the 'non-conventional' sources of energy.

## > International Environmental Conferences and Treaties

| Conference / Treaty (Place)                                             | Year | Important Facts and Exam-useful Points                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stockholm Conference (Sweden)                                           | 1972 | <ul style="list-style-type: none"> <li>It is called the <b>Magna Carta of the Environment</b>.</li> <li>The United Nations Environment Programme (UNEP) was established.</li> <li>It was declared that World Environment Day would be celebrated on <b>5 June</b>.</li> </ul> |
| Montreal Protocol (Canada)                                              | 1987 | <ul style="list-style-type: none"> <li>The <b>first international agreement</b> for the protection of the ozone layer.</li> <li>It was decided to celebrate Ozone Day on <b>16 September</b>.</li> </ul>                                                                      |
| Rio Conference / Earth Summit (Brazil)                                  | 1992 | <ul style="list-style-type: none"> <li><b>Agenda-21</b> for sustainable development was adopted.</li> <li>The <b>UNFCCC</b> on climate change was established.</li> </ul>                                                                                                     |
| Kyoto Protocol / Earth + 5 (Japan)                                      | 1997 | <ul style="list-style-type: none"> <li>Reducing global warming by cutting the emission of greenhouse gases.</li> <li>Decision of a <b>5.2% cut</b> from the 1990 level.</li> </ul>                                                                                            |
| Johannesburg Conference / Rio + 10 / Second Earth Summit (South Africa) | 2002 | <ul style="list-style-type: none"> <li>Agreement on establishing a <b>"World Solidarity Fund"</b> for poverty eradication.</li> <li>Special emphasis on sustainable development.</li> </ul>                                                                                   |
| Cancun Conference (Mexico)                                              | 2010 | <ul style="list-style-type: none"> <li>Decision to establish a <b>"Green Climate Fund" (GCF)</b> of 100 billion dollars.</li> </ul>                                                                                                                                           |
| Rio + 20 Conference / Earth + 20 (Brazil)                               | 2012 | <ul style="list-style-type: none"> <li>The draft <b>"The Future We Want"</b> was presented.</li> <li>The <b>Zero Hunger Challenge</b> (necessary food security for all) was launched.</li> </ul>                                                                              |
| Climate Adaptation Summit (Netherlands)                                 | 2021 | <ul style="list-style-type: none"> <li>A new umbrella fund named the International Fund for Agricultural Development (<b>IFAD</b>) was established.</li> </ul>                                                                                                                |

## > Historical Chronology of the Climate Change Conferences (UNFCCC)

| Historic Conference / Stage     | Year / Date    | Main Exam-useful Provisions and Milestones (Exam Facts)                                                                                                             |
|---------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Earth Summit                    | 1992           | Held in Rio de Janeiro, Brazil; it is called the United Nations Conference on Environment and Development. <b>The UNFCCC, the CBD and Agenda-21</b> were born here. |
| Coming into force of the UNFCCC | 21 March, 1994 | It was ratified by 197 countries of the world. The COP meetings began after this.                                                                                   |

| Historic Conference / Stage | Year / Date | Main Exam-useful Provisions and Milestones (Exam Facts)                                                                     |
|-----------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| COP 1                       | 1995        | Held in Berlin, Germany; the <b>first COP conference</b> towards tackling climate change.                                   |
| COP 3 (Kyoto Protocol)      | 1997        | Kyoto, Japan; legal targets fixed for developed countries to reduce greenhouse gas emissions.                               |
| COP 8 (Delhi Declaration)   | 2002        | <b>New Delhi, India</b> ; emphasis on the technological needs of developing countries and sustainable development.          |
| COP 13                      | 2007        | Bali, Indonesia; formulation of the post-Kyoto Protocol roadmap (the 'Bali Action Plan').                                   |
| COP 16                      | 2010        | Cancun, Mexico; decision to establish the ' <b>Green Climate Fund</b> ' (GCF).                                              |
| COP 17                      | 2011        | Durban, South Africa; blueprint of a new legal agreement for all countries (to come into force from 2020).                  |
| COP 21 (Paris Agreement)    | 2015        | Paris, France; the historic resolve to keep global warming <b>below 2°C (preferably 1.5°C)</b> above pre-industrial levels. |
| COP 22                      | 2016        | Marrakesh, Morocco; discussion on the implementation of the rules and priorities of the Paris Agreement.                    |
| COP 23                      | 2017        | Bonn, Germany (under the presidency of Fiji); the process of finalising the rulebook of the Paris Agreement.                |
| COP 24                      | 2018        | Katowice, Poland; the detailed 'rulebook' for implementing the Paris Agreement was adopted.                                 |
| COP 25                      | 2019        | Madrid, Spain (under the presidency of Chile); ended without any final decision on the rules of the global carbon market.   |

- **History of the Sustainable Development Goals:** These goals (SDGs) were first proposed at the **Rio+20 Conference in 2012**, and have been set globally to be achieved by 2030.

### > Principal Environmental Conventions and Protocols

| Name of the Convention / Protocol | Year of Adoption | Year of Entry into Force | Basic Subject / Regulatory Area (Core Subject)                                           |
|-----------------------------------|------------------|--------------------------|------------------------------------------------------------------------------------------|
| Ramsar Convention                 | 1971             | 1975                     | Related to the conservation of <b>wetlands</b> of international importance.              |
| Bonn Convention (CMS)             | 1979             | 1983                     | Related to the conservation of <b>migratory species</b> of wild animals.                 |
| Vienna Convention                 | 1985             | 1988                     | An inter-governmental framework agreement for the <b>protection of the ozone layer</b> . |

| Name of the Convention / Protocol        | Year of Adoption | Year of Entry into Force | Basic Subject / Regulatory Area (Core Subject)                                                                    |
|------------------------------------------|------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| Montreal Protocol                        | 1987             | 1989                     | For the phased elimination of substances that damage the ozone layer ( <b>ODS/CFCs</b> ).                         |
| Basel Convention                         | 1989             | 1992                     | Related to the transboundary movement and control of <b>hazardous wastes</b> .                                    |
| Convention on Biological Diversity (CBD) | 1992             | 1993                     | Related to the sustainable conservation of biological diversity and the use of genetic resources.                 |
| Kyoto Protocol                           | 1997             | 2005                     | Binding under the UNFCCC for <b>reduction in the emission of greenhouse gases (GHGs)</b> .                        |
| Rotterdam Convention                     | 1998             | 2004                     | Prior Informed Consent (PIC) for certain <b>hazardous chemicals and pesticides</b> in international trade.        |
| Cartagena Protocol                       | 2000             | 2003                     | Supplementary rules related to <b>biosafety</b> under the Convention on Biological Diversity (CBD).               |
| Stockholm Convention                     | 2001             | 2004                     | Protecting human health from the dangers of <b>persistent organic pollutants (POPs)</b> .                         |
| Nagoya Protocol                          | 2010             | 2014                     | Access to genetic resources and the fair and equitable sharing of benefits arising from their use ( <b>ABS</b> ). |
| Minamata Convention                      | 2013             | 2017                     | Protection of human health from the adverse effects of <b>mercury (Hg)</b> and its compounds.                     |
| Kigali Agreement                         | 2016             | 2019                     | Amendment to the Montreal Protocol; for the <b>phase-out of HFC gases</b> that increase global warming.           |

### ➤ Important COP Summits and their Venues

| COP No.  | Venue / Country                     | Special UPPCS Exam-useful Fact (Exam Hit)                                                   |
|----------|-------------------------------------|---------------------------------------------------------------------------------------------|
| COP - 26 | Glasgow (United Kingdom / Scotland) | India announced the 'Panchamrit' targets and net-zero by 2070.                              |
| COP - 27 | Sharm el-Sheikh (Egypt)             | Agreement on establishing the 'Loss and Damage' fund.                                       |
| COP - 28 | Dubai (United Arab Emirates / UAE)  | First review of the 'Global Stocktake' and a call for transitioning away from fossil fuels. |
| COP - 29 | Baku (Azerbaijan)                   | Main discussion on new goals of climate finance.                                            |
|          |                                     |                                                                                             |

| COP No.  | Venue / Country         | Special UPPCS Exam-useful Fact (Exam Hit)                                                 |
|----------|-------------------------|-------------------------------------------------------------------------------------------|
| COP - 30 | Brazil                  | Focused on the conservation of the Amazon forests and new pledges of the Paris Agreement. |
| COP - 33 | India (proposed - 2028) | India has officially offered to host it in 2028.                                          |

## > Important International Environmental Conferences and Conventions

| Conference / Convention | Year | Basic Subject / Objective (Core Function)                                        | Special Exam-useful Fact (Exam Points)                                                                     |
|-------------------------|------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Stockholm Conference    | 1972 | The first summit of the United Nations on the human environment.                 | In memory of this conference, <b>5 June was declared World Environment Day</b> .                           |
| Kyoto Protocol          | 1997 | Reducing the emission of greenhouse gases (control of global warming).           | The concept of ' <b>carbon credit</b> ' developed under it.                                                |
| Cartagena Protocol      | 2000 | Regulation of biosafety and modern biotechnology.                                | A supplementary agreement under the Convention on Biological Diversity (CBD); <b>effective from 2003</b> . |
| Nagoya Protocol         | 2010 | Access to genetic resources and fair and equitable sharing of benefits (ABS).    | Under it, <b>20 'Aichi Targets'</b> were declared for biodiversity conservation.                           |
| Rio+20 Conference       | 2012 | Held in Rio de Janeiro on the completion of 20 years of the Earth Summit (1992). | The main focus was on the ' <b>Green Economy</b> ' and <b>sustainable development</b> .                    |

## > Environmental Institutions

### > Important National Environmental Institutions

| Name of the Institution                                       | Location / Headquarters |
|---------------------------------------------------------------|-------------------------|
| Indian Institute of Forest Management (IIFM)                  | Bhopal (Madhya Pradesh) |
| Wildlife Institute of India (WII)                             | Dehradun (Uttarakhand)  |
| G. B. Pant Institute of Himalayan Environment and Development | Almora (Uttarakhand)    |
| Centre for Ecological Sciences (CES)                          | Bengaluru (Karnataka)   |
| Central Arid Zone Research Institute (CAZRI)                  | Jodhpur (Rajasthan)     |
| Indian Institute of Farming Systems Research                  | Meerut                  |



| Name of the Institution                                            | Location / Headquarters |
|--------------------------------------------------------------------|-------------------------|
| National Bureau of Agriculturally Important Microorganisms (NBAIM) | Mau                     |
| Himalayan Forest Research Institute (HFRI)                         | Shimla                  |
| Rain Forest Research Institute (RFRI)                              | Jorhat                  |
| Tropical Forest Research Institute (TFRI)                          | Jabalpur                |

### > Food Safety and Standards Authority of India (FSSAI)

| Dimension / Point                   | Statutory Structure and Functioning of FSSAI (Core Features)                                                                                                                                 |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nature of the institution           | It is a <b>statutory body</b> .                                                                                                                                                              |
| Establishment and Act               | Established in <b>September 2008</b> under the <b>Food Safety and Standards Act, 2006</b> .                                                                                                  |
| Nodal ministry                      | It functions under the administrative control of the <b>Ministry of Health and Family Welfare</b> , Government of India.                                                                     |
| Headquarters and regional offices   | * <b>Headquarters:</b> New Delhi.<br>* <b>6 regional offices:</b> Delhi, Guwahati, Mumbai, Kolkata, Cochin and Chennai.                                                                      |
| Organisational structure            | It has a Chairperson and <b>22 members</b> , of whom <b>one-third (1/3) must be women</b> .                                                                                                  |
| Main responsibilities and functions | * Food certification in India and framing of science-based standards and guidelines.<br>* Issuing licences to food businesses, conducting safety audits and increasing food safety literacy. |

### > Miscellaneous

### > Principal Breeds of Cattle and Livestock

| Name of the Animal  | Indigenous / Indian Breeds                                                                                                                                                  | Exotic Breeds / Special Facts                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Cow (Cattle)</b> | Gir, Sahiwal, Red Sindhi, Tharparkar, Hariana, Deoni, Ongole, Krishna Valley, Gangatiri, Gaolao, Hallikar, Amritmahal, Khillari, Kangayam, Umblachery, Pulikulam, Alambadi. | Jersey, Holstein Friesian, Brown Swiss, Red Dane, Ayrshire, Guernsey, Ankole, Normande, Simmental. |
|                     |                                                                                                                                                                             |                                                                                                    |

| Name of the Animal | Indigenous / Indian Breeds                                                                                                                                          | Exotic Breeds / Special Facts                                                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Buffalo</b>     | Murrah, Surti, Jaffarabadi, Bhadawari, Nili Ravi, Mehsana, Nagpuri, Toda, Banni, Bargur, Chhattisgarhi, Chilika, Gojri, Kalahandi, Luit (Swamp), Mesra, Marathwadi. | <p>* <b>Bhadawari breed:</b> the fat content in its milk is the highest.</p> <p>* <b>Murrah breed:</b> it is also called 'Black Gold'.</p> |
| <b>Goat</b>        | Jamunapari, Barbari, Beetal, Black Bengal, Sirohi, Jakhrana, Surti, Mehsana, Chegu, Kutchi goat, Attappady Black, Kanni Adu, Berari.                                | Boer, Alpine, Saanen (it is called the queen of milk of the world), Toggenburg, Anglo-Nubian.                                              |

07

# Indian Economy

INDIAN ECONOMY

## > Banking and Money

### > Reserve Bank of India (RBI)

| Dimension / Point                        | Precise Exam-useful Facts and Administrative Structure (Exam Facts)                                                                                                                             |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Establishment and Act                    | On <b>1 April 1935</b> under the provisions of the <b>Reserve Bank of India Act, 1934</b> .                                                                                                     |
| Basis of establishment                   | On the recommendation of the <b>Hilton Young Commission (1926)</b> .                                                                                                                            |
| Central office (history)                 | The initial central office was in <b>Kolkata</b> , which was permanently shifted to <b>Mumbai</b> in <b>1937</b> .                                                                              |
| Nationalisation                          | It was fully nationalised on <b>1 January 1949</b> .                                                                                                                                            |
| Headquarters                             | <b>Mumbai, Maharashtra</b> .                                                                                                                                                                    |
| Four regional zones                      | <ol style="list-style-type: none"> <li><b>Eastern Zone</b>—Kolkata</li> <li><b>Western Zone</b>—Mumbai</li> <li><b>Northern Zone</b>—New Delhi</li> <li><b>Southern Zone</b>—Chennai</li> </ol> |
| Basic objectives and functions           | Controlling credit/currency in the country, regulating the entire banking system and ensuring financial stability.                                                                              |
| Determination of monetary instruments    | To control inflationary pressure in the country, <b>the repo rate and reverse repo rate are determined by the RBI</b> , through which the flow of cash in the banking system is controlled.     |
| Quantitative monetary policy instruments | Bank rate, repo rate, reverse repo rate, Cash Reserve Ratio (CRR), Statutory Liquidity Ratio (SLR), Open Market Operations (OMO), and Marginal Standing Facility (MSF).                         |
| Qualitative monetary policy instruments  | Margin requirement, credit rationing, selective credit control, moral suasion, and direct action.                                                                                               |
| First Governor (British origin)          | <b>Sir Osborne Smith</b> - the first Governor of the RBI.                                                                                                                                       |
| First Indian Governor                    | <b>C. D. Deshmukh (Chintaman Dwarkanath Deshmukh)</b> - the first RBI Governor of independent India (he was also the Governor at the time of nationalisation).                                  |
| Current Governor                         | <b>Shaktikanta Das</b> (the current Governor of the RBI).                                                                                                                                       |

## > Principal Banking Terminology and Instruments

| Banking Term / Instrument              | Basic Banking Meaning and Exam-useful Definition (Core Meaning)                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Liability</b>                       | The money/deposits which the bank has to return to its customers or depositors in the future. E.g., <b>deposits in the bank</b> and various types of savings/small savings schemes.     |
| <b>Bank Rate</b>                       | The interest rate at which the Reserve Bank of India (RBI) provides <b>long-term loans</b> to commercial banks.                                                                         |
| <b>Repo Rate</b>                       | The interest rate at which the Reserve Bank of India (RBI) provides <b>short-term loans</b> to commercial banks for their short-term needs against approved securities.                 |
| <b>Reverse Repo Rate</b>               | The interest rate at which the Reserve Bank of India (RBI) accepts the surplus liquidity/money available with commercial banks (i.e., borrows money from commercial banks).             |
| <b>Cash Reserve Ratio (CRR)</b>        | It is mandatory for commercial banks to keep a fixed part of their total deposits (NDTL) <b>in the form of cash with the Reserve Bank of India (RBI)</b> .                              |
| <b>Statutory Liquidity Ratio (SLR)</b> | It is mandatory for commercial banks to keep a prescribed percentage of their total deposits (in the form of cash, gold or approved securities) in liquid form <b>with themselves</b> . |

## > Four Measures of Money Supply in India: M1, M2, M3, M4 (Money Supply Measures)

| Money Measure | Components                                                                                                                                                 | Liquidity Position and Nickname                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M1</b>     | Currency in circulation (notes and coins) + demand deposits with banks (current and savings accounts) + other deposits with the RBI that can be withdrawn. | * <b>The most liquid money</b> (usable as cash immediately).<br>* It is called " <b>Narrow Money</b> ".                                    |
| <b>M2</b>     | <b>M1</b> + savings deposits of post office savings banks.                                                                                                 | * Slightly less liquid than M1, but still easily convertible into cash.                                                                    |
| <b>M3</b>     | <b>M1</b> + time/long-term deposits with the banking system (fixed deposits - FD).                                                                         | * Its liquidity is less than that of M1 and M2.<br>* It is called " <b>Broad Money</b> ". ( <i>most used in monetary policy analysis</i> ) |
| <b>M4</b>     | <b>M3</b> + total deposits of post office savings organisations (National Savings Certificates, i.e., NSC, are not included in it).                        | * <b>The least liquid measure of money supply</b> (because it includes time deposits that cannot be converted into cash immediately).      |

- **Decreasing order of liquidity: M1 > M2 > M3 > M4** (M1 is the most liquid and M4 the least liquid).

## > Historical Chronology of the Establishment of Banks (Chronological Order)

### > Principal Banks Established before Independence

| Name of the Bank                 | Year of Establishment | Historical and Exam-useful Significance (Exam Facts)                                                                                        |
|----------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Bank of Hindustan                | 1770                  | The <b>first bank</b> established in India on the European pattern (by Alexander and Company).                                              |
| General Bank of Bengal and Bihar | 1773                  | An early attempt of the initial colonial period.                                                                                            |
| General Bank of Bengal           | 1786                  | A banking unit of the early British rule.                                                                                                   |
| Bank of Bengal                   | 1806                  | The first presidency bank (initially formed under the name Bank of Calcutta).                                                               |
| Bank of Bombay                   | 1840                  | The second important presidency bank.                                                                                                       |
| Bank of Madras                   | 1843                  | The third presidency bank. <i>(Note: these three presidency banks were merged to form the Imperial Bank of India in 1921).</i>              |
| Allahabad Bank                   | 1865                  | India's first functioning joint stock bank.                                                                                                 |
| Oudh Commercial Bank             | 1881                  | Was the <b>first commercial bank managed by Indians on the basis of limited liability</b> .                                                 |
| Punjab National Bank (PNB)       | 1894                  | <b>The first bank formed entirely with Indian capital and management</b> (Lala Lajpat Rai was one of its founders; it started from Lahore). |
| People's Bank of India           | 1901                  | An early banking unit of the period of the Swadeshi Movement.                                                                               |
| Bank of India                    | 1906                  | Established by leading businessmen of Bombay, inspired by the Swadeshi Movement.                                                            |
| Canara Bank                      | 1906                  | A swadeshi bank established in Mangalore by Ammembal Subba Rao Pai.                                                                         |
| Indian Bank                      | 1907                  | Established in Madras as part of the Swadeshi Movement.                                                                                     |
| Bank of Baroda (BoB)             | 1908                  | Established by Sayajirao Gaekwad III (Maharaja of Baroda).                                                                                  |
| Central Bank of India            | 1911                  | <b>The first swadeshi commercial bank, wholly owned by Indians.</b>                                                                         |

| Name of the Bank | Year of Establishment | Historical and Exam-useful Significance (Exam Facts)                               |
|------------------|-----------------------|------------------------------------------------------------------------------------|
| Bank of Mysore   | 1913                  | Established through the efforts of the Maharaja of Mysore and Sir M. Visvesvaraya. |

### > Principal Banks Established after Independence

| Name of the Bank                 | Year of Establishment | Regulatory Reform / Current Status                                                                                                                                       |
|----------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State Bank of Hyderabad          | 1941                  | The bank of the Nizam of Hyderabad; after independence it became an associate of SBI under the SBI (Subsidiary Banks) Act, 1959 and was finally <b>merged into SBI</b> . |
| UCO Bank                         | 1943                  | Established by Ghanshyam Das Birla; it was nationalised in 1969.                                                                                                         |
| United Bank of India             | 1950                  | Formed by the merger of some banks of Bengal; (it has now been merged into Punjab National Bank).                                                                        |
| State Bank of India (SBI)        | 1955                  | On the recommendation of the Gorwala Committee, the <b>Imperial Bank of India was nationalised</b> and made the largest bank of the country.                             |
| State Bank of Bikaner and Jaipur | 1963                  | An associate bank formed by the merger of the banks of Bikaner and Jaipur; (now <b>merged into SBI</b> ).                                                                |
| Bank of Maharashtra              | 1935                  | (Established 1935); after independence it came under government ownership in the first phase of nationalisation in 1969.                                                 |
| Union Bank of India              | 1919                  | (Established in 1919, inaugurated by Mahatma Gandhi); nationalised in 1969.                                                                                              |
| Punjab and Sind Bank             | 1908                  | Came under government control in the second phase of nationalisation in 1980 (the group of 6 banks).                                                                     |
| Syndicate Bank                   | 1925                  | (Original name Canara Industrial and Banking Syndicate); nationalised in 1969; (now merged into Canara Bank).                                                            |

### > New Private Sector and Commercial Banks (Post-1994 Reforms)

| Name of the Bank | Year of Establishment | Banking Category / Specific Nature                                                                           |
|------------------|-----------------------|--------------------------------------------------------------------------------------------------------------|
| ICICI Bank       | 1994                  | Established under the new-generation private sector banks on the recommendation of the Narasimham Committee. |

| Name of the Bank      | Year of Establishment | Banking Category / Specific Nature                                                                       |
|-----------------------|-----------------------|----------------------------------------------------------------------------------------------------------|
| HDFC Bank             | 1994                  | A leading new-generation private sector commercial bank (at present the country's largest private bank). |
| Axis Bank             | 1994                  | Its initial name was <b>UTI Bank</b> , which was later renamed Axis Bank.                                |
| IndusInd Bank         | 1994                  | One of the earliest private banks to obtain a licence after the new economic reforms.                    |
| Kotak Mahindra Bank   | 2003                  | The first institution to convert from a Non-Banking Financial Company (NBFC) into a commercial bank.     |
| Yes Bank              | 2004                  | A private sector scheduled commercial bank.                                                              |
| Bandhan Bank          | 2015                  | The first bank to convert from a microfinance institution (MFI) into a full commercial bank.             |
| AU Small Finance Bank | 2017                  | A Rajasthan-based finance company that received the status of a Small Finance Bank (SFB).                |
| IDFC First Bank       | 2018                  | A commercial bank formed by the merger of the infrastructure finance company IDFC and Capital First.     |

## > Financial Institutions

### > Principal Financial Institutions and their Establishment (Financial Institutions Matrix)

| Name of the Financial Institution                             | Date / Year of Establishment | Main Exam-useful Facts (Exam Notes)                                                                                    |
|---------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Reserve Bank of India (RBI)                                   | 1 April 1935                 | Established on the recommendation of the Hilton Young Commission; nationalised on 1 January 1949.                      |
| Industrial Credit and Investment Corporation of India (ICICI) | January 1955                 | An early major financial institution formed with the cooperation of the private sector.                                |
| State Bank of India (SBI)                                     | 1 July 1955                  | On the recommendation of the Gorwala Committee, the <b>Imperial Bank of India (1921)</b> was nationalised and renamed. |
| Life Insurance Corporation of India (LIC)                     | September 1956               | Formed by nationalising about 245 Indian and foreign companies (headquarters: Mumbai).                                 |
|                                                               |                              |                                                                                                                        |

| Name of the Financial Institution                            | Date / Year of Establishment | Main Exam-useful Facts (Exam Notes)                                                                                   |
|--------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Regional Rural Banks (RRB)                                   | 2 October 1975               | Established for rural credit on the recommendation of the M. Narasimham Committee.                                    |
| National Bank for Agriculture and Rural Development (NABARD) | 12 July 1982                 | The apex bank for agriculture and rural development, formed on the recommendation of the <b>Sivaraman Committee</b> . |
| Small Industries Development Bank of India (SIDBI)           | April 1990                   | <b>Headquarters: Lucknow (Uttar Pradesh);</b> extremely important for the UPPCS exam.                                 |

## > Principal Financial and Regulatory Institutions

| Name of the Financial Institution                               | Year of Establishment | Headquarters (HQ) | Main Exam-useful Facts / Nature (Exam Notes)                                                                         |
|-----------------------------------------------------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------|
| Industrial Finance Corporation of India (IFCI)                  | 1948                  | New Delhi         | It is India's <b>first Development Financial Institution (DFI)</b> .                                                 |
| Industrial Development Bank of India (IDBI)                     | 1964                  | Mumbai            | Established to give long-term financial assistance to industries.                                                    |
| Unit Trust of India (UTI)                                       | 1964                  | Mumbai            | The country's first mutual fund institution.                                                                         |
| General Insurance Corporation of India (GIC)                    | 1972                  | Mumbai            | The apex reinsurance provider of non-life (general) insurance.                                                       |
| Deposit Insurance and Credit Guarantee Corporation (DICGC)      | 1978                  | Mumbai            | The institution that <b>insures depositors' money in banks up to ₹5 lakh</b> (a wholly owned subsidiary of the RBI). |
| Export-Import Bank of India (EXIM Bank)                         | 1982                  | Mumbai            | The apex institution for financing foreign trade (import-export).                                                    |
| Securities and Exchange Board of India (SEBI)                   | 1988                  | Mumbai            | The regulatory body of the stock market (statutory status: 1992).                                                    |
| National Housing Bank of India (NHB)                            | 1988                  | New Delhi         | The supreme regulatory body of the housing finance sector.                                                           |
| National Securities Depository Limited (NSDL)                   | 1996                  | Mumbai            | India's first electronic securities depository.                                                                      |
| Insurance Regulatory and Development Authority of India (IRDAI) | 1999                  | Hyderabad         | Regulator of the entire insurance sector (life and non-life) (recommendation of the Malhotra Committee).             |
| Central Depository Services Limited (CDSL)                      | 1999                  | Mumbai            | A leading securities depository of the country.                                                                      |



| Name of the Financial Institution                                   | Year of Establishment | Headquarters (HQ) | Main Exam-useful Facts / Nature (Exam Notes)                                                     |
|---------------------------------------------------------------------|-----------------------|-------------------|--------------------------------------------------------------------------------------------------|
| Credit Guarantee Fund Trust (CGTMSE)                                | 2000                  | Mumbai            | A trust for collateral-free loan facility to micro and small enterprises.                        |
| Clearing Corporation of India (CCIL)                                | 2001                  | Mumbai            | Provider of clearing and settlement facilities for financial market transactions.                |
| Agriculture Insurance Company of India (AIC)                        | 2002                  | New Delhi         | A specialised company of the agriculture and crop insurance sector.                              |
| Pension Fund Regulatory and Development Authority (PFRDA)           | 2003                  | New Delhi         | The supreme regulator of the pension fund sector in the country (statutory status: 2013).        |
| India Infrastructure Finance Company Limited (IIFCL)                | 2006                  | New Delhi         | For giving long-term loans to infrastructure projects.                                           |
| National Payments Corporation of India (NPCI)                       | 2008                  | Mumbai            | The apex body operating retail payments in the country (such as <b>UPI, IMPS, RuPay</b> ).       |
| Micro Units Development and Refinance Agency (MUDRA)                | 2015                  | Mumbai            | An NBFC providing refinance support to micro enterprises.                                        |
| Insolvency and Bankruptcy Board of India (IBBI)                     | 2016                  | New Delhi         | Regulator of the Insolvency and Bankruptcy Code.                                                 |
| National e-Governance Services Limited (NeSL)                       | 2016                  | Mumbai            | India's first <b>Information Utility</b> company.                                                |
| International Financial Services Centres Authority (IFSCA)          | 2020                  | Gandhinagar       | The unified regulator of international financial centres located in <b>GIFT City (Gujarat)</b> . |
| National Bank for Financing Infrastructure and Development (NaBFID) | 2021                  | Mumbai            | A new Development Financial Institution (DFI) for infrastructure development.                    |

## > Poverty

### > Important Committees Related to the Measurement of Poverty

| Name of the Committee | Year of Formation | Main Exam-useful Recommendations and Criteria (Exam Points)                                                                                                                                                                                                                                                |
|-----------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y. K. Alagh Committee | 1979              | <ul style="list-style-type: none"> <li>* For the first time in India, poverty was determined on the basis of <b>nutrients (calorie consumption)</b>.</li> <li>* <b>For rural areas:</b> 2400 kilocalories (per person per day)</li> <li>* <b>For urban areas:</b> 2100 kilocalories were fixed.</li> </ul> |

| Name of the Committee      | Year of Formation | Main Exam-useful Recommendations and Criteria (Exam Points)                                                                                                                                                                                                                                                             |
|----------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D. T. Lakdawala Committee  | 1993              | <ul style="list-style-type: none"> <li>* Recommended making <b>state-specific poverty lines</b> for each state.</li> <li>* Asked for these lines to be updated using the consumer price indices <b>CPI-IW</b> (industrial workers) in urban areas and <b>CPI-AL</b> (agricultural labourers) in rural areas.</li> </ul> |
| Suresh Tendulkar Committee | 2009              | <ul style="list-style-type: none"> <li>* For the first time, monthly per capita consumption expenditure (MPCE) was made the basis instead of calorie consumption.</li> <li>* <b>For rural households:</b> ₹816 (per person per month)</li> <li>* <b>For urban households:</b> ₹1000 was fixed.</li> </ul>               |
| C. Rangarajan Committee    | 2014              | <ul style="list-style-type: none"> <li>* Gave a new criterion by revising consumption expenditure at the all-India level:</li> <li>* <b>For rural areas:</b> ₹972 per month (<b>₹32 per day</b>)</li> <li>* <b>For urban areas:</b> ₹1407 per month (<b>₹47 per day</b>) was fixed.</li> </ul>                          |

## > Unemployment

### > Types and Characteristics of Unemployment

Note: In India the **National Sample Survey Organisation (NSSO)** / **National Statistical Office (NSO)** compiles and publishes data related to employment and unemployment.

| Type of Unemployment            | Basic Meaning and Characteristic (Core Concept)                                                                                                                                                                                       | Special Exam-useful Facts (Exam Facts)                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frictional unemployment         | <ul style="list-style-type: none"> <li>* The transitional period between leaving one job and getting another job.</li> <li>* Unemployment generated due to technological change, etc. (such as retrenchment in companies).</li> </ul> | <ul style="list-style-type: none"> <li>* It is considered "<b>search</b>" or "<b>voluntary</b>" unemployment.</li> <li>* It is mainly found in <b>developed countries</b>.</li> </ul>                         |
| Structural unemployment         | <ul style="list-style-type: none"> <li>* Unemployment generated due to a change in the structure of the economy or the <b>lack of skills/qualifications</b> among workers in line with industrial demand.</li> </ul>                  | <ul style="list-style-type: none"> <li>* <b>This is the unemployment found in the largest amount in India.</b></li> </ul>                                                                                     |
| Disguised / hidden unemployment | <ul style="list-style-type: none"> <li>* When <b>more people than required</b> are engaged in a work or occupation (even if some people are removed, there is no difference in total production).</li> </ul>                          | <ul style="list-style-type: none"> <li>* In it the "<b>marginal productivity of the extra worker is zero</b>."</li> <li>* It is found especially in <b>agriculture and the unorganised sector</b>.</li> </ul> |
| Seasonal unemployment           | <ul style="list-style-type: none"> <li>* Unemployment generated because work is available according to the weather or season and not available in the remaining months/time.</li> </ul>                                               | <ul style="list-style-type: none"> <li>* This too is seen especially in the <b>agriculture sector</b> (except at the time of sowing/harvesting).</li> </ul>                                                   |

| Type of Unemployment         | Basic Meaning and Characteristic (Core Concept)                                                                                                 | Special Exam-useful Facts (Exam Facts)                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cyclical unemployment</b> | * Unemployment generated by a fall in demand due to an economic recession in the market. It ends automatically when demand in the market rises. | * <b>Its figures in India are negligible (zero).</b><br>* It is mainly found in the industrial sectors of <b>developed countries</b> . |

## > Inflation

### > Instruments for Controlling Inflation (Monetary Policy Tools)

| Quantitative Tools                      | Qualitative Tools     |
|-----------------------------------------|-----------------------|
| * Repo rate                             | * Rationing of credit |
| * Reverse repo rate                     | * Margin requirement  |
| * Bank rate                             | * Moral suasion       |
| * Cash Reserve Ratio (CRR)              |                       |
| * Statutory Liquidity Ratio (SLR)       |                       |
| * Open Market Operations (OMO)          |                       |
| * Marginal Standing Facility (MSF rate) |                       |

## > Agriculture

### > Important Revolutions Related to Agriculture and Production (Agricultural Revolutions)

| Name of the Revolution   | Related Production / Sector                      |
|--------------------------|--------------------------------------------------|
| <b>Green Revolution</b>  | Foodgrain production (especially wheat and rice) |
| <b>White Revolution</b>  | Milk production and dairy development            |
| <b>Blue Revolution</b>   | Fish production                                  |
| <b>Yellow Revolution</b> | Oilseed production                               |
| <b>Silver Revolution</b> | Egg and poultry production                       |
| <b>Brown Revolution</b>  | Production of chemical fertilisers               |
| <b>Black Revolution</b>  | Production of petroleum / crude oil              |
| <b>Golden Revolution</b> | Fruit, horticulture and honey production         |

## > Five-Year Plans

### > Main Priority Sectors in the Five-Year Plans (Five-Year Plans Core Matrix)

| Plan                  | Period    | Model / Architect                    | Main Priority and Historic Work (Exam Target)                                                                                                                                                                                                                                          |
|-----------------------|-----------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First Five-Year Plan  | 1951 - 56 | Harrod-Domar model                   | <ul style="list-style-type: none"> <li>* <b>The highest priority was given to the agriculture sector.</b></li> <li>* Large irrigation projects such as Bhakra-Nangal and Damodar Valley were started.</li> </ul>                                                                       |
| Second Five-Year Plan | 1956 - 61 | P. C. Mahalanobis model              | <ul style="list-style-type: none"> <li>* <b>Priority was given to heavy industries and minerals.</b></li> <li>* Large steel plants were established in the country at <b>Bhilai (Russia), Durgapur (Britain) and Rourkela (Germany)</b>, along with hydroelectric projects.</li> </ul> |
| Third Five-Year Plan  | 1961 - 66 | Gadgil Yojana / Sukhamoy Chakravarty | <ul style="list-style-type: none"> <li>* <b>Making both agriculture and industry self-reliant</b> was its main objective.</li> <li>* In <b>1964 the Bokaro Iron and Steel Industry was established with the cooperation of the former Soviet Union (USSR).</b></li> </ul>              |
| Fourth Five-Year Plan | 1969 - 74 | Ashok Rudra and A. S. Manne          | <ul style="list-style-type: none"> <li>* Emphasis was laid on <b>growth with stability</b> and progress towards self-reliance.</li> <li>* The goal of ending inequalities of income and wealth in society was set.</li> </ul>                                                          |
| Fifth Five-Year Plan  | 1974 - 79 | D. P. Dhar model                     | <ul style="list-style-type: none"> <li>* <b>Poverty eradication (Garibi Hatao)</b> and self-reliance / population control were its main objectives.</li> <li>* (Note: due to political change this plan was ended in 1978 itself, 1 year before time).</li> </ul>                      |

## > Principal Schemes

### > Principal Schemes from 2000 to 2013

| Name of the Scheme               | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                    |
|----------------------------------|-----------------------------|------------------------------------------------------------------------------|
| Annapurna Yojana                 | 1 April 2000                | Providing free foodgrains to extremely poor senior citizens.                 |
| Sarva Shiksha Abhiyan            | November 2000               | Universalisation of elementary education for children of 6-14 years.         |
| Pradhan Mantri Gram Sadak Yojana | 25 December 2000            | Connecting rural habitations with all-weather roads (100% centrally funded). |
| Sampoorna Grameen Rozgar Yojana  | 25 September 2001           | Ensuring employment and food security in rural areas.                        |
|                                  |                             |                                                                              |

| Name of the Scheme                                                   | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                                   |
|----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
| <b>Hariyali Yojana</b>                                               | 27 January 2003             | Rural development scheme for water harvesting and watershed management.                     |
| <b>Varishtha Pension Bima Yojana</b>                                 | 14 July 2003                | Assured pension / social security scheme for senior citizens.                               |
| <b>National Horticulture Mission</b>                                 | 2005-06                     | Integrated development of the horticulture sector (now part of MIDH).                       |
| <b>National Rural Health Mission</b>                                 | 12 April 2005               | Providing accessible and quality health services to the rural population.                   |
| <b>Jawaharlal Nehru National Urban Renewal Mission (JNNURM)</b>      | 3 December 2005             | Integrated development of urban infrastructure and basic services.                          |
| <b>National Rural Employment Guarantee Scheme (NREGS)</b>            | 2 February 2006             | Legal guarantee of 100 days of employment (later renamed MGNREGA).                          |
| <b>National Food Security Mission</b>                                | 2007-08                     | Increasing self-sufficiency in the production of wheat, rice and pulses.                    |
| <b>Rashtriya Krishi Vikas Yojana (RKVY)</b>                          | 2007-08                     | Achieving an annual growth rate of 4% in agriculture and allied sectors.                    |
| <b>Rashtriya Swasthya Bima Yojana</b>                                | 1 October 2007              | Providing health insurance to BPL families of the unorganised sector.                       |
| <b>Aam Aadmi Bima Yojana</b>                                         | 2 October 2007              | Social security insurance for rural landless households.                                    |
| <b>Indira Gandhi National Old Age Pension Scheme</b>                 | 19 November 2007            | Financial assistance / pension to the elderly of the BPL category.                          |
| <b>Jawaharlal Nehru National Solar Mission</b>                       | 11 January 2010             | Increasing solar power generation capacity in the country (initial target of 20 gigawatt).  |
| <b>National Rural Livelihood Mission (NRLM)</b>                      | June 2011                   | Removing rural poverty through Self-Help Groups (SHGs).                                     |
| <b>Nirmal Bharat Abhiyan</b>                                         | 1 April 2012                | For increasing sanitation coverage in rural areas (old name Total Sanitation Campaign).     |
| <b>Direct Benefit Transfer Scheme (DBT)</b>                          | 1 January 2013              | Sending the subsidy of government schemes directly into the bank accounts of beneficiaries. |
| <b>Deendayal Antyodaya Yojana–National Urban Livelihoods Mission</b> | 24 September 2013           | Giving urban poor households opportunities for skill development and self-employment.       |

## > Principal Schemes of 2014 and 2015

| Name of the Scheme                             | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                            |
|------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------|
| <b>Pradhan Mantri Jan-Dhan Yojana</b>          | 28 August 2014              | The world's largest campaign of financial inclusion.                                 |
| <b>Make in India</b>                           | 25 September 2014           | Making India a global manufacturing hub.                                             |
| <b>Swachh Bharat Mission</b>                   | 2 October 2014              | Making the country open-defecation free (ODF) and complete cleanliness.              |
| <b>Saansad Adarsh Gram Yojana</b>              | 11 October 2014             | Each MP adopting villages and developing them as model villages.                     |
| <b>Mission Indradhanush (immunisation)</b>     | 25 December 2014            | Ensuring full immunisation coverage of children and pregnant women.                  |
| <b>UJALA Yojana</b>                            | 5 January 2015              | Energy conservation by distributing LED bulbs at low prices.                         |
| <b>Beti Bachao, Beti Padhao Abhiyan</b>        | 22 January 2015             | Improving the sex ratio and ensuring the education of girls (launched from Panipat). |
| <b>Sukanya Samriddhi Yojana</b>                | 22 January 2015             | A small savings account scheme for the secure future of girls.                       |
| <b>Pradhan Mantri Mudra Yojana</b>             | 8 April 2015                | Providing loans to micro enterprises in three categories (Shishu, Kishore, Tarun).   |
| <b>Pradhan Mantri Jeevan Jyoti Bima Yojana</b> | 9 May 2015                  | Voluntary life insurance of ₹2 lakh (for the age group 18-50 years).                 |
| <b>Pradhan Mantri Suraksha Bima Yojana</b>     | 9 May 2015                  | Accidental insurance of ₹2 lakh (for the age group 18-70 years).                     |
| <b>Atal Pension Yojana</b>                     | 9 May 2015                  | Minimum assured pension scheme for workers of the unorganised sector.                |
| <b>Pradhan Mantri Awas Yojana–Urban</b>        | 25 June 2015                | Providing pucca houses to the poor and middle class of urban areas.                  |
| <b>AMRUT Mission</b>                           | 25 June 2015                | Ensuring water supply, sewerage and urban reform in cities.                          |
| <b>Smart City Mission</b>                      | 25 June 2015                | Making 100 cities of the country smart and sustainable.                              |
| <b>Pradhan Mantri Krishi Sinchayee Yojana</b>  | 1 July 2015                 | Goal of 'Har Khet Ko Pani' and 'Per Drop More Crop' (water-use efficiency).          |

| Name of the Scheme                  | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                      |
|-------------------------------------|-----------------------------|--------------------------------------------------------------------------------|
| Digital India Programme             | 1 July 2015                 | Transforming India into a knowledge-based, digitally empowered society.        |
| Pradhan Mantri Kaushal Vikas Yojana | 15 July 2015                | Providing youth with skill training in line with industries.                   |
| National Skill Development Mission  | 15 July 2015                | Speeding up skill training activities and building an institutional framework. |

### > Principal Schemes from 2016 to 2020

| Name of the Scheme                           | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                                   |
|----------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
| Startup India                                | 16 January 2016             | Promoting innovation and new enterprises in the country.                                    |
| Pradhan Mantri Fasal Bima Yojana             | 18 February 2016            | Comprehensive insurance cover to farmers for crop losses from natural disasters.            |
| Stand-Up India                               | 5 April 2016                | Giving loans to SC, ST and women entrepreneurs for greenfield enterprises.                  |
| Pradhan Mantri Ujjwala Yojana                | 1 May 2016                  | Giving free LPG connections to women of BPL households.                                     |
| Pradhan Mantri Awas Yojana–Gramin            | 20 November 2016            | Giving pucca houses to all homeless households and those with kutcha houses in rural areas. |
| Pradhan Mantri Kisan Sampada Yojana          | 2017                        | Development of food processing and mega food park infrastructure.                           |
| Saubhagya Yojana (Sahaj Bijli Har Ghar)      | 11 October 2017             | Ensuring complete electrification of all rural and urban houses of the country.             |
| Deendayal SPARSH Yojana                      | 3 November 2017             | Scholarship to promote philately (stamp collecting) among school children.                  |
| Ayushman Bharat–PM Jan Arogya Yojana         | 23 September 2018           | Cashless secondary and tertiary health insurance cover of up to ₹5 lakh per family.         |
| Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) | 24 February 2019            | Direct income support of ₹6,000 per year to eligible farmer families (3 instalments).       |
| Pradhan Mantri Shram Yogi Maandhan Yojana    | 5 March 2019                | Monthly pension scheme of ₹3,000 for elderly workers of the unorganised sector.             |



| Name of the Scheme                                                   | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                                   |
|----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
| Pradhan Mantri Kisan Urja Suraksha evam Utthan Mahabhiyan (PM-KUSUM) | 19 February 2019            | Helping farmers set up solar pumps and grid-connected solar power plants.                   |
| Jal Jeevan Mission (Rural)                                           | 15 August 2019              | Giving a functional household tap connection (FHTC) to every rural household by 2024.       |
| Pradhan Mantri Kisan Maandhan Yojana                                 | 12 September 2019           | Old-age pension scheme for small and marginal farmers (after the age of 60 years).          |
| National Pension Scheme for Traders and Self-Employed Persons        | 12 September 2019           | Voluntary social security scheme for small shopkeepers and the self-employed.               |
| Atal Bhujal Yojana                                                   | 25 December 2019            | Improving groundwater management in the stressed blocks of identified states.               |
| Pradhan Mantri Garib Kalyan Anna Yojana                              | 26 March 2020               | Giving 5 kg of free foodgrains per person to the poor during the Covid-19 pandemic.         |
| PM SVANidhi Yojana                                                   | 1 June 2020                 | Giving working capital loans to street vendors.                                             |
| Pradhan Mantri Matsya Sampada Yojana                                 | 10 September 2020           | Sustainable development of the fisheries sector in the country through the Blue Revolution. |

### > Principal Schemes from 2021 to 2024

| Name of the Scheme                                     | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                        |
|--------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------|
| Ayushman Bharat Digital Mission                        | 27 September 2021           | Digital Health ID for citizens and an integrated health framework.               |
| Pradhan Mantri Poshan Shakti Nirman Yojana (PM-POSHAN) | 29 September 2021           | The new form of the Mid-Day Meal in government primary schools.                  |
| PM Gati Shakti National Master Plan                    | 13 October 2021             | An integrated plan for multi-modal connectivity and infrastructure development.  |
| PM Ayushman Bharat Health Infrastructure Mission       | 25 October 2021             | Strengthening primary, secondary and tertiary health systems across the country. |
| Pradhan Mantri Schools for Rising India (PM-SHRI)      | 7 September 2022            | Upgrading more than 14,500 schools of the country into modern model schools.     |
|                                                        |                             |                                                                                  |



| Name of the Scheme                                  | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                             |
|-----------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|
| National Logistics Policy                           | 17 September 2022           | Reducing the cost of freight transport and improving last-mile connectivity.          |
| Pradhan Mantri Vishwakarma Yojana                   | 17 September 2023           | Giving skill training and financial assistance to traditional artisans and craftsmen. |
| PM-JANMAN (Janjatiya Adivasi Nyaya Maha Abhiyan)    | 15 November 2023            | Socio-economic welfare of Particularly Vulnerable Tribal Groups (PVTGs).              |
| Pradhan Mantri Kisan Drone Yojana / Namo Drone Didi | 2023                        | Training women SHGs in drone operation for agricultural work (fertiliser spraying).   |
| Pradhan Mantri Surya Ghar: Muft Bijli Yojana        | 13 February 2024            | 300 units of free electricity to 1 crore houses through rooftop solar.                |
| Dharti Aaba Janjatiya Gram Utkarsh Abhiyan          | 2 October 2024              | Development of infrastructure on a saturation basis in tribal-majority villages.      |

### ➤ Recent New Schemes and Expansions (Current Context: 2025–2026)

| Name of the Scheme                                             | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                                                   |
|----------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| Pradhan Mantri Viksit Bharat Rozgar Yojana (PM-VBRY)           | 1 August 2025               | Incentive for providing new employment opportunities to youth in the formal workforce.                      |
| Restructured PM SVANidhi Scheme                                | 1 August 2025               | New expansion for street vendors with a higher loan limit and incentives for digital transactions.          |
| Pradhan Mantri Dhan-Dhaanya Krishi Yojana                      | 11 October 2025             | Promoting the spread of high-yielding, climate-resilient crop varieties and sustainable agriculture.        |
| Pradhan Mantri Surya Ghar: Model Solar Village Initiative      | 2025                        | Development of model villages in every district of the country to demonstrate rural solar power generation. |
| PM Awas Yojana–Urban 2.0                                       | 2025 (implementation)       | The second phase of construction of pucca houses for the urban poor and middle class.                       |
| PM-e-Bus Sewa                                                  | 2025 (expansion)            | Large-scale operation of electric buses to promote public transport in cities.                              |
| Viksit Bharat–Guarantee for Rozgar and Aajeevika (VB-G RAM JI) | 1 July 2026                 | Hundred-per-cent guarantee of employment to rural youth and a skill-development livelihood mission.         |

| Name of the Scheme                  | Exact Date / Year of Launch | Main Exam-useful Objective (Exam Context)                                                           |
|-------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------|
| BHAVYA Rasayan Scheme               | 24 July 2026                | Making India self-reliant at the global level in the manufacturing of petrochemicals and chemicals. |
| Pradhan Mantri Surya Sarovar Yojana | 31 July 2026                | Setting up floating solar power plants on the banks of Amrit Sarovars and on reservoirs.            |

### > Important Committees

### > Important Tax and Banking Reform Committees (Tax & Financial Reform Committees)

| Name of the Committee           | Year of Formation | Main Exam-useful Recommendations and Objectives (Exam Points)                                                                                                                                                                                                  |
|---------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L. K. Jha Committee             | 1977              | <ul style="list-style-type: none"> <li>* Suggestions for reform in indirect taxation.</li> <li>* Recommended strict laws and administrative measures to curb black money and tax evasion.</li> </ul>                                                           |
| Raja Chelliah Committee         | 1991              | <ul style="list-style-type: none"> <li>* Comprehensive reform in the direct and indirect tax system.</li> <li>* Making the direct tax system simple and broad-based and rationalising the tax rates.</li> </ul>                                                |
| Narasimham Committee (I and II) | 1991 and 1998     | <ul style="list-style-type: none"> <li>* Historic recommendations for <b>structural reforms in the financial and banking sector</b>.</li> <li>* Emphasis on bringing transparency into the financial/tax system.</li> </ul>                                    |
| Kelkar Committee                | 2002              | <ul style="list-style-type: none"> <li>* Restructuring direct and indirect taxes and simplifying tax concessions.</li> </ul>                                                                                                                                   |
| Adhia Committee                 | 2016              | <ul style="list-style-type: none"> <li>* Administrative measures for implementing a uniform tax system (<b>GST - Goods and Services Tax</b>) throughout the country.</li> <li>* Simplifying indirect taxes and implementing a single tax structure.</li> </ul> |

### > Important Economic and Financial Committees (Committees Master Matrix)

| Name of the Committee          | Year of Formation | Related Sector / Main Subject (Core Subject)                            |
|--------------------------------|-------------------|-------------------------------------------------------------------------|
| Y. K. Alagh Committee          | 1979              | Determination of the poverty line on the basis of nutrients (calories). |
| Sukhamoy Chakravarty Committee | 1985              | Review of the Indian monetary system and monetary policy.               |

| Name of the Committee        | Year of Formation | Related Sector / Main Subject (Core Subject)                                              |
|------------------------------|-------------------|-------------------------------------------------------------------------------------------|
| Vaghul Committee             | 1987              | Development and institutional framework of the Indian money market.                       |
| M. Narasimham Committee-I    | 1991              | Structural reforms in the financial and banking sector (first phase).                     |
| Raja J. Chelliah Committee   | 1991–93           | Comprehensive reforms in direct and indirect taxes (tax reforms).                         |
| C. Rangarajan Committee      | 1993              | Suggestions on the disinvestment of public sector undertakings (PSUs).                    |
| D. T. Lakdawala Committee    | 1993              | Methodology for determining state-specific poverty lines.                                 |
| R. N. Malhotra Committee     | 1993              | Reforms in the insurance sector and approval of private investment.                       |
| Tarapore Committee-I         | 1997              | Full convertibility of the rupee on the capital account (Capital Account Convertibility). |
| M. Narasimham Committee-II   | 1998              | Second-phase reforms for financial strength in the banking sector.                        |
| J. R. Varma Committee        | 1998              | Risk management in the derivatives financial market.                                      |
| L. C. Gupta Committee        | 1996              | Regulatory framework of the derivatives market and futures trading.                       |
| Prof. Vijay Kelkar Committee | 2002              | Restructuring of direct and indirect taxes (Direct & Indirect Tax Reforms).               |
| Tarapore Committee-II        | 2006              | Roadmap for moving towards full capital account convertibility.                           |
| Raghuram Rajan Committee     | 2008              | Comprehensive structural reforms in the financial sector (Financial Sector Reforms).      |
| Suresh Tendulkar Committee   | 2009              | New estimate of poverty on the basis of monthly per capita consumption expenditure.       |
| Malegam Committee            | 2010              | Regulation of micro-finance institutions.                                                 |
| Damodaran Committee          | 2011              | Improvement of customer service in the banking sector (Customer Service in Banks).        |
| Nachiket Mor Committee       | 2013              | Financial inclusion and universal financial access.                                       |
|                              |                   |                                                                                           |

| Name of the Committee         | Year of Formation | Related Sector / Main Subject (Core Subject)                                                                 |
|-------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| Parthasarathi Shome Committee | 2013              | Reform in tax administration and GAAR (General Anti-Avoidance Rules).                                        |
| Dr. Urjit Patel Committee     | 2014              | Strengthening the monetary policy framework and inflation targeting based on the Consumer Price Index (CPI). |
| P. J. Nayak Committee         | 2014              | Governance of public sector banks and bank boards.                                                           |
| Shanta Kumar Committee        | 2014              | Restructuring of the Food Corporation of India (FCI) and reform in the grain procurement system.             |
| Deepak Mohanty Committee      | 2015              | Medium-term practical action plan for financial inclusion.                                                   |
| N. K. Singh Committee         | 2016              | Review of the FRBM Act and a new framework of fiscal responsibility.                                         |
| Bimal Jalan Committee         | 2018              | Economic Capital Framework of the RBI and transfer of surplus.                                               |

### > Important Curves and Coefficients of Economics

### > Important Curves and Coefficients of Economics (Economic Curves & Indices)

| Name of the Curve / Coefficient | Shape (if applicable) | Field of Measurement / Basic Relationship (Core Logic)                                                                                                              |
|---------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lorenz Curve                    | Graphical line        | A curve that studies the <b>disparity (inequality) of wealth and income</b> in an economy.                                                                          |
| Gini Coefficient                | Value between 0 and 1 | It is the mathematical form of the Lorenz curve, which shows the <b>precise measure of income disparity</b> (0 = complete equality, 1 = complete inequality).       |
| Kuznets Curve                   | Inverted 'U' shape    | It shows that as a country's GDP (per capita income) increases, <b>economic inequality rises at first</b> , but after a certain point inequality begins to decline. |
| Environmental Kuznets Curve     | Inverted 'U' shape    | It shows that in the initial stage of economic growth (GDP) <b>environmental damage increases</b> , but after a peak point the damage begins to decline.            |
| Phillips Curve                  | Negative slope        | Shows the <b>inverse (opposite) relationship between the rates of unemployment and inflation</b> . When inflation rises, unemployment falls.                        |

| Name of the Curve / Coefficient | Shape (if applicable) | Field of Measurement / Basic Relationship (Core Logic)                                                                                                                                            |
|---------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Laffer Curve</b>             | Inverted 'U' shape    | Shows the relationship between tax rates and the total tax revenue obtained from them. <b>At first, raising taxes increases revenue</b> , but after a point raising taxes causes revenue to fall. |

### > Various Deficits of the Economy

### > Various Deficits of the Economy (Budgetary Deficits Formulas)

| Type of Deficit        | Simple Exam-useful Definition and Explanation                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget Deficit</b>  | The excess of total expenditure (revenue + capital expenditure) over total receipts (revenue + capital receipts).                                            |
| <b>Revenue Deficit</b> | The excess of the government's day-to-day administrative revenue expenditure over its ordinary revenue receipts.                                             |
| <b>Fiscal Deficit</b>  | The excess of total expenditure over total receipts excluding the government's borrowings. It shows the government's entire borrowing liability in the year. |
| <b>Primary Deficit</b> | The deficit obtained by <b>subtracting the interest payments</b> made by the government on past loans from the fiscal deficit.                               |

### > Indices

### > Global Multidimensional Poverty Index (MPI)

Note: This index was started in **2010** by the **United Nations Development Programme (UNDP)** and the **Oxford Poverty and Human Development Initiative (OPHI)**. This index measures deprivation in 10 indicators across 3 dimensions:

| Main Dimensions              | The 10 Indicators Included                                                                               |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>1. Health</b>             | (1) Nutrition, (2) Child mortality                                                                       |
| <b>2. Education</b>          | (3) Years of schooling, (4) School attendance                                                            |
| <b>3. Standard of Living</b> | (5) Cooking fuel, (6) Sanitation (toilet), (7) Drinking water, (8) Electricity, (9) Housing, (10) Assets |

## > Economics: Important Terminology

### > Important Terminology Related to Economics (Economic Core Concepts)

| Terminology                         | Simple Exam-useful Definition and Explanation (Core Meaning)                                                                                                                                                        |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gross Domestic Product (GDP)</b> | The total monetary value of all final goods and services produced within the geographical/domestic boundary of a country in a given financial year.                                                                 |
| <b>Inflation</b>                    | A continuous rise in the general prices of goods and services in the market, which indicates a fall in the purchasing power of money.                                                                               |
| <b>Deflation</b>                    | A continuous fall in the general prices of goods and services in the market, which usually occurs during a decline in demand and an economic recession.                                                             |
| <b>Monetary Policy</b>              | The policy of the country's central bank (such as the RBI in India) to influence liquidity through control of the money supply and interest rates in the economy.                                                   |
| <b>Fiscal Policy</b>                | The policy of the country's government to influence and run the economy through taxation, public expenditure (government spending) and public debt.                                                                 |
| <b>Balance of Payments (BoP)</b>    | A systematic record of all kinds of international economic/monetary transactions (imports, exports, financial flows) of the residents of a country with the rest of the world over a given period (usually 1 year). |

## > Census 2011

### > Exact Figures and Administrative Facts of Census 2011

|                                      |                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------|
| <b>Historical sequence</b>           | <b>It is the 15th census of India and the 7th census of independent India.</b>   |
| <b>Census Commissioner</b>           | <b>Dr. C. Chandramouli.</b>                                                      |
| <b>Statutory basis</b>               | Conducted under the provisions of the <b>Census Act, 1948.</b>                   |
| <b>Reference Date</b>                | <b>1 March 2011.</b>                                                             |
| <b>Total population of India</b>     | <b>121.09 crore (about 17.5% of the world's total population).</b>               |
| <b>Gender division</b>               | * <b>Male population:</b> 62.32 crore<br>* <b>Female population:</b> 58.76 crore |
| <b>Decadal growth rate (2001–11)</b> | A growth of <b>17.64%</b> was recorded.                                          |
| <b>Population density</b>            | <b>382 persons per sq km.</b>                                                    |
| <b>Overall sex ratio</b>             | <b>943</b> females per 1000 males.                                               |
| <b>Child sex ratio (0–6 years)</b>   | <b>919</b> females per 1000 males.                                               |

|                                    |                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Historical sequence</b>         | <b>It is the 15th census of India and the 7th census of independent India.</b>                                       |
| <b>Literacy rate (total)</b>       | <b>74.04%</b>                                                                                                        |
| <b>Literacy rate (gender-wise)</b> | * <b>Male literacy:</b> 82.14%<br>* <b>Female literacy:</b> 65.46%                                                   |
| <b>Rural versus urban ratio</b>    | * <b>Rural population:</b> 83.37 crore ( <b>68.84%</b> )<br>* <b>Urban population:</b> 37.71 crore ( <b>31.16%</b> ) |
| <b>Urban growth (fact)</b>         | Compared to 2001, an increase of <b>about 9.1 crore</b> was recorded in the urban population in 2011.                |

### › State-wise Demographic Categories (Top 3 versus Bottom 3)

| <b>Demographic Indicator</b>                 | <b>Top 3 States / UTs (Descending Order)</b>                                                    | <b>Bottom 3 States / UTs (Ascending Order)</b>                                               |
|----------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Total population</b>                      | 1. <b>Uttar Pradesh</b> (19.98 crore)<br>2. Maharashtra (11.24 crore)<br>3. Bihar (10.41 crore) | 1. <b>Sikkim</b> (6.11 lakh)<br>2. Mizoram (10.97 lakh)<br>3. Arunachal Pradesh (13.84 lakh) |
| <b>Population growth rate</b>                | 1. <b>Meghalaya</b> (27.82%)<br>2. Arunachal Pradesh (25.92%)<br>3. Bihar (25.42%)              | 1. <b>Nagaland</b> (–0.58% - negative growth)<br>2. Kerala (4.91%)<br>3. Goa (8.23%)         |
| <b>Population density</b><br>(persons/sq km) | 1. <b>Bihar</b> (1,102)<br>2. West Bengal (1,028)<br>3. Kerala (859)                            | 1. <b>Arunachal Pradesh</b> (17)<br>2. Mizoram (52)<br>3. Sikkim (86)                        |
| <b>Sex ratio</b>                             | 1. <b>Kerala</b> (1,084)<br>2. Tamil Nadu (996)<br>3. Andhra Pradesh (993)                      | 1. <b>Haryana</b> (879)<br>2. Jammu-Kashmir (889)<br>3. Sikkim (890)                         |
| <b>Total literacy rate</b>                   | 1. <b>Kerala</b> (93.91%)<br>2. Lakshadweep (91.85%)<br>3. Mizoram (91.33%)                     | 1. <b>Bihar</b> (61.80%)<br>2. Arunachal Pradesh (65.38%)<br>3. Rajasthan (66.11%)           |
| <b>Female literacy rate</b>                  | 1. <b>Kerala</b> (91.98%)<br>2. Mizoram (89.40%)<br>3. Lakshadweep (87.95%)                     | 1. <b>Rajasthan</b> (52.12%)<br>2. Bihar (51.50%)<br>3. Jharkhand (55.42%)                   |

### › Child Sex Ratio (0-6 Age Group)

| <b>Geographical Unit</b> | <b>Child Sex Ratio Value</b> | <b>Special Exam-useful Fact</b>                      |
|--------------------------|------------------------------|------------------------------------------------------|
| <b>All-India average</b> | <b>919</b>                   | Considerably lower than the overall sex ratio (943). |
|                          |                              |                                                      |

| Geographical Unit | Child Sex Ratio Value          | Special Exam-useful Fact                                   |
|-------------------|--------------------------------|------------------------------------------------------------|
| Highest (first)   | <b>Arunachal Pradesh (972)</b> | The situation is better in tribal areas.                   |
| Highest (second)  | Mizoram (970)                  | The second top state of the North-East.                    |
| Lowest (bottom 1) | <b>Haryana (834)</b>           | The most backward state of the country in child sex ratio. |
| Lowest (bottom 2) | Punjab (846)                   | The second lowest after Haryana.                           |
| Lowest (bottom 3) | Jammu-Kashmir (862)            | A worrying decline in the border hill states.              |

### > Urbanisation Infrastructure

| Urbanisation Parameter                         | Actual Figures / State (Exam Points)                              |
|------------------------------------------------|-------------------------------------------------------------------|
| Urban population (total)                       | 37.71 crore ( <b>31.16%</b> )                                     |
| Rural population (total)                       | 83.37 crore ( <b>68.84%</b> )                                     |
| State with the largest urban population        | <b>Maharashtra</b> (first in terms of number/quantity).           |
| State with the highest urbanisation percentage | <b>Goa (62.17%)</b> (first in terms of ratio/percentage).         |
| State with the lowest urbanisation percentage  | <b>Himachal Pradesh (10.04%)</b> (i.e., 89.96% rural population). |
| Total towns / urban units (2011)               | <b>7,935</b> units.                                               |
| Statutory Towns                                | <b>4,041</b>                                                      |
| Census Towns                                   | <b>3,894</b>                                                      |

### > Religious Census 2011

| Religious Community / Faith | Total Population (in crore / lakh) | Share in Total Population (per cent) |
|-----------------------------|------------------------------------|--------------------------------------|
| Hindu                       | 96.63 crore                        | <b>79.8%</b>                         |
| Muslim                      | 17.22 crore                        | <b>14.2%</b>                         |
| Christian                   | 2.78 crore                         | 2.3%                                 |
| Sikh                        | 2.08 crore                         | 1.7%                                 |



| Religious Community / Faith | Total Population (in crore / lakh) | Share in Total Population (per cent) |
|-----------------------------|------------------------------------|--------------------------------------|
| Buddhist                    | 84.43 lakh                         | 0.7%                                 |
| Jain                        | 44.52 lakh                         | 0.4%                                 |
| Other religions and faiths  | 79.38 lakh                         | 0.7%                                 |
| Religion not stated         | 28.67 lakh                         | 0.2%                                 |

### > Social Categories: Scheduled Castes (SC) and Scheduled Tribes (ST)

| Social Category       | Total Population (in the country) | Per cent of Total Population | First State in terms of Population |
|-----------------------|-----------------------------------|------------------------------|------------------------------------|
| Scheduled Castes (SC) | 20.14 crore                       | 16.63%                       | Uttar Pradesh (highest in number)  |
| Scheduled Tribes (ST) | 10.45 crore                       | 8.63%                        | Madhya Pradesh (highest in number) |

08

# Uttar Pradesh Special

UTTAR PRADESH SPECIAL

## > Uttar Pradesh: Polity and Administrative Facts

## > Uttar Pradesh: Historical Capitals, Naming, Formation and Reorganisation

| Subject / Event                                    | Period / Date                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sequence of the state's capitals                   | <ul style="list-style-type: none"> <li>• <b>Agra</b> (from 1836)</li> <li>• <b>Prayagraj</b> (from 1858)</li> <li>• <b>Lucknow (partially)</b> (from 1921)</li> <li>• <b>Lucknow (fully)</b> (from 1935)</li> </ul>                                                                                                 |
| Change of the state's name                         | <ul style="list-style-type: none"> <li>• North-Western Provinces (<b>1836</b>)</li> <li>• United Provinces of Agra and Oudh (<b>1877</b>)</li> <li>• United Provinces (<b>1937</b>)</li> <li>• Present Uttar Pradesh (<b>24 January 1950</b>) - <i>(that is why UP Day is celebrated on 24 January).</i></li> </ul> |
| Division of the state & formation of 'Uttarakhand' | <b>9 November, 2000</b> (Uttaranchal (present Uttarakhand) was formed by separating <b>13 hill districts</b> of Uttar Pradesh.)                                                                                                                                                                                     |
| National Capital Region (NCR)                      | A total of <b>8 districts</b> (Meerut, Ghaziabad, Gautam Buddh Nagar, Bulandshahr, Hapur, Baghpat, Muzaffarnagar, Shamli.)                                                                                                                                                                                          |

## > State Symbols of Uttar Pradesh

| State Category | Name of the Symbol                                     | Special Description                                  |
|----------------|--------------------------------------------------------|------------------------------------------------------|
| State emblem   | Ganga-Yamuna confluence, a pair of fish, bow and arrow | Arranged within a circle                             |
| State tree     | <b>Ashoka</b>                                          | -                                                    |
| State animal   | <b>Barasingha</b>                                      | Swamp deer                                           |
| State bird     | <b>Sarus crane</b>                                     | -                                                    |
| State flower   | <b>Palash / Tesu</b>                                   | It is also called the ' <b>Flame of the Forest</b> ' |
| State sport    | <b>Hockey</b>                                          | -                                                    |

## > Uttar Pradesh: Polity and Administrative Facts

| Administrative Unit / Indicator                | Number of Seats / Description                                         | Special Exam-useful Fact                                             |
|------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------|
| Form of the state legislature                  | <b>Bicameral</b>                                                      | Both the Legislative Assembly and the Legislative Council exist.     |
| Lok Sabha seats                                | <b>80</b>                                                             | The state with the most Lok Sabha seats in the country.              |
| Rajya Sabha seats                              | <b>31</b>                                                             | The state with the most Rajya Sabha seats in the country.            |
| Total members of the Legislative Assembly (LA) | <b>403</b>                                                            | -                                                                    |
| Total members of the Legislative Council (LC)  | <b>100</b>                                                            | -                                                                    |
| Formation of the first Legislative Assembly    | <b>July 1937</b>                                                      | At the time of the pre-independence United Provinces.                |
| First President's Rule                         | 25 February 1968 to 26 February 1969                                  | The state's first period of President's Rule.                        |
| Assembly Speaker the most times                | <b>Shri Atmaram Govind Kher</b> and <b>Shri Keshari Nath Tripathi</b> | Both leaders were elected Speaker <b>3 times each</b> .              |
| District with the most Assembly seats          | <b>Prayagraj (12 seats)</b>                                           | The number of seats is the highest because of the larger population. |
| Districts with the fewest Assembly seats       | <b>Shravasti, Mahoba, Chitrakoot (2 seats each)</b>                   | -                                                                    |

## > Uttar Pradesh: Geography

### > Geographical Location of Uttar Pradesh

| Geographical Indicator     | Description / Figures                                                        |
|----------------------------|------------------------------------------------------------------------------|
| Total area                 | <b>2,40,928 square kilometres</b> (about <b>7.33%</b> of India's total area) |
| Longitudinal extent        | From <b>77°3' East longitude</b> to <b>84°39' East longitude</b>             |
| East-west length           | <b>650 km</b>                                                                |
| North-south width          | <b>240 km</b>                                                                |
| Total bordering states/UTs | <b>9</b> (8 states and 1 Union Territory - Delhi)                            |

| Geographical Indicator     | Description / Figures                                               |
|----------------------------|---------------------------------------------------------------------|
| States adjoining Sonbhadra | <b>4 states</b> (Madhya Pradesh, Chhattisgarh, Jharkhand and Bihar) |

### > Districts with Distinctive Borders

| District   | Feature of the Geographical Border                                                                              |
|------------|-----------------------------------------------------------------------------------------------------------------|
| Sonbhadra  | The only district in the country bordering <b>four states</b> (Madhya Pradesh, Bihar, Jharkhand, Chhattisgarh). |
| Saharanpur | Borders <b>three states</b> (Haryana, Himachal Pradesh, Uttarakhand).                                           |
| Lalitpur   | Surrounded on <b>three sides by the state of Madhya Pradesh</b> .                                               |

### > Extreme Points of Area and Population

| Indicator / Category          | District                                               | Additional Description             |
|-------------------------------|--------------------------------------------------------|------------------------------------|
| Largest in area               | Lakhimpur Kheri                                        | 7,680 sq km                        |
| Smallest in area              | Hapur                                                  | 660 sq km                          |
| Easternmost district          | Ballia                                                 | On the Bihar border                |
| Westernmost district          | Shamli                                                 | On the Haryana border              |
| Northernmost district         | Saharanpur                                             | On the Himachal/Uttarakhand border |
| Southernmost district         | Sonbhadra                                              | On the Chhattisgarh/MP border      |
| Top 5 most populous districts | Prayagraj > Moradabad > Ghaziabad > Azamgarh > Lucknow | <i>(in decreasing order)</i>       |

### > Districts Adjoining Neighbouring States

| Neighbouring State | Number of Adjoining Districts           | Names of the Districts                                                                            |
|--------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| Madhya Pradesh     | <b>11 districts</b> ( <i>the most</i> ) | Agra, Etawah, Jalaun, Jhansi, Lalitpur, Mahoba, Banda, Chitrakoot, Prayagraj, Mirzapur, Sonbhadra |
| Bihar              | <b>7 districts</b>                      | Maharajganj, Kushinagar, Deoria, Ballia, Ghazipur, Chandauli, Sonbhadra                           |
| Haryana            | <b>6 districts</b>                      | Saharanpur, Shamli, Baghpat, Gautam Buddh Nagar, Aligarh, Mathura                                 |

| Neighbouring State | Number of Adjoining Districts | Names of the Districts |
|--------------------|-------------------------------|------------------------|
| Rajasthan          | 2 districts                   | Agra, Mathura          |

### > Principal Rivers of Uttar Pradesh and the Cities Situated on their Banks

| Name of the River | Principal Towns / Cities Situated on its Banks                                       |
|-------------------|--------------------------------------------------------------------------------------|
| Ganga river       | Bithoor, Varanasi, Prayagraj, Kannauj, Farrukhabad, Ghazipur, Kanpur, Mirzapur, etc. |
| Yamuna river      | Kaushambi, Mathura-Vrindavan, Agra, Etawah, Hamirpur.                                |
| Gomti river       | <b>Lucknow, Jaunpur, Sultanpur</b> ( <i>asked many times</i> )                       |
| Rapti river       | <b>Gorakhpur</b> , Balrampur, Bahraich, Shravasti.                                   |

### > History, Art and Culture of Uttar Pradesh

#### > Folk Dances of Uttar Pradesh

| Name of the Folk Dance | Related Region / District | Main Feature and Occasion (Key Facts)                                                              |
|------------------------|---------------------------|----------------------------------------------------------------------------------------------------|
| Karma dance            | Mirzapur and Sonbhadra    | A dance performed by the <b>Kharwar tribe</b> group.                                               |
| Charkula dance         | Braj region               | Performed on the festival of Holi with <b>burning lamps</b> arranged in several tiers on the head. |
| Raas dance             | Braj region               | Extremely famous in Braj culture.                                                                  |
| Mayur dance            | Braj region               | A special dance performed during the Raslila.                                                      |
| Jhoola dance           | Braj region               | Performed in temples/areas in the <b>month of Shravan</b> (Sawan).                                 |
| Mohaniya dance         | Bundelkhand               | A dance performed during the festival of <b>Deepawali</b> .                                        |
| Rai dance              | Bundelkhand               | By women <b>as thanksgiving</b> after harvesting the crop.                                         |
| Khayal dance           | Bundelkhand               | On the celebration/auspicious occasion of the <b>birth of a son</b> .                              |
| Shaura / Shaira dance  | Bundelkhand               | By the farmers of Bundelkhand <b>while harvesting crops</b> .                                      |
| Ghoda (horse) dance    | Bundelkhand               | Performed by decorated horses on auspicious occasions.                                             |
| Dhuriya dance          | Bundelkhand               | By men of the potter ( <b>Prajapati</b> ) caste <b>dressed as women</b> .                          |

| Name of the Folk Dance | Related Region / District | Main Feature and Occasion (Key Facts)                                                              |
|------------------------|---------------------------|----------------------------------------------------------------------------------------------------|
| Devi dance             | Bundelkhand               | Associated with the traditional folk rituals of Bundelkhand.                                       |
| Pai Danda dance        | Bundelkhand               | A popular dandiya-like dance performed by Ahirs (Yadavs).                                          |
| Kartik Geet dance      | Bundelkhand               | With devotion in the month of Kartik.                                                              |
| Deepawali dance        | Bundelkhand               | At the time of the Deepawali festival.                                                             |
| Dhobiya dance          | Purvanchal                | An artistic portrayal of the livelihood relationship between the <b>washerman and the donkey</b> . |
| Kathghodwa dance       | Purvanchal                | On auspicious occasions the dancer, seated on an artificial (bamboo) mare, dances inside a circle. |
| Natwari dance          | Purvanchal                | A dance performed by people of the <b>Ahir (Yadav) community</b> .                                 |
| Chaular dance          | Mirzapur and Sonbhadra    | In tribal and local society for good rain/crops.                                                   |
| Dharkahari dance       | Sonbhadra                 | By local tribes on special occasions.                                                              |
| Dhedhiya dance         | Fatehpur                  | Performed with a <b>lamp in a perforated earthen pot</b> on the head.                              |
| Chaurasiya dance       | Jaunpur                   | Mainly by the <b>Kahar community</b> of Jaunpur.                                                   |
| Dheevar dance          | General (state level)     | By people of the <b>Kahar caste</b> on auspicious occasions.                                       |
| Nautanki               | Kanpur, Lucknow, Hathras  | It is a folk drama-dance form, famous in the styles of these cities.                               |

## > Folk Songs and Classical Dance of Uttar Pradesh

### A. Principal Folk Songs

| Folk Song | Main Region                                         | Occasion / Special Context                                   |
|-----------|-----------------------------------------------------|--------------------------------------------------------------|
| Kajri     | Purvanchal (especially Mirzapur, Varanasi, Jaunpur) | Related to the <b>rainy season and Sawan</b>                 |
| Birha     | Eastern Uttar Pradesh                               | Based on separation, love and social life                    |
| Alha      | Bundelkhand                                         | Heroic ballad; description of the valour of <b>Alha-Udal</b> |
|           |                                                     |                                                              |

| Folk Song      | Main Region                                 | Occasion / Special Context                                            |
|----------------|---------------------------------------------|-----------------------------------------------------------------------|
| Sohar          | Whole of Uttar Pradesh                      | On the auspicious occasion of <b>birth and childbirth</b>             |
| Banna-Banni    | Various regions                             | Songs sung on the occasion of <b>marriage</b>                         |
| Chaiti         | Eastern Uttar Pradesh                       | A seasonal song sung in the <b>month of Chaitra</b>                   |
| Phag / Phagua  | Braj and Bundelkhand                        | Related to <b>Holi and the spring season</b>                          |
| Rasiya         | Braj region (Mathura, Vrindavan, Agra)      | Related to Radha-Krishna and the <b>romantic sentiment (shringar)</b> |
| Dhola-Maru     | Western Uttar Pradesh (Rajasthan border)    | A folk ballad of love and separation                                  |
| Jogira         | Purvanchal                                  | Especially on the occasion of <b>Holi</b>                             |
| Lachari        | Eastern Uttar Pradesh                       | By women on marriage and social occasions                             |
| Devi Geet      | Purvanchal, Bundelkhand and the whole of UP | Related to the worship and devotion of gods and goddesses             |
| Kaharwa        | Eastern Uttar Pradesh                       | A song style used in folk dance and folk song                         |
| Nautanki songs | Western and Central Uttar Pradesh           | Folk theatre and traditional entertainment                            |

## B. The Only Classical Dance of Uttar Pradesh: Kathak

- **Original gharana:** Lucknow gharana.
- **Royal patronage:** It received special encouragement and patronage in the court of **Wajid Ali Shah**, the last Nawab of Awadh.
- **Technical feature: Ghungroos** have a special place in it. Many kinds of subtle sounds are produced on the stage by the tapping of the feet (footwork).
- **Principal artistes (dancers):** Pandit Birju Maharaj, Lachhu Maharaj, Sitara Devi, Rama Devi, Shambhu Maharaj, etc.

## C. Principal Music Gharanas and Related Artistes

| Music Gharana   | Principal Artistes / Musicians                                                                        |
|-----------------|-------------------------------------------------------------------------------------------------------|
| Lucknow gharana | Ustad Wajid Hussain Khan, Achhan Maharaj, Shambhu Maharaj, Lachhu Maharaj, Birju Maharaj              |
| Banaras gharana | Sitara Devi, Gopi Krishna, Kishan Maharaj, Anokhelal Mishra, Pandit Rajan Mishra, Pandit Sajan Mishra |

| Music Gharana                | Principal Artistes / Musicians                                                          |
|------------------------------|-----------------------------------------------------------------------------------------|
| Rampur-Sahaswan gharana      | Inayat Hussain Khan, Mushtaq Hussain Khan, Nisar Hussain Khan, Rashid Khan              |
| Kairana gharana              | Abdul Karim Khan, Abdul Wahid Khan, Sawai Gandharva, Bhimsen Joshi ( <i>important</i> ) |
| Agra gharana                 | Faiyaz Khan, Vilayat Hussain Khan, Latafat Hussain Khan, Pandit Jagannathbuwa Purohit   |
| Etawah / Imdadkhani gharana  | Imdad Khan, Inayat Khan, Vilayat Khan, Shahid Parvez                                    |
| Atrauli-Jaipur gharana       | Alladiya Khan, Mogubai Kurdikar, Kishori Amonkar                                        |
| Saharanpur gharana (sarangi) | Ustad Bundu Khan, Ustad Ashiq Ali Khan                                                  |

### > Ancient Gupta-period Temples, Stupas and Inscriptions of Uttar Pradesh

| Historical Heritage / Site | Location (Place)                             | Period / Discoverer                                  | Historical Significance and Facts                                                                                                   |
|----------------------------|----------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Dashavatara Temple         | Deogarh village, <b>Lalitpur</b>             | Gupta period (discovered by Captain Charles Strahan) | It was named by Alexander Cunningham. It is an excellent example of the Gupta-period Panchayatana style.                            |
| Bhitargaon Temple          | <b>Kanpur</b> district                       | <b>Sixth century</b> (Gupta period)                  | It is the <b>oldest surviving</b> Hindu temple built of bricks.                                                                     |
| Dhamek Stupa               | Sarnath, <b>Varanasi</b> (13 km away)        | Mauryan/Gupta period                                 | It marks the 'Deer Park' or <b>Rishipattana</b> , where Mahatma Buddha gave his <b>first sermon</b> (Dharmachakrapravartan).        |
| Bhitari Inscription Pillar | Village Bhitari, Saidpur ( <b>Ghazipur</b> ) | Reign of <b>Skandagupta</b> (about 455-467 CE)       | This pillar inscription gives important information about Skandagupta's victory over the Hunas and the history of the Gupta empire. |

### > Ancient / Historical Names of Principal Cities

| Present Name | Ancient / Historical Name                                |
|--------------|----------------------------------------------------------|
| Varanasi     | Kashi                                                    |
| Kannauj      | Kanyakubja Nagar, Mahodayashri ( <i>Mahodaya Nagar</i> ) |
| Gorakhpur    | Ramgram, Gorakshapur                                     |
| Aligarh      | Harigarh, Koil or Kol                                    |



| Present Name                  | Ancient / Historical Name |
|-------------------------------|---------------------------|
| Kanpur                        | Kanhpur                   |
| Kapilavastu (Siddharth Nagar) | Piprahwa                  |
| Lucknow                       | Lakshmanpur, Lakhanpur    |
| Kushinagar                    | Kushinara                 |
| Mathura                       | Madhupuri                 |

## > Uttar Pradesh Environment

## > Wildlife Sanctuaries and Bird Sanctuaries of Uttar Pradesh

### A. Wildlife Sanctuaries

| Name of the Wildlife Sanctuary   | Related District                             | Special Fact (from the UPPCS exam point of view)                |
|----------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| Chandraprabha Wildlife Sanctuary | Chandauli                                    | The first wildlife sanctuary (1957)                             |
| Hastinapur Wildlife Sanctuary    | Muzaffarnagar, Meerut, Hapur, Amroha, Bijnor | The largest wildlife sanctuary                                  |
| Mahavir Swami Wildlife Sanctuary | Lalitpur                                     | The smallest wildlife sanctuary (1977)                          |
| Kishanpur Wildlife Sanctuary     | Lakhimpur Kheri                              | 1972                                                            |
| Chambal Wildlife Sanctuary       | Etawah and Agra                              | 1989                                                            |
| Kaimur Wildlife Sanctuary        | Mirzapur and Sonbhadra                       | -                                                               |
| Katarniaghat Wildlife Sanctuary  | Bahraich                                     | -                                                               |
| Ranipur Wildlife Sanctuary       | Banda                                        | <i>(it has now also received the status of a tiger reserve)</i> |
| Sohagibarwa Wildlife Sanctuary   | Maharajganj                                  | -                                                               |
| Sohelwa Wildlife Sanctuary       | Gonda, Balrampur, Shravasti                  | -                                                               |
| Turtle Wildlife Sanctuary        | Varanasi                                     | -                                                               |

## > Principal Bird Sanctuaries of Uttar Pradesh (Updated 2026)

| S.No. | Name of the Bird Sanctuary                        | Related District          | Year of Establishment | Main Exam-useful Features and Special Facts                                                                                                                                                                                                            |
|-------|---------------------------------------------------|---------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Shaheed Chandrashekhar Azad Bird Sanctuary</b> | <b>Unnao</b>              | <b>1984</b>           | <ul style="list-style-type: none"> <li>Former name: <b>Nawabganj Bird Sanctuary</b>.</li> <li>It is the <b>first (oldest)</b> bird sanctuary of Uttar Pradesh.</li> </ul>                                                                              |
| 2     | <b>Samaspur Bird Sanctuary</b>                    | <b>Raebareli</b>          | 1987                  | <ul style="list-style-type: none"> <li>A system of 6 interconnected lakes situated in the middle Ganga plain.</li> <li>Also included as a Ramsar site.</li> </ul>                                                                                      |
| 3     | <b>Lakh Bahoshi Bird Sanctuary</b>                | <b>Kannauj</b>            | <b>1988</b>           | <ul style="list-style-type: none"> <li>It is the <b>largest bird sanctuary</b> of Uttar Pradesh by area.</li> </ul>                                                                                                                                    |
| 4     | <b>Sandi Bird Sanctuary</b>                       | <b>Hardoi</b>             | 1990                  | <ul style="list-style-type: none"> <li>It is a natural wetland near the Garra river.</li> <li>Included as a Ramsar site.</li> </ul>                                                                                                                    |
| 5     | <b>Bakhira Bird Sanctuary</b>                     | <b>Sant Kabir Nagar</b>   | 1990                  | <ul style="list-style-type: none"> <li>It is also called the 'Bakhira Wildlife Sanctuary'.</li> <li>It is the <b>largest natural freshwater lake</b> in Purvanchal.</li> </ul>                                                                         |
| 6     | <b>Okhla Bird Sanctuary</b>                       | <b>Gautam Buddh Nagar</b> | 1990                  | <ul style="list-style-type: none"> <li>An urban wetland (urban oasis) situated on the Delhi-Noida border.</li> </ul>                                                                                                                                   |
| 7     | <b>Saman Bird Sanctuary</b>                       | <b>Mainpuri</b>           | 1990                  | <ul style="list-style-type: none"> <li>One of the largest habitats and breeding areas of the state bird, the <b>Sarus Crane</b>.</li> </ul>                                                                                                            |
| 8     | <b>Parvati Arga Bird Sanctuary</b>                | <b>Gonda</b>              | 1990                  | <ul style="list-style-type: none"> <li><b>Current affairs update:</b> It has been declared an <b>Eco-Sensitive Zone (ESZ)</b> by the Ministry of Environment, Forest and Climate Change.</li> <li>Formed of two interconnected oxbow lakes.</li> </ul> |
| 9     | <b>Vijay Sagar Bird Sanctuary</b>                 | <b>Mahoba</b>             | 1990                  | <ul style="list-style-type: none"> <li>Situated on the bank of a picturesque lake in Mahoba district of the Bundelkhand region.</li> </ul>                                                                                                             |
| 10    | <b>Patna Bird Sanctuary</b>                       | <b>Etah</b>               | 1991                  | <ul style="list-style-type: none"> <li><b>Ramsar site update (2026):</b> It has recently been included in the Ramsar list.</li> </ul>                                                                                                                  |
| 11    | <b>Jai Prakash Narayan Bird Sanctuary</b>         | <b>Ballia</b>             | 1991                  | <ul style="list-style-type: none"> <li>Popular/local name: <b>Surha Taal Bird Sanctuary</b>.</li> <li><b>Current affairs update (June 2026):</b> It has been declared <b>India's 100th Ramsar site</b>.</li> </ul>                                     |

| S.No. | Name of the Bird Sanctuary          | Related District | Year of Establishment | Main Exam-useful Features and Special Facts                                                                                                                                                                 |
|-------|-------------------------------------|------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12    | Sur Sarovar Bird Sanctuary          | Agra             | 1991                  | <ul style="list-style-type: none"> <li>It is also known as <b>Keetham Lake</b>.</li> <li>India's famous 'Sloth Bear Rescue Facility' is also here.</li> </ul>                                               |
| 13    | Dr. Bhimrao Ambedkar Bird Sanctuary | Pratapgarh       | 2003                  | <ul style="list-style-type: none"> <li>It was formerly called the <b>Bainti Taal Bird Sanctuary</b>.</li> </ul>                                                                                             |
| 14    | Shekha Jheel Bird Sanctuary         | Aligarh          | 2016 (notified)       | <ul style="list-style-type: none"> <li><b>UP's smallest bird sanctuary by area.</b></li> <li><b>Current affairs update (April 2026):</b> It has been designated <b>India's 99th Ramsar site</b>.</li> </ul> |

### > Principal Lakes, Taals and Reservoirs of Uttar Pradesh

| District      | Lake / Taal / Kund / Reservoir Located                                  |
|---------------|-------------------------------------------------------------------------|
| Pilibhit      | <b>Phulhar Lake</b> ( <i>source of the Gomti river</i> )                |
| Sonbhadra     | <b>Govind Ballabh Pant Sagar</b> ( <i>the largest artificial lake</i> ) |
| Agra          | Keetham Lake (Sur Sarovar)                                              |
| Mathura       | Noh Lake, Mansi Ganga Kund-taal, Radha and Shyam Lake                   |
| Unnao         | Nawabganj Lake, Kundra Samandar                                         |
| Kanpur Nagar  | Moti Jheel                                                              |
| Jhansi        | Barua Sagar and Lakshmi Taal                                            |
| Budaun        | Sagar Taal                                                              |
| Meerut        | Ramtaal                                                                 |
| Muzaffarnagar | Shukra Taal                                                             |
| Mahoba        | Bela Sagar Lake, Madan Sagar                                            |
| Fatehpur      | Morai Taal                                                              |
| Raebareli     | Samaspur Lake, Visaidha Taal                                            |
| Pratapgarh    | Bainti or Naiya Lake                                                    |
| Mirzapur      | Tanda Dari Taal and Sirsi Reservoir                                     |
| Varanasi      | Aundhi Taal                                                             |
| Ballia        | Surha Taal                                                              |

| District         | Lake / Taal / Kund / Reservoir Located |
|------------------|----------------------------------------|
| Sultanpur        | Raja ka Bandh, Bhojpur Taal            |
| Gorakhpur        | Ramgarh Taal                           |
| Sant Kabir Nagar | Bakhira Lake                           |
| Ambedkar Nagar   | Darwan Lake                            |
| Gonda            | Parvati and Arga Lake                  |
| Bahraich         | Chitaura Lake                          |
| Lucknow          | Karela Lake                            |
| Kannauj          | Lakh Lake, Bahoshi Lake, Deoria Taal   |
| Shahjahanpur     | Bada Taal (oxbow lake)                 |
| Bareilly         | Leelaur Lake                           |
| Rampur           | Gaur Lake                              |
| Aligarh          | Shekha Lake                            |

### ➤ Updated List of the 13 Ramsar Sites of Uttar Pradesh (2026)

At present Uttar Pradesh has a total of 13 Ramsar Sites. In **2026, 3 new sites were added in UP (Patna Bird Sanctuary, Shekha Jheel Bird Sanctuary and Jai Prakash Narayan Bird Sanctuary / Surha Taal)**, due to which this list has now **increased from 10 to 13**.

| S.No. | Name of the Ramsar Site                                  | Related District           | Year of Declaration | Main Exam-useful Features / Special Facts                                                                                                                    |
|-------|----------------------------------------------------------|----------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Upper Ganga River</b><br>(Brijghat to Narora stretch) | Bulandshahr, Hapur, Amroha | <b>2005</b>         | <ul style="list-style-type: none"> <li>UP's <b>first and largest</b> Ramsar site.</li> <li>A major habitat of the Ganges dolphin and the gharial.</li> </ul> |
| 2     | <b>Nawabganj Bird Sanctuary</b>                          | Unnao                      | 2019                | <ul style="list-style-type: none"> <li>New name: <b>Shaheed Chandrashekhar Azad Bird Sanctuary</b>.</li> <li>A monsoon-fed natural oxbow lake.</li> </ul>    |
| 3     | <b>Sandi Bird Sanctuary</b>                              | Hardoi                     | 2019                | <ul style="list-style-type: none"> <li>A famous resting place for migratory birds (especially the common teal).</li> </ul>                                   |
| 4     | <b>Samaspur Bird Sanctuary</b>                           | Raebareli                  | 2019                | <ul style="list-style-type: none"> <li>Situated in Salon tehsil, part of the middle Ganga plain.</li> </ul>                                                  |
| 5     | <b>Saman Bird Sanctuary</b>                              | Mainpuri                   | 2019                | <ul style="list-style-type: none"> <li>A seasonal oxbow lake.</li> <li>A major breeding site of the <b>Sarus Crane</b>.</li> </ul>                           |

| S.No. | Name of the Ramsar Site            | Related District          | Year of Declaration | Main Exam-useful Features / Special Facts                                                                                                   |
|-------|------------------------------------|---------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | Parvati Arga Bird Sanctuary        | Gonda                     | 2019                | • A unique system of two interconnected oxbow lakes.                                                                                        |
| 7     | Sarsai Nawar Jheel                 | Etawah                    | 2019                | • The densest population and breeding of the <b>Sarus Crane</b> in India take place here.                                                   |
| 8     | Sur Sarovar<br>(Keetham Lake)      | Agra                      | 2020                | • A man-made reservoir situated on the bank of the Yamuna river.                                                                            |
| 9     | Haiderpur Wetland                  | Muzaffarnagar -<br>Bijnor | 2021                | • A <b>man-made wetland</b> formed by the construction of the Madhya Ganga Barrage.<br>• Situated within the Hastinapur Wildlife Sanctuary. |
| 10    | Bakhira Wildlife Sanctuary         | Sant Kabir Nagar          | 2021                | • Uttar Pradesh's <b>largest natural freshwater lake</b> .                                                                                  |
| 11    | Patna Bird Sanctuary               | Etah                      | <b>January 2026</b> | • <b>Latest addition</b> . In the original wildlife sanctuary category it is UP's smallest bird sanctuary.                                  |
| 12    | Shekha Jheel Bird Sanctuary        | Aligarh                   | <b>April 2026</b>   | • <b>Latest addition</b> . <b>UP's smallest Ramsar site</b> by area (only 40 hectares).                                                     |
| 13    | Jai Prakash Narayan Bird Sanctuary | Ballia                    | <b>June 2026</b>    | • Popular name: <b>Surha Taal</b> .<br>• It has the distinction of becoming <b>India's 100th Ramsar site</b> .                              |

## > Uttar Pradesh: Forest Area

### > Top and Bottom Districts in Forest Area

(Note: According to the Forest Status Report, figures are recorded in area in sq km)

| Category (Forest Area)                           | Name of the District                    | Forest Area (sq km)      |
|--------------------------------------------------|-----------------------------------------|--------------------------|
| Minimum forest area <i>(in increasing order)</i> | (1) <b>Sant Ravidas Nagar (Bhadohi)</b> | <b>3.12</b>              |
|                                                  | (2) Mau                                 | 11.00                    |
|                                                  | (3) Mainpuri                            | 13.64                    |
|                                                  | (4) Sant Kabir Nagar                    | 14.00                    |
|                                                  | (5) Deoria                              | 15.21                    |
| Maximum forest area <i>(in decreasing order)</i> | (1) <b>Sonbhadra</b>                    | <b>2540.29 (highest)</b> |
|                                                  | (2) Lakhimpur Kheri                     | 1273.06                  |
|                                                  | (3) Mirzapur                            | 803.73                   |
|                                                  | (4) Pilibhit                            | 687.11                   |

## > Uttar Pradesh: Economy

## > Uttar Pradesh: Agriculture, Economy and Energy Resources

### > Economic and Agricultural Indicators

| Indicator                                      | UP's Figure / Rank                                      | Special Remark (for the Exam)                    |
|------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|
| Contribution to the country's GDP              | About <b>8%</b>                                         | Among the leading states at the national level   |
| Contribution of agriculture to the state's GDP | <b>26%</b>                                              | Strong position of the primary sector            |
| Growth rate of agriculture and allied sector   | <b>10%</b>                                              | -                                                |
| Rank in agricultural exports                   | <b>Third place</b>                                      | -                                                |
| First place in production                      | <b>Milk production</b> and <b>sugarcane cultivation</b> | Uttar Pradesh is number 1 in the country in both |

### > Energy Resources

| Project / Agency / Centre                              | Location / Year                                                           | Special Fact                                      |
|--------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------|
| Formation of the State Electricity Board               | <b>1959</b>                                                               | -                                                 |
| Establishment of the Electricity Regulatory Commission | <b>1998</b>                                                               | -                                                 |
| New and Renewable Energy Development Agency            | <b>1983</b>                                                               | -                                                 |
| Order of total electricity generation                  | <b>Thermal power &gt; Hydro power &gt; Nuclear power &gt; Solar power</b> | <i>Remember the chronology (asked repeatedly)</i> |
| Gas-based projects of NTPC                             | <b>3 projects</b>                                                         | Auraiya, Aonla, Dadri                             |
| Oldest thermal project                                 | <b>Harduaganj Thermal Power Station</b>                                   | Established in 1942                               |
| The only nuclear power station                         | <b>Narora</b> (Bulandshahr)                                               | Established: <b>1991-92</b>                       |

## > Principal Thermal Power Plants of Uttar Pradesh

| Name of the Thermal Power Station / Project | Location (Related District) | Fuel / Specific Type                            |
|---------------------------------------------|-----------------------------|-------------------------------------------------|
| Dadri Thermal Power Station                 | Gautam Buddh Nagar          | <b>Gas-based</b> project                        |
| Aonla Thermal Power Project                 | Bareilly                    | <b>Gas-based</b> project                        |
| Harduaganj Thermal Power Project            | Aligarh                     | One of the oldest thermal projects of the state |
| Anpara Thermal Power Station                | Sonbhadra                   | -                                               |
| Singrauli Thermal Power Project             | Sonbhadra                   | -                                               |
| Meja Thermal Power Station                  | Prayagraj                   | -                                               |
| Tanda Thermal Power Project                 | Ambedkar Nagar              | -                                               |

## > Expressways of Uttar Pradesh (Length & Map Data)

| Name of the Expressway               | Length        | Special Fact / Current Status                                 |
|--------------------------------------|---------------|---------------------------------------------------------------|
| Ganga Expressway                     | <b>594 km</b> | At present the longest under-construction/proposed expressway |
| Gorakhpur-Siliguri Expressway        | 519 km        | Inter-state connectivity                                      |
| Ghaziabad - Kanpur Expressway        | 380 km        | -                                                             |
| Purvanchal Expressway                | 341 km        | -                                                             |
| Agra-Lucknow Expressway              | 302 km        | -                                                             |
| Bundelkhand Expressway               | 296 km        | -                                                             |
| Delhi-Saharanpur-Dehradun Expressway | 210 km        | -                                                             |
| Yamuna Expressway                    | <b>165 km</b> | The first expressway of Uttar Pradesh                         |
| Ghazipur-Ballia-Manjhihat Expressway | 117 km        | -                                                             |
| Delhi-Meerut Expressway              | 96 km         | -                                                             |
| Gorakhpur Link Expressway            | 91 km         | -                                                             |
| Lucknow - Kanpur Expressway          | 63 km         | -                                                             |
| Noida-Greater Noida Expressway       | 25 km         | -                                                             |

## > Principal Industrial Parks and Cities of Uttar Pradesh (Parks & Cities)

| Name of the Park / City           | Location (Place)                 |
|-----------------------------------|----------------------------------|
| Defence Park                      | Kanpur, Agra, Jhansi and Lucknow |
| Aerospace Park                    | Lucknow, Agra, Barabanki         |
| Leather Park                      | Agra                             |
| Leather Technology Park           | Banthar ( <b>Unnao</b> )         |
| Textile and Hosiery Park          | Kanpur                           |
| Mega Food Park                    | Baheri ( <b>Bareilly</b> )       |
| Agro Park                         | Varanasi                         |
| Biotechnology Park                | Lucknow                          |
| Regional Science City             | Lucknow                          |
| Knowledge Park                    | Greater Noida                    |
| Software Technology Park of India | Noida                            |
| Night Safari Park                 | Greater Noida                    |
| Plastic City                      | Dibiyapur ( <b>Auraiya</b> )     |
| Saraswati Hi-Tech City            | Prayagraj                        |
| Tronica City                      | Ghaziabad                        |
| Trans Delhi Signature City        | Ghaziabad                        |
| Wave City                         | Ghaziabad                        |
| NIKIDA City                       | Between Kanpur and Unnao         |
| Trans Ganga City                  | Unnao                            |
| Tourism Theme Park                | Agra                             |

## > Principal Institutions in Uttar Pradesh

### > Principal National Research Institutes

| Name of the Institute                         | Location (City) |
|-----------------------------------------------|-----------------|
| Indian Institute of Vegetable Research (IIVR) | Varanasi        |
|                                               |                 |



| Name of the Institute                                  | Location (City)      |
|--------------------------------------------------------|----------------------|
| Indian Institute of Handloom Technology (IIHT)         | Chaukaghat, Varanasi |
| Indian Institute of Pulses Research (IIPR)             | Kanpur               |
| National Sugar Institute (NSI)                         | Kanpur               |
| Indian Institute of Sugarcane Research (IISR)          | Lucknow              |
| Indian Veterinary Research Institute (IVRI)            | Izzatnagar, Bareilly |
| Indian Grassland and Fodder Research Institute         | Jhansi               |
| Indian Grain Storage Management and Research Institute | Hapur                |

### > Central Universities of Uttar Pradesh

| Name of the University                           | Location  | Year of Establishment / Special Note                      |
|--------------------------------------------------|-----------|-----------------------------------------------------------|
| University of Allahabad                          | Prayagraj | <b>1887</b> ( <i>the oldest university of the state</i> ) |
| Banaras Hindu University (BHU)                   | Varanasi  | 1916                                                      |
| Aligarh Muslim University (AMU)                  | Aligarh   | 1920                                                      |
| Babasaheb Bhimrao Ambedkar University            | Lucknow   | 1996                                                      |
| Rajiv Gandhi National Aviation University        | Raebareli | Established under the 2013 Act                            |
| Rani Lakshmi Bai Central Agricultural University | Jhansi    | 2014                                                      |

### D. Principal Cultural / Historical Institutions of Uttar Pradesh

| Name of the Institution      | Location (City) | Year of Establishment / Special Fact                                                                                                 |
|------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| State Museum                 | Lucknow         | <b>1863</b> ( <i>the oldest museum</i> )                                                                                             |
| College of Arts and Crafts   | Lucknow         | 1911                                                                                                                                 |
| Bharat Kala Bhavan           | Varanasi        | 1920                                                                                                                                 |
| Bhatkhande Music Institute   | Lucknow         | <b>1926</b> ( <i>former name: Marris College of Music; name changed in 1966; received the status of a state university in 2021</i> ) |
| Hindustani Academy           | Prayagraj       | 1927                                                                                                                                 |
| Uttar Pradesh State Archives | Prayagraj       | 1949                                                                                                                                 |

| Name of the Institution                 | Location (City) | Year of Establishment / Special Fact |
|-----------------------------------------|-----------------|--------------------------------------|
| Motilal Nehru Children's Museum         | Lucknow         | 1957                                 |
| State Lalit Kala Akademi                | Lucknow         | 1962                                 |
| Uttar Pradesh Sangeet Natak Akademi     | Lucknow         | 1963                                 |
| Uttar Pradesh Urdu Academy              | Lucknow         | 1972                                 |
| Bhartendu Natya Akademi                 | Lucknow         | 1975                                 |
| Uttar Pradesh Film Corporation          | -               | 1975                                 |
| Uttar Pradesh Hindi Sansthan            | Lucknow         | 1976                                 |
| Uttar Pradesh Sanskrit Sansthanam       | Lucknow         | 1976                                 |
| Ayodhya Shodh Sansthan                  | Ayodhya         | 1988                                 |
| International Ramkatha Museum           | Ayodhya         | 1988                                 |
| National Kathak Institute               | Lucknow         | 1988-89                              |
| Folk Art Museum                         | Lucknow         | 1989                                 |
| Uttar Pradesh Jain Vidya Shodh Sansthan | Lucknow         | 1990                                 |
| District Museum                         | Sultanpur       | 1990                                 |
| Uttar Pradesh Bhasha Sansthan           | Lucknow         | 1994                                 |
| Tribal and Folk Art Culture Museum      | Lucknow         | 1996                                 |
| State Freedom Struggle Museum           | Meerut          | 1997                                 |
| Uttar Pradesh Sindhi Academy            | Lucknow         | 1997                                 |
|                                         |                 |                                      |

| Name of the Institution                   | Location (City) | Year of Establishment / Special Fact |
|-------------------------------------------|-----------------|--------------------------------------|
| Uttar Pradesh Punjabi Academy             | Lucknow         | 1999                                 |
| Museum of Kushan and Gupta Period Culture | Mathura         | The only specialised museum          |
| Aryabhatta Planetarium                    | Rampur          | -                                    |
| Indira Gandhi Planetarium                 | Lucknow         | -                                    |

### › Scheduled Castes (SC) and Scheduled Tribes (ST) in Uttar Pradesh

#### › Principal Tribes of Uttar Pradesh

| Tribe / Group                                                            | Related Main Area of Habitation (District)                                                                                                                                         |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gond group</b> ( <i>Gond, Dhuria, Nayak, Ojha, Pathari, Rajgond</i> ) | Maharajganj, Siddharth Nagar, Basti, Gorakhpur, Deoria, Mau, Azamgarh, Jaunpur, Ballia, Ghazipur, Varanasi, Mirzapur, Sonbhadra, Sant Kabir Nagar, Kushinagar, Chandauli, Bhadohi. |
| <b>Kharwar group</b>                                                     | Deoria, Ballia, Ghazipur, Varanasi, Sonbhadra.                                                                                                                                     |
| <b>Tharu</b>                                                             | Gorakhpur and Balrampur (mainly the <b>Terai region</b> ).                                                                                                                         |
| <b>Buksa / Bhoksa</b>                                                    | Bijnor.                                                                                                                                                                            |
| <b>Mahigir</b>                                                           | Bijnor.                                                                                                                                                                            |
| <b>Saharia</b>                                                           | <b>Lalitpur</b> ( <i>asked repeatedly</i> ).                                                                                                                                       |
| <b>Parahiya, Baiga, Agariya, Patari, Bhuiya / Bhuiyan</b>                | <b>Sonbhadra</b> ( <i>all these are concentrated in Sonbhadra</i> ).                                                                                                               |
| <b>Pankha / Panika</b>                                                   | Sonbhadra and Mirzapur.                                                                                                                                                            |
| <b>Chero</b>                                                             | Sonbhadra and Varanasi.                                                                                                                                                            |

### › Scheduled Castes (SC) and Scheduled Tribes (ST) in Uttar Pradesh

| Indicator / Category                        | UP's Figure / District | Rank at the National Level / Special Note                                                  |
|---------------------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| <b>SC % of total population</b>             | <b>20.69%</b>          | Share of the Dalit population in the total population of the state.                        |
| <b>Rank in the country in SC population</b> | <b>First place</b>     | The highest share, <b>16.6%</b> , of India's total number of Scheduled Castes lives in UP. |

| Indicator / Category                    | UP's Figure / District | Rank at the National Level / Special Note                            |
|-----------------------------------------|------------------------|----------------------------------------------------------------------|
| District with the highest SC population | Sitapur                | -                                                                    |
| District with the lowest SC population  | Baghpat                | -                                                                    |
| ST % of total population                | 0.6%                   | Share of the tribal population in the total population of the state. |
| District with the highest ST population | Sonbhadra              | -                                                                    |
| District with the lowest ST population  | Baghpat                | Baghpat has the lowest population of both SC and ST.                 |
| Districts without Scheduled Tribes      | Ayodhya and Jalaun     | No Scheduled Tribe (ST) lives in these two districts.                |

## > 'Firsts' in Uttar Pradesh (First in UP)

### A. 'Firsts' in Uttar Pradesh

| Category / Post / Place                         | Name of the First Person / Place                            |
|-------------------------------------------------|-------------------------------------------------------------|
| First Governor                                  | Sarojini Naidu ( <i>also India's first woman Governor</i> ) |
| First Chief Minister                            | Govind Ballabh Pant                                         |
| First woman Chief Minister                      | Sucheta Kripalani                                           |
| First Chief Justice of the Allahabad High Court | Sir Walter Morgan                                           |
| First Speaker of the Legislative Assembly       | Purushottam Das Tandon                                      |
| First Information Commissioner                  | Mohammad Asghar                                             |
| First Lokayukta                                 | Justice Vishambhar Dayal                                    |
| First state emergency (President's Rule)        | From February 1968 to February 1969                         |
| Chief Minister the most times                   | <b>Mayawati</b> (4 times in all)                            |
| First Ramsar site                               | Upper Ganga River (2005)                                    |
| First National Park                             | <b>Dudhwa National Park</b>                                 |
| First wildlife sanctuary                        | Chandraprabha Wildlife Sanctuary (Chandauli)                |
| First tiger reserve                             | Dudhwa Tiger Reserve                                        |

| Category / Post / Place          | Name of the First Person / Place                |
|----------------------------------|-------------------------------------------------|
| First zoo                        | Nawab Wajid Ali Shah Zoological Garden, Lucknow |
| First UNESCO World Heritage Site | Agra Fort and the Taj Mahal (included in 1983)  |

## B. Firsts in the Field of Education

| Institution / College             | Place / Name                                | Year of Establishment and Special Fact                                  |
|-----------------------------------|---------------------------------------------|-------------------------------------------------------------------------|
| First university                  | <b>University of Allahabad</b>              | 1887                                                                    |
| First engineering college         | Civil Engineers' (college) at Roorkee       | 1847 ( <i>present Uttarakhand</i> )   <b>BHU</b> (1916)                 |
| First medical college             | <b>Sarojini Naidu Medical College</b>       | 1854 (Agra - former name: Thomason College)                             |
| First agricultural college        | <b>G. B. Pant University of Agriculture</b> | Pantnagar ( <i>then UP, present Uttarakhand</i> )                       |
| First Sanskrit college            | Banaras                                     | <b>1791</b> (through the efforts of Resident Jonathan Duncan)           |
| First university for the disabled | Chitrakoot                                  | <b>Jagadguru Rambhadracharya Handicapped University</b> (India's first) |
| First Urdu Academy                | <b>Lucknow</b>                              | India's first Urdu Academy                                              |

## C. Firsts in History and Politics

| Post / Event                                      | Name / Place                       | Period / Special Description                                      |
|---------------------------------------------------|------------------------------------|-------------------------------------------------------------------|
| First Governor of the United Provinces            | <b>Sir Harcourt Butler</b>         | During 1921-1950                                                  |
| First Governor of Agra                            | Sir C. T. Metcalfe                 | -                                                                 |
| First Lieutenant Governor of the United Provinces | Sir James John Digges<br>La Touche | Agra and Oudh region                                              |
| First Nawab of Awadh                              | <b>Saadat Ali Khan</b>             | Founder of the autonomous state of Awadh                          |
| First session of the Congress in UP               | <b>Allahabad</b>                   | <b>1888</b> (under the presidency of George Yule)                 |
| First Hindi newspaper                             | <b>Banaras Akhbar</b>              | -                                                                 |
| Buddha's first sermon                             | <b>Sarnath</b> (Varanasi)          | In Buddhist history it is called ' <b>Dharmachakrapravartan</b> ' |

D. 'Firsts' in Uttar Pradesh: Awards and Honours

| Category of Award | First Recipient (Male)                                                       | First Recipient (Female)      |
|-------------------|------------------------------------------------------------------------------|-------------------------------|
| Bharat Ratna      | Jawaharlal Nehru and Govind Ballabh Pant (1955)                              | Indira Gandhi (1971)          |
| Padma Vibhushan   | Vijaya Lakshmi Pandit (1962)                                                 | -                             |
| Padma Bhushan     | Amarnath Jha, Shanti Swarup, Maithili Sharan Gupt, S. Narayan Shastri (1954) | Rameshwari Nehru (1955)       |
| Padma Shri        | Akhil Chandra Mitra (1954)                                                   | Mona Chandravati Gupta (1965) |

E. Firsts in Environment and Ecology

| Site / Sanctuary         | Name                                   | Special Fact     |
|--------------------------|----------------------------------------|------------------|
| First Ramsar site        | Upper Ganga River                      | Included in 2005 |
| First National Park      | Dudhwa National Park                   | Lakhimpur Kheri  |
| First wildlife sanctuary | Chandraprabha Wildlife Sanctuary       | Chandauli        |
| First tiger reserve      | Dudhwa Tiger Reserve                   | -                |
| First zoo                | Nawab Wajid Ali Shah Zoological Garden | Lucknow          |

F. Firsts in Infrastructure

| Infrastructure Field        | First Place / Name in Uttar Pradesh | Historical Year / Fact                               |
|-----------------------------|-------------------------------------|------------------------------------------------------|
| First metro service         | Lucknow Metro                       | -                                                    |
| First expressway            | Yamuna Expressway                   | -                                                    |
| First National Highway (NH) | NH-34                               | Starts near Meerut and connects to Loharu in Haryana |
| First railway station       | Kanpur railway station              | 1859                                                 |
| First train operation       | Between Prayagraj and Kanpur        | Run in 1859                                          |
| First biotechnology park    | Bakshi Ka Talab, Lucknow            | -                                                    |
| Asia's first DNA bank       | Biotech Park, Lucknow               | Important at the global level                        |
|                             |                                     |                                                      |

| Infrastructure Field | First Place / Name in Uttar Pradesh | Historical Year / Fact    |
|----------------------|-------------------------------------|---------------------------|
| First Wi-Fi monument | Taj Mahal                           | Wi-Fi facility since 2015 |
| First film centre    | Noida (Gautam Buddh Nagar)          | -                         |

## > Uttar Pradesh: Census (Census 2011)

### A. Principal Statistical Figures

#### > Population

| Category / Indicator   | Total Figures                        | Percentage / Figure of Males | Percentage / Figure of Females |
|------------------------|--------------------------------------|------------------------------|--------------------------------|
| Total population       | 19,98,12,341 (16.51% of the country) | 52.29%                       | 47.71%                         |
| Area-wise population % | -                                    | Rural: 77.73%                | Urban: 22.27%                  |

| Indicator              | UP's Figure / Rate          | Special Note (for UPPCS)                             |
|------------------------|-----------------------------|------------------------------------------------------|
| Population growth rate | 20.23%                      | Decadal growth rate (2001-2011)                      |
| Population density     | 829 persons/km <sup>2</sup> | Considerably higher than the national average (382)  |
| Urbanisation           | 22.3%                       | Rural population is 77.7%                            |
| Sex ratio              | 912                         | Females per 1000 males                               |
| Total literacy rate    | 67.7%                       | Male: 77.3%   Female: 57.2% (the gap is exactly 20%) |
| Area-wise literacy     | -                           | Rural: 65.5%   Urban: 75.1%                          |
| Scheduled Castes (SC)  | 20.7%                       | Share in the total population of the state           |
| Scheduled Tribes (ST)  | 0.6%                        | Share in the total population of the state           |

### B. Decreasing/Increasing Order of Districts (Top 5 District Order)

(Note: > means decreasing order and < means increasing order)

| Category / Indicator | Top 5 Districts (Highest to Lowest)                    | Bottom 5 Districts (Lowest to Highest)                |
|----------------------|--------------------------------------------------------|-------------------------------------------------------|
| Total population     | Prayagraj > Moradabad > Ghaziabad > Azamgarh > Lucknow | Mahoba < Chitrakoot < Hamirpur < Shravasti < Lalitpur |

| Category / Indicator         | Top 5 Districts (Highest to Lowest)                                                                                                                     | Bottom 5 Districts (Lowest to Highest)                                                                                                                                          |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Decadal growth rate</b>   | Gautam Buddh Nagar > Ghaziabad > Shravasti > Bahraich > Balrampur                                                                                       | Kanpur Nagar < Hamirpur < Baghpat < Fatehpur < Deoria                                                                                                                           |
| <b>Population density</b>    | Ghaziabad > Varanasi > Lucknow > Sant Ravidas Nagar > Kanpur Nagar                                                                                      | Lalitpur < Sonbhadra < Hamirpur < Mahoba < Chitrakoot                                                                                                                           |
| <b>Literacy rate</b>         | Gautam Buddh Nagar > Kanpur Nagar > Auraiya > Etawah > Ghaziabad                                                                                        | Shravasti < Bahraich < Balrampur < Budaun < Rampur                                                                                                                              |
| <b>Sex ratio</b>             | Jaunpur > Azamgarh > Deoria > Pratapgarh > Sultanpur                                                                                                    | Gautam Buddh Nagar < Hamirpur/Baghpat < Kanpur Nagar/Banda/Mathura                                                                                                              |
| <b>Rural population</b>      | Prayagraj > Azamgarh > Jaunpur > Sitapur > Gorakhpur                                                                                                    | <i>Minimum rural %:</i> Ghaziabad < Lucknow < Kanpur Nagar < Gautam Buddh Nagar < Meerut                                                                                        |
| <b>Urban population %</b>    | Ghaziabad > Lucknow > Kanpur Nagar > Gautam Buddh Nagar > Meerut                                                                                        | <i>Minimum urban %:</i> Shravasti < Kushinagar < Maharajganj < Sultanpur < Pratapgarh                                                                                           |
| <b>Scheduled Castes (SC)</b> | <i>Population:</i> Sitapur > Prayagraj > Hardoi > Azamgarh > Lakhimpur Kheri<br><i>Percentage (%):</i> Kaushambi > Sitapur > Hardoi > Unnao > Raebareli | <i>Population:</i> Baghpat < Shravasti < Gautam Buddh Nagar < Mahoba < Lalitpur<br><i>Percentage (%):</i> Baghpat < Bareilly < Balrampur < Gautam Buddh Nagar < Rampur/Varanasi |
| <b>Scheduled Tribes (ST)</b> | Sonbhadra > Ballia > Deoria > Kushinagar > Lalitpur                                                                                                     | <i>(Sonbhadra has the highest ST population)</i>                                                                                                                                |

## B. Top Districts with the Highest and Lowest Values (Extreme Points)

| Indicator                  | Highest (Top Districts / Figures)                                      | Lowest (Bottom Districts / Figures)                         |
|----------------------------|------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Total population</b>    | (1) <b>Prayagraj (59,54,351)</b><br>(2) Moradabad (3) Ghaziabad        | (1) <b>Mahoba (8,75,958)</b><br>(2) Chitrakoot (3) Hamirpur |
| <b>Population density</b>  | (1) <b>Ghaziabad (3971)</b><br>(2) Varanasi (3) Lucknow                | (1) <b>Lalitpur (242)</b><br>(2) Sonbhadra (3) Hamirpur     |
| <b>Total literacy rate</b> | (1) <b>Gautam Buddh Nagar (80.12%)</b><br>(2) Kanpur Nagar (3) Auraiya | (1) <b>Shravasti (46.74%)</b><br>(2) Bahraich (3) Balrampur |
|                            |                                                                        |                                                             |



| Indicator           | Highest (Top Districts / Figures) | Lowest (Bottom Districts / Figures)                          |
|---------------------|-----------------------------------|--------------------------------------------------------------|
| Male literacy       | Gautam Buddh Nagar (88.06%)       | Shravasti (57.16%) <i>(the same order as total literacy)</i> |
| Female literacy     | Kanpur Nagar (75.05%)             | Shravasti (34.78%)                                           |
| Decadal growth rate | Gautam Buddh Nagar (49%)          | Kanpur Nagar (9.9%)                                          |
| Sex ratio           | Jaunpur (1024)                    | Gautam Buddh Nagar (851)                                     |

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